

0902.0431: equation evidence

8 rows show a REFINEMENT below, not the OCR reading: 8 verified against the inferred. Each is marked [refined] beside its identifier. The model is unchanged — the original reading is still what ‘latex’ holds; this report chose the refinement for them. Rows: 0902.0431\_EQ0425, 0902.0431\_EQ0515, 0902.0431\_EQ0756, 0902.0431\_EQ0922, 0902.0431\_EQ1032, 0902.0431\_EQ1059 ...

Display equations

| Identifier                                        | Page | Conf. | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                            | Rendered                                                                                                                                                                                                                                                                                                                                                                       | Image                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|---------------------------------------------------|------|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0902.0431_EQ0515 [conf 0.002] [refined: inferred] | 85   | 0.002 | $\begin{aligned} \nu H &= \nu \bigg( \sum_{k=0}^3 \lambda_k H_k \bigg) \\ &= \frac{1}{2} \big( \lambda_0 (-H_0 + H_1 - H_2 + H_3) + \lambda_1 (-H_0 + H_1 + H_2 - H_3) \\ &\quad + \lambda_2 (H_0 + H_1 + H_2 + H_3) + \lambda_3 (-H_0 - H_1 + H_2 + H_3) \big) \\ &= \frac{1}{2} (-\lambda_0 - \lambda_1 + \lambda_2 - \lambda_3) H_0 + \frac{1}{2} (\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3) H_1 \end{aligned}$ | $\nu H = \nu \left( \sum_{k=0}^3 \lambda_k H_k \right)$ $= \frac{1}{2} \left( \lambda_0 (-H_0 + H_1 - H_2 + H_3) + \lambda_1 (-H_0 + H_1 + H_2 - H_3) + \lambda_2 (H_0 + H_1 + H_2 + H_3) + \lambda_3 (-H_0 - H_1 + H_2 + H_3) \right)$ $= \frac{1}{2} (-\lambda_0 - \lambda_1 + \lambda_2 - \lambda_3) H_0 + \frac{1}{2} (\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3) H_1$ | $\begin{array}{cccccccc} \lambda_0 - \lambda_1 & = & 1 & 0 & 0 & 0 & 0 & 0 \\ \lambda_0 - \lambda_2 & = & 1 & 1 & 0 & 0 & 0 & 0 \\ \lambda_0 - \lambda_3 & = & 1 & 1 & 1 & 0 & 0 & 0 \\ \lambda_1 - \lambda_2 & = & 0 & 1 & 0 & 0 & 0 & 0 \\ \lambda_1 - \lambda_3 & = & 0 & 1 & 1 & 0 & 0 & 0 \\ \lambda_2 - \lambda_3 & = & 0 & 0 & 1 & 0 & 0 & 0 \end{array} \quad \begin{array}{cccccccc} \lambda_0 + \lambda_1 & = & 1 & 2 & 2 & 1 & 1 & 1 \\ \lambda_0 + \lambda_2 & = & 1 & 1 & 2 & 1 & 1 & 1 \\ \lambda_0 + \lambda_3 & = & 1 & 1 & 1 & 1 & 1 & 1 \\ \lambda_1 + \lambda_2 & = & 0 & 1 & 2 & 1 & 1 & 1 \\ \lambda_1 + \lambda_3 & = & 0 & 1 & 1 & 1 & 1 & 1 \\ \lambda_2 + \lambda_3 & = & 0 & 0 & 1 & 1 & 1 & 1 \end{array}$ |

| Identifier                        | Page | Conf.  | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Rendered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Image |
|-----------------------------------|------|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 0902.0431_<br>EQ0517 [conf 0.002] | 85   | ■0.002 | $\begin{aligned} & \frac{1}{2} \left( -\lambda_0 - \lambda_1 + \lambda_2 - \lambda_3 \right) + \frac{1}{2} \left( \mu_3 - \mu_1 \right) = 0 \quad \frac{1}{2} \left( \lambda_0 + \lambda_1 + \lambda_2 - \lambda_3 \right) + \frac{1}{2} \left( \mu_3 - \mu_1 \right) = 1 \quad \frac{1}{2} \left( -\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3 \right) + \frac{1}{2} \left( \mu_3 - \mu_1 \right) = 0 \\ & \frac{1}{2} \left( \lambda_0 - \lambda_1 + \lambda_2 + \lambda_3 \right) + \frac{1}{2} \left( \mu_3 - \mu_1 \right) = 1 \quad \frac{1}{2} \left( \lambda_0 + \lambda_1 - \lambda_2 + \lambda_3 \right) + \frac{1}{2} \left( \mu_3 - \mu_1 \right) = 1 \quad \frac{1}{2} \left( -\lambda_0 - \lambda_1 - \lambda_2 + \lambda_3 \right) + \frac{1}{2} \left( \mu_3 - \mu_1 \right) = 0 \\ & \frac{1}{2} \left( \lambda_0 - \lambda_1 - \lambda_2 - \lambda_3 \right) + \frac{1}{2} \left( \mu_3 - \mu_1 \right) = 1 \quad \frac{1}{2} \left( -\lambda_0 + \lambda_1 - \lambda_2 - \lambda_3 \right) + \frac{1}{2} \left( \mu_3 - \mu_1 \right) = 0 \quad \frac{1}{2} \left( \lambda_0 - \lambda_1 + \lambda_2 - \lambda_3 \right) - \frac{1}{2} \left( \mu_1 - \mu_2 \right) = 1 \\ & \frac{1}{2} \left( \lambda_0 - \lambda_1 - \lambda_2 + \lambda_3 \right) - \frac{1}{2} \left( \mu_1 - \mu_2 \right) = 1 \quad \frac{1}{2} \left( \lambda_0 + \lambda_1 + \lambda_2 + \lambda_3 \right) + \frac{1}{2} \left( \mu_1 - \mu_2 \right) = 0 \end{aligned}$ | $\begin{aligned} & \frac{1}{2} \left( -\lambda_0 - \lambda_1 + \lambda_2 - \lambda_3 \right) + \frac{1}{2} \left( \mu_3 - \mu_1 \right) = 0 \quad 0 \quad 1 \quad 0 \quad 1 \quad 0 \\ & \frac{1}{2} \left( \lambda_0 + \lambda_1 + \lambda_2 - \lambda_3 \right) + \frac{1}{2} \left( \mu_3 - \mu_1 \right) = 1 \quad 2 \quad 3 \quad 1 \quad 1 \quad 2 \\ & \frac{1}{2} \left( -\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3 \right) + \frac{1}{2} \left( \mu_3 - \mu_1 \right) = 0 \quad 1 \quad 2 \quad 1 \quad 2 \quad 1 \\ & \frac{1}{2} \left( \lambda_0 - \lambda_1 + \lambda_2 + \lambda_3 \right) + \frac{1}{2} \left( \mu_3 - \mu_1 \right) = 1 \quad 1 \quad 2 \quad 2 \quad 1 \quad 2 \\ & \frac{1}{2} \left( \lambda_0 + \lambda_1 - \lambda_2 + \lambda_3 \right) + \frac{1}{2} \left( \mu_3 - \mu_1 \right) = 1 \quad 2 \quad 2 \quad 1 \quad 2 \quad 1 \\ & \frac{1}{2} \left( -\lambda_0 - \lambda_1 - \lambda_2 + \lambda_3 \right) + \frac{1}{2} \left( \mu_3 - \mu_1 \right) = 0 \quad 0 \quad 0 \quad 0 \quad 1 \quad 0 \\ & \frac{1}{2} \left( \lambda_0 - \lambda_1 - \lambda_2 - \lambda_3 \right) + \frac{1}{2} \left( \mu_3 - \mu_1 \right) = 1 \quad 1 \quad 1 \quad 1 \quad 0 \quad 1 \\ & \frac{1}{2} \left( -\lambda_0 + \lambda_1 - \lambda_2 - \lambda_3 \right) + \frac{1}{2} \left( \mu_3 - \mu_1 \right) = 0 \quad 1 \quad 1 \quad 0 \quad 1 \quad 0 \\ & \frac{1}{2} \left( \lambda_0 - \lambda_1 + \lambda_2 - \lambda_3 \right) - \frac{1}{2} \left( \mu_1 - \mu_2 \right) = 1 \quad 1 \quad 2 \quad 1 \quad 1 \quad 0 \\ & \frac{1}{2} \left( \lambda_0 - \lambda_1 - \lambda_2 + \lambda_3 \right) - \frac{1}{2} \left( \mu_1 - \mu_2 \right) = 1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 0 \\ & \frac{1}{2} \left( \lambda_0 + \lambda_1 + \lambda_2 + \lambda_3 \right) + \frac{1}{2} \left( \mu_1 - \mu_2 \right) = 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 1 \end{aligned}$ |       |

| Identifier                               | Page | Conf.   | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Rendered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Image                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
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| 0902.0431_<br>E01026 <b>[conf 0.018]</b> | 171  | ■ 0.018 | $\begin{array}{l} \backslash left.\backslash begin{array}{\llllllllr\llllll} \backslash lambda_{0}- \\ \backslash lambda_{1}=2 \ \& \ 3 \ \& \ 4 \ \& \ 3 \ \& \ 2 \ \& \ 1 \ \& \ 0 \ \& \ 2 \ \& \ \backslash lambda_{0}+ \\ \backslash lambda_{1}=2 \ \& \ 4 \ \& \ 6 \ \& \ 5 \ \& \ 4 \ \& \ 3 \ \& \ 2 \ \backslash end{array}\backslash right], \\ \backslash begin{array}{\lllllllllll} \backslash lambda_{0}-\backslash lambda_{2}=2 \ \& \ 3 \\ \& \ 4 \ \& \ 3 \ \& \ 2 \ \& \ 1 \ \& \ 1 \ \& \ 2 \ \& \ \backslash lambda_{0}+\backslash lambda_{2}=2 \ \& \ 4 \\ \& \ 6 \ \backslash \backslash \backslash lambda_{0}-\backslash lambda_{3}=2 \ \& \ 3 \ \& \ 4 \ \& \ 3 \ \& \ 2 \ \& \ 2 \ \& \ 1 \\ \& \ 2 \ \& \ \backslash lambda_{0}+\backslash lambda_{3}=2 \ \& \ 4 \ \& \ 6 \ \backslash \backslash \backslash lambda_{1}- \\ \backslash lambda_{2}=0 \ \& \ 0 \ \& \ 0 \ \& \ 0 \ \& \ 0 \ \& \ 0 \ \& \ 1 \ \& \ 0 \ \& \ 4 \ \& \ 2 \ \& \ 1 \\ \backslash \backslash \backslash lambda_{1}-\backslash lambda_{3}=0 \ \& \ 0 \ \& \ 0 \ \& \ 0 \ \& \ 0 \ \& \ 1 \ \& \ 1 \ \& \ 1 \ \& \\ 0 \ \& \ \backslash lambda_{1}+\backslash lambda_{2}=0 \ \& \ 1 \ \& \ 2 \ \backslash \backslash \backslash lambda_{2}- \\ \backslash lambda_{3}=0 \ \& \ 0 \ \& \ 0 \ \& \ 0 \ \& \ 0 \ \& \ 1 \ \& \ 0 \ \& \ 2 \ \& \ 2 \ \& \ 1 \ \backslash \\ \backslash \backslash \backslash lambda_{2}+\backslash lambda_{3}=0 \ \& \ 1 \ \& \ 2 \ \& \ 2 \ \& \ 2 \ \& \ 1 \ \& \ 0 \ \& \ 1 \\ \backslash end{array} \end{array}$                                                                                                                                                                                                                                                                                                        | $\begin{array}{l} \lambda_0 - \lambda_1 = 2 \ 3 \ 4 \ 3 \ 2 \ 1 \ 0 \ 2 \ \lambda_0 + \lambda_1 = 2 \ 4 \ 6 \ 5 \ 4 \ 3 \ 2 \\ \lambda_0 - \lambda_2 = 2 \ 3 \ 4 \ 3 \ 2 \ 1 \ 1 \ 2 \ \lambda_0 + \lambda_2 = 2 \ 4 \ 6 \\ \lambda_0 - \lambda_3 = 2 \ 3 \ 4 \ 3 \ 2 \ 2 \ 1 \ 2 \ \lambda_0 + \lambda_3 = 2 \ 4 \ 6 \\ \lambda_1 - \lambda_2 = 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 1 \ 0 \ \lambda_1 + \lambda_2 = 0 \ 1 \ 2 \\ \lambda_1 - \lambda_3 = 0 \ 0 \ 0 \ 0 \ 0 \ 1 \ 1 \ 0 \ \lambda_1 + \lambda_3 = 0 \ 1 \ 2 \\ \lambda_2 - \lambda_3 = 0 \ 0 \ 0 \ 0 \ 0 \ 1 \ 0 \ 0 \ \lambda_2 + \lambda_3 = 0 \ 1 \ 2 \end{array}$                                                                 | $\begin{array}{l} \lambda_0 - \lambda_1 = 2 \ 3 \ 4 \ 3 \ 2 \ 1 \ 0 \ 2 \ \lambda_0 + \lambda_1 = 2 \ 4 \ 6 \ 5 \ 4 \ 3 \ 2 \ 3 \\ \lambda_0 - \lambda_2 = 2 \ 3 \ 4 \ 3 \ 2 \ 1 \ 1 \ 2 \ \lambda_0 + \lambda_2 = 2 \ 4 \ 6 \ 5 \ 4 \ 3 \ 1 \ 3 \\ \lambda_0 - \lambda_3 = 2 \ 3 \ 4 \ 3 \ 2 \ 2 \ 1 \ 2 \ \lambda_0 + \lambda_3 = 2 \ 4 \ 6 \ 5 \ 4 \ 2 \ 1 \ 3 \\ \lambda_1 - \lambda_2 = 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 1 \ 0 \ \lambda_1 + \lambda_2 = 0 \ 1 \ 2 \ 2 \ 2 \ 2 \ 1 \ 1 \\ \lambda_1 - \lambda_3 = 0 \ 0 \ 0 \ 0 \ 0 \ 1 \ 1 \ 0 \ \lambda_1 + \lambda_3 = 0 \ 1 \ 2 \ 2 \ 2 \ 1 \ 1 \ 1 \\ \lambda_2 - \lambda_3 = 0 \ 0 \ 0 \ 0 \ 0 \ 1 \ 0 \ 0 \ \lambda_2 + \lambda_3 = 0 \ 1 \ 2 \ 2 \ 2 \ 1 \ 0 \ 1 \end{array}$  |
| 0902.0431_<br>E00518 <b>[conf 0.021]</b> | 86   | ■ 0.021 | $\begin{array}{l} \backslash begin{aligned} \ \& \ \frac{1}{2}\backslash left(\backslash lambda_{0}+ \\ \backslash lambda_{1}-\backslash lambda_{2}-\backslash lambda_{3}\backslash right)-\frac{1}{2}\backslash left(\backslash mu_{1}-\backslash mu_{2}\backslash right)=1 \quad 2 \quad 2 \quad \backslash quad \\ 2 \quad \backslash quad 1 \quad \backslash quad 1 \quad \backslash quad c \quad \backslash \backslash \ \& \ \frac{1}{2}\backslash left(- \\ \backslash lambda_{0}+\backslash lambda_{1}-\backslash lambda_{2}+\backslash lambda_{3}\backslash right)- \\ \frac{1}{2}\backslash left(\backslash mu_{1}-\backslash mu_{2}\backslash right)=0 \quad \backslash quad 1 \\ \backslash quad 1 \quad \backslash quad 1 \quad \backslash quad 1 \quad \backslash quad 1 \quad \backslash \backslash \ \& \ \frac{1}{2}\backslash left(- \\ \backslash lambda_{0}+\backslash lambda_{1}+\backslash lambda_{2}-\backslash lambda_{3}\backslash right)- \\ \frac{1}{2}\backslash left(\backslash mu_{1}-\backslash mu_{2}\backslash right)=0 \quad \backslash quad 1 \\ \backslash quad 2 \quad \backslash quad 1 \quad \backslash quad 1 \quad \backslash quad 1 \quad \backslash \backslash \ \& \ \frac{1}{2}\backslash left( \\ \backslash lambda_{0}+\backslash lambda_{1}+\backslash lambda_{2}+\backslash lambda_{3}\backslash right)- \\ \frac{1}{2}\backslash left(\backslash mu_{1}-\backslash mu_{2}\backslash right)=1 \quad \backslash quad 2 \quad \backslash quad 3 \\ \backslash quad 2 \quad \backslash \backslash \ \& \ \frac{1}{2}\backslash left(-\backslash lambda_{0}-\backslash lambda_{1}+ \\ \backslash lambda_{2}+\backslash lambda_{3}\backslash right)-\frac{1}{2}\backslash left(\backslash mu_{1}- \\ \backslash mu_{2}\backslash right)=0 \quad \backslash quad 0 \quad \backslash end{aligned} \end{array}$ | $\begin{array}{l} \frac{1}{2}(\lambda_0 + \lambda_1 - \lambda_2 - \lambda_3) - \frac{1}{2}(\mu_1 - \mu_2) = 1 \quad 2 \quad 2 \quad 2 \quad 1 \quad 1 \quad c \\ \frac{1}{2}(-\lambda_0 + \lambda_1 - \lambda_2 + \lambda_3) - \frac{1}{2}(\mu_1 - \mu_2) = 0 \quad 1 \quad 1 \quad 1 \quad 1 \quad 1 \\ \frac{1}{2}(-\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3) - \frac{1}{2}(\mu_1 - \mu_2) = 0 \quad 1 \quad 2 \quad 1 \quad 1 \quad 1 \\ \frac{1}{2}(\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3) - \frac{1}{2}(\mu_1 - \mu_2) = 1 \quad 2 \quad 3 \quad 2 \\ \frac{1}{2}(-\lambda_0 - \lambda_1 + \lambda_2 + \lambda_3) - \frac{1}{2}(\mu_1 - \mu_2) = 0 \quad 0 \end{array}$ | $\begin{array}{l} \frac{1}{2}(\lambda_0 + \lambda_1 - \lambda_2 - \lambda_3) - \frac{1}{2}(\mu_1 - \mu_2) = 1 \quad 2 \quad 2 \quad 1 \quad 1 \quad 0 \\ \frac{1}{2}(-\lambda_0 + \lambda_1 - \lambda_2 + \lambda_3) - \frac{1}{2}(\mu_1 - \mu_2) = 0 \quad 1 \quad 1 \quad 1 \quad 1 \quad 0 \\ \frac{1}{2}(-\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3) - \frac{1}{2}(\mu_1 - \mu_2) = 0 \quad 1 \quad 2 \quad 1 \quad 1 \quad 0 \\ \frac{1}{2}(\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3) - \frac{1}{2}(\mu_1 - \mu_2) = 1 \quad 2 \quad 3 \quad 2 \quad 2 \quad 1 \\ \frac{1}{2}(-\lambda_0 - \lambda_1 + \lambda_2 + \lambda_3) - \frac{1}{2}(\mu_1 - \mu_2) = 0 \quad 0 \quad 1 \quad 1 \quad 1 \quad 0. \end{array}$ |

| Identifier                                                                                            | Page | Conf.            | LaTeX source                                                                                                                                                                                                                                                                                                                   | Rendered                                                                                                                                                                                                                                                                              | Image                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-------------------------------------------------------------------------------------------------------|------|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div>0902.0431_</div> <div>EQ1032</div> <div>0.022</div> <div>[conf<br/>[refined:<br/>inferred]</div> | 173  | <div>0.022</div> | $\begin{aligned} \kappa\pi H &= \tfrac{1}{2}(-\lambda_0 + \lambda_1 - \lambda_2 + \lambda_3)H_0 + \tfrac{1}{2}(-\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3)H_1 \\ &\quad + \tfrac{1}{2}(\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3)H_2 + \tfrac{1}{2}(-\lambda_0 - \lambda_1 + \lambda_2 + \lambda_3)H_3, \end{aligned}$ | $\kappa\pi H = \tfrac{1}{2}(-\lambda_0 + \lambda_1 - \lambda_2 + \lambda_3)H_0 + \tfrac{1}{2}(-\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3)H_1 + \tfrac{1}{2}(\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3)H_2 + \tfrac{1}{2}(-\lambda_0 - \lambda_1 + \lambda_2 + \lambda_3)H_3,$ | $\begin{aligned} \frac{1}{2}(\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3) + \frac{1}{2}\mu_2 - \frac{2}{3}\nu &= 1 & 2 & 3 & 3 & 2 & 2 & 1 & 1 \\ \frac{1}{2}(\lambda_0 + \lambda_1 - \lambda_2 + \lambda_3) + \frac{1}{2}\mu_2 - \frac{2}{3}\nu &= 1 & 2 & 3 & 3 & 2 & 1 & 1 & 1 \\ \frac{1}{2}(\lambda_0 - \lambda_1 + \lambda_2 + \lambda_3) + \frac{1}{2}\mu_2 - \frac{2}{3}\nu &= 1 & 2 & 3 & 3 & 2 & 1 & 0 & 1 \\ \frac{1}{2}(\lambda_0 - \lambda_1 - \lambda_2 - \lambda_3) + \frac{1}{2}\mu_2 - \frac{2}{3}\nu &= 1 & 1 & 1 & 1 & 0 & 0 & 0 & 0 \\ \frac{1}{2}(\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3) - \frac{1}{2}\mu_2 + \frac{2}{3}\nu &= 1 & 2 & 3 & 2 & 2 & 2 & 1 & 2 \\ \frac{1}{2}(\lambda_0 + \lambda_1 - \lambda_2 + \lambda_3) - \frac{1}{2}\mu_2 + \frac{2}{3}\nu &= 1 & 2 & 3 & 2 & 2 & 1 & 1 & 2 \\ \frac{1}{2}(\lambda_0 - \lambda_1 + \lambda_2 + \lambda_3) - \frac{1}{2}\mu_2 - \frac{2}{3}\nu &= 1 & 2 & 3 & 2 & 2 & 1 & 0 & 2 \\ \frac{1}{2}(\lambda_0 - \lambda_1 - \lambda_2 - \lambda_3) - \frac{1}{2}\mu_2 + \frac{2}{3}\nu &= 1 & 1 & 1 & 0 & 0 & 0 & 0 & 1 \\ \frac{1}{2}(\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3) + \frac{1}{2}\mu_3 - \frac{2}{3}\nu &= 1 & 2 & 3 & 3 & 3 & 2 & 1 & 1 \\ \frac{1}{2}(\lambda_0 + \lambda_1 - \lambda_2 - \lambda_3) + \frac{1}{2}\mu_3 - \frac{2}{3}\nu &= 1 & 1 & 1 & 1 & 1 & 1 & 1 & 0 \\ \frac{1}{2}(\lambda_0 - \lambda_1 + \lambda_2 - \lambda_3) + \frac{1}{2}\mu_3 - \frac{2}{3}\nu &= 1 & 1 & 1 & 1 & 1 & 1 & 0 & 0 \\ \frac{1}{2}(\lambda_0 - \lambda_1 - \lambda_2 + \lambda_3) + \frac{1}{2}\mu_3 - \frac{2}{3}\nu &= 1 & 1 & 1 & 1 & 1 & 0 & 0 & 0 \\ \frac{1}{2}(\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3) - \frac{1}{2}\mu_3 + \frac{2}{3}\nu &= 1 & 3 & 5 & 4 & 3 & 2 & 1 & 3 \\ \frac{1}{2}(\lambda_0 + \lambda_1 - \lambda_2 - \lambda_3) - \frac{1}{2}\mu_3 + \frac{2}{3}\nu &= 1 & 2 & 3 & 2 & 1 & 1 & 1 & 2 \\ \frac{1}{2}(\lambda_0 - \lambda_1 + \lambda_2 - \lambda_3) - \frac{1}{2}\mu_3 + \frac{2}{3}\nu &= 1 & 2 & 3 & 2 & 1 & 1 & 0 & 2 \\ \frac{1}{2}(\lambda_0 - \lambda_1 - \lambda_2 + \lambda_3) - \frac{1}{2}\mu_3 + \frac{2}{3}\nu &= 1 & 2 & 3 & 2 & 1 & 0 & 0 & 2 \end{aligned}$ |

| Identifier                        | Page | Conf. | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        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| 0902.0431_<br>E01033 [conf 0.026] | 174  | 0.026 | $\begin{array}{llllllll} \backslash\mathrm{begin}\{\mathrm{array}\}\{\mathrm{llllllll}\} \backslash\mathrm{lambda}_{\mathrm{0}}+\backslash\mathrm{frac}\{\mathrm{1}\}\{\mathrm{2}\} \backslash\mathrm{mu}_{\mathrm{1}}+ \\ \backslash\mathrm{frac}\{\mathrm{1}\}\{\mathrm{3}\} \backslash\mathrm{nu}+r=2 \& 4 \& 6 \& 4 \& 3 \& 2 \& 1 \& 3 \& \backslash\backslash \\ \backslash\mathrm{lambda}_{\mathrm{0}}+\backslash\mathrm{frac}\{\mathrm{1}\}\{\mathrm{2}\} \backslash\mathrm{mu}_{\mathrm{1}}+\backslash\mathrm{frac}\{\mathrm{1}\}\{\mathrm{3}\} \backslash\mathrm{nu}-r=2 \& \\ 4 \& 5 \& 4 \& 3 \& 2 \& 1 \& 3 \& \backslash\backslash \backslash\mathrm{lambda}_{\mathrm{0}}-\backslash\mathrm{frac}\{\mathrm{1}\}\{\mathrm{2}\} \\ \backslash\mathrm{mu}_{\mathrm{1}}-\backslash\mathrm{frac}\{\mathrm{1}\}\{\mathrm{3}\} \backslash\mathrm{nu}+r=2 \& 3 \& 5 \& 4 \& 3 \& 2 \& 1 \& 2 \\ \backslash\backslash 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\backslash\mathrm{mu}_{\mathrm{1}}-\backslash\mathrm{frac}\{\mathrm{1}\}\{\mathrm{3}\} \backslash\mathrm{nu}+r=0 \& 0 \& 1 \& 1 \& 1 \& 0 \& 0 \& 0 \\ \backslash\backslash \backslash\mathrm{lambda}_{\mathrm{2}}-\backslash\mathrm{frac}\{\mathrm{1}\}\{\mathrm{2}\} \backslash\mathrm{mu}_{\mathrm{1}}-\backslash\mathrm{frac}\{\mathrm{1}\}\{\mathrm{3}\} \backslash\mathrm{nu}-r=0 \\ \& 0 \& 0 \& 1 \& 1 \& 0 \& 0 \& 0 \backslash\backslash \backslash\mathrm{lambda}_{\mathrm{3}}+\backslash\mathrm{frac}\{\mathrm{1}\}\{\mathrm{2}\} \\ \backslash\mathrm{mu}_{\mathrm{1}}+\backslash\mathrm{frac}\{\mathrm{1}\}\{\mathrm{3}\} \backslash\mathrm{nu}+r=0 \& 1 \& 2 \& 1 \& 1 \& 0 \& 0 \& 1 \\ \backslash\backslash \backslash\mathrm{lambda}_{\mathrm{3}}+\backslash\mathrm{frac}\{\mathrm{1}\}\{\mathrm{2}\} \backslash\mathrm{mu}_{\mathrm{1}}+\backslash\mathrm{frac}\{\mathrm{1}\}\{\mathrm{3}\} \backslash\mathrm{nu}-r=0 \\ \& 1 \& 1 \& 1 \& 1 \& 0 \& 0 \& 1 \backslash\backslash \backslash\mathrm{lambda}_{\mathrm{3}}-\backslash\mathrm{frac}\{\mathrm{1}\}\{\mathrm{2}\} \\ \backslash\mathrm{mu}_{\mathrm{1}}-\backslash\mathrm{frac}\{\mathrm{1}\}\{\mathrm{3}\} \backslash\mathrm{nu}+r=0 \& 0 \& 1 \& 1 \& 1 \& 0 \& 0 \& 0 \\ \backslash\backslash \backslash\mathrm{lambda}_{\mathrm{3}}-\backslash\mathrm{frac}\{\mathrm{1}\}\{\mathrm{2}\} \backslash\mathrm{mu}_{\mathrm{1}}-\backslash\mathrm{frac}\{\mathrm{1}\}\{\mathrm{3}\} \backslash\mathrm{nu}-r=0 \\ \& 0 \& 0 \& 1 \& 1 \& 0 \& 0 \& 0 \backslash\backslash \backslash\mathrm{lambda}_{\mathrm{2}}-\backslash\mathrm{frac}\{\mathrm{1}\}\{\mathrm{2}\} \\ \backslash\mathrm{mu}_{\mathrm{1}}-\backslash\mathrm{frac}\{\mathrm{1}\}\{\mathrm{3}\} \backslash\mathrm{nu}+r=0 \& 0 \& 1 \& 1 \& 1 \& 0 \& 0 \& 0 \\ \backslash\backslash \backslash\mathrm{lambda}_{\mathrm{2}}-\backslash\mathrm{frac}\{\mathrm{1}\}\{\mathrm{2}\} \backslash\mathrm{mu}_{\mathrm{1}}-\backslash\mathrm{frac}\{\mathrm{1}\}\{\mathrm{3}\} \backslash\mathrm{nu}-r=0 \\ \& 0 \& 0 \& 1 \& 1 \& 0 \& 0 \& 0 \backslash\backslash \backslash\mathrm{lambda}_{\mathrm{3}}+\backslash\mathrm{frac}\{\mathrm{1}\}\{\mathrm{2}\} \\ \backslash\mathrm{mu}_{\mathrm{1}}+\backslash\mathrm{frac}\{\mathrm{1}\}\{\mathrm{3}\} \backslash\mathrm{nu}+r=0 \& 1 \& 2 \& 1 \& 1 \& 0 \& 0 \& 1 \\ \backslash\backslash \backslash\mathrm{lambda}_{\mathrm{3}}+\backslash\mathrm{frac}\{\mathrm{1}\}\{\mathrm{2}\} \backslash\mathrm{mu}_{\mathrm{1}}+\backslash\mathrm{frac}\{\mathrm{1}\}\{\mathrm{3}\} \backslash\mathrm{nu}-r=0 \\ \& 1 \& 1 \& 1 \& 1 \& 0 \& 0 \& 1 \backslash\backslash \backslash\mathrm{lambda}_{\mathrm{3}}-\backslash\mathrm{frac}\{\mathrm{1}\}\{\mathrm{2}\} \\ \backslash\mathrm{mu}_{\mathrm{1}}-\backslash\mathrm{frac}\{\mathrm{1}\}\{\mathrm{3}\} \backslash\mathrm{nu}+r=0 \& 0 \& 1 \& 1 \& 1 \& 0 \& 0 \& 0 \\ \backslash\backslash \backslash\mathrm{lambda}_{\mathrm{3}}-\backslash\mathrm{frac}\{\mathrm{1}\}\{\mathrm{2}\} \backslash\mathrm{mu}_{\mathrm{1}}-\backslash\mathrm{frac}\{\mathrm{1}\}\{\mathrm{3}\} \backslash\mathrm{nu}-r=0 \\ \& 0 \& 0 \& 1 \& 1 \& 0 \& 0 \& 0 \backslash\backslash \backslash\mathrm{lambda}_{\mathrm{2}}-\backslash\mathrm{frac}\{\mathrm{1}\}\{\mathrm{2}\} \\ \backslash\mathrm{mu}_{\mathrm{1}}-\backslash\mathrm{frac}\{\mathrm{1}\}\{\mathrm{3}\} \backslash\mathrm{nu}+r=0 \& 0 \& 1 \& 1 \& 1 \& 0 \& 0 \& 0 \\ \backslash\backslash \backslash\mathrm{lambda}_{\mathrm{2}}-\backslash\mathrm{frac}\{\mathrm{1}\}\{\mathrm{2}\} \backslash\mathrm{mu}_{\mathrm{1}}-\backslash\mathrm{frac}\{\mathrm{1}\}\{\mathrm{3}\} \backslash\mathrm{nu}-r=0 \\ \& 0 \& 0 \& 1 \& 1 \& 0 \& 0 \& 0 \backslash\backslash \backslash\mathrm{lambda}_{\mathrm{3}}+\backslash\mathrm{frac}\{\mathrm{1}\}\{\mathrm{2}\} \\ \backslash\mathrm{mu}_{\mathrm{1}}+\backslash\mathrm{frac}\{\mathrm{1}\}\{\mathrm{3}\} \backslash\mathrm{nu}+r=0 \& 1 \& 2 \& 1 \& 1 \& 0 \& 0 \& 1 \\ \backslash\backslash \backslash\mathrm{lambda}_{\mathrm{3}}+\backslash\mathrm{frac}\{\mathrm{1}\}\{\mathrm{2}\} \backslash\mathrm{mu}_{\mathrm{1}}+\backslash\mathrm{frac}\{\mathrm{1}\}\{\mathrm{3}\} \backslash\mathrm{nu}-r=0 \\ \& 1 \& 1 \& 1 \& 1 \& 0 \& 0 \& 1 \backslash\backslash \backslash\mathrm{lambda}_{\mathrm{3}}-\backslash\mathrm{frac}\{\mathrm{1}\}\{\mathrm{2}\} \\ \backslash\mathrm{mu}_{\mathrm{1}}-\backslash\mathrm{frac}\{\mathrm{1}\}\{\mathrm{3}\} \backslash\mathrm{nu}+r=0 \& 0 \& 1 \& 1 \& 1 \& 0 \& 0 \& 0 \\ \backslash\backslash \backslash\mathrm{lambda}_{\mathrm{3}}-\backslash\mathrm{frac}\{\mathrm{1}\}\{\mathrm{2}\} \backslash\mathrm{mu}_{\mathrm{1}}-\backslash\mathrm{frac}\{\mathrm{1}\}\{\mathrm{3}\} \backslash\mathrm{nu}-r=0 \\ \& 0 \& 0 \& 1 \& 1 \& 0 \& 0 \& 0 \backslash\backslash \backslash\mathrm{lambda}_{\mathrm{2}}-\backslash\mathrm{frac}\{\mathrm{1}\}\{\mathrm{2}\} \\ \backslash\mathrm{mu}_{\mathrm{1}}-\backslash\mathrm{frac}\{\mathrm{1}\}\{\mathrm{3}\} \backslash\mathrm{nu}+r=0 \& 0 \& 1 \& 1 \& 1 \& 0 \& 0 \& 0 \\ \backslash\backslash \backslash\mathrm{lambda}_{\mathrm{2}}-\backslash\mathrm{frac}\{\mathrm{1}\}\{\mathrm{2}\} \backslash\mathrm{mu}_{\mathrm{1}}-\backslash\mathrm{frac}\{\mathrm{1}\}\{\mathrm{3}\} \backslash\mathrm{nu}-r=0 \\ \& 0 \& 0 \& 1 \& 1 \& 0 \& 0 \& 0 \backslash\backslash \backslash\mathrm{lambda}_{\mathrm{3}}+\backslash\mathrm{frac}\{\mathrm{1}\}\{\mathrm{2}\} \\ \backslash\mathrm{mu}_{\mathrm{1}}+\$ |          |       |

| Identifier                               | Page | Conf.   | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Rendered                                                                                                                                                                                                                                                                                                                                                                        | Image                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|------------------------------------------|------|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0902.0431_<br>EQ1031 <b>[conf 0.036]</b> | 173  | ■ 0.036 | $\begin{aligned} & \backslash \text{begin}\{\text{aligned}\} \ \& \ \backslash \text{lambda}_{\{2\}}+\backslash \text{frac}\{1\}{\{2\}} \ \backslash \text{mu}_{\{1\}}- \\ & \backslash \text{frac}\{2\}{\{3\}} \ \backslash \text{nu}=0 \ \backslash \text{quad} \ 1 \ \backslash \text{quad} \ 1 \ \backslash \text{quad} \ 1 \ \backslash \text{quad} \ 1 \ \backslash \text{quad} \ 1 \ \backslash \text{quad} \ 0 \ \backslash \text{quad} \ 0 \ \backslash \backslash \ \& \ \backslash \text{lambda}_{\{2\}}-\backslash \text{frac}\{1\}{\{2\}} \ \backslash \text{mu}_{\{1\}}+ \\ & \backslash \text{frac}\{2\}{\{3\}} \ \backslash \text{nu}=0 \ \backslash \text{quad} \ 0 \ \backslash \text{quad} \ 1 \ \backslash \text{quad} \ 1 \ \backslash \text{quad} \ 1 \ \backslash \text{quad} \ 1 \ \backslash \text{quad} \ 0 \ \backslash \backslash \ \& \ \backslash \text{lambda}_{\{3\}}+\backslash \text{frac}\{1\}{\{2\}} \ \backslash \text{mu}_{\{1\}}-\backslash \text{frac}\{2\}{\{3\}} \\ & \backslash \text{nu}=0 \ \backslash \text{quad} \ 1 \ \backslash \text{quad} \ 1 \ \backslash \text{quad} \ 1 \ \backslash \text{quad} \ 1 \ \backslash \backslash \ \& \ \backslash \text{lambda}_{\{3\}}- \\ & \backslash \text{frac}\{1\}{\{2\}} \ \backslash \text{mu}_{\{1\}}+\backslash \text{frac}\{2\}{\{3\}} \ \backslash \text{nu}=0 \ \backslash \text{quad} \ 0 \ \backslash \text{quad} \ 1 \\ & \backslash \text{quad} \ 1 \ \backslash \text{quad} \ 1 \ \backslash \text{quad} \ 0 \ \backslash \backslash \ \& \ 0 \ \backslash \text{end}\{\text{aligned}\} \end{aligned}$ | $\begin{aligned} \lambda_2 + \frac{1}{2}\mu_1 - \frac{2}{3}\nu &= 0 & 1 & 1 & 1 & 1 & 1 & 0 & 0 \\ \lambda_2 - \frac{1}{2}\mu_1 + \frac{2}{3}\nu &= 0 & 0 & 1 & 1 & 1 & 1 & 1 & 0 \\ \lambda_3 + \frac{1}{2}\mu_1 - \frac{2}{3}\nu &= 0 & 1 & 1 & 1 & 1 & & & \\ \lambda_3 - \frac{1}{2}\mu_1 + \frac{2}{3}\nu &= 0 & 0 & 1 & 1 & 1 & 0 & & \\ 0 & & & & & & & & \end{aligned}$ | $\begin{aligned} \lambda_2 + \frac{1}{2}\mu_1 - \frac{2}{3}\nu &= 0 & 1 & 1 & 1 & 1 & 1 & 0 & 0 \\ \lambda_2 - \frac{1}{2}\mu_1 + \frac{2}{3}\nu &= 0 & 0 & 1 & 1 & 1 & 1 & 0 & 1 \\ \lambda_3 + \frac{1}{2}\mu_1 - \frac{2}{3}\nu &= 0 & 1 & 1 & 1 & 1 & 0 & 0 & 0 \\ \lambda_3 - \frac{1}{2}\mu_1 + \frac{2}{3}\nu &= 0 & 0 & 1 & 1 & 1 & 0 & 0 & 1 \end{aligned}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 0902.0431_<br>EQ0761 <b>[conf 0.036]</b> | 127  | ■ 0.036 | $\begin{aligned} & \backslash \text{begin}\{\text{aligned}\} \ \& \ \backslash \text{frac}\{1\}{\{2\}}\backslash \text{left}(-\backslash \text{lambda}_{\{0\}}- \\ & \backslash \text{lambda}_{\{1\}}+\backslash \text{lambda}_{\{2\}}-\backslash \text{lambda}_{\{3\}}\backslash \text{right})+\backslash \text{frac}\{1\}{\{2\}} \\ & \backslash \text{mu}_{\{2\}}-\backslash \text{frac}\{2\}{\{3\}} \ \backslash \text{nu}=0 \ \backslash \text{quad} \ 0 \ \backslash \text{quad} \ 1 \ \backslash \text{quad} \ 1 \\ & \backslash \text{quad} \ \textbf{[U+0930]} \ \textbf{[U+094D]} \ 0 \ \backslash \text{quad} \ 0 \ \backslash \backslash \ \& \ \backslash \text{frac}\{1\}{\{2\}}\backslash \text{left}(\backslash \text{lambda}_{\{0\}}+ \\ & \backslash \text{lambda}_{\{1\}}+\backslash \text{lambda}_{\{2\}}-\backslash \text{lambda}_{\{3\}}\backslash \text{right})+ \\ & \backslash \text{frac}\{1\}{\{2\}} \ \backslash \text{mu}_{\{2\}}-\backslash \text{frac}\{2\}{\{3\}} \ \backslash \text{nu}=1 \ \backslash \text{quad} \ 2 \ \backslash \text{quad} \ 3 \\ & \backslash \text{quad} \ 3 \ \backslash \text{quad} \ 2 \ \backslash \text{end}\{\text{aligned}\} \ \backslash \text{quad} \ 1 \ \backslash \text{quad} \ 2 \ . \end{aligned}$                                                                                                                                                                                                                                                                                                                                     | $\begin{aligned} \frac{1}{2}(-\lambda_0 - \lambda_1 + \lambda_2 - \lambda_3) + \frac{1}{2}\mu_2 - \frac{2}{3}\nu &= 0 & 0 & 1 & 1 & \textbf{[U+0930][U+094D]} & 0 & 0 & \\ \frac{1}{2}(\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3) + \frac{1}{2}\mu_2 - \frac{2}{3}\nu &= 1 & 2 & 3 & 3 & 2 & 1 & 2. & \end{aligned}$                                                        | $\begin{aligned} \frac{1}{2}(-\lambda_0 - \lambda_1 + \lambda_2 - \lambda_3) + \frac{1}{2}\mu_2 - \frac{2}{3}\nu &= 0 & 0 & 1 & 1 & 1 & 0 & 0 & 1 \\ \frac{1}{2}(\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3) + \frac{1}{2}\mu_2 - \frac{2}{3}\nu &= 1 & 2 & 3 & 3 & 2 & 1 & 2 \\ \frac{1}{2}(-\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3) + \frac{1}{2}\mu_2 - \frac{2}{3}\nu &= 0 & 1 & 2 & 3 & 2 & 1 & 2 \\ \frac{1}{2}(\lambda_0 - \lambda_1 + \lambda_2 + \lambda_3) + \frac{1}{2}\mu_2 - \frac{2}{3}\nu &= 1 & 1 & 2 & 3 & 2 & 1 & 2 \\ \frac{1}{2}(\lambda_0 + \lambda_1 - \lambda_2 + \lambda_3) + \frac{1}{2}\mu_2 - \frac{2}{3}\nu &= 1 & 2 & 2 & 3 & 2 & 1 & 2 \\ \frac{1}{2}(-\lambda_0 - \lambda_1 - \lambda_2 + \lambda_3) + \frac{1}{2}\mu_2 - \frac{2}{3}\nu &= 0 & 0 & 0 & 1 & 0 & 0 & 1 \\ \frac{1}{2}(\lambda_0 - \lambda_1 - \lambda_2 - \lambda_3) + \frac{1}{2}\mu_2 - \frac{2}{3}\nu &= 1 & 1 & 1 & 1 & 0 & 0 & 1 \\ \frac{1}{2}(-\lambda_0 + \lambda_1 - \lambda_2 - \lambda_3) + \frac{1}{2}\mu_2 - \frac{2}{3}\nu &= 0 & 1 & 1 & 1 & 0 & 0 & 1 \\ \frac{1}{2}(\lambda_0 - \lambda_1 + \lambda_2 - \lambda_3) + \frac{1}{2}\mu_3 - \frac{2}{3}\nu &= 1 & 1 & 2 & 2 & 1 & 0 & 1 \\ \frac{1}{2}(\lambda_0 - \lambda_1 - \lambda_2 + \lambda_3) + \frac{1}{2}\mu_3 + \frac{2}{3}\nu &= 1 & 1 & 1 & 2 & 1 & 0 & 1 \\ \frac{1}{2}(\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3) - \frac{1}{2}\mu_3 - \frac{2}{3}\nu &= 0 & 0 & 0 & 0 & 1 & 1 & 0 \\ \frac{1}{2}(\lambda_0 + \lambda_1 - \lambda_2 - \lambda_3) + \frac{1}{2}\mu_3 - \frac{2}{3}\nu &= 1 & 2 & 2 & 2 & 1 & 0 & 1 \\ \frac{1}{2}(-\lambda_0 + \lambda_1 - \lambda_2 + \lambda_3) + \frac{1}{2}\mu_3 - \frac{2}{3}\nu &= 0 & 1 & 1 & 2 & 1 & 0 & 1 \\ \frac{1}{2}(-\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3) + \frac{1}{2}\mu_3 - \frac{2}{3}\nu &= 0 & 1 & 2 & 2 & 1 & 0 & 1 \\ \frac{1}{2}(\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3) + \frac{1}{2}\mu_3 - \frac{2}{3}\nu &= 1 & 2 & 3 & 4 & 3 & 1 & 2 \\ \frac{1}{2}(-\lambda_0 - \lambda_1 + \lambda_2 + \lambda_3) + \frac{1}{2}\mu_3 - \frac{2}{3}\nu &= 0 & 0 & 1 & 2 & 1 & 0 & 1. \end{aligned}$ |



| Identifier                               | Page | Conf.   | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Rendered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Image |
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| 0902.0431_<br>EQ0757 [conf 0.095]        | 126  | ■ 0.095 | $\begin{aligned} & \lambda_0 - \frac{1}{2}(\mu_2 - \mu_3) = 1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 0 \quad 0 \\ & \lambda_1 - \frac{1}{2}(\mu_2 - \mu_3) = 0 \quad 1 \quad 1 \quad 1 \quad 1 \quad 1 \quad a \\ & \lambda_2 - \frac{1}{2}(\mu_2 - \mu_3) = 0 \quad 0 \quad 1 \quad 1 \quad 1 \\ & \lambda_3 - \frac{1}{2}(\mu_2 - \mu_3) = 0 \quad 0 \quad 0 \\ & 0 \end{aligned}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | $\begin{aligned} \lambda_0 - \frac{1}{2}(\mu_2 - \mu_3) &= 1 & 1 & 1 & 1 & 1 & 0 & 0 \\ \lambda_1 - \frac{1}{2}(\mu_2 - \mu_3) &= 0 & 1 & 1 & 1 & 1 & 0 & 0 \\ \lambda_2 - \frac{1}{2}(\mu_2 - \mu_3) &= 0 & 0 & 1 & 1 & 1 & 0 & 0 \\ \lambda_3 - \frac{1}{2}(\mu_2 - \mu_3) &= 0 & 0 & 0 & 1 & 1 & 0 & 0 \end{aligned}$                                                                                                                                                                                                                                                                                                                                       |       |
| 0902.0431_<br>EQ0922 [refined: inferred] | 153  | ■ 0.102 | $\begin{aligned} & f^{-1} \left( \sum_{i < j < k} x_{ijk} \mathbf{e}_i \wedge \mathbf{e}_j \wedge \mathbf{e}_k \right) = \begin{pmatrix} \begin{pmatrix} h(x_{156}) & h(x_{164}, \bar{x}_{256}) & h(x_{145}, \bar{x}_{356}) \\ h(x_{256}, \bar{x}_{164}) & h(x_{264}) & h(x_{245}, \bar{x}_{364}) \\ h(x_{356}, \bar{x}_{145}) & h(x_{364}, \bar{x}_{245}) & h(x_{345}) \\ h(x_{423}) & h(x_{431}, \bar{x}_{523}) & h(x_{412}, \bar{x}_{623}) \\ h(x_{523}, \bar{x}_{431}) & h(x_{531}) & h(x_{512}, \bar{x}_{631}) \\ h(x_{623}, \bar{x}_{412}) & h(x_{631}, \bar{x}_{512}) & h(x_{612}) \end{pmatrix} \\ \begin{pmatrix} h(x_{123}) & h(x_{456}) \end{pmatrix} \end{pmatrix}, \\ & f^{-1} \left( \sum_{i < j < k} x_{ijk} \mathbf{e}_i \wedge \mathbf{e}_j \wedge \mathbf{e}_k \right) = \begin{pmatrix} \begin{pmatrix} h(x_{156}) & h(x_{164}, \bar{x}_{256}) & h(x_{145}, \bar{x}_{356}) \\ h(x_{256}, \bar{x}_{164}) & h(x_{264}) & h(x_{245}, \bar{x}_{364}) \\ h(x_{356}, \bar{x}_{145}) & h(x_{364}, \bar{x}_{245}) & h(x_{345}) \\ h(x_{423}) & h(x_{431}, \bar{x}_{523}) & h(x_{412}, \bar{x}_{623}) \\ h(x_{523}, \bar{x}_{431}) & h(x_{531}) & h(x_{512}, \bar{x}_{631}) \\ h(x_{623}, \bar{x}_{412}) & h(x_{631}, \bar{x}_{512}) & h(x_{612}) \end{pmatrix} \\ \begin{pmatrix} h(x_{123}) & h(x_{456}) \end{pmatrix} \end{pmatrix}, \end{aligned}$ | $f^{-1} \left( \sum_{i < j < k} x_{ijk} \mathbf{e}_i \wedge \mathbf{e}_j \wedge \mathbf{e}_k \right) = \begin{pmatrix} \begin{pmatrix} h(x_{156}) & h(x_{164}, \bar{x}_{256}) & h(x_{145}, \bar{x}_{356}) \\ h(x_{256}, \bar{x}_{164}) & h(x_{264}) & h(x_{245}, \bar{x}_{364}) \\ h(x_{356}, \bar{x}_{145}) & h(x_{364}, \bar{x}_{245}) & h(x_{345}) \\ h(x_{423}) & h(x_{431}, \bar{x}_{523}) & h(x_{412}, \bar{x}_{623}) \\ h(x_{523}, \bar{x}_{431}) & h(x_{531}) & h(x_{512}, \bar{x}_{631}) \\ h(x_{623}, \bar{x}_{412}) & h(x_{631}, \bar{x}_{512}) & h(x_{612}) \end{pmatrix} \\ \begin{pmatrix} h(x_{123}) & h(x_{456}) \end{pmatrix} \end{pmatrix},$ |       |

| Identifier                                                                                                              | Page | Conf.   | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Rendered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Image                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-------------------------------------------------------------------------------------------------------------------------|------|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div> <div> <div>9982.0431_</div> <div>E01117</div> <div>inferred]</div> </div> <div> <div>refined:</div> </div> </div> | 189  | ■ 0.108 | $\begin{aligned} & (\boldsymbol{X}, \boldsymbol{Y}) = \left( \begin{array}{l} X_1, Y_1 \\ X_2, Y_2 \\ X_3, Y_3 \end{array} \right) = \left( \begin{array}{l} X_1, Y_1 \\ X_2, Y_2 \\ X_3, Y_3 \end{array} \right) \in C, \\ & \langle \boldsymbol{X}, \boldsymbol{Y} \rangle = \langle X_1, Y_1 \rangle + \langle X_2, Y_2 \rangle + \langle X_3, Y_3 \rangle \in C, \\ & \boldsymbol{X} \wedge \boldsymbol{Y} = \left( \begin{array}{l} X_2 \wedge Y_3 - Y_2 \wedge X_3 \\ X_3 \wedge Y_1 - Y_3 \wedge X_1 \\ X_1 \wedge Y_2 - Y_1 \wedge X_2 \end{array} \right) \in (\mathfrak{J}^C)^3, \\ & \boldsymbol{X} \cdot \boldsymbol{Y} = \left( \begin{array}{l} (X_1, Y_1) \quad (X_1, Y_2) \quad (X_1, Y_3) \\ (X_2, Y_1) \quad (X_2, Y_2) \quad (X_2, Y_3) \\ (X_3, Y_1) \quad (X_3, Y_2) \quad (X_3, Y_3) \end{array} \right) - \frac{1}{3}(\boldsymbol{X}, \boldsymbol{Y})E \in \mathfrak{sl}(3, C), \\ & \boldsymbol{X} \vee \boldsymbol{Y} = X_1 \vee Y_1 + X_2 \vee Y_2 + X_3 \vee Y_3 \in \mathfrak{e}_6^C, \end{aligned}$ | $\begin{aligned} & (\boldsymbol{X}, \boldsymbol{Y}) = (X_1, Y_1) + (X_2, Y_2) + (X_3, Y_3) \in C, \\ & \langle \boldsymbol{X}, \boldsymbol{Y} \rangle = \langle X_1, Y_1 \rangle + \langle X_2, Y_2 \rangle + \langle X_3, Y_3 \rangle \in C, \\ & \boldsymbol{X} \wedge \boldsymbol{Y} = \left( \begin{array}{l} X_2 \wedge Y_3 - Y_2 \wedge X_3 \\ X_3 \wedge Y_1 - Y_3 \wedge X_1 \\ X_1 \wedge Y_2 - Y_1 \wedge X_2 \end{array} \right) \in (\mathfrak{J}^C)^3, \\ & \boldsymbol{X} \cdot \boldsymbol{Y} = \left( \begin{array}{l} (X_1, Y_1) \quad (X_1, Y_2) \quad (X_1, Y_3) \\ (X_2, Y_1) \quad (X_2, Y_2) \quad (X_2, Y_3) \\ (X_3, Y_1) \quad (X_3, Y_2) \quad (X_3, Y_3) \end{array} \right) - \frac{1}{3}(\boldsymbol{X}, \boldsymbol{Y})E \in \mathfrak{sl}(3, C), \\ & \boldsymbol{X} \vee \boldsymbol{Y} = X_1 \vee Y_1 + X_2 \vee Y_2 + X_3 \vee Y_3 \in \mathfrak{e}_6^C, \end{aligned}$ | $\begin{aligned} & (\boldsymbol{X}, \boldsymbol{Y}) = (X_1, Y_1) + (X_2, Y_2) + (X_3, Y_3) \in C, \\ & \langle \boldsymbol{X}, \boldsymbol{Y} \rangle = \langle X_1, Y_1 \rangle + \langle X_2, Y_2 \rangle + \langle X_3, Y_3 \rangle \in C, \\ & \boldsymbol{X} \times \boldsymbol{Y} = \left( \begin{array}{l} X_2 \times Y_3 - Y_2 \times X_3 \\ X_3 \times Y_1 - Y_3 \times X_1 \\ X_1 \times Y_2 - Y_1 \times X_2 \end{array} \right) \in (\mathfrak{J}^C)^3, \\ & \boldsymbol{X} \cdot \boldsymbol{Y} = \left( \begin{array}{l} (X_1, Y_1) \quad (X_1, Y_2) \quad (X_1, Y_3) \\ (X_2, Y_1) \quad (X_2, Y_2) \quad (X_2, Y_3) \\ (X_3, Y_1) \quad (X_3, Y_2) \quad (X_3, Y_3) \end{array} \right) - \frac{1}{3}(\boldsymbol{X}, \boldsymbol{Y})E \in \mathfrak{sl}(3, C), \\ & \boldsymbol{X} \vee \boldsymbol{Y} = X_1 \vee Y_1 + X_2 \vee Y_2 + X_3 \vee Y_3 \in \mathfrak{e}_6^C, \end{aligned}$ |

| Identifier                                                                 | Page | Conf.              | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Rendered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Image                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
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| <div>0902.0431_</div> <div>EQ0756</div> <div>[refined:<br/>inferred]</div> | 125  | <div>■</div> 0.119 | $\begin{array}{l} \backslash begin{array}{llllllllllll} \backslash lambda_{\{0\}}-\backslash lambda_{\{1\}}=1 \\ \& \ 0 \ \& \ 0 \ \& \ 0 \ \& \ 0 \ \& \ 0 \ \& \ 0 \ \& \ \backslash lambda_{\{0\}}+\backslash lambda_{\{1\}}=1 \ \& \\ 2 \ \& \ 2 \ \& \ 2 \ \& \ 2 \ \& \ 1 \ \backslash \backslash \ \backslash lambda_{\{0\}}-\backslash lambda_{\{2\}}=1 \ \& \ 1 \ \& \\ 0 \ \& \ 0 \ \& \ 0 \ \& \ 0 \ \& \ 0 \ \& \ \backslash lambda_{\{0\}}+\backslash lambda_{\{2\}}=1 \ \& \ 1 \ \& \ 2 \\ \& \ 2 \ \& \ 2 \ \& \ 1 \ \backslash \backslash \ \backslash lambda_{\{0\}}-\backslash lambda_{\{3\}}=1 \ \& \ 1 \ \& \ 1 \ \& \ 0 \\ \& \ 0 \ \& \ 0 \ \& \ 0 \ \& \ \backslash lambda_{\{0\}}+\backslash lambda_{\{3\}}=1 \ \& \ 1 \ \& \ 1 \ \& \ 2 \ \& \\ 2 \ \& \ 1 \ \backslash \backslash \ \backslash lambda_{\{1\}}-\backslash lambda_{\{2\}}=0 \ \& \ 1 \ \& \ 0 \ \& \ 0 \ \& \ 0 \ \& \\ 0 \ \& \ 0 \ \& \ \backslash lambda_{\{1\}}+\backslash lambda_{\{2\}}=0 \ \& \ 1 \ \& \ 2 \ \& \ 2 \ \& \ 2 \ \& \ 1 \\ \backslash \backslash \ \backslash lambda_{\{1\}}-\backslash lambda_{\{3\}}=0 \ \& \ 1 \ \& \ 1 \ \& \ 0 \ \& \ 0 \ \& \ 0 \ \& \ 0 \\ \& \ \backslash lambda_{\{1\}}+\backslash lambda_{\{3\}}=0 \ \& \ 1 \ \& \ 1 \ \& \ 2 \ \& \ 2 \ \& \ 1 \ \backslash \backslash \\ \backslash lambda_{\{2\}}-\backslash lambda_{\{3\}}=0 \ \& \ 0 \ \& \ 1 \ \& \ 0 \ \& \ 0 \ \& \ 0 \ \& \ 0 \ \& \\ \backslash lambda_{\{2\}}+\backslash lambda_{\{3\}}=0 \ \& \ 0 \ \& \ 1 \ \& \ 2 \ \& \ 2 \ \& \ 1 \ \backslash \backslash \ \& \ \& \\ \backslash lambda_{\{0\}}+\frac{1}{2}\left(\mu_{\{2\}}-\mu_{\{3\}}\right)=1 \\ \& \ 1 \ \& \ 1 \ \& \ 1 \ \& \ 1 \ \& \ 1 \ \& \ 1 \ \& \ \& \ \& \ \backslash \backslash \ \& \ \& \ \backslash lambda_{\{1\}}+ \\ \frac{1}{2}\left(\mu_{\{2\}}-\mu_{\{3\}}\right)=0 \ \& \ 1 \ \& \ 1 \ \& \ 1 \ \& \ 1 \ \& \ 1 \ \& \ 1 \ \& \ 1 \ \& \ 1 \ \& \ 1 \ \& \ \& \ \& \ \backslash \backslash \ \& \ \& \ \backslash lambda_{\{2\}}+\frac{1}{2}\left(\mu_{\{2\}}-\mu_{\{3\}}\right)=0 \ \& \ 0 \ \& \ 1 \ \& \ 1 \ \& \ 1 \ \& \ 1 \ \& \ 1 \ \& \ \& \ \& \\ \& \ \backslash \backslash \ \& \ \& \ \backslash lambda_{\{3\}}+\frac{1}{2}\left(\mu_{\{2\}}-\mu_{\{3\}}\right)=0 \ \& \ 0 \ \& \ 0 \ \& \ 1 \ \& \ 1 \ \& \ 1 \ \& \ 1 \ \& \ \& \ \& \ \end{array}$ | $\begin{array}{llllll} \lambda_0-\lambda_1=1 & 0 & 0 & 0 & 0 & 0 & 0 & \lambda_0+\lambda_1=1 & 2 & 2 & 2 & 2 & 1 & 1 \\ \lambda_0-\lambda_2=1 & 1 & 0 & 0 & 0 & 0 & 0 & \lambda_0+\lambda_2=1 & 1 & 2 & 2 & 2 & 1 & 1 \\ \lambda_0-\lambda_3=1 & 1 & 1 & 0 & 0 & 0 & 0 & \lambda_0+\lambda_3=1 & 1 & 1 & 2 & 2 & 1 & 1 \\ \lambda_1-\lambda_2=0 & 1 & 0 & 0 & 0 & 0 & 0 & \lambda_1+\lambda_2=0 & 1 & 2 & 2 & 2 & 1 & 1 \\ \lambda_1-\lambda_3=0 & 1 & 1 & 0 & 0 & 0 & 0 & \lambda_1+\lambda_3=0 & 1 & 1 & 2 & 2 & 1 & 1 \\ \lambda_2-\lambda_3=0 & 0 & 1 & 0 & 0 & 0 & 0 & \lambda_2+\lambda_3=0 & 0 & 1 & 2 & 2 & 1 & 1 \\ \\ \lambda_0+\frac{1}{2}(\mu_2-\mu_3)=1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ \lambda_1+\frac{1}{2}(\mu_2-\mu_3)=0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ \lambda_2+\frac{1}{2}(\mu_2-\mu_3)=0 & 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ \lambda_3+\frac{1}{2}(\mu_2-\mu_3)=0 & 0 & 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \end{array}$ | $\begin{array}{llllll} \lambda_0-\lambda_1=1 & 0 & 0 & 0 & 0 & 0 & 0 & \lambda_0+\lambda_1=1 & 2 & 2 & 2 & 2 & 1 & 1 \\ \lambda_0-\lambda_2=1 & 1 & 0 & 0 & 0 & 0 & 0 & \lambda_0+\lambda_2=1 & 1 & 2 & 2 & 2 & 1 & 1 \\ \lambda_0-\lambda_3=1 & 1 & 1 & 0 & 0 & 0 & 0 & \lambda_0+\lambda_3=1 & 1 & 1 & 2 & 2 & 1 & 1 \\ \lambda_1-\lambda_2=0 & 1 & 0 & 0 & 0 & 0 & 0 & \lambda_1+\lambda_2=0 & 1 & 2 & 2 & 2 & 1 & 1 \\ \lambda_1-\lambda_3=0 & 1 & 1 & 0 & 0 & 0 & 0 & \lambda_1+\lambda_3=0 & 1 & 1 & 2 & 2 & 1 & 1 \\ \lambda_2-\lambda_3=0 & 0 & 1 & 0 & 0 & 0 & 0 & \lambda_2+\lambda_3=0 & 0 & 1 & 2 & 2 & 1 & 1 \\ \\ \lambda_0+\frac{1}{2}(\mu_2-\mu_3)=1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ \lambda_1+\frac{1}{2}(\mu_2-\mu_3)=0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ \lambda_2+\frac{1}{2}(\mu_2-\mu_3)=0 & 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ \lambda_3+\frac{1}{2}(\mu_2-\mu_3)=0 & 0 & 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \end{array}$ |
| <div>0902.0431_</div> <div>EQ1187</div> <div>[refined:<br/>inferred]</div> | 200  | <div>■</div> 0.183 | $\begin{aligned} H \ \&= \ \nu \backslash Big(\sum_{k=0}^3 \ \backslash lambda_k \\ H_k \backslash Big) \ \backslash \backslash \ \&= \ \backslash tfrac{1}{2} \backslash big(\backslash lambda_0(-H_0 + H_1 - \\ H_2 + H_3) + \backslash lambda_1(-H_0 + H_1 + H_2 - H_3) \ \backslash \backslash \ \&\quad \\ + \backslash lambda_2(H_0 + H_1 + H_2 + H_3) + \backslash lambda_3(-H_0 - \\ H_1 + H_2 + H_3) \backslash big) \ \backslash \backslash \ \&= \ \backslash tfrac{1}{2}(-\backslash lambda_0 - \\ \backslash lambda_1 + \backslash lambda_2 - \backslash lambda_3)H_0 + \backslash tfrac{1}{2}(\backslash lambda_0 + \\ \backslash lambda_1 + \backslash lambda_2 - \backslash lambda_3)H_1 \ \backslash \backslash \ \& \\ \quad + \backslash tfrac{1}{2}(-\backslash lambda_0 + \backslash lambda_1 + \backslash lambda_2 + \backslash lambda_3)H_2 + \backslash tfrac{1}{2}(\backslash lambda_0 - \backslash lambda_1 + \backslash lambda_2 + \backslash lambda_3)H_3, \end{aligned}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | $\begin{aligned} H &= \nu \Big( \sum_{k=0}^3 \lambda_k H_k \\ &= \frac{1}{2}(\lambda_0(-H_0 + H_1 - H_2 + H_3) + \lambda_1(-H_0 + H_1 + H_2 - H_3) \\ &\quad + \lambda_2(H_0 + H_1 + H_2 + H_3) + \lambda_3(-H_0 - H_1 + H_2 + H_3)) \\ &= \frac{1}{2}(-\lambda_0 - \lambda_1 + \lambda_2 - \lambda_3)H_0 + \frac{1}{2}(\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3)H_1 \\ &\quad + \frac{1}{2}(-\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3)H_2 + \frac{1}{2}(\lambda_0 - \lambda_1 + \lambda_2 + \lambda_3)H_3, \end{aligned}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | $\begin{aligned} &= (v,w)(a,*(b\wedge x)) - (a_1,w)(w\wedge a_2,*(b\wedge x)) \\ &\quad + (a_2,w)(w\wedge a_1,*(b\wedge x)) - \frac{3}{5}(*(a\wedge b),x)(v,w) \\ &= \frac{2}{5}(*(a\wedge b),x)(v,w) - (a_1,w)(b\wedge x\wedge w\wedge a_2,e_1\wedge\cdots\wedge e_5) \\ &\quad + (a_2,w)(b\wedge x\wedge w\wedge a_1,e_1\wedge\cdots\wedge e_5), \\ &((b\times *(a\wedge x))v,w) \\ &= (x\wedge a,w\wedge *(b\wedge v)) - \frac{3}{5}(*(a\wedge b),x)(v,w) \\ &= (v,w)(*(a\wedge b),v) + (a_1,w)(b\wedge x\wedge w\wedge a_2,e_1\wedge\cdots\wedge e_5) \\ &\quad - (a_2,w)(b\wedge x\wedge w\wedge a_1,e_1\wedge\cdots\wedge e_5) - \frac{3}{5}(*(a\wedge b),x)(v,w). \end{aligned}$                                                                                                                                                                                                                                                                                                                                   |

| Identifier       | Page | Conf.   | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Rendered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Image                                                                                                                                                                                                                                                                                                                                                                                                                  |
|------------------|------|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0902.0431_E00830 | 138  | ■ 0.186 | $\begin{aligned} \kappa(X, Y, \xi, \eta) = & \kappa \left( \begin{array}{ccc} \xi_1 & x_3 & \bar{x}_2 \\ \bar{x}_3 & \xi_2 & x_1 \\ x_2 & \bar{x}_1 & \xi_3 \end{array} \right), \left( \begin{array}{ccc} \eta_1 & y_3 & \bar{y}_2 \\ \bar{y}_3 & \eta_2 & y_1 \\ y_2 & \bar{y}_1 & \eta_3 \end{array} \right), \xi, \eta \\ = & \left( \left( \begin{array}{ccc} -\xi_1 & 0 & 0 \\ 0 & \xi_2 & x_1 \\ 0 & \bar{x}_1 & \xi_3 \end{array} \right), \left( \begin{array}{ccc} \eta_1 & 0 & 0 \\ 0 & -\eta_2 & -y_1 \\ 0 & -\bar{y}_1 & -\eta_3 \end{array} \right), -\xi, \eta \right), \\ \mu(X, Y, \xi, \eta) = & \left( \left( \begin{array}{ccc} \eta & 0 & 0 \\ 0 & \eta_3 & -y_1 \\ 0 & -\bar{y}_1 & \eta_2 \end{array} \right), \left( \begin{array}{ccc} \xi & 0 & 0 \\ 0 & \xi_3 & -x_1 \\ 0 & -\bar{x}_1 & \xi_2 \end{array} \right), \eta_1, \xi_1 \right). \end{aligned}$ | $\begin{aligned} \kappa(X, Y, \xi, \eta) &= \kappa \left( \begin{pmatrix} \xi_1 & x_3 & \bar{x}_2 \\ \bar{x}_3 & \xi_2 & x_1 \\ x_2 & \bar{x}_1 & \xi_3 \end{pmatrix}, \begin{pmatrix} \eta_1 & y_3 & \bar{y}_2 \\ \bar{y}_3 & \eta_2 & y_1 \\ y_2 & \bar{y}_1 & \eta_3 \end{pmatrix}, \xi, \eta \right) \\ &= \left( \begin{pmatrix} -\xi_1 & 0 & 0 \\ 0 & \xi_2 & x_1 \\ 0 & \bar{x}_1 & \xi_3 \end{pmatrix}, \begin{pmatrix} \eta_1 & 0 & 0 \\ 0 & -\eta_2 & -y_1 \\ 0 & -\bar{y}_1 & -\eta_3 \end{pmatrix}, -\xi, \eta \right), \\ \mu(X, Y, \xi, \eta) &= \left( \begin{pmatrix} \eta & 0 & 0 \\ 0 & \eta_3 & -y_1 \\ 0 & -\bar{y}_1 & \eta_2 \end{pmatrix}, \begin{pmatrix} \xi & 0 & 0 \\ 0 & \xi_3 & -x_1 \\ 0 & -\bar{x}_1 & \xi_2 \end{pmatrix}, \eta_1, \xi_1 \right). \end{aligned}$ |                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 0902.0431_E01041 | 177  | ■ 0.204 | $\begin{aligned} & \left( \alpha_i, \alpha_i \right) = \frac{1}{30}, \quad i = 1, 2, 3, 4, 5, 6, 7, 8, \\ & \left( \alpha_i, \alpha_{i+1} \right) = -\frac{1}{60}, \quad i = 1, 2, \cdots, 6, \quad \left( \alpha_3, \alpha_8 \right) = -\frac{1}{60}, \\ & \left( \alpha_i, \alpha_j \right) = 0, \quad \text{otherwise} \\ & \left( -\mu, -\mu \right) = \frac{1}{30}, \quad \left( -\mu, \alpha_7 \right) = -\frac{1}{60}, \quad \left( -\mu, \alpha_i \right) = 0, \quad i = 1, 2, 3, 4, 5, 6, 8, \end{aligned}$                                                                                                                                                                                                                                                                                                                                                                 | $\begin{aligned} (\alpha_i, \alpha_i) &= \frac{1}{30}, \quad i = 1, 2, 3, 4, 5, 6, 7, 8 \\ (\alpha_i, \alpha_{i+1}) &= -\frac{1}{60}, \quad i = 1, 2, \cdots, 6, \quad (\alpha_3, \alpha_8) = -\frac{1}{60} \\ (\alpha_i, \alpha_j) &= 0, \quad \text{otherwise} \\ (-\mu, -\mu) &= \frac{1}{30}, \quad (-\mu, \alpha_7) = -\frac{1}{60}, \quad (-\mu, \alpha_i) = 0, \quad i = 1, 2, 3, 4, 5, 6, 8, \end{aligned}$                                                                                                                                                                                                                                                                                                                                                                              | $\begin{aligned} (\alpha_i, \alpha_i) &= \frac{1}{30}, \quad i = 1, 2, 3, 4, 5, 6, 7, 8, \\ (\alpha_i, \alpha_{i+1}) &= -\frac{1}{60}, \quad i = 1, 2, \cdots, 6, \quad (\alpha_3, \alpha_8) = -\frac{1}{60}, \\ (\alpha_i, \alpha_j) &= 0, \quad \text{otherwise}, \\ (-\mu, -\mu) &= \frac{1}{30}, \quad (-\mu, \alpha_7) = -\frac{1}{60}, \quad (-\mu, \alpha_i) = 0, \quad i = 1, 2, 3, 4, 5, 6, 8, \end{aligned}$ |
| 0902.0431_E00340 | 54   | ■ 0.226 | $\begin{aligned} \lambda_0 &= \alpha_1 + \alpha_2 + \alpha_3 \\ \lambda_1 &= \alpha_2 + \alpha_3 \\ \lambda_3 &= \alpha_3 \\ \lambda_3 &= \alpha_1 + 2\alpha_2 + 3\alpha_3 + 2\alpha_4, \end{aligned}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | $\begin{aligned} \lambda_0 &= \alpha_1 + \alpha_2 + \alpha_3 \\ \lambda_1 &= \alpha_2 + \alpha_3 \\ \lambda_3 &= \alpha_3 \\ \lambda_3 &= \alpha_1 + 2\alpha_2 + 3\alpha_3 + 2\alpha_4, \end{aligned}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | $\begin{aligned} \lambda_0 &= \alpha_1 + \alpha_2 + \alpha_3 \\ \lambda_1 &= \alpha_2 + \alpha_3 \\ \lambda_3 &= \alpha_3 \\ \lambda_3 &= \alpha_1 + 2\alpha_2 + 3\alpha_3 + 2\alpha_4, \end{aligned}$                                                                                                                                                                                                                 |

| Identifier       | Page | Conf.   | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Rendered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Image |
|------------------|------|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 0902_0431_EQ0937 | 155  | ■ 0.238 | $\begin{aligned} & \text{\texttt{\textbackslash begin{aligned} \& P=(0,0,1,0) \text{\texttt{\textbackslash \& \xrightarrow{f} \boldsymbol{e}_1 \wedge \boldsymbol{e}_2 \wedge \boldsymbol{e}_3} \boldsymbol{e}_1 \wedge \boldsymbol{e}_2 \wedge \boldsymbol{e}_3} \text{\texttt{\textbackslash \& \stackrel{D}{\longrightarrow} D\boldsymbol{e}_1 \wedge \boldsymbol{e}_2 \wedge \boldsymbol{e}_3 + \boldsymbol{e}_1 \wedge D\boldsymbol{e}_2 \wedge \boldsymbol{e}_3 + \boldsymbol{e}_1 \wedge \boldsymbol{e}_2 \wedge D\boldsymbol{e}_3} \\ & + \left(b_{11} + \frac{\nu}{3}\right) \boldsymbol{e}_1 \wedge \boldsymbol{e}_2 \wedge \boldsymbol{e}_3 - \bar{l}_{11} \boldsymbol{e}_4 \wedge \boldsymbol{e}_2 \wedge \boldsymbol{e}_3 - \bar{l}_{12} \boldsymbol{e}_5 \wedge \boldsymbol{e}_2 \wedge \boldsymbol{e}_3 - \bar{l}_{13} \boldsymbol{e}_6 \wedge \boldsymbol{e}_2 \wedge \boldsymbol{e}_3 \\ & + \left(b_{22} + \frac{\nu}{3}\right) \boldsymbol{e}_1 \wedge \boldsymbol{e}_2 \wedge \boldsymbol{e}_3 - \bar{l}_{21} \boldsymbol{e}_4 \wedge \boldsymbol{e}_3 \wedge \boldsymbol{e}_1 - \bar{l}_{22} \boldsymbol{e}_1 \wedge \boldsymbol{e}_5 \wedge \boldsymbol{e}_3 - \bar{l}_{23} \boldsymbol{e}_1 \wedge \boldsymbol{e}_6 \wedge \boldsymbol{e}_3 \\ & + \left(b_{33} + \frac{\nu}{3}\right) \boldsymbol{e}_1 \wedge \boldsymbol{e}_2 \wedge \boldsymbol{e}_3 - \bar{l}_{31} \boldsymbol{e}_4 \wedge \boldsymbol{e}_1 \wedge \boldsymbol{e}_2 - \bar{l}_{32} \boldsymbol{e}_5 \wedge \boldsymbol{e}_1 \wedge \boldsymbol{e}_2 - \bar{l}_{33} \boldsymbol{e}_6 \wedge \boldsymbol{e}_1 \wedge \boldsymbol{e}_2} \end{aligned}$ | $P = (0, 0, 1, 0)$ $\xrightarrow{f} \boldsymbol{e}_1 \wedge \boldsymbol{e}_2 \wedge \boldsymbol{e}_3$ $\xrightarrow{D} D\boldsymbol{e}_1 \wedge \boldsymbol{e}_2 \wedge \boldsymbol{e}_3 + \boldsymbol{e}_1 \wedge D\boldsymbol{e}_2 \wedge \boldsymbol{e}_3 + \boldsymbol{e}_1 \wedge \boldsymbol{e}_2 \wedge D\boldsymbol{e}_3$ $+ \left(b_{11} + \frac{\nu}{3}\right) \boldsymbol{e}_1 \wedge \boldsymbol{e}_2 \wedge \boldsymbol{e}_3 - \bar{l}_{11} \boldsymbol{e}_4 \wedge \boldsymbol{e}_2 \wedge \boldsymbol{e}_3 - \bar{l}_{12} \boldsymbol{e}_5 \wedge \boldsymbol{e}_2 \wedge \boldsymbol{e}_3 - \bar{l}_{13} \boldsymbol{e}_6 \wedge \boldsymbol{e}_2 \wedge \boldsymbol{e}_3$ $+ \left(b_{22} + \frac{\nu}{3}\right) \boldsymbol{e}_1 \wedge \boldsymbol{e}_2 \wedge \boldsymbol{e}_3 - \bar{l}_{21} \boldsymbol{e}_4 \wedge \boldsymbol{e}_3 \wedge \boldsymbol{e}_1 - \bar{l}_{22} \boldsymbol{e}_1 \wedge \boldsymbol{e}_5 \wedge \boldsymbol{e}_3 - \bar{l}_{23} \boldsymbol{e}_1 \wedge \boldsymbol{e}_6 \wedge \boldsymbol{e}_3$ $+ \left(b_{33} + \frac{\nu}{3}\right) \boldsymbol{e}_1 \wedge \boldsymbol{e}_2 \wedge \boldsymbol{e}_3 - \bar{l}_{31} \boldsymbol{e}_4 \wedge \boldsymbol{e}_1 \wedge \boldsymbol{e}_2 - \bar{l}_{32} \boldsymbol{e}_5 \wedge \boldsymbol{e}_1 \wedge \boldsymbol{e}_2 - \bar{l}_{33} \boldsymbol{e}_6 \wedge \boldsymbol{e}_1 \wedge \boldsymbol{e}_2$ $\xrightarrow{f^{-1}} \begin{pmatrix} & \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \\ \begin{pmatrix} -h(\bar{l}_{11}) & -h(\bar{l}_{21}, l_{12}) & -h(\bar{l}_{31}, l_{13}) \\ -h(\bar{l}_{12}, l_{21}) & -h(\bar{l}_{22}) & -h(\bar{l}_{32}, l_{23}) \\ -h(\bar{l}_{13}, l_{31}) & -h(\bar{l}_{23}, l_{32}) & -h(\bar{l}_{33}) \end{pmatrix} \\ h(\nu) \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ -\tau h(L) \\ -i\nu e_1 \\ 0 \end{pmatrix}$ $= \begin{pmatrix} \phi_C(B, C) + \frac{i\nu e_1}{3} & -2\tau h(L) & 0 & h(L) \\ 2h(L) & \tau\phi_C(B, C)\tau - \frac{i\nu e_1}{3} & -\tau h(L) & 0 \\ 0 & h(L) & -i\nu e_1 & 0 \\ -\tau h(L) & 0 & 0 & i\nu e_1 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}$ $= \Phi(\phi_C(B, C), h(L), -\tau h(L), -i\nu e_1)P.$ |       |

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|------------------|------|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0902.0431_E00834 | 139  | ■ 0.239 | $\begin{aligned} & \left( \left( \left( \left( \left( \left( \frac{e}{7} \right) \right)^{\kappa, \mu} \right) \right)^{\kappa, \mu} \right) \right)^{\kappa, \mu} \right)^{\kappa, \mu} \right)^{\kappa, \mu} \right)^{\kappa, \mu} \\ & \left( \left( \left( \left( \left( \left( \left( \frac{e}{7} \right) \right)^{\kappa, \mu} \right) \right)^{\kappa, \mu} \right) \right)^{\kappa, \mu} \right)^{\kappa, \mu} \right)^{\kappa, \mu} \right)^{\kappa, \mu} \right)^{\kappa, \mu} \end{aligned}$ | $((\mathfrak{e}_7)^{\kappa, \mu})_{(0, E_1, 0, 1)} = \{\Phi \in (\mathfrak{e}_7)^{\kappa, \mu} \mid \Phi((0, E_1, 0, 1)) = 0\}$ $= \left\{ \begin{array}{c} \Phi(\phi, A, -\tau A, 0) \in \mathfrak{e}_7 \\ \left  \begin{array}{l} \phi \in \mathfrak{e}_6, \phi E_1 = 0 \\ A \in \mathfrak{J}^C, 2E_1 \times A = \tau A \end{array} \right. \end{array} \right\}.$ | $((\mathfrak{e}_7)^{\kappa, \mu})_{(0, E_1, 0, 1)} = \{\Phi \in (\mathfrak{e}_7)^{\kappa, \mu} \mid \Phi((0, E_1, 0, 1)) = 0\}$ $= \left\{ \Phi(\phi, A, -\tau A, 0) \in \mathfrak{e}_7 \left  \begin{array}{l} \phi \in \mathfrak{e}_6, \phi E_1 = 0 \\ A \in \mathfrak{J}^C, 2E_1 \times A = \tau A \end{array} \right. \right\}.$                                |
| 0902.0431_E00946 | 157  | ■ 0.244 | $\begin{aligned} & \psi(A, \epsilon) \psi(B, 1) P = \psi(A, \epsilon) (f^{-1}(B(fP))) \\ & = f^{-1} \left( Af(f^{-1}(B(fP))) \right) = f^{-1} \left( Af \left( f^{-1} \left( \overline{(\text{Ad} J_3) B} \right) (f\bar{P}) \right) \right) \\ & = f^{-1}((A(\epsilon B))(f\bar{P})) = \psi(A(\epsilon B), \epsilon) P \end{aligned}$                                                                                                                                                                  | $\psi(A, \epsilon) \psi(B, 1) P = \psi(A, \epsilon) (f^{-1}(B(fP)))$ $= f^{-1} \left( Af(f^{-1}(B(fP))) \right) = f^{-1} \left( Af \left( f^{-1} \left( \overline{(\text{Ad} J_3) B} \right) (f\bar{P}) \right) \right)$ $= f^{-1}((A(\epsilon B))(f\bar{P})) = \psi(A(\epsilon B), \epsilon) P$                                                                     | $\psi(A, \epsilon) \psi(B, 1) P = \psi(A, \epsilon) (f^{-1}(B(fP)))$ $= f^{-1} \left( Af(f^{-1}(B(fP))) \right) = f^{-1} \left( Af(f^{-1}(\overline{(\text{Ad} J_3) B})(f\bar{P})) \right)$ $= f^{-1}((A(\epsilon B))(f\bar{P})) = \psi(A(\epsilon B), \epsilon) P.$                                                                                                |
| 0902.0431_E01037 | 176  | ■ 0.252 | $\begin{aligned} & B_8(\tilde{h}, \tilde{h}') = \frac{5}{3} B_7(\Phi(h), \Phi(h')) + 120 r r' \quad (\text{Theorem 5.3.2}) \\ & = \frac{5}{3} 6 \left( 6 \sum_{k=0}^3 \lambda_k \lambda_k' + 3 \sum_{j=1}^3 \mu_j \mu_j' + 4 \nu \nu' \right) + 120 r r' \quad (\text{Theorem 4.6.2}) \\ & = 60 \sum_{k=0}^3 \lambda_k \lambda_k' + 30 \sum_{j=1}^3 \mu_j \mu_j' + 40 \nu \nu' + 120 r r'. \end{aligned}$                                                                                               | $B_8(\tilde{h}, \tilde{h}') = \frac{5}{3} B_7(\Phi(h), \Phi(h')) + 120 r r' \quad (\text{Theorem 5.3.2})$ $= \frac{5}{3} 6 \left( 6 \sum_{k=0}^3 \lambda_k \lambda_k' + 3 \sum_{j=1}^3 \mu_j \mu_j' + 4 \nu \nu' \right) + 120 r r' \quad (\text{Theorem 4.6.2})$ $= 60 \sum_{k=0}^3 \lambda_k \lambda_k' + 30 \sum_{j=1}^3 \mu_j \mu_j' + 40 \nu \nu' + 120 r r'.$  | $B_8(\tilde{h}, \tilde{h}') = \frac{5}{3} B_7(\Phi(h), \Phi(h')) + 120 r r' \quad (\text{Theorem 5.3.2})$ $= \frac{5}{3} 6 \left( 6 \sum_{k=0}^3 \lambda_k \lambda_k' + 3 \sum_{j=1}^3 \mu_j \mu_j' + 4 \nu \nu' \right) + 120 r r' \quad (\text{Theorem 4.6.2})$ $= 60 \sum_{k=0}^3 \lambda_k \lambda_k' + 30 \sum_{j=1}^3 \mu_j \mu_j' + 40 \nu \nu' + 120 r r'.$ |
| 0902.0431_E00365 | 59   | ■ 0.264 | $\begin{aligned} & 1 \longrightarrow \text{Spin}(8) \longrightarrow (F_4)_{E_1} \longrightarrow S^8 \longrightarrow * \\ & \quad \downarrow p' \quad \quad \downarrow p \quad \quad \downarrow = \\ & 1 \longrightarrow SO(8) \longrightarrow SO(9) \longrightarrow S^8 \longrightarrow * \end{aligned}$                                                                                                                                                                                                | $\begin{aligned} & 1 \longrightarrow \text{Spin}(8) \longrightarrow (F_4)_{E_1} \longrightarrow S^8 \longrightarrow * \\ & \quad \downarrow p' \quad \quad \downarrow p \quad \quad \downarrow = \\ & 1 \longrightarrow SO(8) \longrightarrow SO(9) \longrightarrow S^8 \longrightarrow * \end{aligned}$                                                             | $\begin{aligned} & 1 \longrightarrow \text{Spin}(8) \longrightarrow (F_4)_{E_1} \longrightarrow S^8 \longrightarrow * \\ & \quad \downarrow p' \quad \quad \downarrow p \quad \quad \downarrow = \\ & 1 \longrightarrow SO(8) \longrightarrow SO(9) \longrightarrow S^8 \longrightarrow * \end{aligned}$                                                            |

| Identifier       | Page | Conf.   | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Rendered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Image                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|------------------|------|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0902.0431_E00206 | 35   | ■ 0.294 | $\begin{aligned} & \& \kappa\left(\alpha_1, \alpha_2, \alpha_3\right)=\left(\kappa \alpha_1, \kappa \alpha_2, \kappa \alpha_3\right), \\ & \& \pi\left(\alpha_1, \alpha_2, \alpha_3\right)=\left(\kappa \alpha_3, \kappa \alpha_2, \kappa \alpha_1\right), \\ & \& \nu\left(\alpha_1, \alpha_2, \alpha_3\right)=\left(\alpha_2, \alpha_3, \alpha_1\right), \end{aligned}$                                                                                                                                                                   | $\begin{aligned} \kappa\left(\alpha_1, \alpha_2, \alpha_3\right) & =\left(\kappa \alpha_1, \kappa \alpha_3, \kappa \alpha_2\right), \\ \pi\left(\alpha_1, \alpha_2, \alpha_3\right) & =\left(\kappa \alpha_3, \kappa \alpha_2, \kappa \alpha_1\right), \\ \nu\left(\alpha_1, \alpha_2, \alpha_3\right) & =\left(\alpha_2, \alpha_3, \alpha_1\right), \end{aligned}$                                                                                                                                                                                                                                                                                                                          | $\begin{aligned} \kappa\left(\alpha_1, \alpha_2, \alpha_3\right) & =\left(\kappa \alpha_1, \kappa \alpha_3, \kappa \alpha_2\right), \\ \pi\left(\alpha_1, \alpha_2, \alpha_3\right) & =\left(\kappa \alpha_3, \kappa \alpha_2, \kappa \alpha_1\right), \\ \nu\left(\alpha_1, \alpha_2, \alpha_3\right) & =\left(\alpha_2, \alpha_3, \alpha_1\right), \end{aligned}$                                                                                                                                                                                                        |
| 0902.0431_E00655 | 108  | ■ 0.299 | $\begin{aligned} & \& \left(P M \tau h(A, B)^*\right)^*(P N \tau h(A, B)^*)=\tau h(A, B) M^* P^* P N \tau h(A, B)^* \\ & \& =\tau\left(h(A, B) \tau\left(M^* N\right) h(A, B)^*\right), \\ & \& \frac{(P M \tau h(A, B)^*)(h(A, B) Y h(A, B)^*)}{(P M \tau h(A, B)^*) \times(P N \tau h(A, B)^*)}=\frac{\tau(P \tau(M Y) \tau h(A, B)^*)}{\tilde{t} \tilde{P}(M \times N) \tau^t(h(A, B)^*)} \\ & \& =P(\overline{M \times N}) h(A, B)^*=\tau(P(\overline{M \times N}) \tau h(A, B)^*). \end{aligned}$                                      | $\begin{aligned} (P M \tau h(A, B)^*)^*(P N \tau h(A, B)^*) & =\tau h(A, B) M^* P^* P N \tau h(A, B)^* \\ & =\tau\left(h(A, B) \tau\left(M^* N\right) h(A, B)^*\right), \\ \frac{(P M \tau h(A, B)^*)(h(A, B) Y h(A, B)^*)}{(P M \tau h(A, B)^*) \times(P N \tau h(A, B)^*)} & =\frac{\tau(P \tau(M Y) \tau h(A, B)^*)}{\tilde{t} \tilde{P}(M \times N) \tau^t(h(A, B)^*)}, \\ \frac{(P M \tau h(A, B)^*) \times(P N \tau h(A, B)^*)}{(P M \tau h(A, B)^*) \times(P N \tau h(A, B)^*)} & =\frac{\tau(P \tau(M Y) \tau h(A, B)^*)}{\tilde{t} \tilde{P}(M \times N) \tau^t(h(A, B)^*)}, \\ & =P(\overline{M \times N}) h(A, B)^*=\tau(P(\overline{M \times N}) \tau h(A, B)^*). \end{aligned}$ | $\begin{aligned} (P M \tau h(A, B)^*)^*(P N \tau h(A, B)^*) & =\tau h(A, B) M^* P^* P N \tau h(A, B)^* \\ & =\tau\left(h(A, B) \tau\left(M^* N\right) h(A, B)^*\right), \\ (P M \tau h(A, B)^*)(h(A, B) Y h(A, B)^*) & =\tau(P \tau(M Y) \tau h(A, B)^*), \\ \frac{(P M \tau h(A, B)^*) \times(P N \tau h(A, B)^*)}{(P M \tau h(A, B)^*) \times(P N \tau h(A, B)^*)} & =\frac{\tau(P \tau(M Y) \tau h(A, B)^*)}{\tilde{t} \tilde{P}(M \times N) \tau^t(h(A, B)^*)}, \\ & =P(\overline{M \times N}) h(A, B)^*=\tau(P(\overline{M \times N}) \tau h(A, B)^*). \end{aligned}$ |
| 0902.0431_E00101 | 20   | ■ 0.306 | $\begin{aligned} & \& \lambda_1-\lambda_2=\alpha_1, \quad \lambda_1-\lambda_3=2 \alpha_1+3 \alpha_2, \quad \lambda_2-\lambda_3=\alpha_1+3 \alpha_2, \\ & \& \lambda_1=\alpha_1+\alpha_2, \quad \lambda_2=\alpha_2, \quad \lambda_3=\alpha_1+2 \alpha_2. \end{aligned}$                                                                                                                                                                                                                                                                      | $\begin{aligned} \lambda_1-\lambda_2 & =\alpha_1, & \lambda_1-\lambda_3 & =2 \alpha_1+3 \alpha_2, & \lambda_2-\lambda_3 & =\alpha_1+3 \alpha_2, \\ \lambda_1 & =\alpha_1+\alpha_2, & \lambda_2 & =\alpha_2, & & =\lambda_3+2 \alpha_2 \end{aligned}$                                                                                                                                                                                                                                                                                                                                                                                                                                         | $\begin{aligned} \lambda_1-\lambda_2 & =\alpha_1, & \lambda_1-\lambda_3 & =2 \alpha_1+3 \alpha_2, & \lambda_2-\lambda_3 & =\alpha_1+3 \alpha_2, \\ \lambda_1 & =\alpha_1+\alpha_2, & \lambda_2 & =\alpha_2, & & -\lambda_3=\alpha_1+2 \alpha_2. \end{aligned}$                                                                                                                                                                                                                                                                                                             |
| 0902.0431_EQ1154 | 195  | ■ 0.318 | $\begin{aligned} & \& +(\boldsymbol{v}, \boldsymbol{w})(\boldsymbol{u}, \boldsymbol{a})-\sum_{i=1}^3 \sum_{j=1}^3\left(\boldsymbol{u}_i \wedge \boldsymbol{v}_j \wedge \boldsymbol{v}_{j+1}, \boldsymbol{w}\right)\left(\boldsymbol{u}_{i+1} \wedge \boldsymbol{u}_{i+2} \wedge \boldsymbol{v}_{j+2}, \boldsymbol{a}\right) \\ & \& =-(\boldsymbol{u} \wedge \boldsymbol{v}, \boldsymbol{w} \wedge \boldsymbol{a})=-\left(*\left(*(\boldsymbol{u} \wedge \boldsymbol{v}) \wedge \boldsymbol{w}\right), \boldsymbol{a}\right) \end{aligned}$ | $\begin{aligned} & +(\boldsymbol{v}, \boldsymbol{w})(\boldsymbol{u}, \boldsymbol{a})-\sum_{i=1}^3 \sum_{j=1}^3\left(\boldsymbol{u}_i \wedge \boldsymbol{v}_j \wedge \boldsymbol{v}_{j+1}, \boldsymbol{w}\right)\left(\boldsymbol{u}_{i+1} \wedge \boldsymbol{u}_{i+2} \wedge \boldsymbol{v}_{j+2}, \boldsymbol{a}\right) \\ & =-(\boldsymbol{u} \wedge \boldsymbol{v}, \boldsymbol{w} \wedge \boldsymbol{a})=-\left(*\left(*(\boldsymbol{u} \wedge \boldsymbol{v}) \wedge \boldsymbol{w}\right), \boldsymbol{a}\right) \end{aligned}$                                                                                                                                                        | $\begin{aligned} & +(\boldsymbol{v}, \boldsymbol{w})(\boldsymbol{u}, \boldsymbol{a})-\sum_{i=1}^3 \sum_{j=1}^3\left(\boldsymbol{u}_i \wedge \boldsymbol{v}_j \wedge \boldsymbol{v}_{j+1}, \boldsymbol{w}\right)\left(\boldsymbol{u}_{i+1} \wedge \boldsymbol{u}_{i+2} \wedge \boldsymbol{v}_{j+2}, \boldsymbol{a}\right) \\ & =-(\boldsymbol{u} \wedge \boldsymbol{v}, \boldsymbol{w} \wedge \boldsymbol{a})=-\left(*\left(*(\boldsymbol{u} \wedge \boldsymbol{v}) \wedge \boldsymbol{w}\right), \boldsymbol{a}\right). \end{aligned}$                                     |

| Identifier       | Page | Conf.   | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Rendered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Image                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|------------------|------|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0902.0431_EQ1189 | 200  | ■ 0.322 | $\begin{aligned} & ((\boldsymbol{a} \times (\boldsymbol{x} \wedge \boldsymbol{y})) \boldsymbol{v}, \boldsymbol{w}) = (\boldsymbol{x} \wedge \boldsymbol{y} \wedge \boldsymbol{w}, \boldsymbol{v} \wedge \boldsymbol{a}) - \frac{3}{5} (\boldsymbol{a}, \boldsymbol{x} \wedge \boldsymbol{y}) (\boldsymbol{v}, \boldsymbol{w}) \\ & = (\boldsymbol{x}, \boldsymbol{v}) (\boldsymbol{y} \wedge \boldsymbol{w}, \boldsymbol{a}) - (\boldsymbol{y}, \boldsymbol{v}) (\boldsymbol{x} \wedge \boldsymbol{w}, \boldsymbol{a}) + \frac{2}{5} (\boldsymbol{a}, \boldsymbol{x} \wedge \boldsymbol{y}) (\boldsymbol{v}, \boldsymbol{w}) \\ & = (\boldsymbol{x}, \boldsymbol{v}) (*(\boldsymbol{a} \wedge \boldsymbol{y}), \boldsymbol{w}) - (\boldsymbol{y}, \boldsymbol{v}) (*(\boldsymbol{a} \wedge \boldsymbol{x}), \boldsymbol{w}) + \frac{2}{5} (\boldsymbol{a}, \boldsymbol{x} \wedge \boldsymbol{y}) (\boldsymbol{v}, \boldsymbol{w}). \end{aligned}$ | $\begin{aligned} & ((\boldsymbol{a} \times (\boldsymbol{x} \wedge \boldsymbol{y})) \boldsymbol{v}, \boldsymbol{w}) = (\boldsymbol{x} \wedge \boldsymbol{y} \wedge \boldsymbol{w}, \boldsymbol{v} \wedge \boldsymbol{a}) - \frac{3}{5} (\boldsymbol{a}, \boldsymbol{x} \wedge \boldsymbol{y}) (\boldsymbol{v}, \boldsymbol{w}) \\ & = (\boldsymbol{x}, \boldsymbol{v}) (\boldsymbol{y} \wedge \boldsymbol{w}, \boldsymbol{a}) - (\boldsymbol{y}, \boldsymbol{v}) (\boldsymbol{x} \wedge \boldsymbol{w}, \boldsymbol{a}) + \frac{2}{5} (\boldsymbol{a}, \boldsymbol{x} \wedge \boldsymbol{y}) (\boldsymbol{v}, \boldsymbol{w}) \\ & = (\boldsymbol{x}, \boldsymbol{v}) (*(\boldsymbol{a} \wedge \boldsymbol{y}), \boldsymbol{w}) - (\boldsymbol{y}, \boldsymbol{v}) (*(\boldsymbol{a} \wedge \boldsymbol{x}), \boldsymbol{w}) + \frac{2}{5} (\boldsymbol{a}, \boldsymbol{x} \wedge \boldsymbol{y}) (\boldsymbol{v}, \boldsymbol{w}). \end{aligned}$ | $\begin{aligned} & ((\boldsymbol{a} \times (\boldsymbol{x} \wedge \boldsymbol{y})) \boldsymbol{v}, \boldsymbol{w}) = (\boldsymbol{x} \wedge \boldsymbol{y} \wedge \boldsymbol{w}, \boldsymbol{v} \wedge \boldsymbol{a}) - \frac{3}{5} (\boldsymbol{a}, \boldsymbol{x} \wedge \boldsymbol{y}) (\boldsymbol{v}, \boldsymbol{w}) \\ & = (\boldsymbol{x}, \boldsymbol{v}) (\boldsymbol{y} \wedge \boldsymbol{w}, \boldsymbol{a}) - (\boldsymbol{y}, \boldsymbol{v}) (\boldsymbol{x} \wedge \boldsymbol{w}, \boldsymbol{a}) + \frac{2}{5} (\boldsymbol{a}, \boldsymbol{x} \wedge \boldsymbol{y}) (\boldsymbol{v}, \boldsymbol{w}) \\ & = (\boldsymbol{x}, \boldsymbol{v}) (*(\boldsymbol{a} \wedge \boldsymbol{y}), \boldsymbol{w}) - (\boldsymbol{y}, \boldsymbol{v}) (*(\boldsymbol{a} \wedge \boldsymbol{x}), \boldsymbol{w}) + \frac{2}{5} (\boldsymbol{a}, \boldsymbol{x} \wedge \boldsymbol{y}) (\boldsymbol{v}, \boldsymbol{w}). \end{aligned}$ |
| 0902.0431_EQ1157 | 195  | ■ 0.323 | $\begin{aligned} & \& \theta = D \boldsymbol{u} = \sum_{\boldsymbol{I}} \boldsymbol{u}_{\boldsymbol{I}} D \boldsymbol{e}_{\boldsymbol{I}} = 3 \sum_{\boldsymbol{I} \not\supseteq 9} \boldsymbol{u}_{\boldsymbol{I}} \boldsymbol{e}_{\boldsymbol{I}} - 6 \sum_{\boldsymbol{I} \supseteq 9} \boldsymbol{u}_{\boldsymbol{I}} \boldsymbol{e}_{\boldsymbol{I}}, \\ & \& \theta = -{}^t D \boldsymbol{v} = -3 \sum_{\boldsymbol{J} \not\supseteq 9} \boldsymbol{v}_{\boldsymbol{J}} \boldsymbol{e}_{\boldsymbol{J}} + 6 \sum_{\boldsymbol{J} \supseteq 9} \boldsymbol{v}_{\boldsymbol{J}} \boldsymbol{e}_{\boldsymbol{J}}, \end{aligned}$                                                                                                                                                                                                                                                                                                               | $\begin{aligned} 0 &= D\boldsymbol{u} = \sum_I u_I D\boldsymbol{e}_I = 3 \sum_{I \not\supseteq 9} u_I \boldsymbol{e}_I - 6 \sum_{I \supseteq 9} u_I \boldsymbol{e}_I, \\ 0 &= -{}^t D\boldsymbol{v} = -3 \sum_{J \not\supseteq 9} v_J \boldsymbol{e}_J + 6 \sum_{J \supseteq 9} v_J \boldsymbol{e}_J, \end{aligned}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | $\begin{aligned} 0 &= D\boldsymbol{u} = \sum_I u_I D\boldsymbol{e}_I = 3 \sum_{I \not\supseteq 9} u_I \boldsymbol{e}_I - 6 \sum_{I \supseteq 9} u_I \boldsymbol{e}_I, \\ 0 &= -{}^t D\boldsymbol{v} = -3 \sum_{J \not\supseteq 9} v_J \boldsymbol{e}_J + 6 \sum_{J \supseteq 9} v_J \boldsymbol{e}_J, \end{aligned}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 0902.0431_EQ0460 | 75   | ■ 0.325 | $\begin{aligned} & \mathfrak{a} \supset [\mathfrak{a}, \mathfrak{e}_6^C] \supset [\mathfrak{f}_4^C, \widetilde{\mathfrak{J}}_0^C] = \mathfrak{f}_4^C \widetilde{\mathfrak{J}}_0^C \text{ (Lemma 3.2.2)} = \widetilde{\mathfrak{J}}_0^C \text{ (Proposition 2.4.6.(2))}. \\ & \mathfrak{a} \supset [\mathfrak{a}, \mathfrak{e}_6^C] \supset [\mathfrak{f}_4^C, \widetilde{\mathfrak{J}}_0^C] = \mathfrak{f}_4^C \widetilde{\mathfrak{J}}_0^C \text{ (Lemma 3.2.2)} = \widetilde{\mathfrak{J}}_0^C \text{ (Proposition 2.4.6.(2))}. \end{aligned}$                                                                                                                                                                                                                                                                                                                                                                                                  | $\mathfrak{a} \supset [\mathfrak{a}, \mathfrak{e}_6^C] \supset [\mathfrak{f}_4^C, \widetilde{\mathfrak{J}}_0^C] = \mathfrak{f}_4^C \widetilde{\mathfrak{J}}_0^C \text{ (Lemma 3.2.2)} = \widetilde{\mathfrak{J}}_0^C \text{ (Proposition 2.4.6.(2))}.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | $\mathfrak{a} \supset [\mathfrak{a}, \mathfrak{e}_6^C] \supset [\mathfrak{f}_4^C, \widetilde{\mathfrak{J}}_0^C] = \mathfrak{f}_4^C \widetilde{\mathfrak{J}}_0^C \text{ (Lemma 3.2.2)} = \widetilde{\mathfrak{J}}_0^C \text{ (Proposition 2.4.6.(2))}.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

| Identifier       | Page | Conf.  | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Rendered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Image |
|------------------|------|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 0902.0431_E08802 | 134  | ■0.338 | $\left(\begin{array}{c} \left(\begin{array}{ccc} \cos  a_1  a_2 \frac{\sin  a_2 }{ a_2 } a_3 \frac{\sin  a_3 }{ a_3 } & 0 & 0 \\ 0 & a_1 \frac{\sin  a_1 }{ a_1 } \cos  a_2  a_3 \frac{\sin  a_3 }{ a_3 } & 0 \\ 0 & 0 & a_1 \frac{\sin  a_1 }{ a_1 } a_2 \frac{\sin  a_2 }{ a_2 } \cos  a_3  \end{array}\right) \\ a_1 \frac{\sin  a_1 }{ a_1 } \cos  a_2  \cos  a_3  & 0 & 0 \\ 0 & \cos  a_1  a_2 \frac{\sin  a_2 }{ a_2 } \cos  a_3  & 0 \\ 0 & 0 & \cos  a_1  \cos  a_2  a_3 \frac{\sin  a_3 }{ a_3 } \end{array}\right) \\ \cos  a_1  \cos  a_2  \cos  a_3  & a_1 \frac{\sin  a_1 }{ a_1 } a_2 \frac{\sin  a_2 }{ a_2 } a_3 \frac{\sin  a_3 }{ a_3 } & \\ \cos  a_1  a_2 \frac{\sin  a_2 }{ a_2 } \cos  a_3  & 0 & \\ a_1 \frac{\sin  a_1 }{ a_1 } \cos  a_2  \cos  a_3  & 0 & \end{array}\right)$ | $\left(\begin{array}{c} \left(\begin{array}{ccc} \cos  a_1  a_2 \frac{\sin  a_2 }{ a_2 } a_3 \frac{\sin  a_3 }{ a_3 } & 0 & 0 \\ 0 & a_1 \frac{\sin  a_1 }{ a_1 } \cos  a_2  a_3 \frac{\sin  a_3 }{ a_3 } & 0 \\ 0 & 0 & a_1 \frac{\sin  a_1 }{ a_1 } a_2 \frac{\sin  a_2 }{ a_2 } \cos  a_3  \end{array}\right) \\ a_1 \frac{\sin  a_1 }{ a_1 } \cos  a_2  \cos  a_3  & 0 & 0 \\ 0 & \cos  a_1  a_2 \frac{\sin  a_2 }{ a_2 } \cos  a_3  & 0 \\ 0 & 0 & \cos  a_1  \cos  a_2  a_3 \frac{\sin  a_3 }{ a_3 } \end{array}\right) \\ \cos  a_1  \cos  a_2  \cos  a_3  & a_1 \frac{\sin  a_1 }{ a_1 } a_2 \frac{\sin  a_2 }{ a_2 } a_3 \frac{\sin  a_3 }{ a_3 } & \\ \cos  a_1  a_2 \frac{\sin  a_2 }{ a_2 } \cos  a_3  & 0 & \\ a_1 \frac{\sin  a_1 }{ a_1 } \cos  a_2  \cos  a_3  & 0 & \end{array}\right)$ |       |
| 0902.0431_E08875 | 145  | ■0.339 | $\begin{aligned} &\varphi_2(A) \left( \left( \begin{array}{ccc} \xi_1 & x_3 & \bar{x}_2 \\ \bar{x}_3 & \xi_2 & x_1 \\ x_2 & \bar{x}_1 & \xi_3 \end{array} \right), \left( \begin{array}{ccc} \eta_1 & y_3 & \bar{y}_2 \\ \bar{y}_3 & \eta_2 & y_1 \\ y_2 & \bar{y}_1 & \eta_3 \end{array} \right), \xi, \eta \right) \\ &= \left( \left( \begin{array}{ccc} \xi'_1 & x'_3 & \bar{x}'_2 \\ \bar{x}'_3 & \xi'_2 & x'_1 \\ x'_2 & \bar{x}'_1 & \xi'_3 \end{array} \right), \left( \begin{array}{ccc} \eta'_1 & y'_3 & \bar{y}'_2 \\ \bar{y}'_3 & \eta'_2 & y'_1 \\ y'_2 & \bar{y}'_1 & \eta'_3 \end{array} \right), \xi', \eta' \right) \end{aligned}$                                                                                                                                                      | $\varphi_2(A) \left( \left( \begin{array}{ccc} \xi_1 & x_3 & \bar{x}_2 \\ \bar{x}_3 & \xi_2 & x_1 \\ x_2 & \bar{x}_1 & \xi_3 \end{array} \right), \left( \begin{array}{ccc} \eta_1 & y_3 & \bar{y}_2 \\ \bar{y}_3 & \eta_2 & y_1 \\ y_2 & \bar{y}_1 & \eta_3 \end{array} \right), \xi, \eta \right) \\ = \left( \left( \begin{array}{ccc} \xi'_1 & x'_3 & \bar{x}'_2 \\ \bar{x}'_3 & \xi'_2 & x'_1 \\ x'_2 & \bar{x}'_1 & \xi'_3 \end{array} \right), \left( \begin{array}{ccc} \eta'_1 & y'_3 & \bar{y}'_2 \\ \bar{y}'_3 & \eta'_2 & y'_1 \\ y'_2 & \bar{y}'_1 & \eta'_3 \end{array} \right), \xi', \eta' \right),$                                                                                                                                                                                     |       |

| Identifier       | Page | Conf.   | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Rendered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Image                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|------------------|------|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0902.0431_E00400 | 66   | ■ 0.351 | $\backslash\mathrm{beta}_{\{1\}}\mathrm{m}=\backslash\mathrm{operatorname{prm}}\backslash\mathrm{bar}\{r\},\quad\backslash\mathrm{beta}_{\{2\}}\mathrm{m}=\mathrm{r}\mathrm{m}\backslash\mathrm{bar}\{r\}\mathrm{q},\quad\backslash\mathrm{beta}_{\{3\}}\mathrm{m}=\backslash\mathrm{bar}\{q\}\mathrm{r}\mathrm{m}\backslash\mathrm{bar}\{r\}\backslash\mathrm{bar}\{p\},\quad\mathrm{m}\in\boldsymbol{\mathrm{H}}.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | $\beta_1 m = \mathrm{prm} \bar{r}, \quad \beta_2 m = r m \bar{r} q, \quad \beta_3 m = \bar{q} r m \bar{r} \bar{p}, \quad m \in \boldsymbol{H}.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | $\beta_1 m = \mathrm{prm} \bar{r}, \quad \beta_2 m = r m \bar{r} q, \quad \beta_3 m = \bar{q} r m \bar{r} \bar{p}, \quad m \in \boldsymbol{H}.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 0902.0431_E00860 | 143  | ■ 0.354 | $\backslash\mathrm{begin}\{\mathrm{array}\}\{\mathrm{ccccccc}\}\mathrm{1}\&\backslash\mathrm{longrightarrow}\&\backslash\mathrm{operatorname{Spin}}(10)\&\backslash\mathrm{longrightarrow}\&\backslash\mathrm{left}(\backslash\mathrm{left}(\mathrm{E}_{\{7\}}\backslash\mathrm{right})^{\{\kappa,\mu\}}\backslash\mathrm{right})_{\backslash\mathrm{left}(0,\mathrm{E}_{\{1\}},0,1\backslash\mathrm{right})}\&\backslash\mathrm{longrightarrow}\&\mathrm{S}^{\{10\}}\&\backslash\mathrm{longrightarrow}\backslash\&\&\backslash\mathrm{downarrow}\mathrm{p}^{\{\mathrm{prime}\}}\&\&\backslash\mathrm{downarrow}\mathrm{p}\&\&\backslash\mathrm{downarrow}=\&\backslash\&\mathrm{1}\&\backslash\mathrm{longrightarrow}\&\mathrm{S}\mathrm{0}(10)\&\backslash\mathrm{longrightarrow}\&\mathrm{S}\mathrm{0}(11)\&\backslash\mathrm{longrightarrow}\&\mathrm{S}^{\{10\}}\&\backslash\mathrm{longrightarrow}\backslash\mathrm{end}\{\mathrm{array}\}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | $\begin{array}{ccccccc} 1 & \longrightarrow & \mathrm{Spin}(10) & \longrightarrow & ((E_7)^{\kappa,\mu})_{(0,E_1,0,1)} & \longrightarrow & S^{10} \longrightarrow * \\ & & \downarrow p' & & \downarrow p & & \downarrow = \\ 1 & \longrightarrow & SO(10) & \longrightarrow & SO(11) & \longrightarrow & S^{10} \longrightarrow * \end{array}$                                                                                                                                                                                                                                                                                                                                                                     | $\begin{array}{ccccccc} 1 & \longrightarrow & Spin(10) & \longrightarrow & ((E_7)^{\kappa,\mu})_{(0,E_1,0,1)} & \longrightarrow & S^{10} \longrightarrow * \\ & & \downarrow p' & & \downarrow p & & \downarrow = \\ 1 & \longrightarrow & SO(10) & \longrightarrow & SO(11) & \longrightarrow & S^{10} \longrightarrow * \end{array}$                                                                                                                                                                                                                                                                                                                                                                               |
| 0902.0431_E00876 | 145  | ■ 0.358 | $\backslash\mathrm{begin}\{\mathrm{gathered}\}\backslash\mathrm{binom}\{\backslash\mathrm{x}_1^{\{\mathrm{prime}\}}\}\{\mathrm{eta}^{\{\mathrm{prime}\}}\}=A\backslash\mathrm{binom}\{\backslash\mathrm{x}_1\}\{\mathrm{eta}\},\quad\backslash\mathrm{quad}\backslash\mathrm{binom}\{\backslash\mathrm{x}_1^{\{\mathrm{prime}\}}\}\{\mathrm{eta}_{\{1\}}^{\{\mathrm{prime}\}}\}=A\backslash\mathrm{binom}\{\backslash\mathrm{x}_1\}\{\mathrm{eta}_{\{1\}}\},\quad\backslash\mathrm{quad}\backslash\mathrm{binom}\{\mathrm{eta}_{\{2\}}^{\{\mathrm{prime}\}}\}\{\backslash\mathrm{x}_3^{\{\mathrm{prime}\}}\}=A\backslash\mathrm{binom}\{\mathrm{eta}_{\{2\}}\}\{\backslash\mathrm{x}_3\},\quad\backslash\mathrm{quad}\backslash\mathrm{binom}\{\mathrm{eta}_{\{3\}}^{\{\mathrm{prime}\}}\}\{\backslash\mathrm{x}_2^{\{\mathrm{prime}\}}\}=A\backslash\mathrm{binom}\{\mathrm{eta}_{\{3\}}\}\{\backslash\mathrm{x}_2\}\backslash\backslash\backslash\mathrm{binom}\{\mathrm{x}_{\{1\}}^{\{\mathrm{prime}\}}\}\{\mathrm{y}_{\{1\}}^{\{\mathrm{prime}\}}\}=\tau A\backslash\mathrm{binom}\{\mathrm{x}_{\{1\}}\}\{\mathrm{y}_{\{1\}}\},\quad\backslash\mathrm{quad}\backslash\mathrm{binom}\{\mathrm{x}_{\{2\}}^{\{\mathrm{prime}\}}\}\{\mathrm{y}_{\{2\}}^{\{\mathrm{prime}\}}\}=\backslash\mathrm{binom}\{\mathrm{x}_{\{2\}}\}\{\mathrm{y}_{\{2\}}\},\quad\backslash\mathrm{quad}\backslash\mathrm{binom}\{\mathrm{x}_{\{3\}}^{\{\mathrm{prime}\}}\}\{\mathrm{y}_{\{3\}}^{\{\mathrm{prime}\}}\}=\backslash\mathrm{binom}\{\mathrm{x}_{\{3\}}\}\{\mathrm{y}_{\{3\}}\}.\quad\backslash\mathrm{end}\{\mathrm{gathered}\}$ | $\begin{pmatrix} \xi_1' \\ \eta' \end{pmatrix} = A \begin{pmatrix} \xi_1 \\ \eta \end{pmatrix}, \quad \begin{pmatrix} \xi_1' \\ \eta_1' \end{pmatrix} = A \begin{pmatrix} \xi \\ \eta_1 \end{pmatrix}, \quad \begin{pmatrix} \eta_2' \\ \xi_3' \end{pmatrix} = A \begin{pmatrix} \eta_2 \\ \xi_3 \end{pmatrix}, \quad \begin{pmatrix} \eta_3' \\ \xi_2' \end{pmatrix} = A \begin{pmatrix} \eta_3 \\ \xi_2 \end{pmatrix} \\ \begin{pmatrix} x_1' \\ y_1' \end{pmatrix} = \tau A \begin{pmatrix} x_1 \\ y_1 \end{pmatrix}, \quad \begin{pmatrix} x_2' \\ y_2' \end{pmatrix} = \begin{pmatrix} x_2 \\ y_2 \end{pmatrix}, \quad \begin{pmatrix} x_3' \\ y_3' \end{pmatrix} = \begin{pmatrix} x_3 \\ y_3 \end{pmatrix}.$ | $\begin{pmatrix} \xi_1' \\ \eta' \end{pmatrix} = A \begin{pmatrix} \xi_1 \\ \eta \end{pmatrix}, \quad \begin{pmatrix} \xi_1' \\ \eta_1' \end{pmatrix} = A \begin{pmatrix} \xi \\ \eta_1 \end{pmatrix}, \quad \begin{pmatrix} \eta_2' \\ \xi_3' \end{pmatrix} = A \begin{pmatrix} \eta_2 \\ \xi_3 \end{pmatrix}, \quad \begin{pmatrix} \eta_3' \\ \xi_2' \end{pmatrix} = A \begin{pmatrix} \eta_3 \\ \xi_2 \end{pmatrix}, \\ \begin{pmatrix} x_1' \\ y_1' \end{pmatrix} = \tau A \begin{pmatrix} x_1 \\ y_1 \end{pmatrix}, \quad \begin{pmatrix} x_2' \\ y_2' \end{pmatrix} = \begin{pmatrix} x_2 \\ y_2 \end{pmatrix}, \quad \begin{pmatrix} x_3' \\ y_3' \end{pmatrix} = \begin{pmatrix} x_3 \\ y_3 \end{pmatrix}.$ |

[illegible]

| Identifier       | Page | Conf.   | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Rendered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Image                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|------------------|------|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0902.0431_E00093 | 19   | ■ 0.382 | $\begin{aligned} H_{\{1\}} H_{\{1\}} e_{\{2\}} &= H_{\{1\}} e_{\{3\}} = -e_{\{2\}}, \\ &\quad \end{aligned}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | $H_1 H_1 e_2 = H_1 e_3 = -e_2,$                                                                                                                                                                                                                                                                                                                                                                                                                                                              | $\begin{aligned} H_1 H_1 e_2 &= H_1 e_3 = -e_2, & H_1 H_1 e_3 &= -H_1 e_2 = -e_3, \\ H_1 H_1 e_4 &= -H_1 e_5 = -e_4, & H_1 H_1 e_5 &= H_1 e_4 = -e_4, \\ H_1 H_1 e_i &= 0 & \text{otherwise.} \end{aligned}$                                                                                                                                                                                                                                                                                 |
| 0902.0431_E00371 | 61   | ■ 0.388 | $\begin{aligned} &\operatorname{tr}(X) = \operatorname{tr}(\alpha X) \quad (\text{Lemma 2.2.1.(2)}) = \sum_{i=1}^3 \xi_i, \\ (X, X) &= (\alpha X, \alpha X) \quad (\text{Lemma 2.2.4}) = \sum_{i=1}^3 \xi_i^2, \\ \operatorname{tr}(X, X, X) &= \operatorname{tr}(\alpha X, \alpha X, \alpha X) \quad (\text{Lemma 2.2.4}) = \sum_{i=1}^3 \xi_i^3. \end{aligned}$                                                                                                                                                                                                                                                                                                                                                                                                                               | $\begin{aligned} \operatorname{tr}(X) &= \operatorname{tr}(\alpha X) \quad (\text{Lemma 2.2.1.(2)}) = \sum_{i=1}^3 \xi_i, \\ (X, X) &= (\alpha X, \alpha X) \quad (\text{Lemma 2.2.4}) = \sum_{i=1}^3 \xi_i^2, \\ \operatorname{tr}(X, X, X) &= \operatorname{tr}(\alpha X, \alpha X, \alpha X) \quad (\text{Lemma 2.2.4}) = \sum_{i=1}^3 \xi_i^3. \end{aligned}$                                                                                                                            | $\begin{aligned} \operatorname{tr}(X) &= \operatorname{tr}(\alpha X) \quad (\text{Lemma 2.2.1.(2)}) = \sum_{i=1}^3 \xi_i, \\ (X, X) &= (\alpha X, \alpha X) \quad (\text{Lemma 2.2.4}) = \sum_{i=1}^3 \xi_i^2, \\ \operatorname{tr}(X, X, X) &= \operatorname{tr}(\alpha X, \alpha X, \alpha X) \quad (\text{Lemma 2.2.4}) = \sum_{i=1}^3 \xi_i^3. \end{aligned}$                                                                                                                            |
| 0902.0431_E00602 | 100  | ■ 0.396 | $\begin{aligned} &\left. \begin{aligned} &\left( \begin{array}{rrl} \mu_1 & m_3 & \bar{m}_2 \\ \bar{m}_3 & \mu_2 & m_1 \\ m_2 & \bar{m}_1 & \mu_3 \end{array} \right) & \left( \begin{array}{rrl} 0 & a_3 e_4 & -a_2 e_4 \\ -a_3 e_4 & 0 & a_1 e_4 \\ a_2 e_4 & -a_1 e_4 & 0 \end{array} \right) & \left  \begin{array}{l} \mu_i \in \mathbf{R} \\ m_i, a_i \in \mathbf{H} \end{array} \right\} \end{aligned} \right\} \\ &\left( \begin{array}{rrl} \mu_1 & m_3 & \bar{m}_2 \\ \bar{m}_3 & \mu_2 & m_1 \\ m_2 & \bar{m}_1 & \mu_3 \end{array} \right) & \left( \begin{array}{rrl} 0 & a_3 e_4 & -a_2 e_4 \\ -a_3 e_4 & 0 & a_1 e_4 \\ a_2 e_4 & -a_1 e_4 & 0 \end{array} \right) & \left  \begin{array}{l} \mu_i \in \mathbf{R} \\ m_i, a_i \in \mathbf{H} \end{array} \right\} \end{aligned}$ | $\begin{aligned} (\mathfrak{J}^C)_{\tau\gamma} &= \{X \in \mathfrak{J}^C \mid \tau\gamma X = X\} \\ &= \left\{ \left( \begin{array}{rrl} \mu_1 & m_3 & \bar{m}_2 \\ \bar{m}_3 & \mu_2 & m_1 \\ m_2 & \bar{m}_1 & \mu_3 \end{array} \right) + i \left( \begin{array}{rrl} 0 & a_3 e_4 & -a_2 e_4 \\ -a_3 e_4 & 0 & a_1 e_4 \\ a_2 e_4 & -a_1 e_4 & 0 \end{array} \right) \left  \begin{array}{l} \mu_i \in \mathbf{R} \\ m_i, a_i \in \mathbf{H} \end{array} \right\} \right\} \end{aligned}$ | $\begin{aligned} (\mathfrak{J}^C)_{\tau\gamma} &= \{X \in \mathfrak{J}^C \mid \tau\gamma X = X\} \\ &= \left\{ \left( \begin{array}{rrl} \mu_1 & m_3 & \bar{m}_2 \\ \bar{m}_3 & \mu_2 & m_1 \\ m_2 & \bar{m}_1 & \mu_3 \end{array} \right) + i \left( \begin{array}{rrl} 0 & a_3 e_4 & -a_2 e_4 \\ -a_3 e_4 & 0 & a_1 e_4 \\ a_2 e_4 & -a_1 e_4 & 0 \end{array} \right) \left  \begin{array}{l} \mu_i \in \mathbf{R} \\ m_i, a_i \in \mathbf{H} \end{array} \right\} \right\} \end{aligned}$ |

| Identifier       | Page | Conf.   | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Rendered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Image                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|------------------|------|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0902.0431_E00900 | 149  | ■ 0.397 | $\begin{aligned} P = & (X, Y, \xi, \eta) \\ & \stackrel{\chi}{\rightarrow} k\left(gX - \frac{\xi}{2}E\right)J + e_1k\left(g(\gamma Y) - \frac{\eta}{2}E\right)J \\ & \longrightarrow e_1k(T)k\left(gX - \frac{\xi}{2}E\right)J - k(T)k\left(g(\gamma Y) - \frac{\eta}{2}E\right)J \\ & \quad + e_1k\left(gX - \frac{\xi}{2}E\right)J^tk(T) - k\left(g(\gamma Y) - \frac{\eta}{2}E\right)J^tk(T) \\ & = k(-Tg(\gamma Y) - g(\gamma Y)T + \eta T)J + e_1k(T(gX) + (gX)T - \xi T)J \\ & = k(-2gA \circ g(\gamma Y) + \eta gA)J + e_1k(2gA \circ gX - \xi gA)J \\ & = k\left(-2g(\gamma A \times Y) - \frac{1}{2}(A, Y)E + \eta gA\right)J \\ & \quad + e_1k\left(2g(\gamma A \times \gamma X) + \frac{1}{2}(\gamma A, X)E - \xi gA\right)J \text{ (Lemma 3.12.1)} \\ & = \chi\left(\begin{array}{c} -2\gamma A \times Y + \eta A \\ 2A \times X - \xi \gamma A \\ (A, Y) \\ (-\gamma A, X) \end{array}\right) = \chi\left(\left(\begin{array}{cccc} 0 & -2\gamma A & 0 & A \\ 2A & 0 & -\gamma A & 0 \\ 0 & A & 0 & 0 \\ -\gamma A & 0 & 0 & 0 \end{array}\right)\left(\begin{array}{c} X \\ Y \\ \xi \\ \eta \end{array}\right)\right) \\ & = \chi(\Phi(0, A, -\gamma A, 0)P). \end{aligned}$ | $\begin{aligned} P = & (X, Y, \xi, \eta) \\ & \stackrel{\chi}{\rightarrow} k\left(gX - \frac{\xi}{2}E\right)J + e_1k\left(g(\gamma Y) - \frac{\eta}{2}E\right)J \\ & \longrightarrow e_1k(T)k\left(gX - \frac{\xi}{2}E\right)J - k(T)k\left(g(\gamma Y) - \frac{\eta}{2}E\right)J \\ & \quad + e_1k\left(gX - \frac{\xi}{2}E\right)J^tk(T) - k\left(g(\gamma Y) - \frac{\eta}{2}E\right)J^tk(T) \\ & = k(-Tg(\gamma Y) - g(\gamma Y)T + \eta T)J + e_1k(T(gX) + (gX)T - \xi T)J \\ & = k(-2gA \circ g(\gamma Y) + \eta gA)J + e_1k(2gA \circ gX - \xi gA)J \\ & = k\left(-2g(\gamma A \times Y) - \frac{1}{2}(A, Y)E + \eta gA\right)J \\ & \quad + e_1k\left(2g(\gamma A \times \gamma X) + \frac{1}{2}(\gamma A, X)E - \xi gA\right)J \text{ (Lemma 3.12.1)} \\ & = \chi\left(\begin{array}{c} -2\gamma A \times Y + \eta A \\ 2A \times X - \xi \gamma A \\ (A, Y) \end{array}\right) = \chi\left(\left(\begin{array}{cccc} 0 & -2\gamma A & 0 & A \\ 2A & 0 & -\gamma A & 0 \\ 0 & A & 0 & 0 \\ -\gamma A & 0 & 0 & 0 \end{array}\right)\left(\begin{array}{c} X \\ Y \\ \xi \\ \eta \end{array}\right)\right) \\ & = \chi(\Phi(0, A, -\gamma A, 0)P) \end{aligned}$ | $P = (X, Y, \xi, \eta)$ $\stackrel{\chi}{\rightarrow} k\left(gX - \frac{\xi}{2}E\right)J + e_1k\left(g(\gamma Y) - \frac{\eta}{2}E\right)J$ $\longrightarrow e_1k(T)k\left(gX - \frac{\xi}{2}E\right)J - k(T)k\left(g(\gamma Y) - \frac{\eta}{2}E\right)J$ $+ e_1k\left(gX - \frac{\xi}{2}E\right)J^tk(T) - k\left(g(\gamma Y) - \frac{\eta}{2}E\right)J^tk(T)$ $= k(-Tg(\gamma Y) - g(\gamma Y)T + \eta T)J + e_1k(T(gX) + (gX)T - \xi T)J$ $= k(-2gA \circ g(\gamma Y) + \eta gA)J + e_1k(2gA \circ gX - \xi gA)J$ $= k\left(-2g(\gamma A \times Y) - \frac{1}{2}(A, Y)E + \eta gA\right)J$ $+ e_1k\left(2g(\gamma A \times \gamma X) + \frac{1}{2}(\gamma A, X)E - \xi gA\right)J \text{ (Lemma 3.12.1)}$ $= \chi\left(\begin{array}{c} -2\gamma A \times Y + \eta A \\ 2A \times X - \xi \gamma A \\ (A, Y) \\ (-\gamma A, X) \end{array}\right) = \chi\left(\left(\begin{array}{cccc} 0 & -2\gamma A & 0 & A \\ 2A & 0 & -\gamma A & 0 \\ 0 & A & 0 & 0 \\ -\gamma A & 0 & 0 & 0 \end{array}\right)\left(\begin{array}{c} X \\ Y \\ \xi \\ \eta \end{array}\right)\right)$ $= \chi(\Phi(0, A, -\gamma A, 0)P).$ |
| 0902.0431_E01000 | 183  | ■ 0.403 | $S_{\{1\}} \tau S_{\{2\}} - S_{\{2\}} \tau S_{\{1\}} - \frac{1}{8}(S_{\{1\}} \tau S_{\{2\}} - S_{\{2\}} \tau S_{\{1\}}) \in \mathfrak{su}(8),$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | $S_1 \tau S_2 - S_2 \tau S_1 - \frac{1}{8}(S_1 \tau S_2 - S_2 \tau S_1) \in \mathfrak{su}(8),$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | $S_1 \tau S_2 - S_2 \tau S_1 - \frac{1}{8}(S_1 \tau S_2 - S_2 \tau S_1) \in \mathfrak{su}(8),$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 0902.0431_E01030 | 173  | ■ 0.410 | $\begin{aligned} & \lambda_0 + \frac{1}{2}\mu_1 - \frac{2}{3}\nu = 2 \quad 4 \quad 5 \quad 4 \quad 3 \quad 2 \quad 1 \quad 2 \\ & \lambda_0 - \frac{1}{2}\mu_1 + \frac{2}{3}\nu = 2 \quad 3 \quad 5 \quad 4 \\ & \lambda_1 + \frac{1}{2}\mu_1 - \frac{2}{3}\nu = 0 \quad 1 \quad 1 \quad 1 \quad 2 \\ & \lambda_1 - \frac{1}{2}\mu_1 + \frac{2}{3}\nu = 0 \quad 0 \quad 1 \quad 1 \quad 1 \quad 1 \quad 1 \end{aligned}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | $\begin{aligned} & \lambda_0 + \frac{1}{2}\mu_1 - \frac{2}{3}\nu = 2 \quad 4 \quad 5 \quad 4 \quad 3 \quad 2 \quad 1 \quad 2 \\ & \lambda_0 - \frac{1}{2}\mu_1 + \frac{2}{3}\nu = 2 \quad 3 \quad 5 \quad 4 \\ & \lambda_1 + \frac{1}{2}\mu_1 - \frac{2}{3}\nu = 0 \quad 1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 0 \\ & \lambda_1 - \frac{1}{2}\mu_1 + \frac{2}{3}\nu = 0 \quad 0 \quad 1 \quad 1 \quad 1 \quad 1 \quad 1 \end{aligned}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | $\begin{aligned} & \lambda_0 + \frac{1}{2}\mu_1 - \frac{2}{3}\nu = 2 \quad 4 \quad 5 \quad 4 \quad 3 \quad 2 \quad 1 \quad 2 \\ & \lambda_0 - \frac{1}{2}\mu_1 + \frac{2}{3}\nu = 2 \quad 3 \quad 5 \quad 4 \quad 3 \quad 2 \quad 1 \quad 3 \\ & \lambda_1 + \frac{1}{2}\mu_1 - \frac{2}{3}\nu = 0 \quad 1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 0 \\ & \lambda_1 - \frac{1}{2}\mu_1 + \frac{2}{3}\nu = 0 \quad 0 \quad 1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 1 \end{aligned}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

| Identifier       | Page | Conf.  | LaTeX source                                                                                                                                                                                                                                                                                                                                         | Rendered                                                                                                                                                                                                                                                                                                                                             | Image                                                                                                                                                                                                                                                                                                                                                |
|------------------|------|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0902.0431_E00492 | 81   | ■0.411 | $\delta\left(\begin{array}{ccc}\xi_1 & x_{12} & x_{13} \\ x_{21} & \xi_2 & x_{23} \\ x_{31} & x_{32} & \xi_3\end{array}\right)=\left(\begin{array}{ccc}0 & \delta_3x_{12} & \bar{\delta}_2\bar{x}_{13} \\ \bar{\delta}_3\bar{x}_{21} & 0 & \delta_1x_{23} \\ \delta_2x_{31} & \bar{\delta}_1\bar{x}_{32} & 0\end{array}\right).$                     | $\delta\left(\begin{array}{ccc}\xi_1 & x_{12} & x_{13} \\ x_{21} & \xi_2 & x_{23} \\ x_{31} & x_{32} & \xi_3\end{array}\right)=\left(\begin{array}{ccc}0 & \delta_3x_{12} & \bar{\delta}_2\bar{x}_{13} \\ \bar{\delta}_3\bar{x}_{21} & 0 & \delta_1x_{23} \\ \delta_2x_{31} & \bar{\delta}_1\bar{x}_{32} & 0\end{array}\right).$                     | $\delta\left(\begin{array}{ccc}\xi_1 & x_{12} & x_{13} \\ x_{21} & \xi_2 & x_{23} \\ x_{31} & x_{32} & \xi_3\end{array}\right)=\left(\begin{array}{ccc}0 & \delta_3x_{12} & \bar{\delta}_2\bar{x}_{13} \\ \bar{\delta}_3\bar{x}_{21} & 0 & \delta_1x_{23} \\ \delta_2x_{31} & \bar{\delta}_1\bar{x}_{32} & 0\end{array}\right).$                     |
| 0902.0431_E01097 | 186  | ■0.420 | $\begin{aligned} \zeta[(icE),l(S)I]&=\zeta(2l(icS)I) \\ &=\Theta(0,4\lambda\gamma\chi^{-1}(icS),4\chi^{-1}(icS),0,0,0) \\ &=\Theta(0,4\chi^{-1}(cS),-4\lambda\gamma\chi^{-1}(cS),0,0,0)\quad(\text{Lemma 5.8.3}) \\ &=[\Theta(0,0,0,0,2c,-2c),\Theta(0,2\lambda\gamma\chi^{-1}S,2\chi^{-1}S,0,0,0)] \\ &=[\zeta l(icE),\zeta(l(S)I)]. \end{aligned}$ | $\begin{aligned} \zeta[(icE),l(S)I]&=\zeta(2l(icS)I) \\ &=\Theta(0,4\lambda\gamma\chi^{-1}(icS),4\chi^{-1}(icS),0,0,0) \\ &=\Theta(0,4\chi^{-1}(cS),-4\lambda\gamma\chi^{-1}(cS),0,0,0)\quad(\text{Lemma 5.8.3}) \\ &=[\Theta(0,0,0,0,2c,-2c),\Theta(0,2\lambda\gamma\chi^{-1}S,2\chi^{-1}S,0,0,0)] \\ &=[\zeta l(icE),\zeta(l(S)I)]. \end{aligned}$ | $\begin{aligned} \zeta[(icE),l(S)I]&=\zeta(2l(icS)I) \\ &=\Theta(0,4\lambda\gamma\chi^{-1}(icS),4\chi^{-1}(icS),0,0,0) \\ &=\Theta(0,4\chi^{-1}(cS),-4\lambda\gamma\chi^{-1}(cS),0,0,0)\quad(\text{Lemma 5.8.3}) \\ &=[\Theta(0,0,0,0,2c,-2c),\Theta(0,2\lambda\gamma\chi^{-1}S,2\chi^{-1}S,0,0,0)] \\ &=[\zeta l(icE),\zeta(l(S)I)]. \end{aligned}$ |
| 0902.0431_E00694 | 115  | ■0.434 | $\mathfrak{e}_7^C\ni f_r\Phi f_r^{-1}=\left(\begin{array}{cccc}g&r^{-1}l&r^{-2}C&rA\\rk&h&r^{-1}B&r^2D\\r^2c&ra&\nu&r^3\lambda\\r^{-1}b&r^{-2}d&r^{-3}k&\mu\end{array}\right)$                                                                                                                                                                       | $\mathfrak{e}_7^C\ni f_r\Phi f_r^{-1}=\left(\begin{array}{cccc}g&r^{-1}l&r^{-2}C&rA\\rk&h&r^{-1}B&r^2D\\r^2c&ra&\nu&r^3\lambda\\r^{-1}b&r^{-2}d&r^{-3}k&\mu\end{array}\right)$                                                                                                                                                                       | $\mathfrak{e}_7^C\ni f_r\Phi f_r^{-1}=\left(\begin{array}{cccc}g&r^{-1}l&r^{-2}C&rA\\rk&h&r^{-1}B&r^2D\\r^2c&ra&\nu&r^3\lambda\\r^{-1}b&r^{-2}d&r^{-3}k&\mu\end{array}\right)$                                                                                                                                                                       |
| 0902.0431_E00642 | 106  | ■0.434 | $\begin{aligned} &h(a,b)+h(a',b')=h(a+a',b+b'),\quad h(ca,cb)=ch(a,b),c\in\boldsymbol{C}, \\ &h(A,B)+h(A',B')=h(A+A',B+B'),\quad h(cA,cB)=h(A,B),c\in\boldsymbol{C}. \end{aligned}$                                                                                                                                                                  | $\begin{aligned} &h(a,b)+h(a',b')=h(a+a',b+b'),\quad h(ca,cb)=ch(a,b),c\in\boldsymbol{C} \\ &h(A,B)+h(A',B')=h(A+A',B+B'),\quad h(cA,cB)=h(A,B),c\in\boldsymbol{C}. \end{aligned}$                                                                                                                                                                   | $\begin{aligned} &h(a,b)+h(a',b')=h(a+a',b+b'),\quad h(ca,cb)=ch(a,b),c\in\boldsymbol{C}, \\ &h(A,B)+h(A',B')=h(A+A',B+B'),\quad h(cA,cB)=h(A,B),c\in\boldsymbol{C}. \end{aligned}$                                                                                                                                                                  |
| 0902.0431_E00558 | 94   | ■0.435 | $\begin{array}{l} 1\longrightarrow Spin(9)\longrightarrow (E_6)_{E_1}\longrightarrow S^9\longrightarrow * \\ \qquad\qquad\qquad\downarrow p' \qquad\qquad\qquad\downarrow p \qquad\qquad\qquad\downarrow= \\ 1\longrightarrow SO(9)\longrightarrow SO(10)\longrightarrow S^9\longrightarrow * \end{array}$                                           | $\begin{array}{l} 1\longrightarrow Spin(9)\longrightarrow (E_6)_{E_1}\longrightarrow S^9\longrightarrow * \\ \qquad\qquad\qquad\downarrow p' \qquad\qquad\qquad\downarrow p \qquad\qquad\qquad\downarrow= \\ 1\longrightarrow SO(9)\longrightarrow SO(10)\longrightarrow S^9\longrightarrow * \end{array}$                                           | $\begin{array}{l} 1\longrightarrow Spin(9)\longrightarrow (E_6)_{E_1}\longrightarrow S^9\longrightarrow * \\ \qquad\qquad\qquad\downarrow p' \qquad\qquad\qquad\downarrow p \qquad\qquad\qquad\downarrow= \\ 1\longrightarrow SO(9)\longrightarrow SO(10)\longrightarrow S^9\longrightarrow * \end{array}$                                           |

| Identifier       | Page | Conf.   | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Rendered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Image |
|------------------|------|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 0902.0431_E00332 | 53   | ■ 0.438 | $\begin{aligned} & \pm\left(\lambda_k-\lambda_l\right), \pm\left(\lambda_k+\lambda_l\right), \quad 0 \leq k<l \leq 3, \\ & \pm \lambda_0, \quad \pm \lambda_1, \quad \pm \lambda_2, \quad \pm \lambda_3, \\ & \pm \frac{1}{2}\left(-\lambda_0-\lambda_1+\lambda_2-\lambda_3\right), \pm \frac{1}{2}\left(\lambda_0+\lambda_1+\lambda_2-\lambda_3\right), \\ & \pm \frac{1}{2}\left(-\lambda_0+\lambda_1+\lambda_2+\lambda_3\right), \pm \frac{1}{2}\left(\lambda_0-\lambda_1+\lambda_2+\lambda_3\right), \\ & \pm \frac{1}{2}\left(\lambda_0-\lambda_1+\lambda_2-\lambda_3\right), \pm \frac{1}{2}\left(-\lambda_0+\lambda_1+\lambda_2-\lambda_3\right), \\ & \pm \frac{1}{2}\left(\lambda_0+\lambda_1+\lambda_2+\lambda_3\right), \pm \frac{1}{2}\left(-\lambda_0-\lambda_1+\lambda_2+\lambda_3\right) . \end{aligned}$                                                                                                                                                                                                                                                                                                                                                        | $\begin{aligned} & \pm\left(\lambda_k-\lambda_l\right), \pm\left(\lambda_k+\lambda_l\right), \quad 0 \leq k<l \leq 3, \\ & \pm \lambda_0, \quad \pm \lambda_1, \quad \pm \lambda_2, \quad \pm \lambda_3, \\ & \pm \frac{1}{2}\left(-\lambda_0-\lambda_1+\lambda_2-\lambda_3\right), \pm \frac{1}{2}\left(\lambda_0+\lambda_1+\lambda_2-\lambda_3\right), \\ & \pm \frac{1}{2}\left(-\lambda_0+\lambda_1+\lambda_2+\lambda_3\right), \pm \frac{1}{2}\left(\lambda_0-\lambda_1+\lambda_2+\lambda_3\right), \\ & \pm \frac{1}{2}\left(\lambda_0-\lambda_1+\lambda_2-\lambda_3\right), \pm \frac{1}{2}\left(-\lambda_0+\lambda_1+\lambda_2-\lambda_3\right), \\ & \pm \frac{1}{2}\left(\lambda_0+\lambda_1+\lambda_2+\lambda_3\right), \pm \frac{1}{2}\left(-\lambda_0-\lambda_1+\lambda_2+\lambda_3\right) . \end{aligned}$                                                                                                                                                                                                                                                                                                                                |       |
| 0902.0431_EQ1211 | 203  | ■ 0.448 | $\begin{aligned} & \exp (\operatorname{ad}\left\{Z, 0\right\})\left(C, D, \boldsymbol{x} \otimes \boldsymbol{a}, \boldsymbol{b} \otimes \boldsymbol{y}, \boldsymbol{c} \otimes \boldsymbol{z}, \boldsymbol{w} \otimes \boldsymbol{d}\right) \\ & =\left(\exp (\operatorname{ad}(Z)) C, D,\left((\exp Z) \boldsymbol{x}\right) \otimes \boldsymbol{a},\right. \\ & \quad\left.\left((\exp Z) \boldsymbol{b}\right) \otimes \boldsymbol{y},\left((\exp \left(-{}^t Z\right)) \boldsymbol{c} \otimes \boldsymbol{z},\left((\exp \left(-{}^t Z\right)) \boldsymbol{w}\right) \otimes \boldsymbol{d}\right)\right) \\ & =\left(\operatorname{Ad}(\exp Z) C, D,\left((\exp Z) \boldsymbol{x}\right) \otimes \boldsymbol{a},\right. \\ & \quad\left.\left((\exp Z) \boldsymbol{b}\right) \otimes \boldsymbol{y},\left({}^t(\exp Z)^{-1} \boldsymbol{c}\right) \otimes \boldsymbol{z},\left({}^t(\exp Z)^{-1} \boldsymbol{w}\right) \otimes \boldsymbol{d}\right) \\ & =\varphi_1(\exp Z)\left(C, D, \boldsymbol{x} \otimes \boldsymbol{a}, \boldsymbol{b} \otimes \boldsymbol{y}, \boldsymbol{c} \otimes \boldsymbol{z}, \boldsymbol{w} \otimes \boldsymbol{d}\right) . \end{aligned}$ | $\begin{aligned} & \exp (\operatorname{ad}(Z, 0))(C, D, \boldsymbol{x} \otimes \boldsymbol{a}, \boldsymbol{b} \otimes \boldsymbol{y}, \boldsymbol{c} \otimes \boldsymbol{z}, \boldsymbol{w} \otimes \boldsymbol{d}) \\ & =\left(\exp (\operatorname{ad}(Z)) C, D,\left((\exp Z) \boldsymbol{x}\right) \otimes \boldsymbol{a},\right. \\ & \quad\left.\left((\exp Z) \boldsymbol{b}\right) \otimes \boldsymbol{y},\left((\exp \left(-{}^t Z\right)) \boldsymbol{c} \otimes \boldsymbol{z},\left((\exp \left(-{}^t Z\right)) \boldsymbol{w}\right) \otimes \boldsymbol{d}\right)\right) \\ & =\left(\operatorname{Ad}(\exp Z) C, D,\left((\exp Z) \boldsymbol{x}\right) \otimes \boldsymbol{a},\right. \\ & \quad\left.\left((\exp Z) \boldsymbol{b}\right) \otimes \boldsymbol{y},\left({}^t(\exp Z)^{-1} \boldsymbol{c}\right) \otimes \boldsymbol{z},\left({}^t(\exp Z)^{-1} \boldsymbol{w}\right) \otimes \boldsymbol{d}\right) \\ & =\varphi_1(\exp Z)\left(C, D, \boldsymbol{x} \otimes \boldsymbol{a}, \boldsymbol{b} \otimes \boldsymbol{y}, \boldsymbol{c} \otimes \boldsymbol{z}, \boldsymbol{w} \otimes \boldsymbol{d}\right) . \end{aligned}$ |       |

| Identifier       | Page | Conf.  | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Rendered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Image                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|------------------|------|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0902.0431_E01039 | 177  | ■0.452 | $\begin{aligned} H_{\alpha_3} &= \left(0, 0, 0, \frac{1}{60}, 0, 0\right), \\ H_{\alpha_4} &= \left(\Phi\left(\frac{1}{90}(-E_1 + 2E_2 - E_3)^\sim, 0, 0, -\frac{1}{120}\right), 0, 0, -\frac{1}{120}, 0, 0\right), \\ H_{\alpha_5} &= \left(\Phi\left(\frac{1}{60}(H_3 - (E_2 - E_3)^\sim), 0, 0, 0\right), 0, 0, 0, 0, 0\right), \\ H_{\alpha_6} &= \left(\Phi\left(\frac{1}{60}(H_2 - H_3), 0, 0, 0\right), 0, 0, 0, 0, 0\right), \\ H_{\alpha_7} &= \left(\Phi\left(\frac{1}{60}(H_1 - H_2), 0, 0, 0\right), 0, 0, 0, 0, 0\right), \\ H_{\alpha_8} &= \left(\Phi\left(0, 0, 0, \frac{1}{40}\right), 0, 0, -\frac{1}{120}, 0, 0\right). \end{aligned}$ | $\begin{aligned} H_{\alpha_3} &= \left(0, 0, 0, \frac{1}{60}, 0, 0\right), \\ H_{\alpha_4} &= \left(\Phi\left(\frac{1}{90}(-E_1 + 2E_2 - E_3)^\sim, 0, 0, -\frac{1}{120}\right), 0, 0, -\frac{1}{120}, 0, 0\right), \\ H_{\alpha_5} &= \left(\Phi\left(\frac{1}{60}(H_3 - (E_2 - E_3)^\sim), 0, 0, 0\right), 0, 0, 0, 0, 0\right), \\ H_{\alpha_6} &= \left(\Phi\left(\frac{1}{60}(H_2 - H_3), 0, 0, 0\right), 0, 0, 0, 0, 0\right), \\ H_{\alpha_7} &= \left(\Phi\left(\frac{1}{60}(H_1 - H_2), 0, 0, 0\right), 0, 0, 0, 0, 0\right), \\ H_{\alpha_8} &= \left(\Phi\left(0, 0, 0, \frac{1}{40}\right), 0, 0, -\frac{1}{120}, 0, 0\right). \end{aligned}$ |                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 0902.0431_E00939 | 156  | ■0.471 | $\left.\begin{array}{l} \xrightarrow{f^{-1}} \left( \begin{array}{ccc} h(b_{11} - c_{11} - \frac{\nu}{3}) & -h(c_{12}, b_{12}) & -h(c_{13}, b_{13}) \\ -h(\bar{b}_{12}, \bar{c}_{12}) & 0 & 0 \\ -h(\bar{b}_{13}, \bar{c}_{13}) & 0 & 0 \end{array} \right) \\ = \left( \begin{array}{ccc} 0 & 0 & 0 \\ 0 & h(l_{33}) & -h(l_{23}, \bar{l}_{32}) \\ 0 & -h(l_{32}, \bar{l}_{23}) & h(l_{22}) \end{array} \right) \\ 0 \\ -h(\bar{l}_{11}) \end{array} \right).$                                                                                                                                                                                           | $\begin{aligned} & \xrightarrow{f^{-1}} \left( \begin{array}{ccc} h(b_{11} - c_{11} - \frac{\nu}{3}) & -h(c_{12}, b_{12}) & -h(c_{13}, b_{13}) \\ -h(\bar{b}_{12}, \bar{c}_{12}) & 0 & 0 \\ -h(\bar{b}_{13}, \bar{c}_{13}) & 0 & 0 \end{array} \right) \\ & = \left( \begin{array}{ccc} 0 & 0 & 0 \\ 0 & h(l_{33}) & -h(l_{23}, \bar{l}_{32}) \\ 0 & -h(l_{32}, \bar{l}_{23}) & h(l_{22}) \end{array} \right) \\ & 0 \\ & -h(\bar{l}_{11}) \end{aligned} \right).$                                                                                                                                                                                        | $\begin{aligned} & \xrightarrow{f^{-1}} \left( \begin{array}{ccc} h(b_{11} - c_{11} - \frac{\nu}{3}) & -h(c_{12}, b_{12}) & -h(c_{13}, b_{13}) \\ -h(\bar{b}_{12}, \bar{c}_{12}) & 0 & 0 \\ -h(\bar{b}_{13}, \bar{c}_{13}) & 0 & 0 \end{array} \right) \\ & = \left( \begin{array}{ccc} 0 & 0 & 0 \\ 0 & h(l_{33}) & -h(l_{23}, \bar{l}_{32}) \\ 0 & -h(l_{32}, \bar{l}_{23}) & h(l_{22}) \end{array} \right) \\ & 0 \\ & -h(\bar{l}_{11}) \end{aligned} \right).$ |



| Identifier       | Page | Conf. | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Rendered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Image                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
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| 0902.0431_EQ1150 | 194  | 0.504 | $\begin{aligned} & (\boldsymbol{u} \times *(\boldsymbol{v} \wedge \boldsymbol{w}) + \boldsymbol{v} \times *(\boldsymbol{w} \wedge \boldsymbol{u}) + \boldsymbol{w} \times *(\boldsymbol{u} \wedge \boldsymbol{v}))\boldsymbol{x} \\ & = \sum_{j=1}^9 (\boldsymbol{u}_1 \wedge \cdots \wedge \boldsymbol{u}_{j-1} \wedge \boldsymbol{x} \wedge \boldsymbol{u}_{j+1} \wedge \cdots \wedge \boldsymbol{u}_9, \boldsymbol{e}_1 \wedge \cdots \wedge \boldsymbol{e}_9) \boldsymbol{u}_j \\ & \quad - (\boldsymbol{u}_1 \wedge \cdots \wedge \boldsymbol{u}_9, \boldsymbol{e}_1 \wedge \cdots \wedge \boldsymbol{e}_9) \boldsymbol{x} = (\text{ii}) \end{aligned}$ | $(\boldsymbol{u} \times *(\boldsymbol{v} \wedge \boldsymbol{w}) + \boldsymbol{v} \times *(\boldsymbol{w} \wedge \boldsymbol{u}) + \boldsymbol{w} \times *(\boldsymbol{u} \wedge \boldsymbol{v}))\boldsymbol{x}$ $= \sum_{j=1}^9 (\boldsymbol{u}_1 \wedge \cdots \wedge \boldsymbol{u}_{j-1} \wedge \boldsymbol{x} \wedge \boldsymbol{u}_{j+1} \wedge \cdots \wedge \boldsymbol{u}_9, \boldsymbol{e}_1 \wedge \cdots \wedge \boldsymbol{e}_9) \boldsymbol{u}_j$ $- (\boldsymbol{u}_1 \wedge \cdots \wedge \boldsymbol{u}_9, \boldsymbol{e}_1 \wedge \cdots \wedge \boldsymbol{e}_9) \boldsymbol{x} = (\text{ii}).$ |                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 0902.0431_EQ0317 | 51   | 0.506 | $\left(\left(\left[\delta_1, \delta_2\right], \delta_2\right)_4, \delta_2\right)_4 + \left(\delta_1, \left[\delta, \delta_2\right]\right)_4 = 0, \quad \delta, \delta_i \in \mathfrak{f}_4^C.$                                                                                                                                                                                                                                                                                                                                                                                                                                                               | $([\delta, \delta_1], \delta_2)_4 + (\delta_1, [\delta, \delta_2])_4 = 0, \quad \delta, \delta_i \in \mathfrak{f}_4^C.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | $([\delta, \delta_1], \delta_2)_4 + (\delta_1, [\delta, \delta_2])_4 = 0, \quad \delta, \delta_i \in \mathfrak{f}_4^C.$                                                                                                                                                                                                                                                                                                                                        |
| 0902.0431_EQ0593 | 99   | 0.507 | $\begin{aligned} & \left\langle \varphi(A)M, \varphi(A)N \right\rangle = \left\langle k_J(\varphi(A)M), k_J(\varphi(A)N) \right\rangle \quad (\text{Lemma 3.11.1}) \\ & = \left\langle A(k_J(M))^t A, A(k_J(N))^t A \right\rangle = \left\langle k_J(M), k_J(N) \right\rangle \quad (\text{because } A \in SU(6)) \\ & = 2\langle M, N \rangle \end{aligned}$                                                                                                                                                                                                                                                                                                | $2\langle \varphi(A)M, \varphi(A)N \rangle = \langle k_J(\varphi(A)M), k_J(\varphi(A)N) \rangle \quad (\text{Lemma 3.11.1})$ $= \left\langle A(k_J(M))^t A, A(k_J(N))^t A \right\rangle = \langle k_J(M), k_J(N) \rangle \quad (\text{because } A \in SU(6))$ $= 2\langle M, N \rangle$                                                                                                                                                                                                                                                                                                                           | $2\langle \varphi(A)M, \varphi(A)N \rangle = \langle k_J(\varphi(A)M), k_J(\varphi(A)N) \rangle \quad (\text{Lemma 3.11.1})$ $= \langle A(k_J(M))^t A, A(k_J(N))^t A \rangle = \langle k_J(M), k_J(N) \rangle \quad (\text{because } A \in SU(6))$ $= 2\langle M, N \rangle,$                                                                                                                                                                                  |
| 0902.0431_EQ0764 | 128  | 0.507 | $\begin{aligned} & B_7(h, h') = \frac{3}{2} B_6 \left( \sum_{k=0}^3 \lambda_k H_k + \left( \sum_{j=1}^3 \mu_j E_j \right)^{\sim}, \sum_{k=0}^3 \lambda_k' H_k + \left( \sum_{j=1}^3 \mu_j' E_j \right)^{\sim} \right) + 24\nu\nu' \\ & = \frac{3}{2} 12 \left( 2 \sum_{k=0}^3 \lambda_k \lambda_k' + \sum_{j=1}^3 \mu_j \mu_j' \right) + 24\nu\nu' \quad (\text{Theorem 3.6.5}) \\ & = 6 \left( 6 \sum_{k=0}^3 \lambda_k \lambda_k' + 3 \sum_{j=1}^3 \mu_j \mu_j' + 4\nu\nu' \right). \end{aligned}$                                                                                                                                                         | $B_7(h, h') = \frac{3}{2} B_6 \left( \sum_{k=0}^3 \lambda_k H_k + \left( \sum_{j=1}^3 \mu_j E_j \right)^{\sim}, \sum_{k=0}^3 \lambda_k' H_k + \left( \sum_{j=1}^3 \mu_j' E_j \right)^{\sim} \right) + 24\nu\nu'$ $= \frac{3}{2} 12 \left( 2 \sum_{k=0}^3 \lambda_k \lambda_k' + \sum_{j=1}^3 \mu_j \mu_j' \right) + 24\nu\nu' \quad (\text{Theorem 3.6.5})$ $= 6 \left( 6 \sum_{k=0}^3 \lambda_k \lambda_k' + 3 \sum_{j=1}^3 \mu_j \mu_j' + 4\nu\nu' \right).$                                                                                                                                                    | $B_7(h, h') = \frac{3}{2} B_6 \left( \sum_{k=0}^3 \lambda_k H_k + \left( \sum_{j=1}^3 \mu_j E_j \right)^{\sim}, \sum_{k=0}^3 \lambda_k' H_k + \left( \sum_{j=1}^3 \mu_j' E_j \right)^{\sim} \right) + 24\nu\nu'$ $= \frac{3}{2} 12 \left( 2 \sum_{k=0}^3 \lambda_k \lambda_k' + \sum_{j=1}^3 \mu_j \mu_j' \right) + 24\nu\nu' \quad (\text{Theorem 3.6.5})$ $= 6 \left( 6 \sum_{k=0}^3 \lambda_k \lambda_k' + 3 \sum_{j=1}^3 \mu_j \mu_j' + 4\nu\nu' \right).$ |
| 0902.0431_EQ0051 | 13   | 0.507 | $\left( D_2 x \right) y + x \left( D_3^{\prime} y \right) = (\kappa D_1^{\prime})(xy), \quad x, y \in \mathfrak{C},$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | $(D_2 x) y + x (D_3' y) = (\kappa D_1')(xy), \quad x, y \in \mathfrak{C},$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | $(D_2 x) y + x (D_3' y) = (\kappa D_1')(xy), \quad x, y \in \mathfrak{C},$                                                                                                                                                                                                                                                                                                                                                                                     |





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| 0902.0431_EQ0878 | 145  | ■ 0.528 | $\begin{aligned} & \binom{\xi_1'}{\eta'} = \begin{pmatrix} \nu & a \\ -\tau a & -\nu \end{pmatrix} \begin{pmatrix} \xi_1 \\ \eta \end{pmatrix}, \quad \binom{\xi_2'}{\eta_2'} = \begin{pmatrix} \nu & a \\ -\tau a & -\nu \end{pmatrix} \begin{pmatrix} \xi_2 \\ \eta_2 \end{pmatrix}, \\ & \binom{\xi_3'}{\eta_3'} = \begin{pmatrix} \nu & a \\ -\tau a & -\nu \end{pmatrix} \begin{pmatrix} \xi_3 \\ \eta_3 \end{pmatrix}, \quad \binom{\xi_4'}{\eta_4'} = \begin{pmatrix} \nu & a \\ -\tau a & -\nu \end{pmatrix} \begin{pmatrix} \xi_4 \\ \eta_4 \end{pmatrix}, \\ & \binom{x_1'}{y_1'} = \begin{pmatrix} -\nu & \tau a \\ -a & \nu \end{pmatrix} \begin{pmatrix} x_1 \\ y_1 \end{pmatrix}, \quad \binom{x_2'}{y_2'} = \begin{pmatrix} -\nu & \tau a \\ -a & \nu \end{pmatrix} \begin{pmatrix} x_2 \\ y_2 \end{pmatrix}, \quad \binom{x_3'}{y_3'} = \begin{pmatrix} -\nu & \tau a \\ -a & \nu \end{pmatrix} \begin{pmatrix} x_3 \\ y_3 \end{pmatrix}. \end{aligned}$ | $\begin{pmatrix} \xi_1' \\ \eta' \end{pmatrix} = \begin{pmatrix} \nu & a \\ -\tau a & -\nu \end{pmatrix} \begin{pmatrix} \xi_1 \\ \eta \end{pmatrix}, \quad \begin{pmatrix} \xi_2' \\ \eta_2' \end{pmatrix} = \begin{pmatrix} \nu & a \\ -\tau a & -\nu \end{pmatrix} \begin{pmatrix} \xi_2 \\ \eta_2 \end{pmatrix},$ $\begin{pmatrix} \xi_3' \\ \eta_3' \end{pmatrix} = \begin{pmatrix} \nu & a \\ -\tau a & -\nu \end{pmatrix} \begin{pmatrix} \xi_3 \\ \eta_3 \end{pmatrix}, \quad \begin{pmatrix} \xi_4' \\ \eta_4' \end{pmatrix} = \begin{pmatrix} \nu & a \\ -\tau a & -\nu \end{pmatrix} \begin{pmatrix} \xi_4 \\ \eta_4 \end{pmatrix},$ $\begin{pmatrix} x_1' \\ y_1' \end{pmatrix} = \begin{pmatrix} -\nu & \tau a \\ -a & \nu \end{pmatrix} \begin{pmatrix} x_1 \\ y_1 \end{pmatrix}, \quad \begin{pmatrix} x_2' \\ y_2' \end{pmatrix} = \begin{pmatrix} -\nu & \tau a \\ -a & \nu \end{pmatrix} \begin{pmatrix} x_2 \\ y_2 \end{pmatrix}, \quad \begin{pmatrix} x_3' \\ y_3' \end{pmatrix} = \begin{pmatrix} -\nu & \tau a \\ -a & \nu \end{pmatrix} \begin{pmatrix} x_3 \\ y_3 \end{pmatrix}.$ | $\begin{pmatrix} \xi_1' \\ \eta' \end{pmatrix} = \begin{pmatrix} \nu & a \\ -\tau a & -\nu \end{pmatrix} \begin{pmatrix} \xi_1 \\ \eta \end{pmatrix}, \quad \begin{pmatrix} \xi_2' \\ \eta_2' \end{pmatrix} = \begin{pmatrix} \nu & a \\ -\tau a & -\nu \end{pmatrix} \begin{pmatrix} \xi_2 \\ \eta_2 \end{pmatrix},$ $\begin{pmatrix} \xi_3' \\ \eta_3' \end{pmatrix} = \begin{pmatrix} \nu & a \\ -\tau a & -\nu \end{pmatrix} \begin{pmatrix} \xi_3 \\ \eta_3 \end{pmatrix}, \quad \begin{pmatrix} \xi_4' \\ \eta_4' \end{pmatrix} = \begin{pmatrix} \nu & a \\ -\tau a & -\nu \end{pmatrix} \begin{pmatrix} \xi_4 \\ \eta_4 \end{pmatrix},$ $\begin{pmatrix} x_1' \\ y_1' \end{pmatrix} = \begin{pmatrix} -\nu & \tau a \\ -a & \nu \end{pmatrix} \begin{pmatrix} x_1 \\ y_1 \end{pmatrix}, \quad \begin{pmatrix} x_2' \\ y_2' \end{pmatrix} = \begin{pmatrix} -\nu & \tau a \\ -a & \nu \end{pmatrix} \begin{pmatrix} x_2 \\ y_2 \end{pmatrix}, \quad \begin{pmatrix} x_3' \\ y_3' \end{pmatrix} = \begin{pmatrix} -\nu & \tau a \\ -a & \nu \end{pmatrix} \begin{pmatrix} x_3 \\ y_3 \end{pmatrix}.$ |
| 0902.0431_EQ0945 | 157  | ■ 0.528 | $\begin{aligned} & \overline{\Phi(\phi_C(B, C), h(L), -\tau h(L), -ive_1)P} \\ & = \Phi(\overline{\phi_C(B, C)}, \overline{h(L)}, -\tau \overline{h(L)}, -\overline{ive_1})\overline{P} \\ & = \Phi(\phi_C(\overline{B}, \overline{C}), h(\overline{L}^*), -\tau h(\overline{L}^*), -ive_1)\overline{P} \\ & = f^{-1}((\overline{(\text{Ad } J_3)D})(f\overline{P})). \end{aligned}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | $\begin{aligned} \overline{f^{-1}(D(fP))} &= \overline{\Phi(\phi_C(B, C), h(L), -\tau h(L), -ive_1)P} \\ &= \Phi(\overline{\phi_C(B, C)}, \overline{h(L)}, -\tau \overline{h(L)}, -\overline{ive_1})\overline{P} \\ &= \Phi(\phi_C(\overline{B}, \overline{C}), h(\overline{L}^*), -\tau h(\overline{L}^*), -ive_1)\overline{P} \\ &= f^{-1}((\overline{(\text{Ad } J_3)D})(f\overline{P})) \end{aligned}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | $\begin{aligned} \overline{f^{-1}(D(fP))} &= \overline{\Phi(\phi_C(B, C), h(L), -\tau h(L), -ive_1)P} \\ &= \Phi(\overline{\phi_C(B, C)}, \overline{h(L)}, -\tau \overline{h(L)}, -\overline{ive_1})\overline{P} \\ &= \Phi(\phi_C(\overline{B}, \overline{C}), h(\overline{L}^*), -\tau h(\overline{L}^*), -ive_1)\overline{P} \\ &= f^{-1}((\overline{(\text{Ad } J_3)D})(f\overline{P})). \end{aligned}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 0902.0431_EQ1082 | 184  | ■ 0.533 | $\begin{aligned} & \begin{aligned} & \left( \left( gX_1 - \frac{\xi_1}{2}E \right) + ik \left( g(\gamma Y_1) - \frac{\eta_1}{2}E \right) \right) \left( k \left( gX_2 - \frac{\xi_2}{2}E \right) - ik \left( g(\gamma Y_2) - \frac{\eta_2}{2}E \right) \right) J^2 \\ & = -k \left( \left( gX_1 - \frac{\xi_1}{2}E \right) \left( gX_2 - \frac{\xi_2}{2}E \right) + \left( g(\gamma Y_1) - \frac{\eta_1}{2}E \right) \left( g(\gamma Y_2) - \frac{\eta_2}{2}E \right) \right) \\ & \quad - ik \left( \left( g(\gamma Y_1) - \frac{\eta_1}{2}E \right) \left( gX_2 - \frac{\xi_2}{2}E \right) - \left( gX_1 - \frac{\xi_1}{2}E \right) \left( g(\gamma Y_2) - \frac{\eta_2}{2}E \right) \right). \end{aligned} \end{aligned}$                                                                                                                                                                                                                                             | $\begin{aligned} & \left( k \left( gX_1 - \frac{\xi_1}{2}E \right) + ik \left( g(\gamma Y_1) - \frac{\eta_1}{2}E \right) \right) \left( k \left( gX_2 - \frac{\xi_2}{2}E \right) - ik \left( g(\gamma Y_2) - \frac{\eta_2}{2}E \right) \right) J^2 \\ & = -k \left( \left( gX_1 - \frac{\xi_1}{2}E \right) \left( gX_2 - \frac{\xi_2}{2}E \right) + \left( g(\gamma Y_1) - \frac{\eta_1}{2}E \right) \left( g(\gamma Y_2) - \frac{\eta_2}{2}E \right) \right) \\ & \quad - ik \left( \left( g(\gamma Y_1) - \frac{\eta_1}{2}E \right) \left( gX_2 - \frac{\xi_2}{2}E \right) - \left( gX_1 - \frac{\xi_1}{2}E \right) \left( g(\gamma Y_2) - \frac{\eta_2}{2}E \right) \right). \end{aligned}$                                                                                                                                                                                                                                                                                                                                                                                                             | $\begin{aligned} & \left( k \left( gX_1 - \frac{\xi_1}{2}E \right) + ik \left( g(\gamma Y_1) - \frac{\eta_1}{2}E \right) \right) \left( k \left( gX_2 - \frac{\xi_2}{2}E \right) - ik \left( g(\gamma Y_2) - \frac{\eta_2}{2}E \right) \right) J^2 \\ & = -k \left( \left( gX_1 - \frac{\xi_1}{2}E \right) \left( gX_2 - \frac{\xi_2}{2}E \right) + \left( g(\gamma Y_1) - \frac{\eta_1}{2}E \right) \left( g(\gamma Y_2) - \frac{\eta_2}{2}E \right) \right) \\ & \quad - ik \left( \left( g(\gamma Y_1) - \frac{\eta_1}{2}E \right) \left( gX_2 - \frac{\xi_2}{2}E \right) - \left( gX_1 - \frac{\xi_1}{2}E \right) \left( g(\gamma Y_2) - \frac{\eta_2}{2}E \right) \right). \end{aligned}$                                                                                                                                                                                                                                                                                                                                                                                                             |

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| 0902.0431_E09989 | 165  | 0.535 | <pre>\begin{aligned} &amp; \left.=\left(\begin{array}{c} \left[ \Phi, \Phi_1 \right]+P \times Q_1-P_1 \times Q \\ \Phi P_1-\Phi_1 P+r P_1-r_1 P+s Q_1-s_1 Q \\ \Phi Q_1-\Phi_1 Q-r Q_1+r_1 Q+t P_1-t_1 P \\ -\frac{1}{8}\{P, Q_1\}+\frac{1}{8}\{P_1, Q\}+s t_1-s_1 t \\ \frac{1}{4}\{P, P_1\}+2 r s_1-2 r_1 s \\ -\frac{1}{4}\{Q, Q_1\}-2 r t_1+2 r_1 t \end{array}\right),\left(\begin{array}{c} \Phi_2 \\ P_2 \\ Q_2 \\ r_2 \\ s_2 \\ t_2 \end{array}\right)\right)_8 \\ &amp; =\cdots\left(\text { using }([\Phi, \Phi_1], \Phi_2)_7+(\Phi_1, [\Phi, \Phi_2])_7=0 \text { (Lemma 4.5.1.(1))},\right. \\ &amp; \left.-(\Phi, P \times Q)_7=\{\Phi P, Q\} \text { (Lemma 4.5.1.(2)) etc.) } \cdots\right. \\ &amp; \left.\left.-(R_1,[R, R_2])_8.\right.\right) \end{aligned}</pre> | $= \left( \begin{pmatrix} [\Phi, \Phi_1] + P \times Q_1 - P_1 \times Q \\ \Phi P_1 - \Phi_1 P + r P_1 - r_1 P + s Q_1 - s_1 Q \\ \Phi Q_1 - \Phi_1 Q - r Q_1 + r_1 Q + t P_1 - t_1 P \\ -\frac{1}{8}\{P, Q_1\} + \frac{1}{8}\{P_1, Q\} + s t_1 - s_1 t \\ \frac{1}{4}\{P, P_1\} + 2 r s_1 - 2 r_1 s \\ -\frac{1}{4}\{Q, Q_1\} - 2 r t_1 + 2 r_1 t \end{pmatrix}, \begin{pmatrix} \Phi_2 \\ P_2 \\ Q_2 \\ r_2 \\ s_2 \\ t_2 \end{pmatrix} \right)_8$ $= \cdots ( \text{ using } ([\Phi, \Phi_1], \Phi_2)_7 + (\Phi_1, [\Phi, \Phi_2])_7 = 0 \text{ (Lemma 4.5.1.(1))},$ $= -(\Phi, P \times Q)_7 = \{\Phi P, Q\} ( \text{Lemma 4.5.1.(2)) etc.) } \cdots$ | $= \left( \begin{pmatrix} [\Phi, \Phi_1] + P \times Q_1 - P_1 \times Q \\ \Phi P_1 - \Phi_1 P + r P_1 - r_1 P + s Q_1 - s_1 Q \\ \Phi Q_1 - \Phi_1 Q - r Q_1 + r_1 Q + t P_1 - t_1 P \\ -\frac{1}{8}\{P, Q_1\} + \frac{1}{8}\{P_1, Q\} + s t_1 - s_1 t \\ \frac{1}{4}\{P, P_1\} + 2 r s_1 - 2 r_1 s \\ -\frac{1}{4}\{Q, Q_1\} - 2 r t_1 + 2 r_1 t \end{pmatrix}, \begin{pmatrix} \Phi_2 \\ P_2 \\ Q_2 \\ r_2 \\ s_2 \\ t_2 \end{pmatrix} \right)_8$ $= \cdots (\text{using } ([\Phi, \Phi_1], \Phi_2)_7 + (\Phi_1, [\Phi, \Phi_2])_7 = 0 \text{ (Lemma 4.5.1.(1))},$ $(\Phi, P \times Q)_7 = \{\Phi P, Q\} \text{ (Lemma 4.5.1.(2)) etc.)} \cdots$ $= -(R_1, [R, R_2])_8.$ |

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|------------------|------|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0902.0431_E00696 | 115  | 0.535 | $\begin{array}{c} \backslash\begin{array}{cc} \Phi_{-3}=\left(\begin{array}{cccc} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & \kappa & 0 \\ 0 & 0 & 0 & 0 \end{array}\right), & \Phi_3=\left(\begin{array}{cccc} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \lambda \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right), \\ \backslash\begin{array}{cc} \Phi_{-2}=\left(\begin{array}{cccc} 0 & 0 & C & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & d & 0 & 0 \end{array}\right), & \Phi_2=\left(\begin{array}{cccc} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & D \\ c & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right), \\ \backslash\begin{array}{cc} \Phi_{-1}=\left(\begin{array}{cccc} 0 & l & 0 & 0 \\ 0 & 0 & B & 0 \\ 0 & 0 & 0 & 0 \\ b & 0 & 0 & 0 \end{array}\right), & \Phi_1=\left(\begin{array}{cccc} 0 & 0 & 0 & 0 \\ k & 0 & 0 & 0 \\ 0 & a & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right), \\ \Phi_0=\left(\begin{array}{cccc} g & 0 & 0 & 0 \\ 0 & h & 0 & 0 \\ 0 & 0 & \nu & 0 \\ 0 & 0 & 0 & \mu \end{array}\right). \end{array}$ | $\begin{array}{c} \Phi_{-3}=\left(\begin{array}{cccc} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & \kappa & 0 \\ 0 & 0 & 0 & 0 \end{array}\right), \quad \Phi_3=\left(\begin{array}{cccc} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \lambda \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right), \\ \Phi_{-2}=\left(\begin{array}{cccc} 0 & 0 & C & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & d & 0 & 0 \end{array}\right), \quad \Phi_2=\left(\begin{array}{cccc} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & D \\ c & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right), \\ \Phi_{-1}=\left(\begin{array}{cccc} 0 & l & 0 & 0 \\ 0 & 0 & B & 0 \\ 0 & 0 & 0 & 0 \\ b & 0 & 0 & 0 \end{array}\right), \quad \Phi_1=\left(\begin{array}{cccc} 0 & 0 & 0 & 0 \\ k & 0 & 0 & 0 \\ 0 & a & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right), \\ \Phi_0=\left(\begin{array}{cccc} g & 0 & 0 & 0 \\ 0 & h & 0 & 0 \\ 0 & 0 & \nu & 0 \\ 0 & 0 & 0 & \mu \end{array}\right).$ | $\begin{array}{c} \Phi_{-3}=\left(\begin{array}{cccc} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & \kappa & 0 \\ 0 & 0 & 0 & 0 \end{array}\right), \quad \Phi_3=\left(\begin{array}{cccc} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \lambda \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right), \\ \Phi_{-2}=\left(\begin{array}{cccc} 0 & 0 & C & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & d & 0 & 0 \end{array}\right), \quad \Phi_2=\left(\begin{array}{cccc} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & D \\ c & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right), \\ \Phi_{-1}=\left(\begin{array}{cccc} 0 & l & 0 & 0 \\ 0 & 0 & B & 0 \\ 0 & 0 & 0 & 0 \\ b & 0 & 0 & 0 \end{array}\right), \quad \Phi_1=\left(\begin{array}{cccc} 0 & 0 & 0 & 0 \\ k & 0 & 0 & 0 \\ 0 & a & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right), \\ \Phi_0=\left(\begin{array}{cccc} g & 0 & 0 & 0 \\ 0 & h & 0 & 0 \\ 0 & 0 & \nu & 0 \\ 0 & 0 & 0 & \mu \end{array}\right).$ |
| 0902.0431_E01193 | 201  | 0.542 | $\begin{array}{c} \backslash\begin{array}{l} \left( \left( \left( a \wedge c \right) \wedge b \right), d \right) = \left( a \wedge c, b \wedge d \right) \\ = \left( a, b \right) \left( c, d \right) - \left( a_1 \wedge c_1, b \right) \left( a_2 \wedge c_2, d \right) + \left( a_1 \wedge c_2, b \right) \left( a_2 \wedge c_1, d \right) \\ + \left( a_2 \wedge c_1, b \right) \left( a_1 \wedge c_2, d \right) - \left( a_2 \wedge c_2, b \right) \left( a_1 \wedge c_1, d \right) + \left( c, b \right) \left( a, d \right). \end{array}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | $\begin{array}{c} \left( \left( \left( a \wedge c \right) \wedge b \right), d \right) = \left( a \wedge c, b \wedge d \right) \\ = \left( a, b \right) \left( c, d \right) - \left( a_1 \wedge c_1, b \right) \left( a_2 \wedge c_2, d \right) + \left( a_1 \wedge c_2, b \right) \left( a_2 \wedge c_1, d \right) \\ + \left( a_2 \wedge c_1, b \right) \left( a_1 \wedge c_2, d \right) - \left( a_2 \wedge c_2, b \right) \left( a_1 \wedge c_1, d \right) + \left( c, b \right) \left( a, d \right). \end{array}$                                                                                                                                                                                                                                                                                                                                                                                                           | $\begin{array}{c} \left( \left( \left( a \wedge c \right) \wedge b \right), d \right) = \left( a \wedge c, b \wedge d \right) \\ = \left( a, b \right) \left( c, d \right) - \left( a_1 \wedge c_1, b \right) \left( a_2 \wedge c_2, d \right) + \left( a_1 \wedge c_2, b \right) \left( a_2 \wedge c_1, d \right) \\ + \left( a_2 \wedge c_1, b \right) \left( a_1 \wedge c_2, d \right) - \left( a_2 \wedge c_2, b \right) \left( a_1 \wedge c_1, d \right) + \left( c, b \right) \left( a, d \right). \end{array}$                                                                                                                                                                                                                                                                                                                                                                                                           |

| Identifier       | Page | Conf.   | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                                | Rendered                                                                                                                                                                                                                                                                                                                           | Image                                                                                                                                                                                                                                                                                                                              |
|------------------|------|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0902.0431_E00221 | 36   | ■ 0.542 | $\begin{aligned} (X, Y) &= \sum_{i=1}^3 \left( \xi_i \eta_i + 2(x_i, y_i) \right), \\ \operatorname{tr}(X, Y, Z) &= \sum_{i=1}^3 \left( \xi_i \eta_i \zeta_i + R(x_i y_{i+1} z_{i+2} + x_i z_{i+1} y_{i+2}) \right. \\ &\quad \left. + \xi_i((y_{i+1}, z_{i+1}) + (y_{i+2}, z_{i+2})) + \eta_i((z_{i+1}, x_{i+1}) + (z_{i+2}, x_{i+2})) \right. \\ &\quad \left. + \zeta_i((x_{i+1}, y_{i+1}) + (x_{i+2}, y_{i+2})) \right), \end{aligned}$ | $(X, Y) = \sum_{i=1}^3 (\xi_i \eta_i + 2(x_i, y_i)),$ $\operatorname{tr}(X, Y, Z) = \sum_{i=1}^3 (\xi_i \eta_i \zeta_i + R(x_i y_{i+1} z_{i+2} + x_i z_{i+1} y_{i+2})$ $+ \xi_i((y_{i+1}, z_{i+1}) + (y_{i+2}, z_{i+2})) + \eta_i((z_{i+1}, x_{i+1}) + (z_{i+2}, x_{i+2}))$ $+ \zeta_i((x_{i+1}, y_{i+1}) + (x_{i+2}, y_{i+2}))),$ | $(X, Y) = \sum_{i=1}^3 (\xi_i \eta_i + 2(x_i, y_i)),$ $\operatorname{tr}(X, Y, Z) = \sum_{i=1}^3 (\xi_i \eta_i \zeta_i + R(x_i y_{i+1} z_{i+2} + x_i z_{i+1} y_{i+2})$ $+ \xi_i((y_{i+1}, z_{i+1}) + (y_{i+2}, z_{i+2})) + \eta_i((z_{i+1}, x_{i+1}) + (z_{i+2}, x_{i+2}))$ $+ \zeta_i((x_{i+1}, y_{i+1}) + (x_{i+2}, y_{i+2}))),$ |
| 0902.0431_E00960 | 159  | ■ 0.542 | $\begin{aligned} &\beta(P_C + (M + N)) = P_C + Q(M + N) = P_C + (QM + QN) \\ &= \psi(Q, E)(P_C + (M + N)), \quad P_C + (M + N) \in \mathfrak{P}^C \end{aligned}$                                                                                                                                                                                                                                                                            | $\beta(P_C + (M + N)) = P_C + Q(M + N) = P_C + (QM + QN)$ $= \psi(Q, E)(P_C + (M + N)), \quad P_C + (M + N) \in \mathfrak{P}^C$                                                                                                                                                                                                    | $\beta(P_C + (M + N)) = P_C + Q(M + N) = P_C + (QM + QN)$ $= \psi(Q, E)(P_C + (M + N)), \quad P_C + (M + N) \in \mathfrak{P}^C$                                                                                                                                                                                                    |
| 0902.0431_E00005 | 6    | ■ 0.542 | $\begin{aligned} &\overline{x_0 + \sum_{i=1}^7 x_i e_i} = x_0 - \sum_{i=1}^7 x_i e_i, \quad \left( \sum_{i=0}^7 x_i e_i, \sum_{i=0}^7 y_i e_i \right) = \sum_{i=0}^7 x_i y_i, \\ & x  = \sqrt{(x, x)}, \quad R\left(x_0 + \sum_{i=1}^7 x_i e_i\right) = x_0. \end{aligned}$                                                                                                                                                                 | $\overline{x_0 + \sum_{i=1}^7 x_i e_i} = x_0 - \sum_{i=1}^7 x_i e_i, \quad \left( \sum_{i=0}^7 x_i e_i, \sum_{i=0}^7 y_i e_i \right) = \sum_{i=0}^7 x_i y_i$ $ x  = \sqrt{(x, x)}, \quad R\left(x_0 + \sum_{i=1}^7 x_i e_i\right) = x_0.$                                                                                          | $\overline{x_0 + \sum_{i=1}^7 x_i e_i} = x_0 - \sum_{i=1}^7 x_i e_i, \quad \left( \sum_{i=0}^7 x_i e_i, \sum_{i=0}^7 y_i e_i \right) = \sum_{i=0}^7 x_i y_i,$ $ x  = \sqrt{(x, x)}, \quad R\left(x_0 + \sum_{i=1}^7 x_i e_i\right) = x_0.$                                                                                         |
| 0902.0431_E00097 | 19   | ■ 0.544 | $\begin{aligned} &\pm(\lambda_1 - \lambda_2) : \pm(G_{24} + G_{35}) + i(-G_{25} + G_{34}), \\ &\pm(\lambda_1 - \lambda_3) : \pm(G_{26} + G_{37}) + i(-G_{27} + G_{36}), \\ &\pm(\lambda_2 - \lambda_3) : \pm(G_{46} + G_{57}) + i(-G_{47} + G_{56}). \end{aligned}$                                                                                                                                                                         | $\pm(\lambda_1 - \lambda_2) : \pm(G_{24} + G_{35}) + i(-G_{25} + G_{34}),$ $\pm(\lambda_1 - \lambda_3) : \pm(G_{26} + G_{37}) + i(-G_{27} + G_{36}),$ $\pm(\lambda_2 - \lambda_3) : \pm(G_{46} + G_{57}) + i(-G_{47} + G_{56}).$                                                                                                   | $\pm(\lambda_1 - \lambda_2) : \pm(G_{24} + G_{35}) + i(-G_{25} + G_{34}),$ $\pm(\lambda_1 - \lambda_3) : \pm(G_{26} + G_{37}) + i(-G_{27} + G_{36}),$ $\pm(\lambda_2 - \lambda_3) : \pm(G_{46} + G_{57}) + i(-G_{47} + G_{56}).$                                                                                                   |
| 0902.0431_E00949 | 158  | ■ 0.546 | $\begin{aligned} &\mu \begin{pmatrix} M_{11} & M_{12} \\ M_{21} & M_{22} \end{pmatrix} \\ &= \left( \frac{(M_{21} - M_{12})e_1}{2} + i \frac{M_{21} + M_{12}}{2}, \frac{(M_{22} + M_{11})e_1}{2} + i \frac{M_{22} - M_{11}}{2} \right), \end{aligned}$                                                                                                                                                                                      | $\mu \begin{pmatrix} M_{11} & M_{12} \\ M_{21} & M_{22} \end{pmatrix}$ $= \left( \frac{(M_{21} - M_{12})e_1}{2} + i \frac{M_{21} + M_{12}}{2}, \frac{(M_{22} + M_{11})e_1}{2} + i \frac{M_{22} - M_{11}}{2} \right)$                                                                                                               | $\mu \begin{pmatrix} M_{11} & M_{12} \\ M_{21} & M_{22} \end{pmatrix}$ $= \left( \frac{(M_{21} - M_{12})e_1}{2} + i \frac{M_{21} + M_{12}}{2}, \frac{(M_{22} + M_{11})e_1}{2} + i \frac{M_{22} - M_{11}}{2} \right),$                                                                                                              |
| 0902.0431_E00297 | 47   | ■ 0.546 | $x \xrightarrow{G_{i0}} x_i e_0 - x_0 e_i \xrightarrow{G_{j0}(j \neq i)} x_i e_j \xrightarrow{x_j^{-1} G_{0j}} e_0 = 1.$                                                                                                                                                                                                                                                                                                                    | $x \xrightarrow{G_{i0}} x_i e_0 - x_0 e_i \xrightarrow{G_{j0}(j \neq i)} x_i e_j \xrightarrow{x_j^{-1} G_{0j}} e_0 = 1.$                                                                                                                                                                                                           | $x \xrightarrow{G_{i0}} x_i e_0 - x_0 e_i \xrightarrow{G_{j0}(j \neq i)} x_i e_j \xrightarrow{x_j^{-1} G_{0j}} e_0 = 1.$                                                                                                                                                                                                           |



| Identifier       | Page | Conf.   | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Rendered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Image                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|------------------|------|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0902.0431_E09953 | 158  | ■ 0.553 | $\begin{aligned} & \backslash \text{begin}{aligned} \& \ \psi_{*}(D)(\mu \widetilde{M}) \backslash \& = \\ & \Phi \left( \phi_{\boldsymbol{C}}(B, C), h(L), -\tau h(L), -i \nu e_1 \right) (M, N) \\ & \backslash \& = \left( \begin{array}{cccc} \phi_{\boldsymbol{C}}(B, C) + \frac{1}{3} i \nu e_1 & -2 \tau h(L) & 0 & h(L) \\ 2 h(L) & \tau \phi_{\boldsymbol{C}}(B, C) \tau - \frac{1}{3} i \nu e_1 & -\tau h(L) & 0 \\ 0 & h(L) & -i \nu e_1 & 0 \\ -\tau h(L) & 0 & 0 & i \nu e_1 \end{array} \right) \left( \begin{array}{c} M \\ N \\ 0 \\ 0 \end{array} \right) \\ & = \left( \begin{array}{c} \phi_{\boldsymbol{C}}(B, C) M + \frac{1}{3} i \nu e_1 M - 2 \tau h(L) \times N \\ 2 h(l) \times M + \tau \phi_{\boldsymbol{C}}(B, C) \tau N - \frac{1}{3} i \nu e_1 N \\ (h(L), N) \\ -(\tau h(L), M) \end{array} \right) \\ & = \left( \begin{array}{c} -M h(B, C) + N \tau h(L) + \frac{1}{3} i \nu e_1 M \\ -M h(L) - N \tau h(B, C) - \frac{1}{3} i \nu e_1 N \\ 0 \\ 0 \end{array} \right) \end{aligned}$ | $\begin{aligned} & \psi_{*}(D)(\mu \widetilde{M}) \\ & = \Phi \left( \phi_{\boldsymbol{C}}(B, C), h(L), -\tau h(L), -i \nu e_1 \right) (M, N) \\ & = \left( \begin{array}{cccc} \phi_{\boldsymbol{C}}(B, C) + \frac{1}{3} i \nu e_1 & -2 \tau h(L) & 0 & h(L) \\ 2 h(L) & \tau \phi_{\boldsymbol{C}}(B, C) \tau - \frac{1}{3} i \nu e_1 & -\tau h(L) & 0 \\ 0 & h(L) & -i \nu e_1 & 0 \\ -\tau h(L) & 0 & 0 & i \nu e_1 \end{array} \right) \left( \begin{array}{c} M \\ N \\ 0 \\ 0 \end{array} \right) \\ & = \left( \begin{array}{c} \phi_{\boldsymbol{C}}(B, C) M + \frac{1}{3} i \nu e_1 M - 2 \tau h(L) \times N \\ 2 h(l) \times M + \tau \phi_{\boldsymbol{C}}(B, C) \tau N - \frac{1}{3} i \nu e_1 N \\ (h(L), N) \\ -(\tau h(L), M) \end{array} \right) \\ & = \left( \begin{array}{c} -M h(B, C) + N \tau h(L) + \frac{1}{3} i \nu e_1 M \\ -M h(L) - N \tau h(B, C) - \frac{1}{3} i \nu e_1 N \\ 0 \\ 0 \end{array} \right) \end{aligned}$ | $\begin{aligned} & \psi_{*}(D)(\mu \widetilde{M}) \\ & = \Phi \left( \phi_{\boldsymbol{C}}(B, C), h(L), -\tau h(L), -i \nu e_1 \right) (M, N) \\ & = \left( \begin{array}{cccc} \phi_{\boldsymbol{C}}(B, C) + \frac{1}{3} i \nu e_1 & -2 \tau h(L) & 0 & h(L) \\ 2 h(L) & \tau \phi_{\boldsymbol{C}}(B, C) \tau - \frac{1}{3} i \nu e_1 & -\tau h(L) & 0 \\ 0 & h(L) & -i \nu e_1 & 0 \\ -\tau h(L) & 0 & 0 & i \nu e_1 \end{array} \right) \left( \begin{array}{c} M \\ N \\ 0 \\ 0 \end{array} \right) \\ & = \left( \begin{array}{c} \phi_{\boldsymbol{C}}(B, C) M + \frac{1}{3} i \nu e_1 M - 2 \tau h(L) \times N \\ 2 h(l) \times M + \tau \phi_{\boldsymbol{C}}(B, C) \tau N - \frac{1}{3} i \nu e_1 N \\ (h(L), N) \\ -(\tau h(L), M) \end{array} \right) \\ & = \left( \begin{array}{c} -M h(B, C) + N \tau h(L) + \frac{1}{3} i \nu e_1 M \\ -M h(L) - N \tau h(B, C) - \frac{1}{3} i \nu e_1 N \\ 0 \\ 0 \end{array} \right) \end{aligned}$ |
| 0902.0431_E09440 | 72   | ■ 0.554 | $F_{4(4)} = \left( F_4^C \right)^{\tau \gamma}, \quad F_{4(-20)} = \left( F_4^C \right)^{\tau \sigma}.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | $F_{4(4)} = \left( F_4^C \right)^{\tau \gamma}, \quad F_{4(-20)} = \left( F_4^C \right)^{\tau \sigma}.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | $F_{4(4)} = \left( F_4^C \right)^{\tau \gamma}, \quad F_{4(-20)} = \left( F_4^C \right)^{\tau \sigma}.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 0902.0431_E01174 | 197  | ■ 0.554 | $\begin{aligned} & \backslash \text{begin}{gathered} \mathfrak{g}_{-0} = \mathfrak{sl}(5, C) \oplus \mathfrak{sl}(5, C), \\ & \oplus \mathfrak{sl}(5, C), \backslash \& \mathfrak{g}_{-1} = C^5 \otimes \Lambda^2(C^5) = \mathfrak{g}_{-1}, \quad \mathfrak{g}_2 = \Lambda^2(C^5) \otimes C^5 = \mathfrak{g}_{-2}, \\ & \otimes \Lambda^2(C^5) = \mathfrak{g}_{-1}, \quad \mathfrak{g}_2 = \Lambda^2(C^5) \otimes C^5 = \mathfrak{g}_{-2}, \\ & \otimes C^5 = \mathfrak{g}_{-2}, \end{aligned}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | $\begin{aligned} & \mathfrak{g}_0 = \mathfrak{sl}(5, C) \oplus \mathfrak{sl}(5, C), \\ & \mathfrak{g}_1 = C^5 \otimes \Lambda^2(C^5) = \mathfrak{g}_{-1}, \quad \mathfrak{g}_2 = \Lambda^2(C^5) \otimes C^5 = \mathfrak{g}_{-2}, \end{aligned}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | $\begin{aligned} & \mathfrak{g}_0 = \mathfrak{sl}(5, C) \oplus \mathfrak{sl}(5, C), \\ & \mathfrak{g}_1 = C^5 \otimes \Lambda^2(C^5) = \mathfrak{g}_{-1}, \quad \mathfrak{g}_2 = \Lambda^2(C^5) \otimes C^5 = \mathfrak{g}_{-2}, \end{aligned}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 0902.0431_E09037 | 11   | ■ 0.555 | $\begin{aligned} & \backslash \text{begin}{array}{lll} 1 \rightarrow \left( \begin{array}{ccc} 1 & 2 & 3 \\ 1 & 2 & 3 \end{array} \right), \quad \kappa \rightarrow \left( \begin{array}{ccc} 1 & 2 & 3 \\ 2 & 1 & 3 \end{array} \right), \quad \pi \rightarrow \left( \begin{array}{ccc} 1 & 2 & 3 \\ 3 & 2 & 1 \end{array} \right), \\ & \nu \rightarrow \left( \begin{array}{ccc} 1 & 2 & 3 \\ 2 & 3 & 1 \end{array} \right), \quad \nu^2 \rightarrow \left( \begin{array}{ccc} 1 & 2 & 3 \\ 3 & 1 & 2 \end{array} \right), \quad \nu \pi \rightarrow \left( \begin{array}{ccc} 1 & 2 & 3 \\ 1 & 3 & 2 \end{array} \right) \end{aligned}$                                                                                                                                                                                                                                                                                                                                                                             | $\begin{aligned} & 1 \rightarrow \left( \begin{array}{ccc} 1 & 2 & 3 \\ 1 & 2 & 3 \end{array} \right), \quad \kappa \rightarrow \left( \begin{array}{ccc} 1 & 2 & 3 \\ 2 & 1 & 3 \end{array} \right), \quad \pi \rightarrow \left( \begin{array}{ccc} 1 & 2 & 3 \\ 3 & 2 & 1 \end{array} \right), \\ & \nu \rightarrow \left( \begin{array}{ccc} 1 & 2 & 3 \\ 2 & 3 & 1 \end{array} \right), \quad \nu^2 \rightarrow \left( \begin{array}{ccc} 1 & 2 & 3 \\ 3 & 1 & 2 \end{array} \right), \quad \nu \pi \rightarrow \left( \begin{array}{ccc} 1 & 2 & 3 \\ 1 & 3 & 2 \end{array} \right) \end{aligned}$                                                                                                                                                                                                                                                                                                                                               | $\begin{aligned} & 1 \rightarrow \left( \begin{array}{ccc} 1 & 2 & 3 \\ 1 & 2 & 3 \end{array} \right), \quad \kappa \rightarrow \left( \begin{array}{ccc} 1 & 2 & 3 \\ 2 & 1 & 3 \end{array} \right), \quad \pi \rightarrow \left( \begin{array}{ccc} 1 & 2 & 3 \\ 3 & 2 & 1 \end{array} \right), \\ & \nu \rightarrow \left( \begin{array}{ccc} 1 & 2 & 3 \\ 2 & 3 & 1 \end{array} \right), \quad \nu^2 \rightarrow \left( \begin{array}{ccc} 1 & 2 & 3 \\ 3 & 1 & 2 \end{array} \right), \quad \nu \pi \rightarrow \left( \begin{array}{ccc} 1 & 2 & 3 \\ 1 & 3 & 2 \end{array} \right) \end{aligned}$                                                                                                                                                                                                                                                                                                                                               |

| Identifier       | Page | Conf.   | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Rendered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Image                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
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| 0902.0431_E00329 | 52   | ■ 0.556 | $\begin{aligned} & \backslash\mathrm{begin}\{\mathrm{aligned}\} \backslash\mathrm{lambda}_{\{k\}}-\backslash\mathrm{lambda}_{\{l\}} \& :\backslash\mathrm{left}(G_{\{k\}} \\ & \backslash\mathrm{right})-\mathrm{i}\backslash\mathrm{left}(G_{\{k\}}+G_{\{l\}}+G_{\{l\}}+G_{\{k\}} \\ & \backslash\mathrm{right}), \backslash\backslash-\backslash\mathrm{lambda}_{\{k\}}+\backslash\mathrm{lambda}_{\{l\}} \& : \mathrm{i}\backslash\mathrm{left}(G_{\{k\}} \\ & \backslash\mathrm{right})-\backslash\mathrm{left}(G_{\{k\}}+G_{\{l\}}+G_{\{l\}}+G_{\{k\}} \\ & \backslash\mathrm{right}), \backslash\backslash \backslash\mathrm{lambda}_{\{k\}}+\backslash\mathrm{lambda}_{\{l\}} \& :\backslash\mathrm{left}(G_{\{k\}}-G_{\{l\}}+G_{\{k\}} \\ & \backslash\mathrm{right})+\mathrm{i}\backslash\mathrm{left}(G_{\{k\}}+G_{\{l\}}-G_{\{l\}}+G_{\{k\}} \\ & \backslash\mathrm{right}), \backslash\backslash-\backslash\mathrm{lambda}_{\{k\}}-\backslash\mathrm{lambda}_{\{l\}} \& :\backslash\mathrm{left}(G_{\{k\}}-G_{\{l\}}+G_{\{k\}}+G_{\{l\}} \\ & \backslash\mathrm{right})+\backslash\mathrm{left}(G_{\{k\}}+G_{\{l\}}-G_{\{l\}}+G_{\{k\}} \\ & \backslash\mathrm{right}) \ . \ \mathrm{end}\{\mathrm{aligned}\} \end{aligned}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | $\begin{aligned} & \lambda_k-\lambda_l:(G_{kl}+G_{4+k4+l})-i(G_{k4+l}+G_{l4+k}), \\ & -\lambda_k+\lambda_l:i(G_{kl}+G_{4+k4+l})-(G_{k4+l}+G_{l4+k}), \\ & \lambda_k+\lambda_l:(G_{kl}-G_{4+k4+l})+i(G_{k4+l}-G_{l4+k}), \\ & -\lambda_k-\lambda_l:(G_{kl}-G_{4+k4+l})+(G_{k4+l}-G_{l4+k}). \end{aligned}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | $\begin{aligned} & \lambda_k-\lambda_l: (G_{kl}+G_{4+k4+l})-i(G_{k4+l}+G_{l4+k}), \\ & -\lambda_k+\lambda_l: i(G_{kl}+G_{4+k4+l})-(G_{k4+l}+G_{l4+k}), \\ & \lambda_k+\lambda_l: (G_{kl}-G_{4+k4+l})+i(G_{k4+l}-G_{l4+k}), \\ & -\lambda_k-\lambda_l: i(G_{kl}-G_{4+k4+l})+(G_{k4+l}-G_{l4+k}). \end{aligned}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 0902.0431_E01183 | 199  | ■ 0.556 | $\begin{aligned} & \backslash\mathrm{begin}\{\mathrm{aligned}\} \& (\backslash\mathrm{boldsymbol}\{x\} \ \backslash\mathrm{wedge}^*(\backslash\mathrm{boldsymbol}\{a\} \\ & \backslash\mathrm{wedge} \ \backslash\mathrm{boldsymbol}\{y\})+(\backslash\mathrm{boldsymbol}\{y\} \ \backslash\mathrm{wedge}^*(\backslash\mathrm{boldsymbol}\{a\} \\ & \backslash\mathrm{wedge} \ \backslash\mathrm{boldsymbol}\{x\}))-(\backslash\mathrm{boldsymbol}\{x\}, \\ & \backslash\mathrm{boldsymbol}\{y\}) \ \backslash\mathrm{boldsymbol}\{a\}, \ \backslash\mathrm{boldsymbol}\{b\}) \ \backslash\backslash \\ & \& \ \mathrm{quad}=(\backslash\mathrm{boldsymbol}\{b\} \ \backslash\mathrm{wedge} \ \backslash\mathrm{boldsymbol}\{x\}, \ * \\ & \backslash\mathrm{boldsymbol}\{a\} \ \backslash\mathrm{wedge} \ \backslash\mathrm{boldsymbol}\{y\})+(\backslash\mathrm{boldsymbol}\{x\} \\ & \backslash\mathrm{wedge} \ \backslash\mathrm{boldsymbol}\{a\}, \ \backslash\mathrm{boldsymbol}\{y\} \ \backslash\mathrm{wedge} \\ & \backslash\mathrm{boldsymbol}\{b\})-(\backslash\mathrm{boldsymbol}\{x\}, \ \backslash\mathrm{boldsymbol}\{y\})( \\ & \backslash\mathrm{boldsymbol}\{a\}, \ \backslash\mathrm{boldsymbol}\{b\})=0, \ \backslash\backslash \& (*(\backslash\mathrm{boldsymbol}\{a\} \\ & \backslash\mathrm{wedge}^*(\backslash\mathrm{boldsymbol}\{b\} \ \backslash\mathrm{wedge} \ \backslash\mathrm{boldsymbol}\{x\}))-*( \\ & \backslash\mathrm{boldsymbol}\{b\} \ \backslash\mathrm{wedge}^*(\backslash\mathrm{boldsymbol}\{a\} \ \backslash\mathrm{wedge} \\ & \backslash\mathrm{boldsymbol}\{x\}))-(\backslash\mathrm{boldsymbol}\{a\}, \ \backslash\mathrm{boldsymbol}\{b\}) \\ & \backslash\mathrm{boldsymbol}\{x\}, \ \backslash\mathrm{boldsymbol}\{y\}) \ \backslash\backslash \& \ \mathrm{quad}=( \\ & \backslash\mathrm{boldsymbol}\{x\} \ \backslash\mathrm{wedge} \ \backslash\mathrm{boldsymbol}\{b\}, \ \backslash\mathrm{boldsymbol}\{y\} \\ & \backslash\mathrm{wedge} \ \backslash\mathrm{boldsymbol}\{a\})+(\backslash\mathrm{boldsymbol}\{a\} \ \backslash\mathrm{wedge} \\ & \backslash\mathrm{boldsymbol}\{x\}, \ * \ \backslash\mathrm{boldsymbol}\{b\} \ \backslash\mathrm{wedge} \ \backslash\mathrm{boldsymbol}\{y\})-( \\ & \backslash\mathrm{boldsymbol}\{x\}, \ \backslash\mathrm{boldsymbol}\{y\})(\backslash\mathrm{boldsymbol}\{a\}, \\ & \backslash\mathrm{boldsymbol}\{b\})=0 \ . \ \mathrm{end}\{\mathrm{aligned}\} \end{aligned}$ | $\begin{aligned} & (x \wedge *(\boldsymbol{a} \wedge \boldsymbol{y}) + *(\boldsymbol{y} \wedge *(\boldsymbol{a} \wedge \boldsymbol{x})) - (x, \boldsymbol{y})\boldsymbol{a}, \boldsymbol{b}) \\ & = (*\boldsymbol{b} \wedge \boldsymbol{x}, *\boldsymbol{a} \wedge \boldsymbol{y}) + (x \wedge \boldsymbol{a}, \boldsymbol{y} \wedge \boldsymbol{b}) - (x, \boldsymbol{y})(\boldsymbol{a}, \boldsymbol{b}) = 0, \\ & (*(\boldsymbol{a} \wedge *(\boldsymbol{b} \wedge \boldsymbol{x})) - *(\boldsymbol{b} \wedge *(\boldsymbol{a} \wedge \boldsymbol{x})) - (\boldsymbol{a}, \boldsymbol{b})\boldsymbol{x}, \boldsymbol{y}) \\ & = (x \wedge \boldsymbol{b}, \boldsymbol{y} \wedge \boldsymbol{a}) + (*\boldsymbol{a} \wedge \boldsymbol{x}, *\boldsymbol{b} \wedge \boldsymbol{y}) - (x, \boldsymbol{y})(\boldsymbol{a}, \boldsymbol{b}) = 0. \end{aligned}$ | $\begin{aligned} & (\boldsymbol{x} \wedge *(\boldsymbol{a} \wedge \boldsymbol{y}) + *(\boldsymbol{y} \wedge *(\boldsymbol{a} \wedge \boldsymbol{x})) - (\boldsymbol{x}, \boldsymbol{y})\boldsymbol{a}, \boldsymbol{b}) \\ & = (*\boldsymbol{b} \wedge \boldsymbol{x}, *\boldsymbol{a} \wedge \boldsymbol{y}) + (\boldsymbol{x} \wedge \boldsymbol{a}, \boldsymbol{y} \wedge \boldsymbol{b}) - (\boldsymbol{x}, \boldsymbol{y})(\boldsymbol{a}, \boldsymbol{b}) = 0, \\ & (*(\boldsymbol{a} \wedge *(\boldsymbol{b} \wedge \boldsymbol{x})) - *(\boldsymbol{b} \wedge *(\boldsymbol{a} \wedge \boldsymbol{x})) - (\boldsymbol{a}, \boldsymbol{b})\boldsymbol{x}, \boldsymbol{y}) \\ & = (\boldsymbol{x} \wedge \boldsymbol{b}, \boldsymbol{y} \wedge \boldsymbol{a}) + (*\boldsymbol{a} \wedge \boldsymbol{x}, *\boldsymbol{b} \wedge \boldsymbol{y}) - (\boldsymbol{x}, \boldsymbol{y})(\boldsymbol{a}, \boldsymbol{b}) = 0. \end{aligned}$ |

| Identifier           | Page | Conf. | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Rendered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Image                                                                                                                                                                             |
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| 0902.0431_E00370     | 61   | 0.557 | $\begin{aligned} & \begin{aligned} & \eta_1(t)^2 + \eta_2(t)^2 + \eta_3(t)^2 + \\ & \eta_4(t)^2 \quad \&= \xi_1^2 + \left( \frac{\xi_2 + \xi_3}{2} + \frac{\xi_2 - \xi_3}{2} \cos 2t +  x_1  \sin 2t \right)^2 \\ & + \left( \frac{\xi_2 + \xi_3}{2} - \frac{\xi_2 - \xi_3}{2} \cos 2t -  x_1  \sin 2t \right)^2 \\ & + \left( \frac{\xi_2 + \xi_3}{2} + \frac{\xi_2 - \xi_3}{2} \cos 2t +  x_1  \sin 2t \right)^2 \\ & + \left( \frac{\xi_2 + \xi_3}{2} - \frac{\xi_2 - \xi_3}{2} \cos 2t -  x_1  \sin 2t \right)^2 \\ & = \xi_1^2 + 2 \left( \frac{\xi_2 + \xi_3}{2} \right)^2 + 2 \left( \frac{\xi_2 - \xi_3}{2} \cos 2t +  x_1  \sin 2t \right)^2 \\ & + 2 \left( \frac{\xi_2 + \xi_3}{2} - \frac{\xi_2 - \xi_3}{2} \cos 2t -  x_1  \sin 2t \right)^2 \\ & = \xi_1^2 + 2 \left( \frac{\xi_2 + \xi_3}{2} \right)^2 + 2 \left( \left( \frac{\xi_2 - \xi_3}{2} \right)^2 +  x_1 ^2 \right) \sin^2(2t + t_0) \text{ ( for some } t_0 \in \boldsymbol{R} ) \\ & \leq \xi_1^2 + 2 \left( \frac{\xi_2 + \xi_3}{2} \right)^2 + 2 \left( \left( \frac{\xi_2 - \xi_3}{2} \right)^2 +  x_1 ^2 \right) \\ & = \xi_1^2 + \xi_2^2 + \xi_3^2 + 2 x_1 ^2 \end{aligned} \end{aligned}$ | $\begin{aligned} & \eta_1(t)^2 + \eta_2(t)^2 + \eta_3(t)^2 \\ & = \xi_1^2 + \left( \frac{\xi_2 + \xi_3}{2} + \frac{\xi_2 - \xi_3}{2} \cos 2t +  x_1  \sin 2t \right)^2 \\ & + \left( \frac{\xi_2 + \xi_3}{2} - \frac{\xi_2 - \xi_3}{2} \cos 2t -  x_1  \sin 2t \right)^2 \\ & = \xi_1^2 + 2 \left( \frac{\xi_2 + \xi_3}{2} \right)^2 + 2 \left( \frac{\xi_2 - \xi_3}{2} \cos 2t +  x_1  \sin 2t \right)^2 \\ & + 2 \left( \frac{\xi_2 + \xi_3}{2} - \frac{\xi_2 - \xi_3}{2} \cos 2t -  x_1  \sin 2t \right)^2 \\ & = \xi_1^2 + 2 \left( \frac{\xi_2 + \xi_3}{2} \right)^2 + 2 \left( \left( \frac{\xi_2 - \xi_3}{2} \right)^2 +  x_1 ^2 \right) \sin^2(2t + t_0) \text{ ( for some } t_0 \in \boldsymbol{R} ) \\ & \leq \xi_1^2 + 2 \left( \frac{\xi_2 + \xi_3}{2} \right)^2 + 2 \left( \left( \frac{\xi_2 - \xi_3}{2} \right)^2 +  x_1 ^2 \right) \\ & = \xi_1^2 + \xi_2^2 + \xi_3^2 + 2 x_1 ^2 \end{aligned}$ |                                                                                                                                                                                   |
| 0902.0431_E00674 (i) | 112  | 0.558 | $2(P \times R)Q - (P \times Q)R - (R \times Q)P + \frac{3}{8}\{Q, R\}P - \frac{3}{8}\{P, Q\}R = 0.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | $2(P \times R)Q - (P \times Q)R - (R \times Q)P + \frac{3}{8}\{Q, R\}P - \frac{3}{8}\{P, Q\}R = 0.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | $2(P \times R)Q - (P \times Q)R - (R \times Q)P + \frac{3}{8}\{Q, R\}P - \frac{3}{8}\{P, Q\}R = 0.$                                                                               |
| 0902.0431_E01188     | 200  | 0.562 | $\begin{aligned} & (a \times (b \wedge x))v + (b \times (a \wedge x))v = (x, w) * (a \wedge b) - \frac{1}{5}(* (a \wedge b), x)v \\ & = (x \times *(a \wedge b))v. \end{aligned}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | $\begin{aligned} & (a \times (b \wedge x))v + (b \times (a \wedge x))v = (x, w) * (a \wedge b) - \frac{1}{5}(* (a \wedge b), x)v \\ & = (x \times *(a \wedge b))v. \end{aligned}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | $\begin{aligned} & (a \times (b \wedge x))v + (b \times (a \wedge x))v = (x, w) * (a \wedge b) - \frac{1}{5}(* (a \wedge b), x)v \\ & = (x \times *(a \wedge b))v. \end{aligned}$ |



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| 0902.0431_EQ0067     | 15   | ■ 0.573 | $a+\left(\begin{array}{l} m_1 \\ m_2 \\ m_3 \end{array}\right) \cdot$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | $a+\left(\begin{array}{l} m_1 \\ m_2 \\ m_3 \end{array}\right) \cdot$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | $a+\left(\begin{array}{l} m_1 \\ m_2 \\ m_3 \end{array}\right) \cdot$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 0902.0431_EQ1027     | 171  | ■ 0.577 | $\begin{array}{l} \backslash begin{array}{l} \backslash lambda_{\{0\}}+\frac{1}{2}\backslash left( \\ \backslash mu_{\{2\}}-\backslash mu_{\{3\}}\backslash right)=2 \& 4 \& 6 \& 5 \& 3 \& 2 \& 1 \& 3 \backslash \backslash \\ \backslash lambda_{\{1\}}+\frac{1}{2}\backslash left(\backslash mu_{\{2\}}-\backslash mu_{\{3\}}\backslash right)=0 \& 1 \\ \& 2 \& 2 \& 1 \& 1 \& 1 \& 1 \end{array}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | $\begin{array}{l} \lambda_0+\frac{1}{2}(\mu_2-\mu_3)=2 \quad 4 \quad 6 \quad 5 \quad 3 \quad 2 \quad 1 \quad 3 \\ \lambda_1+\frac{1}{2}(\mu_2-\mu_3)=0 \quad 1 \quad 2 \quad 2 \quad 1 \quad 1 \quad 1 \quad 1 \end{array}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | $\begin{array}{l} \lambda_0+\frac{1}{2}(\mu_2-\mu_3)=2 \quad 4 \quad 6 \quad 5 \quad 3 \quad 2 \quad 1 \quad 3 \\ \lambda_1+\frac{1}{2}(\mu_2-\mu_3)=0 \quad 1 \quad 2 \quad 2 \quad 1 \quad 1 \quad 1 \quad 1 \end{array}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 0902.0431_EQ1047     | 178  | ■ 0.578 | $\begin{array}{l} \backslash begin{aligned} \& -\backslash left(\backslash beta_{\{12\}} P, \backslash beta_{\{2\}} P, \\ \backslash beta_{\{32\}} P, b_{\{1\}} P, b_{\{2\}} P, b_{\{3\}} P\backslash right) \backslash \& \\ \backslash quad=\backslash left[\backslash left(\backslash beta_{\{12\}} P, \backslash beta_{\{2\}} P, \backslash beta_{\{32\}} \\ P, b_{\{1\}} P, b_{\{2\}} P, b_{\{3\}} P\backslash right), (0,0,0,1,0,0)\backslash right] \\ \backslash \& \backslash quad=\backslash left(0,-\backslash beta_{\{2\}} P, \backslash beta_{\{32\}} P, 0,-2 b_{\{2\}} \\ P, 2 b_{\{3\}} P\backslash right) \end{array}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | $\begin{array}{l} -(\beta_{12}P, \beta_2P, \beta_{32}P, b_1P, b_2P, b_3P) \\ =[(\beta_{12}P, \beta_2P, \beta_{32}P, b_1P, b_2P, b_3P), (0,0,0,1,0,0)] \\ = (0, -\beta_2P, \beta_{32}P, 0, -2b_2P, 2b_3P) \end{array}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | $\begin{array}{l} -(\beta_{12}P, \beta_2P, \beta_{32}P, b_1P, b_2P, b_3P) \\ =[(\beta_{12}P, \beta_2P, \beta_{32}P, b_1P, b_2P, b_3P), (0,0,0,1,0,0)] \\ = (0, -\beta_2P, \beta_{32}P, 0, -2b_2P, 2b_3P), \end{array}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 0902.0431_EQ0075     | 16   | ■ 0.579 | $\begin{array}{l} \backslash begin{aligned} S_{\{1\}} \& =2 G_{\{12\}}-G_{\{47\}}-G_{\{56\}}, \\ \backslash quad S_{\{2\}}=2 G_{\{13\}}-G_{\{46\}}+G_{\{57\}}, \backslash \backslash S_{\{3\}} \& =2 \\ G_{\{14\}}+G_{\{27\}}+G_{\{36\}}, \backslash quad S_{\{4\}}=2 G_{\{15\}}+G_{\{26\}}-G_{\{37\}}, \\ \backslash \backslash S_{\{5\}} \& =2 G_{\{16\}}-G_{\{25\}}-G_{\{34\}}, \backslash quad S_{\{6\}}=2 \\ G_{\{17\}}-G_{\{24\}}+G_{\{35\}}, \end{array}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | $\begin{array}{l} S_1=2G_{12}-G_{47}-G_{56}, \quad S_2=2G_{13}-G_{46}+G_{57}, \\ S_3=2G_{14}+G_{27}+G_{36}, \quad S_4=2G_{15}+G_{26}-G_{37}, \\ S_5=2G_{16}-G_{25}-G_{34}, \quad S_6=2G_{17}-G_{24}+G_{35}, \end{array}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | $\begin{array}{l} S_1=2G_{12}-G_{47}-G_{56}, \quad S_2=2G_{13}-G_{46}+G_{57}, \\ S_3=2G_{14}+G_{27}+G_{36}, \quad S_4=2G_{15}+G_{26}-G_{37}, \\ S_5=2G_{16}-G_{25}-G_{34}, \quad S_6=2G_{17}-G_{24}+G_{35}, \end{array}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 0902.0431_EQ1176     | 198  | ■ 0.582 | $\begin{array}{l} \backslash begin{aligned} (*(* \backslash boldsymbol{a} \backslash wedge \& \\ *(\backslash boldsymbol{b} \backslash wedge \backslash boldsymbol{c}))), \\ \backslash boldsymbol{x})=(\backslash boldsymbol{a}, *(\backslash boldsymbol{b} \\ \backslash wedge \backslash boldsymbol{c})) \backslash wedge \backslash boldsymbol{x}) \backslash \& = \\ \backslash left(\backslash boldsymbol{a}_{\{1\}}, *(\backslash boldsymbol{b} \backslash wedge \\ \backslash boldsymbol{c})\backslash right)\backslash left(\backslash boldsymbol{a}_{\{2\}}, \\ \backslash boldsymbol{x}\backslash right)-\backslash left(\backslash boldsymbol{a}_{\{2\}}, * \\ \backslash boldsymbol{b} \backslash wedge \backslash boldsymbol{c})\backslash right)\backslash left( \\ \backslash boldsymbol{a}_{\{1\}}, \backslash boldsymbol{x}\backslash right) \backslash \& = \\ \backslash left(\backslash boldsymbol{a}_{\{1\}} \backslash wedge \backslash boldsymbol{b} \backslash wedge \\ \backslash boldsymbol{c}, \backslash boldsymbol{e}_{\{1\}} \backslash wedge \backslash cdots \backslash wedge \\ \backslash boldsymbol{e}_{\{5\}}\backslash right)\backslash left(\backslash boldsymbol{a}_{\{2\}}, \\ \backslash boldsymbol{x}\backslash right)-\backslash left(\backslash boldsymbol{a}_{\{2\}} \backslash wedge \\ \backslash boldsymbol{b} \backslash wedge \backslash boldsymbol{c}, \backslash boldsymbol{e}_{\{1\}} \\ \backslash wedge \backslash cdots \backslash wedge \backslash boldsymbol{e}_{\{5\}}\backslash right)\backslash left( \\ \backslash boldsymbol{a}_{\{1\}}, \backslash boldsymbol{x}\backslash right), \end{array}$ | $\begin{array}{l} (*(*\boldsymbol{a} \wedge *(\boldsymbol{b} \wedge \boldsymbol{c})), \boldsymbol{x}) = (\boldsymbol{a}, *(\boldsymbol{b} \wedge \boldsymbol{c}) \wedge \boldsymbol{x}) \\ = (\boldsymbol{a}_1, *(\boldsymbol{b} \wedge \boldsymbol{c}))(\boldsymbol{a}_2, \boldsymbol{x}) - (\boldsymbol{a}_2, *(\boldsymbol{b} \wedge \boldsymbol{c}))(\boldsymbol{a}_1, \boldsymbol{x}) \\ = (\boldsymbol{a}_1 \wedge \boldsymbol{b} \wedge \boldsymbol{c}, \boldsymbol{e}_1 \wedge \cdots \wedge \boldsymbol{e}_5)(\boldsymbol{a}_2, \boldsymbol{x}) - (\boldsymbol{a}_2 \wedge \boldsymbol{b} \wedge \boldsymbol{c}, \boldsymbol{e}_1 \wedge \cdots \wedge \boldsymbol{e}_5)(\boldsymbol{a}_1, \boldsymbol{x}), \end{array}$ | $\begin{array}{l} (*(*\boldsymbol{a} \wedge *(\boldsymbol{b} \wedge \boldsymbol{c})), \boldsymbol{x}) = (\boldsymbol{a}, *(\boldsymbol{b} \wedge \boldsymbol{c}) \wedge \boldsymbol{x}) \\ = (\boldsymbol{a}_1, *(\boldsymbol{b} \wedge \boldsymbol{c}))(\boldsymbol{a}_2, \boldsymbol{x}) - (\boldsymbol{a}_2, *(\boldsymbol{b} \wedge \boldsymbol{c}))(\boldsymbol{a}_1, \boldsymbol{x}) \\ = (\boldsymbol{a}_1 \wedge \boldsymbol{b} \wedge \boldsymbol{c}, \boldsymbol{e}_1 \wedge \cdots \wedge \boldsymbol{e}_5)(\boldsymbol{a}_2, \boldsymbol{x}) - (\boldsymbol{a}_2 \wedge \boldsymbol{b} \wedge \boldsymbol{c}, \boldsymbol{e}_1 \wedge \cdots \wedge \boldsymbol{e}_5)(\boldsymbol{a}_1, \boldsymbol{x}), \end{array}$ |
| 0902.0431_EQ0728 (b) | 120  | ■ 0.589 | $\begin{array}{l} \backslash begin{aligned} \& W \backslash ni \backslash Phi(0, X, 0,0)(0,0,0,1)=(X, \\ 0,0,0) \backslash \& W \backslash ni \backslash Phi\backslash left(0, E_{\{2\}}, 0,0\backslash right) \\ \backslash left(E_{\{3\}}, 0,0,0\backslash right)=\backslash left(0, E_{\{1\}}, 0,0\backslash right) \backslash \& \\ W \backslash ni \backslash Phi\backslash left(0, E_{\{1\}}, 0,0\backslash right)\backslash left(0, E_{\{1\}}, 0,0 \\ \backslash right)=(0,0,1,0) \backslash \& W \backslash ni \backslash Phi(0,0, Y, 0)(0,0,1,0)=(0, \\ Y, 0,0) \end{array}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | $\begin{array}{l} W \ni \Phi(0, X, 0, 0)(0, 0, 0, 1) = (X, 0, 0, 0) \\ W \ni \Phi(0, E_2, 0, 0)(E_3, 0, 0, 0) = (0, E_1, 0, 0) \\ W \ni \Phi(0, E_1, 0, 0)(0, E_1, 0, 0) = (0, 0, 1, 0) \\ W \ni \Phi(0, 0, Y, 0)(0, 0, 1, 0) = (0, Y, 0, 0) \end{array}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | $\begin{array}{l} W \ni \Phi(0, X, 0, 0)(0, 0, 0, 1) = (X, 0, 0, 0), \\ W \ni \Phi(0, E_2, 0, 0)(E_3, 0, 0, 0) = (0, E_1, 0, 0), \\ W \ni \Phi(0, E_1, 0, 0)(0, E_1, 0, 0) = (0, 0, 1, 0), \\ W \ni \Phi(0, 0, Y, 0)(0, 0, 1, 0) = (0, Y, 0, 0). \end{array}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

| Identifier       | Page | Conf.   | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Rendered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Image                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|------------------|------|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0902.0431_E08688 | 114  | ■ 0.589 | $\begin{aligned} & \left[ \Phi_1, \Phi_2 \right] P = \Phi_1 \Phi_2 P - \Phi_2 \Phi_1 P \\ & = \cdots \text{ ( using the formula of } A \vee B \text{ (Lemma 3.4.1) etc.) } \cdots \\ & = \Phi P \end{aligned}$                                                                                                                                                                                                                                                                                                                                                                                                                                         | $[\Phi_1, \Phi_2] P = \Phi_1 \Phi_2 P - \Phi_2 \Phi_1 P$ $= \cdots \text{ ( using the formula of } A \vee B \text{ (Lemma 3.4.1) etc.) } \cdots$ $= \Phi P$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | $[\Phi_1, \Phi_2] P = \Phi_1 \Phi_2 P - \Phi_2 \Phi_1 P$ $= \cdots \text{ (using the formula of } A \vee B \text{ (Lemma 3.4.1) etc.) } \cdots$ $= \Phi P.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 0902.0431_E08350 | 57   | ■ 0.589 | $\begin{aligned} & R(\left(\left(\alpha_1 x_1\right)\left(\alpha_2 x_2\right)\left(\alpha_3 x_3\right)\right)=R\left(\left(\overline{\alpha_3\left(x_1 x_2\right)}\right) \alpha_3 x_3\right) \\ & =\left(\alpha_3\left(x_1 x_2\right), \alpha_3 x_3\right)=\left(x_1 x_2, x_3\right)=R\left(x_1 x_2 x_3\right) \end{aligned}$                                                                                                                                                                                                                                                                                                                         | $R\left(\left(\alpha_1 x_1\right)\left(\alpha_2 x_2\right)\left(\alpha_3 x_3\right)\right)=R\left(\left(\overline{\alpha_3\left(x_1 x_2\right)}\right) \alpha_3 x_3\right)$ $=\left(\alpha_3\left(x_1 x_2\right), \alpha_3 x_3\right)=\left(x_1 x_2, x_3\right)=R\left(x_1 x_2 x_3\right)$                                                                                                                                                                                                                                                                                                                                                             | $R\left(\left(\alpha_1 x_1\right)\left(\alpha_2 x_2\right)\left(\alpha_3 x_3\right)\right)=R\left(\left(\overline{\alpha_3\left(x_1 x_2\right)}\right) \alpha_3 x_3\right)$ $=\left(\alpha_3\left(x_1 x_2\right), \alpha_3 x_3\right)=\left(x_1 x_2, x_3\right)=R\left(x_1 x_2 x_3\right),$                                                                                                                                                                                                                                                                                                                                                                       |
| 0902.0431_E08870 | 144  | ■ 0.590 | $\begin{array}{l} 1 \rightarrow \text{Spin}(11) \rightarrow (E_7)^{\kappa, \mu} \rightarrow S^{11} \rightarrow * \\ \downarrow p' \quad \downarrow p \quad \downarrow = \\ 1 \rightarrow SO(11) \rightarrow SO(12) \rightarrow S^{11} \rightarrow * \end{array}$                                                                                                                                                                                                                                                                                                                                                                                       | $\begin{array}{l} 1 \rightarrow \text{Spin}(11) \rightarrow (E_7)^{\kappa, \mu} \rightarrow S^{11} \rightarrow * \\ \downarrow p' \quad \downarrow p \quad \downarrow = \\ 1 \rightarrow SO(11) \rightarrow SO(12) \rightarrow S^{11} \rightarrow * \end{array}$                                                                                                                                                                                                                                                                                                                                                                                       | $\begin{array}{ccccccc} 1 & \longrightarrow & Spin(11) & \longrightarrow & (E_7)^{\kappa, \mu} & \longrightarrow & S^{11} \longrightarrow * \\ & & \downarrow p' & & \downarrow p & & \downarrow = \\ 1 & \longrightarrow & SO(11) & \longrightarrow & SO(12) & \longrightarrow & S^{11} \longrightarrow * \end{array}$                                                                                                                                                                                                                                                                                                                                           |
| 0902.0431_E08324 | 52   | ■ 0.591 | $\begin{aligned} & \widetilde{A}_1(1) \widetilde{A}_1(1) E_2 = -\frac{1}{2} \widetilde{A}_1(1) F_1(1) = -\frac{1}{2} (E_2 - E_3), \\ & \widetilde{A}_1(1) \widetilde{A}_1(1) E_3 = \frac{1}{2} \widetilde{A}_1(1) F_1(1) = \frac{1}{2} (E_2 - E_3), \\ & \widetilde{A}_1(1) \widetilde{A}_1(1) F_1(1) = \widetilde{A}_1(1) (E_2 - E_3) = -F_1(1), \\ & \widetilde{A}_1(1) \widetilde{A}_1(1) F_2(e_i) = \frac{1}{2} \widetilde{A}_1(1) F_3(e_i) = -\frac{1}{4} F_2(e_i), \\ & \widetilde{A}_1(1) \widetilde{A}_1(1) F_3(e_i) = -\frac{1}{2} \widetilde{A}_1(1) F_3(\overline{e_i}) = -\frac{1}{4} F_3(e_i), \\ & \text{the others} = 0. \end{aligned}$ | $\begin{aligned} & \widetilde{A}_1(1) \widetilde{A}_1(1) E_2 = -\frac{1}{2} \widetilde{A}_1(1) F_1(1) = -\frac{1}{2} (E_2 - E_3), \\ & \widetilde{A}_1(1) \widetilde{A}_1(1) E_3 = \frac{1}{2} \widetilde{A}_1(1) F_1(1) = \frac{1}{2} (E_2 - E_3), \\ & \widetilde{A}_1(1) \widetilde{A}_1(1) F_1(1) = \widetilde{A}_1(1) (E_2 - E_3) = -F_1(1), \\ & \widetilde{A}_1(1) \widetilde{A}_1(1) F_2(e_i) = \frac{1}{2} \widetilde{A}_1(1) F_3(e_i) = -\frac{1}{4} F_2(e_i), \\ & \widetilde{A}_1(1) \widetilde{A}_1(1) F_3(e_i) = -\frac{1}{2} \widetilde{A}_1(1) F_3(\overline{e_i}) = -\frac{1}{4} F_3(e_i), \\ & \text{the others} = 0. \end{aligned}$ | $\begin{aligned} & \widetilde{A}_1(1) \widetilde{A}_1(1) E_2 = -\frac{1}{2} \widetilde{A}_1(1) F_1(1) = -\frac{1}{2} (E_2 - E_3), \\ & \widetilde{A}_1(1) \widetilde{A}_1(1) E_3 = \frac{1}{2} \widetilde{A}_1(1) F_1(1) = \frac{1}{2} (E_2 - E_3), \\ & \widetilde{A}_1(1) \widetilde{A}_1(1) F_1(1) = \widetilde{A}_1(1) (E_2 - E_3) = -F_1(1), \\ & \widetilde{A}_1(1) \widetilde{A}_1(1) F_2(e_i) = \frac{1}{2} \widetilde{A}_1(1) F_3(\overline{e_i}) = -\frac{1}{4} F_2(e_i), \\ & \widetilde{A}_1(1) \widetilde{A}_1(1) F_3(e_i) = -\frac{1}{2} \widetilde{A}_1(1) F_3(\overline{e_i}) = -\frac{1}{4} F_3(e_i), \\ & \text{the others} = 0. \end{aligned}$ |

| Identifier       | Page | Conf.   | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Rendered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Image                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|------------------|------|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0902.0431_E01083 | 184  | ■ 0.593 | $\begin{aligned} S_1 \tau S_2 - S_2 \tau S_1 &= -k(gX_1 gX_2 - gX_2 gX_1 + g(\gamma Y_1)g(\gamma Y_2) - g(\gamma Y_2)g(\gamma Y_1)) \\ &\quad - ik(g(\gamma Y_1)gX_2 - g(\gamma Y_2)gX_1 - gX_1 g(\gamma Y_2) + gX_2 g(\gamma Y_1)) \\ &\quad - \eta_1 gX_2 + \eta_2 gX_1 + \xi_1 g(\gamma Y_2) - \xi_2 g(\gamma Y_1) + \frac{i}{2}(\xi_1 \eta_2 - \eta_2 \xi_1)E \\ &= k(-[gX_1, gX_2] - [g(\gamma Y_1), g(\gamma Y_2)]) \\ &\quad + ik(g(2\gamma X_1 \times Y_2 - 2\gamma X_2 \times Y_1 + \eta_1 X_2 - \eta_2 X_1 - \xi_1 \gamma Y_2 + \xi_2 \gamma Y_1)) \\ &\quad + \frac{i}{2}((X_1, Y_2) - (X_2, Y_1) + \xi_1 \eta_2 - \xi_2 \eta_1))E \quad (\text{Lemma 3.12.1}) \end{aligned}$                                                                                     | $\begin{aligned} S_1 \tau S_2 - S_2 \tau S_1 &= -k(gX_1 gX_2 - gX_2 gX_1 + g(\gamma Y_1)g(\gamma Y_2) - g(\gamma Y_2)g(\gamma Y_1)) \\ &\quad - ik(g(\gamma Y_1)gX_2 - g(\gamma Y_2)gX_1 - gX_1 g(\gamma Y_2) + gX_2 g(\gamma Y_1)) \\ &\quad - \eta_1 gX_2 + \eta_2 gX_1 + \xi_1 g(\gamma Y_2) - \xi_2 g(\gamma Y_1) + \frac{i}{2}(\xi_1 \eta_2 - \eta_2 \xi_1)E \\ &= k(-[gX_1, gX_2] - [g(\gamma Y_1), g(\gamma Y_2)]) \\ &\quad + ik(g(2\gamma X_1 \times Y_2 - 2\gamma X_2 \times Y_1 + \eta_1 X_2 - \eta_2 X_1 - \xi_1 \gamma Y_2 + \xi_2 \gamma Y_1)) \\ &\quad + \frac{i}{2}((X_1, Y_2) - (X_2, Y_1) + \xi_1 \eta_2 - \xi_2 \eta_1))E \quad (\text{Lemma 3.12.1}) \end{aligned}$                                                                                     | $\begin{aligned} S_1 \tau S_2 - S_2 \tau S_1 &= -k(gX_1 gX_2 - gX_2 gX_1 + g(\gamma Y_1)g(\gamma Y_2) - g(\gamma Y_2)g(\gamma Y_1)) \\ &\quad - ik(g(\gamma Y_1)gX_2 - g(\gamma Y_2)gX_1 - gX_1 g(\gamma Y_2) + gX_2 g(\gamma Y_1)) \\ &\quad - \eta_1 gX_2 + \eta_2 gX_1 + \xi_1 g(\gamma Y_2) - \xi_2 g(\gamma Y_1) + \frac{i}{2}(\xi_1 \eta_2 - \eta_2 \xi_1)E \\ &= k(-[gX_1, gX_2] - [g(\gamma Y_1), g(\gamma Y_2)]) \\ &\quad + ik(g(2\gamma X_1 \times Y_2 - 2\gamma X_2 \times Y_1 + \eta_1 X_2 - \eta_2 X_1 - \xi_1 \gamma Y_2 + \xi_2 \gamma Y_1)) \\ &\quad + \frac{i}{2}((X_1, Y_2) - (X_2, Y_1) + \xi_1 \eta_2 - \xi_2 \eta_1))E \quad (\text{Lemma 3.12.1}) \end{aligned}$                                                                                     |
| 0902.0431_E00678 | 112  | ■ 0.595 | $\begin{aligned} &\left(\begin{array}{c} X \\ \frac{1}{\eta}(X \times X) \\ \frac{1}{\eta^2} \det X \\ \eta \end{array}\right), \quad \left(\begin{array}{c} \frac{1}{\xi}(Y \times Y) \\ Y \\ \xi \\ \frac{1}{\xi^2} \det Y \end{array}\right), \quad \mathbf{i} = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \quad \mathbf{l} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} \\ &\left(\begin{array}{c} X \\ \frac{1}{\eta}(X \times X) \\ \frac{1}{\eta^2} \det X \\ \eta \end{array}\right), \quad \left(\begin{array}{c} \frac{1}{\xi}(Y \times Y) \\ Y \\ \xi \\ \frac{1}{\xi^2} \det Y \end{array}\right), \quad \mathbf{i} = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \quad \mathbf{l} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} \end{aligned}$ | $\begin{aligned} &\left(\begin{array}{c} X \\ \frac{1}{\eta}(X \times X) \\ \frac{1}{\eta^2} \det X \\ \eta \end{array}\right), \quad \left(\begin{array}{c} \frac{1}{\xi}(Y \times Y) \\ Y \\ \xi \\ \frac{1}{\xi^2} \det Y \end{array}\right), \quad \mathbf{i} = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \quad \mathbf{l} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} \\ &\left(\begin{array}{c} X \\ \frac{1}{\eta}(X \times X) \\ \frac{1}{\eta^2} \det X \\ \eta \end{array}\right), \quad \left(\begin{array}{c} \frac{1}{\xi}(Y \times Y) \\ Y \\ \xi \\ \frac{1}{\xi^2} \det Y \end{array}\right), \quad \mathbf{i} = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \quad \mathbf{l} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} \end{aligned}$ | $\begin{aligned} &\left(\begin{array}{c} X \\ \frac{1}{\eta}(X \times X) \\ \frac{1}{\eta^2} \det X \\ \eta \end{array}\right), \quad \left(\begin{array}{c} \frac{1}{\xi}(Y \times Y) \\ Y \\ \xi \\ \frac{1}{\xi^2} \det Y \end{array}\right), \quad \mathbf{i} = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \quad \mathbf{l} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} \\ &\left(\begin{array}{c} X \\ \frac{1}{\eta}(X \times X) \\ \frac{1}{\eta^2} \det X \\ \eta \end{array}\right), \quad \left(\begin{array}{c} \frac{1}{\xi}(Y \times Y) \\ Y \\ \xi \\ \frac{1}{\xi^2} \det Y \end{array}\right), \quad \mathbf{i} = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \quad \mathbf{l} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} \end{aligned}$ |



| Identifier       | Page | Conf.   | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Rendered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Image                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|------------------|------|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0902.0431_E0671  | 111  | ■ 0.606 | $P \times Q = \Phi(\phi, A, B, \nu), \left\{ \begin{array}{l} \phi = -\frac{1}{2}(X \vee W + Z \vee Y) \\ A = -\frac{1}{4}(2Y \times W - \xi Z - \zeta X) \\ B = \frac{1}{4}(2X \times Z - \eta W - \omega Y) \\ \nu = \frac{1}{8}((X, W) + (Z, Y) - 3(\xi\omega + \zeta\eta)). \end{array} \right.$                                                                                                                                                                                                                                                                | $P \times Q = \Phi(\phi, A, B, \nu), \left\{ \begin{array}{l} \phi = -\frac{1}{2}(X \vee W + Z \vee Y) \\ A = -\frac{1}{4}(2Y \times W - \xi Z - \zeta X) \\ B = \frac{1}{4}(2X \times Z - \eta W - \omega Y) \\ \nu = \frac{1}{8}((X, W) + (Z, Y) - 3(\xi\omega + \zeta\eta)). \end{array} \right.$                                                                                                                                                                                                                                                                | $P \times Q = \Phi(\phi, A, B, \nu), \left\{ \begin{array}{l} \phi = -\frac{1}{2}(X \vee W + Z \vee Y) \\ A = -\frac{1}{4}(2Y \times W - \xi Z - \zeta X) \\ B = \frac{1}{4}(2X \times Z - \eta W - \omega Y) \\ \nu = \frac{1}{8}((X, W) + (Z, Y) - 3(\xi\omega + \zeta\eta)). \end{array} \right.$                                                                                                                                                                                                                                                                |
| 0902.0431_E0924  | 153  | ■ 0.608 | $f(h(a)P) = a(fP), \quad a \in C, P \in (\mathfrak{P}_C)^C.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | $f(h(a)P) = a(fP), \quad a \in C, P \in (\mathfrak{P}_C)^C.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | $f(h(a)P) = a(fP), \quad a \in C, P \in (\mathfrak{P}_C)^C.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 0902.0431_E01103 | 187  | ■ 0.608 | $\mathfrak{e}_{8(8)} = \left( \mathfrak{e}_{8(8)} \right)^{\tilde{\lambda}_\gamma} \oplus \left( \mathfrak{e}_{8(8)} \right)_{-\tilde{\lambda}_\gamma}.$                                                                                                                                                                                                                                                                                                                                                                                                            | $\mathfrak{e}_{8(8)} = \left( \mathfrak{e}_{8(8)} \right)^{\tilde{\lambda}_\gamma} \oplus \left( \mathfrak{e}_{8(8)} \right)_{-\tilde{\lambda}_\gamma}.$                                                                                                                                                                                                                                                                                                                                                                                                            | $\mathfrak{e}_{8(8)} = \left( \mathfrak{e}_{8(8)} \right)^{\tilde{\lambda}_\gamma} \oplus \left( \mathfrak{e}_{8(8)} \right)_{-\tilde{\lambda}_\gamma}.$                                                                                                                                                                                                                                                                                                                                                                                                            |
| 0902.0431_E0648  | 107  | ■ 0.608 | $\begin{aligned} & \varphi((A, B), \epsilon) \varphi((C, D), 1) X \\ &= \varphi((A, B), \epsilon) (h(C, D) X h(C, D)^*) \\ &= h(A, B) \overline{h(C, D) X h(C, D)^*} h(A, B)^* \\ &= h(A, B) h(\overline{D}, \overline{C}) \overline{X} h(\overline{D}, \overline{C})^* h(A, B)^* \quad (\text{Lemma 3.13.3.(3)}) \\ &= h(A \overline{D}, B \overline{C}) \overline{X} h(A \overline{D}, B \overline{C})^* = \varphi((A \overline{D}, B \overline{C}), \epsilon) X \\ &= \varphi((A, B) \epsilon(C, D), \epsilon) X, \quad X \in (\mathfrak{I}_C)^C. \end{aligned}$ | $\begin{aligned} & \varphi((A, B), \epsilon) \varphi((C, D), 1) X \\ &= \varphi((A, B), \epsilon) (h(C, D) X h(C, D)^*) \\ &= h(A, B) \overline{h(C, D) X h(C, D)^*} h(A, B)^* \\ &= h(A, B) h(\overline{D}, \overline{C}) \overline{X} h(\overline{D}, \overline{C})^* h(A, B)^* \quad (\text{Lemma 3.13.3.(3)}) \\ &= h(A \overline{D}, B \overline{C}) \overline{X} h(A \overline{D}, B \overline{C})^* = \varphi((A \overline{D}, B \overline{C}), \epsilon) X \\ &= \varphi((A, B) \epsilon(C, D), \epsilon) X, \quad X \in (\mathfrak{I}_C)^C. \end{aligned}$ | $\begin{aligned} & \varphi((A, B), \epsilon) \varphi((C, D), 1) X \\ &= \varphi((A, B), \epsilon) (h(C, D) X h(C, D)^*) \\ &= h(A, B) \overline{h(C, D) X h(C, D)^*} h(A, B)^* \\ &= h(A, B) h(\overline{D}, \overline{C}) \overline{X} h(\overline{D}, \overline{C})^* h(A, B)^* \quad (\text{Lemma 3.13.3.(3)}) \\ &= h(A \overline{D}, B \overline{C}) \overline{X} h(A \overline{D}, B \overline{C})^* = \varphi((A \overline{D}, B \overline{C}), \epsilon) X \\ &= \varphi((A, B) \epsilon(C, D), \epsilon) X, \quad X \in (\mathfrak{I}_C)^C. \end{aligned}$ |
| 0902.0431_E0837  | 54   | ■ 0.622 | $\begin{aligned} & \kappa \pi H = \frac{1}{2} (-\lambda_0 + \lambda_1 - \lambda_2 + \lambda_3) H_0 + \frac{1}{2} (-\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3) H_1 \\ & \quad + \frac{1}{2} (\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3) H_2 + \frac{1}{2} (-\lambda_0 - \lambda_1 + \lambda_2 + \lambda_3) H_3 \end{aligned}$                                                                                                                                                                                                                                | $\begin{aligned} & \kappa \pi H = \frac{1}{2} (-\lambda_0 + \lambda_1 - \lambda_2 + \lambda_3) H_0 + \frac{1}{2} (-\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3) H_1 \\ & \quad + \frac{1}{2} (\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3) H_2 + \frac{1}{2} (-\lambda_0 - \lambda_1 + \lambda_2 + \lambda_3) H_3 \end{aligned}$                                                                                                                                                                                                                                | $\begin{aligned} & \kappa \pi H = \frac{1}{2} (-\lambda_0 + \lambda_1 - \lambda_2 + \lambda_3) H_0 + \frac{1}{2} (-\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3) H_1 \\ & \quad + \frac{1}{2} (\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3) H_2 + \frac{1}{2} (-\lambda_0 - \lambda_1 + \lambda_2 + \lambda_3) H_3, \end{aligned}$                                                                                                                                                                                                                               |
| 0902.0431_E0863  | 143  | ■ 0.623 | $\begin{aligned} & V^{12} = \{P \in \mathfrak{P}^C \mid \kappa P = P, \mu \tau \lambda P = P\} \\ &= \left\{ \left( \begin{pmatrix} 0 & 0 & 0 \\ 0 & \xi & x \\ 0 & \bar{x} & -\tau \xi \end{pmatrix}, \begin{pmatrix} \eta & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, 0, \tau \eta \right) \mid x \in \mathfrak{C}, \xi, \eta \in C \right\} \end{aligned}$                                                                                                                                                                                                  | $\begin{aligned} & V^{12} = \{P \in \mathfrak{P}^C \mid \kappa P = P, \mu \tau \lambda P = P\} \\ &= \left\{ \left( \begin{pmatrix} 0 & 0 & 0 \\ 0 & \xi & x \\ 0 & \bar{x} & -\tau \xi \end{pmatrix}, \begin{pmatrix} \eta & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, 0, \tau \eta \right) \mid x \in \mathfrak{C}, \xi, \eta \in C \right\} \end{aligned}$                                                                                                                                                                                                  | $\begin{aligned} & V^{12} = \{P \in \mathfrak{P}^C \mid \kappa P = P, \mu \tau \lambda P = P\} \\ &= \left\{ \left( \begin{pmatrix} 0 & 0 & 0 \\ 0 & \xi & x \\ 0 & \bar{x} & -\tau \xi \end{pmatrix}, \begin{pmatrix} \eta & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, 0, \tau \eta \right) \mid x \in \mathfrak{C}, \xi, \eta \in C \right\} \end{aligned}$                                                                                                                                                                                                  |

| Identifier                          | Page | Conf.   | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Rendered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Image                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-------------------------------------|------|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0902.0431_E01155                    | 195  | ■ 0.623 | $\mathfrak{e}_{\{8\}}^{\{C\}}=\mathfrak{sl}(9,C)\oplus\mathfrak{q},\quad\mathfrak{q}=\Lambda^3(C^9)\oplus\Lambda^3(C^9).$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | $\mathfrak{e}_8^C=\mathfrak{sl}(9,C)\oplus\mathfrak{q},\quad\mathfrak{q}=\Lambda^3(C^9)\oplus\Lambda^3(C^9).$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | $\mathfrak{e}_8^C=\mathfrak{sl}(9,C)\oplus\mathfrak{q},\quad\mathfrak{q}=\Lambda^3(C^9)\oplus\Lambda^3(C^9).$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 0902.0431_E00319                    | 51   | ■ 0.625 | $B_{\{4\}}(\delta_1,\delta_2)=9\left(\delta_1,\delta_2\right)_4=3\operatorname{tr}\left(\delta_1\delta_2\right),\quad\delta_1,\delta_2\in\mathfrak{f}_4^C.$                                                                                                                                                                                                                                                                                                                                                                                                                                                  | $B_4(\delta_1,\delta_2)=9(\delta_1,\delta_2)_4=3\operatorname{tr}(\delta_1\delta_2),\quad\delta_1,\delta_2\in\mathfrak{f}_4^C.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | $B_4(\delta_1,\delta_2)=9(\delta_1,\delta_2)_4=3\operatorname{tr}(\delta_1\delta_2),\quad\delta_1,\delta_2\in\mathfrak{f}_4^C.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 0902.0431_E00677                    | 112  | ■ 0.625 | $\begin{aligned} \mathfrak{M}^{\{C\}}&=\left\{P\in\mathfrak{P}\mid P\times P=0,P\neq 0\right\} \\ &=\left\{\begin{array}{l} P=(X,Y,\xi,\eta)\in\mathfrak{P}^C \\ P\neq 0 \end{array}\left \begin{array}{l} X\vee Y=0,(X,Y)=3\xi\eta, \\ X\times X=\eta Y,Y\times Y=\xi X \end{array}\right.\right\}. \end{aligned}$                                                                                                                                                                                                                                                                                          | $\mathfrak{M}^C=\left\{P\in\mathfrak{P}^C\mid P\times P=0,P\neq 0\right\} \\ =\left\{\begin{array}{l} P=(X,Y,\xi,\eta)\in\mathfrak{P}^C \\ P\neq 0 \end{array}\left \begin{array}{l} X\vee Y=0,(X,Y)=3\xi\eta, \\ X\times X=\eta Y,Y\times Y=\xi X \end{array}\right.\right\}.$                                                                                                                                                                                                                                                                                                                              | $\mathfrak{M}^C=\left\{P\in\mathfrak{P}^C\mid P\times P=0,P\neq 0\right\} \\ =\left\{\begin{array}{l} P=(X,Y,\xi,\eta)\in\mathfrak{P}^C \\ P\neq 0 \end{array}\left \begin{array}{l} X\vee Y=0,(X,Y)=3\xi\eta, \\ X\times X=\eta Y,Y\times Y=\xi X \end{array}\right.\right\}.$                                                                                                                                                                                                                                                                                                                              |
| 0902.0431_E00472                    | 78   | ■ 0.629 | $\begin{aligned} &=\phi\left(\frac{1}{2}(B,X)A+\frac{1}{6}(A,B)X-2B\times(A\times X)\right)-(A\vee B)\phi X\text{ (Lemma 3.4.1)} \\ &=\frac{1}{2}(B,X)\phi A+\frac{1}{6}(A,B)\phi X-2\phi'B\times(A\times X)-2B\times(\phi A\times X) \\ &\quad-2B\times(A\times\phi X)-\frac{1}{2}(B,\phi X)A-\frac{1}{6}(A,B)\phi X+2B\times(A\times\phi X)\text{ (Lemma 3.4.3)} \\ &=\frac{1}{2}(B,X)\phi A+\frac{1}{6}(\phi A,B)X-2B\times(\phi A\times X) \\ &\quad+\frac{1}{2}(\phi'B,X)A+\frac{1}{6}(A,\phi'B)X-2\phi'B\times(A\times X) \\ &=(\phi A\vee B)X+(A\vee\phi'B)X,\quad X\in\mathfrak{J}^C. \end{aligned}$ | $\begin{aligned} &=\phi\left(\frac{1}{2}(B,X)A+\frac{1}{6}(A,B)X-2B\times(A\times X)\right)-(A\vee B)\phi X\text{ (Lemma 3.4.1)} \\ &=\frac{1}{2}(B,X)\phi A+\frac{1}{6}(A,B)\phi X-2\phi'B\times(A\times X)-2B\times(\phi A\times X) \\ &\quad-2B\times(A\times\phi X)-\frac{1}{2}(B,\phi X)A-\frac{1}{6}(A,B)\phi X+2B\times(A\times\phi X)\text{ (Lemma 3.4.3)} \\ &=\frac{1}{2}(B,X)\phi A+\frac{1}{6}(\phi A,B)X-2B\times(\phi A\times X) \\ &\quad+\frac{1}{2}(\phi'B,X)A+\frac{1}{6}(A,\phi'B)X-2\phi'B\times(A\times X) \\ &=(\phi A\vee B)X+(A\vee\phi'B)X,\quad X\in\mathfrak{J}^C. \end{aligned}$ | $\begin{aligned} &=\phi\left(\frac{1}{2}(B,X)A+\frac{1}{6}(A,B)X-2B\times(A\times X)\right)-(A\vee B)\phi X\text{ (Lemma 3.4.1)} \\ &=\frac{1}{2}(B,X)\phi A+\frac{1}{6}(A,B)\phi X-2\phi'B\times(A\times X)-2B\times(\phi A\times X) \\ &\quad-2B\times(A\times\phi X)-\frac{1}{2}(B,\phi X)A-\frac{1}{6}(A,B)\phi X+2B\times(A\times\phi X)\text{ (Lemma 3.4.3)} \\ &=\frac{1}{2}(B,X)\phi A+\frac{1}{6}(\phi A,B)X-2B\times(\phi A\times X) \\ &\quad+\frac{1}{2}(\phi'B,X)A+\frac{1}{6}(A,\phi'B)X-2\phi'B\times(A\times X) \\ &=(\phi A\vee B)X+(A\vee\phi'B)X,\quad X\in\mathfrak{J}^C. \end{aligned}$ |
| 0902.0431_E00244 (5<br>Rightarrow4) | 40   | ■ 0.631 | $\begin{aligned} &\&(\operatorname{det}(\alpha X))\,\alpha X=(\alpha X\times\alpha X)\times(\alpha X\times\alpha X)\quad(\text{Lemma 2.1.1.}) \\ &\&X=(\alpha X\times\alpha X)\times(\alpha X\times\alpha X)\quad(\text{Lemma 2.1.1.}) \\ &\&\alpha((X\times X)\times(X\times X))=(\operatorname{det} X)\alpha X\quad(\text{Lemma 2.1.1.}) \end{aligned}$                                                                                                                                                                                                                                                    | $\begin{aligned} (\det(\alpha X))\alpha X &= (\alpha X \times \alpha X) \times (\alpha X \times \alpha X) \quad (\text{Lemma 2.1.1.}) \\ &= \alpha((X \times X) \times (X \times X)) = (\det X)\alpha X \quad (\text{Lemma 2.1.1.}) \end{aligned}$                                                                                                                                                                                                                                                                                                                                                           | $\begin{aligned} (\det(\alpha X))\alpha X &= (\alpha X \times \alpha X) \times (\alpha X \times \alpha X) \quad (\text{Lemma 2.1.1.}) \\ &= \alpha((X \times X) \times (X \times X)) = (\det X)\alpha X \quad (\text{Lemma 2.1.1.}) \end{aligned}$                                                                                                                                                                                                                                                                                                                                                           |
| 0902.0431_E01093                    | 185  | ■ 0.633 | $\begin{aligned} &\&{\left[\zeta l(D),\zeta l(S)I\right]} \\ &\&=\left[\Theta\left(\varphi_*D,0,0,0,0,0\right),\Theta\left(0,2\lambda\gamma\chi^{-1}S,2\chi^{-1}S,0,0,0\right)\right] \\ &\&=\Theta\left(0,2\left(\varphi_*D\right)\lambda\gamma\chi^{-1}S,2\left(\varphi_*D\right)\chi^{-1}S,0,0,0\right) \end{aligned}$                                                                                                                                                                                                                                                                                    | $\begin{aligned} &[\zeta l(D),\zeta l(S)I] \\ &= [\Theta(\varphi_*D,0,0,0,0,0),\Theta(0,2\lambda\gamma\chi^{-1}S,2\chi^{-1}S,0,0,0)] \\ &= \Theta(0,2(\varphi_*D)\lambda\gamma\chi^{-1}S,2(\varphi_*D)\chi^{-1}S,0,0,0) \end{aligned}$                                                                                                                                                                                                                                                                                                                                                                       | $\begin{aligned} &[\zeta l(D),\zeta l(S)I] \\ &= [\Theta(\varphi_*D,0,0,0,0,0),\Theta(0,2\lambda\gamma\chi^{-1}S,2\chi^{-1}S,0,0,0)] \\ &= \Theta(0,2(\varphi_*D)\lambda\gamma\chi^{-1}S,2(\varphi_*D)\chi^{-1}S,0,0,0). \end{aligned}$                                                                                                                                                                                                                                                                                                                                                                      |

| Identifier       | Page | Conf.   | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Rendered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Image                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|------------------|------|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0902.0431_E00436 | 72   | ■ 0.637 | $\begin{aligned} \mathfrak{J}\left(3, \mathfrak{C}'\right) &= \left\{X \in M\left(3, \mathfrak{C}'\right) \mid X^*=X\right\}, \\ \mathfrak{J}\left(1,2, \mathfrak{C}\right) &= \left\{X \in M\left(3, \mathfrak{C}\right) \mid I_1 X^* I_1=X\right\}, \end{aligned}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | $\mathfrak{J}\left(3, \mathfrak{C}'\right)=\left\{X \in M\left(3, \mathfrak{C}'\right) \mid X^*=X\right\},$ $\mathfrak{J}\left(1,2, \mathfrak{C}\right)=\left\{X \in M\left(3, \mathfrak{C}\right) \mid I_1 X^* I_1=X\right\},$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | $\mathfrak{J}\left(3, \mathfrak{C}'\right)=\left\{X \in M\left(3, \mathfrak{C}'\right) \mid X^*=X\right\},$ $\mathfrak{J}\left(1,2, \mathfrak{C}\right)=\left\{X \in M\left(3, \mathfrak{C}\right) \mid I_1 X^* I_1=X\right\},$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 0902.0431_E01067 | 181  | ■ 0.642 | $\begin{aligned} \left(E_8\right)^{\widetilde{\lambda} \gamma} &=\left\{\alpha \in E_8 \mid \widetilde{\lambda} \gamma \alpha=\alpha \widetilde{\lambda} \gamma\right\} \\ &=\left\{\alpha \in E_8 \mid \tau \gamma \alpha=\alpha \tau \gamma\right\}=\left(E_8\right)^{\tau \gamma} . \end{aligned}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | $\left(E_8\right)^{\widetilde{\lambda} \gamma}=\left\{\alpha \in E_8 \mid \widetilde{\lambda} \gamma \alpha=\alpha \widetilde{\lambda} \gamma\right\}$ $=\left\{\alpha \in E_8 \mid \tau \gamma \alpha=\alpha \tau \gamma\right\}=\left(E_8\right)^{\tau \gamma} .$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | $\left(E_8\right)^{\widetilde{\lambda} \gamma}=\left\{\alpha \in E_8 \mid \widetilde{\lambda} \gamma \alpha=\alpha \widetilde{\lambda} \gamma\right\}$ $=\left\{\alpha \in E_8 \mid \tau \gamma \alpha=\alpha \tau \gamma\right\}=\left(E_8\right)^{\tau \gamma} .$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 0902.0431_E00811 | 135  | ■ 0.647 | $\begin{aligned} &\left(0, \omega' X, 0,0\right)=-\lambda\left(\omega' X, 0,0,0\right)=-\lambda \alpha(X, 0,0,0) \\ &=-\alpha \lambda(X, 0,0,0)=\alpha(0, X, 0,0)=\left(0, \omega'^{-1} X, 0,0\right) \end{aligned}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | $\left(0, \omega' X, 0,0\right)=-\lambda\left(\omega' X, 0,0,0\right)=-\lambda \alpha(X, 0,0,0)$ $=-\alpha \lambda(X, 0,0,0)=\alpha(0, X, 0,0)=\left(0, \omega'^{-1} X, 0,0\right)$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | $\left(0, \omega' X, 0,0\right)=-\lambda\left(\omega' X, 0,0,0\right)=-\lambda \alpha(X, 0,0,0)$ $=-\alpha \lambda(X, 0,0,0)=\alpha(0, X, 0,0)=\left(0, \omega'^{-1} X, 0,0\right),$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 0902.0431_E00477 | 79   | ■ 0.651 | $\begin{aligned} &\left(\left[\delta, \delta_1\right]+\left[\widetilde{T}, \widetilde{T}_1\right]\right)+\left(\widetilde{\delta T_1}-\widetilde{\delta_1 T}\right), \delta_2+\widetilde{T}_2\right)_6 \quad(\text { Theorem 3.2.2 }) \\ &=\left([\delta, \delta_1], \delta_2\right)_4+\left([\widetilde{T}, \widetilde{T}_1], \delta_2\right)_4+\left(\delta T_1-\delta_1 T, T_2\right) \\ &=-\left(\delta_1,\left[\delta, \delta_2\right]\right)_4+\left(\delta_2 T, T_1\right)+\left(\delta T_1, T_2\right)-\left(\delta_1 T, T_2\right) \quad(\text { Lemma 2.5.2 }) \\ &=-\left(\delta_1,\left[\delta, \delta_2\right]\right)_4-\left(\delta_1 T, T_2\right)-\left(T_1, \delta T_2\right)+\left(T_1, \delta_2 T\right) \\ &=-\left(\delta_1+\widetilde{T}_1,\left[\delta, \delta_2\right]+\left[\widetilde{T}, \widetilde{T}_2\right]\right)+\left(\widetilde{\delta T_2}-\widetilde{\delta_2 T}\right)_6 \\ &=-\left(\delta_1+\widetilde{T}_1,\left[\delta+\widetilde{T}, \delta_2+\widetilde{T}_2\right]\right)_6 \\ &=-\left(\phi_1,\left[\phi, \phi_2\right]\right)_6 \end{aligned}$ | $\left(\left([\delta, \delta_1]+\left[\widetilde{T}, \widetilde{T}_1\right]\right)+\left(\widetilde{\delta T_1}-\widetilde{\delta_1 T}\right), \delta_2+\widetilde{T}_2\right)_6 \quad(\text { Theorem 3.2.2 })$ $=\left([\delta, \delta_1], \delta_2\right)_4+\left([\widetilde{T}, \widetilde{T}_1], \delta_2\right)_4+\left(\delta T_1-\delta_1 T, T_2\right)$ $=-\left(\delta_1,\left[\delta, \delta_2\right]\right)_4+\left(\delta_2 T, T_1\right)+\left(\delta T_1, T_2\right)-\left(\delta_1 T, T_2\right) \quad(\text { Lemma 2.5.2 })$ $=-\left(\delta_1,\left[\delta, \delta_2\right]\right)_4-\left(\delta_1 T, T_2\right)-\left(T_1, \delta T_2\right)+\left(T_1, \delta_2 T\right)$ $=-\left(\delta_1+\widetilde{T}_1,\left[\delta, \delta_2\right]+\left[\widetilde{T}, \widetilde{T}_2\right]\right)+\left(\widetilde{\delta T_2}-\widetilde{\delta_2 T}\right)_6$ $=-\left(\delta_1+\widetilde{T}_1,\left[\delta+\widetilde{T}, \delta_2+\widetilde{T}_2\right]\right)_6$ $=-\left(\phi_1,\left[\phi, \phi_2\right]\right)_6 .$ | $=\left(\left([\delta, \delta_1]+\left[\widetilde{T}, \widetilde{T}_1\right]\right)+\left(\widetilde{\delta T_1}-\widetilde{\delta_1 T}\right), \delta_2+\widetilde{T}_2\right)_6 \quad(\text { Theorem 3.2.2 })$ $=\left([\delta, \delta_1], \delta_2\right)_4+\left([\widetilde{T}, \widetilde{T}_1], \delta_2\right)_4+\left(\delta T_1-\delta_1 T, T_2\right)$ $=-\left(\delta_1,\left[\delta, \delta_2\right]\right)_4+\left(\delta_2 T, T_1\right)+\left(\delta T_1, T_2\right)-\left(\delta_1 T, T_2\right) \quad(\text { Lemma 2.5.2 })$ $=-\left(\delta_1,\left[\delta, \delta_2\right]\right)_4-\left(\delta_1 T, T_2\right)-\left(T_1, \delta T_2\right)+\left(T_1, \delta_2 T\right)$ $=-\left(\delta_1+\widetilde{T}_1,\left[\delta, \delta_2\right]+\left[\widetilde{T}, \widetilde{T}_2\right]\right)+\left(\widetilde{\delta T_2}-\widetilde{\delta_2 T}\right)_6$ $=-\left(\delta_1+\widetilde{T}_1,\left[\delta+\widetilde{T}, \delta_2+\widetilde{T}_2\right]\right)_6$ $=-\left(\phi_1,\left[\phi, \phi_2\right]\right)_6 .$ |

| Identifier       | Page | Conf. | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Rendered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Image                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|------------------|------|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0902.0431_E00522 | 86   | 0.651 | $\begin{aligned} & \& H_{\alpha_1}=\frac{1}{24}\left(H_{\{0\}}-H_{\{1\}}\right), \quad H_{\alpha_2}=\frac{1}{24}\left(H_{\{1\}}-H_{\{2\}}\right), \quad H_{\alpha_3}=\frac{1}{24}\left(H_{\{2\}}-H_{\{3\}}\right), \\ & H_{\alpha_4}=\frac{1}{24}\left(H_{\{3\}}+\left(E_2-E_3\right)^{\sim}\right), \\ & H_{\alpha_5}=\frac{1}{48}\left(\left(-H_0-H_1-H_2+H_3\right)+2\left(E_3-E_1\right)^{\sim}\right), \\ & H_{\alpha_6}=\frac{1}{48}\left(\left(H_0+H_1+H_2+H_3\right)+2\left(E_1-E_2\right)^{\sim}\right). \end{aligned}$ | $\begin{aligned} & H_{\alpha_1}=\frac{1}{24}\left(H_0-H_1\right), \quad H_{\alpha_2}=\frac{1}{24}\left(H_1-H_2\right), \quad H_{\alpha_3}=\frac{1}{24}\left(H_2-H_3\right), \\ & H_{\alpha_4}=\frac{1}{24}\left(H_3+\left(E_2-E_3\right)^{\sim}\right), \\ & H_{\alpha_5}=\frac{1}{48}\left(\left(-H_0-H_1-H_2+H_3\right)+2\left(E_3-E_1\right)^{\sim}\right), \\ & H_{\alpha_6}=\frac{1}{48}\left(\left(H_0+H_1+H_2+H_3\right)+2\left(E_1-E_2\right)^{\sim}\right). \end{aligned}$           | $\begin{aligned} H_{\alpha_1} &= \frac{1}{24}(H_0 - H_1), \quad H_{\alpha_2} = \frac{1}{24}(H_1 - H_2), \quad H_{\alpha_3} = \frac{1}{24}(H_2 - H_3), \\ H_{\alpha_4} &= \frac{1}{24}(H_3 + (E_2 - E_3)^{\sim}), \\ H_{\alpha_5} &= \frac{1}{48}((-H_0 - H_1 - H_2 + H_3) + 2(E_3 - E_1)^{\sim}), \\ H_{\alpha_6} &= \frac{1}{48}((H_0 + H_1 + H_2 + H_3) + 2(E_1 - E_2)^{\sim}). \end{aligned}$                                                                                              |
| 0902.0431_E00024 | 9    | 0.655 | $\begin{aligned} & \& \left[F_{i0}, F_{0k}\right] x = \left(F_{i0} F_{0k} - F_{0k} F_{i0}\right) x = -\frac{1}{4} e_i\left(e_k x\right) + \frac{1}{4} e_k\left(e_i x\right) \\ & = \frac{1}{2} e_k\left(e_i x\right) = F_{ik} x, \end{aligned}$                                                                                                                                                                                                                                                                                  | $\begin{aligned} \left[F_{i0}, F_{0k}\right] x &= \left(F_{i0} F_{0k} - F_{0k} F_{i0}\right) x = -\frac{1}{4} e_i\left(e_k x\right) + \frac{1}{4} e_k\left(e_i x\right) \\ &= \frac{1}{2} e_k\left(e_i x\right) = F_{ik} x, \end{aligned}$                                                                                                                                                                                                                                                    | $\begin{aligned} \left[F_{i0}, F_{0k}\right] x &= \left(F_{i0} F_{0k} - F_{0k} F_{i0}\right) x = -\frac{1}{4} e_i\left(e_k x\right) + \frac{1}{4} e_k\left(e_i x\right) \\ &= \frac{1}{2} e_k\left(e_i x\right) = F_{ik} x, \end{aligned}$                                                                                                                                                                                                                                                    |
| 0902.0431_E00021 | 9    | 0.656 | $\begin{aligned} & \& \left[\kappa D_1, \kappa D_2\right] x = \left(\kappa D_1\right)\left(\kappa D_2 x\right) - \left(\kappa D_2\right)\left(\kappa D_1 x\right) = \overline{D_1\left(\kappa D_2 x\right)} - \overline{D_2\left(\kappa D_1 x\right)} \\ & = \overline{D_1 D_2 \bar{x}} - \overline{D_2 D_1 \bar{x}} = \kappa\left(D_1 D_2 - D_2 D_1\right) x = \kappa\left[D_1, D_2\right] x, \quad x \in \mathfrak{C} \end{aligned}$                                                                                           | $\begin{aligned} \left[\kappa D_1, \kappa D_2\right] x &= \left(\kappa D_1\right)\left(\kappa D_2 x\right) - \left(\kappa D_2\right)\left(\kappa D_1 x\right) = \overline{D_1\left(\kappa D_2 x\right)} - \overline{D_2\left(\kappa D_1 x\right)} \\ &= \overline{D_1 D_2 \bar{x}} - \overline{D_2 D_1 \bar{x}} = \kappa\left(D_1 D_2 - D_2 D_1\right) x = \kappa\left[D_1, D_2\right] x, \quad x \in \mathfrak{C}. \end{aligned}$                                                            | $\begin{aligned} \left[\kappa D_1, \kappa D_2\right] x &= \left(\kappa D_1\right)\left(\kappa D_2 x\right) - \left(\kappa D_2\right)\left(\kappa D_1 x\right) = \overline{D_1\left(\kappa D_2 x\right)} - \overline{D_2\left(\kappa D_1 x\right)} \\ &= \overline{D_1 D_2 \bar{x}} - \overline{D_2 D_1 \bar{x}} = \kappa\left(D_1 D_2 - D_2 D_1\right) x = \kappa\left[D_1, D_2\right] x, \quad x \in \mathfrak{C}. \end{aligned}$                                                            |
| 0902.0431_EQ1122 | 190  | 0.658 | $\begin{aligned} & \& f\left(\Phi\left(\phi, A, B, \nu\right),\left(X, Y, \xi, \eta\right),\left(Z, W, \zeta, \omega\right), r, s, t\right) \\ & =\left(\begin{array}{ccc} \frac{2}{3} \nu & -\frac{1}{2} \xi & \frac{1}{2} \zeta \\ \frac{1}{2} \omega & -\frac{1}{3} \nu-r & t \\ \frac{1}{2} \eta & s & -\frac{1}{3} \nu+r \end{array}\right), \phi,\left(\begin{array}{c} -2 A \\ Z \\ X \end{array}\right),\left(\begin{array}{c} -2 B \\ Y \\ -W \end{array}\right)\right), \end{aligned}$                                 | $\begin{aligned} & f\left(\Phi\left(\phi, A, B, \nu\right),\left(X, Y, \xi, \eta\right),\left(Z, W, \zeta, \omega\right), r, s, t\right) \\ & =\left(\begin{array}{ccc} \frac{2}{3} \nu & -\frac{1}{2} \xi & \frac{1}{2} \zeta \\ \frac{1}{2} \omega & -\frac{1}{3} \nu-r & t \\ \frac{1}{2} \eta & s & -\frac{1}{3} \nu+r \end{array}\right), \phi,\left(\begin{array}{c} -2 A \\ Z \\ X \end{array}\right),\left(\begin{array}{c} -2 B \\ Y \\ -W \end{array}\right)\right), \end{aligned}$ | $\begin{aligned} & f\left(\Phi\left(\phi, A, B, \nu\right),\left(X, Y, \xi, \eta\right),\left(Z, W, \zeta, \omega\right), r, s, t\right) \\ & =\left(\begin{array}{ccc} \frac{2}{3} \nu & -\frac{1}{2} \xi & \frac{1}{2} \zeta \\ \frac{1}{2} \omega & -\frac{1}{3} \nu-r & t \\ \frac{1}{2} \eta & s & -\frac{1}{3} \nu+r \end{array}\right), \phi,\left(\begin{array}{c} -2 A \\ Z \\ X \end{array}\right),\left(\begin{array}{c} -2 B \\ Y \\ -W \end{array}\right)\right), \end{aligned}$ |

| Identifier            | Page | Conf. | LaTeX source                                                                                                                                                                                                                                                                                            | Rendered                                                                                                                                                                                                                                                          | Image                                                                                                                                                                                                                                                              |
|-----------------------|------|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0902.0431_E00546      | 92   | 0.661 | $\begin{aligned} & \xi_2 E_2 + \xi_3 E_3 + F_1(x_1) = \alpha E = \alpha(E \times E) = \tau \alpha \tau E \times \tau \alpha \tau E \\ & = \tau(\xi_2 E_2 + \xi_3 E_3 + F_1(x_1)) \times \tau(\xi_2 E_2 + \xi_3 E_3 + F_1(x_1)) \\ & = (\tau \xi_2 \tau \xi_3 - (\tau x_1, \tau x_1)) E_1 \end{aligned}$ | $\xi_2 E_2 + \xi_3 E_3 + F_1(x_1) = \alpha E = \alpha(E \times E) = \tau \alpha \tau E \times \tau \alpha \tau E$ $= \tau(\xi_2 E_2 + \xi_3 E_3 + F_1(x_1)) \times \tau(\xi_2 E_2 + \xi_3 E_3 + F_1(x_1))$ $= (\tau \xi_2 \tau \xi_3 - (\tau x_1, \tau x_1)) E_1$ | $\xi_2 E_2 + \xi_3 E_3 + F_1(x_1) = \alpha E = \alpha(E \times E) = \tau \alpha \tau E \times \tau \alpha \tau E$ $= \tau(\xi_2 E_2 + \xi_3 E_3 + F_1(x_1)) \times \tau(\xi_2 E_2 + \xi_3 E_3 + F_1(x_1))$ $= (\tau \xi_2 \tau \xi_3 - (\tau x_1, \tau x_1)) E_1.$ |
| 0902.0431_E00675 (ii) | 112  | 0.662 | $2(Q \times R)P - (Q \times P)R - (R \times P)Q + \frac{3}{8}\{P, R\}Q - \frac{3}{8}\{Q, P\}R = 0.$                                                                                                                                                                                                     | $2(Q \times R)P - (Q \times P)R - (R \times P)Q + \frac{3}{8}\{P, R\}Q - \frac{3}{8}\{Q, P\}R = 0.$                                                                                                                                                               | $2(Q \times R)P - (Q \times P)R - (R \times P)Q + \frac{3}{8}\{P, R\}Q - \frac{3}{8}\{Q, P\}R = 0.$                                                                                                                                                                |

| Identifier       | Page | Conf.  | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        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| 0902.0431_EQ1015 | 169  | ■0.662 | \begin{gathered} \pm\left(\lambda_k-\lambda_l\right), \\ \quad \pm\left(\lambda_k+\lambda_l\right), \quad 0 \leq k < l \leq 3, \\ \pm \lambda_k \pm \frac{1}{2}\left(\mu_2-\mu_3\right), \quad 0 \leq k \leq 3, \\ \pm \frac{1}{2}\left(-\lambda_0-\lambda_1+\lambda_2-\lambda_3\right) \pm \frac{1}{2}\left(\mu_3-\mu_1\right), \\ \pm \frac{1}{2}\left(-\lambda_0+\lambda_1+\lambda_2-\lambda_3\right) \pm \frac{1}{2}\left(\mu_3-\mu_1\right), \\ \pm \frac{1}{2}\left(-\lambda_0+\lambda_1+\lambda_2+\lambda_3\right) \pm \frac{1}{2}\left(\mu_3-\mu_1\right), \\ \pm \frac{1}{2}\left(\lambda_0-\lambda_1+\lambda_2+\lambda_3\right) \pm \frac{1}{2}\left(\mu_3-\mu_1\right), \\ \pm \frac{1}{2}\left(\lambda_0-\lambda_1+\lambda_2-\lambda_3\right) \pm \frac{1}{2}\left(\mu_1-\mu_2\right), \\ \pm \frac{1}{2}\left(-\lambda_0+\lambda_1+\lambda_2-\lambda_3\right) \pm \frac{1}{2}\left(\mu_1-\mu_2\right), \\ \pm \frac{1}{2}\left(\lambda_0+\lambda_1+\lambda_2+\lambda_3\right) \pm \frac{1}{2}\left(\mu_1-\mu_2\right), \\ \pm \frac{1}{2}\left(-\lambda_0-\lambda_1+\lambda_2+\lambda_3\right) \pm \frac{1}{2}\left(\mu_1-\mu_2\right), \\ \pm\left(\mu_k+\frac{2}{3} \nu\right), \quad 0 \leq k < l \leq 3, \\ \pm \lambda_k \pm\left(\frac{1}{2} \mu_1-\frac{2}{3} \nu\right), \quad 0 \leq k \leq 3, \\ \pm \frac{1}{2}\left(-\lambda_0-\lambda_1+\lambda_2-\lambda_3\right) \pm\left(\frac{1}{2} \mu_2-\frac{2}{3} \nu\right), \\ \pm \frac{1}{2}\left(\lambda_0+\lambda_1+\lambda_2-\lambda_3\right) \pm\left(\frac{1}{2} \mu_2-\frac{2}{3} \nu\right), \\ \pm \frac{1}{2}\left(-\lambda_0+\lambda_1+\lambda_2+\lambda_3\right) \pm\left(\frac{1}{2} \mu_2-\frac{2}{3} \nu\right), \\ \pm \frac{1}{2}\left(\lambda_0-\lambda_1+\lambda_2+\lambda_3\right) \pm\left(\frac{1}{2} \mu_2-\frac{2}{3} \nu\right), \\ \pm \frac{1}{2}\left(\lambda_0-\lambda_1+\lambda_2-\lambda_3\right) \pm\left(\frac{1}{2} \mu_3-\frac{2}{3} \nu\right), \\ \pm \frac{1}{2}\left(-\lambda_0+\lambda_1+\lambda_2-\lambda_3\right) \pm\left(\frac{1}{2} \mu_3-\frac{2}{3} \nu\right), \\ \pm \frac{1}{2}\left(\lambda_0+\lambda_1+\lambda_2+\lambda_3\right) \pm\left(\frac{1}{2} \mu_3-\frac{2}{3} \nu\right), \\ \pm \frac{1}{2}\left(-\lambda_0-\lambda_1+\lambda_2+\lambda_3\right) \pm\left(\frac{1}{2} \mu_3-\frac{2}{3} \nu\right), \\ \pm\left(\mu_j-\frac{1}{3} \nu+r\right), \quad 1 \leq j \leq 3, \\ \pm \lambda_k \pm\left(\frac{1}{2} \mu_1+\frac{1}{3} \nu\right) \pm r, \quad 0 \leq k \leq 3, \\ \pm \frac{1}{2}\left(\lambda_0+\lambda_1-\lambda_2-\lambda_3\right) \pm\left(\frac{1}{2} \mu_2+\frac{1}{3} \nu\right) \pm r, \\ \pm \frac{1}{2}\left(\lambda_0+\lambda_1-\lambda_2-\lambda_3\right) \pm\left(\frac{1}{2} \mu_2+\frac{1}{3} \nu\right) \pm r, \end{gathered} | $\begin{aligned}&\pm (\lambda_k - \lambda_l), \quad \pm (\lambda_k + \lambda_l), \quad 0 \leq k < l \leq 3,\\&\pm \lambda_k \pm \frac{1}{2} (\mu_2 - \mu_3), \quad 0 \leq k \leq 3,\\&\pm \frac{1}{2} (-\lambda_0 - \lambda_1 + \lambda_2 - \lambda_3) \pm \frac{1}{2} (\mu_3 - \mu_1),\\&\pm \frac{1}{2} (-\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3) \pm \frac{1}{2} (\mu_3 - \mu_1),\\&\pm \frac{1}{2} (-\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3) \pm \frac{1}{2} (\mu_3 - \mu_1),\\&\pm \frac{1}{2} (\lambda_0 - \lambda_1 + \lambda_2 + \lambda_3) \pm \frac{1}{2} (\mu_3 - \mu_1),\\&\pm \frac{1}{2} (\lambda_0 - \lambda_1 + \lambda_2 - \lambda_3) \pm \frac{1}{2} (\mu_1 - \mu_2),\\&\pm \frac{1}{2} (-\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3) \pm \frac{1}{2} (\mu_1 - \mu_2),\\&\pm \frac{1}{2} (\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3) \pm \frac{1}{2} (\mu_1 - \mu_2),\\&\pm \frac{1}{2} (-\lambda_0 - \lambda_1 + \lambda_2 + \lambda_3) \pm \frac{1}{2} (\mu_1 - \mu_2),\\&\pm \left( \mu_k + \frac{2}{3} \nu \right), \quad 0 \leq k < l \leq 3,\\&\pm \lambda_k \pm \left( \frac{1}{2} \mu_1 - \frac{2}{3} \nu \right), \quad 0 \leq k \leq 3,\\&\pm \frac{1}{2} (-\lambda_0 - \lambda_1 + \lambda_2 - \lambda_3) \pm \left( \frac{1}{2} \mu_2 - \frac{2}{3} \nu \right),\\&\pm \frac{1}{2} (\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3) \pm \left( \frac{1}{2} \mu_2 - \frac{2}{3} \nu \right),\\&\pm \frac{1}{2} (-\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3) \pm \left( \frac{1}{2} \mu_2 - \frac{2}{3} \nu \right),\\&\pm \frac{1}{2} (\lambda_0 - \lambda_1 + \lambda_2 + \lambda_3) \pm \left( \frac{1}{2} \mu_2 - \frac{2}{3} \nu \right),\\&\pm \frac{1}{2} (\lambda_0 - \lambda_1 + \lambda_2 - \lambda_3) \pm \left( \frac{1}{2} \mu_3 - \frac{2}{3} \nu \right),\\&\pm \frac{1}{2} (-\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3) \pm \left( \frac{1}{2} \mu_3 - \frac{2}{3} \nu \right),\\&\pm \frac{1}{2} (\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3) \pm \left( \frac{1}{2} \mu_3 - \frac{2}{3} \nu \right),\\&\pm \frac{1}{2} (-\lambda_0 - \lambda_1 + \lambda_2 + \lambda_3) \pm \left( \frac{1}{2} \mu_3 - \frac{2}{3} \nu \right),\\&\pm \left( \mu_j - \frac{1}{3} \nu + r \right), \quad 1 \leq j \leq 3,\\&\pm \lambda_k \pm \left( \frac{1}{2} \mu_1 + \frac{1}{3} \nu \right) \pm r, \quad 0 \leq k \leq 3,\\&\pm \frac{1}{2} (\lambda_0 + \lambda_1 - \lambda_2 - \lambda_3) \pm \left( \frac{1}{2} \mu_2 + \frac{1}{3} \nu \right) \pm r,\\&\pm \frac{1}{2} (\lambda_0 + \lambda_1 - \lambda_2 - \lambda_3) \pm \left( \frac{1}{2} \mu_2 + \frac{1}{3} \nu \right) \pm r,\end{aligned}$ |       |

| Identifier       | Page | Conf.   | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                                   | Rendered                                                                                                                                                                                                                                                                                                                                                                                                         | Image                                                                                                                                                                                                                                                                                                                                                                                                             |
|------------------|------|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0902.0431_E00131 | 25   | ■ 0.662 | $\begin{aligned} & (\alpha x)(\alpha y) = \left( q m \bar{q} + (p a \bar{q}) e_4 \right) \left( q n \bar{q} + (p b \bar{q}) e_4 \right) \\ & = ((q m \bar{q})(q n \bar{q}) - (\overline{p b \bar{q}})(p a \bar{q})) + ((p a \bar{q})(\overline{q n \bar{q}}) + (p b \bar{q})(q m \bar{q})) e_4 \\ & = q(m n - \bar{b} a) \bar{q} + (p(a \bar{n} + b m) \bar{q}) e_4 \\ & = \varphi(p, q) ((m + a e_4)(n + b e_4)) = \alpha(x y) \end{aligned}$ | $\begin{aligned} (\alpha x)(\alpha y) &= (q m \bar{q} + (p a \bar{q}) e_4) (q n \bar{q} + (p b \bar{q}) e_4) \\ &= ((q m \bar{q})(q n \bar{q}) - (\overline{p b \bar{q}})(p a \bar{q})) + ((p a \bar{q})(\overline{q n \bar{q}}) + (p b \bar{q})(q m \bar{q})) e_4 \\ &= q(m n - \bar{b} a) \bar{q} + (p(a \bar{n} + b m) \bar{q}) e_4 \\ &= \varphi(p, q) ((m + a e_4)(n + b e_4)) = \alpha(x y) \end{aligned}$ | $\begin{aligned} (\alpha x)(\alpha y) &= (q m \bar{q} + (p a \bar{q}) e_4) (q n \bar{q} + (p b \bar{q}) e_4) \\ &= ((q m \bar{q})(q n \bar{q}) - (\overline{p b \bar{q}})(p a \bar{q})) + ((p a \bar{q})(\overline{q n \bar{q}}) + (p b \bar{q})(q m \bar{q})) e_4 \\ &= q(m n - \bar{b} a) \bar{q} + (p(a \bar{n} + b m) \bar{q}) e_4 \\ &= \varphi(p, q) ((m + a e_4)(n + b e_4)) = \alpha(x y). \end{aligned}$ |
| 0902.0431_E00734 | 121  | ■ 0.666 | $\left( \left( \Phi_1, \Phi_1 \right), \left( \Phi_1, \Phi_2 \right) \right)_7 + \left( \Phi_1, \Phi_2 \right)_7 = 0, \quad \Phi_i \in \mathfrak{e}_7^C.$                                                                                                                                                                                                                                                                                      | $([\Phi, \Phi_1], \Phi_2)_7 + (\Phi_1, [\Phi, \Phi_2])_7 = 0, \quad \Phi, \Phi_i \in \mathfrak{e}_7^C.$                                                                                                                                                                                                                                                                                                          | $([\Phi, \Phi_1], \Phi_2)_7 + (\Phi_1, [\Phi, \Phi_2])_7 = 0, \quad \Phi, \Phi_i \in \mathfrak{e}_7^C.$                                                                                                                                                                                                                                                                                                           |
| 0902.0431_E00873 | 145  | ■ 0.666 | $\begin{aligned} & \operatorname{Spin}(12)/\{1, \sigma\} \cong SO(12) \\ & \operatorname{Spin}(12)/\{1, -1\} \cong \operatorname{Spin}(12)/\{1, -\sigma\} \cong \operatorname{Ss}(12) \end{aligned}$                                                                                                                                                                                                                                           | $\operatorname{Spin}(12)/\{1, \sigma\} \cong SO(12)$ $\operatorname{Spin}(12)/\{1, -1\} \cong \operatorname{Spin}(12)/\{1, -\sigma\} \cong \operatorname{Ss}(12)$                                                                                                                                                                                                                                                | $\begin{aligned} \operatorname{Spin}(12)/\{1, \sigma\} &\cong SO(12), \\ \operatorname{Spin}(12)/\{1, -1\} &\cong \operatorname{Spin}(12)/\{1, -\sigma\} \cong \operatorname{Ss}(12). \end{aligned}$                                                                                                                                                                                                              |
| 0902.0431_E00649 | 107  | ■ 0.672 | $\varphi_* : \mathfrak{su}(3) \oplus \mathfrak{su}(3) \rightarrow \mathfrak{e}_{6,C}.$                                                                                                                                                                                                                                                                                                                                                         | $\varphi_* : \mathfrak{su}(3) \oplus \mathfrak{su}(3) \rightarrow \mathfrak{e}_{6,C}.$                                                                                                                                                                                                                                                                                                                           | $\varphi_* : \mathfrak{su}(3) \oplus \mathfrak{su}(3) \rightarrow \mathfrak{e}_{6,C}.$                                                                                                                                                                                                                                                                                                                            |
| 0902.0431_E00774 | 130  | ■ 0.675 | $\begin{aligned} & \langle \alpha \dot{X}, \dot{1} \rangle = \langle \alpha \dot{X}, \alpha \dot{1} \rangle = \langle \dot{X}, \dot{1} \rangle = 0 \\ & \langle \alpha \dot{X}, 1 \rangle = \langle \alpha \dot{X}, \alpha 1 \rangle = \langle \dot{X}, 1 \rangle = 0. \end{aligned}$                                                                                                                                                          | $\begin{aligned} \langle \alpha \dot{X}, \dot{1} \rangle &= \langle \alpha \dot{X}, \alpha \dot{1} \rangle = \langle \dot{X}, \dot{1} \rangle = 0 \\ \langle \alpha \dot{X}, 1 \rangle &= \langle \alpha \dot{X}, \alpha 1 \rangle = \langle \dot{X}, 1 \rangle = 0. \end{aligned}$                                                                                                                              | $\begin{aligned} \langle \alpha \dot{X}, \dot{1} \rangle &= \langle \alpha \dot{X}, \alpha \dot{1} \rangle = \langle \dot{X}, \dot{1} \rangle = 0, \\ \langle \alpha \dot{X}, 1 \rangle &= \langle \alpha \dot{X}, \alpha 1 \rangle = \langle \dot{X}, 1 \rangle = 0. \end{aligned}$                                                                                                                              |
| 0902.0431_E00142 | 26   | ■ 0.683 | $\overline{\alpha x} = \alpha \bar{x}, \quad x \in \mathfrak{C}^C.$                                                                                                                                                                                                                                                                                                                                                                            | $\overline{\alpha x} = \alpha \bar{x}, \quad x \in \mathfrak{C}^C.$                                                                                                                                                                                                                                                                                                                                              | $\overline{\alpha x} = \alpha \bar{x}, \quad x \in \mathfrak{C}^C.$                                                                                                                                                                                                                                                                                                                                               |
| 0902.0431_E01206 | 203  | ■ 0.686 | $\begin{aligned} & \left( \operatorname{tr} (C_1 \tau^t C_2) + \operatorname{tr} (D_1 \tau^t D_2) + (x_1, \tau x_2) (a_1, \tau a_2) + (y_1, \tau y_2) (b_1, \tau b_2) \right. \\ & \quad \left. + (z_1, \tau z_2) (c_1, \tau c_2) + (w_1, \tau w_2) (d_1, \tau d_2) \right). \end{aligned}$                                                                                                                                                    | $\begin{aligned} \langle R_1, R_2 \rangle &= 60 \left( \operatorname{tr} (C_1 \tau^t C_2) + \operatorname{tr} (D_1 \tau^t D_2) + (x_1, \tau x_2) (a_1, \tau a_2) + (y_1, \tau y_2) (b_1, \tau b_2) \right. \\ & \quad \left. + (z_1, \tau z_2) (c_1, \tau c_2) + (w_1, \tau w_2) (d_1, \tau d_2) \right). \end{aligned}$                                                                                         | $\begin{aligned} \langle R_1, R_2 \rangle &= 60 \left( \operatorname{tr} (C_1 \tau^t C_2) + \operatorname{tr} (D_1 \tau^t D_2) + (x_1, \tau x_2) (a_1, \tau a_2) + (y_1, \tau y_2) (b_1, \tau b_2) \right. \\ & \quad \left. + (z_1, \tau z_2) (c_1, \tau c_2) + (w_1, \tau w_2) (d_1, \tau d_2) \right). \end{aligned}$                                                                                          |

| Identifier       | Page | Conf. | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Rendered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Image                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|------------------|------|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0902.0431_E00765 | 128  | 0.686 | <pre>\begin{aligned} &amp; \texttt{H_{\alpha_1}}=\frac{1}{36} \ \Phi \texttt{\left(H_{0}-H_{1}, \ 0,0,0\right)}, \ \&amp; \texttt{H_{\alpha_2}}=\\ &amp; \frac{1}{36} \ \Phi \texttt{\left(H_{1}-H_{2}, \ 0,0,0\right)}, \ \&amp; \\ &amp; \texttt{H_{\alpha_3}}=\frac{1}{36} \ \Phi \texttt{\left(H_{2}-H_{3}, \ 0,0,0\right)}, \ \&amp; \texttt{H_{\alpha_4}}=\frac{1}{72} \ \Phi \texttt{\left(\\ &amp; \left(-H_{0}-H_{1}-H_{2}+H_{3}\right)+2\left(E_{3}-E_{1}\right.\\ &amp; \left.\right)^{\sim}, \ 0,0,0\right)}, \ \&amp; \texttt{H_{\alpha_5}}=\\ &amp; \frac{1}{72} \ \Phi \texttt{\left(\left(H_{0}+H_{1}+H_{2}+H_{3}\right)+2\left(E_{1}-E_{2}\right.\\ &amp; \left.\right)^{\sim}, \ 0,0,0\right)}, \ \&amp; \texttt{H_{\alpha_6}}=\frac{1}{54} \ \Phi \texttt{\left((-E_{1}+2E_{2}-E_{3})^{\sim}, \ 0,0,\\ &amp; \frac{3}{2}\right)}, \ \&amp; \texttt{H_{\alpha_7}}=\frac{1}{54} \ \Phi \texttt{\left((E_{1}+E_{2}-2E_{3})^{\sim}, \ 0,0,-\\ &amp; \frac{3}{2}\right)} . \\ &amp; \end{aligned}</pre> | $\begin{aligned} H_{\alpha_1} &= \frac{1}{36} \Phi \left( H_0 - H_1, 0, 0, 0 \right), \\ H_{\alpha_2} &= \frac{1}{36} \Phi \left( H_1 - H_2, 0, 0, 0 \right), \\ H_{\alpha_3} &= \frac{1}{36} \Phi \left( H_2 - H_3, 0, 0, 0 \right), \\ H_{\alpha_4} &= \frac{1}{72} \Phi \left( (-H_0 - H_1 - H_2 + H_3) + 2 \left( E_3 - E_1 \right)^\sim, 0, 0, 0 \right), \\ H_{\alpha_5} &= \frac{1}{72} \Phi \left( (H_0 + H_1 + H_2 + H_3) + 2 \left( E_1 - E_2 \right)^\sim, 0, 0, 0 \right), \\ H_{\alpha_6} &= \frac{1}{54} \Phi \left( (-E_1 + 2E_2 - E_3)^\sim, 0, 0, \frac{3}{2} \right), \\ H_{\alpha_7} &= \frac{1}{54} \Phi \left( (E_1 + E_2 - 2E_3)^\sim, 0, 0, -\frac{3}{2} \right). \end{aligned}$                                                                                                                                                                                                                                     | $\begin{aligned} H_{\alpha_1} &= \frac{1}{36} \Phi(H_0 - H_1, 0, 0, 0), \\ H_{\alpha_2} &= \frac{1}{36} \Phi(H_1 - H_2, 0, 0, 0), \\ H_{\alpha_3} &= \frac{1}{36} \Phi(H_2 - H_3, 0, 0, 0), \\ H_{\alpha_4} &= \frac{1}{72} \Phi((-H_0 - H_1 - H_2 + H_3) + 2(E_3 - E_1)^\sim, 0, 0, 0), \\ H_{\alpha_5} &= \frac{1}{72} \Phi((H_0 + H_1 + H_2 + H_3) + 2(E_1 - E_2)^\sim, 0, 0, 0), \\ H_{\alpha_6} &= \frac{1}{54} \Phi\Big((-E_1 + 2E_2 - E_3)^\sim, 0, 0, \frac{3}{2}\Big), \\ H_{\alpha_7} &= \frac{1}{54} \Phi\Big((E_1 + E_2 - 2E_3)^\sim, 0, 0, -\frac{3}{2}\Big). \end{aligned}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 0902.0431_E00616 | 102  | 0.688 | <pre>\begin{gathered} g(\gamma(X \times Y))=g(\gamma((M+\boldsymbol{a}) \times (N+\boldsymbol{b}))) \ \&amp; \texttt{=g}((M-\boldsymbol{a}) \times (N-\boldsymbol{b})) \\ \texttt{=g}\left(\left(M \times N - \frac{1}{2}(\boldsymbol{a}^* \boldsymbol{b} + \boldsymbol{b}^* \boldsymbol{a})\right) + \frac{1}{2}(\boldsymbol{a} N + \boldsymbol{b} M)\right) \\ \texttt{=}\left(\frac{i}{2}(\boldsymbol{a} N + \boldsymbol{b} M)^* \quad M \times N - \frac{1}{2}(\boldsymbol{a}^* \boldsymbol{b} + \boldsymbol{b}^* \boldsymbol{a}) - \frac{1}{2}(\text{tr}(M \times N) - (\boldsymbol{a}, \boldsymbol{b}))E\right) \\ \texttt{=} \dots \\ \texttt{=}g(M+\boldsymbol{a}) \circ g(N+\boldsymbol{b}) - \left(\frac{1}{4}(M, N) - \frac{1}{2}(\boldsymbol{a}, \boldsymbol{b})\right)E \\ \texttt{=}g(M+\boldsymbol{a}) \circ g(N+\boldsymbol{b}) - \frac{1}{4}(\gamma(M+\boldsymbol{a}), (N+\boldsymbol{b}))E \\ \texttt{=}gX \circ gY - \frac{1}{4}(\gamma X, Y)E. \end{gathered}</pre>                               | $\begin{aligned} g(\gamma(X \times Y)) &= g(\gamma((M + \boldsymbol{a}) \times (N + \boldsymbol{b}))) \\ &= g((M - \boldsymbol{a}) \times (N - \boldsymbol{b})) \\ &= g\left(\left(M \times N - \frac{1}{2}(\boldsymbol{a}^* \boldsymbol{b} + \boldsymbol{b}^* \boldsymbol{a})\right) + \frac{1}{2}(\boldsymbol{a} N + \boldsymbol{b} M)\right) \\ &= \left(\frac{i}{2}(\boldsymbol{a} N + \boldsymbol{b} M)^* \quad M \times N - \frac{1}{2}(\boldsymbol{a}^* \boldsymbol{b} + \boldsymbol{b}^* \boldsymbol{a}) - \frac{1}{2}(\text{tr}(M \times N) - (\boldsymbol{a}, \boldsymbol{b}))E\right) \\ &= \dots \\ &= g(M + \boldsymbol{a}) \circ g(N + \boldsymbol{b}) - \left(\frac{1}{4}(M, N) - \frac{1}{2}(\boldsymbol{a}, \boldsymbol{b})\right)E \\ &= g(M + \boldsymbol{a}) \circ g(N + \boldsymbol{b}) - \frac{1}{4}(\gamma(M + \boldsymbol{a}), (N + \boldsymbol{b}))E \\ &= gX \circ gY - \frac{1}{4}(\gamma X, Y)E. \end{aligned}$ | $\begin{aligned} g(\gamma(X \times Y)) &= g(\gamma((M + \boldsymbol{a}) \times (N + \boldsymbol{b}))) \\ &= g((M - \boldsymbol{a}) \times (N - \boldsymbol{b})) \\ &= g((M \times N - \frac{1}{2}(\boldsymbol{a}^* \boldsymbol{b} + \boldsymbol{b}^* \boldsymbol{a})) + \frac{1}{2}(\boldsymbol{a} N + \boldsymbol{b} M)) \\ &= \left(\frac{1}{2} \text{tr}(M \times N) - \frac{1}{2}(\boldsymbol{a}, \boldsymbol{b}) \quad \frac{i}{2}(\boldsymbol{a} N + \boldsymbol{b} M) \right. \\ &\quad \left. \frac{i}{2}(\boldsymbol{a} N + \boldsymbol{b} M)^* \quad M \times N - \frac{1}{2}(\boldsymbol{a}^* \boldsymbol{b} + \boldsymbol{b}^* \boldsymbol{a}) - \frac{1}{2}(\text{tr}(M \times N) - (\boldsymbol{a}, \boldsymbol{b}))E\right) \\ &= \dots \\ &= g(M + \boldsymbol{a}) \circ g(N + \boldsymbol{b}) - \left(\frac{1}{4}(M, N) - \frac{1}{2}(\boldsymbol{a}, \boldsymbol{b})\right)E \\ &= g(M + \boldsymbol{a}) \circ g(N + \boldsymbol{b}) - \frac{1}{4}(\gamma(M + \boldsymbol{a}), (N + \boldsymbol{b}))E \\ &= gX \circ gY - \frac{1}{4}(\gamma X, Y)E. \end{aligned}$ |



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| 0902.0431_E00095 | 19   | ■ 0.719 | $\mathfrak{h}=\left\{\left(\begin{array}{ccc}\lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3\end{array}\right)\middle \lambda_i\in\boldsymbol{C},\lambda_1+\lambda_2+\lambda_3=0\right\}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |          |       |
| 0902.0431_E00964 | 160  | ■ 0.719 | $E_{\{7\}}^{\{ \}^{\{C\}}}\simeq E_7\times\boldsymbol{R}^{133}.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |          |       |
| 0902.0431_E00197 | 33   | ■ 0.724 | $\begin{aligned} & \widetilde{D}_4=\left\{\left(\alpha_1,\alpha_2,\alpha_3\right)\in SO(8)\times SO(8)\times SO(8)\mid\left(\alpha_1x\right)\left(\alpha_2y\right)=\overline{\alpha_3(\overline{xy})},x,y\in\mathfrak{C}\right\} \\ & \widetilde{D}_4\text{ is a compact group.} \end{aligned}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |          |       |
| 0902.0431_E00532 | 89   | ■ 0.724 | $\begin{aligned} & \left \eta_1(t)\right ^2+\left \eta_2(t)\right ^2+\left \eta_3(t)\right ^2 \\ & =\left \xi_1\right ^2+\left \frac{\xi_2-\xi_3}{2}+\frac{\xi_2+\xi_3}{2} \cos t- q  \sin t+i \nu \sin t\right ^2 \\ & +\left -\frac{\xi_2-\xi_3}{2}+\frac{\xi_2+\xi_3}{2} \cos t- q  \sin t+i \nu \sin t\right ^2 \\ & =\left \xi_1\right ^2+2\left(\frac{\xi_2-\xi_3}{2}\right)^2+2\left(\frac{\xi_2+\xi_3}{2} \cos t- q  \sin t\right)^2+2 \nu^2 \sin ^2 t \\ & =\left \xi_1\right ^2+2\left(\frac{\xi_2-\xi_3}{2}\right)^2+2\left(\left(\frac{\xi_2+\xi_3}{2}\right)^2+ q ^2\right) \sin ^2\left(t+t_0\right)+2 \nu^2 \sin ^2 t \\ & \leq\left \xi_1\right ^2+\left \xi_2\right ^2+\left \xi_3\right ^2+2 q ^2+2 \nu^2 \cos ^2 t_0 \quad\left(\text { for some } t_0 \in \boldsymbol{R}\right) \end{aligned}$ |          |       |

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| 0902.0431_EQ1178 | 199  | 0.725 | $\begin{aligned} &= \sum_{j=1}^5 \sum_{i=1}^6 (-1)^i (\mathbf{a}_1 \wedge \cdots \mathbf{a}_{i-1} \wedge \mathbf{a}_{i+1} \wedge \cdots \wedge \mathbf{a}_6, \mathbf{e}_1 \wedge \cdots \wedge \mathbf{e}_5) a_{ij} \mathbf{e}_j \\ &= - \sum_{j=1}^5 \det \begin{pmatrix} a_{1j} & a_{11} & \cdots & a_{15} \\ a_{2j} & a_{21} & \cdots & a_{25} \\ & & \cdots & \\ a_{6j} & a_{61} & \cdots & a_{65} \end{pmatrix} \mathbf{e}_j = 0. \end{aligned}$                                                                                                                                                                                                                                                                                 | $\begin{aligned} &= \sum_{j=1}^5 \sum_{i=1}^6 (-1)^i (\mathbf{a}_1 \wedge \cdots \mathbf{a}_{i-1} \wedge \mathbf{a}_{i+1} \wedge \cdots \wedge \mathbf{a}_6, \mathbf{e}_1 \wedge \cdots \wedge \mathbf{e}_5) a_{ij} \mathbf{e}_j \\ &= - \sum_{j=1}^5 \det \begin{pmatrix} a_{1j} & a_{11} & \cdots & a_{15} \\ a_{2j} & a_{21} & \cdots & a_{25} \\ & & \cdots & \\ a_{6j} & a_{61} & \cdots & a_{65} \end{pmatrix} \mathbf{e}_j = 0. \end{aligned}$                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 0902.0431_EQ1148 | 194  | 0.732 | $\begin{aligned} &(\mathbf{u} \times \mathbf{v}) \mathbf{x}, \mathbf{y}) = (\mathbf{v} \wedge (\mathbf{u} \wedge \mathbf{x}), \mathbf{y}) + \frac{2}{3} (\mathbf{u}, \mathbf{v}) (\mathbf{x}, \mathbf{y}) \\ &= -(\mathbf{x} \wedge \mathbf{u}, \mathbf{y} \wedge \mathbf{v}) + \frac{2}{3} (\mathbf{u}, \mathbf{v}) (\mathbf{x}, \mathbf{y}) \\ &= (\mathbf{x} \wedge \mathbf{u}_2 \wedge \mathbf{u}_3, \mathbf{v}) (\mathbf{u}_1, \mathbf{y}) - (\mathbf{x} \wedge \mathbf{u}_1 \wedge \mathbf{u}_3, \mathbf{v}) (\mathbf{u}_2, \mathbf{y}) \\ &\quad + (\mathbf{x} \wedge \mathbf{u}_1 \wedge \mathbf{u}_2, \mathbf{v}) (\mathbf{u}_3, \mathbf{y}) - \frac{1}{3} (\mathbf{u}, \mathbf{v}) (\mathbf{x}, \mathbf{y}). \end{aligned}$ | $\begin{aligned} &((\mathbf{u} \times \mathbf{v}) \mathbf{x}, \mathbf{y}) = (\mathbf{v} \wedge (\mathbf{u} \wedge \mathbf{x}), \mathbf{y}) + \frac{2}{3} (\mathbf{u}, \mathbf{v}) (\mathbf{x}, \mathbf{y}) \\ &= -(\mathbf{x} \wedge \mathbf{u}, \mathbf{y} \wedge \mathbf{v}) + \frac{2}{3} (\mathbf{u}, \mathbf{v}) (\mathbf{x}, \mathbf{y}) \\ &= (\mathbf{x} \wedge \mathbf{u}_2 \wedge \mathbf{u}_3, \mathbf{v}) (\mathbf{u}_1, \mathbf{y}) - (\mathbf{x} \wedge \mathbf{u}_1 \wedge \mathbf{u}_3, \mathbf{v}) (\mathbf{u}_2, \mathbf{y}) \\ &\quad + (\mathbf{x} \wedge \mathbf{u}_1 \wedge \mathbf{u}_2, \mathbf{v}) (\mathbf{u}_3, \mathbf{y}) - \frac{1}{3} (\mathbf{u}, \mathbf{v}) (\mathbf{x}, \mathbf{y}). \end{aligned}$ | $\begin{aligned} ((\mathbf{u} \times \mathbf{v}) \mathbf{x}, \mathbf{y}) &= (\mathbf{v} \wedge (\mathbf{u} \wedge \mathbf{x}), \mathbf{y}) + \frac{2}{3} (\mathbf{u}, \mathbf{v}) (\mathbf{x}, \mathbf{y}) \\ &= -(\mathbf{x} \wedge \mathbf{u}, \mathbf{y} \wedge \mathbf{v}) + \frac{2}{3} (\mathbf{u}, \mathbf{v}) (\mathbf{x}, \mathbf{y}) \\ &= (\mathbf{x} \wedge \mathbf{u}_2 \wedge \mathbf{u}_3, \mathbf{v}) (\mathbf{u}_1, \mathbf{y}) - (\mathbf{x} \wedge \mathbf{u}_1 \wedge \mathbf{u}_3, \mathbf{v}) (\mathbf{u}_2, \mathbf{y}) \\ &\quad + (\mathbf{x} \wedge \mathbf{u}_1 \wedge \mathbf{u}_2, \mathbf{v}) (\mathbf{u}_3, \mathbf{y}) - \frac{1}{3} (\mathbf{u}, \mathbf{v}) (\mathbf{x}, \mathbf{y}). \end{aligned}$ |
| 0902.0431_EQ1081 | 183  | 0.733 | $\begin{aligned} &S_1 \tau S_2 = \chi P_1 \tau \chi P_2 = \chi(X_1, Y_1, \xi_1, \eta_1) \tau \chi(X_2, Y_2, \xi_2, \eta_2) \\ &= \left( k \left( g X_1 - \frac{\xi_1}{2} E \right) + i k \left( g(\gamma Y_1) - \frac{\eta_1}{2} E \right) \right) J \tau \left( \left( k \left( g X_2 - \frac{\xi_2}{2} E \right) \right. \right. \\ &\quad \left. \left. + i k \left( g(\gamma Y_2) - \frac{\eta_2}{2} E \right) \right) J \right) \end{aligned}$                                                                                                                                                                                                                                                                                   | $\begin{aligned} S_1 \tau S_2 &= \chi P_1 \tau \chi P_2 = \chi(X_1, Y_1, \xi_1, \eta_1) \tau \chi(X_2, Y_2, \xi_2, \eta_2) \\ &= \left( k \left( g X_1 - \frac{\xi_1}{2} E \right) + i k \left( g(\gamma Y_1) - \frac{\eta_1}{2} E \right) \right) J \tau \left( \left( k \left( g X_2 - \frac{\xi_2}{2} E \right) \right. \right. \\ &\quad \left. \left. + i k \left( g(\gamma Y_2) - \frac{\eta_2}{2} E \right) \right) J \right) \end{aligned}$                                                                                                                                                                                                                                                                                    | $\begin{aligned} S_1 \tau S_2 &= \chi P_1 \tau \chi P_2 = \chi(X_1, Y_1, \xi_1, \eta_1) \tau \chi(X_2, Y_2, \xi_2, \eta_2) \\ &= \left( k \left( g X_1 - \frac{\xi_1}{2} E \right) + i k \left( g(\gamma Y_1) - \frac{\eta_1}{2} E \right) \right) J \tau \left( \left( k \left( g X_2 - \frac{\xi_2}{2} E \right) \right. \right. \\ &\quad \left. \left. + i k \left( g(\gamma Y_2) - \frac{\eta_2}{2} E \right) \right) J \right) \end{aligned}$                                                                                                                                                                                                                                                                                    |

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| 0902.0431_E01096 | 186  | 0.736 | $\begin{aligned} & \left[ \zeta \left( \left( S_1 \right) I, \zeta \left( \left( S_2 \right) I \right) \right) \right] \\ & = \left[ \Theta \left( 0, 2\lambda\gamma\chi^{-1}S_1, 2\chi^{-1}S_1, 0, 0, 0 \right), \Theta \left( 0, 2\lambda\gamma\chi^{-1}S_2, 2\chi^{-1}S_2, 0, 0, 0 \right) \right] \\ & = \Theta \left( 2\lambda\gamma\chi^{-1}S_1 \times 2\chi^{-1}S_2 - 2\lambda\gamma\chi^{-1}S_2 \times 2\chi^{-1}S_1, 0, 0, \right. \\ & \quad \left. \frac{1}{8} \left( -\{2\lambda\gamma\chi^{-1}S_1, 2\chi^{-1}S_2\} + \{2\lambda\gamma\chi^{-1}S_2, 2\chi^{-1}S_1\} \right), \right. \\ & \quad \left. \frac{1}{4} \{2\lambda\gamma\chi^{-1}S_1, 2\lambda\gamma\chi^{-1}S_2\}, -\frac{1}{4} \{2\chi^{-1}S_1, 2\chi^{-1}S_2\} \right), \end{aligned}$ | $\begin{aligned} & [\zeta l(S_1)I, \zeta(l(S_2)I)] \\ & = [\Theta(0, 2\lambda\gamma\chi^{-1}S_1, 2\chi^{-1}S_1, 0, 0, 0), \Theta(0, 2\lambda\gamma\chi^{-1}S_2, 2\chi^{-1}S_2, 0, 0, 0)] \\ & = \Theta\left(2\lambda\gamma\chi^{-1}S_1 \times 2\chi^{-1}S_2 - 2\lambda\gamma\chi^{-1}S_2 \times 2\chi^{-1}S_1, 0, 0, \right. \\ & \quad \left. \frac{1}{8}(-\{2\lambda\gamma\chi^{-1}S_1, 2\chi^{-1}S_2\} + \{2\lambda\gamma\chi^{-1}S_2, 2\chi^{-1}S_1\}), \right. \\ & \quad \left. \frac{1}{4}\{2\lambda\gamma\chi^{-1}S_1, 2\lambda\gamma\chi^{-1}S_2\}, -\frac{1}{4}\{2\chi^{-1}S_1, 2\chi^{-1}S_2\}\right), \end{aligned}$ |       |
| 0902.0431_E00158 | 29   | 0.746 | $\left( \alpha_1 x \right) \left( \alpha_2 y \right) = \alpha_3 (xy), \quad x, y \in \mathfrak{C}.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | $(\alpha_1 x)(\alpha_2 y) = \alpha_3(xy), \quad x, y \in \mathfrak{C}.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |       |
| 0902.0431_E00621 | 103  | 0.747 | $\begin{aligned} & \langle \varphi(A)X, \varphi(A)Y \rangle = \langle g\varphi(A)X, g\varphi(A)Y \rangle \\ & = \langle A(gX)A^*, A(gY)A^* \rangle = \langle gX, gY \rangle = \langle X, Y \rangle. \end{aligned}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | $\begin{aligned} \langle \varphi(A)X, \varphi(A)Y \rangle &= \langle g\varphi(A)X, g\varphi(A)Y \rangle \\ &= \langle A(gX)A^*, A(gY)A^* \rangle = \langle gX, gY \rangle = \langle X, Y \rangle. \end{aligned}$                                                                                                                                                                                                                                                                                                                                                                                                                 |       |
| 0902.0431_E01102 | 187  | 0.751 | $\begin{aligned} & \left( \left( \mathfrak{e}_{\tilde{\lambda}\gamma} \right)_{\tau\tilde{\lambda}} = \left( \left( \mathfrak{e}_8^C \right)_{\tau\gamma} \right)_{\tilde{\lambda}\gamma} = \left( \mathfrak{e}_{8(8)} \right)_{\tilde{\lambda}\gamma} = \left( \mathfrak{e}_{8(8)} \right)^{\tilde{\lambda}\gamma}, \right. \\ & \left. \left( \left( \mathfrak{e}_8^C \right)_{-\tilde{\lambda}\gamma} \right)_{\tau\tilde{\lambda}} = \left( \left( \mathfrak{e}_8^C \right)_{\tau\gamma} \right)_{-\tilde{\lambda}\gamma} = \left( \mathfrak{e}_{8(8)} \right)_{-\tilde{\lambda}\gamma}, \right. \end{aligned}$                                                                                                                                              | $\begin{aligned} ((\mathfrak{e}_8^C)_{\tilde{\lambda}\gamma})_{\tau\tilde{\lambda}} &= ((\mathfrak{e}_8^C)_{\tau\gamma})_{\tilde{\lambda}\gamma} = (\mathfrak{e}_{8(8)})_{\tilde{\lambda}\gamma} = (\mathfrak{e}_{8(8)})^{\tilde{\lambda}\gamma}, \\ ((\mathfrak{e}_8^C)_{-\tilde{\lambda}\gamma})_{\tau\tilde{\lambda}} &= ((\mathfrak{e}_8^C)_{\tau\gamma})_{-\tilde{\lambda}\gamma} = (\mathfrak{e}_{8(8)})_{-\tilde{\lambda}\gamma}, \end{aligned}$                                                                                                                                                                          |       |

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|------------------|------|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0902.0431_E00250 | 41   | 0.754 | $\begin{aligned} & \& (4) \quad \operatorname{tr}(\alpha X) = (\alpha X, E, E) = (\alpha X, \alpha E, \alpha E) = (X, E, E) \text{ (Lemma 2.2.2.(2))} \\ & \operatorname{tr}(X). \text{ Hence } \frac{1}{2}(\operatorname{tr}(X) \operatorname{tr}(Y) - (X, Y)) = (X, Y, E) \text{ (Lemma 2.1.1)} = (\alpha X, \alpha Y, \alpha E) \\ & = (\alpha X, \alpha Y, E) = \frac{1}{2}(\operatorname{tr}(\alpha X) \operatorname{tr}(\alpha Y) - (\alpha X, \alpha Y)) = \frac{1}{2}(\operatorname{tr}(X) \operatorname{tr}(Y) - (\alpha X, \alpha Y)). \text{ There-} \end{aligned}$                                                                                                                                                                                                                                                                                                                                                                                        | $(4) \Rightarrow (2) \operatorname{tr}(\alpha X) = (\alpha X, E, E) = (\alpha X, \alpha E, \alpha E) = (X, E, E) \text{ (Lemma 2.2.2.(2))}$ $= \operatorname{tr}(X). \text{ Hence } \frac{1}{2}(\operatorname{tr}(X) \operatorname{tr}(Y) - (X, Y)) = (X, Y, E) \text{ (Lemma 2.1.1)} = (\alpha X, \alpha Y, \alpha E)$ $= (\alpha X, \alpha Y, E) = \frac{1}{2}(\operatorname{tr}(\alpha X) \operatorname{tr}(\alpha Y) - (\alpha X, \alpha Y)) = \frac{1}{2}(\operatorname{tr}(X) \operatorname{tr}(Y) - (\alpha X, \alpha Y)). \text{ There-}$                                                                                                                                                                                                                                                                                                                                                                                                                 | $(4) \Rightarrow (2) \operatorname{tr}(\alpha X) = (\alpha X, E, E) = (\alpha X, \alpha E, \alpha E) = (X, E, E) \text{ (Lemma 2.2.2.(2))}$ $= \operatorname{tr}(X). \text{ Hence } \frac{1}{2}(\operatorname{tr}(X) \operatorname{tr}(Y) - (X, Y)) = (X, Y, E) \text{ (Lemma 2.1.1)} = (\alpha X, \alpha Y, \alpha E)$ $= (\alpha X, \alpha Y, E) = \frac{1}{2}(\operatorname{tr}(\alpha X) \operatorname{tr}(\alpha Y) - (\alpha X, \alpha Y)) = \frac{1}{2}(\operatorname{tr}(X) \operatorname{tr}(Y) - (\alpha X, \alpha Y)). \text{ There-}$                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 0902.0431_E00911 | 151  | 0.754 | $E_{7,C} = \left\{ \alpha \in \operatorname{Iso}_C \left( (\mathfrak{P}_C)^C \right) \mid \alpha(P \times Q) \alpha^{-1} = \alpha P \times \alpha Q, \langle \alpha P, \alpha Q \rangle = \langle P, Q \rangle \right\}.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | $E_{7,C} = \left\{ \alpha \in \operatorname{Iso}_C \left( (\mathfrak{P}_C)^C \right) \mid \alpha(P \times Q) \alpha^{-1} = \alpha P \times \alpha Q, \langle \alpha P, \alpha Q \rangle = \langle P, Q \rangle \right\}.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | $E_{7,C} = \left\{ \alpha \in \operatorname{Iso}_C \left( (\mathfrak{P}_C)^C \right) \mid \alpha(P \times Q) \alpha^{-1} = \alpha P \times \alpha Q, \langle \alpha P, \alpha Q \rangle = \langle P, Q \rangle \right\}.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 0902.0431_E01107 | 187  | 0.754 | $k^2 B_8(R, R') = B_8(\alpha R, \alpha R') = B_8(R, R'), \quad R, R' \in (\mathfrak{e}_8^C)_{-\widetilde{\lambda}_\gamma},$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | $k^2 B_8(R, R') = B_8(\alpha R, \alpha R') = B_8(R, R'), \quad R, R' \in (\mathfrak{e}_8^C)_{-\widetilde{\lambda}_\gamma},$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | $k^2 B_8(R, R') = B_8(\alpha R, \alpha R') = B_8(R, R'), \quad R, R' \in (\mathfrak{e}_8^C)_{-\widetilde{\lambda}_\gamma},$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 0902.0431_E00448 | 73   | 0.755 | $\phi = \delta + \widetilde{T}, \quad \delta \in \mathfrak{f}_4^C, T \in \mathfrak{J}_0^C.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | $\phi = \delta + \widetilde{T}, \quad \delta \in \mathfrak{f}_4^C, T \in \mathfrak{J}_0^C.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | $\phi = \delta + \widetilde{T}, \quad \delta \in \mathfrak{f}_4^C, T \in \mathfrak{J}_0^C.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 0902.0431_E00318 | 51   | 0.755 | $\begin{aligned} & \& ([\delta, [\widetilde{A}, \widetilde{B}]], [\widetilde{C}, \widetilde{D}])_4 + ([\widetilde{A}, \widetilde{B}], [\delta, [\widetilde{C}, \widetilde{D}]])_4 \\ & = ([\widetilde{\delta A}, \widetilde{B}] + [\widetilde{A}, \widetilde{\delta B}], [\widetilde{C}, \widetilde{D}])_4 + ([\widetilde{A}, \widetilde{B}], [\widetilde{\delta C}, \widetilde{D}] + [\widetilde{C}, \widetilde{\delta D}])_4 \text{ (Lemma 2.4.5.(1))} \\ & = ([\widetilde{\delta A}, \widetilde{B}]C, D) + ([\widetilde{A}, \widetilde{\delta B}]C, D) + ([\widetilde{A}, \widetilde{B}]\delta C, D) + ([\widetilde{A}, \widetilde{B}]C, \delta D) \\ & = (\delta A \circ (B \circ C), D) - (B \circ (\delta A \circ C), D) + (A \circ (\delta B \circ C), D) - (\delta B \circ (A \circ C), D) \\ & \quad + (A \circ (B \circ \delta C), D) - (B \circ (A \circ \delta C), D) + (A \circ (B \circ C), \delta D) - (B \circ (A \circ C), \delta D), \end{aligned}$ | $\begin{aligned} & ([\delta, [\widetilde{A}, \widetilde{B}]], [\widetilde{C}, \widetilde{D}])_4 + ([\widetilde{A}, \widetilde{B}], [\delta, [\widetilde{C}, \widetilde{D}]])_4 \\ & = ([\widetilde{\delta A}, \widetilde{B}] + [\widetilde{A}, \widetilde{\delta B}], [\widetilde{C}, \widetilde{D}])_4 + ([\widetilde{A}, \widetilde{B}], [\widetilde{\delta C}, \widetilde{D}] + [\widetilde{C}, \widetilde{\delta D}])_4 \text{ (Lemma 2.4.5.(1))} \\ & = ([\widetilde{\delta A}, \widetilde{B}]C, D) + ([\widetilde{A}, \widetilde{\delta B}]C, D) + ([\widetilde{A}, \widetilde{B}]\delta C, D) + ([\widetilde{A}, \widetilde{B}]C, \delta D) \\ & = (\delta A \circ (B \circ C), D) - (B \circ (\delta A \circ C), D) + (A \circ (\delta B \circ C), D) - (\delta B \circ (A \circ C), D) \\ & \quad + (A \circ (B \circ \delta C), D) - (B \circ (A \circ \delta C), D) + (A \circ (B \circ C), \delta D) - (B \circ (A \circ C), \delta D) \end{aligned}$ | $\begin{aligned} & ([\delta, [\widetilde{A}, \widetilde{B}]], [\widetilde{C}, \widetilde{D}])_4 + ([\widetilde{A}, \widetilde{B}], [\delta, [\widetilde{C}, \widetilde{D}]])_4 \\ & = ([\widetilde{\delta A}, \widetilde{B}] + [\widetilde{A}, \widetilde{\delta B}], [\widetilde{C}, \widetilde{D}])_4 + ([\widetilde{A}, \widetilde{B}], [\widetilde{\delta C}, \widetilde{D}] + [\widetilde{C}, \widetilde{\delta D}])_4 \text{ (Lemma 2.4.5.(1))} \\ & = ([\widetilde{\delta A}, \widetilde{B}]C, D) + ([\widetilde{A}, \widetilde{\delta B}]C, D) + ([\widetilde{A}, \widetilde{B}]\delta C, D) + ([\widetilde{A}, \widetilde{B}]C, \delta D) \\ & = (\delta A \circ (B \circ C), D) - (B \circ (\delta A \circ C), D) + (A \circ (\delta B \circ C), D) - (\delta B \circ (A \circ C), D) \\ & \quad + (A \circ (B \circ \delta C), D) - (B \circ (A \circ \delta C), D) + (A \circ (B \circ C), \delta D) - (B \circ (A \circ C), \delta D), \end{aligned}$ |

| Identifier       | Page | Conf.   | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Rendered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Image                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|------------------|------|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0902.0431_E01153 | 194  | ■ 0.759 | $\begin{aligned} & \& ((u \times w) v - (v \times w) u, a) \setminus \\ & \& = \left( \left( (u \times w) v_1 \right) \wedge v_2 \wedge v_3, a \right) \wedge v_2 \wedge v_3, a \right) - \left( \left( (u \times w) v_2 \right) \wedge v_1 \wedge v_3, a \right) \\ & \quad + \left( \left( (u \times w) v_3 \right) \wedge v_1 \wedge v_2, a \right) - \left( \left( (v \times w) u_1 \right) \wedge u_2 \wedge u_3, a \right) \\ & \quad + \left( \left( (v \times w) u_2 \right) \wedge u_1 \wedge u_3, a \right) - \left( \left( (v \times w) u_3 \right) \wedge u_1 \wedge u_2, a \right) \\ & = - (u, w) (v, a) + \sum_{i=1}^3 \sum_{j=1}^3 (u_i \wedge u_{i+1} \wedge v_j, w) (u_{i+2} \wedge v_{i+1} \wedge v_{j+2}, a) \end{aligned}$ | $\begin{aligned} & ((u \times w) v - (v \times w) u, a) \\ & = ((u \times w) v_1) \wedge v_2 \wedge v_3, a) - ((u \times w) v_2) \wedge v_1 \wedge v_3, a) \\ & \quad + (((u \times w) v_3) \wedge v_1 \wedge v_2, a) - (((v \times w) u_1) \wedge u_2 \wedge u_3, a) \\ & \quad + (((v \times w) u_2) \wedge u_1 \wedge u_3, a) - (((v \times w) u_3) \wedge u_1 \wedge u_2, a) \\ & = -(u, w)(v, a) + \sum_{i=1}^3 \sum_{j=1}^3 (u_i \wedge u_{i+1} \wedge v_j, w) (u_{i+2} \wedge v_{i+1} \wedge v_{j+2}, a) \end{aligned}$                  | $\begin{aligned} & ((u \times w) v - (v \times w) u, a) \\ & = (((u \times w) v_1) \wedge v_2 \wedge v_3, a) - (((u \times w) v_2) \wedge v_1 \wedge v_3, a) \\ & \quad + (((u \times w) v_3) \wedge v_1 \wedge v_2, a) - (((v \times w) u_1) \wedge u_2 \wedge u_3, a) \\ & \quad + (((v \times w) u_2) \wedge u_1 \wedge u_3, a) - (((v \times w) u_3) \wedge u_1 \wedge u_2, a) \\ & = -(u, w)(v, a) + \sum_{i=1}^3 \sum_{j=1}^3 (u_i \wedge u_{i+1} \wedge v_j, w) (u_{i+2} \wedge v_{i+1} \wedge v_{j+2}, a) \end{aligned}$                |
| 0902.0431_E00134 | 25   | ■ 0.760 | $\alpha = \varphi(1, q) \beta = \varphi(1, q) \varphi(p, 1) = \varphi(p, q), \quad (p, q) \in \operatorname{Sp}(1) \times \operatorname{Sp}(1)$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | $\alpha = \varphi(1, q) \beta = \varphi(1, q) \varphi(p, 1) = \varphi(p, q), \quad (p, q) \in \operatorname{Sp}(1) \times \operatorname{Sp}(1),$                                                                                                                                                                                                                                                                                                                                                                                                | $\alpha = \varphi(1, q) \beta = \varphi(1, q) \varphi(p, 1) = \varphi(p, q), \quad (p, q) \in \operatorname{Sp}(1) \times \operatorname{Sp}(1),$                                                                                                                                                                                                                                                                                                                                                                                                |
| 0902.0431_E00224 | 37   | ■ 0.761 | $\begin{aligned} & E_1 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad E_2 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad E_3 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \\ & F_1(x) = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & x \\ 0 & \bar{x} & 0 \end{pmatrix}, \quad F_2(x) = \begin{pmatrix} 0 & 0 & \bar{x} \\ 0 & 0 & 0 \\ x & 0 & 0 \end{pmatrix}, \quad F_3(x) = \begin{pmatrix} 0 & x & 0 \\ \bar{x} & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}. \end{aligned}$                                                                                                                                                                                                 | $\begin{aligned} & E_1 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad E_2 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad E_3 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \\ & F_1(x) = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & x \\ 0 & \bar{x} & 0 \end{pmatrix}, \quad F_2(x) = \begin{pmatrix} 0 & 0 & \bar{x} \\ 0 & 0 & 0 \\ x & 0 & 0 \end{pmatrix}, \quad F_3(x) = \begin{pmatrix} 0 & x & 0 \\ \bar{x} & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}. \end{aligned}$ | $\begin{aligned} & E_1 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad E_2 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad E_3 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \\ & F_1(x) = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & x \\ 0 & \bar{x} & 0 \end{pmatrix}, \quad F_2(x) = \begin{pmatrix} 0 & 0 & \bar{x} \\ 0 & 0 & 0 \\ x & 0 & 0 \end{pmatrix}, \quad F_3(x) = \begin{pmatrix} 0 & x & 0 \\ \bar{x} & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}. \end{aligned}$ |

| Identifier       | Page | Conf.   | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Rendered                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Image                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|------------------|------|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0902.0431_E00117 | 22   | ■ 0.762 | \begin{aligned} & \left(\alpha e_4\right) \& =a_{\{4\}} a_{\{7\}}=a_{\{4\}} \left(a_{\{3\}} a_{\{4\}} \right. \\ & \left.\right)=-a_{\{4\}} \left(a_{\{4\}} a_{\{3\}} \right)=a_{\{3\}}=\alpha \\ & e_{\{3\}}=\alpha \left(e_{\{4\}} e_{\{7\}} \right), \quad \left(\alpha e_1\right) \& \\ & =a_{\{1\}} \left(a_{\{4\}} a_{\{2\}} \right)=-a_{\{4\}} \left(a_{\{1\}} a_{\{2\}} \right)= \\ & -a_{\{4\}} a_{\{3\}}=a_{\{3\}} a_{\{4\}}=a_{\{7\}} \quad \& = \\ & \alpha e_{\{7\}}=\alpha \left(e_{\{1\}} e_{\{6\}} \right), \quad \left(\alpha e_3\right) \& \\ & =a_{\{3\}} \left(a_{\{6\}} \right)=\left(a_{\{1\}} a_{\{2\}} \right) \left(a_{\{4\}} a_{\{2\}} \right)= \\ & -\left(a_{\{2\}} a_{\{1\}} \right) \left(a_{\{4\}} a_{\{2\}} \right)=-a_{\{2\}} \left(a_{\{1\}} a_{\{4\}} \right) a_{\{2\}} \\ & =-a_{\{2\}} a_{\{5\}} a_{\{2\}}=a_{\{2\}} a_{\{2\}} a_{\{5\}}=-a_{\{5\}}=-\alpha e_{\{5\}}=\alpha \left(e_{\{3\}} e_{\{6\}} \right), \text { etc. } \end{aligned} | $\begin{aligned} (\alpha e_4)(\alpha e_7) &= a_4 a_7 = a_4 (a_3 a_4) = -a_4 (a_4 a_3) = a_3 = \alpha e_3 = \alpha (e_4 e_7), \\ (\alpha e_1)(\alpha e_6) &= a_1 a_6 = a_1 (a_4 a_2) = -a_4 (a_1 a_2) = -a_4 a_3 = a_3 a_4 = a_7 \\ &= \alpha e_7 = \alpha (e_1 e_6), \\ (\alpha e_3)(\alpha e_6) &= a_3 a_6 = (a_1 a_2) (a_4 a_2) = - (a_2 a_1) (a_4 a_2) = -a_2 (a_1 a_4) a_2 \\ &= -a_2 a_5 a_2 = a_2 a_2 a_5 = -a_5 = -\alpha e_5 = \alpha (e_3 e_6), \text{ etc.} \end{aligned}$ | $\begin{aligned} (\alpha e_4)(\alpha e_7) &= a_4 a_7 = a_4 (a_3 a_4) = -a_4 (a_4 a_3) = a_3 = \alpha e_3 = \alpha (e_4 e_7), \\ (\alpha e_1)(\alpha e_6) &= a_1 a_6 = a_1 (a_4 a_2) = -a_4 (a_1 a_2) = -a_4 a_3 = a_3 a_4 = a_7 \\ &= \alpha e_7 = \alpha (e_1 e_6), \\ (\alpha e_3)(\alpha e_6) &= a_3 a_6 = (a_1 a_2) (a_4 a_2) = - (a_2 a_1) (a_4 a_2) = -a_2 (a_1 a_4) a_2 \\ &= -a_2 a_5 a_2 = a_2 a_2 a_5 = -a_5 = -\alpha e_5 = \alpha (e_3 e_6), \text{ etc.} \end{aligned}$ |



| Identifier       | Page | Conf.   | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Rendered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Image                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
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| 0902.0431_E00368 | 60   | ■ 0.766 | $\begin{aligned} & \left\{ \begin{array}{l} \eta_1 = \xi_1 \\ \eta_2 = \frac{\xi_2 + \xi_3}{2} + \frac{\xi_2 - \xi_3}{2} \cos 2 a  + \frac{(a, x_1)}{ a } \sin 2 a  \\ \eta_3 = \frac{\xi_2 + \xi_3}{2} - \frac{\xi_2 - \xi_3}{2} \cos 2 a  - \frac{(a, x_1)}{ a } \sin 2 a , \end{array} \right. \\ & \left\{ \begin{array}{l} y_1 = x_1 - \frac{(\xi_2 - \xi_3)a}{2 a } \sin 2 a  - \frac{2(a, x_1)a}{ a ^2} \sin^2  a  \\ y_2 = x_2 \cos  a  - \frac{\overline{x_3 a}}{ a } \sin  a  \\ y_3 = x_3 \cos  a  + \frac{\overline{a x_2}}{ a } \sin  a  \end{array} \right. \end{aligned}$ | $\left\{ \begin{array}{l} \eta_1 = \xi_1 \\ \eta_2 = \frac{\xi_2 + \xi_3}{2} + \frac{\xi_2 - \xi_3}{2} \cos 2 a  + \frac{(a, x_1)}{ a } \sin 2 a  \\ \eta_3 = \frac{\xi_2 + \xi_3}{2} - \frac{\xi_2 - \xi_3}{2} \cos 2 a  - \frac{(a, x_1)}{ a } \sin 2 a , \end{array} \right.$ $\left\{ \begin{array}{l} y_1 = x_1 - \frac{(\xi_2 - \xi_3)a}{2 a } \sin 2 a  - \frac{2(a, x_1)a}{ a ^2} \sin^2  a  \\ y_2 = x_2 \cos  a  - \frac{\overline{x_3 a}}{ a } \sin  a  \\ y_3 = x_3 \cos  a  + \frac{\overline{a x_2}}{ a } \sin  a  \end{array} \right.$ | $\left\{ \begin{array}{l} \eta_1 = \xi_1 \\ \eta_2 = \frac{\xi_2 + \xi_3}{2} + \frac{\xi_2 - \xi_3}{2} \cos 2 a  + \frac{(a, x_1)}{ a } \sin 2 a  \\ \eta_3 = \frac{\xi_2 + \xi_3}{2} - \frac{\xi_2 - \xi_3}{2} \cos 2 a  - \frac{(a, x_1)}{ a } \sin 2 a , \end{array} \right.$ $\left\{ \begin{array}{l} y_1 = x_1 - \frac{(\xi_2 - \xi_3)a}{2 a } \sin 2 a  - \frac{2(a, x_1)a}{ a ^2} \sin^2  a  \\ y_2 = x_2 \cos  a  - \frac{\overline{x_3 a}}{ a } \sin  a  \\ y_3 = x_3 \cos  a  + \frac{\overline{a x_2}}{ a } \sin  a  \end{array} \right.$ |
| 0902.0431_E01100 | 187  | ■ 0.766 | $\mathfrak{e}_{8(8)} = \left( \mathfrak{e}_8 \right)^{\tau\gamma} = \left\{ R \in \mathfrak{e}_8^C \mid \tau\gamma R = R \right\}$                                                                                                                                                                                                                                                                                                                                                                                                                                                       | $\mathfrak{e}_{8(8)} = (\mathfrak{e}_8^C)^{\tau\gamma} = \{R \in \mathfrak{e}_8^C \mid \tau\gamma R = R\}$                                                                                                                                                                                                                                                                                                                                                                                                                                            | $\mathfrak{e}_{8(8)} = (\mathfrak{e}_8^C)^{\tau\gamma} = \{R \in \mathfrak{e}_8^C \mid \tau\gamma R = R\}$                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 0902.0431_E00816 | 136  | ■ 0.770 | $\alpha = \left( \begin{array}{cccc} \beta & 0 & M & 0 \\ 0 & \delta & 0 & N \\ a & 0 & \mu & 0 \\ 0 & b & 0 & \nu \end{array} \right), \quad \begin{array}{l} \beta, \delta \in \text{Hom}_C(\mathfrak{J}^C), \\ a, b \in \text{Hom}_C(\mathfrak{J}^C, C), \\ M, N \in \mathfrak{J}^C, \\ \mu, \nu \in C. \end{array}$                                                                                                                                                                                                                                                                  | $\alpha = \left( \begin{array}{cccc} \beta & 0 & M & 0 \\ 0 & \delta & 0 & N \\ a & 0 & \mu & 0 \\ 0 & b & 0 & \nu \end{array} \right), \quad \begin{array}{l} \beta, \delta \in \text{Hom}_C(\mathfrak{J}^C), \\ a, b \in \text{Hom}_C(\mathfrak{J}^C, C), \\ M, N \in \mathfrak{J}^C, \\ \mu, \nu \in C. \end{array}$                                                                                                                                                                                                                               | $\alpha = \left( \begin{array}{cccc} \beta & 0 & M & 0 \\ 0 & \delta & 0 & N \\ a & 0 & \mu & 0 \\ 0 & b & 0 & \nu \end{array} \right), \quad \begin{array}{l} \beta, \delta \in \text{Hom}_C(\mathfrak{J}^C), \\ a, b \in \text{Hom}_C(\mathfrak{J}^C, C), \\ M, N \in \mathfrak{J}^C, \\ \mu, \nu \in C. \end{array}$                                                                                                                                                                                                                               |

| Identifier       | Page | Conf. | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Rendered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Image |
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| 0902.0431_E00746 | 123  | 0.770 | $\begin{aligned} & \pm \left( \lambda_k - \lambda_l \right), \quad \pm \left( \lambda_k + \lambda_l \right), \quad 0 \leq k < l \leq 3, \\ & \pm \lambda_k \pm \frac{1}{2} (\mu_2 - \mu_3), \quad 0 \leq k \leq 3, \\ & \pm \frac{1}{2} (-\lambda_0 - \lambda_1 + \lambda_2 - \lambda_3) \pm \frac{1}{2} (\mu_3 - \mu_1), \\ & \pm \frac{1}{2} (\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3) \pm \frac{1}{2} (\mu_3 - \mu_1), \\ & \pm \frac{1}{2} (-\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3) \pm \frac{1}{2} (\mu_3 - \mu_1), \\ & \pm \frac{1}{2} (\lambda_0 - \lambda_1 + \lambda_2 + \lambda_3) \pm \frac{1}{2} (\mu_3 - \mu_1), \\ & \pm \frac{1}{2} (\lambda_0 - \lambda_1 + \lambda_2 - \lambda_3) \pm \frac{1}{2} (\mu_1 - \mu_2), \\ & \pm \frac{1}{2} (-\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3) \pm \frac{1}{2} (\mu_1 - \mu_2), \\ & \pm \frac{1}{2} (\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3) \pm \frac{1}{2} (\mu_1 - \mu_2), \\ & \pm \frac{1}{2} (-\lambda_0 - \lambda_1 + \lambda_2 + \lambda_3) \pm \frac{1}{2} (\mu_1 - \mu_2), \\ & \pm \left( \mu_j + \frac{2}{3} \nu \right), \quad 0 \leq j \leq 3, \\ & \pm \lambda_k \pm \left( \frac{1}{2} \mu_1 - \frac{2}{3} \nu \right), \quad 0 \leq k \leq 3, \\ & \pm \frac{1}{2} (-\lambda_0 - \lambda_1 + \lambda_2 - \lambda_3) \pm \left( \frac{1}{2} \mu_2 - \frac{2}{3} \nu \right), \\ & \pm \frac{1}{2} (\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3) \pm \left( \frac{1}{2} \mu_2 - \frac{2}{3} \nu \right), \end{aligned}$ | $\begin{aligned} & \pm (\lambda_k - \lambda_l), \quad \pm (\lambda_k + \lambda_l), \quad 0 \leq k < l \leq 3, \\ & \pm \lambda_k \pm \frac{1}{2} (\mu_2 - \mu_3), \quad 0 \leq k \leq 3, \\ & \pm \frac{1}{2} (-\lambda_0 - \lambda_1 + \lambda_2 - \lambda_3) \pm \frac{1}{2} (\mu_3 - \mu_1), \\ & \pm \frac{1}{2} (\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3) \pm \frac{1}{2} (\mu_3 - \mu_1), \\ & \pm \frac{1}{2} (-\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3) \pm \frac{1}{2} (\mu_3 - \mu_1), \\ & \pm \frac{1}{2} (-\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3) \pm \frac{1}{2} (\mu_3 - \mu_1), \\ & \pm \frac{1}{2} (\lambda_0 - \lambda_1 + \lambda_2 + \lambda_3) \pm \frac{1}{2} (\mu_3 - \mu_1), \\ & \pm \frac{1}{2} (\lambda_0 - \lambda_1 + \lambda_2 - \lambda_3) \pm \frac{1}{2} (\mu_1 - \mu_2), \\ & \pm \frac{1}{2} (-\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3) \pm \frac{1}{2} (\mu_1 - \mu_2), \\ & \pm \frac{1}{2} (\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3) \pm \frac{1}{2} (\mu_1 - \mu_2), \\ & \pm \frac{1}{2} (-\lambda_0 - \lambda_1 + \lambda_2 + \lambda_3) \pm \frac{1}{2} (\mu_1 - \mu_2), \\ & \pm \left( \mu_j + \frac{2}{3} \nu \right), \quad 0 \leq j \leq 3, \\ & \pm \lambda_k \pm \left( \frac{1}{2} \mu_1 - \frac{2}{3} \nu \right), \quad 0 \leq k \leq 3, \\ & \pm \frac{1}{2} (-\lambda_0 - \lambda_1 + \lambda_2 - \lambda_3) \pm \left( \frac{1}{2} \mu_2 - \frac{2}{3} \nu \right), \\ & \pm \frac{1}{2} (\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3) \pm \left( \frac{1}{2} \mu_2 - \frac{2}{3} \nu \right), \end{aligned}$ |       |

| Identifier       | Page | Conf.   | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                 | Rendered                                                                                                                                                                                                                                                                                                                                                                                                     | Image                                                                                                                                                                                                                                                                                                     |
|------------------|------|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0902.0431_E01119 | 190  | ■ 0.778 | $\mathfrak{e}_{\{8\}^{\{C\}}}=\mathfrak{sl}(3,C)\oplus\mathfrak{e}_{\{6\}^{\{C\}}}\oplus\left(\mathfrak{J}^{\{C\}}\right)^3\oplus\left(\mathfrak{J}^{\{C\}}\right)^3,$                                                                                                                                                                                                                                       | $\mathfrak{e}_8^C=\mathfrak{sl}(3,C)\oplus\mathfrak{e}_6^C\oplus(\mathfrak{J}^C)^3\oplus(\mathfrak{J}^C)^3,$                                                                                                                                                                                                                                                                                                 | $\mathfrak{e}_8^C=\mathfrak{sl}(3,C)\oplus\mathfrak{e}_6^C\oplus(\mathfrak{J}^C)^3\oplus(\mathfrak{J}^C)^3,$                                                                                                                                                                                              |
| 0902.0431_E00856 | 142  | ■ 0.779 | $\begin{aligned} &2\sin^2a\left(E_2,E_3\times X\right)+\tau\xi\sin a\cos a-\left(E_3,Y\right)\sin a\cos a-\eta\cos^2a \\ &=\eta\sin^2a+\left(\tau\xi-\xi\right)\sin a\cos a-\eta\cos^2a \\ &=\frac{1}{2}\left(\tau\xi-\xi\right)\sin 2a-\eta\cos 2a=0. \end{aligned}$                                                                                                                                        | $\begin{aligned} &2\sin^2a\left(E_2,E_3\times X\right)+\tau\xi\sin a\cos a-\left(E_3,Y\right)\sin a\cos a-\eta\cos^2a \\ &=\eta\sin^2a+\left(\tau\xi-\xi\right)\sin a\cos a-\eta\cos^2a \\ &=\frac{1}{2}\left(\tau\xi-\xi\right)\sin 2a-\eta\cos 2a=0. \end{aligned}$                                                                                                                                        | $\begin{aligned} &2\sin^2a\left(E_2,E_3\times X\right)+\tau\xi\sin a\cos a-\left(E_3,Y\right)\sin a\cos a-\eta\cos^2a \\ &=\eta\sin^2a+\left(\tau\xi-\xi\right)\sin a\cos a-\eta\cos^2a \\ &=\frac{1}{2}\left(\tau\xi-\xi\right)\sin 2a-\eta\cos 2a=0. \end{aligned}$                                     |
| 0902.0431_E01077 | 183  | ■ 0.782 | $\begin{aligned} &g\left(2\left(X_1\vee\gamma X_2-X_2\vee\gamma X_1\right)X\right) \\ &=-4gX_2\circ\left(gX_1\circ gX\right)+4gX_1\circ\left(gX_2\circ gX\right) \\ &=-gX_2gX_1gX-gX_2gXgX_1-gX_1gXgX_2-gXgX_1gX_2 \\ &\quad +gX_1gX_2gX+gX_1gXgX_2+gX_2gXgX_1+gXgX_2gX_1 \\ &=\left[gX_1,gX_2\right]gX-gX\left[gX_1,gX_2\right]=g\left(\left(\varphi_*\left[gX_1,gX_2\right]\right)X\right). \end{aligned}$ | $\begin{aligned} &g\left(2\left(X_1\vee\gamma X_2-X_2\vee\gamma X_1\right)X\right) \\ &=-4gX_2\circ\left(gX_1\circ gX\right)+4gX_1\circ\left(gX_2\circ gX\right) \\ &=-gX_2gX_1gX-gX_2gXgX_1-gX_1gXgX_2-gXgX_1gX_2 \\ &\quad +gX_1gX_2gX+gX_1gXgX_2+gX_2gXgX_1+gXgX_2gX_1 \\ &=\left[gX_1,gX_2\right]gX-gX\left[gX_1,gX_2\right]=g\left(\left(\varphi_*\left[gX_1,gX_2\right]\right)X\right). \end{aligned}$ | $\begin{aligned} &g(2(X_1\vee\gamma X_2-X_2\vee\gamma X_1)X) \\ &=-4gX_2\circ(gX_1\circ gX)+4gX_1\circ(gX_2\circ gX) \\ &=-gX_2gX_1gX-gX_2gXgX_1-gX_1gXgX_2-gXgX_1gX_2 \\ &\quad +gX_1gX_2gX+gX_1gXgX_2+gX_2gXgX_1+gXgX_2gX_1 \\ &=[gX_1,gX_2]gX-gX[gX_1,gX_2]=g((\varphi_*[gX_1,gX_2])X). \end{aligned}$ |
| 0902.0431_E01137 | 192  | ■ 0.785 | $\begin{aligned} &\left(\mathfrak{e}_{\{8\}^{\{C\}}}\right)^w=\left\{R\in\mathfrak{e}_{\{8\}^{\{C\}}}\mid wR=R,\tau\tilde{\lambda}R=R\right\} \\ &=\left\{(D,\phi,0,0)\in\mathfrak{e}_{\{8\}^{\{C\}}}\mid D\in\mathfrak{su}(3),\phi\in\mathfrak{e}_6\right\}\cong\mathfrak{su}(3)\oplus\mathfrak{e}_6, \end{aligned}$                                                                                        | $\begin{aligned} (\mathfrak{e}_8)^w &= \left\{R \in \mathfrak{e}_8^C \mid wR = R, \tau\tilde{\lambda}R = R\right\} \\ &= \left\{(D, \phi, 0, 0) \in \mathfrak{e}_8^C \mid D \in \mathfrak{su}(3), \phi \in \mathfrak{e}_6\right\} \cong \mathfrak{su}(3) \oplus \mathfrak{e}_6, \end{aligned}$                                                                                                               | $\begin{aligned} (\mathfrak{e}_8)^w &= \{R \in \mathfrak{e}_8^C \mid wR = R, \tau\tilde{\lambda}R = R\} \\ &= \{(D, \phi, 0, 0) \in \mathfrak{e}_8^C \mid D \in \mathfrak{su}(3), \phi \in \mathfrak{e}_6\} \cong \mathfrak{su}(3) \oplus \mathfrak{e}_6, \end{aligned}$                                  |
| 0902.0431_E00299 | 48   | ■ 0.788 | $\mathfrak{f}_{\{4\}^{\{C\}}}=\mathfrak{d}_{\{4\}^{\{C\}}}\oplus\widetilde{\mathfrak{a}}_{\{1\}^{\{C\}}}\oplus\widetilde{\mathfrak{a}}_{\{2\}^{\{C\}}}\oplus\widetilde{\mathfrak{a}}_{\{3\}^{\{C\}}}=\mathfrak{d}_{\{4\}^{\{C\}}}\oplus\widetilde{\mathfrak{A}}^{\{C\}}.$                                                                                                                                    | $\mathfrak{f}_4^C=\mathfrak{d}_4^C\oplus\widetilde{\mathfrak{A}}_1^C\oplus\widetilde{\mathfrak{A}}_2^C\oplus\widetilde{\mathfrak{A}}_3^C=\mathfrak{d}_4^C\oplus\widetilde{\mathfrak{A}}^C.$                                                                                                                                                                                                                  | $\mathfrak{f}_4^C=\mathfrak{d}_4^C\oplus\widetilde{\mathfrak{A}}_1^C\oplus\widetilde{\mathfrak{A}}_2^C\oplus\widetilde{\mathfrak{A}}_3^C=\mathfrak{d}_4^C\oplus\widetilde{\mathfrak{A}}^C.$                                                                                                               |

| Identifier       | Page | Conf. | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Rendered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Image                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|------------------|------|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0902.0431_E01034 | 176  | 0.789 | $\begin{array}{r} \frac{1}{2}\left(\lambda_0+\lambda_1+\lambda_2+\lambda_3\right)-\frac{1}{2}\mu_3-\frac{1}{3}\nu+r=1 \\ \frac{1}{2}\left(\lambda_0+\lambda_1+\lambda_2+\lambda_3\right)-\frac{1}{2}\mu_3-\frac{1}{3}\nu-r=1 \\ \frac{1}{2}\left(\lambda_0-\lambda_1-\lambda_2+\lambda_3\right)+\frac{1}{2}\mu_3+\frac{1}{3}\nu+r=1 \\ \frac{1}{2}\left(\lambda_0-\lambda_1-\lambda_2+\lambda_3\right)+\frac{1}{2}\mu_3+\frac{1}{3}\nu-r=1 \\ \frac{1}{2}\left(\lambda_0-\lambda_1-\lambda_2+\lambda_3\right)-\frac{1}{2}\mu_3-\frac{1}{3}\nu+r=1 \\ \frac{1}{2}\left(\lambda_0-\lambda_1-\lambda_2+\lambda_3\right)-\frac{1}{2}\mu_3-\frac{1}{3}\nu-r=1 \end{array}$ | $\begin{array}{r} \frac{1}{2}\left(\lambda_0+\lambda_1+\lambda_2+\lambda_3\right)-\frac{1}{2}\mu_3-\frac{1}{3}\nu+r=1 \\ \frac{1}{2}\left(\lambda_0+\lambda_1+\lambda_2+\lambda_3\right)-\frac{1}{2}\mu_3-\frac{1}{3}\nu-r=1 \\ \frac{1}{2}\left(\lambda_0-\lambda_1-\lambda_2+\lambda_3\right)+\frac{1}{2}\mu_3+\frac{1}{3}\nu+r=1 \end{array}$                                                                                                                                                                                                                                                                                                      | $\begin{array}{r} \nu-r=0 \\ 2r=0 \\ \nu+r=0 \end{array}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 0902.0431_E00899 | 149  | 0.790 | $\begin{aligned} & \chi\left(\begin{array}{c} \varphi_*(D)X \\ \gamma\varphi_*(D)\gamma Y \\ 0 \\ 0 \end{array}\right) = \chi\left(\begin{pmatrix} \varphi_*(D)X & 0 & 0 & 0 \\ 0 & \tau\varphi_*(D)\tau & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} X \\ Y \\ \xi \\ \eta \end{pmatrix}\right) \\ & = \chi(\Phi(\varphi_*(D), 0, 0, 0)P). \end{aligned}$                                                                                                                                                                                                                                                                                 | $\begin{aligned} & = \chi\left(\begin{array}{c} \varphi_*(D)X \\ \gamma\varphi_*(D)\gamma Y \\ 0 \\ 0 \end{array}\right) = \chi\left(\begin{pmatrix} \varphi_*(D)X & 0 & 0 & 0 \\ 0 & \tau\varphi_*(D)\tau & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} X \\ Y \\ \xi \\ \eta \end{pmatrix}\right) \\ & = \chi(\Phi(\varphi_*(D), 0, 0, 0)P). \end{aligned}$                                                                                                                                                                                                                                                               | $\begin{aligned} & = \chi\left(\begin{array}{c} \varphi_*(D)X \\ \gamma\varphi_*(D)\gamma Y \\ 0 \\ 0 \end{array}\right) = \chi\left(\begin{pmatrix} \varphi_*(D)X & 0 & 0 & 0 \\ 0 & \tau\varphi_*(D)\tau & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} X \\ Y \\ \xi \\ \eta \end{pmatrix}\right) \\ & = \chi(\Phi(\varphi_*(D), 0, 0, 0)P). \end{aligned}$                                                                                                                                                                                                                                                               |
| 0902.0431_E00631 | 104  | 0.794 | $\begin{aligned} & \varphi(B)(M+\boldsymbol{a}) = g^{-1}(Bg(M+\boldsymbol{a})B^*) \\ & = g^{-1}\left(\begin{pmatrix} p & 0 \\ 0 & D \end{pmatrix} \left(\begin{pmatrix} \frac{1}{2}\mathrm{tr}(M) & i\boldsymbol{a} \\ i\boldsymbol{a}^* & M - \frac{1}{2}\mathrm{tr}(M)E \end{pmatrix} \begin{pmatrix} \bar{p} & 0 \\ 0 & D^* \end{pmatrix}\right)\right) \\ & = g^{-1}\left(\begin{pmatrix} \frac{1}{2}\mathrm{tr}(M) & i\boldsymbol{a} \\ i\boldsymbol{a}^* & M - \frac{1}{2}\mathrm{tr}(M)E \end{pmatrix} \begin{pmatrix} \bar{p} & 0 \\ 0 & D^* \end{pmatrix}\right) \\ & = DMD^* + p\boldsymbol{a}D^* = \beta(M+\boldsymbol{a}). \end{aligned}$                 | $\begin{aligned} & \varphi(B)(M+\boldsymbol{a}) = g^{-1}(Bg(M+\boldsymbol{a})B^*) \\ & = g^{-1}\left(\begin{pmatrix} p & 0 \\ 0 & D \end{pmatrix} \left(\begin{pmatrix} \frac{1}{2}\mathrm{tr}(M) & i\boldsymbol{a} \\ i\boldsymbol{a}^* & M - \frac{1}{2}\mathrm{tr}(M)E \end{pmatrix} \begin{pmatrix} \bar{p} & 0 \\ 0 & D^* \end{pmatrix}\right)\right) \\ & = g^{-1}\left(\begin{pmatrix} \frac{1}{2}\mathrm{tr}(M) & i\boldsymbol{a} \\ i\boldsymbol{a}^* & M - \frac{1}{2}\mathrm{tr}(M)E \end{pmatrix} \begin{pmatrix} \bar{p} & 0 \\ 0 & D^* \end{pmatrix}\right) \\ & = DMD^* + p\boldsymbol{a}D^* = \beta(M+\boldsymbol{a}). \end{aligned}$ | $\begin{aligned} & \varphi(B)(M+\boldsymbol{a}) = g^{-1}(Bg(M+\boldsymbol{a})B^*) \\ & = g^{-1}\left(\begin{pmatrix} p & 0 \\ 0 & D \end{pmatrix} \left(\begin{pmatrix} \frac{1}{2}\mathrm{tr}(M) & i\boldsymbol{a} \\ i\boldsymbol{a}^* & M - \frac{1}{2}\mathrm{tr}(M)E \end{pmatrix} \begin{pmatrix} \bar{p} & 0 \\ 0 & D^* \end{pmatrix}\right)\right) \\ & = g^{-1}\left(\begin{pmatrix} \frac{1}{2}\mathrm{tr}(M) & i\boldsymbol{a} \\ i\boldsymbol{a}^* & M - \frac{1}{2}\mathrm{tr}(M)E \end{pmatrix} \begin{pmatrix} \bar{p} & 0 \\ 0 & D^* \end{pmatrix}\right) \\ & = DMD^* + p\boldsymbol{a}D^* = \beta(M+\boldsymbol{a}). \end{aligned}$ |

| Identifier       | Page | Conf. | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Rendered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Image                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|------------------|------|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0902.0431_EQ0108 | 21   | 0.795 | $\begin{aligned} (\alpha x)(\alpha y) &= (a+A\boldsymbol{m})(b+A\boldsymbol{n}) \quad \&= (a\boldsymbol{b}-\langle A\boldsymbol{m}, A\boldsymbol{n} \rangle + (a\boldsymbol{A}\boldsymbol{m} + \bar{b}A\boldsymbol{m} - \overline{A\boldsymbol{m} \times A\boldsymbol{n}}) \\ &= (ab - \langle \boldsymbol{m}, A^*A\boldsymbol{n} \rangle) + (aA\boldsymbol{n} + \bar{b}A\boldsymbol{m} - \overline{\tilde{A}(\boldsymbol{m} \times \boldsymbol{n})}) \\ &= (ab - \langle \boldsymbol{m}, \boldsymbol{n} \rangle) + A(a\boldsymbol{n} + \bar{b}\boldsymbol{m} - \overline{\boldsymbol{m} \times \boldsymbol{n}}) \\ &= \varphi(A)((a + \boldsymbol{m})(b + \boldsymbol{n})) = \alpha(xy) \end{aligned}$ | $\begin{aligned} (\alpha x)(\alpha y) &= (a + A\boldsymbol{m})(b + A\boldsymbol{n}) \\ &= (ab - \langle A\boldsymbol{m}, A\boldsymbol{n} \rangle) + (aA\boldsymbol{n} + \bar{b}A\boldsymbol{m} - \overline{A\boldsymbol{m} \times A\boldsymbol{n}}) \\ &= (ab - \langle \boldsymbol{m}, A^*A\boldsymbol{n} \rangle) + (aA\boldsymbol{n} + \bar{b}A\boldsymbol{m} - \overline{\tilde{A}(\boldsymbol{m} \times \boldsymbol{n})}) \\ &= (ab - \langle \boldsymbol{m}, \boldsymbol{n} \rangle) + A(a\boldsymbol{n} + \bar{b}\boldsymbol{m} - \overline{\boldsymbol{m} \times \boldsymbol{n}}) \\ &= \varphi(A)((a + \boldsymbol{m})(b + \boldsymbol{n})) = \alpha(xy) \end{aligned}$ | $\begin{aligned} (\alpha x)(\alpha y) &= (a + A\boldsymbol{m})(b + A\boldsymbol{n}) \\ &= (ab - \langle A\boldsymbol{m}, A\boldsymbol{n} \rangle) + (aA\boldsymbol{n} + \bar{b}A\boldsymbol{m} - \overline{A\boldsymbol{m} \times A\boldsymbol{n}}) \\ &= (ab - \langle \boldsymbol{m}, A^*A\boldsymbol{n} \rangle) + (aA\boldsymbol{n} + \bar{b}A\boldsymbol{m} - \overline{\tilde{A}(\boldsymbol{m} \times \boldsymbol{n})}) \\ &= (ab - \langle \boldsymbol{m}, \boldsymbol{n} \rangle) + A(a\boldsymbol{n} + \bar{b}\boldsymbol{m} - \overline{\boldsymbol{m} \times \boldsymbol{n}}) \\ &= \varphi(A)((a + \boldsymbol{m})(b + \boldsymbol{n})) = \alpha(xy). \end{aligned}$ |
| 0902.0431_EQ0882 | 146  | 0.798 | $\alpha P = (X, Y, \xi, \eta), \quad X, Y \quad \text{are diagonal}, \quad \xi > 0.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | $\alpha P = (X, Y, \xi, \eta), \quad X, Y \quad \text{are diagonal}, \quad \xi > 0.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | $\alpha P = (X, Y, \xi, \eta), \quad X, Y \quad \text{are diagonal}, \quad \xi > 0.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 0902.0431_EQ0695 | 115  | 0.803 | $\Phi = \Phi_{-3} + \Phi_{-2} + \Phi_{-1} + \Phi_0 + \Phi_1 + \Phi_2 + \Phi_3, \quad \Phi_i \in \mathfrak{e}_7^C$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | $\Phi = \Phi_{-3} + \Phi_{-2} + \Phi_{-1} + \Phi_0 + \Phi_1 + \Phi_2 + \Phi_3, \quad \Phi_i \in \mathfrak{e}_7^C$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | $\Phi = \Phi_{-3} + \Phi_{-2} + \Phi_{-1} + \Phi_0 + \Phi_1 + \Phi_2 + \Phi_3, \quad \Phi_i \in \mathfrak{e}_7^C,$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 0902.0431_EQ0931 | 154  | 0.803 | $\epsilon A = \overline{(\text{Ad } J_3) A}, \quad J_3 = \begin{pmatrix} 0 & E \\ -E & 0 \end{pmatrix},$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | $\epsilon A = \overline{(\text{Ad } J_3) A}, \quad J_3 = \begin{pmatrix} 0 & E \\ -E & 0 \end{pmatrix},$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | $\epsilon A = \overline{(\text{Ad } J_3) A}, \quad J_3 = \begin{pmatrix} 0 & E \\ -E & 0 \end{pmatrix},$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 0902.0431_EQ0666 | 110  | 0.804 | $\begin{aligned} E_{6(6)} &\simeq \text{Sp}(4)/\boldsymbol{Z}_2 \times \boldsymbol{R}^{42}, \\ E_{6(2)} &\simeq (\text{Sp}(1) \times \text{SU}(6))/\boldsymbol{Z}_2 \times \boldsymbol{R}^{40}, \\ E_{6(-14)} &\simeq (U(1) \times \text{Spin}(10))/\boldsymbol{Z}_4 \times \boldsymbol{R}^{32}, \\ E_{6(-26)} &\simeq F_4 \times \boldsymbol{R}^{26}. \end{aligned}$                                                                                                                                                                                                                                                                                                                                   | $\begin{aligned} E_{6(6)} &\simeq \text{Sp}(4)/\boldsymbol{Z}_2 \times \boldsymbol{R}^{42}, \\ E_{6(2)} &\simeq (\text{Sp}(1) \times \text{SU}(6))/\boldsymbol{Z}_2 \times \boldsymbol{R}^{40}, \\ E_{6(-14)} &\simeq (U(1) \times \text{Spin}(10))/\boldsymbol{Z}_4 \times \boldsymbol{R}^{32}, \\ E_{6(-26)} &\simeq F_4 \times \boldsymbol{R}^{26}. \end{aligned}$                                                                                                                                                                                                                                                                                                            | $\begin{aligned} E_{6(6)} &\simeq \text{Sp}(4)/\boldsymbol{Z}_2 \times \boldsymbol{R}^{42}, \\ E_{6(2)} &\simeq (\text{Sp}(1) \times \text{SU}(6))/\boldsymbol{Z}_2 \times \boldsymbol{R}^{40}, \\ E_{6(-14)} &\simeq (U(1) \times \text{Spin}(10))/\boldsymbol{Z}_4 \times \boldsymbol{R}^{32}, \\ E_{6(-26)} &\simeq F_4 \times \boldsymbol{R}^{26}. \end{aligned}$                                                                                                                                                                                                                                                                                                             |
| 0902.0431_EQ1192 | 200  | 0.804 | $\begin{aligned} (a \times b)c &= -(a_1 \wedge c_1, b)a_1 \wedge c_2 + (a_1 \wedge c_2, b)a_2 \wedge c_1 \\ &\quad + (a_2 \wedge c_1, b)a_1 \wedge c_2 - (a_2 \wedge c_2, b)a_1 \wedge c_1 + \frac{4}{5}(a, b)c. \end{aligned}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | $\begin{aligned} (a \times b)c &= -(a_1 \wedge c_1, b)a_1 \wedge c_2 + (a_1 \wedge c_2, b)a_2 \wedge c_1 \\ &\quad + (a_2 \wedge c_1, b)a_1 \wedge c_2 - (a_2 \wedge c_2, b)a_1 \wedge c_1 + \frac{4}{5}(a, b)c. \end{aligned}$                                                                                                                                                                                                                                                                                                                                                                                                                                                  | $\begin{aligned} (a \times b)c &= -(a_1 \wedge c_1, b)a_1 \wedge c_2 + (a_1 \wedge c_2, b)a_2 \wedge c_1 \\ &\quad + (a_2 \wedge c_1, b)a_1 \wedge c_2 - (a_2 \wedge c_2, b)a_1 \wedge c_1 + \frac{4}{5}(a, b)c. \end{aligned}$                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

| Identifier       | Page | Conf.   | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Rendered                                                                                                                                                                                                                                                                                                                                                                                                                                   | Image                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|------------------|------|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0902.0431_E00059 | 14   | ■ 0.807 | $\begin{array}{c} \backslash\mathrm{begin}\{\mathrm{array}\}\{\mathrm{c}ll\} \backslash\mathrm{lambda} G_{23}+\backslash\mathrm{mu} G_{45}+\backslash\mathrm{nu} G_{67}, \\ \& -\backslash\mathrm{lambda} G_{13}-\backslash\mathrm{mu} G_{46}+\backslash\mathrm{nu} G_{57}, \& \backslash\backslash \backslash\mathrm{lambda} \\ G_{12}+\backslash\mathrm{mu} G_{47}+\backslash\mathrm{nu} G_{56}, \& -\backslash\mathrm{lambda} G_{15}+\backslash\mathrm{mu} \\ G_{26}-\backslash\mathrm{nu} G_{37}, \& \backslash\mathrm{lambda}, \backslash\mathrm{mu}, \backslash\mathrm{nu} \in \backslash\mathrm{boldsymbol}\{R\} \\ \backslash\backslash \backslash\mathrm{lambda} G_{14}-\backslash\mathrm{mu} G_{27}-\backslash\mathrm{nu} G_{36}, \& -\backslash\mathrm{lambda} \\ G_{17}-\backslash\mathrm{mu} G_{24}+\backslash\mathrm{nu} G_{35}, \& \backslash\mathrm{lambda}+\backslash\mathrm{mu}+\backslash\mathrm{nu}=0 \backslash\backslash \& \\ \backslash\mathrm{lambda} G_{16}+\backslash\mathrm{mu} G_{25}+\backslash\mathrm{nu} G_{34} \end{array}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | $\begin{array}{c} \lambda G_{23} + \mu G_{45} + \nu G_{67}, \quad -\lambda G_{13} - \mu G_{46} + \nu G_{57}, \\ \lambda G_{12} + \mu G_{47} + \nu G_{56}, \quad -\lambda G_{15} + \mu G_{26} - \nu G_{37}, \quad \lambda, \mu, \nu \in \boldsymbol{R} \\ \lambda G_{14} - \mu G_{27} - \nu G_{36}, \quad -\lambda G_{17} - \mu G_{24} + \nu G_{35}, \quad \lambda + \mu + \nu = 0 \\ \lambda G_{16} + \mu G_{25} + \nu G_{34} \end{array}$ | $\begin{array}{c} \lambda G_{23} + \mu G_{45} + \nu G_{67}, \quad -\lambda G_{13} - \mu G_{46} + \nu G_{57}, \\ \lambda G_{12} + \mu G_{47} + \nu G_{56}, \quad -\lambda G_{15} + \mu G_{26} - \nu G_{37}, \quad \lambda, \mu, \nu \in \boldsymbol{R} \\ \lambda G_{14} - \mu G_{27} - \nu G_{36}, \quad -\lambda G_{17} - \mu G_{24} + \nu G_{35}, \quad \lambda + \mu + \nu = 0 \\ \lambda G_{16} + \mu G_{25} + \nu G_{34}. \end{array}$ |
| 0902.0431_E01023 | 171  | ■ 0.807 | $\begin{array}{c} \backslash\mathrm{begin}\{\mathrm{aligned}\} \& \{[(\backslash\mathrm{Phi}(h), \ 0,0, \ r, \\ 0), (0,0,0,0,1,0)]= (0,0,0,0,2 \ r, \ 0)\} \backslash\backslash \& \{[(\backslash\mathrm{Phi}(h), \ 0,0, \\ r, \ 0), (0,0,0,0,0,1)]= (0,0,0,0,0, -2 \ r),\} \end{array}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | $\begin{array}{c} [(\Phi(h), 0, 0, r, 0), (0, 0, 0, 0, 1, 0)] = (0, 0, 0, 0, 2r, 0) \\ [(\Phi(h), 0, 0, r, 0), (0, 0, 0, 0, 0, 1)] = (0, 0, 0, 0, 0, -2r), \end{array}$                                                                                                                                                                                                                                                                    | $\begin{array}{c} [(\Phi(h), 0, 0, r, 0), (0, 0, 0, 0, 1, 0)] = (0, 0, 0, 0, 2r, 0), \\ [(\Phi(h), 0, 0, r, 0), (0, 0, 0, 0, 0, 1)] = (0, 0, 0, 0, 0, -2r), \end{array}$                                                                                                                                                                                                                                                                    |
| 0902.0431_E00628 | 104  | ■ 0.808 | $\begin{array}{c} \backslash\mathrm{begin}\{\mathrm{aligned}\} \backslash\mathrm{varphi}(A) \ E \& =\mathrm{g}^{\wedge\{-1\}}\backslash\mathrm{left}(A(\mathrm{g} \ E) \ A^{\wedge\{*\}} \\ \backslash\mathrm{right})=\mathrm{g}^{\wedge\{-1\}}\backslash\mathrm{left}(A\backslash\mathrm{left}(2 \ E_{11}-\backslash\mathrm{frac}\{1\}\{2\} \ E\backslash\mathrm{right}) \\ A^{\wedge\{*\}}\backslash\mathrm{right})=\mathrm{g}^{\wedge\{-1\}}\backslash\mathrm{left}(2 \ A \ E_{11} \ A^{\wedge\{*\}}-\backslash\mathrm{frac}\{1\}\{2\} \ E \\ \backslash\mathrm{right}) \backslash\backslash \& =\mathrm{g}^{\wedge\{-1\}}(g(\backslash\mathrm{alpha} \ E))=\backslash\mathrm{alpha} \ E \end{array}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | $\begin{array}{c} \varphi(A)E = g^{-1}(A(gE)A^*) = g^{-1}\left(A\left(2E_1 - \frac{1}{2}E\right)A^*\right) = g^{-1}\left(2AE_1A^* - \frac{1}{2}E\right) \\ = g^{-1}(g(\alpha E)) = \alpha E \end{array}$                                                                                                                                                                                                                                   | $\begin{array}{c} \varphi(A)E = g^{-1}(A(gE)A^*) = g^{-1}\left(A\left(2E_1 - \frac{1}{2}E\right)A^*\right) = g^{-1}\left(2AE_1A^* - \frac{1}{2}E\right) \\ = g^{-1}(g(\alpha E)) = \alpha E. \end{array}$                                                                                                                                                                                                                                   |
| 0902.0431_E00333 | 53   | ■ 0.813 | $\begin{array}{c} \backslash\mathrm{mathfrak}\{f\}_{4}\{ \ }^{\wedge\{C\}}=\backslash\mathrm{mathfrak}\{d\}_{4}\{ \ }^{\wedge\{C\}} \\ \backslash\mathrm{oplus} \backslash\mathrm{widetilde}\{\backslash\mathrm{mathfrak}\{A\}\}_{1}\{ \ }^{\wedge\{C\}} \backslash\mathrm{oplus} \\ \backslash\mathrm{widetilde}\{\backslash\mathrm{mathfrak}\{A\}\}_{2}\{ \ }^{\wedge\{C\}} \backslash\mathrm{oplus} \backslash\mathrm{widetilde}\{ \\ \backslash\mathrm{mathfrak}\{A\}\}_{3}\{ \ }^{\wedge\{C\}} \end{array}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | $\mathfrak{f}_4^C = \mathfrak{d}_4^C \oplus \widetilde{\mathfrak{A}}_1^C \oplus \widetilde{\mathfrak{A}}_2^C \oplus \widetilde{\mathfrak{A}}_3^C.$                                                                                                                                                                                                                                                                                         | $\mathfrak{f}_4^C = \mathfrak{d}_4^C \oplus \widetilde{\mathfrak{A}}_1^C \oplus \widetilde{\mathfrak{A}}_2^C \oplus \widetilde{\mathfrak{A}}_3^C.$                                                                                                                                                                                                                                                                                          |
| 0902.0431_E00301 | 48   | ■ 0.817 | $\begin{array}{c} \backslash\mathrm{mathfrak}\{a\} \ \backslash\mathrm{supset}\backslash\mathrm{left}[\backslash\mathrm{mathfrak}\{a\}, \backslash\mathrm{mathfrak}\{f\}_{4}\{ \\ \ }^{\wedge\{C\}}\backslash\mathrm{right}] \backslash\mathrm{supset}\backslash\mathrm{left}[\backslash\mathrm{mathfrak}\{d\}_{4}\{ \ }^{\wedge\{C\}}, \\ \backslash\mathrm{widetilde}\{\backslash\mathrm{mathfrak}\{A\}\}_{i}\{ \ }^{\wedge\{C\}}\backslash\mathrm{right}]=\backslash\mathrm{widetilde}\{ \\ \backslash\mathrm{mathfrak}\{A\}\}_{i}\{ \ }^{\wedge\{C\}}, \quad \backslash\mathrm{quad} \ i=1,2,3 \end{array}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | $\mathfrak{a} \supset [\mathfrak{a}, \mathfrak{f}_4^C] \supset [\mathfrak{d}_4^C, \widetilde{\mathfrak{A}}_i^C] = \widetilde{\mathfrak{A}}_i^C, \quad i = 1, 2, 3.$                                                                                                                                                                                                                                                                        | $\mathfrak{a} \supset [\mathfrak{a}, \mathfrak{f}_4^C] \supset [\mathfrak{d}_4^C, \widetilde{\mathfrak{A}}_i^C] = \widetilde{\mathfrak{A}}_i^C, \quad i = 1, 2, 3.$                                                                                                                                                                                                                                                                         |
| 0902.0431_E00315 | 50   | ■ 0.827 | $\begin{array}{c} (\backslash\mathrm{delta},[\backslash\mathrm{widetilde}\{A\}, \backslash\mathrm{widetilde}\{B\}])_{4}=(\backslash\mathrm{delta} \ A, \\ B), \quad \backslash\mathrm{quad} \ \backslash\mathrm{delta} \ \in \backslash\mathrm{mathfrak}\{f\}_{4}\{ \ }^{\wedge\{C\}}, \quad \backslash\mathrm{quad} \ A, \ B \\ \in \backslash\mathrm{mathfrak}\{J\}^{\wedge\{C\}} \end{array}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | $(\delta, [\widetilde{A}, \widetilde{B}])_4 = (\delta A, B), \quad \delta \in \mathfrak{f}_4^C, \quad A, B \in \mathfrak{J}^C.$                                                                                                                                                                                                                                                                                                            | $(\delta, [\widetilde{A}, \widetilde{B}])_4 = (\delta A, B), \quad \delta \in \mathfrak{f}_4^C, \quad A, B \in \mathfrak{J}^C.$                                                                                                                                                                                                                                                                                                             |
| 0902.0431_E00749 | 124  | ■ 0.827 | $\begin{array}{c} \backslash\mathrm{mathfrak}\{h\}=\backslash\mathrm{left}\{\backslash\mathrm{Phi}\backslash\mathrm{left}(\backslash\mathrm{sum}_{k=0}^{\wedge\{3\}} \backslash\mathrm{lambda}_{k} \\ H_{k}+\backslash\mathrm{left}(\backslash\mathrm{sum}_{j=1}^{\wedge\{3\}} \backslash\mathrm{mu}_{j} \ E_{j}\backslash\mathrm{right})^{\wedge\{\mathrm{sim}\}}, \\ 0,0, \ \backslash\mathrm{nu}\backslash\mathrm{right}) \in \backslash\mathrm{mathfrak}\{e\}_{7}\{ \ }^{\wedge\{C\}} \backslash\mathrm{left}\backslash\mathrm{vert}\backslash, \\ \backslash\mathrm{begin}\{\mathrm{array}\}\{l\} \backslash\mathrm{lambda}_{k}, \ \backslash\mathrm{nu} \in C \backslash\backslash \backslash\mathrm{mu}_{jj} \ \in C, \\ \backslash\mathrm{mu}_{1}+\backslash\mathrm{mu}_{2}+\backslash\mathrm{mu}_{3}=0 \end{array}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | $\mathfrak{h} = \left\{ \Phi \left( \sum_{k=0}^3 \lambda_k H_k + \left( \sum_{j=1}^3 \mu_j E_j \right)^{\sim}, 0, 0, \nu \right) \in \mathfrak{e}_7^C \mid \begin{array}{l} \lambda_k, \nu \in C \\ \mu_j \in C, \mu_1 + \mu_2 + \mu_3 = 0 \end{array} \right\}$                                                                                                                                                                           | $\mathfrak{h} = \left\{ \Phi \left( \sum_{k=0}^3 \lambda_k H_k + \left( \sum_{j=1}^3 \mu_j E_j \right)^{\sim}, 0, 0, \nu \right) \in \mathfrak{e}_7^C \mid \begin{array}{l} \lambda_k, \nu \in C \\ \mu_j \in C, \mu_1 + \mu_2 + \mu_3 = 0 \end{array} \right\}$                                                                                                                                                                            |
| 0902.0431_E01076 | 183  | ■ 0.830 | $\begin{array}{c} \backslash\mathrm{begin}\{\mathrm{aligned}\} \& \mathrm{g}\backslash\mathrm{left}(2\backslash\mathrm{left}(X_{11} \ \vee \ \mathrm{gamma} \ X_{22} \\ \backslash\mathrm{right}) \ X\backslash\mathrm{right}) \quad \backslash\mathrm{quad} \ X \in \backslash\mathrm{mathfrak}\{J\}^{\wedge\{C\}} \backslash\backslash \& =\mathrm{g} \\ \backslash\mathrm{left}(\backslash\mathrm{left}(\backslash\mathrm{gamma} \ X_{22}, \ X\backslash\mathrm{right}) \ X_{11}+\backslash\mathrm{frac}\{1\}\{3\} \\ \backslash\mathrm{left}(X_{11}, \ \mathrm{gamma} \ X_{22}\backslash\mathrm{right}) \ X-4 \ \mathrm{gamma} \ X_{22} \ \backslash\mathrm{times} \\ \backslash\mathrm{left}(X_{11} \ \backslash\mathrm{times} \ X\backslash\mathrm{right})\backslash\mathrm{right})(\backslash\mathrm{text} \ \{ \ \mathrm{Lemma} \ 3.4.1 \} ) \\ \backslash\backslash \& =\backslash\mathrm{left}(\backslash\mathrm{gamma} \ X_{22}, \ X\backslash\mathrm{right}) \ \mathrm{g} \ X_{11}+\backslash\mathrm{frac}\{1\}\{3\} \\ \backslash\mathrm{left}(X_{11}, \ \mathrm{gamma} \ X_{22}\backslash\mathrm{right}) \ \mathrm{g} \ X-4 \ \mathrm{g} \ X_{22} \ \backslash\mathrm{circ} \\ \backslash\mathrm{left}(\mathrm{g} \ X_{11} \ \backslash\mathrm{circ} \ \mathrm{g} \ X\backslash\mathrm{right})+\backslash\mathrm{left}(\backslash\mathrm{gamma} \ X_{11}, \ X \\ \backslash\mathrm{right}) \ \mathrm{g} \ X_{22} \backslash\backslash \& \quad \backslash\mathrm{quad}+\backslash\mathrm{left}(\backslash\mathrm{gamma} \ X_{22}, \ \mathrm{gamma} \\ X_{11}, \ \mathrm{gamma} \ X\backslash\mathrm{right}) \ E(\backslash\mathrm{text} \ \{ \ \mathrm{Lemma} \ 3.12.1 \} ), \\ \end{array}$ | $\begin{array}{c} g(2(X_1 \vee \gamma X_2)X) \quad X \in \mathfrak{J}^C \\ = g\left((\gamma X_2, X)X_1 + \frac{1}{3}(X_1, \gamma X_2)X - 4\gamma X_2 \times (X_1 \times X)\right) (\text{Lemma 3.4.1}) \\ = (\gamma X_2, X)gX_1 + \frac{1}{3}(X_1, \gamma X_2)gX - 4gX_2 \circ (gX_1 \circ gX) + (\gamma X_1, X)gX_2 \\ + (\gamma X_2, \gamma X_1, \gamma X)E (\text{Lemma 3.12.1}), \end{array}$                                          | $\begin{array}{c} g(2(X_1 \vee \gamma X_2)X) \quad X \in \mathfrak{J}^C \\ = g((\gamma X_2, X)X_1 + \frac{1}{3}(X_1, \gamma X_2)X - 4\gamma X_2 \times (X_1 \times X)) (\text{Lemma 3.4.1}) \\ = (\gamma X_2, X)gX_1 + \frac{1}{3}(X_1, \gamma X_2)gX - 4gX_2 \circ (gX_1 \circ gX) + (\gamma X_1, X)gX_2 \\ + (\gamma X_2, \gamma X_1, \gamma X)E (\text{Lemma 3.12.1}), \end{array}$                                                      |

| Identifier       | Page | Conf.  | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Rendered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Image                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|------------------|------|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0902.0431_E00963 | 160  | ■0.831 | $\begin{aligned} & \& G_{01}, \quad \quad G_{23}, \quad \quad G_{45}, \quad \quad G_{67}, \quad \quad G_{46}+G_{47}, \quad \quad G_{47}-G_{56}, \\ & G_{24}+G_{35}, \quad \quad G_{25}-G_{34}, \quad \quad G_{26}+G_{37}, \quad \quad G_{27}-G_{36}, \\ & \tilde{A}_l(1), \quad \tilde{A}_l(e_1), \quad \tilde{F}_l(1), \quad \tilde{F}_l(e_1), \quad (E_1-E_2)^\sim, \quad (E_2-E_3)^\sim \\ & \check{F}_l(1), \quad \check{F}_l(e_1), \quad \hat{F}_l(1), \quad \hat{F}_l(e_1), \quad \hat{E}_l, \quad \hat{E}_l, \quad 1, \quad l=1,2,3 \end{aligned}$                                                                                                                                                                                       | $\begin{aligned} & G_{01}, \quad G_{23}, \quad G_{45}, \quad G_{67}, \quad G_{46}+G_{47}, \quad G_{47}-G_{56}, \\ & G_{24}+G_{35}, \quad G_{25}-G_{34}, \quad G_{26}+G_{37}, \quad G_{27}-G_{36}, \\ & \tilde{A}_l(1), \quad \tilde{A}_l(e_1), \quad \tilde{F}_l(1), \quad \tilde{F}_l(e_1), \quad (E_1-E_2)^\sim, \quad (E_2-E_3)^\sim \\ & \check{F}_l(1), \quad \check{F}_l(e_1), \quad \hat{F}_l(1), \quad \hat{F}_l(e_1), \quad \hat{E}_l, \quad \hat{E}_l, \quad 1, \quad l=1,2,3 \end{aligned}$                                                                                                                                                                                                                                          | $\begin{aligned} & G_{01}, \quad G_{23}, \quad G_{45}, \quad G_{67}, \quad G_{46}+G_{47}, \quad G_{47}-G_{56}, \\ & G_{24}+G_{35}, \quad G_{25}-G_{34}, \quad G_{26}+G_{37}, \quad G_{27}-G_{36}, \\ & \tilde{A}_l(1), \quad \tilde{A}_l(e_1), \quad \tilde{F}_l(1), \quad \tilde{F}_l(e_1), \quad (E_1-E_2)^\sim, \quad (E_2-E_3)^\sim \\ & \check{F}_l(1), \quad \check{F}_l(e_1), \quad \hat{F}_l(1), \quad \hat{F}_l(e_1), \quad \hat{E}_l, \quad \hat{E}_l, \quad 1, \quad l=1,2,3 \end{aligned}$                                                                                                                                                                                                                                          |
| 0902.0431_E00573 | 96   | ■0.831 | $\mathfrak{S}(6,C)=\left\{S\in M(6,C)\mid \right. S=-S\left.\right\}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | $\mathfrak{S}(6,C)=\left\{S\in M(6,C)\mid {}^tS=-S\right\}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | $\mathfrak{S}(6,C)=\left\{S\in M(6,C)\mid {}^tS=-S\right\}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 0902.0431_E00714 | 118  | ■0.833 | $\left[\begin{array}{c} \lambda\Phi(\phi,A,B,\nu)\lambda^{-1}=\Phi\left(-{}^t\phi,-B,-A,-\nu\right). \end{array}\right]$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | $\lambda\Phi(\phi,A,B,\nu)\lambda^{-1}=\Phi\left(-{}^t\phi,-B,-A,-\nu\right).$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | $\lambda\Phi(\phi,A,B,\nu)\lambda^{-1}=\Phi\left(-{}^t\phi,-B,-A,-\nu\right).$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 0902.0431_E01088 | 185  | ■0.835 | $\begin{aligned} & \lambda\gamma P_1\times P_2-\lambda\gamma P_2\times P_1 \\ & =\Phi\left(\begin{array}{c} \frac{1}{2}(-X_1\vee\gamma X_2+X_2\vee\gamma X_1-\gamma Y_1\vee Y_2+\gamma Y_2\vee Y_1) \\ \frac{1}{4}(2\gamma X_1\times Y_2-2\gamma X_2\times Y_1+\eta_1X_2-\eta_2X_1+\xi_2\gamma Y_1-\xi_1\gamma Y_2) \\ \frac{1}{4}(2X_2\times\gamma Y_1-2X_1\times\gamma Y_2+\eta_2\gamma X_1-\eta_1\gamma X_2+\xi_1Y_2-\xi_2Y_1) \\ 0 \end{array}\right) \\ & =\Phi\left(\frac{1}{4}\varphi_*D,\frac{1}{4}A,-\frac{1}{4}\gamma A,0\right)\quad(\text{Lemma 5.8.2}) \\ & =\frac{1}{4}\varphi_*(kD+ik(gA)) \\ & =\frac{1}{4}\varphi_*\left(S_1\tau S_2-S_2\tau S_1-\frac{1}{8}\operatorname{tr}(S_1\tau S_2-S_2\tau S_1)E\right). \end{aligned}$ | $\begin{aligned} & \lambda\gamma P_1\times P_2-\lambda\gamma P_2\times P_1 \\ & =\Phi\left(\begin{array}{c} \frac{1}{2}(-X_1\vee\gamma X_2+X_2\vee\gamma X_1-\gamma Y_1\vee Y_2+\gamma Y_2\vee Y_1) \\ \frac{1}{4}(2\gamma X_1\times Y_2-2\gamma X_2\times Y_1+\eta_1X_2-\eta_2X_1+\xi_2\gamma Y_1-\xi_1\gamma Y_2) \\ \frac{1}{4}(2X_2\times\gamma Y_1-2X_1\times\gamma Y_2+\eta_2\gamma X_1-\eta_1\gamma X_2+\xi_1Y_2-\xi_2Y_1) \\ 0 \end{array}\right) \\ & =\Phi\left(\frac{1}{4}\varphi_*D,\frac{1}{4}A,-\frac{1}{4}\gamma A,0\right)\quad(\text{Lemma 5.8.2}) \\ & =\frac{1}{4}\varphi_*(kD+ik(gA)) \\ & =\frac{1}{4}\varphi_*\left(S_1\tau S_2-S_2\tau S_1-\frac{1}{8}\operatorname{tr}(S_1\tau S_2-S_2\tau S_1)E\right). \end{aligned}$ | $\begin{aligned} & \lambda\gamma P_1\times P_2-\lambda\gamma P_2\times P_1 \\ & =\Phi\left(\begin{array}{c} \frac{1}{2}(-X_1\vee\gamma X_2+X_2\vee\gamma X_1-\gamma Y_1\vee Y_2+\gamma Y_2\vee Y_1) \\ \frac{1}{4}(2\gamma X_1\times Y_2-2\gamma X_2\times Y_1+\eta_1X_2-\eta_2X_1+\xi_2\gamma Y_1-\xi_1\gamma Y_2) \\ \frac{1}{4}(2X_2\times\gamma Y_1-2X_1\times\gamma Y_2+\eta_2\gamma X_1-\eta_1\gamma X_2+\xi_1Y_2-\xi_2Y_1) \\ 0 \end{array}\right) \\ & =\Phi\left(\frac{1}{4}\varphi_*D,\frac{1}{4}A,-\frac{1}{4}\gamma A,0\right)\quad(\text{Lemma 5.8.2}) \\ & =\frac{1}{4}\varphi_*(kD+ik(gA)) \\ & =\frac{1}{4}\varphi_*\left(S_1\tau S_2-S_2\tau S_1-\frac{1}{8}\operatorname{tr}(S_1\tau S_2-S_2\tau S_1)E\right). \end{aligned}$ |
| 0902.0431_E00289 | 46   | ■0.835 | $D+\widetilde{A}=0, \quad D\in\mathfrak{d}_4, A\in\mathfrak{M}^-, \operatorname{diag} A=0, \text{ then } D=0, A=0.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | $D+\widetilde{A}=0, \quad D\in\mathfrak{d}_4, A\in\mathfrak{M}^-, \operatorname{diag} A=0, \text{ then } D=0, A=0.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | $D+\widetilde{A}=0, \quad D\in\mathfrak{d}_4, A\in\mathfrak{M}^-, \operatorname{diag} A=0, \text{ then } D=0, A=0.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

| Identifier       | Page | Conf.   | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Rendered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Image                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|------------------|------|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0902.0431_E00548 | 92   | ■ 0.835 | $\begin{aligned} 0 &= \left( \xi E_1 + \xi_2 E_2 + \xi_3 E_3 + F_1(x_1) \right) \times \left( \xi E_1 + \xi_2 E_2 + \xi_3 E_3 + F_1(x_1) \right) \\ &= (\xi_2 \xi_3 - (x_1, x_1)) E_1 + \xi \xi_3 E_2 + \xi \xi_2 E_3 - \xi F_1(x_1). \end{aligned}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | $0 = (\xi E_1 + \xi_2 E_2 + \xi_3 E_3 + F_1(x_1)) \times (\xi E_1 + \xi_2 E_2 + \xi_3 E_3 + F_1(x_1))$ $= (\xi_2 \xi_3 - (x_1, x_1)) E_1 + \xi \xi_3 E_2 + \xi \xi_2 E_3 - \xi F_1(x_1)$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | $0 = (\xi E_1 + \xi_2 E_2 + \xi_3 E_3 + F_1(x_1)) \times (\xi E_1 + \xi_2 E_2 + \xi_3 E_3 + F_1(x_1))$ $= (\xi_2 \xi_3 - (x_1, x_1)) E_1 + \xi \xi_3 E_2 + \xi \xi_2 E_3 - \xi F_1(x_1).$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 0902.0431_E00379 | 63   | ■ 0.836 | $\begin{aligned} \mathfrak{J}_{\sigma} &= \{X \in \mathfrak{J} \mid \sigma X = X\} = \left\{ \begin{pmatrix} \xi_1 & 0 & 0 \\ 0 & \xi_2 & x_1 \\ 0 & \bar{x}_1 & \xi_3 \end{pmatrix} \mid \xi_i \in \mathbf{R}, x_1 \in \mathfrak{C} \right\} \\ &= \{X \in \mathfrak{J} \mid E_1 \circ X = 0\} \oplus \mathfrak{C}_1 \text{ (where } \mathfrak{C}_1 = \{\xi E_1 \mid \xi \in \mathbf{R}\}) \\ &= \mathfrak{J}(2, \mathfrak{C}) \oplus \mathfrak{C}_1, \end{aligned}$ $\mathfrak{J}_{-\sigma} = \{X \in \mathfrak{J} \mid \sigma X = -X\} = \left\{ \begin{pmatrix} 0 & x_3 & \bar{x}_2 \\ \bar{x}_3 & 0 & 0 \\ x_2 & 0 & 0 \end{pmatrix} \mid x_i \in \mathfrak{C} \right\}$ $= \{X \in \mathfrak{J} \mid 2E_1 \circ X = X\}.$ | $\mathfrak{J}_{\sigma} = \{X \in \mathfrak{J} \mid \sigma X = X\} = \left\{ \begin{pmatrix} \xi_1 & 0 & 0 \\ 0 & \xi_2 & x_1 \\ 0 & \bar{x}_1 & \xi_3 \end{pmatrix} \mid \xi_i \in \mathbf{R}, x_1 \in \mathfrak{C} \right\}$ $= \{X \in \mathfrak{J} \mid E_1 \circ X = 0\} \oplus \mathfrak{C}_1 \text{ (where } \mathfrak{C}_1 = \{\xi E_1 \mid \xi \in \mathbf{R}\})$ $= \mathfrak{J}(2, \mathfrak{C}) \oplus \mathfrak{C}_1,$ $\mathfrak{J}_{-\sigma} = \{X \in \mathfrak{J} \mid \sigma X = -X\} = \left\{ \begin{pmatrix} 0 & x_3 & \bar{x}_2 \\ \bar{x}_3 & 0 & 0 \\ x_2 & 0 & 0 \end{pmatrix} \mid x_i \in \mathfrak{C} \right\}$ $= \{X \in \mathfrak{J} \mid 2E_1 \circ X = X\}.$ | $\mathfrak{J}_{\sigma} = \{X \in \mathfrak{J} \mid \sigma X = X\} = \left\{ \begin{pmatrix} \xi_1 & 0 & 0 \\ 0 & \xi_2 & x_1 \\ 0 & \bar{x}_1 & \xi_3 \end{pmatrix} \mid \xi_i \in \mathbf{R}, x_1 \in \mathfrak{C} \right\}$ $= \{X \in \mathfrak{J} \mid E_1 \circ X = 0\} \oplus \mathfrak{C}_1 \text{ (where } \mathfrak{C}_1 = \{\xi E_1 \mid \xi \in \mathbf{R}\})$ $= \mathfrak{J}(2, \mathfrak{C}) \oplus \mathfrak{C}_1,$ $\mathfrak{J}_{-\sigma} = \{X \in \mathfrak{J} \mid \sigma X = -X\} = \left\{ \begin{pmatrix} 0 & x_3 & \bar{x}_2 \\ \bar{x}_3 & 0 & 0 \\ x_2 & 0 & 0 \end{pmatrix} \mid x_i \in \mathfrak{C} \right\}$ $= \{X \in \mathfrak{J} \mid 2E_1 \circ X = X\}.$ |
| 0902.0431_E00189 | 32   | ■ 0.837 | $\begin{aligned} \alpha e_1((\alpha e_3)b_0) &= a_1(a_3b_0) = a_1(a_2(a_1b_0)) = -a_1(a_1(a_2b_0)) = a_2b_0 \\ &= \alpha(e_2)b_0 = \alpha(e_1e_3)b_0, \\ \alpha e_2((\alpha e_5)b_0) &= a_2(a_5b_0) = a_2(a_1(a_4b_0)) = -a_1(a_2(a_4b_0)) = -a_1(a_6b_0) \\ &= a_6(a_1b_0) = a_7b_0 = \alpha(e_7)b_0 = \alpha(e_2e_5)b_0. \end{aligned}$                                                                                                                                                                                                                                                                                                                                                                                       | $(\alpha e_1)((\alpha e_3)b_0) = a_1(a_3b_0) = a_1(a_2(a_1b_0)) = -a_1(a_1(a_2b_0)) = a_2b_0$ $= \alpha(e_2)b_0 = \alpha(e_1e_3)b_0,$ $(\alpha e_2)((\alpha e_5)b_0) = a_2(a_5b_0) = a_2(a_1(a_4b_0)) = -a_1(a_2(a_4b_0)) = -a_1(a_6b_0)$ $= a_6(a_1b_0) = a_7b_0 = \alpha(e_7)b_0 = \alpha(e_2e_5)b_0.$                                                                                                                                                                                                                                                                                                                                                                                     | $(\alpha e_1)((\alpha e_3)b_0) = a_1(a_3b_0) = a_1(a_2(a_1b_0)) = -a_1(a_1(a_2b_0)) = a_2b_0$ $= \alpha(e_2)b_0 = \alpha(e_1e_3)b_0,$ $(\alpha e_2)((\alpha e_5)b_0) = a_2(a_5b_0) = a_2(a_1(a_4b_0)) = -a_1(a_2(a_4b_0)) = -a_1(a_6b_0)$ $= a_6(a_1b_0) = a_7b_0 = \alpha(e_7)b_0 = \alpha(e_2e_5)b_0.$                                                                                                                                                                                                                                                                                                                                                                                     |
| 0902.0431_E00643 | 106  | ■ 0.840 | $\tau h(A, B) = h(B, A), \quad \overline{h(A, B)} = h(\bar{A}, \bar{B}), \quad h(A, B)^* = h(B^*, A^*).$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | $\tau h(A, B) = h(B, A), \quad \overline{h(A, B)} = h(\bar{A}, \bar{B}), \quad h(A, B)^* = h(B^*, A^*).$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | $\tau h(A, B) = h(B, A), \quad \overline{h(A, B)} = h(\bar{A}, \bar{B}), \quad h(A, B)^* = h(B^*, A^*).$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

| Identifier        | Page | Conf. | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Rendered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Image |
|-------------------|------|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 9902.0431_EQ06603 | 101  | 0.841 | $\begin{aligned} & \begin{aligned} & \left( M + iF(\boldsymbol{ae}_4) \mid M \in \mathfrak{J}(3, \boldsymbol{H}), \boldsymbol{a} \in \boldsymbol{H}^3 \right) \\ & = \mathfrak{J}_{\boldsymbol{H}} \oplus i\boldsymbol{H}^3, \\ & (\mathfrak{J}^C)_{-\tau\gamma} = \{X \in \mathfrak{J}^C \mid \tau\gamma X = -X\} \\ & = \left\{ i \begin{pmatrix} \mu_1 & m_3 & \bar{m}_2 \\ \bar{m}_3 & \mu_2 & m_1 \\ m_2 & \bar{m}_1 & \mu_3 \end{pmatrix} + \begin{pmatrix} 0 & a_3e_4 & -a_2e_4 \\ -a_3e_4 & 0 & a_1e_4 \\ a_2e_4 & -a_1e_4 & 0 \end{pmatrix} \middle  \begin{matrix} \mu_i \in \boldsymbol{R} \\ m_i, a_i \in \boldsymbol{H} \end{matrix} \right\} \\ & = \{iM + F(\boldsymbol{ae}_4) \mid M \in \mathfrak{J}(3, \boldsymbol{H}), \boldsymbol{a} \in \boldsymbol{H}^3\} \\ & = i\mathfrak{J}_{\boldsymbol{H}} \oplus \boldsymbol{H}^3, \end{aligned} \end{aligned}$ | $\begin{aligned} & = \{M + iF(\boldsymbol{ae}_4) \mid M \in \mathfrak{J}(3, \boldsymbol{H}), \boldsymbol{a} \in \boldsymbol{H}^3\} \\ & = \mathfrak{J}_{\boldsymbol{H}} \oplus i\boldsymbol{H}^3, \\ & (\mathfrak{J}^C)_{-\tau\gamma} = \{X \in \mathfrak{J}^C \mid \tau\gamma X = -X\} \\ & = \left\{ i \begin{pmatrix} \mu_1 & m_3 & \bar{m}_2 \\ \bar{m}_3 & \mu_2 & m_1 \\ m_2 & \bar{m}_1 & \mu_3 \end{pmatrix} + \begin{pmatrix} 0 & a_3e_4 & -a_2e_4 \\ -a_3e_4 & 0 & a_1e_4 \\ a_2e_4 & -a_1e_4 & 0 \end{pmatrix} \middle  \begin{matrix} \mu_i \in \boldsymbol{R} \\ m_i, a_i \in \boldsymbol{H} \end{matrix} \right\} \\ & = \{iM + F(\boldsymbol{ae}_4) \mid M \in \mathfrak{J}(3, \boldsymbol{H}), \boldsymbol{a} \in \boldsymbol{H}^3\} \\ & = i\mathfrak{J}_{\boldsymbol{H}} \oplus \boldsymbol{H}^3, \end{aligned}$ |       |
| 9902.0431_EQ06348 | 56   | 0.842 | $\begin{aligned} & \left( \alpha_1, \alpha_1 \right) = \left( \alpha_2, \alpha_2 \right) = \frac{1}{9}, \quad \left( \alpha_3, \alpha_3 \right) = \left( \alpha_4, \alpha_4 \right) = \frac{1}{18}, \\ & \left( \alpha_1, \alpha_2 \right) = -\frac{1}{18}, \quad \left( \alpha_2, \alpha_3 \right) = -\frac{1}{18}, \quad \left( \alpha_3, \alpha_4 \right) = -\frac{1}{36}, \\ & \left( \alpha_1, \alpha_3 \right) = \left( \alpha_1, \alpha_4 \right) = \left( \alpha_2, \alpha_4 \right) = 0, \\ & \left( -\mu, -\mu \right) = \frac{1}{9}, \quad \left( -\mu, \alpha_1 \right) = -\frac{1}{18}, \quad \left( -\mu, \alpha_i \right) = 0, \quad i = 2, 3, 4, \end{aligned}$                                                                                                                                                                                             | $\begin{aligned} & \left( \alpha_1, \alpha_1 \right) = \left( \alpha_2, \alpha_2 \right) = \frac{1}{9}, \quad \left( \alpha_3, \alpha_3 \right) = \left( \alpha_4, \alpha_4 \right) = \frac{1}{18} \\ & \left( \alpha_1, \alpha_2 \right) = -\frac{1}{18}, \quad \left( \alpha_2, \alpha_3 \right) = -\frac{1}{18}, \quad \left( \alpha_3, \alpha_4 \right) = -\frac{1}{36} \\ & \left( \alpha_1, \alpha_3 \right) = \left( \alpha_1, \alpha_4 \right) = \left( \alpha_2, \alpha_4 \right) = 0, \\ & \left( -\mu, -\mu \right) = \frac{1}{9}, \quad \left( -\mu, \alpha_1 \right) = -\frac{1}{18}, \quad \left( -\mu, \alpha_i \right) = 0, \quad i = 2, 3, 4 \end{aligned}$                                                                                                                                                       |       |
| 9902.0431_EQ0637  | 105  | 0.844 | $\begin{aligned} & \left( \phi X, X, X \right) = 0, \left( \phi X, Y \right) + \left( X, \phi Y \right) = 0 \\ & = \left\{ \delta + i\tilde{T} \mid \delta \in \mathfrak{f}_{4, \boldsymbol{C}}, T \in (\mathfrak{J}_{\boldsymbol{C}})_0 \right\} \end{aligned}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | $\begin{aligned} \mathfrak{e}_{6, \boldsymbol{C}} &= \left\{ \phi \in \text{Hom}_{\boldsymbol{C}} \left( (\mathfrak{J}_{\boldsymbol{C}})^C \right) \mid (\phi X, X, X) = 0, \langle \phi X, Y \rangle + \langle X, \phi Y \rangle = 0 \right\} \\ &= \left\{ \delta + i\tilde{T} \mid \delta \in \mathfrak{f}_{4, \boldsymbol{C}}, T \in (\mathfrak{J}_{\boldsymbol{C}})_0 \right\} \end{aligned}$                                                                                                                                                                                                                                                                                                                                                                                                                                 |       |

| Identifier           | Page | Conf.   | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                           | Rendered                                                                                                                                                                                                                                                                                                                                                                                        | Image                                                                                                                                                                                                                                                                                                                                                                                           |
|----------------------|------|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0902.0431_E00686     | 114  | ■ 0.844 | $\begin{aligned} \phi &= \left[ \phi_1, \phi_2 \right] + 2A_1 \vee B_2 - 2A_2 \vee B_1 \\ A &= \left( \phi_1 + \frac{2}{3}\nu_1 \right) A_2 - \left( \phi_2 + \frac{2}{3}\nu_2 \right) A_1 \\ B &= - \left( {}^t\phi_1 + \frac{2}{3}\nu_1 \right) B_2 + \left( {}^t\phi_2 + \frac{2}{3}\nu_2 \right) B_1 \\ \nu &= (A_1, B_2) - (B_1, A_2). \end{aligned}$                                                                             | $\begin{cases} \phi = [\phi_1, \phi_2] + 2A_1 \vee B_2 - 2A_2 \vee B_1 \\ A = \left( \phi_1 + \frac{2}{3}\nu_1 \right) A_2 - \left( \phi_2 + \frac{2}{3}\nu_2 \right) A_1 \\ B = - \left( {}^t\phi_1 + \frac{2}{3}\nu_1 \right) B_2 + \left( {}^t\phi_2 + \frac{2}{3}\nu_2 \right) B_1 \\ \nu = (A_1, B_2) - (B_1, A_2). \end{cases}$                                                           | $\begin{cases} \phi = [\phi_1, \phi_2] + 2A_1 \vee B_2 - 2A_2 \vee B_1 \\ A = \left( \phi_1 + \frac{2}{3}\nu_1 \right) A_2 - \left( \phi_2 + \frac{2}{3}\nu_2 \right) A_1 \\ B = - \left( {}^t\phi_1 + \frac{2}{3}\nu_1 \right) B_2 + \left( {}^t\phi_2 + \frac{2}{3}\nu_2 \right) B_1 \\ \nu = (A_1, B_2) - (B_1, A_2). \end{cases}$                                                           |
| 0902.0431_E00509     | 83   | ■ 0.847 | $\begin{aligned} &\left[ h, \widetilde{F}_{23}(a) \right] = \left[ h_\delta, \widetilde{F}_{23}(a) \right] + \left[ \widetilde{H}, \widetilde{F}_{23}(a) \right] \\ &= (h_\delta F_{23}(a))^\sim + \frac{1}{2} [H, F_{23}(a)]^\sim \quad (\text{Lemma 3.6.3}) \\ &= \widetilde{F}_{23}(h_\delta a) + \frac{1}{2} [H, F_{23}(a)]^\sim \\ &= \left( \lambda_k + \frac{1}{2}(\mu_2 - \mu_3) \right) \widetilde{F}_{23}(a), \end{aligned}$ | $\begin{aligned} [h, \widetilde{F}_{23}(a)] &= [h_\delta, \widetilde{F}_{23}(a)] + [\widetilde{H}, \widetilde{F}_{23}(a)] \\ &= (h_\delta F_{23}(a))^\sim + \frac{1}{2} [H, F_{23}(a)]^\sim \quad (\text{Lemma 3.6.3}) \\ &= \widetilde{F}_{23}(h_\delta a) + \frac{1}{2} [H, F_{23}(a)]^\sim \\ &= \left( \lambda_k + \frac{1}{2}(\mu_2 - \mu_3) \right) \widetilde{F}_{23}(a), \end{aligned}$ | $\begin{aligned} [h, \widetilde{F}_{23}(a)] &= [h_\delta, \widetilde{F}_{23}(a)] + [\widetilde{H}, \widetilde{F}_{23}(a)] \\ &= (h_\delta F_{23}(a))^\sim + \frac{1}{2} [H, F_{23}(a)]^\sim \quad (\text{Lemma 3.6.3}) \\ &= \widetilde{F}_{23}(h_\delta a) + \frac{1}{2} [H, F_{23}(a)]^\sim \\ &= \left( \lambda_k + \frac{1}{2}(\mu_2 - \mu_3) \right) \widetilde{F}_{23}(a), \end{aligned}$ |
| 0902.0431_E00091     | 18   | ■ 0.848 | $\begin{aligned} &\left[ H_1, [H_1, S_1] \right] = [H_1, S_2] = -S_1, \quad [H_1, [H_1, S_2]] = [H_1, -S_1] = -S_2, \\ &\left[ H_1, [H_1, S_3] \right] = [H_1, -S_4] = -S_3, \quad [H_1, [H_1, S_4]] = [H_1, S_3] = -S_4, \\ &\left[ H_1, [H_1, S_5] \right] = 0, \quad [H_1, [H_1, S_6]] = 0 \end{aligned}$                                                                                                                           | $\begin{aligned} [H_1, [H_1, S_1]] &= [H_1, S_2] = -S_1, & [H_1, [H_1, S_2]] &= [H_1, -S_1] = -S_2, \\ [H_1, [H_1, S_3]] &= [H_1, -S_4] = -S_3, & [H_1, [H_1, S_4]] &= [H_1, S_3] = -S_4, \\ [H_1, [H_1, S_5]] &= 0, & [H_1, [H_1, S_6]] &= 0 \end{aligned}$                                                                                                                                    | $\begin{aligned} [H_1, [H_1, S_1]] &= [H_1, S_2] = -S_1, & [H_1, [H_1, S_2]] &= [H_1, -S_1] = -S_2, \\ [H_1, [H_1, S_3]] &= [H_1, -S_4] = -S_3, & [H_1, [H_1, S_4]] &= [H_1, S_3] = -S_4, \\ [H_1, [H_1, S_5]] &= 0, & [H_1, [H_1, S_6]] &= 0 \end{aligned}$                                                                                                                                    |
| 0902.0431_E00683 (2) | 113  | ■ 0.848 | $\begin{aligned} &\left[ \alpha P, \alpha Q \right] \alpha P = \frac{8}{3} ((\alpha P \times \alpha P) \alpha Q - (\alpha P \times \alpha Q) \alpha P) \quad (\text{Lemma 4.1.1.(2)}) \\ &= \frac{8}{3} (\alpha(P \times P) Q - \alpha(P \times Q) P) = \{P, Q\} \alpha P. \end{aligned}$                                                                                                                                              | $\begin{aligned} \{\alpha P, \alpha Q\} \alpha P &= \frac{8}{3} ((\alpha P \times \alpha P) \alpha Q - (\alpha P \times \alpha Q) \alpha P) \quad (\text{Lemma 4.1.1.(2)}) \\ &= \frac{8}{3} (\alpha(P \times P) Q - \alpha(P \times Q) P) = \{P, Q\} \alpha P. \end{aligned}$                                                                                                                  | $\begin{aligned} \{\alpha P, \alpha Q\} \alpha P &= \frac{8}{3} ((\alpha P \times \alpha P) \alpha Q - (\alpha P \times \alpha Q) \alpha P) \quad (\text{Lemma 4.1.1.(2)}) \\ &= \frac{8}{3} (\alpha(P \times P) Q - \alpha(P \times Q) P) = \{P, Q\} \alpha P. \end{aligned}$                                                                                                                  |
| 0902.0431_E00423     | 69   | ■ 0.848 | $\beta X = B X B^* \quad \text{or} \quad \beta X = B \bar{X} B^*, \quad X \in \mathfrak{J}_C.$                                                                                                                                                                                                                                                                                                                                         | $\beta X = B X B^* \quad \text{or} \quad \beta X = B \bar{X} B^*, \quad X \in \mathfrak{J}_C.$                                                                                                                                                                                                                                                                                                  | $\beta X = B X B^* \quad \text{or} \quad \beta X = B \bar{X} B^*, \quad X \in \mathfrak{J}_C.$                                                                                                                                                                                                                                                                                                  |
| 0902.0431_E00995     | 165  | ■ 0.852 | $E_8^C = \{ \alpha \in \text{Iso}_C(\mathfrak{e}_8^C) \mid \alpha[R_1, R_2] = [\alpha R_1, \alpha R_2] \}.$                                                                                                                                                                                                                                                                                                                            | $E_8^C = \{ \alpha \in \text{Iso}_C(\mathfrak{e}_8^C) \mid \alpha[R_1, R_2] = [\alpha R_1, \alpha R_2] \}.$                                                                                                                                                                                                                                                                                     | $E_8^C = \{ \alpha \in \text{Iso}_C(\mathfrak{e}_8^C) \mid \alpha[R_1, R_2] = [\alpha R_1, \alpha R_2] \}.$                                                                                                                                                                                                                                                                                     |



| Identifier       | Page | Conf.   | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Rendered                                                                                                                                                                                                                                                                                                                                                                                                                   | Image                                                                                                                                                                                                                                                                                                                                                                                                                      |
|------------------|------|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0902.0431_E00772 | 130  | ■ 0.859 | $\begin{aligned} & \widetilde{\alpha} P \times \widetilde{\alpha} Q = (\alpha X, \tau \alpha \tau Y, \xi, \eta) \times (\alpha Z, \tau \alpha \tau W, \zeta, \omega) \\ & = \cdots \left( \text{using } \alpha(X \vee Y) \alpha^{-1} = \alpha X \vee \tau \alpha \tau Y, (\alpha \phi \alpha^{-1})' = (\tau \alpha \tau) \phi' (\tau \alpha^{-1} \tau) \text{ etc.} \right) \cdots \\ & = \widetilde{\alpha} (P \times Q) \widetilde{\alpha}^{-1} \end{aligned}$                                                                                                                                               | $\widetilde{\alpha} P \times \widetilde{\alpha} Q = (\alpha X, \tau \alpha \tau Y, \xi, \eta) \times (\alpha Z, \tau \alpha \tau W, \zeta, \omega)$ $= \cdots \left( \text{using } \alpha(X \vee Y) \alpha^{-1} = \alpha X \vee \tau \alpha \tau Y, (\alpha \phi \alpha^{-1})' = (\tau \alpha \tau) \phi' (\tau \alpha^{-1} \tau) \text{ etc.} \right) \cdots$ $= \widetilde{\alpha} (P \times Q) \widetilde{\alpha}^{-1}$ | $\widetilde{\alpha} P \times \widetilde{\alpha} Q = (\alpha X, \tau \alpha \tau Y, \xi, \eta) \times (\alpha Z, \tau \alpha \tau W, \zeta, \omega)$ $= \cdots \left( \text{using } \alpha(X \vee Y) \alpha^{-1} = \alpha X \vee \tau \alpha \tau Y, (\alpha \phi \alpha^{-1})' = (\tau \alpha \tau) \phi' (\tau \alpha^{-1} \tau) \text{ etc.} \right) \cdots$ $= \widetilde{\alpha} (P \times Q) \widetilde{\alpha}^{-1}$ |
| 0902.0431_E00908 | 151  | ■ 0.860 | $\alpha_1 \left( a_1 \right)^{-1} \alpha_2 \left( a_2 \right)^{-1} \alpha_3 \left( a_3 \right)^{-1} \beta \alpha \mathbf{i} = \mathbf{i},$                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | $\alpha_1 \left( a_1 \right)^{-1} \alpha_2 \left( a_2 \right)^{-1} \alpha_3 \left( a_3 \right)^{-1} \beta \alpha \mathbf{i} = \mathbf{i},$                                                                                                                                                                                                                                                                                 | $\alpha_1 \left( a_1 \right)^{-1} \alpha_2 \left( a_2 \right)^{-1} \alpha_3 \left( a_3 \right)^{-1} \beta \alpha \mathbf{i} = \mathbf{i},$                                                                                                                                                                                                                                                                                 |
| 0902.0431_E01078 | 183  | ■ 0.862 | $\begin{aligned} & \left[ \varphi_* \left( (S_1 \tau S_2 - S_2 \tau S_1) - \frac{1}{8} \text{tr} (S_1 \tau S_2 - S_2 \tau S_1) E \right) \right. \\ & \quad \left. = 4 \left( \lambda \gamma \chi^{-1} S_1 \times \chi^{-1} S_2 - \lambda \gamma \chi^{-1} S_2 \times \chi^{-1} S_1 \right) \right] \\ & \quad \left[ \varphi_* \left( (S_1 \tau S_2 - S_2 \tau S_1) - \frac{1}{8} \text{tr} (S_1 \tau S_2 - S_2 \tau S_1) E \right) \right. \\ & \quad \left. = 4 \left( \lambda \gamma \chi^{-1} S_1 \times \chi^{-1} S_2 - \lambda \gamma \chi^{-1} S_2 \times \chi^{-1} S_1 \right) \right] \end{aligned}$ | $\varphi_* \left( (S_1 \tau S_2 - S_2 \tau S_1) - \frac{1}{8} \text{tr} (S_1 \tau S_2 - S_2 \tau S_1) E \right)$ $= 4 \left( \lambda \gamma \chi^{-1} S_1 \times \chi^{-1} S_2 - \lambda \gamma \chi^{-1} S_2 \times \chi^{-1} S_1 \right).$                                                                                                                                                                               | $\varphi_* \left( (S_1 \tau S_2 - S_2 \tau S_1) - \frac{1}{8} \text{tr} (S_1 \tau S_2 - S_2 \tau S_1) E \right)$ $= 4 \left( \lambda \gamma \chi^{-1} S_1 \times \chi^{-1} S_2 - \lambda \gamma \chi^{-1} S_2 \times \chi^{-1} S_1 \right).$                                                                                                                                                                               |
| 0902.0431_E01060 | 180  | ■ 0.863 | $\mathfrak{W}_1 = \left\{ R \in \mathfrak{e}_8^C \mid R \times R = 0, \langle R, R \rangle = 4 \right\}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | $\mathfrak{W}_1 = \left\{ R \in \mathfrak{e}_8^C \mid R \times R = 0, \langle R, R \rangle = 4 \right\}$                                                                                                                                                                                                                                                                                                                   | $\mathfrak{W}_1 = \left\{ R \in \mathfrak{e}_8^C \mid R \times R = 0, \langle R, R \rangle = 4 \right\}$                                                                                                                                                                                                                                                                                                                   |
| 0902.0431_E00535 | 90   | ■ 0.868 | $\begin{aligned} & \xi_1 \xi_2 \xi_3 = \det(\alpha X) = \det X = 1, \quad (\text{hence } \xi_i > 0, i = 1, 2, 3), \\ & \xi_1^2 + \xi_2^2 + \xi_3^2 = \langle \alpha X, \alpha Y \rangle = \langle X, Y \rangle = 3. \end{aligned}$                                                                                                                                                                                                                                                                                                                                                                             | $\xi_1 \xi_2 \xi_3 = \det(\alpha X) = \det X = 1, \quad (\text{hence } \xi_i > 0, i = 1, 2, 3),$ $\xi_1^2 + \xi_2^2 + \xi_3^2 = \langle \alpha X, \alpha Y \rangle = \langle X, Y \rangle = 3.$                                                                                                                                                                                                                            | $\xi_1 \xi_2 \xi_3 = \det(\alpha X) = \det X = 1, \quad (\text{hence } \xi_i > 0, i = 1, 2, 3),$ $\xi_1^2 + \xi_2^2 + \xi_3^2 = \langle \alpha X, \alpha Y \rangle = \langle X, Y \rangle = 3.$                                                                                                                                                                                                                            |
| 0902.0431_E00955 | 159  | ■ 0.868 | $\begin{aligned} & f(P_C + (M + N)) = f(P_C) + \mu^{-1}(M + N), \\ & P_C + (M + N) \in (\mathfrak{P}_C)^C \oplus (M(3, \mathbf{C})^C \oplus M(3, \mathbf{C})^C) = \mathfrak{P}^C. \end{aligned}$                                                                                                                                                                                                                                                                                                                                                                                                               | $f(P_C + (M + N)) = f(P_C) + \mu^{-1}(M + N),$ $P_C + (M + N) \in (\mathfrak{P}_C)^C \oplus (M(3, \mathbf{C})^C \oplus M(3, \mathbf{C})^C) = \mathfrak{P}^C.$                                                                                                                                                                                                                                                              | $f(P_C + (M + N)) = f(P_C) + \mu^{-1}(M + N),$ $P_C + (M + N) \in (\mathfrak{P}_C)^C \oplus (M(3, \mathbf{C})^C \oplus M(3, \mathbf{C})^C) = \mathfrak{P}^C.$                                                                                                                                                                                                                                                              |



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| 0902.0431_E00708       | 117  | ■ 0.882 | $\begin{aligned} & \{[\ ] \& \Phi, P \times P\} \setminus = \& \left[ \Phi(\phi, A, B, \nu), \Phi\left(-X \vee Y, -\frac{1}{2}(Y \times Y - \xi X), \frac{1}{2}(X \times X - \eta Y), \frac{1}{4}((X, Y) - 3\xi\eta)\right) \right] \\ & = \cdots \text{(using } \phi(X \times Y) = \phi'X \times Y + X \times \phi'Y \text{ (Lemma 3.4.3.(1)), } [\phi, A \vee B] \\ & \quad = \phi A \vee B + A \vee \phi' B \text{ (Lemma 3.4.4.(1)), the formula about } A \vee B \text{ (Lemma 3.4.1) etc.)} \cdots \\ & = 2\Phi P \times P. \end{aligned}$                                                                                                                                | $[\Phi, P \times P]$ $= \left[ \Phi(\phi, A, B, \nu), \Phi\left(-X \vee Y, -\frac{1}{2}(Y \times Y - \xi X), \frac{1}{2}(X \times X - \eta Y), \frac{1}{4}((X, Y) - 3\xi\eta)\right) \right]$ $= \cdots \text{( using } \phi(X \times Y) = \phi'X \times Y + X \times \phi'Y \text{ (Lemma 3.4.3.(1)), } [\phi, A \vee B]$ $= \phi A \vee B + A \vee \phi' B \text{ (Lemma 3.4.4.(1)), the formula about } A \vee B \text{ (Lemma 3.4.1) etc.)} \cdots$ $= 2\Phi P \times P.$                                                                                                                                                                                     | $[\Phi, P \times P]$ $= \left[ \Phi(\phi, A, B, \nu), \Phi\left(-X \vee Y, -\frac{1}{2}(Y \times Y - \xi X), \frac{1}{2}(X \times X - \eta Y), \frac{1}{4}((X, Y) - 3\xi\eta)\right) \right]$ $= \cdots \text{(using } \phi(X \times Y) = \phi'X \times Y + X \times \phi'Y \text{ (Lemma 3.4.3.(1)), } [\phi, A \vee B]$ $= \phi A \vee B + A \vee \phi' B \text{ (Lemma 3.4.4.(1)), the formula about } A \vee B \text{ (Lemma 3.4.1) etc.)} \cdots$ $= 2\Phi P \times P.$                                                                                                                                                                                       |
| 0902.0431_EQ1121       | 190  | ■ 0.884 | $\left\{ \begin{array}{l} D = \left[ D_1, D_2 \right] + \frac{1}{4} \boldsymbol{X}_1 \cdot \boldsymbol{Y}_2 - \frac{1}{4} \boldsymbol{X}_2 \cdot \boldsymbol{Y}_1 \\ \phi = [\phi_1, \phi_2] + \frac{1}{2} \boldsymbol{X}_1 \vee \boldsymbol{Y}_2 - \frac{1}{2} \boldsymbol{X}_2 \vee \boldsymbol{Y}_1 \\ \boldsymbol{X} = \phi_1 \boldsymbol{X}_2 - \phi_2 \boldsymbol{X}_1 + D_1 \boldsymbol{X}_2 - D_2 \boldsymbol{X}_1 - \boldsymbol{Y}_1 \times \boldsymbol{Y}_2 \\ \boldsymbol{Y} = -{}^t \phi_1 \boldsymbol{Y}_2 + {}^t \phi_2 \boldsymbol{Y}_1 - {}^t D_1 \boldsymbol{Y}_2 + {}^t D_2 \boldsymbol{Y}_1 + \boldsymbol{X}_1 \times \boldsymbol{X}_2, \end{array} \right.$ | $\left\{ \begin{array}{l} D = [D_1, D_2] + \frac{1}{4} \boldsymbol{X}_1 \cdot \boldsymbol{Y}_2 - \frac{1}{4} \boldsymbol{X}_2 \cdot \boldsymbol{Y}_1 \\ \phi = [\phi_1, \phi_2] + \frac{1}{2} \boldsymbol{X}_1 \vee \boldsymbol{Y}_2 - \frac{1}{2} \boldsymbol{X}_2 \vee \boldsymbol{Y}_1 \\ \boldsymbol{X} = \phi_1 \boldsymbol{X}_2 - \phi_2 \boldsymbol{X}_1 + D_1 \boldsymbol{X}_2 - D_2 \boldsymbol{X}_1 - \boldsymbol{Y}_1 \times \boldsymbol{Y}_2 \\ \boldsymbol{Y} = -{}^t \phi_1 \boldsymbol{Y}_2 + {}^t \phi_2 \boldsymbol{Y}_1 - {}^t D_1 \boldsymbol{Y}_2 + {}^t D_2 \boldsymbol{Y}_1 + \boldsymbol{X}_1 \times \boldsymbol{X}_2 \end{array} \right.$ | $\left\{ \begin{array}{l} D = [D_1, D_2] + \frac{1}{4} \boldsymbol{X}_1 \cdot \boldsymbol{Y}_2 - \frac{1}{4} \boldsymbol{X}_2 \cdot \boldsymbol{Y}_1 \\ \phi = [\phi_1, \phi_2] + \frac{1}{2} \boldsymbol{X}_1 \vee \boldsymbol{Y}_2 - \frac{1}{2} \boldsymbol{X}_2 \vee \boldsymbol{Y}_1 \\ \boldsymbol{X} = \phi_1 \boldsymbol{X}_2 - \phi_2 \boldsymbol{X}_1 + D_1 \boldsymbol{X}_2 - D_2 \boldsymbol{X}_1 - \boldsymbol{Y}_1 \times \boldsymbol{Y}_2 \\ \boldsymbol{Y} = -{}^t \phi_1 \boldsymbol{Y}_2 + {}^t \phi_2 \boldsymbol{Y}_1 - {}^t D_1 \boldsymbol{Y}_2 + {}^t D_2 \boldsymbol{Y}_1 + \boldsymbol{X}_1 \times \boldsymbol{X}_2, \end{array} \right.$ |
| 0902.0431_EQ0821 (iii) | 137  | ■ 0.887 | $\left\{ \begin{array}{l} 2\delta(X \times X) \times N = a(X)\beta X \\ \delta(X \times X) \times \delta(X \times X) = a(X)(\det X)M \\ (\beta X, \delta(X \times X)) + \det X = 3a(X)b(X \times X) \quad (\text{use (i)}) \end{array} \right.$                                                                                                                                                                                                                                                                                                                                                                                                                                 | $\left\{ \begin{array}{l} 2\delta(X \times X) \times N = a(X)\beta X \\ \delta(X \times X) \times \delta(X \times X) = a(X)(\det X)M \\ (\beta X, \delta(X \times X)) + \det X = 3a(X)b(X \times X) \quad (\text{use (i)}) \end{array} \right.$                                                                                                                                                                                                                                                                                                                                                                                                                   | $\left\{ \begin{array}{l} 2\delta(X \times X) \times N = a(X)\beta X \\ \delta(X \times X) \times \delta(X \times X) = a(X)(\det X)M \\ (\beta X, \delta(X \times X)) + \det X = 3a(X)b(X \times X) \quad (\text{use (i)}). \end{array} \right.$                                                                                                                                                                                                                                                                                                                                                                                                                   |



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| 0902.0431_E00172 (ii) | 30   | ■ 0.890 | $\overline{\bar{x}}(\bar{y}p) = (\overline{xy})p, \quad x, y \in \mathfrak{C}.$                                                                                                                                                                                                                                                                                                                    | $\bar{x}(\bar{y}p) = (\overline{xy})p, \quad x, y \in \mathfrak{C}.$                                                                                                                                                                                                                                                                                                           | $\bar{x}(\bar{y}p) = (\overline{xy})p, \quad x, y \in \mathfrak{C}.$                                                                                                                                                                                                                                                                                                           |
| 0902.0431_E00235      | 39   | ■ 0.891 | $\begin{aligned} \{ \}^{\{t\}} \alpha^{-1}(Y \times Y) \times \{ \}^{\{t\}} \alpha^{-1}(Y \times Y) &= (\alpha Y \times \alpha Y) \times (\alpha Y \times \alpha Y) \quad (\text{Lemma 2.2.2.(4)}) \\ &= (\det \alpha Y) \alpha Y \quad (\text{Lemma 2.1.1.(3)}) = (\det Y) \alpha Y = \alpha((\det Y) Y) \\ &= \alpha((Y \times Y) \times (Y \times Y)), \quad Y \in \mathfrak{J}. \end{aligned}$ | $\begin{aligned} {}^t\alpha^{-1}(Y \times Y) \times {}^t\alpha^{-1}(Y \times Y) &= (\alpha Y \times \alpha Y) \times (\alpha Y \times \alpha Y) \quad (\text{Lemma 2.2.2.(4)}) \\ &= (\det \alpha Y) \alpha Y \quad (\text{Lemma 2.1.1.(3)}) = (\det Y) \alpha Y = \alpha((\det Y) Y) \\ &= \alpha((Y \times Y) \times (Y \times Y)), \quad Y \in \mathfrak{J}. \end{aligned}$ | $\begin{aligned} {}^t\alpha^{-1}(Y \times Y) \times {}^t\alpha^{-1}(Y \times Y) &= (\alpha Y \times \alpha Y) \times (\alpha Y \times \alpha Y) \quad (\text{Lemma 2.2.2.(4)}) \\ &= (\det \alpha Y) \alpha Y \quad (\text{Lemma 2.1.1.(3)}) = (\det Y) \alpha Y = \alpha((\det Y) Y) \\ &= \alpha((Y \times Y) \times (Y \times Y)), \quad Y \in \mathfrak{J}. \end{aligned}$ |
| 0902.0431_E00087      | 18   | ■ 0.892 | $\begin{array}{l} \{\left[H_{11}, L_{12}\right]=2 L_{21},\} \\ \& \{\left[H_{11}, L_{21}\right]=-2 L_{12},\} \& \{\left[H_{11}, L_{13}\right]=L_{31},\} \\ \& \{\left[H_{11}, L_{31}\right]=-L_{13},\} \& \{\left[H_{11}, L_{23}\right]=-L_{32},\} \\ \& \{\left[H_{11}, L_{32}\right]=L_{23}\} . \end{array}$                                                                                     | $\begin{array}{l} \left[H_{11}, L_{12}\right]=2 L_{21}, \quad \left[H_{11}, L_{21}\right]=-2 L_{12}, \quad \left[H_{11}, L_{13}\right]=L_{31}, \\ \left[H_{11}, L_{31}\right]=-L_{13}, \quad \left[H_{11}, L_{23}\right]=-L_{32}, \quad \left[H_{11}, L_{32}\right]=L_{23}. \end{array}$                                                                                       | $\begin{array}{l} \left[H_{11}, L_{12}\right]=2 L_{21}, \quad \left[H_{11}, L_{21}\right]=-2 L_{12}, \quad \left[H_{11}, L_{13}\right]=L_{31}, \\ \left[H_{11}, L_{31}\right]=-L_{13}, \quad \left[H_{11}, L_{23}\right]=-L_{32}, \quad \left[H_{11}, L_{32}\right]=L_{23}. \end{array}$                                                                                       |
| 0902.0431_E00417      | 68   | ■ 0.894 | $\begin{aligned} (X+M) \times (Y+N) &\&= \left(X \times Y - \frac{1}{2}(M^*N + N^*M)\right) - \frac{1}{2}(MY + NX + \overline{M \times N}), \\ (X+M, Y+N) &= (X, Y) + 2(M, N), \\ w(X+M) &= X + \omega_1 M, \quad \left(\omega_1 = -\frac{1}{2} + \frac{\sqrt{3}}{2}e_1\right), \end{aligned}$                                                                                                     | $\begin{aligned} (X+M) \times (Y+N) &= \left(X \times Y - \frac{1}{2}(M^*N + N^*M)\right) - \frac{1}{2}(MY + NX + \overline{M \times N}), \\ (X+M, Y+N) &= (X, Y) + 2(M, N), \\ w(X+M) &= X + \omega_1 M, \quad \left(\omega_1 = -\frac{1}{2} + \frac{\sqrt{3}}{2}e_1\right), \end{aligned}$                                                                                   | $\begin{aligned} (X+M) \times (Y+N) &= \left(X \times Y - \frac{1}{2}(M^*N + N^*M)\right) - \frac{1}{2}(MY + NX + \overline{M \times N}), \\ (X+M, Y+N) &= (X, Y) + 2(M, N), \\ w(X+M) &= X + \omega_1 M, \quad \left(\omega_1 = -\frac{1}{2} + \frac{\sqrt{3}}{2}e_1\right), \end{aligned}$                                                                                   |
| 0902.0431_E00690 (i)  | 114  | ■ 0.894 | $\begin{cases} \Phi P \times P = 0, & P \in \mathfrak{M}^C \\ \{\Phi P, Q\} + \{P, \Phi Q\} = 0, & P, Q \in \mathfrak{P}^C. \end{cases}$                                                                                                                                                                                                                                                           | $\begin{cases} \Phi P \times P = 0, & P \in \mathfrak{M}^C \\ \{\Phi P, Q\} + \{P, \Phi Q\} = 0, & P, Q \in \mathfrak{P}^C. \end{cases}$                                                                                                                                                                                                                                       | $\begin{cases} \Phi P \times P = 0, & P \in \mathfrak{M}^C \\ \{\Phi P, Q\} + \{P, \Phi Q\} = 0, & P, Q \in \mathfrak{P}^C. \end{cases}$                                                                                                                                                                                                                                       |
| 0902.0431_E00084      | 17   | ■ 0.895 | $\mathfrak{g}_2 = \mathfrak{su}(3) \oplus \mathfrak{S} \quad (\text{Theorem 1.5.1}).$                                                                                                                                                                                                                                                                                                              | $\mathfrak{g}_2 = \mathfrak{su}(3) \oplus \mathfrak{S} \quad (\text{Theorem 1.5.1}).$                                                                                                                                                                                                                                                                                          | $\mathfrak{g}_2 = \mathfrak{su}(3) \oplus \mathfrak{S} \quad (\text{Theorem 1.5.1}).$                                                                                                                                                                                                                                                                                          |

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| 0902.0431_E00738 | 122  | 0.896 | $\begin{aligned} & \left( \Phi, P \times Q \right)_7 = \left( \Phi \begin{pmatrix} \phi \\ A \\ B \\ \nu \end{pmatrix}, \Phi \begin{pmatrix} -\frac{1}{2}(X \vee W + Z \vee Y) \\ -\frac{1}{4}(2Y \times W - \xi Z - \zeta X) \\ \frac{1}{4}(2X \times Z - \eta W - \omega Y) \\ \frac{1}{8}((X, W) + (Z, Y) - 3(\xi \omega + \zeta \eta)) \end{pmatrix} \right)_7 \\ & = (\phi, X \vee W + Z \vee Y)_6 - (A, 2X \times Z - \eta W - \omega Y) + (2Y \times W - \xi Z - \zeta X, B) \\ & \quad - \frac{1}{3}\nu((X, W) + (Z, Y) - 3(\xi \omega + \zeta \eta)) \\ & = (\phi X, W) + (\phi Z, Y) - 2(A, X, Z) + \eta(A, W) + \omega(A, Y) + 2(Y, W, B) \\ & \quad - \xi(Z, B) - \zeta(X, B) - \frac{1}{3}\nu(X, W) - \frac{1}{3}\nu(Z, Y) + \nu(\xi \omega + \zeta \eta) \\ & = \left\{ \begin{pmatrix} \phi X - \frac{1}{3}\nu X + 2B \times Y + \eta A \\ 2A \times X - {}^t\phi Y + \frac{1}{3}\nu Y + \xi B \\ (A, Y) + \nu \xi \\ (B, X) - \nu \eta \end{pmatrix}, \begin{pmatrix} Z \\ W \\ \zeta \\ \omega \end{pmatrix} \right\} = \{\Phi P, Q\}. \end{aligned}$ | $\begin{aligned} & \left( \Phi, P \times Q \right)_7 = \left( \Phi \begin{pmatrix} \phi \\ A \\ B \\ \nu \end{pmatrix}, \Phi \begin{pmatrix} -\frac{1}{2}(X \vee W + Z \vee Y) \\ -\frac{1}{4}(2Y \times W - \xi Z - \zeta X) \\ \frac{1}{4}(2X \times Z - \eta W - \omega Y) \\ \frac{1}{8}((X, W) + (Z, Y) - 3(\xi \omega + \zeta \eta)) \end{pmatrix} \right)_7 \\ & = (\phi, X \vee W + Z \vee Y)_6 - (A, 2X \times Z - \eta W - \omega Y) + (2Y \times W - \xi Z - \zeta X, B) \\ & \quad - \frac{1}{3}\nu((X, W) + (Z, Y) - 3(\xi \omega + \zeta \eta)) \\ & = (\phi X, W) + (\phi Z, Y) - 2(A, X, Z) + \eta(A, W) + \omega(A, Y) + 2(Y, W, B) \\ & \quad - \xi(Z, B) - \zeta(X, B) - \frac{1}{3}\nu(X, W) - \frac{1}{3}\nu(Z, Y) + \nu(\xi \omega + \zeta \eta) \\ & = \left\{ \begin{pmatrix} \phi X - \frac{1}{3}\nu X + 2B \times Y + \eta A \\ 2A \times X - {}^t\phi Y + \frac{1}{3}\nu Y + \xi B \\ (A, Y) + \nu \xi \\ (B, X) - \nu \eta \end{pmatrix}, \begin{pmatrix} Z \\ W \\ \zeta \\ \omega \end{pmatrix} \right\} = \{\Phi P, Q\}. \end{aligned}$ |                                                                                                                                                                                                                                                                                                        |
| 0902.0431_E00122 | 23   | 0.896 | $\alpha e_1 = e_1 \quad \text{or} \quad \alpha e_1 = -e_1.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | $\alpha e_1 = e_1 \quad \text{or} \quad \alpha e_1 = -e_1.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | $\alpha e_1 = e_1 \quad \text{or} \quad \alpha e_1 = -e_1.$                                                                                                                                                                                                                                            |
| 0902.0431_E00587 | 98   | 0.896 | $\begin{aligned} & \left( \phi \in \text{Hom}_C((\mathfrak{H})^C) \mid (\phi M, M, M) = 0, \langle \phi M, N \rangle + \langle M, \phi N \rangle = 0 \right) \\ & = \left\{ \delta + i\tilde{T} \mid \delta \in \mathfrak{f}_{4, \mathbf{H}}, T \in (\mathfrak{H})_0 \right\} \end{aligned}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | $\begin{aligned} \mathfrak{e}_{6, \mathbf{H}} &= \left\{ \phi \in \text{Hom}_C \left( (\mathfrak{H})^C \right) \mid (\phi M, M, M) = 0, \langle \phi M, N \rangle + \langle M, \phi N \rangle = 0 \right\} \\ &= \left\{ \delta + i\tilde{T} \mid \delta \in \mathfrak{f}_{4, \mathbf{H}}, T \in (\mathfrak{H})_0 \right\} \end{aligned}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | $\begin{aligned} \mathfrak{e}_{6, \mathbf{H}} &= \{ \phi \in \text{Hom}_C((\mathfrak{H})^C) \mid (\phi M, M, M) = 0, \langle \phi M, N \rangle + \langle M, \phi N \rangle = 0 \} \\ &= \{ \delta + i\tilde{T} \mid \delta \in \mathfrak{f}_{4, \mathbf{H}}, T \in (\mathfrak{H})_0 \}, \end{aligned}$ |
| 0902.0431_E00615 | 102  | 0.896 | $\langle gX, gY \rangle = \langle X, Y \rangle, \quad X, Y \in \mathfrak{J}^C.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | $\langle gX, gY \rangle = \langle X, Y \rangle, \quad X, Y \in \mathfrak{J}^C.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | $\langle gX, gY \rangle = \langle X, Y \rangle, \quad X, Y \in \mathfrak{J}^C.$                                                                                                                                                                                                                        |

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|------------------|------|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0902.0431_E00307 | 49   | ■ 0.899 | $\begin{aligned} & \backslash begin{aligned} \& =\delta(A \circ(B \circ X)-B \circ(A \circ X))-(A \circ(B \circ \delta X)-B \circ(A \circ \delta X)) \\ & \backslash circ X))-(A \circ(B \circ \delta X)-B \circ(A \circ \delta X)) \backslash \& =\delta A \circ(B \circ X)+A \circ(\delta B \circ X)+A \circ(B \circ \delta X)-\delta B \circ(A \circ X) \\ & B \circ \delta X)+A \circ(B \circ \delta X)-\delta B \circ(A \circ X) \backslash \& -B \circ(\delta A \circ X)-B \circ(A \circ \delta X)-A \circ(B \circ \delta X)+B \circ(A \circ \delta X) \\ & \backslash circ X) \backslash \& -B \circ(\delta A \circ X)-B \circ(A \circ \delta X)-A \circ(B \circ \delta X)+B \circ(A \circ \delta X) \\ & \backslash delta X)-A \circ(B \circ \delta X)+B \circ(A \circ \delta X) \backslash \& =\delta A \circ(B \circ X)-B \circ(\delta A \circ X)+A \circ(\delta B \circ X)-\delta B \circ(A \circ X) \\ & \backslash delta X) \backslash \& =\delta A \circ(B \circ X)-B \circ(\delta A \circ X)+A \circ(\delta B \circ X)-\delta B \circ(A \circ X) \\ & A \circ \delta X)+A \circ(\delta B \circ X)-\delta B \circ(A \circ X) \backslash \& =[\widetilde{\delta A}, \widetilde{B}] X+[\widetilde{A}, \widetilde{\delta B}] X, \quad X \in \mathfrak{J}^C . \end{aligned}$ | $\begin{aligned} & =\delta(A \circ(B \circ X)-B \circ(A \circ X))-(A \circ(B \circ \delta X)-B \circ(A \circ \delta X)) \\ & =\delta A \circ(B \circ X)+A \circ(\delta B \circ X)+A \circ(B \circ \delta X)-\delta B \circ(A \circ X) \\ & -B \circ(\delta A \circ X)-B \circ(A \circ \delta X)-A \circ(B \circ \delta X)+B \circ(A \circ \delta X) \\ & =\delta A \circ(B \circ X)-B \circ(\delta A \circ X)+A \circ(\delta B \circ X)-\delta B \circ(A \circ X) \\ & =[\widetilde{\delta A}, \widetilde{B}] X+[\widetilde{A}, \widetilde{\delta B}] X, \quad X \in \mathfrak{J}^C . \end{aligned}$ | $\begin{aligned} & =\delta(A \circ(B \circ X)-B \circ(A \circ X))-(A \circ(B \circ \delta X)-B \circ(A \circ \delta X)) \\ & =\delta A \circ(B \circ X)+A \circ(\delta B \circ X)+A \circ(B \circ \delta X)-\delta B \circ(A \circ X) \\ & -B \circ(\delta A \circ X)-B \circ(A \circ \delta X)-A \circ(B \circ \delta X)+B \circ(A \circ \delta X) \\ & =\delta A \circ(B \circ X)-B \circ(\delta A \circ X)+A \circ(\delta B \circ X)-\delta B \circ(A \circ X) \\ & =[\widetilde{\delta A}, \widetilde{B}] X+[\widetilde{A}, \widetilde{\delta B}] X, \quad X \in \mathfrak{J}^C . \end{aligned}$ |
| 0902.0431_E01108 | 188  | ■ 0.902 | $\left(\mathfrak{e}_8^C\right)^{\tilde{\lambda}} \gamma=\left\{\sum_{k, l}\left[R_k, R_l\right] \mid R_k, R_l \in\left(\mathfrak{e}_8^C\right)_{-\tilde{\lambda} \gamma}\right\} .$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | $\left(\mathfrak{e}_8^C\right)^{\tilde{\lambda} \gamma}=\left\{\sum_{k, l}\left[R_k, R_l\right] \mid R_k, R_l \in\left(\mathfrak{e}_8^C\right)_{-\tilde{\lambda} \gamma}\right\} .$                                                                                                                                                                                                                                                                                                                                                                                                                  | $\left(\mathfrak{e}_8^C\right)^{\tilde{\lambda} \gamma}=\left\{\sum_{k, l}\left[R_k, R_l\right] \mid R_k, R_l \in\left(\mathfrak{e}_8^C\right)_{-\tilde{\lambda} \gamma}\right\} .$                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 0902.0431_E00372 | 62   | ■ 0.903 | $\begin{aligned} & \backslash left\{\backslash begin{array}{l} \xi_1+\xi_2+\xi_3= \\ & \backslash operatorname{tr}(X) \backslash \backslash \xi_1\{ \}^2+\xi_2\{ \}^2+ \\ & \xi_3\{ \}^2=(X, X) \backslash \backslash \xi_1\{ \}^3+\xi_2\{ \}^3+ \\ & \xi_3\{ \}^3=\operatorname{tr}(X, X, X) \backslash end{array} \\ & \backslash right. \end{aligned}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | $\begin{cases} \xi_1+\xi_2+\xi_3=\operatorname{tr}(X) \\ \xi_1^2+\xi_2^2+\xi_3^2=(X, X) \\ \xi_1^3+\xi_2^3+\xi_3^3=\operatorname{tr}(X, X, X) \end{cases}$                                                                                                                                                                                                                                                                                                                                                                                                                                           | $\begin{cases} \xi_1+\xi_2+\xi_3=\operatorname{tr}(X) \\ \xi_1^2+\xi_2^2+\xi_3^2=(X, X) \\ \xi_1^3+\xi_2^3+\xi_3^3=\operatorname{tr}(X, X, X) . \end{cases}$                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 0902.0431_E00023 | 9    | ■ 0.904 | $\begin{aligned} & \backslash begin{array}{ll} \{ \backslash left[F_{i j}, F_{j k} \backslash right]=F_{i k}, \\ & \& i, j, k \text { are distinct } \} \backslash \backslash \{ \backslash left[F_{i j}, \\ & F_{k l} \backslash right]=0, \& i, j, k, l \text { are distinct. } \} \\ & \backslash end{array} \end{aligned}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | $\begin{aligned} & \left[F_{i j}, F_{j k}\right]=F_{i k}, \quad i, j, k \text { are distinct} \\ & \left[F_{i j}, F_{k l}\right]=0, \quad i, j, k, l \text { are distinct.} \end{aligned}$                                                                                                                                                                                                                                                                                                                                                                                                           | $\begin{aligned} & \left[F_{i j}, F_{j k}\right]=F_{i k}, \quad i, j, k \text { are distinct,} \\ & \left[F_{i j}, F_{k l}\right]=0, \quad i, j, k, l \text { are distinct.} \end{aligned}$                                                                                                                                                                                                                                                                                                                                                                                                          |
| 0902.0431_E01106 | 187  | ■ 0.904 | $\alpha R=k R, \quad R \in\left(\mathfrak{e}_8^C\right)_{-\tilde{\lambda} \gamma} .$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | $\alpha R=k R, \quad R \in\left(\mathfrak{e}_8^C\right)_{-\tilde{\lambda} \gamma} .$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | $\alpha R=k R, \quad R \in\left(\mathfrak{e}_8^C\right)_{-\tilde{\lambda} \gamma} .$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 0902.0431_E00947 | 157  | ■ 0.904 | $\sum(A \boldsymbol{a} \wedge A \boldsymbol{b} \wedge A \boldsymbol{c})=\overline{\boldsymbol{a}} \wedge \overline{\boldsymbol{b}} \wedge \overline{\boldsymbol{c}}, \quad \boldsymbol{a}, \boldsymbol{b}, \boldsymbol{c} \in C^6$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | $\sum(A \boldsymbol{a} \wedge A \boldsymbol{b} \wedge A \boldsymbol{c})=\overline{\boldsymbol{a}} \wedge \overline{\boldsymbol{b}} \wedge \overline{\boldsymbol{c}}, \quad \boldsymbol{a}, \boldsymbol{b}, \boldsymbol{c} \in C^6$                                                                                                                                                                                                                                                                                                                                                                   | $\sum(A \boldsymbol{a} \wedge A \boldsymbol{b} \wedge A \boldsymbol{c})=\overline{\boldsymbol{a}} \wedge \overline{\boldsymbol{b}} \wedge \overline{\boldsymbol{c}}, \quad \boldsymbol{a}, \boldsymbol{b}, \boldsymbol{c} \in C^6$                                                                                                                                                                                                                                                                                                                                                                   |
| 0902.0431_E01009 | 168  | ■ 0.905 | $U(248)=U\left(\mathfrak{e}_8^C\right)=\left\{\alpha \in \operatorname{Iso}_C\left(\mathfrak{e}_8^C\right) \mid\left\langle\alpha R_1, \alpha R_2\right\rangle=\left\langle R_1, R_2\right\rangle\right\} .$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | $U(248)=U\left(\mathfrak{e}_8^C\right)=\left\{\alpha \in \operatorname{Iso}_C\left(\mathfrak{e}_8^C\right) \mid\left\langle\alpha R_1, \alpha R_2\right\rangle=\left\langle R_1, R_2\right\rangle\right\} .$                                                                                                                                                                                                                                                                                                                                                                                         | $U(248)=U\left(\mathfrak{e}_8^C\right)=\left\{\alpha \in \operatorname{Iso}_C\left(\mathfrak{e}_8^C\right) \mid\left\langle\alpha R_1, \alpha R_2\right\rangle=\left\langle R_1, R_2\right\rangle\right\} .$                                                                                                                                                                                                                                                                                                                                                                                         |
| 0902.0431_E00981 | 163  | ■ 0.905 | $\begin{aligned} & \backslash begin{aligned} \& \mathfrak{a} \ni\left[1, \backslash left[1_{-},[1, R] \backslash right] \backslash right]=\left[1, \backslash left[1_{-},(\theta, P,-Q, 0, 2 s,-2 t) \backslash right] \backslash right] \backslash \& \quad=\left[1,(\theta, 0, P,-2 s, 0, 0)\right]=-(\theta, 0, P, 0, 0, 0)=-P_{-} . \end{aligned}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | $\begin{aligned} & \mathfrak{a} \ni\left[1, \left[1_{-},\left[1, R\right]\right]\right]=\left[1,\left[1_{-},\left(\left(0, P,-Q, 0, 2 s,-2 t\right)\right)\right]\right] \\ & =\left[1,\left(\left(0, 0, P,-2 s, 0, 0\right)\right)\right]=-\left(\left(0, 0, P, 0, 0, 0\right)\right)=-P_{-} \end{aligned}$                                                                                                                                                                                                                                                                                         | $\begin{aligned} & \mathfrak{a} \ni\left[1, \left[1_{-},\left[1, R\right]\right]\right]=\left[1,\left[1_{-},\left(\left(0, P,-Q, 0, 2 s,-2 t\right)\right)\right]\right] \\ & =\left[1,\left(\left(0, 0, P,-2 s, 0, 0\right)\right)\right]=-\left(\left(0, 0, P, 0, 0, 0\right)\right)=-P_{-} . \end{aligned}$                                                                                                                                                                                                                                                                                       |
| 0902.0431_E00459 | 75   | ■ 0.908 | $\mathfrak{a} \ni\left[\delta_1, \delta+\widetilde{T}\right]=\left[\delta_1, \delta\right]+\widetilde{\delta_1 T} \text { (Theorem 3.2.2), }$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | $\mathfrak{a} \ni\left[\delta_1, \delta+\widetilde{T}\right]=\left[\delta_1, \delta\right]+\widetilde{\delta_1 T} \text { (Theorem 3.2.2), }$                                                                                                                                                                                                                                                                                                                                                                                                                                                        | $\mathfrak{a} \ni\left[\delta_1, \delta+\widetilde{T}\right]=\left[\delta_1, \delta\right]+\widetilde{\delta_1 T} \text { (Theorem 3.2.2), }$                                                                                                                                                                                                                                                                                                                                                                                                                                                        |







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| 0902.0431_E00508 | 83   | ■ 0.928 | $\begin{aligned} \{[h, S]\} &= \left[h_{\delta} + \widetilde{H}, S\right] - [S, \widetilde{H}] \\ &= \alpha(h_{\delta})S - \widetilde{S\widetilde{H}} \text{ (Lemma 3.6.3.(1))} \\ &= \alpha(h_{\delta})S = (\pm\lambda_k \pm \lambda_l)S \end{aligned}$                                                                                                                                                                                                                                                                                                                                                               | $\begin{aligned} [h, S] &= \left[h_{\delta} + \widetilde{H}, S\right] = [h_{\delta}, S] - [S, \widetilde{H}] \\ &= \alpha(h_{\delta})S - \widetilde{S\widetilde{H}} \text{ (Lemma 3.6.3.(1))} \\ &= \alpha(h_{\delta})S = (\pm\lambda_k \pm \lambda_l)S \end{aligned}$                                                                                                                                                                                                                                                                                                                                                 | $\begin{aligned} [h, S] &= [h_{\delta} + \widetilde{H}, S] = [h_{\delta}, S] - [S, \widetilde{H}] \\ &= \alpha(h_{\delta})S - \widetilde{S\widetilde{H}} \text{ (Lemma 3.6.3.(1))} \\ &= \alpha(h_{\delta})S = (\pm\lambda_k \pm \lambda_l)S. \end{aligned}$                                                                                                                                                                                                                                                                                                                                                           |
| 0902.0431_E00665 | 110  | ■ 0.935 | $E_{6(6)} \cong \left(E_6^C\right)^{\tau\gamma}, E_{6(2)} \cong \left(E_6^C\right)^{\tau\lambda\gamma}, E_{6(-14)} \cong \left(E_6^C\right)^{\tau\lambda\sigma}, E_{6(-26)} \cong \left(E_6^C\right)^{\tau}.$                                                                                                                                                                                                                                                                                                                                                                                                          | $E_{6(6)} \cong \left(E_6^C\right)^{\tau\gamma}, E_{6(2)} \cong \left(E_6^C\right)^{\tau\lambda\gamma}, E_{6(-14)} \cong \left(E_6^C\right)^{\tau\lambda\sigma}, E_{6(-26)} \cong \left(E_6^C\right)^{\tau}.$                                                                                                                                                                                                                                                                                                                                                                                                          | $E_{6(6)} \cong \left(E_6^C\right)^{\tau\gamma}, E_{6(2)} \cong \left(E_6^C\right)^{\tau\lambda\gamma}, E_{6(-14)} \cong \left(E_6^C\right)^{\tau\lambda\sigma}, E_{6(-26)} \cong \left(E_6^C\right)^{\tau}.$                                                                                                                                                                                                                                                                                                                                                                                                          |
| 0902.0431_E01052 | 179  | ■ 0.936 | $\left\{\beta_2 P, \beta_3 Q\right\} = \left\{\beta_2 P, \beta_2 Q\right\}, \quad P, Q \in \mathfrak{P}^C,$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | $\{\beta_2 P, \beta_3 Q\} = \{\beta_2 P, \beta_2 Q\}, \quad P, Q \in \mathfrak{P}^C,$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | $\{\beta_2 P, \beta_3 Q\} = \{\beta_2 P, \beta_2 Q\}, \quad P, Q \in \mathfrak{P}^C,$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 0902.0431_E00512 | 84   | ■ 0.938 | $\begin{aligned} \nu h_{\delta} &= \frac{1}{2}(-\lambda_0 - \lambda_1 + \lambda_2 - \lambda_3)H_0 + \frac{1}{2}(\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3)H_1 \\ &+ \frac{1}{2}(-\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3)H_2 + \frac{1}{2}(\lambda_0 - \lambda_1 + \lambda_2 + \lambda_3)H_3, \\ \kappa\pi h_{\delta} &= \frac{1}{2}(-\lambda_0 + \lambda_1 - \lambda_2 + \lambda_3)H_0 + \frac{1}{2}(-\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3)H_1 \\ &+ \frac{1}{2}(\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3)H_2 + \frac{1}{2}(-\lambda_0 - \lambda_1 + \lambda_2 + \lambda_3)H_3 \end{aligned}$ | $\begin{aligned} \nu h_{\delta} &= \frac{1}{2}(-\lambda_0 - \lambda_1 + \lambda_2 - \lambda_3)H_0 + \frac{1}{2}(\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3)H_1 \\ &+ \frac{1}{2}(-\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3)H_2 + \frac{1}{2}(\lambda_0 - \lambda_1 + \lambda_2 + \lambda_3)H_3, \\ \kappa\pi h_{\delta} &= \frac{1}{2}(-\lambda_0 + \lambda_1 - \lambda_2 + \lambda_3)H_0 + \frac{1}{2}(-\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3)H_1 \\ &+ \frac{1}{2}(\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3)H_2 + \frac{1}{2}(-\lambda_0 - \lambda_1 + \lambda_2 + \lambda_3)H_3 \end{aligned}$ | $\begin{aligned} \nu h_{\delta} &= \frac{1}{2}(-\lambda_0 - \lambda_1 + \lambda_2 - \lambda_3)H_0 + \frac{1}{2}(\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3)H_1 \\ &+ \frac{1}{2}(-\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3)H_2 + \frac{1}{2}(\lambda_0 - \lambda_1 + \lambda_2 + \lambda_3)H_3, \\ \kappa\pi h_{\delta} &= \frac{1}{2}(-\lambda_0 + \lambda_1 - \lambda_2 + \lambda_3)H_0 + \frac{1}{2}(-\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3)H_1 \\ &+ \frac{1}{2}(\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3)H_2 + \frac{1}{2}(-\lambda_0 - \lambda_1 + \lambda_2 + \lambda_3)H_3 \end{aligned}$ |

| Identifier       | Page | Conf.   | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Rendered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Image                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
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| 0902.0431_E00582 | 97   | ■ 0.938 | $\begin{aligned} & \operatorname{det}(k_J(M)) \\ &= \operatorname{det} \begin{pmatrix} 0 & \xi_1 & -n_3 & m_3 & n_2 & \tau(m_2) \\ -\xi_1 & 0 & -\tau(m_3) & -\tau(n_3) & -m_2 & \tau(n_2) \\ n_3 & \tau(m_3) & 0 & \xi_2 & -n_1 & m_1 \\ -m_3 & \tau(n_3) & -\xi_2 & 0 & -\tau(m_1) & -\tau(n_1) \\ -n_2 & m_2 & n_1 & \tau(m_1) & 0 & \xi_3 \\ -\tau(m_2) & -\tau(n_2) & -m_1 & \tau(n_1) & -\xi_3 & 0 \end{pmatrix} \\ &= (\xi_1 \xi_2 \xi_3 - \xi_1 n_1 \tau(n_1) - \xi_1 m_1 \tau(m_1) - n_3 \tau(n_3) \xi_3 - n_3 m_2 \tau(n_1) - n_3 \tau(n_2) \tau(m_1) \\ &\quad - m_3 \tau(m_3) \xi_3 + m_3 m_2 m_1 - m_3 \tau(n_2) n_1 - n_2 \tau(m_3) \tau(n_1) - n_2 \tau(n_3) m_1 - n_2 \tau(n_2) \xi_2 \\ &\quad + \tau(m_2) \tau(m_3) \tau(m_1) - \tau(m_2) \tau(n_3) n_1 - \tau(m_2) m_2 \xi_2)^2 \end{aligned}$ | $\det(k_J(M)) = \det \begin{pmatrix} 0 & \xi_1 & -n_3 & m_3 & n_2 & \tau(m_2) \\ -\xi_1 & 0 & -\tau(m_3) & -\tau(n_3) & -m_2 & \tau(n_2) \\ n_3 & \tau(m_3) & 0 & \xi_2 & -n_1 & m_1 \\ -m_3 & \tau(n_3) & -\xi_2 & 0 & -\tau(m_1) & -\tau(n_1) \\ -n_2 & m_2 & n_1 & \tau(m_1) & 0 & \xi_3 \\ -\tau(m_2) & -\tau(n_2) & -m_1 & \tau(n_1) & -\xi_3 & 0 \end{pmatrix}$ $= (\xi_1 \xi_2 \xi_3 - \xi_1 n_1 \tau(n_1) - \xi_1 m_1 \tau(m_1) - n_3 \tau(n_3) \xi_3 - n_3 m_2 \tau(n_1) - n_3 \tau(n_2) \tau(m_1) \\ - m_3 \tau(m_3) \xi_3 + m_3 m_2 m_1 - m_3 \tau(n_2) n_1 - n_2 \tau(m_3) \tau(n_1) - n_2 \tau(n_3) m_1 - n_2 \tau(n_2) \xi_2 \\ + \tau(m_2) \tau(m_3) \tau(m_1) - \tau(m_2) \tau(n_3) n_1 - \tau(m_2) m_2 \xi_2)^2$ | $\det(k_J(M)) = \det \begin{pmatrix} 0 & \xi_1 & -n_3 & m_3 & n_2 & \tau(m_2) \\ -\xi_1 & 0 & -\tau(m_3) & -\tau(n_3) & -m_2 & \tau(n_2) \\ n_3 & \tau(m_3) & 0 & \xi_2 & -n_1 & m_1 \\ -m_3 & \tau(n_3) & -\xi_2 & 0 & -\tau(m_1) & -\tau(n_1) \\ -n_2 & m_2 & n_1 & \tau(m_1) & 0 & \xi_3 \\ -\tau(m_2) & -\tau(n_2) & -m_1 & \tau(n_1) & -\xi_3 & 0 \end{pmatrix}$ $= (\xi_1 \xi_2 \xi_3 - \xi_1 n_1 \tau(n_1) - \xi_1 m_1 \tau(m_1) - n_3 \tau(n_3) \xi_3 - n_3 m_2 \tau(n_1) - n_3 \tau(n_2) \tau(m_1) \\ - m_3 \tau(m_3) \xi_3 + m_3 m_2 m_1 - m_3 \tau(n_2) n_1 - n_2 \tau(m_3) \tau(n_1) - n_2 \tau(n_3) m_1 - n_2 \tau(n_2) \xi_2 \\ + \tau(m_2) \tau(m_3) \tau(m_1) - \tau(m_2) \tau(n_3) n_1 - \tau(m_2) m_2 \xi_2)^2.$ |
| 0902.0431_E00892 | 148  | ■ 0.939 | $\left( \Phi(\phi_1, A_1, -\gamma A_1, 0), \Phi(\phi_2, A_2, -\gamma A_2, 0) \right) = \Phi(\phi, A, -\gamma A, 0),$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | $[\Phi(\phi_1, A_1, -\gamma A_1, 0), \Phi(\phi_2, A_2, -\gamma A_2, 0)] = \Phi(\phi, A, -\gamma A, 0),$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | $[\Phi(\phi_1, A_1, -\gamma A_1, 0), \Phi(\phi_2, A_2, -\gamma A_2, 0)] = \Phi(\phi, A, -\gamma A, 0),$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 0902.0431_E00737 | 122  | ■ 0.939 | $\begin{aligned} & \dots \left( \text{using } [\phi, A \vee B] = \phi A \vee B + A \vee \phi' B \text{ (Lemma 3.4.4) etc.} \right) \dots \\ &= -(\Phi_1, [\Phi, \Phi_2])_7. \end{aligned}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | $= \dots \left( \text{using } [\phi, A \vee B] = \phi A \vee B + A \vee \phi' B \text{ (Lemma 3.4.4) etc.} \right) \dots$ $= -(\Phi_1, [\Phi, \Phi_2])_7$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | $= \dots \left( \text{using } ([\phi, A \vee B] = \phi A \vee B + A \vee \phi' B \text{ (Lemma 3.4.4) etc.}) \dots \right.$ $= -(\Phi_1, [\Phi, \Phi_2])_7.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 0902.0431_E01073 | 182  | ■ 0.940 | $Jl(S)I = l(S)JI = -l(S)IJ,$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | $Jl(S)I = l(S)JI = -l(S)IJ$ ${}^t(l(S)I) = {}^tI^t(l(S)) = Il(\tau^t S) = -Il(\tau S) = -l(S)I$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | $Jl(S)I = l(S)JI = -l(S)IJ,$ ${}^t(l(S)I) = {}^tI^t(l(S)) = Il(\tau^t S) = -Il(\tau S) = -l(S)I.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |



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| 0902.0431_E00413 | 68   | ■0.941 | $\left(\begin{array}{lll} \xi_1 & a_3 & \bar{a}_2 \\ \bar{a}_3 & \xi_2 & a_1 \\ a_2 & \bar{a}_1 & \xi_3 \end{array}\right) + (\boldsymbol{m}_1, \boldsymbol{m}_2, \boldsymbol{m}_3)$                                                                                                                                                                                                             | $\left(\begin{array}{ccc} \xi_1 & a_3 & \bar{a}_2 \\ \bar{a}_3 & \xi_2 & a_1 \\ a_2 & \bar{a}_1 & \xi_3 \end{array}\right) + (\boldsymbol{m}_1, \boldsymbol{m}_2, \boldsymbol{m}_3)$                                                                                                                | $\left(\begin{array}{ccc} \xi_1 & a_3 & \bar{a}_2 \\ \bar{a}_3 & \xi_2 & a_1 \\ a_2 & \bar{a}_1 & \xi_3 \end{array}\right) + (\boldsymbol{m}_1, \boldsymbol{m}_2, \boldsymbol{m}_3)$                                                                                                                |
| 0902.0431_E00017 | 8    | ■0.942 | $\left(\begin{array}{l} \{ 2 F_{\{ 0 1 \}} = G_{\{ 0 1 \}} + G_{\{ 2 3 \}} + G_{\{ 4 5 \}} + G_{\{ 6 7 \}} \\ \{ 2 F_{\{ 2 3 \}} = G_{\{ 0 1 \}} + G_{\{ 2 3 \}} - G_{\{ 4 5 \}} - G_{\{ 6 7 \}} \\ \{ 2 F_{\{ 4 5 \}} = G_{\{ 0 1 \}} - G_{\{ 2 3 \}} + G_{\{ 4 5 \}} - G_{\{ 6 7 \}} \\ \{ 2 F_{\{ 6 7 \}} = G_{\{ 0 1 \}} - G_{\{ 2 3 \}} - G_{\{ 4 5 \}} + G_{\{ 6 7 \}} \end{array}\right)$ | $\left\{\begin{array}{l} 2F_{01} = G_{01} + G_{23} + G_{45} + G_{67} \\ 2F_{23} = G_{01} + G_{23} - G_{45} - G_{67} \\ 2F_{45} = G_{01} - G_{23} + G_{45} - G_{67} \\ 2F_{67} = G_{01} - G_{23} - G_{45} + G_{67}, \end{array}\right\}$                                                             | $\left\{\begin{array}{l} 2F_{02} = G_{02} - G_{13} - G_{46} + G_{57} \\ 2F_{13} = -G_{02} + G_{13} - G_{46} + G_{57} \\ 2F_{46} = -G_{02} - G_{13} + G_{46} + G_{57} \\ 2F_{57} = G_{02} + G_{13} + G_{46} + G_{57}, \end{array}\right\}$                                                           |
| 0902.0431_E00927 | 153  | ■0.942 | $D = \left(\begin{array}{cc} B & L \\ -L^* & C \end{array}\right) + \frac{\nu}{3} \left(\begin{array}{cc} E & 0 \\ 0 & -E \end{array}\right), \quad B, C \in \mathfrak{su}(3), L \in M(3, C), \nu \in e_1 \boldsymbol{R}.$                                                                                                                                                                       | $D = \left(\begin{array}{cc} B & L \\ -L^* & C \end{array}\right) + \frac{\nu}{3} \left(\begin{array}{cc} E & 0 \\ 0 & -E \end{array}\right), \quad B, C \in \mathfrak{su}(3), L \in M(3, C), \nu \in e_1 \boldsymbol{R}.$                                                                          | $D = \left(\begin{array}{cc} B & L \\ -L^* & C \end{array}\right) + \frac{\nu}{3} \left(\begin{array}{cc} E & 0 \\ 0 & -E \end{array}\right), \quad B, C \in \mathfrak{su}(3), L \in M(3, C), \nu \in e_1 \boldsymbol{R}.$                                                                          |
| 0902.0431_E00421 | 69   | ■0.942 | $\varphi(A, 1)X = AXA^*, \quad \varphi(A, \epsilon)X = A\bar{X}A^*, \quad X \in \mathfrak{J}_C.$                                                                                                                                                                                                                                                                                                 | $\varphi(A, 1)X = AXA^*, \quad \varphi(A, \epsilon)X = A\bar{X}A^*, \quad X \in \mathfrak{J}_C.$                                                                                                                                                                                                    | $\varphi(A, 1)X = AXA^*, \quad \varphi(A, \epsilon)X = A\bar{X}A^*, \quad X \in \mathfrak{J}_C.$                                                                                                                                                                                                    |
| 0902.0431_E00257 | 42   | ■0.943 | $\frac{f_4}{\delta} = \left(\begin{array}{l} \delta(X \circ Y) = \delta X \circ Y + X \circ \delta Y \\ \text{tr}(\delta X, Y, Z) + \text{tr}(X, \delta Y, Z) + \text{tr}(X, Y, \delta Z) = 0 \\ (\delta X, Y) + (X, \delta Y) = 0 \end{array}\right)$                                                                                                                                           | $f_4 = \left\{\delta \in \text{Hom}_{\boldsymbol{R}}(\mathfrak{J}) \mid \begin{array}{l} \delta(X \circ Y) = \delta X \circ Y + X \circ \delta Y \\ \text{tr}(\delta X, Y, Z) + \text{tr}(X, \delta Y, Z) + \text{tr}(X, Y, \delta Z) = 0 \\ (\delta X, Y) + (X, \delta Y) = 0 \end{array}\right\}$ | $f_4 = \left\{\delta \in \text{Hom}_{\boldsymbol{R}}(\mathfrak{J}) \mid \begin{array}{l} \delta(X \circ Y) = \delta X \circ Y + X \circ \delta Y \\ \text{tr}(\delta X, Y, Z) + \text{tr}(X, \delta Y, Z) + \text{tr}(X, Y, \delta Z) = 0 \\ (\delta X, Y) + (X, \delta Y) = 0 \end{array}\right\}$ |

| Identifier           | Page | Conf.   | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                                           | Rendered                                                                                                                                                                                                                                                                                                                                                                                                                           | Image                                                                                                                                                                                                                                                                                                                                                                                                                               |
|----------------------|------|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0902.0431_E00431     | 71   | ■ 0.943 | $\begin{aligned} F_4^C &= \{\alpha \in \text{Iso}_C(\mathfrak{J}^C) \mid \alpha(X \circ Y) = \alpha X \circ \alpha Y\} \\ &= \{\alpha \in \text{Iso}_C(\mathfrak{J}^C) \mid \det(\alpha X) = \det X, (\alpha X, \alpha Y) = (X, Y)\} \\ &= \{\alpha \in \text{Iso}_C(\mathfrak{J}^C) \mid \det(\alpha X) = \det X, \alpha E = E\} \\ &= \{\alpha \in \text{Iso}_C(\mathfrak{J}^C) \mid \alpha(X \times Y) = \alpha X \times \alpha Y\}. \end{aligned}$ | $F_4^C = \{\alpha \in \text{Iso}_C(\mathfrak{J}^C) \mid \alpha(X \circ Y) = \alpha X \circ \alpha Y\}$ $= \{\alpha \in \text{Iso}_C(\mathfrak{J}^C) \mid \det(\alpha X) = \det X, (\alpha X, \alpha Y) = (X, Y)\}$ $= \{\alpha \in \text{Iso}_C(\mathfrak{J}^C) \mid \det(\alpha X) = \det X, \alpha E = E\}$ $= \{\alpha \in \text{Iso}_C(\mathfrak{J}^C) \mid \alpha(X \times Y) = \alpha X \times \alpha Y\}.$                  | $F_4^C = \{\alpha \in \text{Iso}_C(\mathfrak{J}^C) \mid \alpha(X \circ Y) = \alpha X \circ \alpha Y\}$ $= \{\alpha \in \text{Iso}_C(\mathfrak{J}^C) \mid \det(\alpha X) = \det X, (\alpha X, \alpha Y) = (X, Y)\}$ $= \{\alpha \in \text{Iso}_C(\mathfrak{J}^C) \mid \det(\alpha X) = \det X, \alpha E = E\}$ $= \{\alpha \in \text{Iso}_C(\mathfrak{J}^C) \mid \alpha(X \times Y) = \alpha X \times \alpha Y\}.$                   |
| 0902.0431_E00645     | 106  | ■ 0.946 | $\epsilon(A, B) = (\bar{B}, \bar{A}),$                                                                                                                                                                                                                                                                                                                                                                                                                 | $\epsilon(A, B) = (\bar{B}, \bar{A}),$                                                                                                                                                                                                                                                                                                                                                                                             | $\epsilon(A, B) = (\bar{B}, \bar{A}),$                                                                                                                                                                                                                                                                                                                                                                                              |
| 0902.0431_E01049 (i) | 179  | ■ 0.948 | $\beta_1(P \times Q) = \beta_2 P \times \beta_3 Q, \quad \beta_2 P, \beta_3 Q = \{P, Q\},$ $\beta_1(P \times Q) = \beta_2 P \times \beta_3 Q, \quad \beta_2 P, \beta_3 Q = \{P, Q\},$                                                                                                                                                                                                                                                                  | $\beta_1(P \times Q) = \beta_2 P \times \beta_3 Q, \quad \beta_2 P, \beta_3 Q = \{P, Q\},$                                                                                                                                                                                                                                                                                                                                         | $\beta_1(P \times Q) = \beta_2 P \times \beta_3 Q, \quad \beta_2 P, \beta_3 Q = \{P, Q\},$                                                                                                                                                                                                                                                                                                                                          |
| 0902.0431_E00435     | 72   | ■ 0.948 | $F_4^C \simeq (F_4^C \cap U(\mathfrak{J}^C)) \times \mathbf{R}^d = F_4 \times \mathbf{R}^d,$                                                                                                                                                                                                                                                                                                                                                           | $F_4^C \simeq (F_4^C \cap U(\mathfrak{J}^C)) \times \mathbf{R}^d = F_4 \times \mathbf{R}^d,$                                                                                                                                                                                                                                                                                                                                       | $F_4^C \simeq (F_4^C \cap U(\mathfrak{J}^C)) \times \mathbf{R}^d = F_4 \times \mathbf{R}^d,$                                                                                                                                                                                                                                                                                                                                        |
| 0902.0431_E00105     | 20   | ■ 0.950 | $\begin{aligned} & \left( \frac{1}{8} \right) \left( \frac{1}{8} \right) (1+1) = \frac{1}{4} \\ & \left( \frac{1}{24} \right) \left( \frac{1}{24} \right) (1+4+1) = \frac{1}{12} \\ & \left( \frac{1}{8} \right) \left( \frac{1}{24} \right) (-1-2) = -\frac{1}{8} \\ & (-\mu, -\mu) = \frac{1}{4}, \quad (-\mu, \alpha_1) = -\frac{1}{8}, \quad (-\mu, \alpha_2) = 0. \end{aligned}$                                                                  | $(\alpha_1, \alpha_1) = B_2(H_{\alpha_1}, H_{\alpha_1}) = 8 \frac{1}{8} \frac{1}{8} (1+1) = \frac{1}{4}$ $(\alpha_2, \alpha_2) = B_2(H_{\alpha_2}, H_{\alpha_2}) = 8 \frac{1}{24} \frac{1}{24} (1+4+1) = \frac{1}{12}$ $(\alpha_1, \alpha_2) = B_2(H_{\alpha_1}, H_{\alpha_2}) = 8 \frac{1}{8} \frac{1}{24} (-1-2) = -\frac{1}{8}$ $(-\mu, -\mu) = \frac{1}{4}, \quad (-\mu, \alpha_1) = -\frac{1}{8}, \quad (-\mu, \alpha_2) = 0$ | $(\alpha_1, \alpha_1) = B_2(H_{\alpha_1}, H_{\alpha_1}) = 8 \frac{1}{8} \frac{1}{8} (1+1) = \frac{1}{4}$ $(\alpha_2, \alpha_2) = B_2(H_{\alpha_2}, H_{\alpha_2}) = 8 \frac{1}{24} \frac{1}{24} (1+4+1) = \frac{1}{12}$ $(\alpha_1, \alpha_2) = B_2(H_{\alpha_1}, H_{\alpha_2}) = 8 \frac{1}{8} \frac{1}{24} (-1-2) = -\frac{1}{8}$ $(-\mu, -\mu) = \frac{1}{4}, \quad (-\mu, \alpha_1) = -\frac{1}{8}, \quad (-\mu, \alpha_2) = 0.$ |
| 0902.0431_E00272     | 44   | ■ 0.950 | $\begin{aligned} & \left( \frac{1}{8} \right) \left( \frac{1}{8} \right) (1+1) = \frac{1}{4} \\ & \left( \frac{1}{24} \right) \left( \frac{1}{24} \right) (1+4+1) = \frac{1}{12} \\ & \left( \frac{1}{8} \right) \left( \frac{1}{24} \right) (-1-2) = -\frac{1}{8} \\ & (-\mu, -\mu) = \frac{1}{4}, \quad (-\mu, \alpha_1) = -\frac{1}{8}, \quad (-\mu, \alpha_2) = 0. \end{aligned}$                                                                  | $\begin{cases} \tilde{A}_i(a)E_i = 0 \\ \tilde{A}_i(a)E_{i+1} = -\frac{1}{2}F_i(a) \\ \tilde{A}_i(a)E_{i+2} = \frac{1}{2}F_i(a), \end{cases} \quad \begin{cases} \tilde{A}_i(a)F_i(x) = (a, x)(E_{i+1} - E_{i+2}) \\ \tilde{A}_i(a)F_{i+1}(x) = \frac{1}{2}F_{i+2}(\overline{ax}) \\ \tilde{A}_i(a)F_{i+2}(x) = -\frac{1}{2}F_{i+1}(\overline{x\overline{a}}). \end{cases}$                                                        | $\begin{cases} \tilde{A}_i(a)E_i = 0 \\ \tilde{A}_i(a)E_{i+1} = -\frac{1}{2}F_i(a) \\ \tilde{A}_i(a)E_{i+2} = \frac{1}{2}F_i(a), \end{cases} \quad \begin{cases} \tilde{A}_i(a)F_i(x) = (a, x)(E_{i+1} - E_{i+2}) \\ \tilde{A}_i(a)F_{i+1}(x) = \frac{1}{2}F_{i+2}(\overline{ax}) \\ \tilde{A}_i(a)F_{i+2}(x) = -\frac{1}{2}F_{i+1}(\overline{x\overline{a}}). \end{cases}$                                                         |
| 0902.0431_E00473     | 78   | ■ 0.951 | $\left( \phi_1, \phi_2 \right)_6 = (\delta_1, \delta_2)_4 + (T_1, T_2),$                                                                                                                                                                                                                                                                                                                                                                               | $(\phi_1, \phi_2)_6 = (\delta_1, \delta_2)_4 + (T_1, T_2),$                                                                                                                                                                                                                                                                                                                                                                        | $(\phi_1, \phi_2)_6 = (\delta_1, \delta_2)_4 + (T_1, T_2),$                                                                                                                                                                                                                                                                                                                                                                         |

| Identifier       | Page | Conf.   | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Rendered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Image                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
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| 0902.0431_E00726 | 120  | ■ 0.951 | $\begin{aligned} & \backslash[ \begin{aligned} & \mathfrak{a} \ni \left[ \Phi(0, A, B, \nu), \Phi \left( 0, 0, 0, -\frac{3}{2} \right) \right] = \Phi(0, A, -B, 0), \\ & \Phi(0, A, -B, 0), \backslash \backslash \mathfrak{a} \ni \left[ \Phi(0, A, -B, 0), \Phi \left( 0, 0, B_1, 0 \right) \right] = \Phi(2A \vee B_1, 0, 0, (A, B_1)), \\ & \Phi \left( 2A \vee B_1, 0, 0, (A, B_1) \right), \Phi(\phi, 0, 0, 0) \right] = \Phi(2[A \vee B_1 \cdot \phi], 0, 0, 0), \end{aligned} \end{aligned}$                                                                   | $\begin{aligned} & \mathfrak{a} \ni \left[ \Phi(0, A, B, \nu), \Phi \left( 0, 0, 0, -\frac{3}{2} \right) \right] = \Phi(0, A, -B, 0), \\ & \mathfrak{a} \ni [\Phi(0, A, -B, 0), \Phi(0, 0, B_1, 0)] = \Phi(2A \vee B_1, 0, 0, (A, B_1)), \\ & \mathfrak{a} \ni [\Phi(2A \vee B_1, 0, 0, (A, B_1)), \Phi(\phi, 0, 0, 0)] = \Phi(2[A \vee B_1 \cdot \phi], 0, 0, 0), \end{aligned}$                                                                                                                                        | $\begin{aligned} & \mathfrak{a} \ni \left[ \Phi(0, A, B, \nu), \Phi \left( 0, 0, 0, -\frac{3}{2} \right) \right] = \Phi(0, A, -B, 0), \\ & \mathfrak{a} \ni [\Phi(0, A, -B, 0), \Phi(0, 0, B_1, 0)] = \Phi(2A \vee B_1, 0, 0, (A, B_1)), \\ & \mathfrak{a} \ni [\Phi(2A \vee B_1, 0, 0, (A, B_1)), \Phi(\phi, 0, 0, 0)] = \Phi(2[A \vee B_1 \cdot \phi], 0, 0, 0), \end{aligned}$                                                                                                                          |
| 0902.0431_E00323 | 51   | ■ 0.953 | $B_4(\delta, \delta) = \text{tr} \left( \left( \text{ad} \tilde{A}_1(1) \right)^2 \right) = (-1) \times 7 \times 2 + \left( -\frac{1}{4} \right) \times 8 \times 2 = -18.$                                                                                                                                                                                                                                                                                                                                                                                             | $B_4(\delta, \delta) = \text{tr} \left( \left( \text{ad} \tilde{A}_1(1) \right)^2 \right) = (-1) \times 7 \times 2 + \left( -\frac{1}{4} \right) \times 8 \times 2 = -18.$                                                                                                                                                                                                                                                                                                                                               | $B_4(\delta, \delta) = \text{tr}((\text{ad} \tilde{A}_1(1))^2) = (-1) \times 7 \times 2 + \left( -\frac{1}{4} \right) \times 8 \times 2 = -18.$                                                                                                                                                                                                                                                                                                                                                            |
| 0902.0431_E00161 | 29   | ■ 0.953 | $\left( \alpha_1 x \right) \left( \alpha_2 y \right) = xy, \quad x, y \in \mathfrak{C}.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | $(\alpha_1 x)(\alpha_2 y) = xy, \quad x, y \in \mathfrak{C}.$                                                                                                                                                                                                                                                                                                                                                                                                                                                            | $(\alpha_1 x)(\alpha_2 y) = xy, \quad x, y \in \mathfrak{C}.$                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 0902.0431_E00489 | 80   | ■ 0.954 | $\begin{aligned} & \begin{aligned} & = 2 \left( \phi' A \times A, A \right) \text{ (Lemma 3.4.3.(1))} = 2(\phi' A, A \times A) = -2(A, \phi(A \times A)) \\ & = -2(\phi, (A \times A) \vee A)_6. \end{aligned} \end{aligned}$                                                                                                                                                                                                                                                                                                                                          | $\begin{aligned} & = 2(\phi' A \times A, A) \text{ (Lemma 3.4.3.(1))} = 2(\phi' A, A \times A) = -2(A, \phi(A \times A)) \\ & = -2(\phi, (A \times A) \vee A)_6. \end{aligned}$                                                                                                                                                                                                                                                                                                                                          | $\begin{aligned} & = 2(\phi' A \times A, A) \text{ (Lemma 3.4.3.(1))} = 2(\phi' A, A \times A) = -2(A, \phi(A \times A)) \\ & = -2(\phi, (A \times A) \vee A)_6. \end{aligned}$                                                                                                                                                                                                                                                                                                                            |
| 0902.0431_E00178 | 31   | ■ 0.954 | $(a \times \bar{a})(ay \bar{a}) = a(xy) \bar{a}, \quad x, y \in \mathfrak{C}.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | $(ax\bar{a})(ay\bar{a}) = a(xy)\bar{a}, \quad x, y \in \mathfrak{C}.$                                                                                                                                                                                                                                                                                                                                                                                                                                                    | $(ax\bar{a})(ay\bar{a}) = a(xy)\bar{a}, \quad x, y \in \mathfrak{C}.$                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 0902.0431_E00343 | 55   | ■ 0.954 | $B_4(H, H') = 18 \sum_{k=0}^3 \lambda_k \lambda_k', \quad H = \sum_{k=0}^3 \lambda_k H_k, H' = \sum_{k=0}^3 \lambda_k' H_k \in \mathfrak{h}_R.$                                                                                                                                                                                                                                                                                                                                                                                                                        | $B_4(H, H') = 18 \sum_{k=0}^3 \lambda_k \lambda_k', \quad H = \sum_{k=0}^3 \lambda_k H_k, H' = \sum_{k=0}^3 \lambda_k' H_k \in \mathfrak{h}_R.$                                                                                                                                                                                                                                                                                                                                                                          | $B_4(H, H') = 18 \sum_{k=0}^3 \lambda_k \lambda_k', \quad H = \sum_{k=0}^3 \lambda_k H_k, H' = \sum_{k=0}^3 \lambda_k' H_k \in \mathfrak{h}_R.$                                                                                                                                                                                                                                                                                                                                                            |
| 0902.0431_E00495 | 81   | ■ 0.956 | $\begin{aligned} & \begin{aligned} & R \left( \sum_{k=1}^3 ((\sigma_{ik} x_{ik}) y_{ki} + \bar{y}_{ki} (\bar{\sigma}_{ik} \bar{x}_{ik})) + \sum_{k=1}^3 (x_{ik} (\sigma_{ki} y_{ki}) + (\bar{\sigma}_{ki} \bar{y}_{ki}) \bar{x}_{ik}) \right) \\ & = 2R \left( \sum_k ((\sigma_{ik} x_{ik}) y_{ki} + x_{ik} (\sigma_{ki} y_{ki})) \right) \\ & = 2 \sum_k ((\sigma_{ik} x_{ik}, \bar{y}_{ki}) + (x_{ik}, \bar{\sigma}_{ki} \bar{y}_{ki})) \\ & = 2 \sum_k ((x_{ik}, -\sigma_{ik} \bar{y}_{ki}) + (x_{ik}, \sigma_{ik} \bar{y}_{ki})) = 0. \end{aligned} \end{aligned}$ | $\begin{aligned} & R \left( \sum_{k=1}^3 ((\sigma_{ik} x_{ik}) y_{ki} + \bar{y}_{ki} (\bar{\sigma}_{ik} \bar{x}_{ik})) + \sum_{k=1}^3 (x_{ik} (\sigma_{ki} y_{ki}) + (\bar{\sigma}_{ki} \bar{y}_{ki}) \bar{x}_{ik}) \right) \\ & = 2R(\sum_k ((\sigma_{ik} x_{ik}) y_{ki} + x_{ik} (\sigma_{ki} y_{ki}))) \\ & = 2 \sum_k ((\sigma_{ik} x_{ik}, \bar{y}_{ki}) + (x_{ik}, \bar{\sigma}_{ki} \bar{y}_{ki})) \\ & = 2 \sum_k ((x_{ik}, -\sigma_{ik} \bar{y}_{ki}) + (x_{ik}, \sigma_{ik} \bar{y}_{ki})) = 0. \end{aligned}$ | $\begin{aligned} & R(\sum_{k=1}^3 ((\sigma_{ik} x_{ik}) y_{ki} + \bar{y}_{ki} (\bar{\sigma}_{ik} \bar{x}_{ik})) + \sum_{k=1}^3 (x_{ik} (\sigma_{ki} y_{ki}) + (\bar{\sigma}_{ki} \bar{y}_{ki}) \bar{x}_{ik})) \\ & = 2R(\sum_k ((\sigma_{ik} x_{ik}) y_{ki} + x_{ik} (\sigma_{ki} y_{ki}))) \\ & = 2 \sum_k ((\sigma_{ik} x_{ik}, \bar{y}_{ki}) + (x_{ik}, \bar{\sigma}_{ki} \bar{y}_{ki})) \\ & = 2 \sum_k ((x_{ik}, -\sigma_{ik} \bar{y}_{ki}) + (x_{ik}, \sigma_{ik} \bar{y}_{ki})) = 0. \end{aligned}$ |

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|------------------|------|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-------|
| 0902.0431_E00226 | 38   | ■ 0.957 | $\left\{ \begin{array}{l} E_{-i} \times E_i = 0 \\ E_i \times F_i(x) = -\frac{1}{2}F_i(x) \\ F_i(x) \times F_i(y) = -(x,y)E_i, \end{array} \right. \quad \left\{ \begin{array}{l} E_i \times E_{i+1} = \frac{1}{2}E_{i+2} \\ E_i \times F_j(x) = 0, \quad i \neq j \\ F_i(x) \times F_{i+1}(y) = \frac{1}{2}F_{i+2}(\overline{xy}), \end{array} \right.$                                                                                                                                                                                                                                                                                          |          |       |
| 0902.0431_E00862 | 143  | ■ 0.957 | $\begin{aligned} & \alpha(t) \left( \begin{pmatrix} \xi_1 & x_3 & \bar{x}_2 \\ \bar{x}_3 & \xi_2 & x_1 \\ x_2 & \bar{x}_1 & \xi_3 \end{pmatrix}, \begin{pmatrix} \eta_1 & y_3 & \bar{y}_2 \\ \bar{y}_3 & \eta_2 & y_1 \\ y_2 & \bar{y}_1 & \eta_3 \end{pmatrix}, \xi, \eta \right) \\ & = \left( \begin{pmatrix} e^{2it}\xi_1 & e^{it}x_3 & e^{it}\bar{x}_2 \\ e^{it}\bar{x}_3 & \xi_2 & x_1 \\ e^{it}x_2 & \bar{x}_1 & \xi_3 \end{pmatrix}, \begin{pmatrix} e^{-2it}\eta_1 & e^{-it}y_3 & e^{-it}\bar{y}_2 \\ e^{-it}\bar{y}_3 & \eta_2 & y_1 \\ e^{-it}y_2 & \bar{y}_1 & \eta_3 \end{pmatrix}, e^{-2it}\xi, e^{2it}\eta \right), \end{aligned}$ |          |       |
| 0902.0431_E00216 | 36   | ■ 0.960 | $X=X(\xi, x)=\left(\begin{array}{ccc} \xi_1 & x_3 & \bar{x}_2 \\ \bar{x}_3 & \xi_2 & x_1 \\ x_2 & \bar{x}_1 & \xi_3 \end{array}\right), \quad \xi_i \in \boldsymbol{R}, x_i \in \mathfrak{C}.$                                                                                                                                                                                                                                                                                                                                                                                                                                                    |          |       |
| 0902.0431_E00164 | 29   | ■ 0.960 | $(\alpha_1 x)(\alpha_2 y) = \overline{\alpha_3(\overline{xy})}, \quad x, y \in \mathfrak{C}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |          |       |
| 0902.0431_E00656 | 108  | ■ 0.960 | $\langle \alpha X, \alpha M \rangle = 0 = \langle X, M \rangle, \quad \langle \alpha M, \alpha N \rangle = (\tau \alpha M, \alpha N) = (P \tau M h(A, B)^*, P N \tau h(A, B)^*) = (\tau M, N) = \langle M, N \rangle,$                                                                                                                                                                                                                                                                                                                                                                                                                            |          |       |
| 0902.0431_E01197 | 201  | ■ 0.960 | $[(C, 0), (C, D, g_1, g_2, g_{-2}, g_{-1})] = (0, 0, [C, g_1], [C, g_2], [C, g_{-2}], [C, g_{-1}]) \in \mathfrak{q} \cap \mathfrak{a} = \{0\},$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |          |       |



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| 0902.0431_E00494 | 81   | ■ 0.964 | $\begin{array}{lll} \sigma_{23}=\delta_1, & \sigma_{31}=\delta_2, & \sigma_{12}=\delta_3 \\ \sigma_{32}=\kappa\delta_1, & \sigma_{13}=\kappa\delta_2, & \sigma_{21}=\kappa\delta_3. \end{array}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | $\begin{array}{lll} \sigma_{23}=\delta_1, & \sigma_{31}=\delta_2, & \sigma_{12}=\delta_3 \\ \sigma_{32}=\kappa\delta_1, & \sigma_{13}=\kappa\delta_2, & \sigma_{21}=\kappa\delta_3. \end{array}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | $\begin{array}{lll} \sigma_{23}=\delta_1, & \sigma_{31}=\delta_2, & \sigma_{12}=\delta_3, \\ \sigma_{32}=\kappa\delta_1, & \sigma_{13}=\kappa\delta_2, & \sigma_{21}=\kappa\delta_3. \end{array}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 0902.0431_E00760 | 127  | ■ 0.964 | $\begin{array}{l} \lambda_0-\frac{1}{2}\mu_1+\frac{2}{3}\nu=1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 0 \\ \lambda_1-\frac{1}{2}\mu_1+\frac{2}{3}\nu=0 \quad 1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 0 \\ \lambda_2-\frac{1}{2}\mu_1+\frac{2}{3}\nu=0 \quad 0 \quad 1 \quad 1 \quad 1 \quad 1 \quad 0 \\ \lambda_3-\frac{1}{2}\mu_1+\frac{2}{3}\nu=0 \quad 0 \quad 0 \quad 1 \quad 1 \quad 1 \quad 0 \\ \lambda_0+\frac{1}{2}\mu_1-\frac{2}{3}\nu=1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 0 \quad 1 \\ \lambda_1+\frac{1}{2}\mu_1-\frac{2}{3}\nu=0 \quad 1 \quad 1 \quad 1 \quad 1 \quad 0 \quad 1 \\ \lambda_2+\frac{1}{2}\mu_1-\frac{2}{3}\nu=0 \quad 0 \quad 1 \quad 1 \quad 1 \quad 0 \quad 1 \\ \lambda_3+\frac{1}{2}\mu_1-\frac{2}{3}\nu=0 \quad 0 \quad 0 \quad 1 \quad 1 \quad 0 \quad 1 \end{array}$ | $\begin{array}{l} \lambda_0-\frac{1}{2}\mu_1+\frac{2}{3}\nu=1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 0 \\ \lambda_1-\frac{1}{2}\mu_1+\frac{2}{3}\nu=0 \quad 1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 0 \\ \lambda_2-\frac{1}{2}\mu_1+\frac{2}{3}\nu=0 \quad 0 \quad 1 \quad 1 \quad 1 \quad 1 \quad 0 \\ \lambda_3-\frac{1}{2}\mu_1+\frac{2}{3}\nu=0 \quad 0 \quad 0 \quad 1 \quad 1 \quad 1 \quad 0 \\ \lambda_0+\frac{1}{2}\mu_1-\frac{2}{3}\nu=1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 0 \quad 1 \\ \lambda_1+\frac{1}{2}\mu_1-\frac{2}{3}\nu=0 \quad 1 \quad 1 \quad 1 \quad 1 \quad 0 \quad 1 \\ \lambda_2+\frac{1}{2}\mu_1-\frac{2}{3}\nu=0 \quad 0 \quad 1 \quad 1 \quad 1 \quad 0 \quad 1 \\ \lambda_3+\frac{1}{2}\mu_1-\frac{2}{3}\nu=0 \quad 0 \quad 0 \quad 1 \quad 1 \quad 0 \quad 1 \end{array}$ | $\begin{array}{l} \lambda_0-\frac{1}{2}\mu_1+\frac{2}{3}\nu=1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 0 \\ \lambda_1-\frac{1}{2}\mu_1+\frac{2}{3}\nu=0 \quad 1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 0 \\ \lambda_2-\frac{1}{2}\mu_1+\frac{2}{3}\nu=0 \quad 0 \quad 1 \quad 1 \quad 1 \quad 1 \quad 0 \\ \lambda_3-\frac{1}{2}\mu_1+\frac{2}{3}\nu=0 \quad 0 \quad 0 \quad 1 \quad 1 \quad 1 \quad 0 \\ \lambda_0+\frac{1}{2}\mu_1-\frac{2}{3}\nu=1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 0 \quad 1 \\ \lambda_1+\frac{1}{2}\mu_1-\frac{2}{3}\nu=0 \quad 1 \quad 1 \quad 1 \quad 1 \quad 0 \quad 1 \\ \lambda_2+\frac{1}{2}\mu_1-\frac{2}{3}\nu=0 \quad 0 \quad 1 \quad 1 \quad 1 \quad 0 \quad 1 \\ \lambda_3+\frac{1}{2}\mu_1-\frac{2}{3}\nu=0 \quad 0 \quad 0 \quad 1 \quad 1 \quad 0 \quad 1 \end{array}$ |
| 0902.0431_E00433 | 71   | ■ 0.965 | $F_4=\left\{\alpha\in F_4\mid \tau\alpha=\alpha\tau\right\}.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | $F_4=\left\{\alpha\in F_4^C\mid \tau\alpha=\alpha\tau\right\}.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | $F_4=\left\{\alpha\in F_4^C\mid \tau\alpha=\alpha\tau\right\}.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 0902.0431_E00713 | 118  | ■ 0.965 | $\alpha\in E_7 \quad \text{if and only if} \quad \tau\lambda\alpha=\alpha\tau\lambda.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | $\alpha\in E_7 \quad \text{if and only if} \quad \tau\lambda\alpha=\alpha\tau\lambda.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | $\alpha\in E_7 \quad \text{if and only if} \quad \tau\lambda\alpha=\alpha\tau\lambda.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 0902.0431_E00511 | 84   | ■ 0.965 | $\begin{array}{l} \left[h,\widetilde{F}_{31}(a)\right]=\left(h_{\delta}F_{31}(a)\right)^{\sim}+\frac{1}{2}\left[\widetilde{H},\widetilde{F}_{31}(a)\right]=\widetilde{F}_{31}\left((\nu h_{\delta})a\right)+\frac{1}{2}\left(\mu_3-\mu_1\right)\widetilde{F}_{31}(a), \\ \left[h,\widetilde{F}_{12}(a)\right]=\left(h_{\delta}F_{12}(a)\right)^{\sim}+\frac{1}{2}\left[\widetilde{H},\widetilde{F}_{12}(a)\right]=\widetilde{F}_{12}\left((\kappa\pi h_{\delta})a\right)+\frac{1}{2}\left(\mu_1-\mu_2\right)\widetilde{F}_{12}(a), \end{array}$                                                                                                                                                                                                                                                                     | $\begin{array}{l} \left[h,\widetilde{F}_{31}(a)\right]=\left(h_{\delta}F_{31}(a)\right)^{\sim}+\frac{1}{2}\left[\widetilde{H},\widetilde{F}_{31}(a)\right]=\widetilde{F}_{31}\left((\nu h_{\delta})a\right)+\frac{1}{2}\left(\mu_3-\mu_1\right)\widetilde{F}_{31}(a), \\ \left[h,\widetilde{F}_{12}(a)\right]=\left(h_{\delta}F_{12}(a)\right)^{\sim}+\frac{1}{2}\left[\widetilde{H},\widetilde{F}_{12}(a)\right]=\widetilde{F}_{12}\left((\kappa\pi h_{\delta})a\right)+\frac{1}{2}\left(\mu_1-\mu_2\right)\widetilde{F}_{12}(a), \end{array}$                                                                                                                                                                                                                                                                     | $\begin{array}{l} \left[h,\widetilde{F}_{31}(a)\right]=\left(h_{\delta}F_{31}(a)\right)^{\sim}+\frac{1}{2}\left[\widetilde{H},\widetilde{F}_{31}(a)\right]=\widetilde{F}_{31}\left((\nu h_{\delta})a\right)+\frac{1}{2}\left(\mu_3-\mu_1\right)\widetilde{F}_{31}(a), \\ \left[h,\widetilde{F}_{12}(a)\right]=\left(h_{\delta}F_{12}(a)\right)^{\sim}+\frac{1}{2}\left[\widetilde{H},\widetilde{F}_{12}(a)\right]=\widetilde{F}_{12}\left((\kappa\pi h_{\delta})a\right)+\frac{1}{2}\left(\mu_1-\mu_2\right)\widetilde{F}_{12}(a), \end{array}$                                                                                                                                                                                                                                                                     |

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| 0902.0431_E00906 | 151  | ■ 0.966 | $\begin{gathered} \gamma \varphi(D) \gamma Y = g^{-1}(D(g(\gamma Y))D^*) \text{ is real diagonal,} \\ \varphi(D)X = \varphi(D) \left( \frac{1}{\xi} Y \times Y \right) = \frac{1}{\xi} (\gamma \varphi(D) \gamma Y \times \gamma \varphi(D) \gamma Y) \text{ is real diagonal.} \end{gathered}$ | $\gamma \varphi(D) \gamma Y = g^{-1}(D(g(\gamma Y))D^*) \text{ is real diagonal,}$ $\varphi(D)X = \varphi(D) \left( \frac{1}{\xi} Y \times Y \right) = \frac{1}{\xi} (\gamma \varphi(D) \gamma Y \times \gamma \varphi(D) \gamma Y) \text{ is real diagonal.}$ | $\gamma \varphi(D) \gamma Y = g^{-1}(D(g(\gamma Y))D^*) \text{ is real diagonal,}$ $\varphi(D)X = \varphi(D) \left( \frac{1}{\xi} Y \times Y \right) = \frac{1}{\xi} (\gamma \varphi(D) \gamma Y \times \gamma \varphi(D) \gamma Y) \text{ is real diagonal.}$ |
| 0902.0431_E00191 | 33   | ■ 0.966 | $\widetilde{\alpha} e_i = b_i, \quad i = 0, 1, \dots, 7$                                                                                                                                                                                                                                        | $\widetilde{\alpha} e_i = b_i, \quad i = 0, 1, \dots, 7$                                                                                                                                                                                                       | $\widetilde{\alpha} e_i = b_i, \quad i = 0, 1, \dots, 7$                                                                                                                                                                                                       |
| 0902.0431_E00614 | 102  | ■ 0.967 | $\begin{aligned} gX \circ gY &= g(\gamma(X \times Y)) + \frac{1}{4}(\gamma X, Y)E, \quad X, Y \in \mathfrak{J}^C \\ (gX, gY) &= (\gamma X, Y) \end{aligned}$                                                                                                                                    | $gX \circ gY = g(\gamma(X \times Y)) + \frac{1}{4}(\gamma X, Y)E, \quad X, Y \in \mathfrak{J}^C$ $(gX, gY) = (\gamma X, Y)$                                                                                                                                    | $gX \circ gY = g(\gamma(X \times Y)) + \frac{1}{4}(\gamma X, Y)E, \quad X, Y \in \mathfrak{J}^C.$ $(gX, gY) = (\gamma X, Y),$                                                                                                                                  |
| 0902.0431_E00730 | 120  | ■ 0.969 | $W \ni \Phi(0, 0, X_1, 0)(X, 0, 0, 3\eta) = (0, 0, 0, (X_1, X)),$                                                                                                                                                                                                                               | $W \ni \Phi(0, 0, X_1, 0)(X, 0, 0, 3\eta) = (0, 0, 0, (X_1, X)),$                                                                                                                                                                                              | $W \ni \Phi(0, 0, X_1, 0)(X, 0, 0, 3\eta) = (0, 0, 0, (X_1, X)),$                                                                                                                                                                                              |
| 0902.0431_E01004 | 167  | ■ 0.969 | $\begin{aligned} &\langle R_1, R_2 \rangle \\ &= (\tau \lambda \Phi_1 \lambda^{-1} \tau, \Phi_2)_7 + \langle P_1, P_2 \rangle + \langle Q_1, Q_2 \rangle + 8(\tau r_1) r_2 + 4(\tau s_1) s_2 + 4(\tau t_1) t_2. \end{aligned}$                                                                  | $\langle R_1, R_2 \rangle$ $= (\tau \lambda \Phi_1 \lambda^{-1} \tau, \Phi_2)_7 + \langle P_1, P_2 \rangle + \langle Q_1, Q_2 \rangle + 8(\tau r_1) r_2 + 4(\tau s_1) s_2 + 4(\tau t_1) t_2.$                                                                  | $\langle R_1, R_2 \rangle$ $= (\tau \lambda \Phi_1 \lambda^{-1} \tau, \Phi_2)_7 + \langle P_1, P_2 \rangle + \langle Q_1, Q_2 \rangle + 8(\tau r_1) r_2 + 4(\tau s_1) s_2 + 4(\tau t_1) t_2.$                                                                  |

| Identifier       | Page | Conf.   | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Rendered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Image                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
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| 0902.0431_E00595 | 99   | ■ 0.969 | $\begin{aligned} & 2\langle \tau^t \varphi(p, A) \tau(M + \mathbf{a}), N + \mathbf{b} \rangle \quad M + \mathbf{a}, N + \mathbf{b} \in \mathfrak{J}^C \\ &= 2\langle M + \mathbf{a}, \varphi(p, A)(N + \mathbf{b}) \rangle \\ &= 2\langle M + \mathbf{a}, k_J^{-1}(A(k_J(N))^t A) + p\mathbf{b}k^{-1}(\tau^t A) \rangle \\ &= 2\langle M, k_J^{-1}(A(k_J(N))^t A) \rangle + 4\langle \mathbf{a}, p\mathbf{b}k^{-1}(\tau^t A) \rangle \\ &= \langle k_J M, A(k_J(N))^t A \rangle + 2\langle k(\bar{p}\mathbf{a}), (k\mathbf{b})\tau^t A \rangle \\ &= \langle \tau^t A(k_J(M))\tau A, k_J(N) \rangle + 2\langle k(\bar{p}\mathbf{a})A, k\mathbf{b} \rangle \\ &= 2\langle k_J^{-1}(\tau^t A(k_J(M))\tau A), N \rangle + 4\langle \bar{p}\mathbf{a}k^{-1}(A), \mathbf{b} \rangle \\ &= 2\langle k_J^{-1}(\tau^t A(k_J(M))\tau A + \bar{p}\mathbf{a}k^{-1}(A)), N + \mathbf{b} \rangle \\ &= 2\langle \varphi(\bar{p}, \tau^t A)(M + \mathbf{a}), N + \mathbf{b} \rangle, \end{aligned}$ | $\begin{aligned} & 2\langle \tau^t \varphi(p, A) \tau(M + \mathbf{a}), N + \mathbf{b} \rangle \quad M + \mathbf{a}, N + \mathbf{b} \in \mathfrak{J}^C \\ &= 2\langle M + \mathbf{a}, \varphi(p, A)(N + \mathbf{b}) \rangle \\ &= 2\langle M + \mathbf{a}, k_J^{-1}(A(k_J(N))^t A) + p\mathbf{b}k^{-1}(\tau^t A) \rangle \\ &= 2\langle M, k_J^{-1}(A(k_J(N))^t A) \rangle + 4\langle \mathbf{a}, p\mathbf{b}k^{-1}(\tau^t A) \rangle \\ &= \langle k_J M, A(k_J(N))^t A \rangle + 2\langle k(\bar{p}\mathbf{a}), (k\mathbf{b})\tau^t A \rangle \\ &= \langle \tau^t A(k_J(M))\tau A, k_J(N) \rangle + 2\langle k(\bar{p}\mathbf{a})A, k\mathbf{b} \rangle \\ &= 2\langle k_J^{-1}(\tau^t A(k_J(M))\tau A), N \rangle + 4\langle \bar{p}\mathbf{a}k^{-1}(A), \mathbf{b} \rangle \\ &= 2\langle k_J^{-1}(\tau^t A(k_J(M))\tau A + \bar{p}\mathbf{a}k^{-1}(A)), N + \mathbf{b} \rangle \\ &= 2\langle \varphi(\bar{p}, \tau^t A)(M + \mathbf{a}), N + \mathbf{b} \rangle, \end{aligned}$ | $\begin{aligned} & 2\langle \tau^t \varphi(p, A) \tau(M + \mathbf{a}), N + \mathbf{b} \rangle \quad M + \mathbf{a}, N + \mathbf{b} \in \mathfrak{J}^C \\ &= 2\langle M + \mathbf{a}, \varphi(p, A)(N + \mathbf{b}) \rangle \\ &= 2\langle M + \mathbf{a}, k_J^{-1}(A(k_J(N))^t A) + p\mathbf{b}k^{-1}(\tau^t A) \rangle \\ &= 2\langle M, k_J^{-1}(A(k_J(N))^t A) \rangle + 4\langle \mathbf{a}, p\mathbf{b}k^{-1}(\tau^t A) \rangle \\ &= \langle k_J M, A(k_J(N))^t A \rangle + 2\langle k(\bar{p}\mathbf{a}), (k\mathbf{b})\tau^t A \rangle \\ &= \langle \tau^t A(k_J(M))\tau A, k_J(N) \rangle + 2\langle k(\bar{p}\mathbf{a})A, k\mathbf{b} \rangle \\ &= 2\langle k_J^{-1}(\tau^t A(k_J(M))\tau A), N \rangle + 4\langle \bar{p}\mathbf{a}k^{-1}(A), \mathbf{b} \rangle \\ &= 2\langle k_J^{-1}(\tau^t A(k_J(M))\tau A + \bar{p}\mathbf{a}k^{-1}(A)), N + \mathbf{b} \rangle \\ &= 2\langle \varphi(\bar{p}, \tau^t A)(M + \mathbf{a}), N + \mathbf{b} \rangle, \end{aligned}$ |
| 0902.0431_E00253 | 41   | ■ 0.969 | $0(27)=0(\frac{J}{\alpha})=\left\{\alpha\in\mathrm{Iso}_{\mathbf{R}}(\mathfrak{J})\mid(\alpha X,\alpha Y)=(X,Y)\right\}\quad\text{(Lemma 2.2.4.(2)).}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | $O(27) = O(\mathfrak{J}) = \{\alpha \in \mathrm{Iso}_{\mathbf{R}}(\mathfrak{J}) \mid (\alpha X, \alpha Y) = (X, Y)\} \quad (\text{Lemma 2.2.4.(2)}).$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | $O(27) = O(\mathfrak{J}) = \{\alpha \in \mathrm{Iso}_{\mathbf{R}}(\mathfrak{J}) \mid (\alpha X, \alpha Y) = (X, Y)\} \quad (\text{Lemma 2.2.4.(2)}).$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 0902.0431_E00997 | 166  | ■ 0.970 | $(R \times R)R_1 = \Theta(R)^2 R_1 + \frac{1}{30} B_8(R, R_1) R, \quad R_1 \in \mathfrak{e}_8^C,$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | $(R \times R)R_1 = \Theta(R)^2 R_1 + \frac{1}{30} B_8(R, R_1) R, \quad R_1 \in \mathfrak{e}_8^C,$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | $(R \times R)R_1 = \Theta(R)^2 R_1 + \frac{1}{30} B_8(R, R_1) R, \quad R_1 \in \mathfrak{e}_8^C,$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

| Identifier       | Page | Conf.   | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Rendered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Image                                                                                                                                                                                                                       |
|------------------|------|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0902.0431_E01016 | 170  | ■ 0.971 | $\begin{gathered} \pm \frac{1}{2}(\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3) \pm \left(\frac{1}{2}\mu_2 + \frac{1}{3}\nu\right) \pm r, \\ \pm \frac{1}{2}(\lambda_0 + \lambda_1 - \lambda_2 + \lambda_3) \pm \left(\frac{1}{2}\mu_2 + \frac{1}{3}\nu\right) \pm r, \\ \pm \frac{1}{2}(\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3) \pm \left(\frac{1}{2}\mu_3 + \frac{1}{3}\nu\right) \pm r \\ \pm \frac{1}{2}(\lambda_0 + \lambda_1 - \lambda_2 - \lambda_3) \pm \left(\frac{1}{2}\mu_3 + \frac{1}{3}\nu\right) \pm r, \\ \pm \frac{1}{2}(\lambda_0 - \lambda_1 + \lambda_2 + \lambda_3) \pm \left(\frac{1}{2}\mu_3 + \frac{1}{3}\nu\right) \pm r, \\ \pm \frac{1}{2}(\lambda_0 - \lambda_1 - \lambda_2 + \lambda_3) \pm \left(\frac{1}{2}\mu_3 + \frac{2}{3}\nu\right) \pm r, \\ \pm 2r, \\ \pm \nu \pm r \end{gathered}$ | $\begin{gathered} \pm \frac{1}{2}(\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3) \pm \left(\frac{1}{2}\mu_2 + \frac{1}{3}\nu\right) \pm r, \\ \pm \frac{1}{2}(\lambda_0 + \lambda_1 - \lambda_2 + \lambda_3) \pm \left(\frac{1}{2}\mu_2 + \frac{1}{3}\nu\right) \pm r, \\ \pm \frac{1}{2}(\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3) \pm \left(\frac{1}{2}\mu_3 + \frac{1}{3}\nu\right) \pm r \\ \pm \frac{1}{2}(\lambda_0 + \lambda_1 - \lambda_2 - \lambda_3) \pm \left(\frac{1}{2}\mu_3 + \frac{1}{3}\nu\right) \pm r, \\ \pm \frac{1}{2}(\lambda_0 - \lambda_1 + \lambda_2 + \lambda_3) \pm \left(\frac{1}{2}\mu_3 + \frac{1}{3}\nu\right) \pm r, \\ \pm \frac{1}{2}(\lambda_0 - \lambda_1 - \lambda_2 + \lambda_3) \pm \left(\frac{1}{2}\mu_3 + \frac{2}{3}\nu\right) \pm r, \\ \pm 2r, \\ \pm \nu \pm r \end{gathered}$ |                                                                                                                                                                                                                             |
| 0902.0431_E00230 | 38   | ■ 0.971 | $\begin{aligned} & (\operatorname{tr}(\alpha X) \wedge - \\ & \operatorname{tr}(X)) X^2 + \frac{1}{2} \left( \operatorname{tr}(X)^2 - \operatorname{tr}(\alpha X)^2 + \operatorname{tr}((\alpha X)^2) - \operatorname{tr}(X^2) \right) X \\ & = (\det X - \det(\alpha X)) E \end{aligned}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | $(\operatorname{tr}(\alpha X) - \operatorname{tr}(X))X^2 + \frac{1}{2} \left( \operatorname{tr}(X)^2 - \operatorname{tr}(\alpha X)^2 + \operatorname{tr}((\alpha X)^2) - \operatorname{tr}(X^2) \right) X$ $= (\det X - \det(\alpha X)) E$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | $(\operatorname{tr}(\alpha X) - \operatorname{tr}(X))X^2 + \frac{1}{2}(\operatorname{tr}(X)^2 - \operatorname{tr}(\alpha X)^2 + \operatorname{tr}((\alpha X)^2) - \operatorname{tr}(X^2))X$ $= (\det X - \det(\alpha X))E.$ |
| 0902.0431_E00944 | 157  | ■ 0.971 | $\overline{f^{-1}(A(fP))}=f^{-1}\left(\left(\overline{\left(\operatorname{Ad} J_3\right) A} f \bar{P}\right)\right), \quad A \in S U(6), P \in \left(\mathfrak{P}_C\right)^C .$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | $\overline{f^{-1}(A(fP))}=f^{-1}\left(\left(\overline{\left(\operatorname{Ad} J_3\right) A} f \bar{P}\right)\right), \quad A \in S U(6), P \in \left(\mathfrak{P}_C\right)^C .$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | $\overline{f^{-1}(A(fP))}=f^{-1}(\overline{((\operatorname{Ad} J_3) A f \bar{P})}), \quad A \in S U(6), P \in \left(\mathfrak{P}_C\right)^C .$                                                                              |
| 0902.0431_E00979 | 163  | ■ 0.974 | $\mathfrak{a} \ni \left[\Phi_1,(\Phi, P, Q, r, s, t)\right]=\left(\left[\Phi_1, \Phi\right], \Phi_1 P, \Phi_1 Q, 0,0,0\right),$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | $\mathfrak{a} \ni \left[\Phi_1,(\Phi, P, Q, r, s, t)\right]=\left(\left[\Phi_1, \Phi\right], \Phi_1 P, \Phi_1 Q, 0,0,0\right),$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | $\mathfrak{a} \ni \left[\Phi_1,(\Phi, P, Q, r, s, t)\right]=\left(\left[\Phi_1, \Phi\right], \Phi_1 P, \Phi_1 Q, 0,0,0\right),$                                                                                             |

| Identifier       | Page | Conf.   | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Rendered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Image                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|------------------|------|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0902.0431_E08751 | 124  | ■ 0.975 | $\begin{aligned} & \left[ \Phi(h_\delta + \tilde{H}, 0, 0, \nu), \Phi(0, E_j, 0, 0) \right] \\ &= \Phi\left(0, \left(h_\delta + \tilde{H} + \frac{2}{3}\nu\right)E_j, 0, 0\right) = \left(\mu_j + \frac{2}{3}\nu\right)\Phi(0, E_j, 0, 0), \\ & \left[ \Phi\left((h_\delta + \tilde{H}, 0, 0, \nu), \Phi(0, 0, E_j, 0)\right) = \Phi\left(0, 0, \left((h_\delta + \tilde{H})' - \frac{2}{3}\nu\right)E_j, 0\right) \right. \\ & \quad \left. = \Phi\left(0, 0, \left(h_\delta - \tilde{H} - \frac{2}{3}\nu\right)E_j, 0\right) = \left(-\mu_j - \frac{2}{3}\nu\right)\Phi(0, 0, E_j, 0). \right. \end{aligned}$ | $\begin{aligned} & [\Phi(h_\delta + \tilde{H}, 0, 0, \nu), \Phi(0, E_j, 0, 0)] \\ &= \Phi\left(0, \left(h_\delta + \tilde{H} + \frac{2}{3}\nu\right)E_j, 0, 0\right) = \left(\mu_j + \frac{2}{3}\nu\right)\Phi(0, E_j, 0, 0), \\ & \left[ \Phi\left((h_\delta + \tilde{H}, 0, 0, \nu), \Phi(0, 0, E_j, 0)\right) = \Phi\left(0, 0, \left((h_\delta + \tilde{H})' - \frac{2}{3}\nu\right)E_j, 0\right) \right. \\ & \quad \left. = \Phi\left(0, 0, \left(h_\delta - \tilde{H} - \frac{2}{3}\nu\right)E_j, 0\right) = \left(-\mu_j - \frac{2}{3}\nu\right)\Phi(0, 0, E_j, 0). \right. \end{aligned}$ | $\begin{aligned} & [\Phi(h_\delta + \tilde{H}, 0, 0, \nu), \Phi(0, E_j, 0, 0)] \\ &= \Phi\left(0, \left(h_\delta + \tilde{H} + \frac{2}{3}\nu\right)E_j, 0, 0\right) = \left(\mu_j + \frac{2}{3}\nu\right)\Phi(0, E_j, 0, 0), \\ & [\Phi((h_\delta + \tilde{H}, 0, 0, \nu), \Phi(0, 0, E_j, 0))] = \Phi\left(0, 0, \left((h_\delta + \tilde{H})' - \frac{2}{3}\nu\right)E_j, 0\right) \\ &= \Phi\left(0, 0, \left(h_\delta - \tilde{H} - \frac{2}{3}\nu\right)E_j, 0\right) = \left(-\mu_j - \frac{2}{3}\nu\right)\Phi(0, 0, E_j, 0). \end{aligned}$ |
| 0902.0431_E08894 | 148  | ■ 0.975 | $\mathfrak{S}(8, \boldsymbol{C}) = \{S \in M(8, \boldsymbol{C}) \mid {}^t S = -S\},$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | $\mathfrak{S}(8, \boldsymbol{C}) = \{S \in M(8, \boldsymbol{C}) \mid {}^t S = -S\},$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | $\mathfrak{S}(8, \boldsymbol{C}) = \{S \in M(8, \boldsymbol{C}) \mid {}^t S = -S\},$                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 0902.0431_E08608 | 101  | ■ 0.976 | $\begin{aligned} & \boldsymbol{P}_3 \in \mathfrak{J}(4, \boldsymbol{H}) \mid P^2 = P, \operatorname{tr}(P) = 1 \\ &= \{AE_1 A^* \mid A \in \operatorname{Sp}(4), E_1 = \operatorname{diag}(1, 0, 0, 0) \in M(4, \boldsymbol{H})\}. \end{aligned}$                                                                                                                                                                                                                                                                                                                                                               | $\begin{aligned} \boldsymbol{HP}_3 &= \{P \in \mathfrak{J}(4, \boldsymbol{H}) \mid P^2 = P, \operatorname{tr}(P) = 1\} \\ &= \{AE_1 A^* \mid A \in \operatorname{Sp}(4), E_1 = \operatorname{diag}(1, 0, 0, 0) \in M(4, \boldsymbol{H})\}. \end{aligned}$                                                                                                                                                                                                                                                                                                                                          | $\begin{aligned} \boldsymbol{HP}_3 &= \{P \in \mathfrak{J}(4, \boldsymbol{H}) \mid P^2 = P, \operatorname{tr}(P) = 1\} \\ &= \{AE_1 A^* \mid A \in \operatorname{Sp}(4), E_1 = \operatorname{diag}(1, 0, 0, 0) \in M(4, \boldsymbol{H})\}. \end{aligned}$                                                                                                                                                                                                                                                                                            |
| 0902.0431_E08813 | 136  | ■ 0.976 | $\widetilde{\iota}(\alpha) = \iota \alpha \iota^{-1}, \quad \alpha \in E_7.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | $\widetilde{\iota}(\alpha) = \iota \alpha \iota^{-1}, \quad \alpha \in E_7.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | $\widetilde{\iota}(\alpha) = \iota \alpha \iota^{-1}, \quad \alpha \in E_7.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 0902.0431_E08414 | 68   | ■ 0.976 | $M \times N = \left( \begin{array}{ccc} \boldsymbol{m}_2 \times \boldsymbol{n}_3 & \boldsymbol{m}_3 \times \boldsymbol{n}_1 & \boldsymbol{m}_1 \times \boldsymbol{n}_2 \\ + & + & + \\ \boldsymbol{n}_2 \times \boldsymbol{m}_3 & \boldsymbol{n}_3 \times \boldsymbol{m}_1 & \boldsymbol{n}_1 \times \boldsymbol{m}_2 \end{array} \right) \in M(3, \boldsymbol{C}),$                                                                                                                                                                                                                                            | $M \times N = \left( \begin{array}{ccc} \boldsymbol{m}_2 \times \boldsymbol{n}_3 & \boldsymbol{m}_3 \times \boldsymbol{n}_1 & \boldsymbol{m}_1 \times \boldsymbol{n}_2 \\ + & + & + \\ \boldsymbol{n}_2 \times \boldsymbol{m}_3 & \boldsymbol{n}_3 \times \boldsymbol{m}_1 & \boldsymbol{n}_1 \times \boldsymbol{m}_2 \end{array} \right) \in M(3, \boldsymbol{C}),$                                                                                                                                                                                                                               | $M \times N = \left( \begin{array}{ccc} \boldsymbol{m}_2 \times \boldsymbol{n}_3 & \boldsymbol{m}_3 \times \boldsymbol{n}_1 & \boldsymbol{m}_1 \times \boldsymbol{n}_2 \\ + & + & + \\ \boldsymbol{n}_2 \times \boldsymbol{m}_3 & \boldsymbol{n}_3 \times \boldsymbol{m}_1 & \boldsymbol{n}_1 \times \boldsymbol{m}_2 \end{array} \right) \in M(3, \boldsymbol{C}),$                                                                                                                                                                                 |
| 0902.0431_E01088 | 167  | ■ 0.976 | $\begin{aligned} & E_8 = \{\alpha \in E_8^C \mid \langle \alpha R_1, \alpha R_2 \rangle = \langle R_1, R_2 \rangle\} \\ &= \{\alpha \in E_8^C \mid \tau \tilde{\lambda} \alpha = \alpha \tau \tilde{\lambda}\}. \end{aligned}$                                                                                                                                                                                                                                                                                                                                                                                  | $\begin{aligned} E_8 &= \{\alpha \in E_8^C \mid \langle \alpha R_1, \alpha R_2 \rangle = \langle R_1, R_2 \rangle\} \\ &= \{\alpha \in E_8^C \mid \tau \tilde{\lambda} \alpha = \alpha \tau \tilde{\lambda}\}. \end{aligned}$                                                                                                                                                                                                                                                                                                                                                                      | $\begin{aligned} E_8 &= \{\alpha \in E_8^C \mid \langle \alpha R_1, \alpha R_2 \rangle = \langle R_1, R_2 \rangle\} \\ &= \{\alpha \in E_8^C \mid \tau \tilde{\lambda} \alpha = \alpha \tau \tilde{\lambda}\}. \end{aligned}$                                                                                                                                                                                                                                                                                                                        |
| 0902.0431_E08088 | 18   | ■ 0.976 | $B_2(D_1, D_2) = 4 \operatorname{tr}(D_1 D_2), \quad D_i \in \mathfrak{g}_2^C.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | $B_2(D_1, D_2) = 4 \operatorname{tr}(D_1 D_2), \quad D_i \in \mathfrak{g}_2^C.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | $B_2(D_1, D_2) = 4 \operatorname{tr}(D_1 D_2), \quad D_i \in \mathfrak{g}_2^C.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

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|------------------|------|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0902.0431_E00669 | 111  | ■ 0.977 | $\begin{aligned} (P, Q) &= (X, Z) + (Y, W) + \xi \zeta + \eta \omega, \\ \langle P, Q \rangle &= \langle X, Z \rangle + \langle Y, W \rangle + (\tau \xi) \zeta + (\tau \eta) \omega, \\ \{P, Q\} &= (X, W) - (Z, Y) + \xi \omega - \zeta \eta, \end{aligned}$                                                                  | $\begin{aligned} (P, Q) &= (X, Z) + (Y, W) + \xi \zeta + \eta \omega, \\ \langle P, Q \rangle &= \langle X, Z \rangle + \langle Y, W \rangle + (\tau \xi) \zeta + (\tau \eta) \omega, \\ \{P, Q\} &= (X, W) - (Z, Y) + \xi \omega - \zeta \eta, \end{aligned}$                                                                 | $\begin{aligned} (P, Q) &= (X, Z) + (Y, W) + \xi \zeta + \eta \omega, \\ \langle P, Q \rangle &= \langle X, Z \rangle + \langle Y, W \rangle + (\tau \xi) \zeta + (\tau \eta) \omega, \\ \{P, Q\} &= (X, W) - (Z, Y) + \xi \omega - \zeta \eta, \end{aligned}$                                                                  |
| 0902.0431_E00684 | 113  | ■ 0.977 | $\mathfrak{e}_7^C = \left\{ \Phi(\phi, A, B, \nu) \in \text{Hom}_C(\mathfrak{P}^C) \mid \phi \in \mathfrak{e}_6^C, A, B \in \mathfrak{J}^C, \nu \in C \right\}.$                                                                                                                                                                | $\mathfrak{e}_7^C = \left\{ \Phi(\phi, A, B, \nu) \in \text{Hom}_C(\mathfrak{P}^C) \mid \phi \in \mathfrak{e}_6^C, A, B \in \mathfrak{J}^C, \nu \in C \right\}.$                                                                                                                                                               | $\mathfrak{e}_7^C = \left\{ \Phi(\phi, A, B, \nu) \in \text{Hom}_C(\mathfrak{P}^C) \mid \phi \in \mathfrak{e}_6^C, A, B \in \mathfrak{J}^C, \nu \in C \right\}.$                                                                                                                                                                |
| 0902.0431_E00948 | 157  | ■ 0.979 | $((X, Y, \xi, \eta), (M, N)) = (X + M, Y + N, \xi, \eta).$                                                                                                                                                                                                                                                                      | $((X, Y, \xi, \eta), (M, N)) = (X + M, Y + N, \xi, \eta).$                                                                                                                                                                                                                                                                     | $((X, Y, \xi, \eta), (M, N)) = (X + M, Y + N, \xi, \eta).$                                                                                                                                                                                                                                                                      |
| 0902.0431_E00994 | 19   | ■ 0.979 | $\begin{array}{l} f_*(A) = \varepsilon A - \bar{\varepsilon}^t A, \quad \varepsilon = \frac{1}{2}(1 + ie_1) \\ \varphi_*(D)(a + \mathbf{m}) = D\mathbf{m}, \quad a + \mathbf{m} \in \mathbf{C}^C \oplus (\mathbf{C}^3)^C = \mathfrak{C}^C \end{array}$                                                                          | $\begin{array}{l} f_*(A) = \varepsilon A - \bar{\varepsilon}^t A, \quad \varepsilon = \frac{1}{2}(1 + ie_1) \\ \varphi_*(D)(a + \mathbf{m}) = D\mathbf{m}, \quad a + \mathbf{m} \in \mathbf{C}^C \oplus (\mathbf{C}^3)^C = \mathfrak{C}^C \end{array}$                                                                         | $\begin{array}{l} f_*(A) = \varepsilon A - \bar{\varepsilon}^t A, \quad \varepsilon = \frac{1}{2}(1 + ie_1), \\ \varphi_*(D)(a + \mathbf{m}) = D\mathbf{m}, \quad a + \mathbf{m} \in \mathbf{C}^C \oplus (\mathbf{C}^3)^C = \mathfrak{C}^C, \end{array}$                                                                        |
| 0902.0431_E00381 | 63   | ■ 0.979 | $\begin{aligned} \alpha E_2 &= \alpha(-F_2(1) \times F_2(1)) = -\alpha F_2(1) \times \alpha F_2(1) \\ &= -(F_2(x_2) + F_3(x_3)) \times (F_2(x_2) + F_3(x_3)) \quad (\text{for some } x_2, x_3 \in \mathfrak{C}) \\ &= (x_2, x_2)E_2 + (x_3, x_3)E_3 - F_1(\overline{x_2 x_3}) \in \mathfrak{J}(2, \mathfrak{C}). \end{aligned}$ | $\begin{aligned} \alpha E_2 &= \alpha(-F_2(1) \times F_2(1)) = -\alpha F_2(1) \times \alpha F_2(1) \\ &= -(F_2(x_2) + F_3(x_3)) \times (F_2(x_2) + F_3(x_3)) \quad (\text{for some } x_2, x_3 \in \mathfrak{C}) \\ &= (x_2, x_2)E_2 + (x_3, x_3)E_3 - F_1(\overline{x_2 x_3}) \in \mathfrak{J}(2, \mathfrak{C}) \end{aligned}$ | $\begin{aligned} \alpha E_2 &= \alpha(-F_2(1) \times F_2(1)) = -\alpha F_2(1) \times \alpha F_2(1) \\ &= -(F_2(x_2) + F_3(x_3)) \times (F_2(x_2) + F_3(x_3)) \quad (\text{for some } x_2, x_3 \in \mathfrak{C}) \\ &= (x_2, x_2)E_2 + (x_3, x_3)E_3 - F_1(\overline{x_2 x_3}) \in \mathfrak{J}(2, \mathfrak{C}). \end{aligned}$ |
| 0902.0431_E01171 | 197  | ■ 0.979 | $\begin{aligned} &\exp(\text{ad}(X, 0, 0))(D, \mathbf{u}, \mathbf{v}) = (\exp(\text{ad}(X)D, (\exp X)\mathbf{u}, (\exp(-^t X))\mathbf{v})) \\ &= (\text{Ad}(\exp X)D, (\exp X)\mathbf{u}, {}^t(\exp X)^{-1}\mathbf{v}) \\ &= \varphi(\exp X)(D, \mathbf{u}, \mathbf{v}). \end{aligned}$                                         | $\begin{aligned} \exp(\text{ad}(X, 0, 0))(D, \mathbf{u}, \mathbf{v}) &= (\exp(\text{ad}(X)D, (\exp X)\mathbf{u}, (\exp(-^t X))\mathbf{v})) \\ &= (\text{Ad}(\exp X)D, (\exp X)\mathbf{u}, {}^t(\exp X)^{-1}\mathbf{v}) \\ &= \varphi(\exp X)(D, \mathbf{u}, \mathbf{v}). \end{aligned}$                                        | $\begin{aligned} \exp(\text{ad}(X, 0, 0))(D, \mathbf{u}, \mathbf{v}) &= (\exp(\text{ad}(X)D, (\exp X)\mathbf{u}, (\exp(-^t X))\mathbf{v})) \\ &= (\text{Ad}(\exp X)D, (\exp X)\mathbf{u}, {}^t(\exp X)^{-1}\mathbf{v}) \\ &= \varphi(\exp X)(D, \mathbf{u}, \mathbf{v}). \end{aligned}$                                         |
| 0902.0431_E00632 | 105  | ■ 0.979 | $\alpha = \varphi(A)\beta = \varphi(A)\varphi(B) = \varphi(AB), \quad AB \in \text{Sp}(4),$                                                                                                                                                                                                                                     | $\alpha = \varphi(A)\beta = \varphi(A)\varphi(B) = \varphi(AB), \quad AB \in \text{Sp}(4),$                                                                                                                                                                                                                                    | $\alpha = \varphi(A)\beta = \varphi(A)\varphi(B) = \varphi(AB), \quad AB \in \text{Sp}(4),$                                                                                                                                                                                                                                     |

| Identifier       | Page | Conf.   | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Rendered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Image                                                                                                                                                                                                                                                                                                                                                                                                    |
|------------------|------|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0902.0431_E00599 | 100  | ■ 0.980 | $\begin{aligned} & (\alpha \boldsymbol{a})^* (\alpha \boldsymbol{b}) = (\boldsymbol{p} \boldsymbol{a} k^{-1} (\tau^t A))^* (\boldsymbol{p} \boldsymbol{b} k^{-1} (\tau^t A)) \\ & = k^{-1} (A) \boldsymbol{a}^* \boldsymbol{b} k^{-1} (\tau^t A), \\ & \tau \alpha \tau (\boldsymbol{a}^* \boldsymbol{b}) = \tau \left( k_J^{-1} \left( A (k_J \tau (\boldsymbol{a}^* \boldsymbol{b}))^t A \right) \right) \\ & = -\tau k^{-1} \left( J \tau \left( A (k (\tau (\boldsymbol{a}^* \boldsymbol{b}))) J^t A \right) \right) \\ & = -\tau k^{-1} \left( J \tau \left( A J \tau k (\boldsymbol{a}^* \boldsymbol{b})^t A \right) \right) \\ & = -\tau k^{-1} \left( J \tau A J k (\boldsymbol{a}^* \boldsymbol{b}) \tau^t A \right) \\ & = -k^{-1} \left( A \tau k (\boldsymbol{a}^* \boldsymbol{b}) J^t A J \right) = k^{-1} (A) \boldsymbol{a}^* \boldsymbol{b} k^{-1} (\tau^t A), \\ & (\alpha \boldsymbol{a}) (\alpha N) = (\boldsymbol{p} \boldsymbol{a} k^{-1} (\tau^t A)) k_J^{-1} \left( A (k_J (N))^t A \right) \\ & = -\boldsymbol{p} \boldsymbol{a} k^{-1} (\tau^t A) k^{-1} \left( A (k(N)) J^t A J \right) \\ & = -\boldsymbol{p} \boldsymbol{a} N k^{-1} (J^t A J) = \boldsymbol{p} \boldsymbol{a} N \tau k^{-1} (\tau^t A), \\ & \tau \alpha \tau (\boldsymbol{a} N) = \tau (p \tau (\boldsymbol{a} N) k^{-1} (\tau^t A)) = \boldsymbol{p} \boldsymbol{a} N \tau k^{-1} (\tau^t A). \end{aligned}$ | $\begin{aligned} & (\alpha \boldsymbol{a})^* (\alpha \boldsymbol{b}) = (\boldsymbol{p} \boldsymbol{a} k^{-1} (\tau^t A))^* (\boldsymbol{p} \boldsymbol{b} k^{-1} (\tau^t A)) \\ & = k^{-1} (A) \boldsymbol{a}^* \boldsymbol{b} k^{-1} (\tau^t A), \\ & \tau \alpha \tau (\boldsymbol{a}^* \boldsymbol{b}) = \tau \left( k_J^{-1} \left( A (k_J \tau (\boldsymbol{a}^* \boldsymbol{b}))^t A \right) \right) \\ & = -\tau k^{-1} \left( J \tau \left( A (k (\tau (\boldsymbol{a}^* \boldsymbol{b}))) J^t A \right) \right) \\ & = -\tau k^{-1} \left( J \tau \left( A J \tau k (\boldsymbol{a}^* \boldsymbol{b})^t A \right) \right) \\ & = -\tau k^{-1} \left( J \tau A J k (\boldsymbol{a}^* \boldsymbol{b}) \tau^t A \right) \\ & = -k^{-1} \left( A \tau k (\boldsymbol{a}^* \boldsymbol{b}) J^t A J \right) = k^{-1} (A) \boldsymbol{a}^* \boldsymbol{b} k^{-1} (\tau^t A), \\ & (\alpha \boldsymbol{a}) (\alpha N) = (\boldsymbol{p} \boldsymbol{a} k^{-1} (\tau^t A)) k_J^{-1} \left( A (k_J (N))^t A \right) \\ & = -\boldsymbol{p} \boldsymbol{a} k^{-1} (\tau^t A) k^{-1} \left( A (k(N)) J^t A J \right) \\ & = -\boldsymbol{p} \boldsymbol{a} N k^{-1} (J^t A J) = \boldsymbol{p} \boldsymbol{a} N \tau k^{-1} (\tau^t A), \\ & \tau \alpha \tau (\boldsymbol{a} N) = \tau (p \tau (\boldsymbol{a} N) k^{-1} (\tau^t A)) = \boldsymbol{p} \boldsymbol{a} N \tau k^{-1} (\tau^t A). \end{aligned}$ |                                                                                                                                                                                                                                                                                                                                                                                                          |
| 0902.0431_E01089 | 185  | ■ 0.980 | $\begin{aligned} & \left( \mathfrak{e}_8 \right)^{\tilde{\lambda} \gamma} = \left\{ \Theta \in \Theta \left( \mathfrak{e}_8^C \right) \mid \tilde{\lambda} \gamma \Theta = \Theta \tilde{\lambda} \gamma \right\} \\ & = \left\{ \Theta (\Phi, P, -\lambda \gamma P, 0, s, -s) \mid \Phi \in (\mathfrak{e}_7)^{\lambda \gamma}, P \in (\mathfrak{P}^C)_{\lambda \gamma}, s \in \boldsymbol{R} \right\} \end{aligned}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | $\begin{aligned} & (\mathfrak{e}_8)^{\tilde{\lambda} \gamma} = \left\{ \Theta \in \Theta \left( \mathfrak{e}_8^C \right) \mid \tilde{\lambda} \gamma \Theta = \Theta \tilde{\lambda} \gamma \right\} \\ & = \left\{ \Theta (\Phi, P, -\lambda \gamma P, 0, s, -s) \mid \Phi \in (\mathfrak{e}_7)^{\lambda \gamma}, P \in (\mathfrak{P}^C)_{\lambda \gamma}, s \in \boldsymbol{R} \right\} \end{aligned}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | $\begin{aligned} & (\mathfrak{e}_8)^{\tilde{\lambda} \gamma} = \left\{ \Theta \in \Theta \left( \mathfrak{e}_8^C \right) \mid \tilde{\lambda} \gamma \Theta = \Theta \tilde{\lambda} \gamma \right\} \\ & = \left\{ \Theta (\Phi, P, -\lambda \gamma P, 0, s, -s) \mid \Phi \in (\mathfrak{e}_7)^{\lambda \gamma}, P \in (\mathfrak{P}^C)_{\lambda \gamma}, s \in \boldsymbol{R} \right\} \end{aligned}$ |

| Identifier       | Page | Conf.   | LaTeX source                                                                                                                                                                                                                                                                                                                                                                      | Rendered                                                                                                                                                                                                                                                                                                                                                         | Image                                                                                                                                                                                                                                                                                                                                                              |
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| 0902.0431_E00506 | 83   | ■ 0.980 | $\mathfrak{h}=\left\{\begin{array}{l} h=h_{\delta}+\widetilde{H} \in \mathfrak{e}_6^C \\ \text{\textit{H}} \in \mathfrak{e}_6^C \end{array}\right\} \quad \left\{ \begin{array}{l} h_{\delta}=\sum_{k=0}^3 \lambda_k H_k=-\sum_{k=0}^3 \lambda_k i G_{k4+k}, \lambda_k \in C \\ \text{\textit{H}}=\sum_{j=1}^3 \mu_j E_j, \mu_j \in C, \mu_1+\mu_2+\mu_3=0 \end{array} \right\},$ | $\mathfrak{h}=\left\{ \begin{array}{l} h=h_{\delta}+\widetilde{H} \in \mathfrak{e}_6^C \\ \text{\textit{H}}=\sum_{j=1}^3 \mu_j E_j, \mu_j \in C, \mu_1+\mu_2+\mu_3=0 \end{array} \right\},$                                                                                                                                                                      | $\mathfrak{h}=\left\{ \begin{array}{l} h=h_{\delta}+\widetilde{H} \in \mathfrak{e}_6^C \\ \text{\textit{H}}=\sum_{j=1}^3 \mu_j E_j, \mu_j \in C, \mu_1+\mu_2+\mu_3=0 \end{array} \right\},$                                                                                                                                                                        |
| 0902.0431_E00831 | 139  | ■ 0.980 | $\begin{aligned} \left(E_7\right)^{\kappa, \mu} &=\left\{\alpha \in E_7 \mid \kappa \alpha=\alpha \kappa, \mu \alpha=\alpha \mu\right\}, \\ \left(\left(E_7\right)^{\kappa, \mu}\right)_{(0, E_1, 0, 1)} &=\left\{\alpha \in\left(E_7\right)^{\kappa, \mu} \mid \alpha\left(0, E_1, 0, 1\right)=\left(0, E_1, 0, 1\right)\right\} . \end{aligned}$                                | $\begin{aligned} \left(E_7\right)^{\kappa, \mu} &=\left\{\alpha \in E_7 \mid \kappa \alpha=\alpha \kappa, \mu \alpha=\alpha \mu\right\}, \\ \left(\left(E_7\right)^{\kappa, \mu}\right)_{(0, E_1, 0, 1)} &=\left\{\alpha \in\left(E_7\right)^{\kappa, \mu} \mid \alpha\left(0, E_1, 0, 1\right)=\left(0, E_1, 0, 1\right)\right\} . \end{aligned}$               | $\begin{aligned} \left(E_7\right)^{\kappa, \mu} &=\left\{\alpha \in E_7 \mid \kappa \alpha=\alpha \kappa, \mu \alpha=\alpha \mu\right\}, \\ \left(\left(E_7\right)^{\kappa, \mu}\right)_{(0, E_1, 0, 1)} &=\left\{\alpha \in\left(E_7\right)^{\kappa, \mu} \mid \alpha\left(0, E_1, 0, 1\right)=\left(0, E_1, 0, 1\right)\right\} . \end{aligned}$                 |
| 0902.0431_E01132 | 191  | ■ 0.981 | $\begin{aligned} \operatorname{tr}\left(A D_{\{1\}} \tau^t A\right) &=\operatorname{tr}\left(A D_{\{1\}} \tau^t A\right) \\ \operatorname{tr}\left(A D_{\{1\}} \tau^t A\right) &=\operatorname{tr}\left(A D_{\{1\}} \tau^t A\right) \\ \operatorname{tr}\left(A D_{\{1\}} \tau^t A\right) &=\operatorname{tr}\left(A D_{\{1\}} \tau^t A\right) \end{aligned}$                     | $\operatorname{tr}\left(A D_1 \tau^t A\left(\tau^t\left(A D_2 \tau^t A\right)\right)\right)=\operatorname{tr}\left(A D_1\left(\tau^t D_2\right) A^{-1}\right)=\operatorname{tr}\left(D_1\left(\tau^t D_2\right)\right),$ $\langle A \boldsymbol{X}, A \boldsymbol{Y}\rangle=\langle \boldsymbol{X}, \boldsymbol{Y}\rangle,$                                      | $\operatorname{tr}\left(A D_1 \tau^t A\left(\tau^t\left(A D_2 \tau^t A\right)\right)\right)=\operatorname{tr}\left(A D_1\left(\tau^t D_2\right) A^{-1}\right)=\operatorname{tr}\left(D_1\left(\tau^t D_2\right)\right),$ $\langle A \boldsymbol{X}, A \boldsymbol{Y}\rangle=\langle \boldsymbol{X}, \boldsymbol{Y}\rangle,$                                        |
| 0902.0431_E00384 | 64   | ■ 0.981 | $\begin{aligned} \alpha \sigma X &=\alpha \sigma\left(X_1+X_2\right) \quad X_1 \in \mathfrak{J}_{\sigma}, \quad X_2 \in \mathfrak{J}_{-\sigma} \\ &=\alpha\left(X_1-X_2\right)=\alpha X_1-\alpha X_2=\sigma\left(\alpha X_1\right)+\sigma\left(\alpha X_2\right) \\ &=\sigma \alpha\left(X_1+X_2\right)=\sigma \alpha X, \quad X \in \mathfrak{J} . \end{aligned}$                | $\begin{aligned} \alpha \sigma X &=\alpha \sigma\left(X_1+X_2\right) \quad X_1 \in \mathfrak{J}_{\sigma}, \quad X_2 \in \mathfrak{J}_{-\sigma} \\ &=\alpha\left(X_1-X_2\right)=\alpha X_1-\alpha X_2=\sigma\left(\alpha X_1\right)+\sigma\left(\alpha X_2\right) \\ &=\sigma \alpha\left(X_1+X_2\right)=\sigma \alpha X, \quad X \in \mathfrak{J} \end{aligned}$ | $\begin{aligned} \alpha \sigma X &=\alpha \sigma\left(X_1+X_2\right) \quad X_1 \in \mathfrak{J}_{\sigma}, \quad X_2 \in \mathfrak{J}_{-\sigma} \\ &=\alpha\left(X_1-X_2\right)=\alpha X_1-\alpha X_2=\sigma\left(\alpha X_1\right)+\sigma\left(\alpha X_2\right) \\ &=\sigma \alpha\left(X_1+X_2\right)=\sigma \alpha X, \quad X \in \mathfrak{J} . \end{aligned}$ |
| 0902.0431_E00641 | 106  | ■ 0.981 | $\begin{aligned} h(a, b) &=\frac{a+b}{2}+i \frac{a-b}{2} e_1=\iota a+\bar{\iota} b, \\ h(A, B) &=\frac{A+B}{2}+i \frac{A-B}{2} e_1=\iota A+\bar{\iota} B, \end{aligned}$                                                                                                                                                                                                          | $\begin{aligned} h(a, b) &=\frac{a+b}{2}+i \frac{a-b}{2} e_1=\iota a+\bar{\iota} b, \\ h(A, B) &=\frac{A+B}{2}+i \frac{A-B}{2} e_1=\iota A+\bar{\iota} B, \end{aligned}$                                                                                                                                                                                         | $\begin{aligned} h(a, b) &=\frac{a+b}{2}+i \frac{a-b}{2} e_1=\iota a+\bar{\iota} b, \\ h(A, B) &=\frac{A+B}{2}+i \frac{A-B}{2} e_1=\iota A+\bar{\iota} B, \end{aligned}$                                                                                                                                                                                           |
| 0902.0431_E00474 | 78   | ■ 0.981 | $\left(\left[\phi, \phi_1\right], \phi_2\right)_6+\left(\phi_1,\left[\phi, \phi_2\right]\right)_6=0, \quad \phi, \phi_i \in \mathfrak{e}_6^C .$                                                                                                                                                                                                                                   | $\left(\left[\phi, \phi_1\right], \phi_2\right)_6+\left(\phi_1,\left[\phi, \phi_2\right]\right)_6=0, \quad \phi, \phi_i \in \mathfrak{e}_6^C .$                                                                                                                                                                                                                  | $\left(\left[\phi, \phi_1\right], \phi_2\right)_6+\left(\phi_1,\left[\phi, \phi_2\right]\right)_6=0, \quad \phi, \phi_i \in \mathfrak{e}_6^C .$                                                                                                                                                                                                                    |
| 0902.0431_E01173 | 197  | ■ 0.982 | $\mathfrak{e}_8^C=\mathfrak{g}_0 \oplus \mathfrak{g}_1 \oplus \mathfrak{g}_2 \oplus \mathfrak{g}_{-2} \oplus \mathfrak{g}_{-1},$                                                                                                                                                                                                                                                  | $\mathfrak{e}_8^C=\mathfrak{g}_0 \oplus \mathfrak{g}_1 \oplus \mathfrak{g}_2 \oplus \mathfrak{g}_{-2} \oplus \mathfrak{g}_{-1},$                                                                                                                                                                                                                                 | $\mathfrak{e}_8^C=\mathfrak{g}_0 \oplus \mathfrak{g}_1 \oplus \mathfrak{g}_2 \oplus \mathfrak{g}_{-2} \oplus \mathfrak{g}_{-1},$                                                                                                                                                                                                                                   |



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| 0902.0431_E00123 | 24   | ■0.985 | $\overline{\omega_{\{1\}} \overline{\boldsymbol{m}}} = \overline{\gamma_{\{1\}}(w \boldsymbol{m})} = \overline{\gamma_{\{1\}}(w \boldsymbol{m})} = \overline{\gamma_{\{1\}}(w \boldsymbol{m})}$<br>for all $\boldsymbol{m} \in \boldsymbol{C}^3$ ,                                                                                                                                                                                              | $\omega_1 \overline{\boldsymbol{m}} = w(\gamma_1 \boldsymbol{m}) = \gamma_1(w \boldsymbol{m}) = \overline{\omega_1 \boldsymbol{m}} = \overline{\omega_1 \boldsymbol{m}}$ for all $\boldsymbol{m} \in \boldsymbol{C}^3$ ,                                                                                                                                                                                               | $\omega_1 \overline{\boldsymbol{m}} = w(\gamma_1 \boldsymbol{m}) = \gamma_1(w \boldsymbol{m}) = \overline{\omega_1 \boldsymbol{m}} = \overline{\omega_1 \boldsymbol{m}}$ for all $\boldsymbol{m} \in \boldsymbol{C}^3$ ,                                                                                                                                                                     |
| 0902.0431_E00740 | 122  | ■0.985 | $B_{\{7\}}(\Phi_1, \Phi_2) = k(\Phi_1, \Phi_2)_7 = k' \operatorname{tr}(\Phi_1 \Phi_2)$                                                                                                                                                                                                                                                                                                                                                         | $B_7(\Phi_1, \Phi_2) = k(\Phi_1, \Phi_2)_7 = k' \operatorname{tr}(\Phi_1 \Phi_2)$ .                                                                                                                                                                                                                                                                                                                                    | $B_7(\Phi_1, \Phi_2) = k(\Phi_1, \Phi_2)_7 = k' \operatorname{tr}(\Phi_1 \Phi_2)$ .                                                                                                                                                                                                                                                                                                          |
| 0902.0431_E01007 | 167  | ■0.985 | $\begin{gathered} \sqrt{2} \left[ \widetilde{E}_1, \widetilde{F}_2(e_i) \right], \quad \sqrt{2} \left[ \widetilde{E}_1, \widetilde{F}_3(e_i) \right], \quad \sqrt{2} \left[ \widetilde{E}_3, \widetilde{F}_1(e_i) \right], \quad 0 \leq i \leq 7, \\ \frac{1}{\sqrt{2}} \left[ \widetilde{F}_1(e_i), \widetilde{F}_1(e_j) \right], \quad 0 \leq i < j \leq 7 \end{gathered}$                                                                    | $\sqrt{2} \left[ \widetilde{E}_1, \widetilde{F}_2(e_i) \right], \quad \sqrt{2} \left[ \widetilde{E}_1, \widetilde{F}_3(e_i) \right], \quad \sqrt{2} \left[ \widetilde{E}_3, \widetilde{F}_1(e_i) \right], \quad 0 \leq i \leq 7, \\ \frac{1}{\sqrt{2}} \left[ \widetilde{F}_1(e_i), \widetilde{F}_1(e_j) \right], \quad 0 \leq i < j \leq 7$                                                                           | $\sqrt{2} [\widetilde{E}_1, \widetilde{F}_2(e_i)], \quad \sqrt{2} [\widetilde{E}_1, \widetilde{F}_3(e_i)], \quad \sqrt{2} [\widetilde{E}_3, \widetilde{F}_1(e_i)], \quad 0 \leq i \leq 7, \\ \frac{1}{\sqrt{2}} [\widetilde{F}_1(e_i), \widetilde{F}_1(e_j)], \quad 0 \leq i < j \leq 7$                                                                                                     |
| 0902.0431_E01214 | 204  | ■0.985 | $\operatorname{Ker} \varphi = \{(E, E), (\zeta E, \zeta^2 E), (\zeta^2 E, \zeta^4 E), (\zeta^3 E, \zeta E), (\zeta^4 E, \zeta^3 E)\} = \boldsymbol{Z}_5$ .                                                                                                                                                                                                                                                                                      | $\operatorname{Ker} \varphi = \{(E, E), (\zeta E, \zeta^2 E), (\zeta^2 E, \zeta^4 E), (\zeta^3 E, \zeta E), (\zeta^4 E, \zeta^3 E)\} = \boldsymbol{Z}_5$ .                                                                                                                                                                                                                                                             | $\operatorname{Ker} \varphi = \{(E, E), (\zeta E, \zeta^2 E), (\zeta^2 E, \zeta^4 E), (\zeta^3 E, \zeta E), (\zeta^4 E, \zeta^3 E)\} = \boldsymbol{Z}_5$ .                                                                                                                                                                                                                                   |
| 0902.0431_E00780 | 131  | ■0.985 | $\beta_1(X \times X) = \beta X \times \beta X = \tau \beta \tau(X \times X)$ ,                                                                                                                                                                                                                                                                                                                                                                  | $\beta_1(X \times X) = \beta X \times \beta X = \tau \beta \tau(X \times X)$ ,                                                                                                                                                                                                                                                                                                                                         | $\beta_1(X \times X) = \beta X \times \beta X = \tau \beta \tau(X \times X)$ ,                                                                                                                                                                                                                                                                                                               |
| 0902.0431_E00152 | 28   | ■0.985 | $\begin{aligned} G_{\{2\}} \cap O(8) &\simeq \left( G_{\{2\}} \cap O(8) \right) \times \boldsymbol{R}^d \\ &= \left( (G_2^C)^{\tau \gamma} \right)^\gamma \times \boldsymbol{R}^d = \left( (G_2^C)^\tau \right)^\gamma \times \boldsymbol{R}^d = (G_2)^\gamma \times \boldsymbol{R}^d \\ &= (\operatorname{Sp}(1) \times \operatorname{Sp}(1)) / \boldsymbol{Z}_2 \times \boldsymbol{R}^d \text{ (Theorem 1.10.1)}, \quad d = 8. \end{aligned}$ | $\begin{aligned} G_{2(2)} &\simeq (G_{2(2)} \cap O(8)) \times \boldsymbol{R}^d \\ &= \left( (G_2^C)^{\tau \gamma} \right)^\gamma \times \boldsymbol{R}^d = \left( (G_2^C)^\tau \right)^\gamma \times \boldsymbol{R}^d = (G_2)^\gamma \times \boldsymbol{R}^d \\ &= (\operatorname{Sp}(1) \times \operatorname{Sp}(1)) / \boldsymbol{Z}_2 \times \boldsymbol{R}^d \text{ (Theorem 1.10.1)}, \quad d = 8. \end{aligned}$ | $\begin{aligned} G_{2(2)} &\simeq (G_{2(2)} \cap O(8)) \times \boldsymbol{R}^d \\ &= ((G_2^C)^{\tau \gamma})^\gamma \times \boldsymbol{R}^d = ((G_2^C)^\tau)^\gamma \times \boldsymbol{R}^d = (G_2)^\gamma \times \boldsymbol{R}^d \\ &= (\operatorname{Sp}(1) \times \operatorname{Sp}(1)) / \boldsymbol{Z}_2 \times \boldsymbol{R}^d \text{ (Theorem 1.10.1)}, \quad d = 8. \end{aligned}$ |

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| 0902.0431_E08529 | 88   | ■ 0.985 | $\begin{aligned} & \left\{ \begin{array}{l} \eta_1 = \xi_1 \\ \eta_2 = \frac{\xi_2 - \xi_3}{2} + \frac{\xi_2 + \xi_3}{2} \cos  a  + i \frac{(a, x_1)}{ a } \sin  a  \\ \eta_3 = -\frac{\xi_2 - \xi_3}{2} + \frac{\xi_2 + \xi_3}{2} \cos  a  + i \frac{(a, x_1)}{ a } \sin  a , \end{array} \right. \\ & \left\{ \begin{array}{l} y_1 = x_1 + i \frac{(\xi_2 + \xi_3)a}{2 a } \sin  a  - \frac{2(a, x_1)a}{ a ^2} \sin^2 \frac{ a }{2} \\ y_2 = x_2 \cos \frac{ a }{2} + i \frac{\overline{x_3 a}}{ a } \sin \frac{ a }{2} \\ y_3 = x_3 \cos \frac{ a }{2} + i \frac{\overline{a x_2}}{ a } \sin \frac{ a }{2} \end{array} \right. \end{aligned}$ | $\left\{ \begin{array}{l} \eta_1 = \xi_1 \\ \eta_2 = \frac{\xi_2 - \xi_3}{2} + \frac{\xi_2 + \xi_3}{2} \cos  a  + i \frac{(a, x_1)}{ a } \sin  a  \\ \eta_3 = -\frac{\xi_2 - \xi_3}{2} + \frac{\xi_2 + \xi_3}{2} \cos  a  + i \frac{(a, x_1)}{ a } \sin  a  \end{array} \right.$ $\left\{ \begin{array}{l} y_1 = x_1 + i \frac{(\xi_2 + \xi_3)a}{2 a } \sin  a  - \frac{2(a, x_1)a}{ a ^2} \sin^2 \frac{ a }{2} \\ y_2 = x_2 \cos \frac{ a }{2} + i \frac{\overline{x_3 a}}{ a } \sin \frac{ a }{2} \\ y_3 = x_3 \cos \frac{ a }{2} + i \frac{\overline{a x_2}}{ a } \sin \frac{ a }{2} \end{array} \right.$ | $\left\{ \begin{array}{l} \eta_1 = \xi_1 \\ \eta_2 = \frac{\xi_2 - \xi_3}{2} + \frac{\xi_2 + \xi_3}{2} \cos  a  + i \frac{(a, x_1)}{ a } \sin  a  \\ \eta_3 = -\frac{\xi_2 - \xi_3}{2} + \frac{\xi_2 + \xi_3}{2} \cos  a  + i \frac{(a, x_1)}{ a } \sin  a , \end{array} \right.$ $\left\{ \begin{array}{l} y_1 = x_1 + i \frac{(\xi_2 + \xi_3)a}{2 a } \sin  a  - \frac{2(a, x_1)a}{ a ^2} \sin^2 \frac{ a }{2} \\ y_2 = x_2 \cos \frac{ a }{2} + i \frac{\overline{x_3 a}}{ a } \sin \frac{ a }{2} \\ y_3 = x_3 \cos \frac{ a }{2} + i \frac{\overline{a x_2}}{ a } \sin \frac{ a }{2} \end{array} \right.$ |
| 0902.0431_E08314 | 50   | ■ 0.986 | $\begin{aligned} & \begin{aligned} & \left( [\widetilde{A}, \widetilde{B}]C, D \right) = (A \circ (B \circ C), D) - (B \circ (A \circ C), D) \\ & = (B \circ C, A \circ D) - (A \circ C, B \circ D) \\ & = (C \circ (D \circ A), B) - (D \circ (C \circ A), B) = ([\widetilde{C}, \widetilde{D}]A, B). \end{aligned} \end{aligned}$                                                                                                                                                                                                                                                                                                              | $([\widetilde{A}, \widetilde{B}]C, D) = (A \circ (B \circ C), D) - (B \circ (A \circ C), D)$ $= (B \circ C, A \circ D) - (A \circ C, B \circ D)$ $= (C \circ (D \circ A), B) - (D \circ (C \circ A), B) = ([\widetilde{C}, \widetilde{D}]A, B).$                                                                                                                                                                                                                                                                                                                                                             | $([\widetilde{A}, \widetilde{B}]C, D) = (A \circ (B \circ C), D) - (B \circ (A \circ C), D)$ $= (B \circ C, A \circ D) - (A \circ C, B \circ D)$ $= (C \circ (D \circ A), B) - (D \circ (C \circ A), B) = ([\widetilde{C}, \widetilde{D}]A, B).$                                                                                                                                                                                                                                                                                                                                                              |
| 0902.0431_E08449 | 73   | ■ 0.986 | $\dim_C(\mathfrak{e}_6^C) = 52 + 26 = 78.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | $\dim_C(\mathfrak{e}_6^C) = 52 + 26 = 78.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | $\dim_C(\mathfrak{e}_6^C) = 52 + 26 = 78.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 0902.0431_E01090 | 185  | ■ 0.986 | $\zeta(l(D) + l(S)I + l(icE)) = \Theta(\varphi_* D, 2\lambda\gamma\chi^{-1}S, 2\chi^{-1}S, 0, 2c, -2c),$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | $\zeta(l(D) + l(S)I + l(icE)) = \Theta(\varphi_* D, 2\lambda\gamma\chi^{-1}S, 2\chi^{-1}S, 0, 2c, -2c),$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | $\zeta(l(D) + l(S)I + l(icE)) = \Theta(\varphi_* D, 2\lambda\gamma\chi^{-1}S, 2\chi^{-1}S, 0, 2c, -2c),$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 0902.0431_E08718 | 118  | ■ 0.987 | $\left\{ \begin{array}{l} \phi = [\phi_1, \phi_2] - 2A_1 \vee \tau A_2 + 2A_2 \vee \tau A_1 \\ A = \left( \phi_1 + \frac{2}{3}\nu_1 \right) A_2 - \left( \phi_2 + \frac{2}{3}\nu_2 \right) A_1 \\ \nu = \langle A_1, A_2 \rangle - \langle A_2, A_1 \rangle. \end{array} \right.$                                                                                                                                                                                                                                                                                                                                                                | $\left\{ \begin{array}{l} \phi = [\phi_1, \phi_2] - 2A_1 \vee \tau A_2 + 2A_2 \vee \tau A_1 \\ A = \left( \phi_1 + \frac{2}{3}\nu_1 \right) A_2 - \left( \phi_2 + \frac{2}{3}\nu_2 \right) A_1 \\ \nu = \langle A_1, A_2 \rangle - \langle A_2, A_1 \rangle. \end{array} \right.$                                                                                                                                                                                                                                                                                                                            | $\left\{ \begin{array}{l} \phi = [\phi_1, \phi_2] - 2A_1 \vee \tau A_2 + 2A_2 \vee \tau A_1 \\ A = \left( \phi_1 + \frac{2}{3}\nu_1 \right) A_2 - \left( \phi_2 + \frac{2}{3}\nu_2 \right) A_1 \\ \nu = \langle A_1, A_2 \rangle - \langle A_2, A_1 \rangle. \end{array} \right.$                                                                                                                                                                                                                                                                                                                             |
| 0902.0431_E08055 | 13   | ■ 0.988 | $\mathfrak{g}_2 = \{D \in \mathfrak{D}_4 \mid \nu D = D, \pi D = D\}$ $= \{D \in \mathfrak{D}_4 \mid \lambda D = D, \lambda \in \mathfrak{S}_3\}$ $= \{D \in \mathfrak{b}_3 \mid \pi D = D\}.$                                                                                                                                                                                                                                                                                                                                                                                                                                                   | $\mathfrak{g}_2 = \{D \in \mathfrak{D}_4 \mid \nu D = D, \pi D = D\}$ $= \{D \in \mathfrak{D}_4 \mid \lambda D = D, \lambda \in \mathfrak{S}_3\}$ $= \{D \in \mathfrak{b}_3 \mid \pi D = D\}.$                                                                                                                                                                                                                                                                                                                                                                                                               | $\mathfrak{g}_2 = \{D \in \mathfrak{D}_4 \mid \nu D = D, \pi D = D\}$ $= \{D \in \mathfrak{D}_4 \mid \lambda D = D, \lambda \in \mathfrak{S}_3\}$ $= \{D \in \mathfrak{b}_3 \mid \pi D = D\}.$                                                                                                                                                                                                                                                                                                                                                                                                                |

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| 0902.0431_EQ1152 | 194  | ■ 0.988 | $\text{ (ii) } \begin{aligned} &x_{\{k\}}\widetilde{u}_{\{j\}k}\boldsymbol{u}_{\{j\}}-(\operatorname{det}U)\boldsymbol{x}=\sum_{i,j,k}x_{\{k\}}\widetilde{u}_{\{j\}k}u_{\{j\}i}\boldsymbol{e}_{\{i\}}-(\operatorname{det}U)\boldsymbol{x} \\ &=\sum_{j,k}x_{\{k\}}(\operatorname{det}U)\boldsymbol{\delta}_{\{k\}i}-(\operatorname{det}U)\boldsymbol{x}=\boldsymbol{0} . \end{aligned}$ | $\begin{aligned} \text{(ii)} \quad &= \sum_{j,k} x_k \widetilde{u}_{jk} \boldsymbol{u}_j - (\det U) \boldsymbol{x} = \sum_{i,j,k} x_k \widetilde{u}_{jk} u_{ji} \boldsymbol{e}_i - (\det U) \boldsymbol{x} \\ &= \sum_{j,k} x_k (\det U) \delta_{ki} \boldsymbol{e}_i - (\det U) \boldsymbol{x} = \boldsymbol{0}. \end{aligned}$ | $\begin{aligned} \text{(ii)} \quad &= \sum_{j,k} x_k \widetilde{u}_{jk} \boldsymbol{u}_j - (\det U) \boldsymbol{x} = \sum_{i,j,k} x_k \widetilde{u}_{jk} u_{ji} \boldsymbol{e}_i - (\det U) \boldsymbol{x} \\ &= \sum_{j,k} x_k (\det U) \delta_{ki} \boldsymbol{e}_i - (\det U) \boldsymbol{x} = \boldsymbol{0}. \end{aligned}$ |
| 0902.0431_EQ1022 | 170  | ■ 0.988 | $[(\Phi(h),\,\,0,0,\,\,r,\,\,0,0), (0,0,\,\,Q,\,\,0,0,0)]=(0,0,(\Phi(h)-r)Q,0,0,0),$                                                                                                                                                                                                                                                                                                    | $[(\Phi(h),0,0,r,0,0),(0,0,Q,0,0,0)]=(0,0,(\Phi(h)-r)Q,0,0,0),$                                                                                                                                                                                                                                                                  | $[(\Phi(h),0,0,r,0,0),(0,0,Q,0,0,0)]=(0,0,(\Phi(h)-r)Q,0,0,0),$                                                                                                                                                                                                                                                                  |
| 0902.0431_EQ1066 | 181  | ■ 0.988 | $\widetilde{\lambda}\gamma(\Phi,P,Q,,s,t)=\left(\lambda\gamma\Phi\gamma\lambda^{-1},\lambda\gamma Q,-\lambda\gamma P,-r,-t,-s\right).$                                                                                                                                                                                                                                                  | $\tilde{\lambda}\gamma(\Phi,P,Q,,s,t)=\left(\lambda\gamma\Phi\gamma\lambda^{-1},\lambda\gamma Q,-\lambda\gamma P,-r,-t,-s\right).$                                                                                                                                                                                               | $\tilde{\lambda}\gamma(\Phi,P,Q,,s,t)=\left(\lambda\gamma\Phi\gamma\lambda^{-1},\lambda\gamma Q,-\lambda\gamma P,-r,-t,-s\right).$                                                                                                                                                                                               |
| 0902.0431_EQ0033 | 10   | ■ 0.988 | $\pi L_{e_i}=\pi(2F_{i0})=-2\pi F_{0i}=-2G_{0i}(\text{ Lemma 1.3.1 })=2G_{i0}=T_{e_i},$                                                                                                                                                                                                                                                                                                 | $\pi L_{e_i}=\pi(2F_{i0})=-2\pi F_{0i}=-2G_{0i}(\text{ Lemma 1.3.1 })=2G_{i0}=T_{e_i},$                                                                                                                                                                                                                                          | $\pi L_{e_i}=\pi(2F_{i0})=-2\pi F_{0i}=-2G_{0i}(\text{ Lemma 1.3.1 })=2G_{i0}=T_{e_i},$                                                                                                                                                                                                                                          |
| 0902.0431_EQ0762 | 128  | ■ 0.989 | $\mathfrak{h}_{\boldsymbol{R}}=\left\{\Phi\left(\sum_{k=0}^3\lambda_kH_k+\left(\sum_{j=1}^3\mu_jE_j\right)^{\sim},0,0,\nu\right)\in\mathfrak{e}_7^C\left \begin{array}{l}\lambda_k,\nu\in\boldsymbol{R}\\\mu_j\in\boldsymbol{R},\mu_1+\mu_2+\mu_3=0\end{array}\right.\right\}$                                                                                                          | $\mathfrak{h}_{\boldsymbol{R}}=\left\{\Phi\left(\sum_{k=0}^3\lambda_kH_k+\left(\sum_{j=1}^3\mu_jE_j\right)^{\sim},0,0,\nu\right)\in\mathfrak{e}_7^C\left \begin{array}{l}\lambda_k,\nu\in\boldsymbol{R}\\\mu_j\in\boldsymbol{R},\mu_1+\mu_2+\mu_3=0\end{array}\right.\right\}$                                                   | $\mathfrak{h}_{\boldsymbol{R}}=\left\{\Phi\left(\sum_{k=0}^3\lambda_kH_k+\left(\sum_{j=1}^3\mu_jE_j\right)^{\sim},0,0,\nu\right)\in\mathfrak{e}_7^C\left \begin{array}{l}\lambda_k,\nu\in\boldsymbol{R}\\\mu_j\in\boldsymbol{R},\mu_1+\mu_2+\mu_3=0\end{array}\right.\right\}$                                                   |
| 0902.0431_EQ0269 | 44   | ■ 0.989 | $\begin{aligned} ([A,X],XX) &= (AX,XX)-(XA,XX) \\ &= (A,X(XX))-(A,(XX)X) \quad (\text{ Lemma 2.3.5 }) \\ &= (A,[X,XX])=(A,aE)=\frac{1}{2}\operatorname{tr}(Aa+\overline{a}A^*) \\ &= \frac{1}{2}\operatorname{tr}(Aa-aA)=\frac{1}{2}(\operatorname{tr}(A)a-a\operatorname{tr}(A))=0 . \end{aligned}$                                                                                    | $\begin{aligned} ([A,X],XX) &= (AX,XX)-(XA,XX) \\ &= (A,X(XX))-(A,(XX)X) \quad (\text{ Lemma 2.3.5 }) \\ &= (A,[X,XX])=(A,aE)=\frac{1}{2}\operatorname{tr}(Aa+\overline{a}A^*) \\ &= \frac{1}{2}\operatorname{tr}(Aa-aA)=\frac{1}{2}(\operatorname{tr}(A)a-a\operatorname{tr}(A))=0. \end{aligned}$                              | $\begin{aligned} ([A,X],XX) &= (AX,XX)-(XA,XX) \\ &= (A,X(XX))-(A,(XX)X) \quad (\text{ Lemma 2.3.5 }) \\ &= (A,[X,XX])=(A,aE)=\frac{1}{2}\operatorname{tr}(Aa+\overline{a}A^*) \\ &= \frac{1}{2}\operatorname{tr}(Aa-aA)=\frac{1}{2}(\operatorname{tr}(A)a-a\operatorname{tr}(A))=0. \end{aligned}$                              |
| 0902.0431_EQ0893 | 148  | ■ 0.989 | $\left\{\begin{array}{l}\phi=\left[\phi_1,\phi_2\right]-2A_1\vee\gamma A_2+2A_2\vee\gamma A_1,\\ A=\phi_1A_2-\phi_2A_1.\end{array}\right.$                                                                                                                                                                                                                                              | $\left\{\begin{array}{l}\phi=\left[\phi_1,\phi_2\right]-2A_1\vee\gamma A_2+2A_2\vee\gamma A_1,\\ A=\phi_1A_2-\phi_2A_1.\end{array}\right.$                                                                                                                                                                                       | $\left\{\begin{array}{l}\phi=\left[\phi_1,\phi_2\right]-2A_1\vee\gamma A_2+2A_2\vee\gamma A_1,\\ A=\phi_1A_2-\phi_2A_1.\end{array}\right.$                                                                                                                                                                                       |
| 0902.0431_EQ0267 | 43   | ■ 0.989 | $\left\{\begin{array}{l}([A,X],Y)+(X,[A,Y])=0,\quad X,Y\in\mathfrak{J}\\ \operatorname{tr}([A,X],Y,Z)+\operatorname{tr}(X,[A,Y],Z)+\operatorname{tr}(X,Y,[A,Z])=0,\quad X,Y,Z\in\mathfrak{J}\end{array}\right.$                                                                                                                                                                         | $\left\{\begin{array}{l}([A,X],Y)+(X,[A,Y])=0,\quad X,Y\in\mathfrak{J}\\ \operatorname{tr}([A,X],Y,Z)+\operatorname{tr}(X,[A,Y],Z)+\operatorname{tr}(X,Y,[A,Z])=0,\quad X,Y,Z\in\mathfrak{J}\end{array}\right.$                                                                                                                  | $\left\{\begin{array}{l}([A,X],Y)+(X,[A,Y])=0,\quad X,Y\in\mathfrak{J},\\ \operatorname{tr}([A,X],Y,Z)+\operatorname{tr}(X,[A,Y],Z)+\operatorname{tr}(X,Y,[A,Z])=0,\quad X,Y,Z\in\mathfrak{J}.\end{array}\right.$                                                                                                                |
| 0902.0431_EQ0195 | 33   | ■ 0.990 | $\alpha(xy)=(\widetilde{\alpha}x)\overline{(\widetilde{\alpha}y)},\quad x,y\in\mathfrak{C}.$                                                                                                                                                                                                                                                                                            | $\alpha(xy)=(\widetilde{\alpha}x)\overline{(\widetilde{\alpha}y)},\quad x,y\in\mathfrak{C}.$                                                                                                                                                                                                                                     | $\alpha(xy)=(\widetilde{\alpha}x)\overline{(\widetilde{\alpha}y)},\quad x,y\in\mathfrak{C}.$                                                                                                                                                                                                                                     |

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| 0902.0431_E00409 | 67   | ■ 0.990 | $\begin{aligned} & G_{ij}, \quad 0 \leq i < j \leq 3, 4 \leq i < j \leq 7, \\ & \widetilde{A}_k(e_j), \quad 0 \leq j \leq 3, k = 1, 2, 3 \end{aligned}$                                                                                                                                                          | $G_{ij}, \quad 0 \leq i < j \leq 3, 4 \leq i < j \leq 7, \\ \widetilde{A}_k(e_j), \quad 0 \leq j \leq 3, k = 1, 2, 3$                                                                                                                                                          | $G_{ij}, \quad 0 \leq i < j \leq 3, 4 \leq i < j \leq 7, \\ \widetilde{A}_k(e_j), \quad 0 \leq j \leq 3, k = 1, 2, 3$                                                                       |
| 0902.0431_E00508 | 98   | ■ 0.990 | $\mathfrak{e}_{\{6\}} = \mathfrak{f}_{\{4\}} + i \left( \widetilde{\mathfrak{J}}_{\{H\}} \right)_0.$                                                                                                                                                                                                             | $\mathfrak{e}_{6,H} = \mathfrak{f}_{4,H} \oplus i \left( \widetilde{\mathfrak{J}}_H \right)_0.$                                                                                                                                                                                | $\mathfrak{e}_{6,H} = \mathfrak{f}_{4,H} \oplus i \left( \widetilde{\mathfrak{J}}_H \right)_0.$                                                                                             |
| 0902.0431_E00748 | 124  | ■ 0.990 | $\mathfrak{e}_{\{7\}}^{\{C\}} = \mathfrak{e}_{\{6\}}^{\{C\}} \oplus \mathfrak{J}^C \oplus \mathfrak{J}^C \oplus C.$                                                                                                                                                                                              | $\mathfrak{e}_7^C = \mathfrak{e}_6^C \oplus \mathfrak{J}^C \oplus \mathfrak{J}^C \oplus C.$                                                                                                                                                                                    | $\mathfrak{e}_7^C = \mathfrak{e}_6^C \oplus \mathfrak{J}^C \oplus \mathfrak{J}^C \oplus C.$                                                                                                 |
| 0902.0431_E00750 | 124  | ■ 0.990 | $\begin{aligned} & [h, S] = \left[ \Phi \left( h_\delta + \widetilde{H}, 0, 0, \nu \right), \Phi(S, 0, 0, 0) \right] \\ & = \Phi \left( \left[ h_\delta + \widetilde{H}, S \right], 0, 0, 0 \right) = \Phi \left( \alpha \left( h_\delta + \widetilde{H} \right) S, 0, 0, 0 \right) = \alpha(h)S. \end{aligned}$ | $[h, S] = \left[ \Phi \left( h_\delta + \widetilde{H}, 0, 0, \nu \right), \Phi(S, 0, 0, 0) \right] \\ = \Phi \left( \left[ h_\delta + \widetilde{H}, S \right], 0, 0, 0 \right) = \Phi \left( \alpha \left( h_\delta + \widetilde{H} \right) S, 0, 0, 0 \right) = \alpha(h)S.$ | $[h, S] = [\Phi(h_\delta + \widetilde{H}, 0, 0, \nu), \Phi(S, 0, 0, 0)] \\ = \Phi([h_\delta + \widetilde{H}, S], 0, 0, 0) = \Phi(\alpha(h_\delta + \widetilde{H})S, 0, 0, 0) = \alpha(h)S.$ |
| 0902.0431_E00969 | 161  | ■ 0.990 | $E_{\{7(7)\}} \cong \left( E_{\{7\}}^{\{C\}} \right)^{\tau \gamma}, \quad \text{quad} \quad E_{\{7(-5)\}} \cong \left( E_{\{7\}}^{\{C\}} \right)^{\tau \lambda \sigma}, \quad \text{quad} \quad E_{\{7(-25)\}} \cong \left( E_{\{7\}}^{\{C\}} \right)^{\tau}.$                                                   | $E_{7(7)} \cong (E_7^C)^{\tau \gamma}, \quad E_{7(-5)} \cong (E_7^C)^{\tau \lambda \sigma}, \quad E_{7(-25)} \cong (E_7^C)^{\tau}.$                                                                                                                                            | $E_{7(7)} \cong (E_7^C)^{\tau \gamma}, \quad E_{7(-5)} \cong (E_7^C)^{\tau \lambda \sigma}, \quad E_{7(-25)} \cong (E_7^C)^{\tau}.$                                                         |
| 0902.0431_E00293 | 47   | ■ 0.990 | $\widetilde{A}X = A \circ X, \quad X \in \mathfrak{J}^C.$                                                                                                                                                                                                                                                        | $\widetilde{A}X = A \circ X, \quad X \in \mathfrak{J}^C.$                                                                                                                                                                                                                      | $\widetilde{A}X = A \circ X, \quad X \in \mathfrak{J}^C.$                                                                                                                                   |

| Identifier       | Page | Conf.   | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Rendered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Image                                                                                                                                                                                                                                    |
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| 0902.0431_E00018 | 9    | ■ 0.990 | $\begin{gathered} \left\{ \begin{array}{l} 2F_{\phantom{0}3} = G_{\phantom{0}3} + G_{12} + G_{47} + G_{56} \\ 2F_{12} = G_{\phantom{0}3} + G_{12} - G_{47} - G_{56} \\ 2F_{47} = G_{\phantom{0}3} - G_{12} + G_{47} - G_{56} \\ 2F_{56} = G_{\phantom{0}3} - G_{12} - G_{47} + G_{56}, \end{array} \right\} \left\{ \begin{array}{l} 2F_{04} = G_{04} - G_{15} + G_{26} - G_{37} \\ 2F_{15} = -G_{04} + G_{15} + G_{26} - G_{37} \\ 2F_{26} = G_{04} + G_{15} + G_{26} + G_{37} \\ 2F_{37} = -G_{04} - G_{15} + G_{26} + G_{37}, \end{array} \right\} \\ \left\{ \begin{array}{l} 2F_{05} = G_{05} + G_{14} - G_{27} - G_{36} \\ 2F_{14} = G_{05} + G_{14} + G_{27} + G_{36} \\ 2F_{27} = -G_{05} + G_{14} + G_{27} - G_{36} \\ 2F_{36} = -G_{05} + G_{14} - G_{27} + G_{36}, \end{array} \right\} \left\{ \begin{array}{l} 2F_{06} = G_{06} - G_{17} - G_{24} + G_{35} \\ 2F_{17} = -G_{06} + G_{17} - G_{24} + G_{35} \\ 2F_{24} = -G_{06} - G_{17} + G_{24} + G_{35} \\ 2F_{35} = G_{06} + G_{17} + G_{24} + G_{35}, \end{array} \right\} \\ \left\{ \begin{array}{l} 2F_{07} = G_{07} + G_{16} + G_{25} + G_{34} \\ 2F_{16} = G_{07} + G_{16} - G_{25} - G_{34} \\ 2F_{25} = G_{07} - G_{16} + G_{25} - G_{34} \\ 2F_{34} = G_{07} - G_{16} - G_{25} + G_{34}. \end{array} \right.$ | $\left\{ \begin{array}{l} 2F_{03} = G_{03} + G_{12} + G_{47} + G_{56} \\ 2F_{12} = G_{03} + G_{12} - G_{47} - G_{56} \\ 2F_{47} = G_{03} - G_{12} + G_{47} - G_{56} \\ 2F_{56} = G_{03} - G_{12} - G_{47} + G_{56}, \end{array} \right\} \left\{ \begin{array}{l} 2F_{04} = G_{04} - G_{15} + G_{26} - G_{37} \\ 2F_{15} = -G_{04} + G_{15} + G_{26} - G_{37} \\ 2F_{26} = G_{04} + G_{15} + G_{26} + G_{37} \\ 2F_{37} = -G_{04} - G_{15} + G_{26} + G_{37}, \end{array} \right\} \\ \left\{ \begin{array}{l} 2F_{05} = G_{05} + G_{14} - G_{27} - G_{36} \\ 2F_{14} = G_{05} + G_{14} + G_{27} + G_{36} \\ 2F_{27} = -G_{05} + G_{14} + G_{27} - G_{36} \\ 2F_{36} = -G_{05} + G_{14} - G_{27} + G_{36}, \end{array} \right\} \left\{ \begin{array}{l} 2F_{06} = G_{06} - G_{17} - G_{24} + G_{35} \\ 2F_{17} = -G_{06} + G_{17} - G_{24} + G_{35} \\ 2F_{24} = -G_{06} - G_{17} + G_{24} + G_{35} \\ 2F_{35} = G_{06} + G_{17} + G_{24} + G_{35}, \end{array} \right\} \\ \left\{ \begin{array}{l} 2F_{07} = G_{07} + G_{16} + G_{25} + G_{34} \\ 2F_{16} = G_{07} + G_{16} - G_{25} - G_{34} \\ 2F_{25} = G_{07} - G_{16} + G_{25} - G_{34} \\ 2F_{34} = G_{07} - G_{16} - G_{25} + G_{34}. \end{array} \right.$ | $\left\{ \begin{array}{l} 2F_{07} = G_{07} + G_{16} + G_{25} + G_{34} \\ 2F_{16} = G_{07} + G_{16} - G_{25} - G_{34} \\ 2F_{25} = G_{07} - G_{16} + G_{25} - G_{34} \\ 2F_{34} = G_{07} - G_{16} - G_{25} + G_{34}. \end{array} \right.$ |
| 0902.0431_E00722 | 119  | ■ 0.991 | $\mathfrak{e}_{\{7\}}^{\{ \}^{\{C\}}} = \mathfrak{e}_{\{6\}}^{\{ \}^{\{C\}}} \oplus \mathfrak{e}_{\{N\}}^{\{ \}^{\{C\}}},$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | $\mathfrak{e}_7^C = \mathfrak{e}_6^C \oplus \mathfrak{N}^C,$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | $\mathfrak{e}_7^C = \mathfrak{e}_6^C \oplus \mathfrak{N}^C,$                                                                                                                                                                             |
| 0902.0431_E01001 | 166  | ■ 0.991 | $\widetilde{\lambda} = \lambda \lambda' = \lambda' \lambda.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | $\widetilde{\lambda} = \lambda \lambda' = \lambda' \lambda.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | $\widetilde{\lambda} = \lambda \lambda' = \lambda' \lambda.$                                                                                                                                                                             |
| 0902.0431_E00162 | 29   | ■ 0.991 | $(x\,p)(\overline{p}\,y) = x\,y, \quad x, y \in \mathfrak{C}.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | $(xp)(\overline{p}y) = xy, \quad x, y \in \mathfrak{C}.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | $(xp)(\overline{p}y) = xy, \quad x, y \in \mathfrak{C}.$                                                                                                                                                                                 |

| Identifier       | Page | Conf.   | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Rendered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Image                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
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| 0902.0431_E00296 | 47   | ■ 0.991 | $\begin{array}{ll} \{\left[\widetilde{E}_{_i}, \right. \\ \left. \widetilde{E}_{_j}\right]=0, \} \& \{\left[\widetilde{E}_{_i}, \right. \\ \left. \widetilde{F}_{_i}(a)\right]=0, \} \quad \& \{\left[\right. \\ \left. \widetilde{E}_{_i}, \widetilde{F}_{_{i+1}}(a)\right]= \\ \frac{1}{2} \widetilde{A}_{_{i+1}}(a), \} \& \{\left[\right. \\ \left. \widetilde{E}_{_i}, \widetilde{F}_{_{i+2}}(a)\right]= \\ \frac{1}{2} \widetilde{A}_{_{i+2}}(a), \} \quad \& \{\left[\right. \\ \left. \widetilde{F}_{_i}(a), \widetilde{F}_{_i}(b)\right] \in \mathfrak{d}_4^C, \\ \left. \widetilde{D}, \widetilde{A}_i(a)\right] = \widetilde{A}_i(D_i a), \\ \left. \widetilde{A}_i(a), \widetilde{A}_i(b)\right] \in \mathfrak{d}_4^C, \\ \left. \widetilde{F}_1(e_i), \widetilde{F}_1(e_j)\right] = G_{ij}. \end{array}$ | $\begin{array}{ll} \left[\widetilde{E}_i, \widetilde{E}_j\right] = 0, & \left[\widetilde{E}_i, \widetilde{F}_i(a)\right] = 0, \\ \left[\widetilde{E}_i, \widetilde{F}_{i+1}(a)\right] = -\frac{1}{2}\widetilde{A}_{i+1}(a), & \left[\widetilde{E}_i, \widetilde{F}_{i+2}(a)\right] = \frac{1}{2}\widetilde{A}_{i+2}(a), \\ \left[\widetilde{F}_i(a), \widetilde{F}_i(b)\right] \in \mathfrak{d}_4^C, & \left[\widetilde{F}_i(a), \widetilde{F}_{i+1}(b)\right] = -\frac{1}{2}\widetilde{A}_{i+2}(\overline{ab}), \\ \left[\widetilde{D}, \widetilde{A}_i(a)\right] = \widetilde{A}_i(D_i a), & \text{where } D = (D_1, D_2, D_3) \in \mathfrak{d}_4^C, \\ \left[\widetilde{A}_i(a), \widetilde{A}_i(b)\right] \in \mathfrak{d}_4^C, & \left[\widetilde{A}_i(a), \widetilde{A}_{i+1}(b)\right] = -\frac{1}{2}\widetilde{A}_{i+2}(\overline{ab}), \\ \left[\widetilde{F}_1(e_i), \widetilde{F}_1(e_j)\right] = G_{ij}. & \left[\widetilde{A}_i(a), \widetilde{A}_{i+2}(b)\right] = \frac{1}{2}\widetilde{A}_{i+1}(\overline{ba}). \end{array}$ | $\begin{array}{ll} [\widetilde{E}_i, \widetilde{E}_j] = 0, & [\widetilde{E}_i, \widetilde{F}_i(a)] = 0, \\ [\widetilde{E}_i, \widetilde{F}_{i+1}(a)] = -\frac{1}{2}\widetilde{A}_{i+1}(a), & [\widetilde{E}_i, \widetilde{F}_{i+2}(a)] = \frac{1}{2}\widetilde{A}_{i+2}(a), \\ [\widetilde{F}_i(a), \widetilde{F}_i(b)] \in \mathfrak{d}_4^C, & [\widetilde{F}_i(a), \widetilde{F}_{i+1}(b)] = -\frac{1}{2}\widetilde{A}_{i+2}(\overline{ab}), \\ [D, \widetilde{A}_i(a)] = \widetilde{A}_i(D_i a), & \text{where } D = (D_1, D_2, D_3) \in \mathfrak{d}_4^C, \\ [\widetilde{A}_i(a), \widetilde{A}_i(b)] \in \mathfrak{d}_4^C, & [\widetilde{A}_i(a), \widetilde{A}_{i+1}(b)] = -\frac{1}{2}\widetilde{A}_{i+2}(\overline{ab}), \\ [\widetilde{F}_1(e_i), \widetilde{F}_1(e_j)] = G_{ij}. & [\widetilde{A}_i(a), \widetilde{A}_{i+2}(b)] = \frac{1}{2}\widetilde{A}_{i+1}(\overline{ba}). \end{array}$ |
| 0902.0431_E01151 | 194  | ■ 0.991 | $\begin{array}{l} \& \left(\boldsymbol{u}_{_1} \wedge \cdots \wedge \boldsymbol{u}_{_j-1} \wedge \boldsymbol{x} \wedge \boldsymbol{u}_{_j+1} \wedge \cdots \wedge \boldsymbol{u}_9, e_1 \wedge \cdots \wedge e_9\right) = \sum_{k=1}^9 \widetilde{u}_{jk} x_k, \\ \& \left(\boldsymbol{u}_{_1} \wedge \cdots \wedge \boldsymbol{u}_9, \boldsymbol{e}_1 \wedge \cdots \wedge \boldsymbol{e}_9\right) = \det U, \end{array}$                                                                                                                                                                                                                                                                                                                                                                                            | $\begin{array}{l} \& \left(\boldsymbol{u}_{_1} \wedge \cdots \wedge \boldsymbol{u}_{_j-1} \wedge \boldsymbol{x} \wedge \boldsymbol{u}_{_j+1} \wedge \cdots \wedge \boldsymbol{u}_9, e_1 \wedge \cdots \wedge e_9\right) = \sum_{k=1}^9 \widetilde{u}_{jk} x_k, \\ \& \left(\boldsymbol{u}_{_1} \wedge \cdots \wedge \boldsymbol{u}_9, \boldsymbol{e}_1 \wedge \cdots \wedge \boldsymbol{e}_9\right) = \det U, \end{array}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | $\begin{array}{l} (\boldsymbol{u}_1 \wedge \cdots \wedge \boldsymbol{u}_{j-1} \wedge \boldsymbol{x} \wedge \boldsymbol{u}_{j+1} \wedge \cdots \wedge \boldsymbol{u}_9, e_1 \wedge \cdots \wedge e_9) = \sum_{k=1}^9 \widetilde{u}_{jk} x_k, \\ (\boldsymbol{u}_1 \wedge \cdots \wedge \boldsymbol{u}_9, e_1 \wedge \cdots \wedge e_9) = \det U, \end{array}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |



| Identifier       | Page | Conf.  | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Rendered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Image                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
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| 0902.0431_E00936 | 155  | ■0.992 | =\left(\begin{array}{rrrrrr} b_{11} & b_{12} & b_{13} & l_{11} & l_{12} & l_{13} \\ & \bar{b}_{12} & b_{22} & b_{23} & l_{21} & l_{22} & l_{23} \\ & \bar{b}_{13} & b_{23} & b_{33} & l_{31} & l_{32} & l_{33} \\ & \bar{l}_{11} & \bar{l}_{21} & \bar{l}_{31} & c_{11} & c_{12} & c_{13} \\ & \bar{l}_{12} & \bar{l}_{22} & \bar{l}_{32} & \bar{c}_{12} & c_{22} & c_{23} \\ & \bar{l}_{13} & \bar{l}_{23} & \bar{l}_{33} & \bar{c}_{13} & \bar{c}_{23} & c_{33} \end{array}\right)+\frac{\nu}{3}\left(\begin{array}{cc} E & 0 \\ 0 & -E \end{array}\right)\in\mathfrak{su}(6), | $= \begin{pmatrix} b_{11} & b_{12} & b_{13} & l_{11} & l_{12} & l_{13} \\ -\bar{b}_{12} & b_{22} & b_{23} & l_{21} & l_{22} & l_{23} \\ -\bar{b}_{13} & b_{23} & b_{33} & l_{31} & l_{32} & l_{33} \\ -\bar{l}_{11} & -\bar{l}_{21} & -\bar{l}_{31} & c_{11} & c_{12} & c_{13} \\ -\bar{l}_{12} & -\bar{l}_{22} & -\bar{l}_{32} & -\bar{c}_{12} & c_{22} & c_{23} \\ -\bar{l}_{13} & -\bar{l}_{23} & -\bar{l}_{33} & -\bar{c}_{13} & -\bar{c}_{23} & c_{33} \end{pmatrix} + \frac{\nu}{3} \begin{pmatrix} E & 0 \\ 0 & -E \end{pmatrix} \in \mathfrak{su}(6),$ | $= \begin{pmatrix} b_{11} & b_{12} & b_{13} & l_{11} & l_{12} & l_{13} \\ -\bar{b}_{12} & b_{22} & b_{23} & l_{21} & l_{22} & l_{23} \\ -\bar{b}_{13} & b_{23} & b_{33} & l_{31} & l_{32} & l_{33} \\ -\bar{l}_{11} & -\bar{l}_{21} & -\bar{l}_{31} & c_{11} & c_{12} & c_{13} \\ -\bar{l}_{12} & -\bar{l}_{22} & -\bar{l}_{32} & -\bar{c}_{12} & c_{22} & c_{23} \\ -\bar{l}_{13} & -\bar{l}_{23} & -\bar{l}_{33} & -\bar{c}_{13} & -\bar{c}_{23} & c_{33} \end{pmatrix} + \frac{\nu}{3} \begin{pmatrix} E & 0 \\ 0 & -E \end{pmatrix} \in \mathfrak{su}(6),$ |
| 0902.0431_E00994 | 165  | ■0.992 | $B_8(1,1)=56 \times 2+4 \times 2=120$ .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | $B_8(1,1)=56 \times 2+4 \times 2=120.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | $B_8(1,1)=56 \times 2+4 \times 2=120.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 0902.0431_E00998 | 166  | ■0.992 | $\mathfrak{W}^C=\left\{R\in\mathfrak{e}_8^C\mid R\times R=0,R\neq 0\right\}.$<br>$\mid R\times R=0, R\neq 0\right\}.$                                                                                                                                                                                                                                                                                                                                                                                                                                                            | $\mathfrak{W}^C=\left\{R\in\mathfrak{e}_8^C\mid R\times R=0,R\neq 0\right\}.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | $\mathfrak{W}^C=\{R\in\mathfrak{e}_8^C\mid R\times R=0,R\neq 0\}.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 0902.0431_E00034 | 11   | ■0.992 | $\begin{aligned} &\& \pi T_{e_i}=\pi\left(2G_{i0}\right)=2F_{i0}=L_{e_i}, \\ &\& \pi R_a=\pi\left(T_a-L_a\right)=L_a-T_a=-R_a. \end{aligned}$                                                                                                                                                                                                                                                                                                                                                                                                                                    | $\begin{aligned} \pi T_{e_i} &= \pi\left(2G_{i0}\right)=2F_{i0}=L_{e_i}, \\ \pi R_a &= \pi\left(T_a-L_a\right)=L_a-T_a=-R_a. \end{aligned}$                                                                                                                                                                                                                                                                                                                                                                                                                    | $\begin{aligned} \pi T_{e_i} &= \pi\left(2G_{i0}\right)=2F_{i0}=L_{e_i}, \\ \pi R_a &= \pi\left(T_a-L_a\right)=L_a-T_a=-R_a. \end{aligned}$                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 0902.0431_E00065 | 14   | ■0.992 | $\begin{aligned} x&=x_0+x_1\,e_1+x_2\,e_2+x_3\,e_3+x_4\,e_4+x_5\,e_5+x_6\,e_6+x_7\,e_7 \\ &\quad\left(x_i\in\boldsymbol{R}\right)\quad\&=\\ &\quad\left(x_0+x_1\,e_1\right)+\left(x_2+x_3\,e_1\right)e_2+\left(x_4+x_5\,e_1\right)e_4+\left(x_6+x_7\,e_1\right)e_6, \end{aligned}$                                                                                                                                                                                                                                                                                               | $\begin{aligned} x&=x_0+x_1e_1+x_2e_2+x_3e_3+x_4e_4+x_5e_5+x_6e_6+x_7e_7\quad(x_i\in\boldsymbol{R}) \\ &=(x_0+x_1e_1)+(x_2+x_3e_1)e_2+(x_4+x_5e_1)e_4+(x_6+x_7e_1)e_6, \end{aligned}$                                                                                                                                                                                                                                                                                                                                                                          | $\begin{aligned} x&=x_0+x_1e_1+x_2e_2+x_3e_3+x_4e_4+x_5e_5+x_6e_6+x_7e_7\quad(x_i\in\boldsymbol{R}) \\ &=(x_0+x_1e_1)+(x_2+x_3e_1)e_2+(x_4+x_5e_1)e_4+(x_6+x_7e_1)e_6, \end{aligned}$                                                                                                                                                                                                                                                                                                                                                                          |
| 0902.0431_E00012 | 136  | ■0.992 | $\iota(X,Y,\xi,\eta)=(-iX,iY,-i\xi,i\eta).$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | $\iota(X,Y,\xi,\eta)=(-iX,iY,-i\xi,i\eta).$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | $\iota(X,Y,\xi,\eta)=(-iX,iY,-i\xi,i\eta).$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 0902.0431_E01046 | 178  | ■0.992 | $0=[(\beta_1\Phi,0,0,a_1\Phi,0,0),(0,0,0,0,1,0)]= (0,0,0,0,2a_1\Phi,0),$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | $0=[(\beta_1\Phi,0,0,a_1\Phi,0,0),(0,0,0,0,1,0)]= (0,0,0,0,2a_1\Phi,0),$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | $0=[(\beta_1\Phi,0,0,a_1\Phi,0,0),(0,0,0,0,1,0)]= (0,0,0,0,2a_1\Phi,0),$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 0902.0431_E00835 | 139  | ■0.993 | $(A,Y)=-\left(A,\kappa_1Y\right),\quad Y\in\mathfrak{J}^C.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | $(A,Y)=-\left(A,\kappa_1Y\right),\quad Y\in\mathfrak{J}^C.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | $(A,Y)=-\left(A,\kappa_1Y\right),\quad Y\in\mathfrak{J}^C.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 0902.0431_E01019 | 170  | ■0.993 | $[(\Phi(h),0,0,r,0,0),(0,P,0,0,0,0)]=(0,(\Phi(h)+r)P,0,0,0,0),$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | $[(\Phi(h),0,0,r,0,0),(0,P,0,0,0,0)]=(0,(\Phi(h)+r)P,0,0,0,0),$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | $[(\Phi(h),0,0,r,0,0),(0,P,0,0,0,0)]=(0,(\Phi(h)+r)P,0,0,0,0),$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

| Identifier       | Page | Conf.   | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Rendered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Image                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
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| 0902.0431_E08443 | 73   | ■ 0.993 | $\begin{aligned} E_6^C &= \{\alpha \in \operatorname{Iso}_C(\mathfrak{J}^C) \mid \det(\alpha X) = \det X, \\ &\quad \langle \alpha X, \alpha Y, \alpha Z \rangle = \langle X, Y, Z \rangle, \langle \alpha X, \alpha Y \rangle = \langle X, Y \rangle\} \\ E_6 &= \{\alpha \in \operatorname{Iso}_C(\mathfrak{J}^C) \mid \det(\alpha X) = \det X, \langle \alpha X, \alpha Y \rangle = \langle X, Y \rangle\} \\ &= \{\alpha \in \operatorname{Iso}_C(\mathfrak{J}^C) \mid \langle \alpha X, \alpha Y, \alpha Z \rangle = \langle X, Y, Z \rangle, \langle \alpha X, \alpha Y \rangle = \langle X, Y \rangle\} \\ &= \{\alpha \in \operatorname{Iso}_C(\mathfrak{J}^C) \mid \alpha X \times \alpha Y = {}^t\alpha^{-1}(X \times Y), \langle \alpha X, \alpha Y \rangle = \langle X, Y \rangle\} \\ &= \{\alpha \in \operatorname{Iso}_C(\mathfrak{J}^C) \mid \alpha X \times \alpha Y = \tau\alpha\tau(X \times Y), \langle \alpha X, \alpha Y \rangle = \langle X, Y \rangle\}. \end{aligned}$ | $\begin{aligned} E_6^C &= \{\alpha \in \operatorname{Iso}_C(\mathfrak{J}^C) \mid \det(\alpha X) = \det X\} \\ &= \{\alpha \in \operatorname{Iso}_C(\mathfrak{J}^C) \mid \langle \alpha X, \alpha Y, \alpha Z \rangle = \langle X, Y, Z \rangle\}, \\ E_6 &= \{\alpha \in \operatorname{Iso}_C(\mathfrak{J}^C) \mid \det(\alpha X) = \det X, \langle \alpha X, \alpha Y \rangle = \langle X, Y \rangle\} \\ &= \{\alpha \in \operatorname{Iso}_C(\mathfrak{J}^C) \mid \langle \alpha X, \alpha Y, \alpha Z \rangle = \langle X, Y, Z \rangle, \langle \alpha X, \alpha Y \rangle = \langle X, Y \rangle\} \\ &= \{\alpha \in \operatorname{Iso}_C(\mathfrak{J}^C) \mid \alpha X \times \alpha Y = {}^t\alpha^{-1}(X \times Y), \langle \alpha X, \alpha Y \rangle = \langle X, Y \rangle\} \\ &= \{\alpha \in \operatorname{Iso}_C(\mathfrak{J}^C) \mid \alpha X \times \alpha Y = \tau\alpha\tau(X \times Y), \langle \alpha X, \alpha Y \rangle = \langle X, Y \rangle\}. \end{aligned}$ | $\begin{aligned} E_6^C &= \{\alpha \in \operatorname{Iso}_C(\mathfrak{J}^C) \mid \det(\alpha X) = \det X\} \\ &= \{\alpha \in \operatorname{Iso}_C(\mathfrak{J}^C) \mid \langle \alpha X, \alpha Y, \alpha Z \rangle = \langle X, Y, Z \rangle\}, \\ E_6 &= \{\alpha \in \operatorname{Iso}_C(\mathfrak{J}^C) \mid \det(\alpha X) = \det X, \langle \alpha X, \alpha Y \rangle = \langle X, Y \rangle\} \\ &= \{\alpha \in \operatorname{Iso}_C(\mathfrak{J}^C) \mid \langle \alpha X, \alpha Y, \alpha Z \rangle = \langle X, Y, Z \rangle, \langle \alpha X, \alpha Y \rangle = \langle X, Y \rangle\} \\ &= \{\alpha \in \operatorname{Iso}_C(\mathfrak{J}^C) \mid \alpha X \times \alpha Y = {}^t\alpha^{-1}(X \times Y), \langle \alpha X, \alpha Y \rangle = \langle X, Y \rangle\} \\ &= \{\alpha \in \operatorname{Iso}_C(\mathfrak{J}^C) \mid \alpha X \times \alpha Y = \tau\alpha\tau(X \times Y), \langle \alpha X, \alpha Y \rangle = \langle X, Y \rangle\}. \end{aligned}$ |
| 0902.0431_E08789 | 117  | ■ 0.993 | $\begin{aligned} &(\exp t\Phi)(P \times Q)(\exp t\Phi)^{-1} \\ &= (\exp t(\operatorname{ad} \Phi))(P \times Q) \quad ((\operatorname{ad} \Phi)\Phi_1 = [\Phi, \Phi_1], \Phi_1 \in \mathfrak{e}_7^C) \\ &= \sum_{n=0}^{\infty} \frac{1}{n!} (t(\operatorname{ad} \Phi))^n (P \times Q) \\ &= \sum_{n=0}^{\infty} \frac{t^n}{n!} \left( \sum_{k+l=n} \frac{n!}{k!l!} \Phi^k P \times \Phi^l Q \right) \quad (\text{Proposition 4.3.2}) \\ &= \sum_{n=0}^{\infty} \left( \sum_{k+l=n} \frac{t^k t^l}{k!l!} \Phi^k P \times \Phi^l Q \right) \\ &= \left( \sum_{k=0}^{\infty} \frac{1}{k!} (t\Phi)^k P \right) \times \left( \sum_{l=0}^{\infty} \frac{1}{l!} (t\Phi)^l Q \right) \\ &= (\exp t\Phi)P \times (\exp t\Phi)Q. \end{aligned}$                                                                                                                                                                                                                                                          | $\begin{aligned} &(\exp t\Phi)(P \times Q)(\exp t\Phi)^{-1} \\ &= (\exp t(\operatorname{ad} \Phi))(P \times Q) \quad ((\operatorname{ad} \Phi)\Phi_1 = [\Phi, \Phi_1], \Phi_1 \in \mathfrak{e}_7^C) \\ &= \sum_{n=0}^{\infty} \frac{1}{n!} (t(\operatorname{ad} \Phi))^n (P \times Q) \\ &= \sum_{n=0}^{\infty} \frac{t^n}{n!} \left( \sum_{k+l=n} \frac{n!}{k!l!} \Phi^k P \times \Phi^l Q \right) \quad (\text{Proposition 4.3.2}) \\ &= \sum_{n=0}^{\infty} \left( \sum_{k+l=n} \frac{t^k t^l}{k!l!} \Phi^k P \times \Phi^l Q \right) \\ &= \left( \sum_{k=0}^{\infty} \frac{1}{k!} (t\Phi)^k P \right) \times \left( \sum_{l=0}^{\infty} \frac{1}{l!} (t\Phi)^l Q \right) \\ &= (\exp t\Phi)P \times (\exp t\Phi)Q. \end{aligned}$                                                                                                                                                                                                                                                    | $\begin{aligned} &(\exp t\Phi)(P \times Q)(\exp t\Phi)^{-1} \\ &= (\exp t(\operatorname{ad} \Phi))(P \times Q) \quad ((\operatorname{ad} \Phi)\Phi_1 = [\Phi, \Phi_1], \Phi_1 \in \mathfrak{e}_7^C) \\ &= \sum_{n=0}^{\infty} \frac{1}{n!} (t(\operatorname{ad} \Phi))^n (P \times Q) \\ &= \sum_{n=0}^{\infty} \frac{t^n}{n!} \left( \sum_{k+l=n} \frac{n!}{k!l!} \Phi^k P \times \Phi^l Q \right) \quad (\text{Proposition 4.3.2}) \\ &= \sum_{n=0}^{\infty} \left( \sum_{k+l=n} \frac{t^k t^l}{k!l!} \Phi^k P \times \Phi^l Q \right) \\ &= \left( \sum_{k=0}^{\infty} \frac{1}{k!} (t\Phi)^k P \right) \times \left( \sum_{l=0}^{\infty} \frac{1}{l!} (t\Phi)^l Q \right) \\ &= (\exp t\Phi)P \times (\exp t\Phi)Q. \end{aligned}$                                                                                                                                                                                                                                                    |
| 0902.0431_E08555 | 93   | ■ 0.993 | $\beta\alpha_{23}(t_0)X = E_2 - E_3 \in S^8.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | $\beta\alpha_{23}(t_0)X = E_2 - E_3 \in S^8.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | $\beta\alpha_{23}(t_0)X = E_2 - E_3 \in S^8.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

| Identifier            | Page | Conf.   | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Rendered                                                                                                                                                                                                                                                         | Image                                                                                                                                                                                                                                                            |
|-----------------------|------|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0902.0431_E00777      | 130  | ■ 0.993 | $\backslash\mathrm{mathfrak{M}}^{\mathrm{C}}\backslash\mathrm{ni}\backslash\alpha(X,X\times X,\backslash\mathrm{operatorname{det}}X,1)=\backslash\mathrm{left}(\backslash\beta X,\backslash\beta_{\mathrm{1}}(X\times X)\backslash\mathrm{times}X),\backslash\mathrm{operatorname{det}}X,1\backslash\mathrm{right})$ ,                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | $\mathfrak{M}^C \ni \alpha(X, X \times X, \det X, 1) = (\beta X, \beta_1(X \times X), \det X, 1),$                                                                                                                                                               | $\mathfrak{M}^C \ni \alpha(X, X \times X, \det X, 1) = (\beta X, \beta_1(X \times X), \det X, 1),$                                                                                                                                                               |
| 0902.0431_E00841 (ii) | 12   | ■ 0.994 | $\backslash\mathrm{left}(\mathrm{L}_{\mathrm{b}}\mathrm{x}\backslash\mathrm{right})\mathrm{y}+\mathrm{x}\backslash\mathrm{left}(\mathrm{R}_{\mathrm{b}}\mathrm{y}\backslash\mathrm{right})=\mathrm{T}_{\mathrm{b}}(\mathrm{x}\mathrm{y}),\quad \mathrm{x},\mathrm{y}\backslash\mathrm{in}\backslash\mathrm{mathfrak{C}}.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | $(L_b x)y+x(R_b y)=T_b(xy),\quad x,y\in\mathfrak{C}.$                                                                                                                                                                                                            | $(L_b x)y+x(R_b y)=T_b(xy),\quad x,y\in\mathfrak{C}.$                                                                                                                                                                                                            |
| 0902.0431_E00077      | 16   | ■ 0.994 | $\mathrm{B}=\mathrm{D}+\sum_{\mathrm{i}=1}^{\mathrm{6}}\mathrm{x}_{\mathrm{i}}\mathrm{S}_{\mathrm{i}},\quad \mathrm{D}\backslash\mathrm{in}\backslash\mathrm{mathfrak{s u}}(3),\mathrm{x}_{\mathrm{i}}\backslash\mathrm{in}\backslash\mathrm{boldsymbol{R}}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | $B=D+\sum_{i=1}^6x_iS_i,\quad D\in\mathfrak{su}(3),x_i\in\boldsymbol{R}$                                                                                                                                                                                         | $B=D+\sum_{i=1}^6x_iS_i,\quad D\in\mathfrak{su}(3),x_i\in\boldsymbol{R}.$                                                                                                                                                                                        |
| 0902.0431_E00996      | 166  | ■ 0.994 | $\backslash\mathrm{operatorname{Aut}}\backslash\mathrm{left}(\backslash\mathrm{mathfrak{e}}_{\mathrm{8}}^{\mathrm{C}}\backslash\mathrm{right})=\backslash\mathrm{operatorname{Inn}}\backslash\mathrm{left}(\backslash\mathrm{mathfrak{e}}_{\mathrm{8}}^{\mathrm{C}}\backslash\mathrm{right}).$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | $\mathrm{Aut}\left(\mathfrak{e}_8^C\right)=\mathrm{Inn}\left(\mathfrak{e}_8^C\right).$                                                                                                                                                                           | $\mathrm{Aut}(\mathfrak{e}_8^C)=\mathrm{Inn}(\mathfrak{e}_8^C).$                                                                                                                                                                                                 |
| 0902.0431_E00999      | 166  | ■ 0.994 | $\mathrm{E}_{\mathrm{8}}^{\mathrm{C}}\backslash\mathrm{left}(\mathrm{E}_{\mathrm{8}}^{\mathrm{C}}\backslash\mathrm{right})_{\mathrm{1}_{-}}\backslash\mathrm{simeq}\backslash\mathrm{mathfrak{W}}^{\mathrm{C}},$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | $E_8^C/(E_8^C)_{1_-}\simeq\mathfrak{W}^C,$                                                                                                                                                                                                                       | $E_8^C/(E_8^C)_{1_-}\simeq\mathfrak{W}^C,$                                                                                                                                                                                                                       |
| 0902.0431_E00562      | 94   | ■ 0.994 | $\backslash\mathrm{begin}{gathered}\backslash\mathrm{mathfrak{E}}_{\mathrm{1}}^{\mathrm{C}}=\backslash\mathrm{left}\backslash\{\mathrm{x_i}\mathrm{E}_{\mathrm{1}}\backslash\mathrm{mid}\mathrm{x_i}\backslash\mathrm{in}\mathrm{C}\backslash\mathrm{right}\},\quad\backslash\mathrm{quad}\backslash\mathrm{mathfrak{J}}^{\mathrm{C}}(2,\backslash\mathrm{mathfrak{C}})^{\mathrm{C}}=\backslash\mathrm{left}\backslash\{X\backslash\mathrm{in}\backslash\mathrm{mathfrak{J}}^{\mathrm{C}}\backslash\mathrm{mid}4\mathrm{E}_{\mathrm{1}}\backslash\mathrm{times}\backslash\mathrm{left}(\mathrm{E}_{\mathrm{1}}\backslash\mathrm{times}X\backslash\mathrm{right})=X\backslash\mathrm{right}\},\quad\backslash\backslash\backslash\mathrm{left}(\backslash\mathrm{mathfrak{J}}^{\mathrm{C}}\backslash\mathrm{right})_{-\sigma}=\backslash\mathrm{left}\backslash\{X\backslash\mathrm{in}\backslash\mathrm{mathfrak{J}}^{\mathrm{C}}\backslash\mathrm{mid}\mathrm{E}_{\mathrm{1}}\backslash\mathrm{times}X=0,\backslash\mathrm{left}\backslash\mathrm{angle}\mathrm{E}_{\mathrm{1}},X\backslash\mathrm{right}\backslash\mathrm{angle}=0\backslash\mathrm{right}\}.\backslash\mathrm{end}{gathered}$ | $\mathfrak{E}_1^C=\{\xi E_1\mid\xi\in C\},\quad\mathfrak{J}(2,\mathfrak{C})^C=\{X\in\mathfrak{J}^C\mid 4E_1\times(E_1\times X)=X\},\quad(\mathfrak{J}^C)_{-\sigma}=\{X\in\mathfrak{J}^C\mid E_1\times X=0,\langle E_1,X\rangle=0\}.$                             | $\mathfrak{E}_1^C=\{\xi E_1\mid\xi\in C\},\quad\mathfrak{J}(2,\mathfrak{C})^C=\{X\in\mathfrak{J}^C\mid 4E_1\times(E_1\times X)=X\},\quad(\mathfrak{J}^C)_{-\sigma}=\{X\in\mathfrak{J}^C\mid E_1\times X=0,\langle E_1,X\rangle=0\}.$                             |
| 0902.0431_E00810      | 135  | ■ 0.994 | $\backslash\alpha=\backslash\mathrm{left}(\backslash\mathrm{begin}{array}{cccc}\backslash\omega^{\backslash\mathrm{prime}}1&\&\&\&\&\&\&\&\backslash\omega^{\backslash\mathrm{prime}-1}1&\&\&\&\&\&\&\&\backslash\omega^{\backslash\mathrm{prime}}=1,\backslash\omega\backslash\mathrm{text}\{\mathrm{or}\}\backslash\omega^{\mathrm{2}}.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | $\alpha=\left(\begin{array}{cccc}\omega'1&0&0&0\\0&\omega'^{-1}1&0&0\\0&0&1&0\\0&0&0&1\end{array}\right),\quad\omega'=1,\omega\text{ or }\omega^2.$                                                                                                              | $\alpha=\left(\begin{array}{cccc}\omega'1&0&0&0\\0&\omega'^{-1}1&0&0\\0&0&1&0\\0&0&0&1\end{array}\right),\quad\omega'=1,\omega\text{ or }\omega^2.$                                                                                                              |
| 0902.0431_E00012      | 8    | ■ 0.994 | $\backslash\mathrm{begin}{aligned}(\backslash\alpha\mathrm{x},\backslash\alpha\mathrm{y})&=\mathrm{frac}{1}{2}((\backslash\alpha\mathrm{x})(\overline{\backslash\alpha\mathrm{y}})+(\backslash\alpha\mathrm{y})(\overline{\backslash\alpha\mathrm{x}}))=\mathrm{frac}{1}{2}((\backslash\alpha\mathrm{x})(\backslash\alpha\mathrm{y})+(\backslash\alpha\mathrm{y})(\backslash\alpha\mathrm{x}))\backslash\backslash&=\backslash\alpha\backslash\mathrm{left}(\mathrm{frac}{1}{2}(\mathrm{x}\backslash\mathrm{bar}{\mathrm{y}}+\mathrm{y}\backslash\mathrm{bar}{\mathrm{x}})\backslash\mathrm{right})=\backslash\alpha((\mathrm{x},\mathrm{y}))=(\mathrm{x},\mathrm{y}).\backslash\mathrm{end}{aligned}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | $(\alpha x,\alpha y)=\frac{1}{2}((\alpha x)(\overline{\alpha y})+(\alpha y)(\overline{\alpha x}))=\frac{1}{2}((\alpha x)(\alpha \overline{y})+(\alpha y)(\alpha \overline{x}))=\alpha\left(\frac{1}{2}(x\overline{y}+y\overline{x})\right)=\alpha((x,y))=(x,y).$ | $(\alpha x,\alpha y)=\frac{1}{2}((\alpha x)(\overline{\alpha y})+(\alpha y)(\overline{\alpha x}))=\frac{1}{2}((\alpha x)(\alpha \overline{y})+(\alpha y)(\alpha \overline{x}))=\alpha\left(\frac{1}{2}(x\overline{y}+y\overline{x})\right)=\alpha((x,y))=(x,y).$ |
| 0902.0431_E00412      | 68   | ■ 0.994 | $\backslash\mathrm{left}(\backslash\mathrm{begin}{array}{ccc}\backslash\mathrm{x_i}_{\mathrm{1}}&\&\backslash\mathrm{x}_{\mathrm{3}}&\&\backslash\mathrm{bar}{\mathrm{x}}_{\mathrm{2}}\backslash\backslash\backslash\mathrm{bar}{\mathrm{x}}_{\mathrm{3}}&\&\backslash\mathrm{x_i}_{\mathrm{2}}&\&\backslash\mathrm{x}_{\mathrm{1}}\backslash\backslash\backslash\mathrm{x}_{\mathrm{2}}&\&\backslash\mathrm{bar}{\mathrm{x}}_{\mathrm{1}}&\&\backslash\mathrm{x_i}_{\mathrm{3}}\backslash\mathrm{end}{array}\backslash\mathrm{right}),\quad\backslash\mathrm{quad}\backslash\mathrm{x_i}_{\mathrm{i}}\backslash\mathrm{in}\backslash\mathrm{boldsymbol{R}},\mathrm{x}_{\mathrm{i}}=\mathrm{a}_{\mathrm{i}}+\backslash\mathrm{boldsymbol{m}}_{\mathrm{i}}\backslash\mathrm{in}\backslash\mathrm{boldsymbol{C}}\backslash\mathrm{oplus}\backslash\mathrm{boldsymbol{C}}^{\mathrm{3}}=\backslash\mathrm{mathfrak{C}}$                                                                                                                                                                                                                                                                              | $\left(\begin{array}{ccc}\xi_1&x_3&\bar{x}_2\\\bar{x}_3&\xi_2&x_1\\x_2&\bar{x}_1&\xi_3\end{array}\right),\quad\xi_i\in\boldsymbol{R},x_i=a_i+\boldsymbol{m}_i\in\boldsymbol{C}\oplus\boldsymbol{C}^3=\mathfrak{C}$                                               | $\left(\begin{array}{ccc}\xi_1&x_3&\bar{x}_2\\\bar{x}_3&\xi_2&x_1\\x_2&\bar{x}_1&\xi_3\end{array}\right),\quad\xi_i\in\boldsymbol{R},x_i=a_i+\boldsymbol{m}_i\in\boldsymbol{C}\oplus\boldsymbol{C}^3=\mathfrak{C}$                                               |
| 0902.0431_E00673      | 112  | ■ 0.994 | $\backslash\mathrm{begin}{aligned}&=\backslash\Phi\backslash\mathrm{left}(-X\backslash\mathrm{vee}Y,-\mathrm{frac}{1}{2}(Y\backslash\mathrm{times}Y-\backslash\mathrm{x_i}X),\mathrm{frac}{1}{2}(X\backslash\mathrm{times}X-\backslash\mathrm{eta}Y),\mathrm{frac}{1}{4}((X,Y)-3\backslash\mathrm{x_i}\backslash\mathrm{eta})\backslash\mathrm{right})(Z,W,\backslash\mathrm{zeta},\backslash\mathrm{omega}),\quad\backslash\backslash&-\mathrm{frac}{3}{8}((X,W)-(Z,Y)+\backslash\mathrm{x_i}\backslash\mathrm{omega}-\backslash\mathrm{zeta}\backslash\mathrm{eta})(X,Y,\backslash\mathrm{x_i},\backslash\mathrm{eta})\backslash\backslash&=(P\backslash\mathrm{times}P)Q-\mathrm{frac}{3}{8}\{P,Q\}P.\backslash\mathrm{end}{aligned}$                                                                                                                                                                                                                                                                                                                                                                                                                                                         | $=\Phi\left(-X\vee Y,-\frac{1}{2}(Y\times Y-\xi X),\frac{1}{2}(X\times X-\eta Y),\frac{1}{4}((X,Y)-3\xi\eta)\right)(Z,W,\zeta,\omega),\\-\frac{3}{8}((X,W)-(Z,Y)+\xi\omega-\zeta\eta)(X,Y,\xi,\eta)\\=(P\times P)Q-\frac{3}{8}\{P,Q\}P.$                         | $=\Phi\left(-X\vee Y,-\frac{1}{2}(Y\times Y-\xi X),\frac{1}{2}(X\times X-\eta Y),\frac{1}{4}((X,Y)-3\xi\eta)\right)(Z,W,\zeta,\omega),\\-\frac{3}{8}((X,W)-(Z,Y)+\xi\omega-\zeta\eta)(X,Y,\xi,\eta)\\=(P\times P)Q-\frac{3}{8}\{P,Q\}P.$                         |

| Identifier            | Page | Conf.   | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Rendered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Image                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-----------------------|------|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0902.0431_E00136      | 25   | ■ 0.995 | $\begin{array}{cc} 2 G_{12}-G_{47}-G_{56}, & \\ -G_{47}+G_{56}, & \backslash \backslash 2 G_{13}-G_{46}-G_{57}, & \\ G_{46}+G_{57}, & \backslash \backslash 2 G_{23}-G_{45}-G_{67}, & \\ -G_{45}+G_{67} \end{array}$                                                                                                                                                                                                                                                                                                     | $\begin{array}{cc} 2G_{12}-G_{47}-G_{56}, & -G_{47}+G_{56}, \\ 2G_{13}-G_{46}-G_{57}, & G_{46}+G_{57}, \\ 2G_{23}-G_{45}-G_{67}, & -G_{45}+G_{67} \end{array}$                                                                                                                                                                                                                                                                                                                                                        | $\begin{array}{cc} 2G_{12}-G_{47}-G_{56}, & -G_{47}+G_{56}, \\ 2G_{13}-G_{46}-G_{57}, & G_{46}+G_{57}, \\ 2G_{23}-G_{45}-G_{67}, & -G_{45}+G_{67} \end{array}$                                                                                                                                                                                                                                                                                                                                                        |
| 0902.0431_E00282      | 45   | ■ 0.995 | $\left(D_{\{i\}} x, y\right)+\left(x, D_{\{i\}} y\right)=0, \quad \text {quad} \quad x, y \in \mathfrak{C} .$                                                                                                                                                                                                                                                                                                                                                                                                            | $(D_i x, y) + (x, D_i y) = 0, \quad x, y \in \mathfrak{C}.$                                                                                                                                                                                                                                                                                                                                                                                                                                                           | $(D_i x, y) + (x, D_i y) = 0, \quad x, y \in \mathfrak{C}.$                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 0902.0431_E01149 (i)  | 194  | ■ 0.995 | $\begin{aligned} & (\boldsymbol{u} \times \boldsymbol{v}) \boldsymbol{x} = \boldsymbol{x} \wedge \boldsymbol{u}_2 \wedge \boldsymbol{u}_3, \boldsymbol{v}) \boldsymbol{u}_1 + (\boldsymbol{u}_1 \wedge \boldsymbol{x} \wedge \boldsymbol{u}_3, \boldsymbol{v}) \boldsymbol{u}_2 \\ & + (\boldsymbol{u}_1 \wedge \boldsymbol{u}_2 \wedge \boldsymbol{x}, \boldsymbol{v}) \boldsymbol{u}_3 - \frac{1}{3}(\boldsymbol{u}, \boldsymbol{v}) \boldsymbol{x}. \end{aligned}$                                                    | $\begin{aligned} & (\boldsymbol{u} \times \boldsymbol{v}) \boldsymbol{x} = (\boldsymbol{x} \wedge \boldsymbol{u}_2 \wedge \boldsymbol{u}_3, \boldsymbol{v}) \boldsymbol{u}_1 + (\boldsymbol{u}_1 \wedge \boldsymbol{x} \wedge \boldsymbol{u}_3, \boldsymbol{v}) \boldsymbol{u}_2 \\ & + (\boldsymbol{u}_1 \wedge \boldsymbol{u}_2 \wedge \boldsymbol{x}, \boldsymbol{v}) \boldsymbol{u}_3 - \frac{1}{3}(\boldsymbol{u}, \boldsymbol{v}) \boldsymbol{x}. \end{aligned}$                                                | $\begin{aligned} & (\boldsymbol{u} \times \boldsymbol{v}) \boldsymbol{x} = (\boldsymbol{x} \wedge \boldsymbol{u}_2 \wedge \boldsymbol{u}_3, \boldsymbol{v}) \boldsymbol{u}_1 + (\boldsymbol{u}_1 \wedge \boldsymbol{x} \wedge \boldsymbol{u}_3, \boldsymbol{v}) \boldsymbol{u}_2 \\ & + (\boldsymbol{u}_1 \wedge \boldsymbol{u}_2 \wedge \boldsymbol{x}, \boldsymbol{v}) \boldsymbol{u}_3 - \frac{1}{3}(\boldsymbol{u}, \boldsymbol{v}) \boldsymbol{x}. \end{aligned}$                                                |
| 0902.0431_E00292      | 46   | ■ 0.995 | $\begin{aligned} & \mathfrak{e}_6^C = \{ \phi \in \operatorname{Hom}_{\mathfrak{C}} \mid \phi(X, X, X) = 0 \} \\ & = \{ \phi \in \operatorname{Hom}_{\mathfrak{C}} \mid -{}^t \phi(X \times Y) = \phi X \times Y + X \times \phi Y \}, \\ & \mathfrak{f}_4^C = \{ \delta \in \operatorname{Hom}_{\mathfrak{C}} \mid \delta(X \circ Y) = \delta X \circ Y + X \circ \delta Y \} \\ & = \{ \delta \in \operatorname{Hom}_{\mathfrak{C}} \mid \delta(X \times Y) = \delta X \times Y + X \times \delta Y \}. \end{aligned}$ | $\begin{aligned} \mathfrak{e}_6^C &= \{ \phi \in \operatorname{Hom}_{\mathfrak{C}} \mid (\phi X, X, X) = 0 \} \\ &= \{ \phi \in \operatorname{Hom}_{\mathfrak{C}} \mid -{}^t \phi(X \times Y) = \phi X \times Y + X \times \phi Y \}, \\ \mathfrak{f}_4^C &= \{ \delta \in \operatorname{Hom}_{\mathfrak{C}} \mid \delta(X \circ Y) = \delta X \circ Y + X \circ \delta Y \} \\ &= \{ \delta \in \operatorname{Hom}_{\mathfrak{C}} \mid \delta(X \times Y) = \delta X \times Y + X \times \delta Y \}. \end{aligned}$ | $\begin{aligned} \mathfrak{e}_6^C &= \{ \phi \in \operatorname{Hom}_{\mathfrak{C}} \mid (\phi X, X, X) = 0 \} \\ &= \{ \phi \in \operatorname{Hom}_{\mathfrak{C}} \mid -{}^t \phi(X \times Y) = \phi X \times Y + X \times \phi Y \}, \\ \mathfrak{f}_4^C &= \{ \delta \in \operatorname{Hom}_{\mathfrak{C}} \mid \delta(X \circ Y) = \delta X \circ Y + X \circ \delta Y \} \\ &= \{ \delta \in \operatorname{Hom}_{\mathfrak{C}} \mid \delta(X \times Y) = \delta X \times Y + X \times \delta Y \}. \end{aligned}$ |
| 0902.0431_E00822 (iv) | 137  | ■ 0.995 | $\begin{aligned} & a(X) \operatorname{det} X = a(X) (\operatorname{det} X) (M, N) = (\delta(X \times X) \times \delta(X \times X), N) \\ & = (\delta(X \times X), \delta(X \times X) \times N) = \frac{1}{2} a(X) (\delta(X \times X), \beta X) \\ & = \frac{1}{2} a(X) (3a(X) b(X \times X) - \operatorname{det} X). \end{aligned}$                                                                                                                                                                                     | $\begin{aligned} a(X) \operatorname{det} X &= a(X) (\operatorname{det} X) (M, N) = (\delta(X \times X) \times \delta(X \times X), N) \\ &= (\delta(X \times X), \delta(X \times X) \times N) = \frac{1}{2} a(X) (\delta(X \times X), \beta X) \\ &= \frac{1}{2} a(X) (3a(X) b(X \times X) - \operatorname{det} X). \end{aligned}$                                                                                                                                                                                     | $\begin{aligned} a(X) \operatorname{det} X &= a(X) (\operatorname{det} X) (M, N) = (\delta(X \times X) \times \delta(X \times X), N) \\ &= (\delta(X \times X), \delta(X \times X) \times N) = \frac{1}{2} a(X) (\delta(X \times X), \beta X) \\ &= \frac{1}{2} a(X) (3a(X) b(X \times X) - \operatorname{det} X). \end{aligned}$                                                                                                                                                                                     |
| 0902.0431_E01042      | 178  | ■ 0.995 | $\left(E_8^C\right)_{1,1^-,1_-}=\left\{\alpha \in E_8^C \mid \alpha 1=1, \alpha 1^-=1^-, \alpha 1_-=1_-\right\} .$                                                                                                                                                                                                                                                                                                                                                                                                       | $\left(E_8^C\right)_{1,1^-,1_-}=\left\{\alpha \in E_8^C \mid \alpha 1=1, \alpha 1^-=1^-, \alpha 1_-=1_-\right\} .$                                                                                                                                                                                                                                                                                                                                                                                                    | $\left(E_8^C\right)_{1,1^-,1_-}=\left\{\alpha \in E_8^C \mid \alpha 1=1, \alpha 1^-=1^-, \alpha 1_-=1_-\right\} .$                                                                                                                                                                                                                                                                                                                                                                                                    |

| Identifier       | Page | Conf.   | LaTeX source                                                                                                                                                                                                                                                                                                      | Rendered                                                                                                                                                                                                                                                                                                          | Image                                                                                                                                                                                                                                                                                                             |
|------------------|------|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0902.0431_E00387 | 64   | ■ 0.995 | $\gamma\left(\begin{array}{ccc}\xi_1 & x_3 & \bar{x}_2 \\ \bar{x}_3 & \xi_2 & x_1 \\ x_2 & \bar{x}_1 & \xi_3\end{array}\right)=\left(\begin{array}{ccc}\xi_1 & \gamma x_3 & \overline{\gamma x_2} \\ \overline{\gamma x_3} & \xi_2 & \gamma x_1 \\ \gamma x_2 & \overline{\gamma x_1} & \xi_3\end{array}\right).$ | $\gamma\left(\begin{array}{ccc}\xi_1 & x_3 & \bar{x}_2 \\ \bar{x}_3 & \xi_2 & x_1 \\ x_2 & \bar{x}_1 & \xi_3\end{array}\right)=\left(\begin{array}{ccc}\xi_1 & \gamma x_3 & \overline{\gamma x_2} \\ \overline{\gamma x_3} & \xi_2 & \gamma x_1 \\ \gamma x_2 & \overline{\gamma x_1} & \xi_3\end{array}\right).$ | $\gamma\left(\begin{array}{ccc}\xi_1 & x_3 & \bar{x}_2 \\ \bar{x}_3 & \xi_2 & x_1 \\ x_2 & \bar{x}_1 & \xi_3\end{array}\right)=\left(\begin{array}{ccc}\xi_1 & \gamma x_3 & \overline{\gamma x_2} \\ \overline{\gamma x_3} & \xi_2 & \gamma x_1 \\ \gamma x_2 & \overline{\gamma x_1} & \xi_3\end{array}\right).$ |
| 0902.0431_E00744 | 123  | ■ 0.995 | $\Phi_0\Phi_0(X,Y,\xi,\eta)=\Phi_0\left(-\frac{X}{3},\frac{Y}{3},\xi,-\eta\right)=\left(\frac{X}{9},\frac{Y}{9},\xi,\eta\right),$                                                                                                                                                                                 | $\Phi_0\Phi_0(X,Y,\xi,\eta)=\Phi_0\left(-\frac{X}{3},\frac{Y}{3},\xi,-\eta\right)=\left(\frac{X}{9},\frac{Y}{9},\xi,\eta\right),$                                                                                                                                                                                 | $\Phi_0\Phi_0(X,Y,\xi,\eta)=\Phi_0\left(-\frac{X}{3},\frac{Y}{3},\xi,-\eta\right)=\left(\frac{X}{9},\frac{Y}{9},\xi,\eta\right),$                                                                                                                                                                                 |
| 0902.0431_E01109 | 188  | ■ 0.995 | $S\,0(16),\quad S\,s(16)\,.$                                                                                                                                                                                                                                                                                      | $SO(16),\quad Ss(16).$                                                                                                                                                                                                                                                                                            | $SO(16),\quad Ss(16).$                                                                                                                                                                                                                                                                                            |
| 0902.0431_E01124 | 190  | ■ 0.995 | $\tau\widetilde{\lambda}(D,\phi,\boldsymbol{X},\boldsymbol{Y})=(-\tau^tD,-\tau^t\phi\tau,-\tau\boldsymbol{Y},-\tau\boldsymbol{X}).$                                                                                                                                                                               | $\tau\widetilde{\lambda}(D,\phi,\boldsymbol{X},\boldsymbol{Y})=(-\tau^tD,-\tau^t\phi\tau,-\tau\boldsymbol{Y},-\tau\boldsymbol{X}).$                                                                                                                                                                               | $\tau\widetilde{\lambda}(D,\phi,\boldsymbol{X},\boldsymbol{Y})=(-\tau^tD,-\tau^t\phi\tau,-\tau\boldsymbol{Y},-\tau\boldsymbol{X}).$                                                                                                                                                                               |
| 0902.0431_E01044 | 178  | ■ 0.995 | $\alpha=\left(\begin{array}{cccccc}\beta_1 & \beta_{12} & \beta_{13} & 0 & 0 & 0 \\ \beta_{21} & \beta_2 & \beta_{23} & 0 & 0 & 0 \\ \beta_{31} & \beta_{32} & \beta_3 & 0 & 0 & 0 \\ a_1 & b_1 & c_1 & 1 & 0 & 0 \\ a_2 & b_2 & c_2 & 0 & 1 & 0 \\ a_3 & b_3 & c_3 & 0 & 0 & 1\end{array}\right),$               | $\alpha=\left(\begin{array}{cccccc}\beta_1 & \beta_{12} & \beta_{13} & 0 & 0 & 0 \\ \beta_{21} & \beta_2 & \beta_{23} & 0 & 0 & 0 \\ \beta_{31} & \beta_{32} & \beta_3 & 0 & 0 & 0 \\ a_1 & b_1 & c_1 & 1 & 0 & 0 \\ a_2 & b_2 & c_2 & 0 & 1 & 0 \\ a_3 & b_3 & c_3 & 0 & 0 & 1\end{array}\right),$               | $\alpha=\left(\begin{array}{cccccc}\beta_1 & \beta_{12} & \beta_{13} & 0 & 0 & 0 \\ \beta_{21} & \beta_2 & \beta_{23} & 0 & 0 & 0 \\ \beta_{31} & \beta_{32} & \beta_3 & 0 & 0 & 0 \\ a_1 & b_1 & c_1 & 1 & 0 & 0 \\ a_2 & b_2 & c_2 & 0 & 1 & 0 \\ a_3 & b_3 & c_3 & 0 & 0 & 1\end{array}\right),$               |
| 0902.0431_E00410 | 68   | ■ 0.995 | $w\left(\begin{array}{ccc}\xi_1 & x_3 & \bar{x}_2 \\ \bar{x}_3 & \xi_2 & x_1 \\ x_2 & \bar{x}_1 & \xi_3\end{array}\right)=\left(\begin{array}{ccc}\xi_1 & \omega x_3 & \overline{\omega x_2} \\ \overline{\omega x_3} & \xi_2 & \omega x_1 \\ \omega x_2 & \overline{\omega x_1} & \xi_3\end{array}\right).$      | $w\left(\begin{array}{ccc}\xi_1 & x_3 & \bar{x}_2 \\ \bar{x}_3 & \xi_2 & x_1 \\ x_2 & \bar{x}_1 & \xi_3\end{array}\right)=\left(\begin{array}{ccc}\xi_1 & \omega x_3 & \overline{\omega x_2} \\ \overline{\omega x_3} & \xi_2 & \omega x_1 \\ \omega x_2 & \overline{\omega x_1} & \xi_3\end{array}\right).$      | $w\left(\begin{array}{ccc}\xi_1 & x_3 & \bar{x}_2 \\ \bar{x}_3 & \xi_2 & x_1 \\ x_2 & \bar{x}_1 & \xi_3\end{array}\right)=\left(\begin{array}{ccc}\xi_1 & \omega x_3 & \overline{\omega x_2} \\ \overline{\omega x_3} & \xi_2 & \omega x_1 \\ \omega x_2 & \overline{\omega x_1} & \xi_3\end{array}\right).$      |
| 0902.0431_E01134 | 191  | ■ 0.995 | $(\mathrm{ad}\,\phi')(D,\phi,\boldsymbol{X},\boldsymbol{Y})=(0,(\mathrm{ad}\,\phi')\phi,\phi'\boldsymbol{X},-{}^t\phi'\boldsymbol{Y}),$                                                                                                                                                                           | $(\mathrm{ad}\,\phi')(D,\phi,\boldsymbol{X},\boldsymbol{Y})=(0,(\mathrm{ad}\,\phi')\phi,\phi'\boldsymbol{X},-{}^t\phi'\boldsymbol{Y}),$                                                                                                                                                                           | $(\mathrm{ad}\,\phi')(D,\phi,\boldsymbol{X},\boldsymbol{Y})=(0,(\mathrm{ad}\,\phi')\phi,\phi'\boldsymbol{X},-{}^t\phi'\boldsymbol{Y}),$                                                                                                                                                                           |
| 0902.0431_E00493 | 81   | ■ 0.996 | $\delta(X\circ Y)=\delta X\circ Y+X\circ\delta Y,\quad X,Y\in(\mathfrak{M}^r)^C.$                                                                                                                                                                                                                                 | $\delta(X\circ Y)=\delta X\circ Y+X\circ\delta Y,\quad X,Y\in(\mathfrak{M}^r)^C.$                                                                                                                                                                                                                                 | $\delta(X\circ Y)=\delta X\circ Y+X\circ\delta Y,\quad X,Y\in(\mathfrak{M}^r)^C.$                                                                                                                                                                                                                                 |



| Identifier       | Page | Conf.   | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                   | Rendered                                                                                                                                                                                                                                                                                                                                           | Image                                                                                                                                                                                                                                                                                                                                              |
|------------------|------|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0902.0431_E00571 | 95   | ■ 0.996 | J=\left(\begin{array}{ccc} J & \theta & \theta \\ \theta & J & \theta \\ \theta & \theta & J \end{array}\right) \in M(6, C), \quad J=\left(\begin{array}{cc} \theta & 1 \\ -1 & \theta \end{array}\right)                                                                                                                                                                                      | $J = \begin{pmatrix} J & 0 & 0 \\ 0 & J & 0 \\ 0 & 0 & J \end{pmatrix} \in M(6, C), \quad J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$                                                                                                                                                                                                       | $J = \begin{pmatrix} J & 0 & 0 \\ 0 & J & 0 \\ 0 & 0 & J \end{pmatrix} \in M(6, C), \quad J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$                                                                                                                                                                                                       |
| 0902.0431_E00638 | 105  | ■ 0.996 | \operatorname{dim}\left(\mathfrak{e}_{6,C}, \boldsymbol{\mathrm{C}}\right)=8+8=16 .                                                                                                                                                                                                                                                                                                            | $\dim(\mathfrak{e}_{6,C}) = 8 + 8 = 16.$                                                                                                                                                                                                                                                                                                           | $\dim(\mathfrak{e}_{6,C}) = 8 + 8 = 16.$                                                                                                                                                                                                                                                                                                           |
| 0902.0431_E00647 | 107  | ■ 0.996 | \begin{aligned} \operatorname{det}(\alpha X) &= (\operatorname{det}(h(A, B))) (\operatorname{det} X) \\ \left(\operatorname{det}\left(h(A, B)^*\right)\right) &= \operatorname{det} X, \\ \angle \alpha X, \alpha Y &= \angle h(A, B) X h(A, B)^*, h(A, B) Y h(A, B)^* \\ &= (\tau h(A, B) \tau X \tau h(A, B)^*, h(A, B) Y h(A, B)^*) \\ &= (\tau X, Y) = \langle X, Y \rangle. \end{aligned} | $\begin{aligned} \det(\alpha X) &= (\det(h(A, B))) (\det X) (\det(h(A, B)^*)) = \det X, \\ \langle \alpha X, \alpha Y \rangle &= \langle h(A, B) X h(A, B)^*, h(A, B) Y h(A, B)^* \rangle \\ &= (\tau h(A, B) \tau X \tau h(A, B)^*, h(A, B) Y h(A, B)^*) \\ &= (\tau X, Y) = \langle X, Y \rangle. \end{aligned}$                                 | $\begin{aligned} \det(\alpha X) &= (\det(h(A, B))) (\det X) (\det(h(A, B)^*)) = \det X, \\ \langle \alpha X, \alpha Y \rangle &= \langle h(A, B) X h(A, B)^*, h(A, B) Y h(A, B)^* \rangle \\ &= (\tau h(A, B) \tau X \tau h(A, B)^*, h(A, B) Y h(A, B)^*) \\ &= (\tau X, Y) = \langle X, Y \rangle. \end{aligned}$                                 |
| 0902.0431_E00687 | 114  | ■ 0.996 | \operatorname{dim}_{\mathbf{C}}\left(\mathfrak{e}_{7}^{\mathbf{C}}\right)=78+27 \times 2+1=133 .                                                                                                                                                                                                                                                                                               | $\dim_C(\mathfrak{e}_7^C) = 78 + 27 \times 2 + 1 = 133.$                                                                                                                                                                                                                                                                                           | $\dim_C(\mathfrak{e}_7^C) = 78 + 27 \times 2 + 1 = 133.$                                                                                                                                                                                                                                                                                           |
| 0902.0431_E00807 | 135  | ■ 0.996 | \begin{aligned} &\left(\theta, \theta, \theta, \theta^3\right) \eta=\varphi_1(\theta)(\theta, \theta, \theta, \eta)=\varphi_1(\theta) \alpha(\theta, \theta, 1, 0) \\ &=\alpha \varphi_1(\theta)(\theta, \theta, 1, 0)=\alpha\left(\theta, \theta, \theta^3, 0\right)=\left(\theta, \theta, \theta, \theta^3 \eta\right), \end{aligned}                                                        | $\begin{aligned} (0, 0, 0, \theta^{-3} \eta) &= \varphi_1(\theta)(0, 0, 0, \eta) = \varphi_1(\theta) \alpha(0, 0, 1, 0) \\ &= \alpha \varphi_1(\theta)(0, 0, 1, 0) = \alpha(0, 0, \theta^3, 0) = (0, 0, 0, \theta^3 \eta), \end{aligned}$                                                                                                          | $\begin{aligned} (0, 0, 0, \theta^{-3} \eta) &= \varphi_1(\theta)(0, 0, 0, \eta) = \varphi_1(\theta) \alpha(0, 0, 1, 0) \\ &= \alpha \varphi_1(\theta)(0, 0, 1, 0) = \alpha(0, 0, \theta^3, 0) = (0, 0, 0, \theta^3 \eta), \end{aligned}$                                                                                                          |
| 0902.0431_E00933 | 154  | ■ 0.996 | \psi(A, 1) P=f^{-1}(A(f P)), \quad \psi(A, \epsilon) P=f^{-1}(A(f \bar{P})), \quad P \in\left(\mathfrak{P}_C\right)^C .                                                                                                                                                                                                                                                                        | $\psi(A, 1) P = f^{-1}(A(f P)), \quad \psi(A, \epsilon) P = f^{-1}(A(f \bar{P})), \quad P \in (\mathfrak{P}_C)^C .$                                                                                                                                                                                                                                | $\psi(A, 1) P = f^{-1}(A(f P)), \quad \psi(A, \epsilon) P = f^{-1}(A(f \bar{P})), \quad P \in (\mathfrak{P}_C)^C .$                                                                                                                                                                                                                                |
| 0902.0431_E00977 | 162  | ■ 0.996 | \begin{aligned} \Phi &= (\Phi, \theta, \theta, \theta, \theta), \quad P^-= (\theta, P, \theta, \theta, \theta), \\ Q_- &= (\theta, \theta, Q, \theta, \theta), \quad r= (\theta, \theta, \theta, r, \theta), \\ s^- &= (\theta, \theta, \theta, \theta, s), \quad t_- = (\theta, \theta, \theta, \theta, t). \end{aligned}                                                                     | $\begin{aligned} \Phi &= (\Phi, \theta, \theta, \theta, \theta), \quad P^- = (\theta, P, \theta, \theta, \theta), \\ Q_- &= (\theta, \theta, Q, \theta, \theta), \quad r = (\theta, \theta, \theta, r, \theta), \\ s^- &= (\theta, \theta, \theta, \theta, s), \quad t_- = (\theta, \theta, \theta, \theta, t). \end{aligned}$                     | $\begin{aligned} \Phi &= (\Phi, \theta, \theta, \theta, \theta), \quad P^- = (\theta, P, \theta, \theta, \theta), \\ Q_- &= (\theta, \theta, Q, \theta, \theta), \quad r = (\theta, \theta, \theta, r, \theta), \\ s^- &= (\theta, \theta, \theta, \theta, s), \quad t_- = (\theta, \theta, \theta, \theta, t). \end{aligned}$                     |
| 0902.0431_E01098 | 186  | ■ 0.996 | \begin{aligned} &\zeta\left[l\left(i c_1 E\right), l\left(i c_2 E\right)\right]=\zeta l\left[i c_1 E, i c_2 E\right]=\zeta 0=0 \\ &=\Theta\left(0,0,0,0,2 c_1,-2 c_1\right), \Theta\left(0,0,0,0,2 c_2,-2 c_2\right)) \\ &=\left[\zeta l\left(i c_1 E\right), \zeta l\left(i c_2 E\right)\right] . \end{aligned}                                                                               | $\begin{aligned} \zeta[l(ic_1 E), l(ic_2 E)] &= \zeta l[ic_1 E, ic_2 E] = \zeta 0 = 0 \\ &= \Theta(0, 0, 0, 0, 2c_1, -2c_1), \Theta(0, 0, 0, 0, 2c_2, -2c_2)) \\ &= [\zeta l(ic_1 E), \zeta l(ic_2 E)]. \end{aligned}$                                                                                                                             | $\begin{aligned} \zeta[l(ic_1 E), l(ic_2 E)] &= \zeta l[ic_1 E, ic_2 E] = \zeta 0 = 0 \\ &= \Theta(0, 0, 0, 0, 2c_1, -2c_1), \Theta(0, 0, 0, 0, 2c_2, -2c_2)) \\ &= [\zeta l(ic_1 E), \zeta l(ic_2 E)]. \end{aligned}$                                                                                                                             |
| 0902.0431_E01186 | 199  | ■ 0.996 | \begin{aligned} &((\boldsymbol{a} \times *(\boldsymbol{b} \wedge \boldsymbol{x})) \boldsymbol{v}, \boldsymbol{w}) \\ &= (\boldsymbol{v} \wedge \boldsymbol{a}, \boldsymbol{w} \wedge *(\boldsymbol{b} \wedge \boldsymbol{x})) - \frac{3}{5}(*(\boldsymbol{a} \wedge \boldsymbol{b}), \boldsymbol{x})(\boldsymbol{v}, \boldsymbol{w}) \end{aligned}                                             | $\begin{aligned} ((\boldsymbol{a} \times *(\boldsymbol{b} \wedge \boldsymbol{x})) \boldsymbol{v}, \boldsymbol{w}) \\ = (\boldsymbol{v} \wedge \boldsymbol{a}, \boldsymbol{w} \wedge *(\boldsymbol{b} \wedge \boldsymbol{x})) - \frac{3}{5}(*(\boldsymbol{a} \wedge \boldsymbol{b}), \boldsymbol{x})(\boldsymbol{v}, \boldsymbol{w}) \end{aligned}$ | $\begin{aligned} ((\boldsymbol{a} \times *(\boldsymbol{b} \wedge \boldsymbol{x})) \boldsymbol{v}, \boldsymbol{w}) \\ = (\boldsymbol{v} \wedge \boldsymbol{a}, \boldsymbol{w} \wedge *(\boldsymbol{b} \wedge \boldsymbol{x})) - \frac{3}{5}(*(\boldsymbol{a} \wedge \boldsymbol{b}), \boldsymbol{x})(\boldsymbol{v}, \boldsymbol{w}) \end{aligned}$ |

| Identifier       | Page | Conf.   | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                          | Rendered                                                                                                                                                                                                                                                                                                                                                                                                              | Image                                                                                                                                                                                                                                                                                                                                                                                                                 |
|------------------|------|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0902.0431_E00724 | 119  | ■ 0.996 | $\begin{aligned} \mathfrak{a} \supset & \left\{ \mathfrak{a}, \mathfrak{e}_7^C \right\} \supset \left[ \Phi \left( \mathfrak{e}_6^C, 0, 0, 0 \right), \Phi \left( 0, \mathfrak{J}^C, 0, 0 \right) \right] \\ & = \Phi \left( 0, \mathfrak{e}_6^C \mathfrak{J}^C, 0, 0 \right) = \Phi \left( 0, \mathfrak{J}^C, 0, 0 \right) \text{ (Proposition 3.3.2.(3)).} \end{aligned}$                                           | $\mathfrak{a} \supset [\mathfrak{a}, \mathfrak{e}_7^C] \supset [\Phi(\mathfrak{e}_6^C, 0, 0, 0), \Phi(0, \mathfrak{J}^C, 0, 0)]$ $= \Phi(0, \mathfrak{e}_6^C \mathfrak{J}^C, 0, 0) = \Phi(0, \mathfrak{J}^C, 0, 0) \text{ (Proposition 3.3.2.(3)).}$                                                                                                                                                                  | $\mathfrak{a} \supset [\mathfrak{a}, \mathfrak{e}_7^C] \supset [\Phi(\mathfrak{e}_6^C, 0, 0, 0), \Phi(0, \mathfrak{J}^C, 0, 0)]$ $= \Phi(0, \mathfrak{e}_6^C \mathfrak{J}^C, 0, 0) = \Phi(0, \mathfrak{J}^C, 0, 0) \text{ (Proposition 3.3.2.(3)).}$                                                                                                                                                                  |
| 0902.0431_E00063 | 14   | ■ 0.997 | $\begin{aligned} \lambda_{23} + \lambda_{45} + \lambda_{67} = 0, \quad & -\lambda_{13} - \lambda_{46} + \lambda_{57} = 0, \\ \lambda_{12} + \lambda_{47} + \lambda_{56} = 0, \quad & -\lambda_{15} + \lambda_{26} - \lambda_{37} = 0, \\ \lambda_{14} - \lambda_{27} - \lambda_{36} = 0, \quad & -\lambda_{17} - \lambda_{24} + \lambda_{35} = 0, \\ & \lambda_{16} + \lambda_{25} + \lambda_{34} = 0, \end{aligned}$ | $\begin{aligned} \lambda_{23} + \lambda_{45} + \lambda_{67} = 0, \quad & -\lambda_{13} - \lambda_{46} + \lambda_{57} = 0, \\ \lambda_{12} + \lambda_{47} + \lambda_{56} = 0, \quad & -\lambda_{15} + \lambda_{26} - \lambda_{37} = 0, \\ \lambda_{14} - \lambda_{27} - \lambda_{36} = 0, \quad & -\lambda_{17} - \lambda_{24} + \lambda_{35} = 0, \\ & \lambda_{16} + \lambda_{25} + \lambda_{34} = 0, \end{aligned}$ | $\begin{aligned} \lambda_{23} + \lambda_{45} + \lambda_{67} = 0, \quad & -\lambda_{13} - \lambda_{46} + \lambda_{57} = 0, \\ \lambda_{12} + \lambda_{47} + \lambda_{56} = 0, \quad & -\lambda_{15} + \lambda_{26} - \lambda_{37} = 0, \\ \lambda_{14} - \lambda_{27} - \lambda_{36} = 0, \quad & -\lambda_{17} - \lambda_{24} + \lambda_{35} = 0, \\ & \lambda_{16} + \lambda_{25} + \lambda_{34} = 0, \end{aligned}$ |
| 0902.0431_E00264 | 43   | ■ 0.997 | $(X, Y) = \frac{1}{2} \operatorname{tr} \left( X Y + Y^* X^* \right) .$                                                                                                                                                                                                                                                                                                                                               | $(X, Y) = \frac{1}{2} \operatorname{tr} (XY + Y^* X^*) .$                                                                                                                                                                                                                                                                                                                                                             | $(X, Y) = \frac{1}{2} \operatorname{tr} (XY + Y^* X^*) .$                                                                                                                                                                                                                                                                                                                                                             |
| 0902.0431_E00335 | 53   | ■ 0.997 | $\begin{aligned} H \left( e_k + i e_{4+k} \right) &= - \sum_{j=0}^3 \lambda_k i G_{j4+j} \left( e_k + i e_{4+k} \right) \\ &= - i \lambda_k \left( - e_{4+k} + i e_k \right) = \lambda_k \left( e_k + i e_{4+k} \right), \end{aligned}$                                                                                                                                                                               | $\begin{aligned} H \left( e_k + i e_{4+k} \right) &= - \sum_{j=0}^3 \lambda_k i G_{j4+j} \left( e_k + i e_{4+k} \right) \\ &= - i \lambda_k \left( - e_{4+k} + i e_k \right) = \lambda_k \left( e_k + i e_{4+k} \right), \end{aligned}$                                                                                                                                                                               | $\begin{aligned} H \left( e_k + i e_{4+k} \right) &= - \sum_{j=0}^3 \lambda_k i G_{j4+j} \left( e_k + i e_{4+k} \right) \\ &= - i \lambda_k \left( - e_{4+k} + i e_k \right) = \lambda_k \left( e_k + i e_{4+k} \right), \end{aligned}$                                                                                                                                                                               |
| 0902.0431_E00081 | 146  | ■ 0.997 | $\alpha X = \left( \begin{array}{ccc} \xi_1 & 0 & 0 \\ 0 & \xi_2 & 0 \\ 0 & 0 & \xi_3 \end{array} \right), \quad \xi_i \in C.$                                                                                                                                                                                                                                                                                        | $\alpha X = \left( \begin{array}{ccc} \xi_1 & 0 & 0 \\ 0 & \xi_2 & 0 \\ 0 & 0 & \xi_3 \end{array} \right), \quad \xi_i \in C.$                                                                                                                                                                                                                                                                                        | $\alpha X = \left( \begin{array}{ccc} \xi_1 & 0 & 0 \\ 0 & \xi_2 & 0 \\ 0 & 0 & \xi_3 \end{array} \right), \quad \xi_i \in C.$                                                                                                                                                                                                                                                                                        |
| 0902.0431_E01057 | 180  | ■ 0.997 | $-21_- = \left[ \left( \Phi, P, Q, r, s, t \right), 1_- \right] = \left( 0, 0, -P, s, 0, -2r \right),$                                                                                                                                                                                                                                                                                                                | $-21_- = [(\Phi, P, Q, r, s, t), 1_-] = (0, 0, -P, s, 0, -2r),$                                                                                                                                                                                                                                                                                                                                                       | $-21_- = [(\Phi, P, Q, r, s, t), 1_-] = (0, 0, -P, s, 0, -2r),$                                                                                                                                                                                                                                                                                                                                                       |
| 0902.0431_E00081 | 17   | ■ 0.997 | $\begin{aligned} H_1 &= -G_{23} + G_{45}, \quad L_{12} = G_{24} + G_{35}, \quad L_{21} = -G_{25} + G_{34}, \quad L_{13} = G_{26} + G_{37}, \\ L_{21} &= -G_{27} + G_{36}, \quad L_{23} = G_{46} + G_{57}, \quad L_{32} = -G_{47} + G_{56}. \end{aligned}$                                                                                                                                                             | $\begin{aligned} H_1 &= -G_{23} + G_{45}, \quad H_2 = -G_{45} + G_{67} \\ L_{12} &= G_{24} + G_{35}, \quad L_{21} = -G_{25} + G_{34}, \quad L_{13} = G_{26} + G_{37}, \\ L_{21} &= -G_{27} + G_{36}, \quad L_{23} = G_{46} + G_{57}, \quad L_{32} = -G_{47} + G_{56}. \end{aligned}$                                                                                                                                  | $\begin{aligned} H_1 &= -G_{23} + G_{45}, \quad H_2 = -G_{45} + G_{67} \\ L_{12} &= G_{24} + G_{35}, \quad L_{21} = -G_{25} + G_{34}, \quad L_{13} = G_{26} + G_{37}, \\ L_{21} &= -G_{27} + G_{36}, \quad L_{23} = G_{46} + G_{57}, \quad L_{32} = -G_{47} + G_{56}. \end{aligned}$                                                                                                                                  |

| Identifier       | Page | Conf.   | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Rendered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Image                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|------------------|------|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0902.0431_E00624 | 104  | ■ 0.997 | $\begin{aligned} & (g(\alpha E))^2 = g(\alpha E) \circ g(\alpha E) = g(\gamma(\alpha E \times \alpha E)) + \frac{1}{4}(\gamma \alpha E, \alpha E)E \text{ (Lemma 3.12.1)} \\ & = g(\gamma \tau \alpha \tau E) + \frac{1}{4}\langle \tau \gamma \alpha E, \alpha E \rangle E = g(\alpha \tau \gamma E) + \frac{1}{4}\langle \alpha \tau \gamma E, \alpha E \rangle E \\ & = g(\alpha E) + \frac{1}{4}\langle \alpha E, \alpha E \rangle E = g(\alpha E) + \frac{1}{4}\langle E, E \rangle E = g(\alpha E) + \frac{3}{4}E. \end{aligned}$ | $\begin{aligned} & (g(\alpha E))^2 = g(\alpha E) \circ g(\alpha E) = g(\gamma(\alpha E \times \alpha E)) + \frac{1}{4}(\gamma \alpha E, \alpha E)E \text{ (Lemma 3.12.1)} \\ & = g(\gamma \tau \alpha \tau E) + \frac{1}{4}\langle \tau \gamma \alpha E, \alpha E \rangle E = g(\alpha \tau \gamma E) + \frac{1}{4}\langle \alpha \tau \gamma E, \alpha E \rangle E \\ & = g(\alpha E) + \frac{1}{4}\langle \alpha E, \alpha E \rangle E = g(\alpha E) + \frac{1}{4}\langle E, E \rangle E = g(\alpha E) + \frac{3}{4}E. \end{aligned}$ | $\begin{aligned} (g(\alpha E))^2 &= g(\alpha E) \circ g(\alpha E) = g(\gamma(\alpha E \times \alpha E)) + \frac{1}{4}(\gamma \alpha E, \alpha E)E \text{ (Lemma 3.12.1)} \\ &= g(\gamma \tau \alpha \tau E) + \frac{1}{4}\langle \tau \gamma \alpha E, \alpha E \rangle E = g(\alpha \tau \gamma E) + \frac{1}{4}\langle \alpha \tau \gamma E, \alpha E \rangle E \\ &= g(\alpha E) + \frac{1}{4}\langle \alpha E, \alpha E \rangle E = g(\alpha E) + \frac{1}{4}\langle E, E \rangle E = g(\alpha E) + \frac{3}{4}E. \end{aligned}$ |
| 0902.0431_E00374 | 62   | ■ 0.997 | $F_4 / \operatorname{Spin}(9) \simeq \mathfrak{C}P_2.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | $F_4 / \operatorname{Spin}(9) \simeq \mathfrak{C}P_2.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | $F_4 / \operatorname{Spin}(9) \simeq \mathfrak{C}P_2.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 0902.0431_E00434 | 71   | ■ 0.997 | $F_4^{\mathcal{C}} \simeq F_4 \times \boldsymbol{R}^{52}.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | $F_4^{\mathcal{C}} \simeq F_4 \times \boldsymbol{R}^{52}.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | $F_4^{\mathcal{C}} \simeq F_4 \times \boldsymbol{R}^{52}.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 0902.0431_E00717 | 118  | ■ 0.997 | $\left[ \Phi(\phi_1, A_1, -\tau A_1, \nu_1), \Phi(\phi_2, A_2, -\tau A_2, \nu_2) \right] = \Phi(\phi, A, -\tau A, \nu),$                                                                                                                                                                                                                                                                                                                                                                                                                | $[\Phi(\phi_1, A_1, -\tau A_1, \nu_1), \Phi(\phi_2, A_2, -\tau A_2, \nu_2)] = \Phi(\phi, A, -\tau A, \nu),$                                                                                                                                                                                                                                                                                                                                                                                                                             | $[\Phi(\phi_1, A_1, -\tau A_1, \nu_1), \Phi(\phi_2, A_2, -\tau A_2, \nu_2)] = \Phi(\phi, A, -\tau A, \nu),$                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 0902.0431_E00914 | 152  | ■ 0.997 | $\dim \mathfrak{e}_{7,C} = 16 + 18 + 1 = 35.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | $\dim \mathfrak{e}_{7,C} = 16 + 18 + 1 = 35.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | $\dim \mathfrak{e}_{7,C} = 16 + 18 + 1 = 35.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 0902.0431_E00208 | 35   | ■ 0.997 | $\begin{aligned} & \operatorname{Spin}(7) \cap \kappa \alpha = \alpha, \\ & G_2 \cap \kappa \alpha = \alpha, \lambda \in \mathfrak{S}_3 \\ & \lambda \alpha = \alpha, \nu \alpha = \alpha \\ & \nu \alpha = \alpha \\ & \pi \alpha = \alpha. \end{aligned}$                                                                                                                                                                                                                                                                             | $\begin{aligned} \operatorname{Spin}(7) &= \{ \alpha \in \operatorname{Spin}(8) \mid \kappa \alpha = \alpha \}, \\ G_2 &= \{ \alpha \in \operatorname{Spin}(8) \mid \lambda \alpha = \alpha, \lambda \in \mathfrak{S}_3 \} \\ &= \{ \alpha \in \operatorname{Spin}(8) \mid \pi \alpha = \alpha, \nu \alpha = \alpha \} \\ &= \{ \alpha \in \operatorname{Spin}(8) \mid \nu \alpha = \alpha \} \\ &= \{ \alpha \in \operatorname{Spin}(7) \mid \pi \alpha = \alpha \}. \end{aligned}$                                                    | $\begin{aligned} \operatorname{Spin}(7) &= \{ \alpha \in \operatorname{Spin}(8) \mid \kappa \alpha = \alpha \}, \\ G_2 &= \{ \alpha \in \operatorname{Spin}(8) \mid \lambda \alpha = \alpha, \lambda \in \mathfrak{S}_3 \} \\ &= \{ \alpha \in \operatorname{Spin}(8) \mid \pi \alpha = \alpha, \nu \alpha = \alpha \} \\ &= \{ \alpha \in \operatorname{Spin}(8) \mid \nu \alpha = \alpha \} \\ &= \{ \alpha \in \operatorname{Spin}(7) \mid \pi \alpha = \alpha \}. \end{aligned}$                                                 |



| Identifier            | Page | Conf.   | LaTeX source                                                                                                                                                                                                                                                                                                                                          | Rendered                                                                                                                                                                                                                                                                                                               | Image                                                                                                                                                                                                                                                                                                                  |
|-----------------------|------|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0902.0431_E00183      | 31   | ■ 0.998 | $\widetilde{B}_3=\{\widetilde{\alpha}\in SO(8)\mid (\alpha x)(\widetilde{\alpha}y)=\widetilde{\alpha}(xy),x,y\in \mathfrak{C}\text{ for some }\alpha\in SO(7)\}.$                                                                                                                                                                                     | $\widetilde{B}_3=\{\widetilde{\alpha}\in SO(8)\mid (\alpha x)(\widetilde{\alpha}y)=\widetilde{\alpha}(xy),x,y\in \mathfrak{C}\text{ for some }\alpha\in SO(7)\}.$                                                                                                                                                      | $\widetilde{B}_3=\{\widetilde{\alpha}\in SO(8)\mid (\alpha x)(\widetilde{\alpha}y)=\widetilde{\alpha}(xy),x,y\in \mathfrak{C}\text{ for some }\alpha\in SO(7)\}.$                                                                                                                                                      |
| 0902.0431_E00255      | 41   | ■ 0.998 | $\begin{aligned} E_{6(-26)}&=\left\{\alpha\in\operatorname{Iso}_{\boldsymbol{R}}(\mathfrak{J})\mid\det(\alpha X)=\det X\right.\\&\quad\left.\mid\operatorname{det}(\alpha X)=\operatorname{det}X\right\}\\&=\left\{\alpha\in\operatorname{Iso}_{\boldsymbol{R}}(\mathfrak{J})\mid\alpha^{-1}(X\times Y)=\alpha X\times\alpha Y\right\}.\end{aligned}$ | $E_{6(-26)}=\{\alpha\in\operatorname{Iso}_{\boldsymbol{R}}(\mathfrak{J})\mid\det(\alpha X)=\det X\}\\=\left\{\alpha\in\operatorname{Iso}_{\boldsymbol{R}}(\mathfrak{J})\mid\alpha^{-1}(X\times Y)=\alpha X\times\alpha Y\right\}.$                                                                                     | $E_{6(-26)}=\{\alpha\in\operatorname{Iso}_{\boldsymbol{R}}(\mathfrak{J})\mid\det(\alpha X)=\det X\}\\=\{\alpha\in\operatorname{Iso}_{\boldsymbol{R}}(\mathfrak{J})\mid\alpha^{-1}(X\times Y)=\alpha X\times\alpha Y\}.$                                                                                                |
| 0902.0431_E00617 (i)  | 103  | ■ 0.998 | $(g\,X,\,g\,Y)=(\gamma X,\,Y),\quad X,\,Y\in\mathfrak{J}^C.$                                                                                                                                                                                                                                                                                          | $(gX,gY)=(\gamma X,Y),\quad X,Y\in\mathfrak{J}^C.$                                                                                                                                                                                                                                                                     | $(gX,gY)=(\gamma X,Y),\quad X,Y\in\mathfrak{J}^C.$                                                                                                                                                                                                                                                                     |
| 0902.0431_E00635      | 105  | ■ 0.998 | $E_{6,\boldsymbol{C}}=\left\{\alpha\in\operatorname{Iso}_C\left((\mathfrak{J}_C)^C\right)\mid\det(\alpha X)=\det X,\langle\alpha X,\alpha Y\rangle=\langle X,Y\rangle\right\}.$                                                                                                                                                                       | $E_{6,\boldsymbol{C}}=\left\{\alpha\in\operatorname{Iso}_C\left((\mathfrak{J}_C)^C\right)\mid\det(\alpha X)=\det X,\langle\alpha X,\alpha Y\rangle=\langle X,Y\rangle\right\}.$                                                                                                                                        | $E_{6,\boldsymbol{C}}=\{\alpha\in\operatorname{Iso}_C((\mathfrak{J}_C)^C)\mid\det(\alpha X)=\det X,\langle\alpha X,\alpha Y\rangle=\langle X,Y\rangle\}.$                                                                                                                                                              |
| 0902.0431_E00965      | 160  | ■ 0.998 | $E_7\left\{ \right\} ^{\left\{ C \right\} }\simeq \left( E_7\left\{ \right\} ^{\left\{ C \right\} }\cap U\left( \mathfrak{P}^C \right) \right) \times \boldsymbol{R}^d = E_7 \times \boldsymbol{R}^d, \quad d = 133.$                                                                                                                                 | $E_7^C\simeq\left(E_7^C\cap U\left(\mathfrak{P}^C\right)\right)\times\boldsymbol{R}^d=E_7\times\boldsymbol{R}^d,\quad d=133.$                                                                                                                                                                                          | $E_7^C\simeq\left(E_7^C\cap U\left(\mathfrak{P}^C\right)\right)\times\boldsymbol{R}^d=E_7\times\boldsymbol{R}^d,\quad d=133.$                                                                                                                                                                                          |
| 0902.0431_E01113      | 188  | ■ 0.998 | $\left(E_8\right)^{\sigma}\cong Ss(16).$                                                                                                                                                                                                                                                                                                              | $(E_8)^{\sigma}\cong Ss(16).$                                                                                                                                                                                                                                                                                          | $(E_8)^{\sigma}\cong Ss(16).$                                                                                                                                                                                                                                                                                          |
| 0902.0431_E01116      | 189  | ■ 0.998 | $\left(\mathfrak{J}^C\right)^3=\left\{\boldsymbol{X}=\begin{pmatrix}X_1\\X_2\\X_3\end{pmatrix}\middle X_i\in\mathfrak{J}^C\right\}.$                                                                                                                                                                                                                  | $\left(\mathfrak{J}^C\right)^3=\left\{\boldsymbol{X}=\begin{pmatrix}X_1\\X_2\\X_3\end{pmatrix}\middle X_i\in\mathfrak{J}^C\right\}.$                                                                                                                                                                                   | $\left(\mathfrak{J}^C\right)^3=\left\{\boldsymbol{X}=\begin{pmatrix}X_1\\X_2\\X_3\end{pmatrix}\middle X_i\in\mathfrak{J}^C\right\}.$                                                                                                                                                                                   |
| 0902.0431_E01138      | 192  | ■ 0.998 | $\begin{gathered}\left(\boldsymbol{x}_1\wedge\cdots\wedge\boldsymbol{x}_k,\boldsymbol{y}_1\wedge\cdots\wedge\boldsymbol{y}_k\right)=\det\left(\left(\boldsymbol{x}_i,\boldsymbol{y}_j\right)\right),\quad k\geq1,\\(a,b)=ab,\quad a,b\in\Lambda^0\left(C^n\right)=C.\end{gathered}$                                                                   | $\left(\boldsymbol{x}_1\wedge\cdots\wedge\boldsymbol{x}_k,\boldsymbol{y}_1\wedge\cdots\wedge\boldsymbol{y}_k\right)=\det\left(\left(\boldsymbol{x}_i,\boldsymbol{y}_j\right)\right),\quad k\geq1,\\(a,b)=ab,\quad a,b\in\Lambda^0\left(C^n\right)=C.$                                                                  | $\left(\boldsymbol{x}_1\wedge\cdots\wedge\boldsymbol{x}_k,\boldsymbol{y}_1\wedge\cdots\wedge\boldsymbol{y}_k\right)=\det\left(\left(\boldsymbol{x}_i,\boldsymbol{y}_j\right)\right),\quad k\geq1,\\(a,b)=ab,\quad a,b\in\Lambda^0\left(C^n\right)=C.$                                                                  |
| 0902.0431_E00148      | 27   | ■ 0.998 | $(m+ae_4')(n+be_4')=(mn+\overline{b}a)+(a\overline{n}+bm)e_4'.$                                                                                                                                                                                                                                                                                       | $(m+ae_4')(n+be_4')=(mn+\overline{b}a)+(a\overline{n}+bm)e_4'.$                                                                                                                                                                                                                                                        | $(m+ae_4')(n+be_4')=(mn+\overline{b}a)+(a\overline{n}+bm)e_4'.$                                                                                                                                                                                                                                                        |
| 0902.0431_E00691 (ii) | 114  | ■ 0.998 | $\Phi=\left(\begin{array}{cccc}g&l&C&A\\k&h&B&D\\c&a&\nu&\lambda\\b&d&\kappa&\mu\end{array}\right),\quad\begin{array}{l}g,h,k,l\in\operatorname{Hom}_C\left(\mathfrak{J}^C\right),\\a,b,c,d\in\operatorname{Hom}_C\left(\mathfrak{J}^C,C\right),\\A,B,C,D\in\mathfrak{J}^C,\\ \nu,\mu,\kappa,\lambda\in C.\end{array}$                                | $\Phi=\left(\begin{array}{cccc}g&l&C&A\\k&h&B&D\\c&a&\nu&\lambda\\b&d&\kappa&\mu\end{array}\right),\quad\begin{array}{l}g,h,k,l\in\operatorname{Hom}_C\left(\mathfrak{J}^C\right),\\a,b,c,d\in\operatorname{Hom}_C\left(\mathfrak{J}^C,C\right),\\A,B,C,D\in\mathfrak{J}^C,\\ \nu,\mu,\kappa,\lambda\in C.\end{array}$ | $\Phi=\left(\begin{array}{cccc}g&l&C&A\\k&h&B&D\\c&a&\nu&\lambda\\b&d&\kappa&\mu\end{array}\right),\quad\begin{array}{l}g,h,k,l\in\operatorname{Hom}_C\left(\mathfrak{J}^C\right),\\a,b,c,d\in\operatorname{Hom}_C\left(\mathfrak{J}^C,C\right),\\A,B,C,D\in\mathfrak{J}^C,\\ \nu,\mu,\kappa,\lambda\in C.\end{array}$ |

| Identifier            | Page | Conf.   | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Rendered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Image                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-----------------------|------|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0902.0431_E00745      | 123  | ■ 0.998 | $\operatorname{tr}\left(\Phi_0^2\right)=\frac{1}{9}\times 27\times 2+1+1=8$ .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | $\operatorname{tr}\left(\Phi_0^2\right)=\frac{1}{9}\times 27\times 2+1+1=8$ .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | $\operatorname{tr}\left(\Phi_0^2\right)=\frac{1}{9}\times 27\times 2+1+1=8$ .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 0902.0431_E00025      | 10   | ■ 0.998 | $\begin{aligned} & \left[F_{i0}, F_{kl}\right] x = \left(F_{i0} F_{kl} - F_{kl} F_{i0}\right) x = \frac{1}{4} e_i\left(e_l\left(e_k x\right)\right) - \frac{1}{4} e_l\left(e_k\left(e_i x\right)\right) \\ & = -\frac{1}{4} e_l\left(e_i\left(e_k x\right)\right) + \frac{1}{4} e_l\left(e_i\left(e_k x\right)\right) = 0, \\ & \left[F_{ij}, F_{jk}\right] x = \left(F_{ij} F_{jk} - F_{jk} F_{ij}\right) x = \frac{1}{4} e_j\left(e_i\left(e_k\left(e_j x\right)\right)\right) - \frac{1}{4} e_k\left(e_j\left(e_i\left(e_j x\right)\right)\right) \\ & = \frac{1}{4} e_i\left(e_k\left(e_j\left(e_j x\right)\right)\right) + \frac{1}{4} e_k\left(e_i x\right) = -\frac{1}{4} e_i\left(e_k x\right) + \frac{1}{4} e_k\left(e_i x\right) \\ & = \frac{1}{2} e_k\left(e_i x\right) = F_{ik} x, \\ & \left[F_{ij}, F_{kl}\right] x = \left(F_{ij} F_{kl} - F_{kl} F_{ij}\right) x = \frac{1}{4} e_j\left(e_i\left(e_l\left(e_k x\right)\right)\right) - \frac{1}{4} e_l\left(e_k\left(e_j\left(e_i x\right)\right)\right) \\ & = \frac{1}{4} e_j\left(e_k\left(e_i\left(e_l x\right)\right)\right) - \frac{1}{4} e_j\left(e_l\left(e_k\left(e_i x\right)\right)\right) = \cdots = 0. \end{aligned}$ | $\begin{aligned} & \left[F_{i0}, F_{kl}\right] x = \left(F_{i0} F_{kl} - F_{kl} F_{i0}\right) x = \frac{1}{4} e_i\left(e_l\left(e_k x\right)\right) - \frac{1}{4} e_l\left(e_k\left(e_i x\right)\right) \\ & = -\frac{1}{4} e_l\left(e_i\left(e_k x\right)\right) + \frac{1}{4} e_l\left(e_i\left(e_k x\right)\right) = 0, \\ & \left[F_{ij}, F_{jk}\right] x = \left(F_{ij} F_{jk} - F_{jk} F_{ij}\right) x = \frac{1}{4} e_j\left(e_i\left(e_k\left(e_j x\right)\right)\right) - \frac{1}{4} e_k\left(e_j\left(e_i\left(e_j x\right)\right)\right) \\ & = \frac{1}{4} e_i\left(e_k\left(e_j\left(e_j x\right)\right)\right) + \frac{1}{4} e_k\left(e_i x\right) = -\frac{1}{4} e_i\left(e_k x\right) + \frac{1}{4} e_k\left(e_i x\right) \\ & = \frac{1}{2} e_k\left(e_i x\right) = F_{ik} x, \\ & \left[F_{ij}, F_{kl}\right] x = \left(F_{ij} F_{kl} - F_{kl} F_{ij}\right) x = \frac{1}{4} e_j\left(e_i\left(e_l\left(e_k x\right)\right)\right) - \frac{1}{4} e_l\left(e_k\left(e_j\left(e_i x\right)\right)\right) \\ & = \frac{1}{4} e_j\left(e_k\left(e_i\left(e_l x\right)\right)\right) - \frac{1}{4} e_j\left(e_l\left(e_k\left(e_i x\right)\right)\right) = \cdots = 0. \end{aligned}$ | $\begin{aligned} \left[F_{i0}, F_{kl}\right] x &= \left(F_{i0} F_{kl} - F_{kl} F_{i0}\right) x = \frac{1}{4} e_i\left(e_l\left(e_k x\right)\right) - \frac{1}{4} e_l\left(e_k\left(e_i x\right)\right) \\ &= -\frac{1}{4} e_l\left(e_i\left(e_k x\right)\right) + \frac{1}{4} e_l\left(e_i\left(e_k x\right)\right) = 0, \\ \left[F_{ij}, F_{jk}\right] x &= \left(F_{ij} F_{jk} - F_{jk} F_{ij}\right) x = \frac{1}{4} e_j\left(e_i\left(e_k\left(e_j x\right)\right)\right) - \frac{1}{4} e_k\left(e_j\left(e_i\left(e_j x\right)\right)\right) \\ &= \frac{1}{4} e_i\left(e_k\left(e_j\left(e_j x\right)\right)\right) + \frac{1}{4} e_k\left(e_i x\right) = -\frac{1}{4} e_i\left(e_k x\right) + \frac{1}{4} e_k\left(e_i x\right) \\ &= \frac{1}{2} e_k\left(e_i x\right) = F_{ik} x, \\ \left[F_{ij}, F_{kl}\right] x &= \left(F_{ij} F_{kl} - F_{kl} F_{ij}\right) x = \frac{1}{4} e_j\left(e_i\left(e_l\left(e_k x\right)\right)\right) - \frac{1}{4} e_l\left(e_k\left(e_j\left(e_i x\right)\right)\right) \\ &= \frac{1}{4} e_j\left(e_k\left(e_i\left(e_l x\right)\right)\right) - \frac{1}{4} e_j\left(e_l\left(e_k\left(e_i x\right)\right)\right) = \cdots = 0. \end{aligned}$ |
| 0902.0431_E00047 (iv) | 12   | ■ 0.998 | $x(Dy)=D(x\,y),\quad \text{quad}\quad x,\,y\in\mathfrak{C}.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | $x(Dy)=D(xy),\quad x,y\in\mathfrak{C}.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | $x(Dy)=D(xy),\quad x,y\in\mathfrak{C}.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 0902.0431_E00138      | 26   | ■ 0.998 | $\alpha=1\quad\text{quad}\quad\text{\texttt{\{ or \}}}\quad\text{quad}\quad\alpha=\gamma.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | $\alpha=1\quad\text{or}\quad\alpha=\gamma.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | $\alpha=1\quad\text{or}\quad\alpha=\gamma.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 0902.0431_E00330      | 53   | ■ 0.998 | $\mathfrak{D}_{4\{ \}^{\mathfrak{C}}}\ni D\rightarrowtail(D,\nu D,\kappa\pi D)\in\mathfrak{d}_4^{\mathfrak{C}}\subset\mathfrak{f}_4^{\mathfrak{C}}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | $\mathfrak{D}_4^{\mathfrak{C}}\ni D\rightarrowtail(D,\nu D,\kappa\pi D)\in\mathfrak{d}_4^{\mathfrak{C}}\subset\mathfrak{f}_4^{\mathfrak{C}}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | $\mathfrak{D}_4^{\mathfrak{C}}\ni D\rightarrowtail(D,\nu D,\kappa\pi D)\in\mathfrak{d}_4^{\mathfrak{C}}\subset\mathfrak{f}_4^{\mathfrak{C}}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 0902.0431_E00457      | 75   | ■ 0.998 | $\phi=\frac{\phi-\phi^*}{2}+i\frac{\phi+\phi^*}{2i},\quad\text{quad}\quad\frac{\phi-\phi^*}{2},\frac{\phi+\phi^*}{2i}\in\mathfrak{e}_6.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | $\phi=\frac{\phi-\phi^*}{2}+i\frac{\phi+\phi^*}{2i},\quad\frac{\phi-\phi^*}{2},\frac{\phi+\phi^*}{2i}\in\mathfrak{e}_6.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | $\phi=\frac{\phi-\phi^*}{2}+i\frac{\phi+\phi^*}{2i},\quad\frac{\phi-\phi^*}{2},\frac{\phi+\phi^*}{2i}\in\mathfrak{e}_6.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 0902.0431_E00499      | 81   | ■ 0.998 | $\widetilde{T}X=\frac{1}{2}\left(TX+XT^*\right),\quad\text{quad}\quad\text{\texttt{\{ where \}}}\quad\text{quad}\quad T^{\wedge\{*\}}=\{\,\,\}^{\wedge\{t\}}\bar{\phantom{T}}\,.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | $\widetilde{T}X=\frac{1}{2}(TX+XT^*),\quad\text{where}\quad T^*={}^t\bar{T}.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | $\widetilde{T}X=\frac{1}{2}(TX+XT^*),\quad\text{where}\quad T^*={}^t\bar{T}.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

| Identifier       | Page | Conf.   | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Rendered                                                                                                                                                                                                                                                                                                                                                                                                            | Image                                                                                                                                                                                                                                                                                                                                                                                                               |
|------------------|------|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0902.0431_E00513 | 84   | ■ 0.998 | $\begin{aligned} & \backslash\mathrm{alpha}_{1}=\backslash\mathrm{lambda}_{0}-\backslash\mathrm{lambda}_{1}, \\ & \backslash\mathrm{quad} \backslash\mathrm{alpha}_{2}=\backslash\mathrm{lambda}_{1}-\backslash\mathrm{lambda}_{2}, \backslash\mathrm{quad} \\ & \backslash\mathrm{alpha}_{3}=\backslash\mathrm{lambda}_{2}-\backslash\mathrm{lambda}_{3}, \backslash\mathrm{quad} \backslash\mathrm{alpha}_{4}= \\ & \backslash\mathrm{lambda}_{3}+\backslash\mathrm{frac}{1}{2}\backslash\mathrm{left}(\backslash\mathrm{mu}_{2}-\backslash\mathrm{mu}_{3}\backslash\mathrm{right}), \backslash\mathrm{quad} \\ & \backslash\mathrm{alpha}_{5}=\backslash\mathrm{frac}{1}{2}\backslash\mathrm{left}(-\backslash\mathrm{lambda}_{0}-\backslash\mathrm{lambda}_{1}- \\ & \backslash\mathrm{lambda}_{2}+\backslash\mathrm{lambda}_{3}\backslash\mathrm{right})+\backslash\mathrm{frac}{1}{2}\backslash\mathrm{left}(\backslash\mathrm{mu}_{3}- \\ & \backslash\mathrm{mu}_{1}\backslash\mathrm{right}), \backslash\mathrm{quad} \backslash\mathrm{alpha}_{6}=\backslash\mathrm{frac}{1}{2}\backslash\mathrm{left} \\ & \backslash\mathrm{lambda}_{0}+\backslash\mathrm{lambda}_{1}+\backslash\mathrm{lambda}_{2}+\backslash\mathrm{lambda}_{3}\backslash\mathrm{right})+ \\ & \backslash\mathrm{frac}{1}{2}\backslash\mathrm{left}(\backslash\mathrm{mu}_{1}-\backslash\mathrm{mu}_{2}\backslash\mathrm{right}) \end{aligned}$ | $\begin{aligned} \alpha_1 &= \lambda_0 - \lambda_1, & \alpha_2 &= \lambda_1 - \lambda_2, & \alpha_3 &= \lambda_2 - \lambda_3, \\ \alpha_4 &= \lambda_3 + \frac{1}{2}(\mu_2 - \mu_3), \\ \alpha_5 &= \frac{1}{2}(-\lambda_0 - \lambda_1 - \lambda_2 + \lambda_3) + \frac{1}{2}(\mu_3 - \mu_1), \\ \alpha_6 &= \frac{1}{2}(\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3) + \frac{1}{2}(\mu_1 - \mu_2) \end{aligned}$ | $\begin{aligned} \alpha_1 &= \lambda_0 - \lambda_1, & \alpha_2 &= \lambda_1 - \lambda_2, & \alpha_3 &= \lambda_2 - \lambda_3, \\ \alpha_4 &= \lambda_3 + \frac{1}{2}(\mu_2 - \mu_3), \\ \alpha_5 &= \frac{1}{2}(-\lambda_0 - \lambda_1 - \lambda_2 + \lambda_3) + \frac{1}{2}(\mu_3 - \mu_1), \\ \alpha_6 &= \frac{1}{2}(\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3) + \frac{1}{2}(\mu_1 - \mu_2) \end{aligned}$ |
| 0902.0431_E00679 | 113  | ■ 0.998 | $\begin{aligned} & E_{\{7\}^C} \equiv \left\{ \alpha \in \mathrm{Iso}_C(\mathfrak{P}^C) \mid \alpha(P \times Q) \alpha^{-1} = \alpha P \times \alpha Q \right\} \\ & \mid \alpha(P \times Q) \alpha^{-1} = \alpha P \times \alpha Q, \langle \alpha P, \alpha Q \rangle = \langle P, Q \rangle \} \\ & E_{\{7\}} \equiv \left\{ \alpha \in \mathrm{Iso}_C(\mathfrak{P}^C) \mid \alpha(P \times Q) \alpha^{-1} = \alpha P \times \alpha Q, \langle \alpha P, \alpha Q \rangle = \langle P, Q \rangle \right\} \end{aligned}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | $\begin{aligned} E_7^C &= \{ \alpha \in \mathrm{Iso}_C(\mathfrak{P}^C) \mid \alpha(P \times Q) \alpha^{-1} = \alpha P \times \alpha Q \}, \\ E_7 &= \{ \alpha \in \mathrm{Iso}_C(\mathfrak{P}^C) \mid \alpha(P \times Q) \alpha^{-1} = \alpha P \times \alpha Q, \langle \alpha P, \alpha Q \rangle = \langle P, Q \rangle \}. \end{aligned}$                                                                       | $\begin{aligned} E_7^C &= \{ \alpha \in \mathrm{Iso}_C(\mathfrak{P}^C) \mid \alpha(P \times Q) \alpha^{-1} = \alpha P \times \alpha Q \}, \\ E_7 &= \{ \alpha \in \mathrm{Iso}_C(\mathfrak{P}^C) \mid \alpha(P \times Q) \alpha^{-1} = \alpha P \times \alpha Q, \langle \alpha P, \alpha Q \rangle = \langle P, Q \rangle \}. \end{aligned}$                                                                       |
| 0902.0431_E00959 | 159  | ■ 0.998 | $\alpha P = f^{-1}(A(fP)) \quad \text{or} \quad \alpha P = f^{-1}(A(f\bar{P})), \quad P \in (\mathfrak{P}_C)^C$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | $\alpha P = f^{-1}(A(fP)) \quad \text{or} \quad \alpha P = f^{-1}(A(f\bar{P})), \quad P \in (\mathfrak{P}_C)^C$                                                                                                                                                                                                                                                                                                     | $\alpha P = f^{-1}(A(fP)) \quad \text{or} \quad \alpha P = f^{-1}(A(f\bar{P})), \quad P \in (\mathfrak{P}_C)^C$                                                                                                                                                                                                                                                                                                     |
| 0902.0431_E00985 | 164  | ■ 0.998 | $\mathfrak{e}_{\{8\}^C} \cong \Theta(\mathfrak{e}_8^C) = \{ \Theta(R) \mid R \in \mathfrak{e}_8^C \}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | $\mathfrak{e}_8^C \cong \Theta(\mathfrak{e}_8^C) = \{ \Theta(R) \mid R \in \mathfrak{e}_8^C \}$                                                                                                                                                                                                                                                                                                                     | $\mathfrak{e}_8^C \cong \Theta(\mathfrak{e}_8^C) = \{ \Theta(R) \mid R \in \mathfrak{e}_8^C \}$                                                                                                                                                                                                                                                                                                                     |
| 0902.0431_E01128 | 191  | ■ 0.998 | $w(D, \phi, \boldsymbol{X}, \boldsymbol{Y}) = (D, \phi, \omega \boldsymbol{X}, \omega^2 \boldsymbol{Y}),$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | $w(D, \phi, \boldsymbol{X}, \boldsymbol{Y}) = (D, \phi, \omega \boldsymbol{X}, \omega^2 \boldsymbol{Y}),$                                                                                                                                                                                                                                                                                                           | $w(D, \phi, \boldsymbol{X}, \boldsymbol{Y}) = (D, \phi, \omega \boldsymbol{X}, \omega^2 \boldsymbol{Y}),$                                                                                                                                                                                                                                                                                                           |
| 0902.0431_E00007 | 7    | ■ 0.998 | $\begin{aligned} & (ax)y + x(ya) = a(xy) + (xy)a, \\ & (xa)y + (xy)a = x(ay) + x(ya), \\ & (ax)y + (xa)y = a(xy) + x(ay). \end{aligned}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | $\begin{aligned} (ax)y + x(ya) &= a(xy) + (xy)a, \\ (xa)y + (xy)a &= x(ay) + x(ya), \\ (ax)y + (xa)y &= a(xy) + x(ay). \end{aligned}$                                                                                                                                                                                                                                                                               | $\begin{aligned} (ax)y + x(ya) &= a(xy) + (xy)a, \\ (xa)y + (xy)a &= x(ay) + x(ya), \\ (ax)y + (xa)y &= a(xy) + x(ay). \end{aligned}$                                                                                                                                                                                                                                                                               |
| 0902.0431_E00147 | 27   | ■ 0.998 | $G_{\{2\}^C} \simeq (G_{\{2\}^C} \cap U(\mathfrak{C}^C)) \times \boldsymbol{R}^d = G_2 \times \boldsymbol{R}^d,$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | $G_2^C \simeq (G_2^C \cap U(\mathfrak{C}^C)) \times \boldsymbol{R}^d = G_2 \times \boldsymbol{R}^d,$                                                                                                                                                                                                                                                                                                                | $G_2^C \simeq (G_2^C \cap U(\mathfrak{C}^C)) \times \boldsymbol{R}^d = G_2 \times \boldsymbol{R}^d,$                                                                                                                                                                                                                                                                                                                |

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| 0902_0431_EQ1029 | 172  | 0.998 | $\begin{aligned} & \lambda_2 + \frac{1}{2}(\mu_2 - \mu_3) = 0 \\ & \lambda_3 + \frac{1}{2}(\mu_2 - \mu_3) = 0 \\ & \lambda_0 - \frac{1}{2}(\mu_2 - \mu_3) = 2 \\ & \lambda_1 - \frac{1}{2}(\mu_2 - \mu_3) = 0 \\ & \lambda_2 - \frac{1}{2}(\mu_2 - \mu_3) = 0 \\ & \lambda_3 - \frac{1}{2}(\mu_2 - \mu_3) = 0 \\ & \frac{1}{2}(\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3) + \frac{1}{2}(\mu_3 - \mu_1) = 1 \\ & \frac{1}{2}(\lambda_0 + \lambda_1 - \lambda_2 + \lambda_3) + \frac{1}{2}(\mu_3 - \mu_1) = 1 \\ & \frac{1}{2}(\lambda_0 - \lambda_1 + \lambda_2 + \lambda_3) + \frac{1}{2}(\mu_3 - \mu_1) = 1 \\ & \frac{1}{2}(\lambda_0 - \lambda_1 - \lambda_2 - \lambda_3) + \frac{1}{2}(\mu_3 - \mu_1) = 1 \\ & \frac{1}{2}(\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3) - \frac{1}{2}(\mu_3 - \mu_1) = 1 \\ & \frac{1}{2}(\lambda_0 + \lambda_1 - \lambda_2 + \lambda_3) - \frac{1}{2}(\mu_3 - \mu_1) = 1 \\ & \frac{1}{2}(\lambda_0 - \lambda_1 + \lambda_2 + \lambda_3) - \frac{1}{2}(\mu_3 - \mu_1) = 1 \\ & \frac{1}{2}(\lambda_0 - \lambda_1 - \lambda_2 - \lambda_3) + \frac{1}{2}(\mu_1 - \mu_2) = 1 \\ & \frac{1}{2}(\lambda_0 - \lambda_1 - \lambda_2 + \lambda_3) + \frac{1}{2}(\mu_1 - \mu_2) = 1 \\ & \frac{1}{2}(\lambda_0 + \lambda_1 - \lambda_2 - \lambda_3) + \frac{1}{2}(\mu_1 - \mu_2) = 1 \\ & \frac{1}{2}(\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3) + \frac{1}{2}(\mu_1 - \mu_2) = 1 \\ & \frac{1}{2}(\lambda_0 - \lambda_1 + \lambda_2 - \lambda_3) - \frac{1}{2}(\mu_1 - \mu_2) = 1 \\ & \frac{1}{2}(\lambda_0 - \lambda_1 - \lambda_2 + \lambda_3) - \frac{1}{2}(\mu_1 - \mu_2) = 1 \\ & \frac{1}{2}(\lambda_0 + \lambda_1 - \lambda_2 - \lambda_3) - \frac{1}{2}(\mu_1 - \mu_2) = 1 \\ & \frac{1}{2}(\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3) - \frac{1}{2}(\mu_1 - \mu_2) = 1 \\ & \mu_1 + \frac{2}{3}\nu = 0 \\ & \mu_2 + \frac{2}{3}\nu = 0 \\ & -\mu_3 - \frac{2}{3}\nu = 0 \end{aligned}$ | $\begin{aligned} \lambda_2 + \frac{1}{2}(\mu_2 - \mu_3) &= 0 & 1 & 2 & 2 & 1 & 1 & 0 & 1 \\ \lambda_3 + \frac{1}{2}(\mu_2 - \mu_3) &= 0 & 1 & 2 & 2 & 1 & 0 & 0 & 1 \\ \lambda_0 - \frac{1}{2}(\mu_2 - \mu_3) &= 2 & 3 & 4 & 3 & 3 & 2 & 1 & 2 \\ \lambda_1 - \frac{1}{2}(\mu_2 - \mu_3) &= 0 & 0 & 0 & 0 & 1 & 1 & 1 & 0 \\ \lambda_2 - \frac{1}{2}(\mu_2 - \mu_3) &= 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 \\ \lambda_3 - \frac{1}{2}(\mu_2 - \mu_3) &= 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ \frac{1}{2}(\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3) + \frac{1}{2}(\mu_3 - \mu_1) &= 1 & 1 & 2 & 2 & 2 & 2 & 1 & 1 \\ \frac{1}{2}(\lambda_0 + \lambda_1 - \lambda_2 + \lambda_3) + \frac{1}{2}(\mu_3 - \mu_1) &= 1 & 1 & 2 & 2 & 2 & 1 & 1 & 1 \\ \frac{1}{2}(\lambda_0 - \lambda_1 + \lambda_2 + \lambda_3) + \frac{1}{2}(\mu_3 - \mu_1) &= 1 & 1 & 2 & 2 & 2 & 1 & 0 & 1 \\ \frac{1}{2}(\lambda_0 - \lambda_1 - \lambda_2 - \lambda_3) + \frac{1}{2}(\mu_3 - \mu_1) &= 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{1}{2}(\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3) - \frac{1}{2}(\mu_3 - \mu_1) &= 1 & 3 & 4 & 3 & 2 & 2 & 1 & 2 \\ \frac{1}{2}(\lambda_0 + \lambda_1 - \lambda_2 + \lambda_3) - \frac{1}{2}(\mu_3 - \mu_1) &= 1 & 3 & 4 & 3 & 2 & 1 & 1 & 2 \\ \frac{1}{2}(\lambda_0 - \lambda_1 + \lambda_2 + \lambda_3) - \frac{1}{2}(\mu_3 - \mu_1) &= 1 & 3 & 4 & 3 & 2 & 1 & 0 & 2 \\ \frac{1}{2}(\lambda_0 - \lambda_1 - \lambda_2 - \lambda_3) - \frac{1}{2}(\mu_3 - \mu_1) &= 1 & 2 & 2 & 1 & 0 & 0 & 0 & 1 \\ \frac{1}{2}(\lambda_0 - \lambda_1 + \lambda_2 - \lambda_3) + \frac{1}{2}(\mu_1 - \mu_2) &= 1 & 2 & 2 & 1 & 1 & 1 & 0 & 1 \\ \frac{1}{2}(\lambda_0 - \lambda_1 - \lambda_2 + \lambda_3) + \frac{1}{2}(\mu_1 - \mu_2) &= 1 & 2 & 2 & 1 & 1 & 0 & 0 & 1 \\ \frac{1}{2}(\lambda_0 + \lambda_1 - \lambda_2 - \lambda_3) + \frac{1}{2}(\mu_1 - \mu_2) &= 1 & 2 & 2 & 1 & 1 & 1 & 1 & 1 \\ \frac{1}{2}(\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3) + \frac{1}{2}(\mu_1 - \mu_2) &= 1 & 3 & 4 & 3 & 3 & 2 & 1 & 2 \\ \frac{1}{2}(\lambda_0 - \lambda_1 + \lambda_2 - \lambda_3) - \frac{1}{2}(\mu_1 - \mu_2) &= 1 & 1 & 2 & 2 & 1 & 1 & 0 & 1 \\ \frac{1}{2}(\lambda_0 - \lambda_1 - \lambda_2 + \lambda_3) - \frac{1}{2}(\mu_1 - \mu_2) &= 1 & 1 & 2 & 2 & 1 & 0 & 0 & 1 \\ \frac{1}{2}(\lambda_0 + \lambda_1 - \lambda_2 - \lambda_3) - \frac{1}{2}(\mu_1 - \mu_2) &= 1 & 1 & 2 & 2 & 1 & 1 & 1 & 1 \\ \frac{1}{2}(\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3) - \frac{1}{2}(\mu_1 - \mu_2) &= 1 & 2 & 4 & 4 & 3 & 2 & 1 & 2 \\ \mu_1 + \frac{2}{3}\nu &= 0 & 1 & 1 & 0 & 0 & 0 & 0 & 1 \\ \mu_2 + \frac{2}{3}\nu &= 0 & 0 & 1 & 1 & 0 & 0 & 0 & 1 \\ -\mu_3 - \frac{2}{3}\nu &= 0 & 1 & 1 & 1 & 0 & 0 & 0 & 0 \end{aligned}$ |       |



| Identifier           | Page | Conf.   | LaTeX source                                                                                                                                                                                                                                                                                                                                           | Rendered                                                                                                                                                                                                                                                                                                                                               | Image                                                                                                                                                                                                                                                                                                                                                  |
|----------------------|------|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0902.0431_E00560     | 94   | ■ 0.999 | $\phi(\theta)\left(\begin{array}{ccc}\xi_1 & x_3 & \bar{x}_2 \\ \bar{x}_3 & \xi_2 & x_1 \\ x_2 & \bar{x}_1 & \xi_3\end{array}\right)=\left(\begin{array}{ccc}\theta^4\xi_1 & \theta x_3 & \theta\bar{x}_2 \\ \theta\bar{x}_3 & \theta^{-2}\xi_2 & \theta^{-2}x_1 \\ \theta x_2 & \theta^{-2}\bar{x}_1 & \theta^{-2}\xi_3\end{array}\right).$           | $\phi(\theta)\left(\begin{array}{ccc}\xi_1 & x_3 & \bar{x}_2 \\ \bar{x}_3 & \xi_2 & x_1 \\ x_2 & \bar{x}_1 & \xi_3\end{array}\right)=\left(\begin{array}{ccc}\theta^4\xi_1 & \theta x_3 & \theta\bar{x}_2 \\ \theta\bar{x}_3 & \theta^{-2}\xi_2 & \theta^{-2}x_1 \\ \theta x_2 & \theta^{-2}\bar{x}_1 & \theta^{-2}\xi_3\end{array}\right).$           | $\phi(\theta)\left(\begin{array}{ccc}\xi_1 & x_3 & \bar{x}_2 \\ \bar{x}_3 & \xi_2 & x_1 \\ x_2 & \bar{x}_1 & \xi_3\end{array}\right)=\left(\begin{array}{ccc}\theta^4\xi_1 & \theta x_3 & \theta\bar{x}_2 \\ \theta\bar{x}_3 & \theta^{-2}\xi_2 & \theta^{-2}x_1 \\ \theta x_2 & \theta^{-2}\bar{x}_1 & \theta^{-2}\xi_3\end{array}\right).$           |
| 0902.0431_E00578     | 96   | ■ 0.999 | $k\left((a+be_2)+i(c+de_2)\right)=0, \quad a,b,c,d\in\boldsymbol{C},$                                                                                                                                                                                                                                                                                  | $k\left((a+be_2)+i(c+de_2)\right)=0, \quad a,b,c,d\in\boldsymbol{C},$                                                                                                                                                                                                                                                                                  | $k\left((a+be_2)+i(c+de_2)\right)=0, \quad a,b,c,d\in\boldsymbol{C},$                                                                                                                                                                                                                                                                                  |
| 0902.0431_E00607     | 101  | ■ 0.999 | $A(P\circ Q)A^*=APA^*\circ AQA^*, \quad A\in\mathrm{Sp}(4), P,Q\in\mathfrak{J}(4,\boldsymbol{H}).$                                                                                                                                                                                                                                                     | $A(P\circ Q)A^*=APA^*\circ AQA^*, \quad A\in\mathrm{Sp}(4), P,Q\in\mathfrak{J}(4,\boldsymbol{H}).$                                                                                                                                                                                                                                                     | $A(P\circ Q)A^*=APA^*\circ AQA^*, \quad A\in\mathrm{Sp}(4), P,Q\in\mathfrak{J}(4,\boldsymbol{H}).$                                                                                                                                                                                                                                                     |
| 0902.0431_E00693     | 114  | ■ 0.999 | $rf_r(P\times Q)f_r^{-1}=f_rP\times f_rQ, \quad P,Q\in\mathfrak{P}^C.$                                                                                                                                                                                                                                                                                 | $rf_r(P\times Q)f_r^{-1}=f_rP\times f_rQ, \quad P,Q\in\mathfrak{P}^C.$                                                                                                                                                                                                                                                                                 | $rf_r(P\times Q)f_r^{-1}=f_rP\times f_rQ, \quad P,Q\in\mathfrak{P}^C.$                                                                                                                                                                                                                                                                                 |
| 0902.0431_E00720     | 118  | ■ 0.999 | $\Phi\in\mathfrak{e}_7 \quad \text{if and only if} \quad \tau\lambda\Phi\lambda^{-1}\tau=\Phi.$                                                                                                                                                                                                                                                        | $\Phi\in\mathfrak{e}_7 \quad \text{if and only if} \quad \tau\lambda\Phi\lambda^{-1}\tau=\Phi.$                                                                                                                                                                                                                                                        | $\Phi\in\mathfrak{e}_7 \quad \text{if and only if} \quad \tau\lambda\Phi\lambda^{-1}\tau=\Phi.$                                                                                                                                                                                                                                                        |
| 0902.0431_E00836 (i) | 139  | ■ 0.999 | $(E_1,\phi X)-\frac{1}{3}\nu(E_1,X)=-\nu(E_1,X).$                                                                                                                                                                                                                                                                                                      | $(E_1,\phi X)-\frac{1}{3}\nu(E_1,X)=-\nu(E_1,X).$                                                                                                                                                                                                                                                                                                      | $(E_1,\phi X)-\frac{1}{3}\nu(E_1,X)=-\nu(E_1,X).$                                                                                                                                                                                                                                                                                                      |
| 0902.0431_E00883     | 147  | ■ 0.999 | $\tau\gamma(X,Y,\xi,\eta)=(\tau\gamma X,\tau\gamma Y,\tau\xi,\tau\eta),$                                                                                                                                                                                                                                                                               | $\tau\gamma(X,Y,\xi,\eta)=(\tau\gamma X,\tau\gamma Y,\tau\xi,\tau\eta),$                                                                                                                                                                                                                                                                               | $\tau\gamma(X,Y,\xi,\eta)=(\tau\gamma X,\tau\gamma Y,\tau\xi,\tau\eta),$                                                                                                                                                                                                                                                                               |
| 0902.0431_E00962     | 160  | ■ 0.999 | $\gamma_1:\mathfrak{P}^C\rightarrow\mathfrak{P}^C,\gamma_1(P_C+(M+N))=\overline{P_C}+(\bar{M}+\bar{N}), \quad P_C+(M+N)\in\mathfrak{P}^C.$                                                                                                                                                                                                             | $\gamma_1:\mathfrak{P}^C\rightarrow\mathfrak{P}^C,\gamma_1(P_C+(M+N))=\overline{P_C}+(\bar{M}+\bar{N}), \quad P_C+(M+N)\in\mathfrak{P}^C.$                                                                                                                                                                                                             | $\gamma_1:\mathfrak{P}^C\rightarrow\mathfrak{P}^C,\gamma_1(P_C+(M+N))=\overline{P_C}+(\bar{M}+\bar{N}), \quad P_C+(M+N)\in\mathfrak{P}^C.$                                                                                                                                                                                                             |
| 0902.0431_E00974     | 162  | ■ 0.999 | $\left\{\begin{array}{l}\Phi=[\Phi_1,\Phi_2]+P_1\times Q_2-P_2\times Q_1\\P=\Phi_1P_2-\Phi_2P_1+r_1P_2-r_2P_1+s_1Q_2-s_2Q_1\\Q=\Phi_1Q_2-\Phi_2Q_1-r_1Q_2+r_2Q_1+t_1P_2-t_2P_1\\r=-\frac{1}{8}\{P_1,Q_2\}+\frac{1}{8}\{P_2,Q_1\}+s_1t_2-s_2t_1\\s=\frac{1}{4}\{P_1,P_2\}+2r_1s_2-2r_2s_1\\t=-\frac{1}{4}\{Q_1,Q_2\}-2r_1t_2+2r_2t_1\end{array}\right.$ | $\left\{\begin{array}{l}\Phi=[\Phi_1,\Phi_2]+P_1\times Q_2-P_2\times Q_1\\P=\Phi_1P_2-\Phi_2P_1+r_1P_2-r_2P_1+s_1Q_2-s_2Q_1\\Q=\Phi_1Q_2-\Phi_2Q_1-r_1Q_2+r_2Q_1+t_1P_2-t_2P_1\\r=-\frac{1}{8}\{P_1,Q_2\}+\frac{1}{8}\{P_2,Q_1\}+s_1t_2-s_2t_1\\s=\frac{1}{4}\{P_1,P_2\}+2r_1s_2-2r_2s_1\\t=-\frac{1}{4}\{Q_1,Q_2\}-2r_1t_2+2r_2t_1\end{array}\right.$ | $\left\{\begin{array}{l}\Phi=[\Phi_1,\Phi_2]+P_1\times Q_2-P_2\times Q_1\\P=\Phi_1P_2-\Phi_2P_1+r_1P_2-r_2P_1+s_1Q_2-s_2Q_1\\Q=\Phi_1Q_2-\Phi_2Q_1-r_1Q_2+r_2Q_1+t_1P_2-t_2P_1\\r=-\frac{1}{8}\{P_1,Q_2\}+\frac{1}{8}\{P_2,Q_1\}+s_1t_2-s_2t_1\\s=\frac{1}{4}\{P_1,P_2\}+2r_1s_2-2r_2s_1\\t=-\frac{1}{4}\{Q_1,Q_2\}-2r_1t_2+2r_2t_1\end{array}\right.$ |
| 0902.0431_E00987     | 164  | ■ 0.999 | $(R_1,R_2)_8=(\Phi_1,\Phi_2)_7-\{Q_1,P_2\}+\{P_1,Q_2\}-8r_1r_2-4t_1s_2-s_1t_2,$                                                                                                                                                                                                                                                                        | $(R_1,R_2)_8=(\Phi_1,\Phi_2)_7-\{Q_1,P_2\}+\{P_1,Q_2\}-8r_1r_2-4t_1s_2-s_1t_2,$                                                                                                                                                                                                                                                                        | $(R_1,R_2)_8=(\Phi_1,\Phi_2)_7-\{Q_1,P_2\}+\{P_1,Q_2\}-8r_1r_2-4t_1s_2-s_1t_2,$                                                                                                                                                                                                                                                                        |
| 0902.0431_E01054     | 179  | ■ 0.999 | $\beta(P\times Q)\beta^{-1}=\beta P\times\beta Q,$                                                                                                                                                                                                                                                                                                     | $\beta(P\times Q)\beta^{-1}=\beta P\times\beta Q,$                                                                                                                                                                                                                                                                                                     | $\beta(P\times Q)\beta^{-1}=\beta P\times\beta Q,$                                                                                                                                                                                                                                                                                                     |
| 0902.0431_E01099     | 186  | ■ 0.999 | $\mathrm{Spin}(16), \quad \mathrm{SO}(16), \quad \mathrm{Ss}(16), \quad \mathrm{SO}(16)/\boldsymbol{Z}_2.$                                                                                                                                                                                                                                             | $\mathrm{Spin}(16), \quad \mathrm{SO}(16), \quad \mathrm{Ss}(16), \quad \mathrm{SO}(16)/\boldsymbol{Z}_2.$                                                                                                                                                                                                                                             | $\mathrm{Spin}(16), \quad \mathrm{SO}(16), \quad \mathrm{Ss}(16), \quad \mathrm{SO}(16)/\boldsymbol{Z}_2.$                                                                                                                                                                                                                                             |
| 0902.0431_E01136     | 192  | ■ 0.999 | $\varphi(A,\alpha)=\varphi_1(A)\varphi_2(\alpha).$                                                                                                                                                                                                                                                                                                     | $\varphi(A,\alpha)=\varphi_1(A)\varphi_2(\alpha).$                                                                                                                                                                                                                                                                                                     | $\varphi(A,\alpha)=\varphi_1(A)\varphi_2(\alpha).$                                                                                                                                                                                                                                                                                                     |

| Identifier       | Page | Conf.   | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                       | Rendered                                                                                                                                                                                                                                                                                                                                                                                           | Image                                                                                                                                                                                                                                                                                                                                                                                              |
|------------------|------|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0902.0431_E01216 | 204  | ■ 0.999 | $\begin{aligned} E_{8(8)} &= \left\{ \alpha \in \operatorname{Iso}_R(\mathfrak{e}_{8(8)}) \mid \alpha[R_1, R_2] = [\alpha R_1, \alpha R_2] \right\}, \\ E_{8(-24)} &= \left\{ \alpha \in \operatorname{Iso}_R(\mathfrak{e}_{8(-24)}) \mid \alpha[R_1, R_2] = [\alpha R_1, \alpha R_2] \right\}. \end{aligned}$                                                                                     | $E_{8(8)} = \left\{ \alpha \in \operatorname{Iso}_R(\mathfrak{e}_{8(8)}) \mid \alpha[R_1, R_2] = [\alpha R_1, \alpha R_2] \right\},$ $E_{8(-24)} = \left\{ \alpha \in \operatorname{Iso}_R(\mathfrak{e}_{8(-24)}) \mid \alpha[R_1, R_2] = [\alpha R_1, \alpha R_2] \right\}.$                                                                                                                      | $E_{8(8)} = \{ \alpha \in \operatorname{Iso}_R(\mathfrak{e}_{8(8)}) \mid \alpha[R_1, R_2] = [\alpha R_1, \alpha R_2] \},$ $E_{8(-24)} = \{ \alpha \in \operatorname{Iso}_R(\mathfrak{e}_{8(-24)}) \mid \alpha[R_1, R_2] = [\alpha R_1, \alpha R_2] \}.$                                                                                                                                            |
| 0902.0431_E01218 | 204  | ■ 0.999 | $\begin{aligned} E_{8(8)} &\simeq Ss(16) \times \boldsymbol{R}^{128}, \\ E_{8(-24)} &\simeq (SU(2) \times E_7) / \boldsymbol{Z}_2 \times \boldsymbol{R}^{112}. \end{aligned}$                                                                                                                                                                                                                      | $E_{8(8)} \simeq Ss(16) \times \boldsymbol{R}^{128},$ $E_{8(-24)} \simeq (SU(2) \times E_7) / \boldsymbol{Z}_2 \times \boldsymbol{R}^{112}.$                                                                                                                                                                                                                                                       | $E_{8(8)} \simeq Ss(16) \times \boldsymbol{R}^{128},$ $E_{8(-24)} \simeq (SU(2) \times E_7) / \boldsymbol{Z}_2 \times \boldsymbol{R}^{112}.$                                                                                                                                                                                                                                                       |
| 0902.0431_E00092 | 19   | ■ 0.999 | $B_2(H_1, H_1) = \operatorname{tr}((\operatorname{ad} H_1)^2) = (-4) \times 2 + (-1) \times 8 = -16.$                                                                                                                                                                                                                                                                                              | $B_2(H_1, H_1) = \operatorname{tr}((\operatorname{ad} H_1)^2) = (-4) \times 2 + (-1) \times 8 = -16.$                                                                                                                                                                                                                                                                                              | $B_2(H_1, H_1) = \operatorname{tr}((\operatorname{ad} H_1)^2) = (-4) \times 2 + (-1) \times 8 = -16.$                                                                                                                                                                                                                                                                                              |
| 0902.0431_E00165 | 29   | ■ 0.999 | $\begin{array}{ll} \left( \alpha_2 x \right) \left( \alpha_3 y \right) = \overline{\alpha_1(\overline{xy})}, & x, y \in \mathfrak{C}, \\ \left( \alpha_3 x \right) \left( \alpha_1 y \right) = \overline{\alpha_2(\overline{xy})}, & x, y \in \mathfrak{C}. \end{array}$                                                                                                                           | $\begin{array}{ll} (\alpha_2 x)(\alpha_3 y) = \overline{\alpha_1(\overline{xy})}, & x, y \in \mathfrak{C}, \\ (\alpha_3 x)(\alpha_1 y) = \overline{\alpha_2(\overline{xy})}, & x, y \in \mathfrak{C}. \end{array}$                                                                                                                                                                                 | $\begin{array}{ll} (\alpha_2 x)(\alpha_3 y) = \overline{\alpha_1(\overline{xy})}, & x, y \in \mathfrak{C}, \\ (\alpha_3 x)(\alpha_1 y) = \overline{\alpha_2(\overline{xy})}, & x, y \in \mathfrak{C}. \end{array}$                                                                                                                                                                                 |
| 0902.0431_E00450 | 74   | ■ 0.999 | $\delta + \widetilde{T} = 0, \quad \delta \in \mathfrak{f}_4^C, T \in \mathfrak{J}_0^C \quad \text{implies} \quad \delta = 0, T = 0.$                                                                                                                                                                                                                                                              | $\delta + \widetilde{T} = 0, \quad \delta \in \mathfrak{f}_4^C, T \in \mathfrak{J}_0^C \quad \text{implies} \quad \delta = 0, T = 0.$                                                                                                                                                                                                                                                              | $\delta + \widetilde{T} = 0, \quad \delta \in \mathfrak{f}_4^C, T \in \mathfrak{J}_0^C \quad \text{implies} \quad \delta = 0, T = 0.$                                                                                                                                                                                                                                                              |
| 0902.0431_E00541 | 91   | ■ 0.999 | $\begin{aligned} \left( \mathfrak{J}^C \right)_\sigma &= \{ X \in \mathfrak{J}^C \mid \sigma X = X \} \\ &= \{ X \in \mathfrak{J}^C \mid 4E_1 \times (E_1 \times X) = X \} \oplus \mathfrak{E}_1^C, \\ \left( \mathfrak{J}^C \right)_{-\sigma} &= \{ X \in \mathfrak{J}^C \mid \sigma X = -X \} \\ &= \{ X \in \mathfrak{J}^C \mid E_1 \times X = 0, \langle E_1, X \rangle = 0 \}, \end{aligned}$ | $\begin{aligned} \left( \mathfrak{J}^C \right)_\sigma &= \{ X \in \mathfrak{J}^C \mid \sigma X = X \} \\ &= \{ X \in \mathfrak{J}^C \mid 4E_1 \times (E_1 \times X) = X \} \oplus \mathfrak{E}_1^C, \\ \left( \mathfrak{J}^C \right)_{-\sigma} &= \{ X \in \mathfrak{J}^C \mid \sigma X = -X \} \\ &= \{ X \in \mathfrak{J}^C \mid E_1 \times X = 0, \langle E_1, X \rangle = 0 \}, \end{aligned}$ | $\begin{aligned} \left( \mathfrak{J}^C \right)_\sigma &= \{ X \in \mathfrak{J}^C \mid \sigma X = X \} \\ &= \{ X \in \mathfrak{J}^C \mid 4E_1 \times (E_1 \times X) = X \} \oplus \mathfrak{E}_1^C, \\ \left( \mathfrak{J}^C \right)_{-\sigma} &= \{ X \in \mathfrak{J}^C \mid \sigma X = -X \} \\ &= \{ X \in \mathfrak{J}^C \mid E_1 \times X = 0, \langle E_1, X \rangle = 0 \}, \end{aligned}$ |
| 0902.0431_E00201 | 34   | ■ 0.999 | $\left( \alpha_1 x \right) \left( \alpha_2 y \right) = \overline{\alpha_3(\overline{xy})}. \quad x, y \in \mathfrak{C}.$                                                                                                                                                                                                                                                                           | $(\alpha_1 x)(\alpha_2 y) = \overline{\alpha_3(\overline{xy})}. \quad x, y \in \mathfrak{C}.$                                                                                                                                                                                                                                                                                                      | $(\alpha_1 x)(\alpha_2 y) = \overline{\alpha_3(\overline{xy})}. \quad x, y \in \mathfrak{C}.$                                                                                                                                                                                                                                                                                                      |
| 0902.0431_E00002 | 5    | ■ 0.999 | $\gamma, \quad \sigma, \quad \iota, \quad \kappa, \quad \lambda, \quad \mu, \quad \tau, \quad v, \quad \chi, \quad g, \quad w.$                                                                                                                                                                                                                                                                    | $\gamma, \quad \sigma, \quad \iota, \quad \kappa, \quad \lambda, \quad \mu, \quad \tau, \quad v, \quad \chi, \quad g, \quad w.$                                                                                                                                                                                                                                                                    | $\gamma, \quad \sigma, \quad \iota, \quad \kappa, \quad \lambda, \quad \mu, \quad \tau, \quad v, \quad \chi, \quad g, \quad w.$                                                                                                                                                                                                                                                                    |
| 0902.0431_E00004 | 6    | ■ 0.999 | $e_i^2 = -1, \quad i \neq 0, \quad e_i e_j = -e_j e_i, \quad i \neq j, i \neq 0, j \neq 0,$                                                                                                                                                                                                                                                                                                        | $e_i^2 = -1, \quad i \neq 0, \quad e_i e_j = -e_j e_i, \quad i \neq j, i \neq 0, j \neq 0,$                                                                                                                                                                                                                                                                                                        | $e_i^2 = -1, \quad i \neq 0, \quad e_i e_j = -e_j e_i, \quad i \neq j, i \neq 0, j \neq 0,$                                                                                                                                                                                                                                                                                                        |



| Identifier       | Page | Conf.   | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Rendered                                                                                                                                                                                                                                                                                                                                         | Image                                                                                                                                                                                                                                                                                                                                            |
|------------------|------|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0902.0431_E09362 | 58   | ■ 0.999 | $\begin{array}{l} \backslash begin{array}{l} \backslash alpha E=E, \& \backslash alpha E_{\{1\}}=E_{\{1\}}, \\ \& \backslash alpha \backslash left(E_{\{2\}}-E_{\{3\}}\backslash right)=A, \backslash \backslash \backslash alpha F_{\{1\}} \\ \backslash left(e_{\{i\}}\backslash right)=X_{\{i\}}, \& \backslash alpha F_{\{2\}}\backslash left(e_{\{i\}} \\ \backslash right)=Y_{\{i\}}, \& \backslash alpha F_{\{3\}}\backslash left(e_{\{i\}}\backslash right)=Z_{\{i\}}, \\ \backslash quad i=0,1, \backslash cdots, 7 \backslash end{array} \end{array}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | $\begin{array}{l} \alpha E = E, \qquad \alpha E_1 = E_1, \qquad \alpha (E_2 - E_3) = A, \\ \alpha F_1(e_i) = X_i, \quad \alpha F_2(e_i) = Y_i, \quad \alpha F_3(e_i) = Z_i, \quad i = 0, 1, \cdots, 7 \end{array}$                                                                                                                               | $\begin{array}{l} \alpha E = E, \qquad \alpha E_1 = E_1, \qquad \alpha (E_2 - E_3) = A, \\ \alpha F_1(e_i) = X_i, \quad \alpha F_2(e_i) = Y_i, \quad \alpha F_3(e_i) = Z_i, \quad i = 0, 1, \cdots, 7 \end{array}$                                                                                                                               |
| 0902.0431_E09426 | 70   | ■ 0.999 | $\backslash varphi(P, A)((X+M) \backslash times(Y+N))=\backslash varphi(P, A)(X+M) \backslash times \backslash varphi(P, A)(Y+N),$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | $\varphi(P,A)((X+M)\times(Y+N))=\varphi(P,A)(X+M)\times\varphi(P,A)(Y+N),$                                                                                                                                                                                                                                                                       | $\varphi(P,A)((X+M)\times(Y+N))=\varphi(P,A)(X+M)\times\varphi(P,A)(Y+N),$                                                                                                                                                                                                                                                                       |
| 0902.0431_E09438 | 72   | ■ 0.999 | $\begin{array}{l} \backslash begin{aligned} \backslash mathfrak{J}\backslash left(3, \backslash mathfrak{C}\wedge\backslash prime\backslash right) \& \backslash cong\backslash left\{X \in \backslash mathfrak{J}\}\backslash left(3, \\ \backslash mathfrak{C}\wedge\backslash C\backslash right) \backslash mid \backslash tau \backslash gamma X=X\backslash right \\ \backslash \}=\backslash left(\backslash mathfrak{J}\backslash left(3, \backslash mathfrak{C}\wedge\backslash C\backslash right) \\ \backslash right)_{\backslash tau \backslash gamma}, \backslash \backslash \backslash mathfrak{J}\{1,2, \backslash mathfrak{C}\} \\ \& \backslash cong\backslash left\{X \in \backslash mathfrak{J}\}\backslash left(3, \backslash mathfrak{C}\wedge\backslash C \\ \backslash right) \backslash mid \backslash tau \backslash sigma X=X\backslash right\backslash \}=\backslash left(\backslash mathfrak{J} \\ \backslash left(3, \backslash mathfrak{C}\wedge\backslash C\backslash right)\backslash right)_{\backslash tau \backslash sigma} \\ \backslash end{aligned} \end{array}$ | $\begin{array}{l} \mathfrak{J}(3, \mathfrak{C}') \cong \{X \in \mathfrak{J}(3, \mathfrak{C}^C) \mid \tau\gamma X = X\} = (\mathfrak{J}(3, \mathfrak{C}^C))_{\tau\gamma}, \\ \mathfrak{J}(1, 2, \mathfrak{C}) \cong \{X \in \mathfrak{J}(3, \mathfrak{C}^C) \mid \tau\sigma X = X\} = (\mathfrak{J}(3, \mathfrak{C}^C))_{\tau\sigma} \end{array}$ | $\begin{array}{l} \mathfrak{J}(3, \mathfrak{C}') \cong \{X \in \mathfrak{J}(3, \mathfrak{C}^C) \mid \tau\gamma X = X\} = (\mathfrak{J}(3, \mathfrak{C}^C))_{\tau\gamma}, \\ \mathfrak{J}(1, 2, \mathfrak{C}) \cong \{X \in \mathfrak{J}(3, \mathfrak{C}^C) \mid \tau\sigma X = X\} = (\mathfrak{J}(3, \mathfrak{C}^C))_{\tau\sigma} \end{array}$ |
| 0902.0431_E09483 | 79   | ■ 0.999 | $\backslash left[\backslash phi,\backslash left[\backslash phi, \backslash widetilde{A}_{\{1\}}\backslash left(e_{\{i\}} \\ \backslash right)\backslash right]\backslash right]=\backslash left[\backslash phi,-\backslash frac{1}{2} \\ \backslash widetilde{F}_{\{1\}}\backslash left(e_{\{i\}}\backslash right)\backslash right]=\backslash frac{1}{4} \\ \backslash widetilde{A}_{\{1\}}\backslash left(e_{\{i\}}\backslash right),$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | $\left[\phi,\left[\phi,\widetilde{A}_1(e_i)\right]\right]=\left[\phi,-\frac{1}{2}\widetilde{F}_1(e_i)\right]=\frac{1}{4}\widetilde{A}_1(e_i),$                                                                                                                                                                                                   | $[\phi,[\phi,\widetilde{A}_1(e_i)]]=\left[\phi,-\frac{1}{2}\widetilde{F}_1(e_i)\right]=\frac{1}{4}\widetilde{A}_1(e_i),$                                                                                                                                                                                                                         |
| 0902.0431_E09487 | 80   | ■ 0.999 | $\backslash operatorname{tr}(\backslash phi \backslash phi)=1 \backslash times 2+\backslash frac{1}{4} \backslash times 8 \backslash times 2=6 \; .$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | $\mathrm{tr}(\phi\phi)=1\times2+\frac{1}{4}\times8\times2=6.$                                                                                                                                                                                                                                                                                    | $\mathrm{tr}(\phi\phi)=1\times2+\frac{1}{4}\times8\times2=6.$                                                                                                                                                                                                                                                                                    |
| 0902.0431_E09597 | 99   | ■ 0.999 | $(M+\backslash boldsymbol{a}) \backslash times (N+\backslash boldsymbol{b})=\backslash left(M \backslash times N-\backslash frac{1}{2})\backslash left(\backslash boldsymbol{a}\wedge\backslash * \backslash boldsymbol{b}+\backslash boldsymbol{b}\wedge\backslash * \backslash boldsymbol{a}\backslash right) \backslash right)-\backslash frac{1}{2}(\backslash boldsymbol{a} \backslash N+\backslash boldsymbol{b} \backslash M) \; .$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | $(M+\boldsymbol{a})\times(N+\boldsymbol{b})=\left(M\times N-\frac{1}{2}(\boldsymbol{a}^*\boldsymbol{b}+\boldsymbol{b}^*\boldsymbol{a})\right)-\frac{1}{2}(\boldsymbol{a}N+\boldsymbol{b}M).$                                                                                                                                                     | $(M+\boldsymbol{a})\times(N+\boldsymbol{b})=\left(M\times N-\frac{1}{2}(\boldsymbol{a}^*\boldsymbol{b}+\boldsymbol{b}^*\boldsymbol{a})\right)-\frac{1}{2}(\boldsymbol{a}N+\boldsymbol{b}M).$                                                                                                                                                     |
| 0902.0431_E09604 | 101  | ■ 0.999 | $\backslash mathfrak{J}\wedge\backslash C=\backslash left(\backslash mathfrak{J}\wedge\backslash C\backslash right)_{\backslash \tau \backslash gamma} \backslash oplus \backslash left(\backslash mathfrak{J}\wedge\backslash C\backslash right)_{\backslash -\backslash \tau \backslash gamma} \backslash oplus i \backslash left(\backslash mathfrak{J}\wedge\backslash C\backslash right)_{\backslash \tau \backslash gamma} \backslash oplus i \backslash left(\backslash mathfrak{J}\wedge\backslash C\backslash right)_{\backslash \tau \backslash gamma} \; .$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | $\mathfrak{J}^C=(\mathfrak{J}^C)_{\tau\gamma}\oplus(\mathfrak{J}^C)_{-\tau\gamma}=(\mathfrak{J}^C)_{\tau\gamma}\oplus i(\mathfrak{J}^C)_{\tau\gamma}.$                                                                                                                                                                                           | $\mathfrak{J}^C=(\mathfrak{J}^C)_{\tau\gamma}\oplus(\mathfrak{J}^C)_{-\tau\gamma}=(\mathfrak{J}^C)_{\tau\gamma}\oplus i(\mathfrak{J}^C)_{\tau\gamma}.$                                                                                                                                                                                           |
| 0902.0431_E09668 | 111  | ■ 0.999 | $\backslash mathfrak{P}\wedge\backslash C=\backslash mathfrak{J}\wedge\backslash C \backslash oplus \backslash mathfrak{J}\wedge\backslash C \backslash oplus C \backslash oplus C \; .$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | $\mathfrak{P}^C=\mathfrak{J}^C\oplus\mathfrak{J}^C\oplus C\oplus C.$                                                                                                                                                                                                                                                                             | $\mathfrak{P}^C=\mathfrak{J}^C\oplus\mathfrak{J}^C\oplus C\oplus C.$                                                                                                                                                                                                                                                                             |
| 0902.0431_E09672 | 112  | ■ 0.999 | $\begin{array}{l} \backslash begin{aligned} \& (P \backslash times Q) P \backslash \backslash = \& \backslash Phi \backslash left(- \\ \backslash frac{1}{2}(X \backslash vee W+Z \backslash vee Y),-\backslash frac{1}{4}(2 Y \backslash times W- \\ \backslash xi Z-\backslash zeta X), \backslash frac{1}{4}(2 X \backslash times Z-\backslash eta W-\backslash omega \\ Y),\backslash right. \backslash \backslash \& \backslash left.\backslash frac{1}{8}((X, W)+(Z, Y)-3(\xi\omega+\zeta\eta)) \\ \backslash omega+\backslash zeta \backslash eta))\backslash right)(X, Y, \backslash xi, \backslash eta) \backslash end{aligned} \end{array}$                                                                                                                                                                                                                                                                                                                                                                                                                                                 | $\begin{array}{l} (P\times Q)P \\ =\Phi\left(-\frac{1}{2}(X\vee W+Z\vee Y),-\frac{1}{4}(2Y\times W-\xi Z-\zeta X),\frac{1}{4}(2X\times Z-\eta W-\omega Y),\right. \\ \left.\frac{1}{8}((X,W)+(Z,Y)-3(\xi\omega+\zeta\eta))\right)(X,Y,\xi,\eta) \end{array}$                                                                                     | $\begin{array}{l} (P\times Q)P \\ =\Phi\left(-\frac{1}{2}(X\vee W+Z\vee Y),-\frac{1}{4}(2Y\times W-\xi Z-\zeta X),\frac{1}{4}(2X\times Z-\eta W-\omega Y),\right. \\ \left.\frac{1}{8}((X,W)+(Z,Y)-3(\xi\omega+\zeta\eta))\right)(X,Y,\xi,\eta) \end{array}$                                                                                     |
| 0902.0431_E09715 | 118  | ■ 0.999 | $\backslash mathfrak{e}_{\{7\}}=\backslash left\{\backslash Phi(\backslash phi, A,-\backslash tau A, \backslash nu) \backslash mid \backslash phi \in \backslash mathfrak{e}_{\{6\}}, A \in \backslash mathfrak{J}\wedge\backslash C, \backslash nu \in i \backslash boldsymbol{R}\backslash right\},$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | $\mathfrak{e}_7=\left\{\Phi(\phi,A,-\tau A,\nu)\mid\phi\in\mathfrak{e}_6,A\in\mathfrak{J}^C,\nu\in i\boldsymbol{R}\right\},$                                                                                                                                                                                                                     | $\mathfrak{e}_7=\left\{\Phi(\phi,A,-\tau A,\nu)\mid\phi\in\mathfrak{e}_6,A\in\mathfrak{J}^C,\nu\in i\boldsymbol{R}\right\},$                                                                                                                                                                                                                     |



| Identifier       | Page | Conf.   | LaTeX source                                                                                                                                                                                                                                                                                                                                             | Rendered                                                                                                                                                                                                                                                                                                       | Image                                                                                                                                                                                                                                                                                                          |
|------------------|------|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0902.0431_E00243 | 40   | ■ 0.999 | $\begin{aligned} & 3 \operatorname{det}(\alpha X) = \operatorname{tr}(\alpha X, \alpha X, \alpha X) - \frac{3}{2} \operatorname{tr}(\alpha X)(\alpha X, \alpha X) + \frac{1}{2} \operatorname{tr}(\alpha X)^3 \\ & = \operatorname{tr}(X, X, X) - \frac{3}{2} \operatorname{tr}(X)(X, X) + \frac{1}{2} \operatorname{tr}(X)^3 = 3 \det X. \end{aligned}$ | $3 \det(\alpha X) = \operatorname{tr}(\alpha X, \alpha X, \alpha X) - \frac{3}{2} \operatorname{tr}(\alpha X)(\alpha X, \alpha X) + \frac{1}{2} \operatorname{tr}(\alpha X)^3$ $= \operatorname{tr}(X, X, X) - \frac{3}{2} \operatorname{tr}(X)(X, X) + \frac{1}{2} \operatorname{tr}(X)^3 = 3 \det X.$        | $3 \det(\alpha X) = \operatorname{tr}(\alpha X, \alpha X, \alpha X) - \frac{3}{2} \operatorname{tr}(\alpha X)(\alpha X, \alpha X) + \frac{1}{2} \operatorname{tr}(\alpha X)^3$ $= \operatorname{tr}(X, X, X) - \frac{3}{2} \operatorname{tr}(X)(X, X) + \frac{1}{2} \operatorname{tr}(X)^3 = 3 \det X.$        |
| 0902.0431_E00706 | 116  | ■ 0.999 | $\begin{aligned} \Phi &= \left( \begin{array}{cccc} \phi - \frac{1}{3}\nu & 2B & 0 & A \\ 2A & \phi' + \frac{1}{3}\nu & B & 0 \\ 0 & A & \nu & 0 \\ B & 0 & 0 & -\nu \end{array} \right) \\ &= \Phi(\phi, A, B, \nu), \quad \phi \in \mathfrak{e}_6^C, A, B \in \mathfrak{J}^C, \nu \in C. \end{aligned}$                                                | $\Phi = \left( \begin{array}{cccc} \phi - \frac{1}{3}\nu & 2B & 0 & A \\ 2A & \phi' + \frac{1}{3}\nu & B & 0 \\ 0 & A & \nu & 0 \\ B & 0 & 0 & -\nu \end{array} \right)$ $= \Phi(\phi, A, B, \nu), \quad \phi \in \mathfrak{e}_6^C, A, B \in \mathfrak{J}^C, \nu \in C.$                                       | $\Phi = \begin{pmatrix} \phi - \frac{1}{3}\nu & 2B & 0 & A \\ 2A & \phi' + \frac{1}{3}\nu & B & 0 \\ 0 & A & \nu & 0 \\ B & 0 & 0 & -\nu \end{pmatrix}$ $= \Phi(\phi, A, B, \nu), \quad \phi \in \mathfrak{e}_6^C, A, B \in \mathfrak{J}^C, \nu \in C.$                                                        |
| 0902.0431_E01071 | 182  | ■ 0.999 | $\begin{aligned} B &= l(D') + l(S)I, & D' \in \mathfrak{u}(8), S \in \mathfrak{S}(8, C) \\ &= l(D) + l(S)I + l(icE), & D \in \mathfrak{su}(8), S \in \mathfrak{S}(8, C), c \in \mathbf{R}. \end{aligned}$                                                                                                                                                | $B = l(D') + l(S)I, \quad D' \in \mathfrak{u}(8), S \in \mathfrak{S}(8, C)$ $= l(D) + l(S)I + l(icE), \quad D \in \mathfrak{su}(8), S \in \mathfrak{S}(8, C), c \in \mathbf{R}.$                                                                                                                               | $B = l(D') + l(S)I, \quad D' \in \mathfrak{u}(8), S \in \mathfrak{S}(8, C)$ $= l(D) + l(S)I + l(icE), \quad D \in \mathfrak{su}(8), S \in \mathfrak{S}(8, C), c \in \mathbf{R}.$                                                                                                                               |
| 0902.0431_E01110 | 188  | ■ 0.999 | $\begin{array}{l} \omega_1 \quad 2\omega_1 \quad \omega_2 \quad 2\omega_2 \quad \omega_3 \quad \omega_4 \quad \omega_5 \quad \omega_6 \quad \omega_7 \quad \omega_8 \quad \cdots \\ 16 \quad 135 \quad 120 \quad 5304 \quad 560 \quad 1820 \quad 4368 \quad 8008 \quad 128 \quad 128 \quad \cdots \end{array}$                                           | $\begin{array}{l} \omega_1 \quad 2\omega_1 \quad \omega_2 \quad 2\omega_2 \quad \omega_3 \quad \omega_4 \quad \omega_5 \quad \omega_6 \quad \omega_7 \quad \omega_8 \quad \cdots \\ 16 \quad 135 \quad 120 \quad 5304 \quad 560 \quad 1820 \quad 4368 \quad 8008 \quad 128 \quad 128 \quad \cdots \end{array}$ | $\begin{array}{l} \omega_1 \quad 2\omega_1 \quad \omega_2 \quad 2\omega_2 \quad \omega_3 \quad \omega_4 \quad \omega_5 \quad \omega_6 \quad \omega_7 \quad \omega_8 \quad \cdots \\ 16 \quad 135 \quad 120 \quad 5304 \quad 560 \quad 1820 \quad 4368 \quad 8008 \quad 128 \quad 128 \quad \cdots \end{array}$ |
| 0902.0431_E00006 | 7    | ■ 1.000 | $\{x, y, a\} = \{y, a, x\} = \{a, x, y\} = -\{y, x, a\} = -\{x, a, y\} = -\{a, y, x\}.$                                                                                                                                                                                                                                                                  | $\{x, y, a\} = \{y, a, x\} = \{a, x, y\} = -\{y, x, a\} = -\{x, a, y\} = -\{a, y, x\}.$                                                                                                                                                                                                                        | $\{x, y, a\} = \{y, a, x\} = \{a, x, y\} = -\{y, x, a\} = -\{x, a, y\} = -\{a, y, x\}.$                                                                                                                                                                                                                        |
| 0902.0431_E00011 | 8    | ■ 1.000 | $\alpha 1 = 1, \quad \overline{\alpha e_i} = -\alpha e_i, \quad i = 1, 2, \dots, 7.$                                                                                                                                                                                                                                                                     | $\alpha 1 = 1, \quad \overline{\alpha e_i} = -\alpha e_i, \quad i = 1, 2, \dots, 7.$                                                                                                                                                                                                                           | $\alpha 1 = 1, \quad \overline{\alpha e_i} = -\alpha e_i, \quad i = 1, 2, \dots, 7.$                                                                                                                                                                                                                           |
| 0902.0431_E00019 | 9    | ■ 1.000 | $\begin{aligned} (\kappa D)x &= \overline{Dx}, \quad x \in \mathfrak{C}, \\ \pi(G_{ij}) &= F_{ij}, \quad i, j = 0, 1, \dots, 7, i \neq j, \\ \nu &= \pi\kappa. \end{aligned}$                                                                                                                                                                            | $(\kappa D)x = \overline{Dx}, \quad x \in \mathfrak{C},$ $\pi(G_{ij}) = F_{ij}, \quad i, j = 0, 1, \dots, 7, i \neq j,$ $\nu = \pi\kappa.$                                                                                                                                                                     | $(\kappa D)x = \overline{Dx}, \quad x \in \mathfrak{C},$ $\pi(G_{ij}) = F_{ij}, \quad i, j = 0, 1, \dots, 7, i \neq j,$ $\nu = \pi\kappa.$                                                                                                                                                                     |
| 0902.0431_E00026 | 10   | ■ 1.000 | $L_a x = ax, \quad R_a x = xa, \quad T_a x = ax + xa = (L_a + R_a)x, \quad x \in \mathfrak{C}.$                                                                                                                                                                                                                                                          | $L_a x = ax, \quad R_a x = xa, \quad T_a x = ax + xa = (L_a + R_a)x, \quad x \in \mathfrak{C}.$                                                                                                                                                                                                                | $L_a x = ax, \quad R_a x = xa, \quad T_a x = ax + xa = (L_a + R_a)x, \quad x \in \mathfrak{C}.$                                                                                                                                                                                                                |
| 0902.0431_E00032 | 10   | ■ 1.000 | $\begin{aligned} L_{e_i} x &= e_i x = 2F_{i0}x, \quad x \in \mathfrak{C}, \\ T_{e_i} x &= e_i x + x e_i = \begin{cases} 2e_i, & x = 1 \\ -2, & x = e_i \\ 0, & x = e_j, \quad j \neq 0, i. \end{cases} \end{aligned}$                                                                                                                                    | $L_{e_i} x = e_i x = 2F_{i0}x, \quad x \in \mathfrak{C},$ $T_{e_i} x = e_i x + x e_i = \begin{cases} 2e_i, & x = 1 \\ -2, & x = e_i \\ 0, & x = e_j, \quad j \neq 0, i. \end{cases}$                                                                                                                           | $L_{e_i} x = e_i x = 2F_{i0}x, \quad x \in \mathfrak{C},$ $T_{e_i} x = e_i x + x e_i = \begin{cases} 2e_i, & x = 1 \\ -2, & x = e_i \\ 0, & x = e_j, \quad j \neq 0, i. \end{cases}$                                                                                                                           |

| Identifier            | Page | Conf.   | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Rendered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Image                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-----------------------|------|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0902.0431_E00072      | 15   | ■ 1.000 | $\varphi_*(D)(a+\boldsymbol{m})=D\ \boldsymbol{m},\quad a+\boldsymbol{m}\in C\oplus C^3=\mathfrak{C}.$<br>$\quad a+\boldsymbol{m}\in C\oplus C^3=\mathfrak{C}.$                                                                                                                                                                                                                                                                                                                                                | $\varphi_*(D)(a+\boldsymbol{m})=D\boldsymbol{m},\quad a+\boldsymbol{m}\in C\oplus C^3=\mathfrak{C}.$                                                                                                                                                                                                                                                                                                                                                                                                           | $\varphi_*(D)(a+\boldsymbol{m})=D\boldsymbol{m},\quad a+\boldsymbol{m}\in C\oplus C^3=\mathfrak{C}.$                                                                                                                                                                                                                                                                                                                                                                                                           |
| 0902.0431_E00089      | 18   | ■ 1.000 | $B_2\!\left(D_1,D_2\right)=k\operatorname{tr}\!\left(D_1D_2\right).$                                                                                                                                                                                                                                                                                                                                                                                                                                           | $B_2(D_1,D_2)=k\operatorname{tr}(D_1D_2).$                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | $B_2(D_1,D_2)=k\operatorname{tr}(D_1D_2).$                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 0902.0431_E00090      | 18   | ■ 1.000 | $\begin{array}{ll} \left[H_1,\left[H_1,L_{12}\right]\right]=\left[H_1,2L_{21}\right]=-4L_{12}, & \left[H_1,\left[H_1,L_{21}\right]\right]=\left[H_1,-2L_{12}\right]=-4L_{21}, \\ \left[H_1,\left[H_1,L_{13}\right]\right]=\left[H_1,L_{31}\right]=-L_{13}, & \left[H_1,\left[H_1,L_{31}\right]\right]=\left[H_1,-L_{13}\right]=-L_{31}, \\ \left[H_1,\left[H_1,L_{23}\right]\right]=\left[H_1,-L_{32}\right]=-L_{23}, & \left[H_1,\left[H_1,L_{32}\right]\right]=\left[H_1,L_{23}\right]=-L_{32}, \end{array}$ | $\begin{array}{ll} \left[H_1,\left[H_1,L_{12}\right]\right]=\left[H_1,2L_{21}\right]=-4L_{12}, & \left[H_1,\left[H_1,L_{21}\right]\right]=\left[H_1,-2L_{12}\right]=-4L_{21}, \\ \left[H_1,\left[H_1,L_{13}\right]\right]=\left[H_1,L_{31}\right]=-L_{13}, & \left[H_1,\left[H_1,L_{31}\right]\right]=\left[H_1,-L_{13}\right]=-L_{31}, \\ \left[H_1,\left[H_1,L_{23}\right]\right]=\left[H_1,-L_{32}\right]=-L_{23}, & \left[H_1,\left[H_1,L_{32}\right]\right]=\left[H_1,L_{23}\right]=-L_{32}, \end{array}$ | $\begin{array}{ll} \left[H_1,\left[H_1,L_{12}\right]\right]=\left[H_1,2L_{21}\right]=-4L_{12}, & \left[H_1,\left[H_1,L_{21}\right]\right]=\left[H_1,-2L_{12}\right]=-4L_{21}, \\ \left[H_1,\left[H_1,L_{13}\right]\right]=\left[H_1,L_{31}\right]=-L_{13}, & \left[H_1,\left[H_1,L_{31}\right]\right]=\left[H_1,-L_{13}\right]=-L_{31}, \\ \left[H_1,\left[H_1,L_{23}\right]\right]=\left[H_1,-L_{32}\right]=-L_{23}, & \left[H_1,\left[H_1,L_{32}\right]\right]=\left[H_1,L_{23}\right]=-L_{32}, \end{array}$ |
| 0902.0431_E00119      | 23   | ■ 1.000 | $w(a+\boldsymbol{m})=a+\omega_1\boldsymbol{m},\quad a+\boldsymbol{m}\in C\oplus C^3=\mathfrak{C}.$<br>$\quad a+\boldsymbol{m}\in C\oplus C^3=\mathfrak{C}.$                                                                                                                                                                                                                                                                                                                                                    | $w(a+\boldsymbol{m})=a+\omega_1\boldsymbol{m},\quad a+\boldsymbol{m}\in C\oplus C^3=\mathfrak{C}.$                                                                                                                                                                                                                                                                                                                                                                                                             | $w(a+\boldsymbol{m})=a+\omega_1\boldsymbol{m},\quad a+\boldsymbol{m}\in C\oplus C^3=\mathfrak{C}.$                                                                                                                                                                                                                                                                                                                                                                                                             |
| 0902.0431_E00146      | 27   | ■ 1.000 | $G_2\{ \ }^{\mathfrak{C}}\simeq G_2\ \times\ \boldsymbol{R}^{14}.$                                                                                                                                                                                                                                                                                                                                                                                                                                             | $G_2^C\simeq G_2\times\boldsymbol{R}^{14}.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | $G_2^C\simeq G_2\times\boldsymbol{R}^{14}.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 0902.0431_E00176      | 31   | ■ 1.000 | $(x\,\overline{a})(a\,y\,a)=(x\,y)\,a,\quad x,\,y\in\mathfrak{C}.$                                                                                                                                                                                                                                                                                                                                                                                                                                             | $(x\overline{a})(aya)=(xy)a,\quad x,y\in\mathfrak{C}.$                                                                                                                                                                                                                                                                                                                                                                                                                                                         | $(x\overline{a})(aya)=(xy)a,\quad x,y\in\mathfrak{C}.$                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 0902.0431_E00181 (i)  | 31   | ■ 1.000 | $(\alpha\,x)(\widetilde{\alpha}\,y)=\alpha^{\prime}(x\,y),\quad x,\,y\in\mathfrak{C}.$                                                                                                                                                                                                                                                                                                                                                                                                                         | $(\alpha x)(\widetilde{\alpha}y)=\alpha'(xy),\quad x,y\in\mathfrak{C}.$                                                                                                                                                                                                                                                                                                                                                                                                                                        | $(\alpha x)(\widetilde{\alpha}y)=\alpha'(xy),\quad x,y\in\mathfrak{C}.$                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 0902.0431_E00185      | 32   | ■ 1.000 | $\begin{array}{l} \left(a_2,a_6\right)=\left(a_2b_0,a_6b_0\right)=\left(a_2b_0,a_2\left(a_4b_0\right)\right)=\left(b_0,a_4b_0\right)=\left(1,a_4\right)=0, \\ \left(a_3,a_7\right)=\left(a_3b_0,a_7b_0\right)=\left(a_2\left(a_1b_0\right),a_6\left(a_1b_0\right)\right)=\left(a_1\left(a_2b_0\right),a_1\left(a_6b_0\right)\right) \\ \qquad\qquad\qquad=\left(a_2b_0,a_6b_0\right)=\left(a_2b_0,a_2\left(a_4b_0\right)\right)=\left(b_0,a_4b_0\right)=\left(1,a_4\right)=0. \end{array}$                     | $\begin{array}{l} \left(a_2,a_6\right)=\left(a_2b_0,a_6b_0\right)=\left(a_2b_0,a_2\left(a_4b_0\right)\right)=\left(b_0,a_4b_0\right)=\left(1,a_4\right)=0, \\ \left(a_3,a_7\right)=\left(a_3b_0,a_7b_0\right)=\left(a_2\left(a_1b_0\right),a_6\left(a_1b_0\right)\right)=\left(a_1\left(a_2b_0\right),a_1\left(a_6b_0\right)\right) \\ \qquad\qquad\qquad=\left(a_2b_0,a_6b_0\right)=\left(a_2b_0,a_2\left(a_4b_0\right)\right)=\left(b_0,a_4b_0\right)=\left(1,a_4\right)=0. \end{array}$                     | $\begin{array}{l} \left(a_2,a_6\right)=\left(a_2b_0,a_6b_0\right)=\left(a_2b_0,a_2\left(a_4b_0\right)\right)=\left(b_0,a_4b_0\right)=\left(1,a_4\right)=0, \\ \left(a_3,a_7\right)=\left(a_3b_0,a_7b_0\right)=\left(a_2\left(a_1b_0\right),a_6\left(a_1b_0\right)\right)=\left(a_1\left(a_2b_0\right),a_1\left(a_6b_0\right)\right) \\ \qquad\qquad\qquad=\left(a_2b_0,a_6b_0\right)=\left(a_2b_0,a_2\left(a_4b_0\right)\right)=\left(b_0,a_4b_0\right)=\left(1,a_4\right)=0. \end{array}$                     |
| 0902.0431_E00192 (ii) | 33   | ■ 1.000 | $(\alpha\,x)(\widetilde{\alpha}\,y)=\widetilde{\alpha}(x\,y),\quad x,\,y\in\mathfrak{C}.$                                                                                                                                                                                                                                                                                                                                                                                                                      | $(\alpha x)(\widetilde{\alpha}y)=\widetilde{\alpha}(xy),\quad x,y\in\mathfrak{C}.$                                                                                                                                                                                                                                                                                                                                                                                                                             | $(\alpha x)(\widetilde{\alpha}y)=\widetilde{\alpha}(xy),\quad x,y\in\mathfrak{C}.$                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 0902.0431_E00194      | 33   | ■ 1.000 | $(\alpha\,x)(\widetilde{\alpha}\,y)=\widetilde{\alpha}(x\,y),\quad x,\,y\in\mathfrak{C}.$                                                                                                                                                                                                                                                                                                                                                                                                                      | $(\alpha x)(\widetilde{\alpha}y)=\widetilde{\alpha}(xy),\quad x,y\in\mathfrak{C}.$                                                                                                                                                                                                                                                                                                                                                                                                                             | $(\alpha x)(\widetilde{\alpha}y)=\widetilde{\alpha}(xy),\quad x,y\in\mathfrak{C}.$                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 0902.0431_E00200      | 34   | ■ 1.000 | $\left(\alpha_1,\alpha_2,\alpha_3\right)a=\alpha_1a,\quad a\in S^7.$                                                                                                                                                                                                                                                                                                                                                                                                                                           | $(\alpha_1,\alpha_2,\alpha_3)a=\alpha_1a,\quad a\in S^7.$                                                                                                                                                                                                                                                                                                                                                                                                                                                      | $(\alpha_1,\alpha_2,\alpha_3)a=\alpha_1a,\quad a\in S^7.$                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

| Identifier       | Page | Conf.   | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Rendered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Image                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|------------------|------|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0902.0431_E00222 | 37   | ■ 1.000 | $\begin{aligned} (X, Y, Z) &= \sum_{i=1}^3 \left( \frac{1}{2} (\xi_i \eta_{i+1} \zeta_{i+2} + \xi_i \eta_{i+2} \zeta_{i+1}) + R(x_i y_{i+1} z_{i+2} + x_i z_{i+1} y_{i+2}) \right. \\ &\quad \left. - (\xi_i (y_i, z_i) + \eta_i (z_i, x_i) + \zeta_i (x_i, y_i)) \right), \\ \det X &= \xi_1 \xi_2 \xi_3 + 2R(x_1 x_2 x_3) - \xi_1 x_1 \bar{x}_1 - \xi_2 x_2 \bar{x}_2 - \xi_3 x_3 \bar{x}_3. \end{aligned}$                                                                                                                                                                    | $(X, Y, Z) = \sum_{i=1}^3 \left( \frac{1}{2} (\xi_i \eta_{i+1} \zeta_{i+2} + \xi_i \eta_{i+2} \zeta_{i+1}) + R(x_i y_{i+1} z_{i+2} + x_i z_{i+1} y_{i+2}) \right. \\ \left. - (\xi_i (y_i, z_i) + \eta_i (z_i, x_i) + \zeta_i (x_i, y_i)) \right) \\ \det X = \xi_1 \xi_2 \xi_3 + 2R(x_1 x_2 x_3) - \xi_1 x_1 \bar{x}_1 - \xi_2 x_2 \bar{x}_2 - \xi_3 x_3 \bar{x}_3$                                                                                                                                                               | $(X, Y, Z) = \sum_{i=1}^3 \left( \frac{1}{2} (\xi_i \eta_{i+1} \zeta_{i+2} + \xi_i \eta_{i+2} \zeta_{i+1}) + R(x_i y_{i+1} z_{i+2} + x_i z_{i+1} y_{i+2}) \right. \\ \left. - (\xi_i (y_i, z_i) + \eta_i (z_i, x_i) + \zeta_i (x_i, y_i)) \right), \\ \det X = \xi_1 \xi_2 \xi_3 + 2R(x_1 x_2 x_3) - \xi_1 x_1 \bar{x}_1 - \xi_2 x_2 \bar{x}_2 - \xi_3 x_3 \bar{x}_3.$                                                                                                                                                             |
| 0902.0431_E00237 | 39   | ■ 1.000 | $\begin{aligned} &3 \operatorname{det} \left( \begin{matrix} \alpha^{-1} X \\ \alpha^{-1} X \\ \alpha^{-1} X \end{matrix} \right) = \left( \begin{matrix} \alpha^{-1} X \\ \alpha^{-1} X \\ \alpha^{-1} X \end{matrix} \right) \times \left( \begin{matrix} \alpha^{-1} X \\ \alpha^{-1} X \\ \alpha^{-1} X \end{matrix} \right) \\ &= \left( \begin{matrix} \alpha^{-1} X \\ \alpha^{-1} X \\ \alpha^{-1} X \end{matrix} \right) \times \left( \begin{matrix} \alpha^{-1} X \\ \alpha^{-1} X \\ \alpha^{-1} X \end{matrix} \right) = (X, X \times X) = 3 \det X, \end{aligned}$ | $3 \det \left( \begin{matrix} \alpha^{-1} X \\ \alpha^{-1} X \\ \alpha^{-1} X \end{matrix} \right) = \left( \begin{matrix} \alpha^{-1} X \\ \alpha^{-1} X \\ \alpha^{-1} X \end{matrix} \right) \times \left( \begin{matrix} \alpha^{-1} X \\ \alpha^{-1} X \\ \alpha^{-1} X \end{matrix} \right) \\ = \left( \begin{matrix} \alpha^{-1} X \\ \alpha^{-1} X \\ \alpha^{-1} X \end{matrix} \right) \times \left( \begin{matrix} \alpha^{-1} X \\ \alpha^{-1} X \\ \alpha^{-1} X \end{matrix} \right) = (X, X \times X) = 3 \det X,$ | $3 \det \left( \begin{matrix} \alpha^{-1} X \\ \alpha^{-1} X \\ \alpha^{-1} X \end{matrix} \right) = \left( \begin{matrix} \alpha^{-1} X \\ \alpha^{-1} X \\ \alpha^{-1} X \end{matrix} \right) \times \left( \begin{matrix} \alpha^{-1} X \\ \alpha^{-1} X \\ \alpha^{-1} X \end{matrix} \right) \\ = \left( \begin{matrix} \alpha^{-1} X \\ \alpha^{-1} X \\ \alpha^{-1} X \end{matrix} \right) \times \left( \begin{matrix} \alpha^{-1} X \\ \alpha^{-1} X \\ \alpha^{-1} X \end{matrix} \right) = (X, X \times X) = 3 \det X,$ |
| 0902.0431_E00252 | 41   | ■ 1.000 | $\operatorname{tr}(\alpha X, \alpha Y, \alpha Z) = \operatorname{tr}(X, Y, Z).$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | $\operatorname{tr}(\alpha X, \alpha Y, \alpha Z) = \operatorname{tr}(X, Y, Z).$                                                                                                                                                                                                                                                                                                                                                                                                                                                    | $\operatorname{tr}(\alpha X, \alpha Y, \alpha Z) = \operatorname{tr}(X, Y, Z).$                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 0902.0431_E00256 | 42   | ■ 1.000 | $\begin{aligned} &\mathfrak{e}_{6(-26)} = \{ \phi \in \operatorname{Hom}_{\mathbf{R}}(\mathfrak{J}) \mid (\phi X, X, X) = 0 \} \\ &= \{ \phi \in \operatorname{Hom}_{\mathbf{R}}(\mathfrak{J}) \mid (\phi X, Y, Z) + (X, \phi Y, Z) + (X, Y, \phi Z) = 0 \} \\ &= \{ \phi \in \operatorname{Hom}_{\mathbf{R}}(\mathfrak{J}) \mid -{}^t \phi(X \times Y) = \phi X \times Y + X \times \phi Y \}. \end{aligned}$                                                                                                                                                                   | $\mathfrak{e}_{6(-26)} = \{ \phi \in \operatorname{Hom}_{\mathbf{R}}(\mathfrak{J}) \mid (\phi X, X, X) = 0 \} \\ = \{ \phi \in \operatorname{Hom}_{\mathbf{R}}(\mathfrak{J}) \mid (\phi X, Y, Z) + (X, \phi Y, Z) + (X, Y, \phi Z) = 0 \} \\ = \{ \phi \in \operatorname{Hom}_{\mathbf{R}}(\mathfrak{J}) \mid -{}^t \phi(X \times Y) = \phi X \times Y + X \times \phi Y \}.$                                                                                                                                                      | $\mathfrak{e}_{6(-26)} = \{ \phi \in \operatorname{Hom}_{\mathbf{R}}(\mathfrak{J}) \mid (\phi X, X, X) = 0 \} \\ = \{ \phi \in \operatorname{Hom}_{\mathbf{R}}(\mathfrak{J}) \mid (\phi X, Y, Z) + (X, \phi Y, Z) + (X, Y, \phi Z) = 0 \} \\ = \{ \phi \in \operatorname{Hom}_{\mathbf{R}}(\mathfrak{J}) \mid -{}^t \phi(X \times Y) = \phi X \times Y + X \times \phi Y \}.$                                                                                                                                                      |
| 0902.0431_E00260 | 42   | ■ 1.000 | $\widetilde{A}X = \frac{1}{2}[A, X], \quad X \in \mathfrak{J}.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | $\widetilde{A}X = \frac{1}{2}[A, X], \quad X \in \mathfrak{J}.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | $\widetilde{A}X = \frac{1}{2}[A, X], \quad X \in \mathfrak{J}.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 0902.0431_E00265 | 43   | ■ 1.000 | $(X Y, Z) = (Y Z, X) = (Z X, Y) = (X, Y Z) = (Y, Z X) = (Z, X Y).$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | $(XY, Z) = (YZ, X) = (ZX, Y) = (X, YZ) = (Y, ZX) = (Z, XY).$                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | $(XY, Z) = (YZ, X) = (ZX, Y) = (X, YZ) = (Y, ZX) = (Z, XY).$                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 0902.0431_E00266 | 43   | ■ 1.000 | $\begin{aligned} (XY, Z) &= R(XY, Z) = \frac{1}{2} R(\operatorname{tr}((XY)Z + Z^*(Y^* X^*))) \\ &= \frac{1}{2} R \left( \sum_{i,j,k} ((x_{ij} y_{jk}) z_{ki} + \bar{z}_{ji} (\bar{y}_{kj} \bar{x}_{ik})) \right) \\ &= \frac{1}{2} R \left( \sum_{i,j,k} ((y_{jk} z_{ki}) x_{ij} + \bar{x}_{ik} (\bar{z}_{ji} \bar{y}_{kj})) \right) \\ &= \frac{1}{2} R(\operatorname{tr}((YZ)X + X^*(Z^* Y^*))) = (YZ, X). \end{aligned}$                                                                                                                                                     | $(XY, Z) = R(XY, Z) = \frac{1}{2} R(\operatorname{tr}((XY)Z + Z^*(Y^* X^*))) \\ = \frac{1}{2} R \left( \sum_{i,j,k} ((x_{ij} y_{jk}) z_{ki} + \bar{z}_{ji} (\bar{y}_{kj} \bar{x}_{ik})) \right) \\ = \frac{1}{2} R \left( \sum_{i,j,k} ((y_{jk} z_{ki}) x_{ij} + \bar{x}_{ik} (\bar{z}_{ji} \bar{y}_{kj})) \right) \\ = \frac{1}{2} R(\operatorname{tr}((YZ)X + X^*(Z^* Y^*))) = (YZ, X).$                                                                                                                                         | $(XY, Z) = R(XY, Z) = \frac{1}{2} R(\operatorname{tr}((XY)Z + Z^*(Y^* X^*))) \\ = \frac{1}{2} R \left( \sum_{i,j,k} ((x_{ij} y_{jk}) z_{ki} + \bar{z}_{ji} (\bar{y}_{kj} \bar{x}_{ik})) \right) \\ = \frac{1}{2} R \left( \sum_{i,j,k} ((y_{jk} z_{ki}) x_{ij} + \bar{x}_{ik} (\bar{z}_{ji} \bar{y}_{kj})) \right) \\ = \frac{1}{2} R(\operatorname{tr}((YZ)X + X^*(Z^* Y^*))) = (YZ, X).$                                                                                                                                         |

| Identifier           | Page | Conf.   | LaTeX source                                                                                                                                                                                                                                                                                                                                   | Rendered                                                                                                                                                                                                                                                                                                                            | Image                                                                                                                                                                                                                                                                                                                               |
|----------------------|------|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0902.0431_E00283     | 45   | ■ 1.000 | $\left(D_1x\right)y+x\left(D_2y\right)=\overline{\overline{\left(D_3\right)\left(\overline{xy}\right)}},\quad x,y\in\mathfrak{C}.$                                                                                                                                                                                                             | $(D_1x)y+x(D_2y)=\overline{D_3(\overline{xy})},\quad x,y\in\mathfrak{C}.$                                                                                                                                                                                                                                                           | $(D_1x)y+x(D_2y)=\overline{D_3(\overline{xy})},\quad x,y\in\mathfrak{C}.$                                                                                                                                                                                                                                                           |
| 0902.0431_E00295 (2) | 47   | ■ 1.000 | $\begin{aligned} & \{[\widetilde{A}],\widetilde{B}]=\left[\left(A-\frac{1}{3}\operatorname{tr}(A)E\right)^{\sim},\left(B-\frac{1}{3}\operatorname{tr}(B)E\right)^{\sim}\right]\in e_6^C, \\ & [\widetilde{A},\widetilde{B}]E=\widetilde{A}(\widetilde{B}E)-\widetilde{B}(\widetilde{A}E)=A\circ B-B\circ A=0. \end{aligned}$                   | $[\widetilde{A},\widetilde{B}]=\left[\left(A-\frac{1}{3}\operatorname{tr}(A)E\right)^{\sim},\left(B-\frac{1}{3}\operatorname{tr}(B)E\right)^{\sim}\right]\in e_6^C,$ $[\widetilde{A},\widetilde{B}]E=\widetilde{A}(\widetilde{B}E)-\widetilde{B}(\widetilde{A}E)=A\circ B-B\circ A=0.$                                              | $[\widetilde{A},\widetilde{B}]=\left[\left(A-\frac{1}{3}\operatorname{tr}(A)E\right)^{\sim},\left(B-\frac{1}{3}\operatorname{tr}(B)E\right)^{\sim}\right]\in e_6^C,$ $[\widetilde{A},\widetilde{B}]E=\widetilde{A}(\widetilde{B}E)-\widetilde{B}(\widetilde{A}E)=A\circ B-B\circ A=0.$                                              |
| 0902.0431_E00304     | 49   | ■ 1.000 | $\emptyset\not\in\left[\widetilde{A}_{\scriptscriptstyle{1}}(1),\widetilde{A}_{\scriptscriptstyle{1}}(e_1)\right]\in\mathfrak{d}_4^C\cap\mathfrak{a}$                                                                                                                                                                                          | $0\neq\left[\widetilde{A}_1(1),\widetilde{A}_1(e_1)\right]\in\mathfrak{d}_4^C\cap\mathfrak{a}$                                                                                                                                                                                                                                      | $0\neq[\widetilde{A}_1(1),\widetilde{A}_1(e_1)]\in\mathfrak{d}_4^C\cap\mathfrak{a}$                                                                                                                                                                                                                                                 |
| 0902.0431_E00311     | 50   | ■ 1.000 | $F_{\scriptscriptstyle{3}}(1)+F_{\scriptscriptstyle{1}}\left(z_{\scriptscriptstyle{1}}\right)\in W.$                                                                                                                                                                                                                                           | $F_3(1)+F_1(z_1)\in W.$                                                                                                                                                                                                                                                                                                             | $F_3(1)+F_1(z_1)\in W.$                                                                                                                                                                                                                                                                                                             |
| 0902.0431_E00320     | 51   | ■ 1.000 | $B_{\scriptscriptstyle{4}}\left(\delta_{\scriptscriptstyle{1}},\delta_{\scriptscriptstyle{2}}\right)=k\left(\delta_{\scriptscriptstyle{1}},\delta_{\scriptscriptstyle{2}}\right)_{\scriptscriptstyle{4}}=k^{\prime}\operatorname{tr}\left(\delta_{\scriptscriptstyle{1}}\delta_{\scriptscriptstyle{2}}\right).$                                | $B_4(\delta_1,\delta_2)=k(\delta_1,\delta_2)_4=k'\operatorname{tr}(\delta_1\delta_2).$                                                                                                                                                                                                                                              | $B_4(\delta_1,\delta_2)=k(\delta_1,\delta_2)_4=k'\operatorname{tr}(\delta_1\delta_2).$                                                                                                                                                                                                                                              |
| 0902.0431_E00321     | 51   | ■ 1.000 | $\begin{aligned} & (\delta,\delta)_4=\left(\widetilde{A}_1(1),\widetilde{A}_1(1)\right)_4=-2\left(\widetilde{A}_1(1),\left[\widetilde{E}_3,\widetilde{F}_1(1)\right]\right)_4 \\ & \qquad\qquad=-2\left(\widetilde{A}_1(1)E_3,F_1(1)\right)=-(F_1(1),F_1(1))=-2. \end{aligned}$                                                                | $(\delta,\delta)_4=\left(\widetilde{A}_1(1),\widetilde{A}_1(1)\right)_4=-2\left(\widetilde{A}_1(1),\left[\widetilde{E}_3,\widetilde{F}_1(1)\right]\right)_4$ $=-2\left(\widetilde{A}_1(1)E_3,F_1(1)\right)=-(F_1(1),F_1(1))=-2.$                                                                                                    | $(\delta,\delta)_4=\left(\widetilde{A}_1(1),\widetilde{A}_1(1)\right)_4=-2\left(\widetilde{A}_1(1),\left[\widetilde{E}_3,\widetilde{F}_1(1)\right]\right)_4$ $=-2\left(\widetilde{A}_1(1)E_3,F_1(1)\right)=-(F_1(1),F_1(1))=-2.$                                                                                                    |
| 0902.0431_E00325     | 52   | ■ 1.000 | $\operatorname{tr}(\delta\delta)=\operatorname{tr}\left(\left(\widetilde{A}_1(1)\right)^2\right)=\left(-\frac{1}{2}\right)\times 2-1+\left(-\frac{1}{4}\right)\times 8\times 2=-6.$                                                                                                                                                            | $\operatorname{tr}(\delta\delta)=\operatorname{tr}\left(\left(\widetilde{A}_1(1)\right)^2\right)=\left(-\frac{1}{2}\right)\times 2-1+\left(-\frac{1}{4}\right)\times 8\times 2=-6.$                                                                                                                                                 | $\operatorname{tr}(\delta\delta)=\operatorname{tr}((\widetilde{A}_1(1))^2)=\left(-\frac{1}{2}\right)\times 2-1+\left(-\frac{1}{4}\right)\times 8\times 2=-6.$                                                                                                                                                                       |
| 0902.0431_E00349     | 57   | ■ 1.000 | $\varphi\left(\alpha_1,\alpha_2,\alpha_3\right)\left(\begin{array}{ccc}\xi_1&x_3&\overline{x_2}\\\overline{x_3}&\xi_2&x_1\\x_2&\overline{x_1}&\xi_3\end{array}\right)=\left(\begin{array}{ccc}\xi_1&\alpha_3x_3&\overline{\alpha_2x_2}\\\overline{\alpha_3x_3}&\xi_2&\alpha_1x_1\\\alpha_2x_2&\overline{\alpha_1x_1}&\xi_3\end{array}\right).$ | $\varphi(\alpha_1,\alpha_2,\alpha_3)\left(\begin{array}{ccc}\xi_1&x_3&\overline{x_2}\\\overline{x_3}&\xi_2&x_1\\x_2&\overline{x_1}&\xi_3\end{array}\right)=\left(\begin{array}{ccc}\xi_1&\alpha_3x_3&\overline{\alpha_2x_2}\\\overline{\alpha_3x_3}&\xi_2&\alpha_1x_1\\\alpha_2x_2&\overline{\alpha_1x_1}&\xi_3\end{array}\right).$ | $\varphi(\alpha_1,\alpha_2,\alpha_3)\left(\begin{array}{ccc}\xi_1&x_3&\overline{x_2}\\\overline{x_3}&\xi_2&x_1\\x_2&\overline{x_1}&\xi_3\end{array}\right)=\left(\begin{array}{ccc}\xi_1&\alpha_3x_3&\overline{\alpha_2x_2}\\\overline{\alpha_3x_3}&\xi_2&\alpha_1x_1\\\alpha_2x_2&\overline{\alpha_1x_1}&\xi_3\end{array}\right).$ |
| 0902.0431_E00354     | 57   | ■ 1.000 | $\left(\alpha_1x\right)\left(\alpha_2y\right)=\overline{\alpha_3\left(\overline{xy}\right)},\quad x,y\in\mathfrak{C}.$                                                                                                                                                                                                                         | $(\alpha_1x)(\alpha_2y)=\overline{\alpha_3(\overline{xy})},\quad x,y\in\mathfrak{C}.$                                                                                                                                                                                                                                               | $(\alpha_1x)(\alpha_2y)=\overline{\alpha_3(\overline{xy})},\quad x,y\in\mathfrak{C}.$                                                                                                                                                                                                                                               |
| 0902.0431_E00361     | 58   | ■ 1.000 | $\begin{array}{l} Y_i=-2Z_0\circ X_i,\quad i=4,5,6,7, \\ Z_i=-2X_i\circ Y_0,\quad i=1,3,5,6,7. \end{array}$                                                                                                                                                                                                                                    | $Y_i=-2Z_0\circ X_i,\quad i=4,5,6,7,$ $Z_i=-2X_i\circ Y_0,\quad i=1,3,5,6,7.$                                                                                                                                                                                                                                                       | $Y_i=-2Z_0\circ X_i,\quad i=4,5,6,7,$ $Z_i=-2X_i\circ Y_0,\quad i=1,3,5,6,7.$                                                                                                                                                                                                                                                       |
| 0902.0431_E00364     | 59   | ■ 1.000 | $p:\left(F_{\scriptscriptstyle{4}}\right)_{E_{\scriptscriptstyle{1}}}\rightarrow SO(9).$                                                                                                                                                                                                                                                       | $p:(F_4)_{E_1}\rightarrow SO(9).$                                                                                                                                                                                                                                                                                                   | $p:(F_4)_{E_1}\rightarrow SO(9).$                                                                                                                                                                                                                                                                                                   |

| Identifier       | Page | Conf.   | LaTeX source                                                                                                                                                                                                                                                                                                                                                                 | Rendered                                                                                                                                                                                                                                                                                                                                                                     | Image                                                                                                                                                                                                                                                                                                                                                                        |
|------------------|------|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0902.0431_E00393 | 65   | ■ 1.000 | $\begin{aligned} F_{4, \boldsymbol{H}} &= \{\alpha \in \operatorname{Iso}_{\boldsymbol{R}}(\mathfrak{J}_{\boldsymbol{H}}) \mid \alpha(M \circ N) = \alpha M \circ \alpha N\} \\ &= \{\alpha \in \operatorname{Iso}_{\boldsymbol{R}}(\mathfrak{J}_{\boldsymbol{H}}) \mid \alpha(M \times N) = \alpha M \times \alpha N\}. \end{aligned}$                                      | $F_{4, \boldsymbol{H}} = \{\alpha \in \operatorname{Iso}_{\boldsymbol{R}}(\mathfrak{J}_{\boldsymbol{H}}) \mid \alpha(M \circ N) = \alpha M \circ \alpha N\} \\ = \{\alpha \in \operatorname{Iso}_{\boldsymbol{R}}(\mathfrak{J}_{\boldsymbol{H}}) \mid \alpha(M \times N) = \alpha M \times \alpha N\}.$                                                                      | $F_{4, \boldsymbol{H}} = \{\alpha \in \operatorname{Iso}_{\boldsymbol{R}}(\mathfrak{J}_{\boldsymbol{H}}) \mid \alpha(M \circ N) = \alpha M \circ \alpha N\} \\ = \{\alpha \in \operatorname{Iso}_{\boldsymbol{R}}(\mathfrak{J}_{\boldsymbol{H}}) \mid \alpha(M \times N) = \alpha M \times \alpha N\}.$                                                                      |
| 0902.0431_E00402 | 66   | ■ 1.000 | $\alpha = \varphi(A) \beta = \varphi(A) \varphi(B) = \varphi(AB), \quad AB \in \operatorname{Sp}(3).$                                                                                                                                                                                                                                                                        | $\alpha = \varphi(A) \beta = \varphi(A) \varphi(B) = \varphi(AB), \quad AB \in \operatorname{Sp}(3).$                                                                                                                                                                                                                                                                        | $\alpha = \varphi(A) \beta = \varphi(A) \varphi(B) = \varphi(AB), \quad AB \in \operatorname{Sp}(3).$                                                                                                                                                                                                                                                                        |
| 0902.0431_E00447 | 73   | ■ 1.000 | $\mathfrak{e}_6^C = \{\phi \in \operatorname{Hom}_C(\mathfrak{J}^C) \mid (\phi X, X, X) = 0\}.$                                                                                                                                                                                                                                                                              | $\mathfrak{e}_6^C = \{\phi \in \operatorname{Hom}_C(\mathfrak{J}^C) \mid (\phi X, X, X) = 0\}.$                                                                                                                                                                                                                                                                              | $\mathfrak{e}_6^C = \{\phi \in \operatorname{Hom}_C(\mathfrak{J}^C) \mid (\phi X, X, X) = 0\}.$                                                                                                                                                                                                                                                                              |
| 0902.0431_E00468 | 77   | ■ 1.000 | $\left(\phi^{\prime} X, Y\right)=-\left(X, \phi Y\right), \quad X, Y \in \mathfrak{J}^C.$                                                                                                                                                                                                                                                                                    | $\left(\phi^{\prime} X, Y\right)=-\left(X, \phi Y\right), \quad X, Y \in \mathfrak{J}^C.$                                                                                                                                                                                                                                                                                    | $\left(\phi^{\prime} X, Y\right)=-\left(X, \phi Y\right), \quad X, Y \in \mathfrak{J}^C.$                                                                                                                                                                                                                                                                                    |
| 0902.0431_E00469 | 77   | ■ 1.000 | $\phi(X \times Y) = \phi^{\prime} X \times Y + X \times \phi^{\prime} Y, \quad X, Y \in \mathfrak{J}^C.$                                                                                                                                                                                                                                                                     | $\phi(X \times Y) = \phi^{\prime} X \times Y + X \times \phi^{\prime} Y, \quad X, Y \in \mathfrak{J}^C.$                                                                                                                                                                                                                                                                     | $\phi(X \times Y) = \phi^{\prime} X \times Y + X \times \phi^{\prime} Y, \quad X, Y \in \mathfrak{J}^C.$                                                                                                                                                                                                                                                                     |
| 0902.0431_E00478 | 79   | ■ 1.000 | $\begin{aligned} (\phi, A \vee B)_6 &= \left(\delta + \widetilde{T}, [\widetilde{A}, \widetilde{B}] + \left(A \circ B - \frac{1}{3}(A, B)E\right)^{\sim}\right)_6 \\ &= (\delta, [\widetilde{A}, \widetilde{B}])_4 + \left(T, A \circ B - \frac{1}{3}(A, B)E\right) \\ &= (\delta A, B) + (\widetilde{T}A, B) = ((\delta + \widetilde{T})A, B) = (\phi A, B). \end{aligned}$ | $\begin{aligned} (\phi, A \vee B)_6 &= \left(\delta + \widetilde{T}, [\widetilde{A}, \widetilde{B}] + \left(A \circ B - \frac{1}{3}(A, B)E\right)^{\sim}\right)_6 \\ &= (\delta, [\widetilde{A}, \widetilde{B}])_4 + \left(T, A \circ B - \frac{1}{3}(A, B)E\right) \\ &= (\delta A, B) + (\widetilde{T}A, B) = ((\delta + \widetilde{T})A, B) = (\phi A, B). \end{aligned}$ | $\begin{aligned} (\phi, A \vee B)_6 &= \left(\delta + \widetilde{T}, [\widetilde{A}, \widetilde{B}] + \left(A \circ B - \frac{1}{3}(A, B)E\right)^{\sim}\right)_6 \\ &= (\delta, [\widetilde{A}, \widetilde{B}])_4 + \left(T, A \circ B - \frac{1}{3}(A, B)E\right) \\ &= (\delta A, B) + (\widetilde{T}A, B) = ((\delta + \widetilde{T})A, B) = (\phi A, B). \end{aligned}$ |

| Identifier       | Page | Conf. | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Rendered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Image |
|------------------|------|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 0902.0431_E00504 | 82   | 1.000 | $\begin{aligned} & \pm (\lambda_k - \lambda_l), \quad \pm (\lambda_k + \lambda_l), \quad 0 \leq k < l \leq 3, \\ & \pm \lambda_k \pm \frac{1}{2} (\mu_2 - \mu_3), \quad 0 \leq k \leq 3, \\ & \pm \frac{1}{2} (-\lambda_0 - \lambda_1 + \lambda_2 - \lambda_3) \pm \frac{1}{2} (\mu_3 - \mu_1), \\ & \pm \frac{1}{2} (\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3) \pm \frac{1}{2} (\mu_3 - \mu_1), \\ & \pm \frac{1}{2} (-\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3) \pm \frac{1}{2} (\mu_3 - \mu_1), \\ & \pm \frac{1}{2} (\lambda_0 - \lambda_1 + \lambda_2 + \lambda_3) \pm \frac{1}{2} (\mu_3 - \mu_1), \\ & \pm \frac{1}{2} (\lambda_0 - \lambda_1 + \lambda_2 - \lambda_3) \pm \frac{1}{2} (\mu_1 - \mu_2), \\ & \pm \frac{1}{2} (-\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3) \pm \frac{1}{2} (\mu_1 - \mu_2), \\ & \pm \frac{1}{2} (\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3) \pm \frac{1}{2} (\mu_1 - \mu_2), \\ & \pm \frac{1}{2} (-\lambda_0 - \lambda_1 + \lambda_2 + \lambda_3) \pm \frac{1}{2} (\mu_1 - \mu_2), \end{aligned}$ | $\begin{aligned} & \pm (\lambda_k - \lambda_l), \quad \pm (\lambda_k + \lambda_l), \quad 0 \leq k < l \leq 3, \\ & \pm \lambda_k \pm \frac{1}{2} (\mu_2 - \mu_3), \quad 0 \leq k \leq 3, \\ & \pm \frac{1}{2} (-\lambda_0 - \lambda_1 + \lambda_2 - \lambda_3) \pm \frac{1}{2} (\mu_3 - \mu_1), \\ & \pm \frac{1}{2} (\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3) \pm \frac{1}{2} (\mu_3 - \mu_1), \\ & \pm \frac{1}{2} (-\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3) \pm \frac{1}{2} (\mu_3 - \mu_1), \\ & \pm \frac{1}{2} (\lambda_0 - \lambda_1 + \lambda_2 + \lambda_3) \pm \frac{1}{2} (\mu_3 - \mu_1), \\ & \pm \frac{1}{2} (\lambda_0 - \lambda_1 + \lambda_2 - \lambda_3) \pm \frac{1}{2} (\mu_1 - \mu_2), \\ & \pm \frac{1}{2} (-\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3) \pm \frac{1}{2} (\mu_1 - \mu_2), \\ & \pm \frac{1}{2} (\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3) \pm \frac{1}{2} (\mu_1 - \mu_2), \\ & \pm \frac{1}{2} (-\lambda_0 - \lambda_1 + \lambda_2 + \lambda_3) \pm \frac{1}{2} (\mu_1 - \mu_2), \end{aligned}$ |       |
| 0902.0431_E00530 | 89   | 1.000 | $\alpha X = \left( \begin{array}{ccc} \xi_1 & 0 & 0 \\ 0 & \xi_2 & 0 \\ 0 & 0 & \xi_3 \end{array} \right), \quad \xi_i \in C.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | $\alpha X = \left( \begin{array}{ccc} \xi_1 & 0 & 0 \\ 0 & \xi_2 & 0 \\ 0 & 0 & \xi_3 \end{array} \right), \quad \xi_i \in C.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |       |
| 0902.0431_E00531 | 89   | 1.000 | $0 \neq x_1 = p + i q, \quad p, q \in \mathfrak{C}.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | $0 \neq x_1 = p + i q, \quad p, q \in \mathfrak{C}.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |       |
| 0902.0431_E00569 | 95   | 1.000 | $k \left( a + b e_2 \right) = \left( \begin{array}{cc} a' & b' \\ -\tau b' & \tau a' \end{array} \right), \quad a, b \in \boldsymbol{C}.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | $k \left( a + b e_2 \right) = \left( \begin{array}{cc} a' & b' \\ -\tau b' & \tau a' \end{array} \right), \quad a, b \in C.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |       |

| Identifier           | Page | Conf.   | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Rendered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Image                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|----------------------|------|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0902.0431_E00572     | 96   | ■ 1.000 | $\begin{aligned} & \kappa\left(M_{\{1\}}+i M_{\{2\}}\right) \&= \kappa\left(M_{\{1\}}\right)+i \kappa\left(M_{\{2\}}\right), \&\& \\ & M_{\{1\}}, M_{\{2\}} \in M(3, \boldsymbol{H}), \&\& \kappa\left(\boldsymbol{a}_{\{1\}}+i \boldsymbol{a}_{\{2\}}\right) \&= \kappa\left(\boldsymbol{a}_{\{1\}}\right)+i \kappa\left(\boldsymbol{a}_{\{2\}}\right), \&\& \\ & \boldsymbol{a}_{\{1\}}, \boldsymbol{a}_{\{2\}} \in \boldsymbol{H}^3 . \end{aligned}$                                                                                                                                                                                                                                                          | $\begin{aligned} k\left(M_1+i M_2\right) &= k\left(M_1\right)+i k\left(M_2\right), \quad M_1, M_2 \in M(3, \boldsymbol{H}), \\ k\left(\boldsymbol{a}_1+i \boldsymbol{a}_2\right) &= k\left(\boldsymbol{a}_1\right)+i k\left(\boldsymbol{a}_2\right), \quad \boldsymbol{a}_1, \boldsymbol{a}_2 \in \boldsymbol{H}^3 . \end{aligned}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | $\begin{aligned} k\left(M_1+i M_2\right) &= k\left(M_1\right)+i k\left(M_2\right), \quad M_1, M_2 \in M(3, \boldsymbol{H}), \\ k\left(\boldsymbol{a}_1+i \boldsymbol{a}_2\right) &= k\left(\boldsymbol{a}_1\right)+i k\left(\boldsymbol{a}_2\right), \quad \boldsymbol{a}_1, \boldsymbol{a}_2 \in \boldsymbol{H}^3 . \end{aligned}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 0902.0431_E00594     | 99   | ■ 1.000 | $\begin{aligned} & \varphi(p, A)(M+\boldsymbol{a})=k_J^{-1}\left(A\left(k_J(M)\right)^t A\right)+p \boldsymbol{a} k^{-1}\left(\tau^t A\right), \\ & M+\boldsymbol{a} \in\left(\mathfrak{J}_H\right)^C \oplus\left(\boldsymbol{H}^3\right)^C=\mathfrak{J}^C . \end{aligned}$                                                                                                                                                                                                                                                                                                                                                                                                                                      | $\begin{aligned} \varphi(p, A)(M+\boldsymbol{a}) &= k_J^{-1}\left(A\left(k_J(M)\right)^t A\right)+p \boldsymbol{a} k^{-1}\left(\tau^t A\right), \\ M+\boldsymbol{a} &\in\left(\mathfrak{J}_H\right)^C \oplus\left(\boldsymbol{H}^3\right)^C=\mathfrak{J}^C . \end{aligned}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | $\begin{aligned} \varphi(p, A)(M+\boldsymbol{a}) &= k_J^{-1}\left(A\left(k_J(M)\right)^t A\right)+p \boldsymbol{a} k^{-1}\left(\tau^t A\right), \\ M+\boldsymbol{a} &\in\left(\mathfrak{J}_H\right)^C \oplus\left(\boldsymbol{H}^3\right)^C=\mathfrak{J}^C . \end{aligned}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 0902.0431_E00640     | 106  | ■ 1.000 | $E_{\{6, \boldsymbol{C}\}} / F_{\{4, \boldsymbol{C}\}}} \simeq E I V_C .$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | $E_{6, C} / F_{4, C} \simeq E I V_C .$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | $E_{6, C} / F_{4, C} \simeq E I V_C .$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 0902.0431_E00652     | 108  | ■ 1.000 | $\begin{aligned} & (X+M) \times(Y+N) \&= \left(X \times Y-\frac{1}{2}\left(M^* N+N^* M\right)\right)-\frac{1}{2}(M Y+N X+\overline{M \times N}), \\ & \frac{1}{2}(M Y+N X+\overline{M \times N}) \&\& \frac{1}{2}(M Y+N X+\overline{M \times N}) \end{aligned}$                                                                                                                                                                                                                                                                                                                                                                                                                                                  | $\begin{aligned} (X+M) \times(Y+N) &= \left(X \times Y-\frac{1}{2}\left(M^* N+N^* M\right)\right)-\frac{1}{2}(M Y+N X+\overline{M \times N}) \\ \langle X+M, Y+N\rangle &= \langle X, Y\rangle+2\langle M, N\rangle \end{aligned}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | $\begin{aligned} (X+M) \times(Y+N) &= \left(X \times Y-\frac{1}{2}\left(M^* N+N^* M\right)\right)-\frac{1}{2}(M Y+N X+\overline{M \times N}), \\ \langle X+M, Y+N\rangle &= \langle X, Y\rangle+2\langle M, N\rangle \end{aligned}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 0902.0431_E00658     | 108  | ■ 1.000 | $\beta(X+M)=X+P M=\varphi(P, E, E)(X+M), \quad X+M \in\left(\mathfrak{J}_C\right)^C \oplus M(3, C)^C=\mathfrak{J}^C .$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | $\beta(X+M)=X+P M=\varphi(P, E, E)(X+M), \quad X+M \in\left(\mathfrak{J}_C\right)^C \oplus M(3, C)^C=\mathfrak{J}^C .$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | $\beta(X+M)=X+P M=\varphi(P, E, E)(X+M), \quad X+M \in\left(\mathfrak{J}_C\right)^C \oplus M(3, C)^C=\mathfrak{J}^C .$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 0902.0431_E00664     | 110  | ■ 1.000 | $\begin{aligned} & E_{\{6(6)\}} \&= \left\{\alpha \in \operatorname{Iso}_{\boldsymbol{R}}\left(\mathfrak{J}\left(3, \mathfrak{C}'\right)\right) \mid \det (\alpha X)=\det X\right\}, \\ & \operatorname{Iso}_{\left\{\boldsymbol{R}\right\}}\left(\mathfrak{J}\left(3, \mathfrak{C}'\right)\right) \mid \det (\alpha X)=\det X, \langle\alpha X, \alpha Y\rangle_{\gamma}=\langle X, Y\rangle_{\gamma}\}, \\ & \left(3, \mathfrak{C}'\right) \mid \det (\alpha X)=\det X, \langle\alpha X, \alpha Y\rangle_{\sigma}=\langle X, Y\rangle_{\sigma}\}, \\ & \operatorname{Iso}_{\left\{\boldsymbol{R}\right\}}\left(\mathfrak{J}\left(3, \mathfrak{C}'\right)\right) \mid \det (\alpha X)=\det X\} . \end{aligned}$ | $\begin{aligned} E_{6(6)} &= \left\{\alpha \in \operatorname{Iso}_{\boldsymbol{R}}\left(\mathfrak{J}\left(3, \mathfrak{C}'\right)\right) \mid \det (\alpha X)=\det X\right\}, \\ E_{6(2)} &= \left\{\alpha \in \operatorname{Iso}_C\left(\mathfrak{J}\left(3, \mathfrak{C}^C\right)\right) \mid \det (\alpha X)=\det X, \langle\alpha X, \alpha Y\rangle_{\gamma}=\langle X, Y\rangle_{\gamma}\right\}, \\ E_{6(-14)} &= \left\{\alpha \in \operatorname{Iso}_C\left(\mathfrak{J}\left(3, \mathfrak{C}^C\right)\right) \mid \det (\alpha X)=\det X, \langle\alpha X, \alpha Y\rangle_{\sigma}=\langle X, Y\rangle_{\sigma}\right\}, \\ E_{6(-26)} &= \left\{\alpha \in \operatorname{Iso}_{\boldsymbol{R}}\left(\mathfrak{J}\left(3, \mathfrak{C}\right)\right) \mid \det (\alpha X)=\det X\right\} . \end{aligned}$ | $\begin{aligned} E_{6(6)} &= \left\{\alpha \in \operatorname{Iso}_{\boldsymbol{R}}\left(\mathfrak{J}\left(3, \mathfrak{C}'\right)\right) \mid \det (\alpha X)=\det X\right\}, \\ E_{6(2)} &= \left\{\alpha \in \operatorname{Iso}_C\left(\mathfrak{J}\left(3, \mathfrak{C}^C\right)\right) \mid \det (\alpha X)=\det X, \langle\alpha X, \alpha Y\rangle_{\gamma}=\langle X, Y\rangle_{\gamma}\right\}, \\ E_{6(-14)} &= \left\{\alpha \in \operatorname{Iso}_C\left(\mathfrak{J}\left(3, \mathfrak{C}^C\right)\right) \mid \det (\alpha X)=\det X, \langle\alpha X, \alpha Y\rangle_{\sigma}=\langle X, Y\rangle_{\sigma}\right\}, \\ E_{6(-26)} &= \left\{\alpha \in \operatorname{Iso}_{\boldsymbol{R}}\left(\mathfrak{J}\left(3, \mathfrak{C}\right)\right) \mid \det (\alpha X)=\det X\right\} . \end{aligned}$ |
| 0902.0431_E00682 (2) | 113  | ■ 1.000 | $\{\alpha P, \alpha Q\}=\{P, Q\}, \quad P, Q \in \mathfrak{P}^C .$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | $\{\alpha P, \alpha Q\}=\{P, Q\}, \quad P, Q \in \mathfrak{P}^C .$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | $\{\alpha P, \alpha Q\}=\{P, Q\}, \quad P, Q \in \mathfrak{P}^C .$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 0902.0431_E00692     | 114  | ■ 1.000 | $f_{\{r\}}(X, Y, \xi, \eta)=\left(X, r Y, r^2 \xi, r^{-1} \eta\right),$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | $f_r(X, Y, \xi, \eta)=\left(X, r Y, r^2 \xi, r^{-1} \eta\right),$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | $f_r(X, Y, \xi, \eta)=\left(X, r Y, r^2 \xi, r^{-1} \eta\right),$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 0902.0431_E00712     | 118  | ■ 1.000 | $\left\{\alpha^{-1}\right\}=\lambda \alpha \lambda^{-1} .$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | ${}^t \alpha^{-1}=\lambda \alpha \lambda^{-1} .$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | ${}^t \alpha^{-1}=\lambda \alpha \lambda^{-1} .$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

| Identifier       | Page | Conf.   | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Rendered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Image                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|------------------|------|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0902.0431_E00743 | 122  | ■ 1.000 | $B_{\{7\}}\!\left(\!\!\Phi_{\{0\}},\,\,\Phi_{\{0\}}\!\!\right)\!=\!\operatorname{tr}\!\left(\!\!\left(\!\!\operatorname{ad}\!\!\Phi_{\{0\}}\!\!\right)^{\!2}\!\!\right)\!=\!\frac{4}{9}\, \times 27 \times 2\!=\!24\,\,.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | $B_7(\Phi_0,\Phi_0)=\operatorname{tr}\left((\operatorname{ad}\Phi_0)^2\right)=\frac{4}{9}\times 27\times 2=24.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | $B_7(\Phi_0,\Phi_0)=\operatorname{tr}((\operatorname{ad}\Phi_0)^2)=\frac{4}{9}\times 27\times 2=24.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 0902.0431_E00747 | 124  | ■ 1.000 | $\begin{aligned} &\pm \frac{1}{2}(-\lambda_0+\lambda_1+\lambda_2+\lambda_3)\pm\left(\frac{1}{2}\mu_2-\frac{2}{3}\nu\right), \\ &\pm \frac{1}{2}(\lambda_0-\lambda_1+\lambda_2+\lambda_3)\pm\left(\frac{1}{2}\mu_2-\frac{2}{3}\nu\right), \\ &\pm \frac{1}{2}(\lambda_0-\lambda_1+\lambda_2-\lambda_3)\pm\left(\frac{1}{2}\mu_3-\frac{2}{3}\nu\right), \\ &\pm \frac{1}{2}(-\lambda_0+\lambda_1+\lambda_2-\lambda_3)\pm\left(\frac{1}{2}\mu_3-\frac{2}{3}\nu\right), \\ &\pm \frac{1}{2}(\lambda_0+\lambda_1+\lambda_2+\lambda_3)\pm\left(\frac{1}{2}\mu_3-\frac{2}{3}\nu\right), \\ &\pm \frac{1}{2}(-\lambda_0-\lambda_1+\lambda_2+\lambda_3)\pm\left(\frac{1}{2}\mu_3-\frac{2}{3}\nu\right) \end{aligned}$ | $\begin{aligned} &\pm \frac{1}{2}(-\lambda_0+\lambda_1+\lambda_2+\lambda_3)\pm\left(\frac{1}{2}\mu_2-\frac{2}{3}\nu\right), \\ &\pm \frac{1}{2}(\lambda_0-\lambda_1+\lambda_2+\lambda_3)\pm\left(\frac{1}{2}\mu_2-\frac{2}{3}\nu\right), \\ &\pm \frac{1}{2}(\lambda_0-\lambda_1+\lambda_2-\lambda_3)\pm\left(\frac{1}{2}\mu_3-\frac{2}{3}\nu\right), \\ &\pm \frac{1}{2}(-\lambda_0+\lambda_1+\lambda_2-\lambda_3)\pm\left(\frac{1}{2}\mu_3-\frac{2}{3}\nu\right), \\ &\pm \frac{1}{2}(\lambda_0+\lambda_1+\lambda_2+\lambda_3)\pm\left(\frac{1}{2}\mu_3-\frac{2}{3}\nu\right), \\ &\pm \frac{1}{2}(-\lambda_0-\lambda_1+\lambda_2+\lambda_3)\pm\left(\frac{1}{2}\mu_3-\frac{2}{3}\nu\right) \end{aligned}$ | $\begin{aligned} &\pm \frac{1}{2}(-\lambda_0+\lambda_1+\lambda_2+\lambda_3)\pm\left(\frac{1}{2}\mu_2-\frac{2}{3}\nu\right), \\ &\pm \frac{1}{2}(\lambda_0-\lambda_1+\lambda_2+\lambda_3)\pm\left(\frac{1}{2}\mu_2-\frac{2}{3}\nu\right), \\ &\pm \frac{1}{2}(\lambda_0-\lambda_1+\lambda_2-\lambda_3)\pm\left(\frac{1}{2}\mu_3-\frac{2}{3}\nu\right), \\ &\pm \frac{1}{2}(-\lambda_0+\lambda_1+\lambda_2-\lambda_3)\pm\left(\frac{1}{2}\mu_3-\frac{2}{3}\nu\right), \\ &\pm \frac{1}{2}(\lambda_0+\lambda_1+\lambda_2+\lambda_3)\pm\left(\frac{1}{2}\mu_3-\frac{2}{3}\nu\right), \\ &\pm \frac{1}{2}(-\lambda_0-\lambda_1+\lambda_2+\lambda_3)\pm\left(\frac{1}{2}\mu_3-\frac{2}{3}\nu\right) \end{aligned}$ |
| 0902.0431_E00799 | 133  | ■ 1.000 | $\xi=\cos r_{\{1\}}\cos r_{\{2\}}\cos r_{\{3\}}\,\,.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | $\xi=\cos r_1\cos r_2\cos r_3.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | $\xi=\cos r_1\cos r_2\cos r_3.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 0902.0431_E00829 | 138  | ■ 1.000 | $\begin{aligned} &\kappa\left(\begin{array}{c} X\\ Y\\ \xi\\ \eta \end{array}\right)=\left(\begin{array}{c} -\kappa_1X\\ \kappa_1Y\\ -\xi\\ \eta \end{array}\right),\quad \kappa_1X=(E_1,X)-4E_1\times(E_1\times X), \\ &\mu\left(\begin{array}{c} X\\ Y\\ \xi\\ \eta \end{array}\right)=\left(\begin{array}{c} 2E_1\times Y+\eta E_1\\ 2E_1\times X+\xi E_1\\ (E_1,Y)\\ (E_1,X) \end{array}\right). \end{aligned}$                                                                                                                                                                                                                                                                                          | $\begin{aligned} &\kappa\left(\begin{array}{c} X\\ Y\\ \xi\\ \eta \end{array}\right)=\left(\begin{array}{c} -\kappa_1X\\ \kappa_1Y\\ -\xi\\ \eta \end{array}\right),\quad \kappa_1X=(E_1,X)-4E_1\times(E_1\times X), \\ &\mu\left(\begin{array}{c} X\\ Y\\ \xi\\ \eta \end{array}\right)=\left(\begin{array}{c} 2E_1\times Y+\eta E_1\\ 2E_1\times X+\xi E_1\\ (E_1,Y)\\ (E_1,X) \end{array}\right). \end{aligned}$                                                                                                                                                                                                                                                                                          | $\begin{aligned} &\kappa\left(\begin{array}{c} X\\ Y\\ \xi\\ \eta \end{array}\right)=\left(\begin{array}{c} -\kappa_1X\\ \kappa_1Y\\ -\xi\\ \eta \end{array}\right),\quad \kappa_1X=(E_1,X)-4E_1\times(E_1\times X), \\ &\mu\left(\begin{array}{c} X\\ Y\\ \xi\\ \eta \end{array}\right)=\left(\begin{array}{c} 2E_1\times Y+\eta E_1\\ 2E_1\times X+\xi E_1\\ (E_1,Y)\\ (E_1,X) \end{array}\right). \end{aligned}$                                                                                                                                                                                                                                                                                          |
| 0902.0431_E00840 | 140  | ■ 1.000 | $\phi(\nu)\left(\begin{array}{ccc} \xi_1 & x_3 & \bar{x}_2 \\ \bar{x}_3 & \xi_2 & x_1 \\ x_2 & \bar{x}_1 & \xi_3 \end{array}\right)=\frac{\nu}{3}\left(\begin{array}{ccc} 4\xi_1 & x_3 & \bar{x}_2 \\ \bar{x}_3 & -2\xi_2 & -2x_1 \\ x_2 & -2\bar{x}_1 & -2\xi_3 \end{array}\right)$                                                                                                                                                                                                                                                                                                                                                                                                                         | $\phi(\nu)\left(\begin{array}{ccc} \xi_1 & x_3 & \bar{x}_2 \\ \bar{x}_3 & \xi_2 & x_1 \\ x_2 & \bar{x}_1 & \xi_3 \end{array}\right)=\frac{\nu}{3}\left(\begin{array}{ccc} 4\xi_1 & x_3 & \bar{x}_2 \\ \bar{x}_3 & -2\xi_2 & -2x_1 \\ x_2 & -2\bar{x}_1 & -2\xi_3 \end{array}\right)$                                                                                                                                                                                                                                                                                                                                                                                                                         | $\phi(\nu)\left(\begin{array}{ccc} \xi_1 & x_3 & \bar{x}_2 \\ \bar{x}_3 & \xi_2 & x_1 \\ x_2 & \bar{x}_1 & \xi_3 \end{array}\right)=\frac{\nu}{3}\left(\begin{array}{ccc} 4\xi_1 & x_3 & \bar{x}_2 \\ \bar{x}_3 & -2\xi_2 & -2x_1 \\ x_2 & -2\bar{x}_1 & -2\xi_3 \end{array}\right)$                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 0902.0431_E00841 | 140  | ■ 1.000 | $\left(\mathfrak{e}_{\{7\}}\right)^{\!\sigma}\!\cong\mathfrak{a}_{\{1\}}\oplus\left(\mathfrak{e}_{\{7\}}\right)^{\!\kappa,\mu}\!\left(\mathfrak{kappa},\,\mu\right)\,\,.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | $\left(\mathfrak{e}_7\right)^{\sigma}\cong\mathfrak{a}_1\oplus\left(\mathfrak{e}_7\right)^{\kappa,\mu}.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | $\left(\mathfrak{e}_7\right)^{\sigma}\cong\mathfrak{a}_1\oplus\left(\mathfrak{e}_7\right)^{\kappa,\mu}.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

| Identifier            | Page | Conf.   | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                | Rendered                                                                                                                                                                                                                                                                                                                                                                     | Image                                                                                                                                                                                                                                                                                                                                                                        |
|-----------------------|------|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0902.0431_E00852      | 141  | ■ 1.000 | $\begin{aligned} V^{\{1\}} &= \left\{ P \in \mathfrak{P} \mid \kappa P = P, \mu\tau\lambda P = P, P \times (0, E_1, 0, 1, 0) = 0 \right\} \\ &= \left\{ \left( \begin{pmatrix} 0 & 0 & 0 \\ 0 & \xi & x \\ 0 & \bar{x} & -\tau\xi \end{pmatrix}, \begin{pmatrix} \eta & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, 0, \tau\eta \right) \mid x \in \mathfrak{C}, \xi \in C, \eta \in i\mathbf{R} \right\} \end{aligned}$ | $V^{11} = \{P \in \mathfrak{P}^C \mid \kappa P = P, \mu\tau\lambda P = P, P \times (0, E_1, 0, 1, 0) = 0\}$ $= \left\{ \left( \begin{pmatrix} 0 & 0 & 0 \\ 0 & \xi & x \\ 0 & \bar{x} & -\tau\xi \end{pmatrix}, \begin{pmatrix} \eta & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, 0, \tau\eta \right) \mid x \in \mathfrak{C}, \xi \in C, \eta \in i\mathbf{R} \right\}$ | $V^{11} = \{P \in \mathfrak{P}^C \mid \kappa P = P, \mu\tau\lambda P = P, P \times (0, E_1, 0, 1, 0) = 0\}$ $= \left\{ \left( \begin{pmatrix} 0 & 0 & 0 \\ 0 & \xi & x \\ 0 & \bar{x} & -\tau\xi \end{pmatrix}, \begin{pmatrix} \eta & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, 0, \tau\eta \right) \mid x \in \mathfrak{C}, \xi \in C, \eta \in i\mathbf{R} \right\}$ |
| 0902.0431_E00872      | 145  | ■ 1.000 | $z(\operatorname{Spin}(12)) = \{1, -1, \sigma, -\sigma\} \cong \{1, -1\} \times \{1, \sigma\} \cong \mathbf{Z}_2 \times \mathbf{Z}_2.$                                                                                                                                                                                                                                                                                      | $z(\operatorname{Spin}(12)) = \{1, -1, \sigma - \sigma\} \cong \{1, -1\} \times \{1, \sigma\} \cong \mathbf{Z}_2 \times \mathbf{Z}_2.$                                                                                                                                                                                                                                       | $z(\operatorname{Spin}(12)) = \{1, -1, \sigma - \sigma\} \cong \{1, -1\} \times \{1, \sigma\} \cong \mathbf{Z}_2 \times \mathbf{Z}_2.$                                                                                                                                                                                                                                       |
| 0902.0431_E00909      | 151  | ■ 1.000 | $w(X, Y, \xi, \eta) = (wX, wY, \xi, \eta).$                                                                                                                                                                                                                                                                                                                                                                                 | $w(X, Y, \xi, \eta) = (wX, wY, \xi, \eta).$                                                                                                                                                                                                                                                                                                                                  | $w(X, Y, \xi, \eta) = (wX, wY, \xi, \eta).$                                                                                                                                                                                                                                                                                                                                  |
| 0902.0431_E00923      | 153  | ■ 1.000 | $\begin{aligned} h(a, b) &= \frac{a+b}{2} + i \frac{(b-a)e_1}{2}, \quad a, b \in \mathbf{C}, \\ h(a + be_1) &= a + bi, \quad a, b \in \mathbf{R}. \end{aligned}$                                                                                                                                                                                                                                                            | $h(a, b) = \frac{a+b}{2} + i \frac{(b-a)e_1}{2}, \quad a, b \in \mathbf{C},$ $h(a + be_1) = a + bi, \quad a, b \in \mathbf{R}.$                                                                                                                                                                                                                                              | $h(a, b) = \frac{a+b}{2} + i \frac{(b-a)e_1}{2}, \quad a, b \in \mathbf{C},$ $h(a + be_1) = a + bi, \quad a, b \in \mathbf{R}.$                                                                                                                                                                                                                                              |
| 0902.0431_E00957      | 159  | ■ 1.000 | $\psi(Q, A)P = f^{-1}((Q, A)(fP)), \quad P \in \mathfrak{P}^C.$                                                                                                                                                                                                                                                                                                                                                             | $\psi(Q, A)P = f^{-1}((Q, A)(fP)), \quad P \in \mathfrak{P}^C.$                                                                                                                                                                                                                                                                                                              | $\psi(Q, A)P = f^{-1}((Q, A)(fP)), \quad P \in \mathfrak{P}^C.$                                                                                                                                                                                                                                                                                                              |
| 0902.0431_E00988      | 164  | ■ 1.000 | $([R, R_1], R_2)_8 + (R_1, [R, R_2])_8 = 0, \quad R, R_i \in \mathfrak{e}_8^C.$                                                                                                                                                                                                                                                                                                                                             | $([R, R_1], R_2)_8 + (R_1, [R, R_2])_8 = 0, \quad R, R_i \in \mathfrak{e}_8^C.$                                                                                                                                                                                                                                                                                              | $([R, R_1], R_2)_8 + (R_1, [R, R_2])_8 = 0, \quad R, R_i \in \mathfrak{e}_8^C.$                                                                                                                                                                                                                                                                                              |
| 0902.0431_E01003      | 167  | ■ 1.000 | $\langle R_1, R_2 \rangle = -\frac{1}{15} B_8 \left( \tau \tilde{\lambda} R_1, R_2 \right).$                                                                                                                                                                                                                                                                                                                                | $\langle R_1, R_2 \rangle = -\frac{1}{15} B_8 \left( \tau \tilde{\lambda} R_1, R_2 \right).$                                                                                                                                                                                                                                                                                 | $\langle R_1, R_2 \rangle = -\frac{1}{15} B_8 (\tau \tilde{\lambda} R_1, R_2).$                                                                                                                                                                                                                                                                                              |
| 0902.0431_E01013      | 168  | ■ 1.000 | $E_8^C \simeq E_8 \times \mathbf{R}^{248}.$                                                                                                                                                                                                                                                                                                                                                                                 | $E_8^C \simeq E_8 \times \mathbf{R}^{248}.$                                                                                                                                                                                                                                                                                                                                  | $E_8^C \simeq E_8 \times \mathbf{R}^{248}.$                                                                                                                                                                                                                                                                                                                                  |
| 0902.0431_E01050 (ii) | 179  | ■ 1.000 | $\left\{ \beta_2 P, \beta_2 Q \right\} = \{P, Q\}.$                                                                                                                                                                                                                                                                                                                                                                         | $\{\beta_2 P, \beta_2 Q\} = \{P, Q\}.$                                                                                                                                                                                                                                                                                                                                       | $\{\beta_2 P, \beta_2 Q\} = \{P, Q\}.$                                                                                                                                                                                                                                                                                                                                       |
| 0902.0431_E01053      | 179  | ■ 1.000 | $\beta_1 \Phi = \beta \Phi \beta^{-1}.$                                                                                                                                                                                                                                                                                                                                                                                     | $\beta_1 \Phi = \beta \Phi \beta^{-1}.$                                                                                                                                                                                                                                                                                                                                      | $\beta_1 \Phi = \beta \Phi \beta^{-1}.$                                                                                                                                                                                                                                                                                                                                      |
| 0902.0431_E01144      | 193  | ■ 1.000 | $(\mathbf{u} \times \mathbf{v})\mathbf{x} = *(\mathbf{v} \wedge *(\mathbf{u} \wedge \mathbf{x})) + (-1)^{n-k} \frac{n-k}{n} (\mathbf{u}, \mathbf{v})\mathbf{x}, \quad \mathbf{x} \in C^n.$                                                                                                                                                                                                                                  | $(\mathbf{u} \times \mathbf{v})\mathbf{x} = *(\mathbf{v} \wedge *(\mathbf{u} \wedge \mathbf{x})) + (-1)^{n-k} \frac{n-k}{n} (\mathbf{u}, \mathbf{v})\mathbf{x}, \quad \mathbf{x} \in C^n.$                                                                                                                                                                                   | $(\mathbf{u} \times \mathbf{v})\mathbf{x} = *(\mathbf{v} \wedge *(\mathbf{u} \wedge \mathbf{x})) + (-1)^{n-k} \frac{n-k}{n} (\mathbf{u}, \mathbf{v})\mathbf{x}, \quad \mathbf{x} \in C^n.$                                                                                                                                                                                   |
| 0902.0431_E01145      | 193  | ■ 1.000 | $\mathfrak{e}_8^C = \mathfrak{sl}(9, C) \oplus \Lambda^3(C^9) \oplus \Lambda^3(C^9),$                                                                                                                                                                                                                                                                                                                                       | $\mathfrak{e}_8^C = \mathfrak{sl}(9, C) \oplus \Lambda^3(C^9) \oplus \Lambda^3(C^9),$                                                                                                                                                                                                                                                                                        | $\mathfrak{e}_8^C = \mathfrak{sl}(9, C) \oplus \Lambda^3(C^9) \oplus \Lambda^3(C^9),$                                                                                                                                                                                                                                                                                        |
| 0902.0431_E01147      | 193  | ■ 1.000 | $\begin{aligned} D &= [D_1, D_2] + \mathbf{u}_1 \times \mathbf{v}_2 - \mathbf{u}_2 \times \mathbf{v}_1 \\ \mathbf{u} &= D_1 \mathbf{u}_2 - D_2 \mathbf{u}_1 + *(\mathbf{v}_1 \wedge \mathbf{v}_2) \\ \mathbf{v} &= -{}^t D_1 \mathbf{v}_2 + {}^t D_2 \mathbf{v}_1 - *(\mathbf{u}_1 \wedge \mathbf{u}_2), \end{aligned}$                                                                                                     | $\begin{cases} D = [D_1, D_2] + \mathbf{u}_1 \times \mathbf{v}_2 - \mathbf{u}_2 \times \mathbf{v}_1 \\ \mathbf{u} = D_1 \mathbf{u}_2 - D_2 \mathbf{u}_1 + *(\mathbf{v}_1 \wedge \mathbf{v}_2) \\ \mathbf{v} = -{}^t D_1 \mathbf{v}_2 + {}^t D_2 \mathbf{v}_1 - *(\mathbf{u}_1 \wedge \mathbf{u}_2), \end{cases}$                                                             | $\begin{cases} D = [D_1, D_2] + \mathbf{u}_1 \times \mathbf{v}_2 - \mathbf{u}_2 \times \mathbf{v}_1 \\ \mathbf{u} = D_1 \mathbf{u}_2 - D_2 \mathbf{u}_1 + *(\mathbf{v}_1 \wedge \mathbf{v}_2) \\ \mathbf{v} = -{}^t D_1 \mathbf{v}_2 + {}^t D_2 \mathbf{v}_1 - *(\mathbf{u}_1 \wedge \mathbf{u}_2), \end{cases}$                                                             |

| Identifier       | Page | Conf.   | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Rendered                                                                                                                                                                                                                                                                                                                                                                                                                   | Image                                                                                                                                                                                                                                                                                                                                                                                         |
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| 0902.0431_E01160 | 196  | ■ 1.000 | $\begin{aligned} &0 \neq \operatorname{ad}(T) \operatorname{ad}(S_{37}) \operatorname{ad}(S_{27}) \operatorname{ad}(S_{17}) \operatorname{ad}(S_{36}) \operatorname{ad}(S_{25}) \operatorname{ad}(S_{14}) R \\ &= (-E_{34}, 0, 0) \in \mathfrak{sl}(9, C) \cap \mathfrak{a}. \end{aligned}$                                                                                                                                                                                       | $0 \neq \operatorname{ad}(T) \operatorname{ad}(S_{37}) \operatorname{ad}(S_{27}) \operatorname{ad}(S_{17}) \operatorname{ad}(S_{36}) \operatorname{ad}(S_{25}) \operatorname{ad}(S_{14}) R \\ = (-E_{34}, 0, 0) \in \mathfrak{sl}(9, C) \cap \mathfrak{a}.$                                                                                                                                                                | $0 \neq \operatorname{ad}(T) \operatorname{ad}(S_{37}) \operatorname{ad}(S_{27}) \operatorname{ad}(S_{17}) \operatorname{ad}(S_{36}) \operatorname{ad}(S_{25}) \operatorname{ad}(S_{14}) R \\ = (-E_{34}, 0, 0) \in \mathfrak{sl}(9, C) \cap \mathfrak{a}.$                                                                                                                                   |
| 0902.0431_E01165 | 196  | ■ 1.000 | $\left\langle R_1, R_2 \right\rangle = -B_8 \left( R_1, \tau \widetilde{\lambda} R_2 \right).$                                                                                                                                                                                                                                                                                                                                                                                    | $\langle R_1, R_2 \rangle = -B_8 \left( R_1, \tau \widetilde{\lambda} R_2 \right).$                                                                                                                                                                                                                                                                                                                                        | $\langle R_1, R_2 \rangle = -B_8(R_1, \tau \widetilde{\lambda} R_2).$                                                                                                                                                                                                                                                                                                                         |
| 0902.0431_E01168 | 197  | ■ 1.000 | $w_3(D, \boldsymbol{u}, \boldsymbol{v}) = \left( D, \omega \boldsymbol{u}, \omega^2 \boldsymbol{v} \right),$                                                                                                                                                                                                                                                                                                                                                                      | $w_3(D, \boldsymbol{u}, \boldsymbol{v}) = \left( D, \omega \boldsymbol{u}, \omega^2 \boldsymbol{v} \right),$                                                                                                                                                                                                                                                                                                               | $w_3(D, \boldsymbol{u}, \boldsymbol{v}) = (D, \omega \boldsymbol{u}, \omega^2 \boldsymbol{v}),$                                                                                                                                                                                                                                                                                               |
| 0902.0431_E01180 | 199  | ■ 1.000 | $(*(\boldsymbol{x} \wedge \boldsymbol{y}) \wedge \boldsymbol{z}), \boldsymbol{v}) = (\boldsymbol{x} \wedge \boldsymbol{y}, \boldsymbol{z} \wedge \boldsymbol{v}) = (\boldsymbol{x}, \boldsymbol{z})(\boldsymbol{y}, \boldsymbol{v}) - (\boldsymbol{y}, \boldsymbol{z})(\boldsymbol{x}, \boldsymbol{v}).$                                                                                                                                                                          | $(*(\boldsymbol{x} \wedge \boldsymbol{y}) \wedge \boldsymbol{z}), \boldsymbol{v}) = (\boldsymbol{x} \wedge \boldsymbol{y}, \boldsymbol{z} \wedge \boldsymbol{v}) = (\boldsymbol{x}, \boldsymbol{z})(\boldsymbol{y}, \boldsymbol{v}) - (\boldsymbol{y}, \boldsymbol{z})(\boldsymbol{x}, \boldsymbol{v}).$                                                                                                                   | $(*(\boldsymbol{x} \wedge \boldsymbol{y}) \wedge \boldsymbol{z}), \boldsymbol{v}) = (\boldsymbol{x} \wedge \boldsymbol{y}, \boldsymbol{z} \wedge \boldsymbol{v}) = (\boldsymbol{x}, \boldsymbol{z})(\boldsymbol{y}, \boldsymbol{v}) - (\boldsymbol{y}, \boldsymbol{z})(\boldsymbol{x}, \boldsymbol{v}).$                                                                                      |
| 0902.0431_E01201 | 202  | ■ 1.000 | $\begin{aligned} &\& B_8 \left( R_1, R_2 \right) = 60 \left( \operatorname{tr}(C_1 C_2) + \operatorname{tr}(D_1 D_2) - (\boldsymbol{x}_1, \boldsymbol{w}_2)(\boldsymbol{a}_1, \boldsymbol{d}_2) - (\boldsymbol{x}_2, \boldsymbol{w}_1)(\boldsymbol{a}_2, \boldsymbol{d}_1) \right. \\ &\quad \left. - (\boldsymbol{y}_1, \boldsymbol{z}_2)(\boldsymbol{b}_1, \boldsymbol{c}_2) - (\boldsymbol{y}_2, \boldsymbol{z}_1)(\boldsymbol{b}_2, \boldsymbol{c}_1) \right), \end{aligned}$ | $B_8(R_1, R_2) = 60 \left( \operatorname{tr}(C_1 C_2) + \operatorname{tr}(D_1 D_2) - (\boldsymbol{x}_1, \boldsymbol{w}_2)(\boldsymbol{a}_1, \boldsymbol{d}_2) - (\boldsymbol{x}_2, \boldsymbol{w}_1)(\boldsymbol{a}_2, \boldsymbol{d}_1) \right. \\ \left. - (\boldsymbol{y}_1, \boldsymbol{z}_2)(\boldsymbol{b}_1, \boldsymbol{c}_2) - (\boldsymbol{y}_2, \boldsymbol{z}_1)(\boldsymbol{b}_2, \boldsymbol{c}_1) \right),$ | $B_8(R_1, R_2) = 60(\operatorname{tr}(C_1 C_2) + \operatorname{tr}(D_1 D_2) - (\boldsymbol{x}_1, \boldsymbol{w}_2)(\boldsymbol{a}_1, \boldsymbol{d}_2) - (\boldsymbol{x}_2, \boldsymbol{w}_1)(\boldsymbol{a}_2, \boldsymbol{d}_1) \\ - (\boldsymbol{y}_1, \boldsymbol{z}_2)(\boldsymbol{b}_1, \boldsymbol{c}_2) - (\boldsymbol{y}_2, \boldsymbol{z}_1)(\boldsymbol{b}_2, \boldsymbol{c}_1)),$ |
| 0902.0431_E00001 | 5    | ■ 1.000 | $\tau(u + i v) = u - i v.$                                                                                                                                                                                                                                                                                                                                                                                                                                                        | $\tau(u + i v) = u - i v.$                                                                                                                                                                                                                                                                                                                                                                                                 | $\tau(u + i v) = u - i v.$                                                                                                                                                                                                                                                                                                                                                                    |
| 0902.0431_E00008 | 7    | ■ 1.000 | $G_2 = \{ \alpha \in \operatorname{Iso}_{\boldsymbol{R}}(\mathfrak{C}) \mid \alpha(xy) = (\alpha x)(\alpha y) \}.$                                                                                                                                                                                                                                                                                                                                                                | $G_2 = \{ \alpha \in \operatorname{Iso}_{\boldsymbol{R}}(\mathfrak{C}) \mid \alpha(xy) = (\alpha x)(\alpha y) \}.$                                                                                                                                                                                                                                                                                                         | $G_2 = \{ \alpha \in \operatorname{Iso}_{\boldsymbol{R}}(\mathfrak{C}) \mid \alpha(xy) = (\alpha x)(\alpha y) \}.$                                                                                                                                                                                                                                                                            |
| 0902.0431_E00009 | 7    | ■ 1.000 | $(\alpha x, \alpha y) = (x, y), \quad x, y \in \mathfrak{C}.$                                                                                                                                                                                                                                                                                                                                                                                                                     | $(\alpha x, \alpha y) = (x, y), \quad x, y \in \mathfrak{C}.$                                                                                                                                                                                                                                                                                                                                                              | $(\alpha x, \alpha y) = (x, y), \quad x, y \in \mathfrak{C}.$                                                                                                                                                                                                                                                                                                                                 |
| 0902.0431_E00010 | 7    | ■ 1.000 | $\overline{\alpha x} = \alpha \bar{x}.$                                                                                                                                                                                                                                                                                                                                                                                                                                           | $\overline{\alpha x} = \alpha \bar{x}.$                                                                                                                                                                                                                                                                                                                                                                                    | $\overline{\alpha x} = \alpha \bar{x}.$                                                                                                                                                                                                                                                                                                                                                       |
| 0902.0431_E00013 | 8    | ■ 1.000 | $O(8) = O(\mathfrak{C}) = \{ \alpha \in \operatorname{Iso}_{\boldsymbol{R}}(\mathfrak{C}) \mid (\alpha x, \alpha y) = (x, y) \}.$                                                                                                                                                                                                                                                                                                                                                 | $O(8) = O(\mathfrak{C}) = \{ \alpha \in \operatorname{Iso}_{\boldsymbol{R}}(\mathfrak{C}) \mid (\alpha x, \alpha y) = (x, y) \}.$                                                                                                                                                                                                                                                                                          | $O(8) = O(\mathfrak{C}) = \{ \alpha \in \operatorname{Iso}_{\boldsymbol{R}}(\mathfrak{C}) \mid (\alpha x, \alpha y) = (x, y) \}.$                                                                                                                                                                                                                                                             |
| 0902.0431_E00014 | 8    | ■ 1.000 | $\mathfrak{D}_4 = \mathfrak{so}(8) = \mathfrak{so}(\mathfrak{C}) = \{ D \in \operatorname{Hom}_{\boldsymbol{R}}(\mathfrak{C}) \mid (Dx, y) + (x, Dy) = 0 \}$                                                                                                                                                                                                                                                                                                                      | $\mathfrak{D}_4 = \mathfrak{so}(8) = \mathfrak{so}(\mathfrak{C}) = \{ D \in \operatorname{Hom}_{\boldsymbol{R}}(\mathfrak{C}) \mid (Dx, y) + (x, Dy) = 0 \}$                                                                                                                                                                                                                                                               | $\mathfrak{D}_4 = \mathfrak{so}(8) = \mathfrak{so}(\mathfrak{C}) = \{ D \in \operatorname{Hom}_{\boldsymbol{R}}(\mathfrak{C}) \mid (Dx, y) + (x, Dy) = 0 \}$                                                                                                                                                                                                                                  |
| 0902.0431_E00015 | 8    | ■ 1.000 | $G_{ij} e_j = e_i, \quad G_{ij} e_i = -e_j, \quad G_{ij} e_k = 0, \quad k \neq i, j.$                                                                                                                                                                                                                                                                                                                                                                                             | $G_{ij} e_j = e_i, \quad G_{ij} e_i = -e_j, \quad G_{ij} e_k = 0, \quad k \neq i, j.$                                                                                                                                                                                                                                                                                                                                      | $G_{ij} e_j = e_i, \quad G_{ij} e_i = -e_j, \quad G_{ij} e_k = 0, \quad k \neq i, j.$                                                                                                                                                                                                                                                                                                         |
| 0902.0431_E00016 | 8    | ■ 1.000 | $F_{ij} x = \frac{1}{2} e_i (\bar{e}_j x), \quad x \in \mathfrak{C}.$                                                                                                                                                                                                                                                                                                                                                                                                             | $F_{ij} x = \frac{1}{2} e_i (\bar{e}_j x), \quad x \in \mathfrak{C}.$                                                                                                                                                                                                                                                                                                                                                      | $F_{ij} x = \frac{1}{2} e_i (\bar{e}_j x), \quad x \in \mathfrak{C}.$                                                                                                                                                                                                                                                                                                                         |
| 0902.0431_E00020 | 9    | ■ 1.000 | $\kappa, \pi, \nu \in \operatorname{Aut}(\mathfrak{D}_4).$                                                                                                                                                                                                                                                                                                                                                                                                                        | $\kappa, \pi, \nu \in \operatorname{Aut}(\mathfrak{D}_4).$                                                                                                                                                                                                                                                                                                                                                                 | $\kappa, \pi, \nu \in \operatorname{Aut}(\mathfrak{D}_4).$                                                                                                                                                                                                                                                                                                                                    |
| 0902.0431_E00022 | 9    | ■ 1.000 | $[\pi G_{ij}, \pi G_{kl}] = \pi [G_{ij}, G_{kl}],$                                                                                                                                                                                                                                                                                                                                                                                                                                | $[\pi G_{ij}, \pi G_{kl}] = \pi [G_{ij}, G_{kl}],$                                                                                                                                                                                                                                                                                                                                                                         | $[\pi G_{ij}, \pi G_{kl}] = \pi [G_{ij}, G_{kl}],$                                                                                                                                                                                                                                                                                                                                            |

| Identifier             | Page | Conf.   | LaTeX source                                                                                                                                                                                                                                | Rendered                                                                                                                                                                               | Image                                                                                                                                                                                  |
|------------------------|------|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0902.0431_E00027 (1)   | 10   | ■ 1.000 | $L_{\{a\}}, R_{\{a\}}, T_{\{a\}} \in \mathfrak{D}_{\{4\}}.$                                                                                                                                                                                 | $L_a, R_a, T_a \in \mathfrak{D}_4.$                                                                                                                                                    | $L_a, R_a, T_a \in \mathfrak{D}_4.$                                                                                                                                                    |
| 0902.0431_E00028 (2)   | 10   | ■ 1.000 | $\kappa L_{\{a\}}=-R_{\{a\}}, \quad \text{\quad} \kappa R_{\{a\}}=-L_{\{a\}}, \quad \text{\quad} \kappa T_{\{a\}}=-T_{\{a\}}.$                                                                                                              | $\kappa L_a = -R_a, \quad \kappa R_a = -L_a, \quad \kappa T_a = -T_a.$                                                                                                                 | $\kappa L_a = -R_a, \quad \kappa R_a = -L_a, \quad \kappa T_a = -T_a.$                                                                                                                 |
| 0902.0431_E00029 (3)   | 10   | ■ 1.000 | $\pi L_{\{a\}}=T_{\{a\}}, \quad \text{\quad} \pi R_{\{a\}}=-R_{\{a\}}, \quad \text{\quad} \pi T_{\{a\}}=L_{\{a\}}.$                                                                                                                         | $\pi L_a = T_a, \quad \pi R_a = -R_a, \quad \pi T_a = L_a.$                                                                                                                            | $\pi L_a = T_a, \quad \pi R_a = -R_a, \quad \pi T_a = L_a.$                                                                                                                            |
| 0902.0431_E00030 (4)   | 10   | ■ 1.000 | $\nu L_{\{a\}}=R_{\{a\}}, \quad \text{\quad} \nu R_{\{a\}}=-T_{\{a\}}, \quad \text{\quad} \nu T_{\{a\}}=-L_{\{a\}}.$                                                                                                                        | $\nu L_a = R_a, \quad \nu R_a = -T_a, \quad \nu T_a = -L_a.$                                                                                                                           | $\nu L_a = R_a, \quad \nu R_a = -T_a, \quad \nu T_a = -L_a.$                                                                                                                           |
| 0902.0431_E00031       | 10   | ■ 1.000 | $L_{\{e_{\{i\}}\}}=2\,F_{\{i\}\,0}, \quad \text{\quad} T_{\{e_{\{i\}}\}}=2\,G_{\{i\}\,0}.$                                                                                                                                                  | $L_{e_i} = 2F_{i0}, \quad T_{e_i} = 2G_{i0}.$                                                                                                                                          | $L_{e_i} = 2F_{i0}, \quad T_{e_i} = 2G_{i0}.$                                                                                                                                          |
| 0902.0431_E00035       | 11   | ■ 1.000 | $D=L_{\{a\}}+\sum_{\{i\}}\left[L_{\{b_{\{i\}}\}},L_{\{c_{\{i\}}\}}\right], \quad \text{\quad} a,b_{\{i\}},c_{\{i\}} \in \mathfrak{C}_{\{0\}}.$                                                                                              | $D = L_a + \sum_i [L_{b_i}, L_{c_i}], \quad a, b_i, c_i \in \mathfrak{C}_0.$                                                                                                           | $D = L_a + \sum_i [L_{b_i}, L_{c_i}], \quad a, b_i, c_i \in \mathfrak{C}_0.$                                                                                                           |
| 0902.0431_E00036       | 11   | ■ 1.000 | $\kappa^2=1, \quad \text{\quad} \pi^2=1, \quad \text{\quad} \nu^3=1, \quad \text{\quad} \nu=\pi\kappa.$                                                                                                                                     | $\kappa^2 = 1, \quad \pi^2 = 1, \quad \nu^3 = 1, \quad \nu = \pi\kappa.$                                                                                                               | $\kappa^2 = 1, \quad \pi^2 = 1, \quad \nu^3 = 1, \quad \nu = \pi\kappa.$                                                                                                               |
| 0902.0431_E00038       | 11   | ■ 1.000 | $\left(D_{\{1\}}\,x\right)\,y+x\left(D_{\{2\}}\,y\right)=D_{\{3\}}(x\,y), \quad \text{\quad} x,y \in \mathfrak{C}.$                                                                                                                         | $(D_1x)y+x(D_2y)=D_3(xy), \quad x,y \in \mathfrak{C}.$                                                                                                                                 | $(D_1x)y+x(D_2y)=D_3(xy), \quad x,y \in \mathfrak{C}.$                                                                                                                                 |
| 0902.0431_E00039       | 11   | ■ 1.000 | $D_{\{2\}}=\nu D_{\{1\}}, \quad \text{\quad} D_{\{3\}}=\pi D_{\{1\}}.$                                                                                                                                                                      | $D_2 = \nu D_1, \quad D_3 = \pi D_1.$                                                                                                                                                  | $D_2 = \nu D_1, \quad D_3 = \pi D_1.$                                                                                                                                                  |
| 0902.0431_E00040 (i)   | 11   | ■ 1.000 | $\left(L_{\{a\}}\,x\right)\,y+x\left(R_{\{a\}}\,y\right)=T_{\{a\}}(x\,y), \quad \text{\quad} x,y \in \mathfrak{C}.$                                                                                                                         | $(L_ax)y+x(R_ay)=T_a(xy), \quad x,y \in \mathfrak{C}.$                                                                                                                                 | $(L_ax)y+x(R_ay)=T_a(xy), \quad x,y \in \mathfrak{C}.$                                                                                                                                 |
| 0902.0431_E00042       | 12   | ■ 1.000 | $\left(L_{\{a\}}\,L_{\{b\}}\,x\right)\,y+\left(L_{\{b\}}\,x\right)\,\left(R_{\{a\}}\,y\right)+\left(L_{\{a\}}\,x\right)\,\left(R_{\{b\}}\,y\right)+x\left(R_{\{a\}}\,R_{\{b\}}\,y\right)=T_{\{a\}}\,T_{\{b\}}(x\,y).$                       | $(L_aL_bx)y+(L_bx)(R_ay)+(L_ax)(R_by)+x(R_aR_by)=T_aT_b(xy).$                                                                                                                          | $(L_aL_bx)y+(L_bx)(R_ay)+(L_ax)(R_by)+x(R_aR_by)=T_aT_b(xy).$                                                                                                                          |
| 0902.0431_E00043 (iii) | 12   | ■ 1.000 | $\left(\left(L_{\{a\}},L_{\{b\}}\right)x\right)\,y+x\left(\left(R_{\{a\}},R_{\{b\}}\right)y\right)=\left[T_{\{a\}},T_{\{b\}}\right](x\,y).$                                                                                                 | $([L_a,L_b]x)y+x([R_a,R_b]y)=[T_a,T_b](xy).$                                                                                                                                           | $([L_a,L_b]x)y+x([R_a,R_b]y)=[T_a,T_b](xy).$                                                                                                                                           |
| 0902.0431_E00044       | 12   | ■ 1.000 | $D_{\{1\}}=L_{\{a\}}+\sum\left[L_{\{b\}},L_{\{c\}}\right], \quad \text{\quad} a,b,c \in \mathfrak{C}_{\{0\}}$                                                                                                                               | $D_1 = L_a + \sum [L_b, L_c], \quad a, b, c \in \mathfrak{C}_0$                                                                                                                        | $D_1 = L_a + \sum [L_b, L_c], \quad a, b, c \in \mathfrak{C}_0$                                                                                                                        |
| 0902.0431_E00045       | 12   | ■ 1.000 | $\left(D_{\{1\}}\,x\right)\,y+x\left(D_{\{2\}}\,y\right)=D_{\{3\}}(x\,y), \quad \text{\quad} x,y \in \mathfrak{C}.$                                                                                                                         | $(D_1x)y+x(D_2y)=D_3(xy), \quad x,y \in \mathfrak{C}.$                                                                                                                                 | $(D_1x)y+x(D_2y)=D_3(xy), \quad x,y \in \mathfrak{C}.$                                                                                                                                 |
| 0902.0431_E00046       | 12   | ■ 1.000 | $x\left(D_{\{2\}}\,y\right)=D_{\{3\}}(x\,y), \quad \text{\quad} x,y \in \mathfrak{C}.$                                                                                                                                                      | $x(D_2y)=D_3(xy), \quad x,y \in \mathfrak{C}.$                                                                                                                                         | $x(D_2y)=D_3(xy), \quad x,y \in \mathfrak{C}.$                                                                                                                                         |
| 0902.0431_E00048       | 12   | ■ 1.000 | $x(y\,p)=(x\,y)\,p, \quad \text{\quad} \text{\texttt{\text{for all}}}\,x,y \in \mathfrak{C}.$                                                                                                                                               | $x(y p) = (xy)p, \quad \text{\texttt{\text{for all}}}\,x,y \in \mathfrak{C}.$                                                                                                          | $x(y p) = (xy)p, \quad \text{\texttt{\text{for all}}}\,x,y \in \mathfrak{C}.$                                                                                                          |
| 0902.0431_E00049       | 12   | ■ 1.000 | $\left(D_{\{1\}}\,x\right)\,y+x\left(D_{\{2\}}\,y\right)=\left(\kappa D_{\{3\}}\right)(x\,y), \quad \text{\quad} x,y \in \mathfrak{C}$                                                                                                      | $(D_1x)y+x(D_2y)=(\kappa D_3)(xy), \quad x,y \in \mathfrak{C}$                                                                                                                         | $(D_1x)y+x(D_2y)=(\kappa D_3)(xy), \quad x,y \in \mathfrak{C}$                                                                                                                         |
| 0902.0431_E00052       | 13   | ■ 1.000 | $\begin{aligned} & \& D_{\{3\}}^{\prime}=\nu D_{\{2\}}=\nu \nu D_{\{1\}}=\nu^{-1} D_{\{1\}}=\kappa \pi D_{\{1\}}=\kappa \kappa D_{\{3\}}=D_{\{3\}} \\ & D_{\{1\}}^{\prime}=\kappa \pi D_{\{2\}}=\nu^{-1} D_{\{2\}}=D_{\{1\}} \end{aligned}$ | $\begin{aligned} D_3' &= \nu D_2 = \nu \nu D_1 = \nu^{-1} D_1 = \kappa \pi D_1 = \kappa \kappa D_3 = D_3 \\ D_1' &= \kappa \pi D_2 = \nu^{-1} D_2 = D_1 \end{aligned}$                 | $\begin{aligned} D_3' &= \nu D_2 = \nu \nu D_1 = \nu^{-1} D_1 = \kappa \pi D_1 = \kappa \kappa D_3 = D_3, \\ D_1' &= \kappa \pi D_2 = \nu^{-1} D_2 = D_1. \end{aligned}$               |
| 0902.0431_E00053       | 13   | ■ 1.000 | $\mathfrak{g}_{\{2\}}=\left\{D \in \operatorname{Hom}_{\{\boldsymbol{R}\}}\left(\mathfrak{C}\right) \mid D(x\,y)=(D\,x)\,y+x(D\,y)\right\}.$                                                                                                | $\mathfrak{g}_2 = \{D \in \operatorname{Hom}_{\boldsymbol{R}}(\mathfrak{C}) \mid D(xy) = (Dx)y + x(Dy)\}.$                                                                             | $\mathfrak{g}_2 = \{D \in \operatorname{Hom}_{\boldsymbol{R}}(\mathfrak{C}) \mid D(xy) = (Dx)y + x(Dy)\}.$                                                                             |
| 0902.0431_E00054       | 13   | ■ 1.000 | $\begin{aligned} & \mathfrak{b}_{\{3\}}=\left\{D \in \mathfrak{D}_{\{4\}} \mid D1=0\right\} \\ & =\left\{D \in \mathfrak{D}_{\{4\}} \mid \kappa D=D\right\}=\left\{D \in \mathfrak{D}_{\{4\}} \mid \nu D=\pi D\right\}. \end{aligned}$      | $\begin{aligned} \mathfrak{b}_3 &= \{D \in \mathfrak{D}_4 \mid D1 = 0\} \\ &= \{D \in \mathfrak{D}_4 \mid \kappa D = D\} = \{D \in \mathfrak{D}_4 \mid \nu D = \pi D\}. \end{aligned}$ | $\begin{aligned} \mathfrak{b}_3 &= \{D \in \mathfrak{D}_4 \mid D1 = 0\} \\ &= \{D \in \mathfrak{D}_4 \mid \kappa D = D\} = \{D \in \mathfrak{D}_4 \mid \nu D = \pi D\}. \end{aligned}$ |
| 0902.0431_E00056       | 13   | ■ 1.000 | $D\,x \in \mathfrak{C}_{\{0\}}, \quad \text{\quad} x \in \mathfrak{C}.$                                                                                                                                                                     | $Dx \in \mathfrak{C}_0, \quad x \in \mathfrak{C}.$                                                                                                                                     | $Dx \in \mathfrak{C}_0, \quad x \in \mathfrak{C}.$                                                                                                                                     |
| 0902.0431_E00057       | 13   | ■ 1.000 | $x\,y+y\,x=-(x\,\overline{y}+y\,\overline{x})=-2(x,y), \quad \text{\quad} x,y \in \mathfrak{C}_{\{0\}}.$                                                                                                                                    | $xy+yx=-(x\overline{y}+y\overline{x})=-2(x,y), \quad x,y \in \mathfrak{C}_0.$                                                                                                          | $xy+yx=-(x\overline{y}+y\overline{x})=-2(x,y), \quad x,y \in \mathfrak{C}_0.$                                                                                                          |
| 0902.0431_E00058       | 13   | ■ 1.000 | $(D\,x,y)+(x,D\,y)=0, \quad \text{\quad} x,y \in \mathfrak{C}.$                                                                                                                                                                             | $(Dx,y)+(x,Dy)=0, \quad x,y \in \mathfrak{C}.$                                                                                                                                         | $(Dx,y)+(x,Dy)=0, \quad x,y \in \mathfrak{C}.$                                                                                                                                         |
| 0902.0431_E00060       | 14   | ■ 1.000 | $\operatorname{dim}\mathfrak{g}_{\{2\}}=14.$                                                                                                                                                                                                | $\dim \mathfrak{g}_2 = 14.$                                                                                                                                                            | $\dim \mathfrak{g}_2 = 14.$                                                                                                                                                            |

| Identifier                  | Page | Conf.   | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Rendered                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Image                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-----------------------------|------|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0902.0431_E00061            | 14   | ■ 1.000 | $\sum_{0<i<j} \lambda_{ij} F_{ij} = \sum_{0<i<j} \lambda_{ij} G_{ij}$<br>$\lambda_{ij} F_{ij} = \sum_{0<i<j} \lambda_{ij} G_{ij}$ .                                                                                                                                                                                                                                                                                                                                           | $\sum_{0<i<j} \lambda_{ij} F_{ij} = \sum_{0<i<j} \lambda_{ij} G_{ij}$ .                                                                                                                                                                                                                                                                                                                                                                                                       | $\sum_{0<i<j} \lambda_{ij} F_{ij} = \sum_{0<i<j} \lambda_{ij} G_{ij}$ .                                                                                                                                                                                                                                                                                                                                                                                                       |
| 0902.0431_E00062            | 14   | ■ 1.000 | $\sum_{0<i<j} \lambda_{ij} e_j e_i = 0$ .                                                                                                                                                                                                                                                                                                                                                                                                                                     | $\sum_{0<i<j} \lambda_{ij} e_j e_i = 0$ .                                                                                                                                                                                                                                                                                                                                                                                                                                     | $\sum_{0<i<j} \lambda_{ij} e_j e_i = 0$ .                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 0902.0431_E00064            | 14   | ■ 1.000 | $\boldsymbol{C} = \left\{ \boldsymbol{x}_0 + \boldsymbol{x}_1 e_1 \mid \boldsymbol{x}_i \in \boldsymbol{R} \right\}$ .                                                                                                                                                                                                                                                                                                                                                        | $\boldsymbol{C} = \{x_0 + x_1 e_1 \mid x_i \in \boldsymbol{R}\}$ .                                                                                                                                                                                                                                                                                                                                                                                                            | $\boldsymbol{C} = \{x_0 + x_1 e_1 \mid x_i \in \boldsymbol{R}\}$ .                                                                                                                                                                                                                                                                                                                                                                                                            |
| 0902.0431_E00066            | 15   | ■ 1.000 | $x = a + m_1 e_2 + m_2 e_4 + m_3 e_6$ , $\quad a, m_i \in \boldsymbol{C}$ .                                                                                                                                                                                                                                                                                                                                                                                                   | $x = a + m_1 e_2 + m_2 e_4 + m_3 e_6, \quad a, m_i \in \boldsymbol{C}$ .                                                                                                                                                                                                                                                                                                                                                                                                      | $x = a + m_1 e_2 + m_2 e_4 + m_3 e_6, \quad a, m_i \in \boldsymbol{C}$ .                                                                                                                                                                                                                                                                                                                                                                                                      |
| 0902.0431_E00070            | 15   | ■ 1.000 | $\boldsymbol{C} \oplus \boldsymbol{C}^3 = \mathfrak{C}$ .                                                                                                                                                                                                                                                                                                                                                                                                                     | $\boldsymbol{C} \oplus \boldsymbol{C}^3 = \mathfrak{C}$ .                                                                                                                                                                                                                                                                                                                                                                                                                     | $\boldsymbol{C} \oplus \boldsymbol{C}^3 = \mathfrak{C}$ .                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 0902.0431_E00071            | 15   | ■ 1.000 | $\left( \mathfrak{g}_2 \right)_{e_1} = \{ D \in \mathfrak{g}_2 \mid D e_1 = 0 \}$ .                                                                                                                                                                                                                                                                                                                                                                                           | $(\mathfrak{g}_2)_{e_1} = \{ D \in \mathfrak{g}_2 \mid D e_1 = 0 \}$ .                                                                                                                                                                                                                                                                                                                                                                                                        | $(\mathfrak{g}_2)_{e_1} = \{ D \in \mathfrak{g}_2 \mid D e_1 = 0 \}$ .                                                                                                                                                                                                                                                                                                                                                                                                        |
| 0902.0431_E00074            | 16   | ■ 1.000 | $\begin{array}{l} \varphi_*(e_1(E_{11} - E_{22})) = -G_{23} + G_{45}, \quad \varphi_*(e_1(E_{22} - E_{33})) = -G_{45} + G_{67}, \\ \varphi_*(E_{12} - E_{21}) = G_{24} + G_{35}, \quad \varphi_*(e_1(E_{12} + E_{21})) = -G_{25} + G_{34}, \\ \varphi_*(E_{13} - E_{31}) = G_{26} + G_{37}, \quad \varphi_*(e_1(E_{13} + E_{31})) = -G_{27} + G_{36}, \\ \varphi_*(E_{23} - E_{32}) = G_{46} + G_{57}, \quad \varphi_*(e_1(E_{23} + E_{32})) = -G_{47} + G_{56}. \end{array}$ | $\begin{array}{l} \varphi_*(e_1(E_{11} - E_{22})) = -G_{23} + G_{45}, \quad \varphi_*(e_1(E_{22} - E_{33})) = -G_{45} + G_{67}, \\ \varphi_*(E_{12} - E_{21}) = G_{24} + G_{35}, \quad \varphi_*(e_1(E_{12} + E_{21})) = -G_{25} + G_{34}, \\ \varphi_*(E_{13} - E_{31}) = G_{26} + G_{37}, \quad \varphi_*(e_1(E_{13} + E_{31})) = -G_{27} + G_{36}, \\ \varphi_*(E_{23} - E_{32}) = G_{46} + G_{57}, \quad \varphi_*(e_1(E_{23} + E_{32})) = -G_{47} + G_{56}. \end{array}$ | $\begin{array}{l} \varphi_*(e_1(E_{11} - E_{22})) = -G_{23} + G_{45}, \quad \varphi_*(e_1(E_{22} - E_{33})) = -G_{45} + G_{67}, \\ \varphi_*(E_{12} - E_{21}) = G_{24} + G_{35}, \quad \varphi_*(e_1(E_{12} + E_{21})) = -G_{25} + G_{34}, \\ \varphi_*(E_{13} - E_{31}) = G_{26} + G_{37}, \quad \varphi_*(e_1(E_{13} + E_{31})) = -G_{27} + G_{36}, \\ \varphi_*(E_{23} - E_{32}) = G_{46} + G_{57}, \quad \varphi_*(e_1(E_{23} + E_{32})) = -G_{47} + G_{56}. \end{array}$ |
| 0902.0431_E00078            | 16   | ■ 1.000 | $-2x_1e_2 - 2x_2e_3 - 2x_3e_4 - 2x_4e_5 - 2x_5e_6 - 2x_6e_7 = 0$ .                                                                                                                                                                                                                                                                                                                                                                                                            | $-2x_1e_2 - 2x_2e_3 - 2x_3e_4 - 2x_4e_5 - 2x_5e_6 - 2x_6e_7 = 0$ .                                                                                                                                                                                                                                                                                                                                                                                                            | $-2x_1e_2 - 2x_2e_3 - 2x_3e_4 - 2x_4e_5 - 2x_5e_6 - 2x_6e_7 = 0$ .                                                                                                                                                                                                                                                                                                                                                                                                            |
| 0902.0431_E00079            | 16   | ■ 1.000 | $\overline{x_1 + ix_2} = \bar{x}_1 + i\bar{x}_2, \quad \tau(x_1 + ix_2) = x_1 - ix_2, \quad x_i \in \mathfrak{C}$ .                                                                                                                                                                                                                                                                                                                                                           | $\overline{x_1 + ix_2} = \bar{x}_1 + i\bar{x}_2, \quad \tau(x_1 + ix_2) = x_1 - ix_2, \quad x_i \in \mathfrak{C}$ .                                                                                                                                                                                                                                                                                                                                                           | $\overline{x_1 + ix_2} = \bar{x}_1 + i\bar{x}_2, \quad \tau(x_1 + ix_2) = x_1 - ix_2, \quad x_i \in \mathfrak{C}$ .                                                                                                                                                                                                                                                                                                                                                           |
| 0902.0431_E00080            | 16   | ■ 1.000 | $\tau(xy) = (\tau x)(\tau y), \quad x, y \in \mathfrak{C}^C$ .                                                                                                                                                                                                                                                                                                                                                                                                                | $\tau(xy) = (\tau x)(\tau y), \quad x, y \in \mathfrak{C}^C$ .                                                                                                                                                                                                                                                                                                                                                                                                                | $\tau(xy) = (\tau x)(\tau y), \quad x, y \in \mathfrak{C}^C$ .                                                                                                                                                                                                                                                                                                                                                                                                                |
| 0902.0431_E00083 (cdots ii) | 17   | ■ 1.000 | $-x_1S_6 + x_2S_5 \in W \quad \cdots (i) \quad \text{and} \quad -x_1S_5 - x_2S_6 \in W$                                                                                                                                                                                                                                                                                                                                                                                       | $-x_1S_6 + x_2S_5 \in W \quad \cdots (i) \quad \text{and} \quad -x_1S_5 - x_2S_6 \in W$                                                                                                                                                                                                                                                                                                                                                                                       | $-x_1S_6 + x_2S_5 \in W \quad \cdots (i) \quad \text{and} \quad -x_1S_5 - x_2S_6 \in W$                                                                                                                                                                                                                                                                                                                                                                                       |
| 0902.0431_E00085            | 17   | ■ 1.000 | $\mathfrak{a} \ni [D', D + S] = [D', D] + [D', S], \quad [D', S] \in \mathfrak{S}$                                                                                                                                                                                                                                                                                                                                                                                            | $\mathfrak{a} \ni [D', D + S] = [D', D] + [D', S], \quad [D', S] \in \mathfrak{S}$                                                                                                                                                                                                                                                                                                                                                                                            | $\mathfrak{a} \ni [D', D + S] = [D', D] + [D', S], \quad [D', S] \in \mathfrak{S}$                                                                                                                                                                                                                                                                                                                                                                                            |
| 0902.0431_E00096            | 19   | ■ 1.000 | $\pm(\lambda_1 - \lambda_2), \quad \pm(\lambda_1 - \lambda_3), \quad \pm(\lambda_2 - \lambda_3), \quad \pm\lambda_1, \quad \pm\lambda_2 \quad \pm\lambda_3$                                                                                                                                                                                                                                                                                                                   | $\pm(\lambda_1 - \lambda_2), \quad \pm(\lambda_1 - \lambda_3), \quad \pm(\lambda_2 - \lambda_3), \quad \pm\lambda_1, \quad \pm\lambda_2 \quad \pm\lambda_3$                                                                                                                                                                                                                                                                                                                   | $\pm(\lambda_1 - \lambda_2), \quad \pm(\lambda_1 - \lambda_3), \quad \pm(\lambda_2 - \lambda_3), \quad \pm\lambda_1, \quad \pm\lambda_2 \quad \pm\lambda_3$                                                                                                                                                                                                                                                                                                                   |
| 0902.0431_E00099            | 20   | ■ 1.000 | $\alpha_1 = \lambda_1 - \lambda_2, \quad \alpha_2 = \lambda_2$                                                                                                                                                                                                                                                                                                                                                                                                                | $\alpha_1 = \lambda_1 - \lambda_2, \quad \alpha_2 = \lambda_2$                                                                                                                                                                                                                                                                                                                                                                                                                | $\alpha_1 = \lambda_1 - \lambda_2, \quad \alpha_2 = \lambda_2$                                                                                                                                                                                                                                                                                                                                                                                                                |
| 0902.0431_E00100            | 20   | ■ 1.000 | $\mu = 2\alpha_1 + 3\alpha_3$                                                                                                                                                                                                                                                                                                                                                                                                                                                 | $\mu = 2\alpha_1 + 3\alpha_3$                                                                                                                                                                                                                                                                                                                                                                                                                                                 | $\mu = 2\alpha_1 + 3\alpha_3$                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 0902.0431_E00102            | 20   | ■ 1.000 | $\mathfrak{h}_{\boldsymbol{R}} = \{-i\lambda_1 G_{23} - i\lambda_2 G_{45} - i\lambda_3 G_{67} \mid \lambda_i \in \boldsymbol{R}, \lambda_1 + \lambda_2 + \lambda_3 = 0\}$                                                                                                                                                                                                                                                                                                     | $\mathfrak{h}_{\boldsymbol{R}} = \{-i\lambda_1 G_{23} - i\lambda_2 G_{45} - i\lambda_3 G_{67} \mid \lambda_i \in \boldsymbol{R}, \lambda_1 + \lambda_2 + \lambda_3 = 0\}$                                                                                                                                                                                                                                                                                                     | $\mathfrak{h}_{\boldsymbol{R}} = \{-i\lambda_1 G_{23} - i\lambda_2 G_{45} - i\lambda_3 G_{67} \mid \lambda_i \in \boldsymbol{R}, \lambda_1 + \lambda_2 + \lambda_3 = 0\}$                                                                                                                                                                                                                                                                                                     |
| 0902.0431_E00103            | 20   | ■ 1.000 | $B_2(H, H') = 8 \sum_{k=1}^3 \lambda_k \lambda'_k,$                                                                                                                                                                                                                                                                                                                                                                                                                           | $B_2(H, H') = 8 \sum_{k=1}^3 \lambda_k \lambda'_k,$                                                                                                                                                                                                                                                                                                                                                                                                                           | $B_2(H, H') = 8 \sum_{k=1}^3 \lambda_k \lambda'_k,$                                                                                                                                                                                                                                                                                                                                                                                                                           |

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|------------------|------|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0902.0431_E00104 | 20   | ■ 1.000 | $H_{\alpha_1}=-\frac{1}{8}iG_{23}+\frac{1}{8}iG_{45},\quad H_{\alpha_2}=\frac{1}{24}iG_{23}-\frac{1}{12}iG_{45}+\frac{1}{24}iG_{67}.$                                 | $H_{\alpha_1}=-\frac{1}{8}iG_{23}+\frac{1}{8}iG_{45},\quad H_{\alpha_2}=\frac{1}{24}iG_{23}-\frac{1}{12}iG_{45}+\frac{1}{24}iG_{67}.$                                 | $H_{\alpha_1}=-\frac{1}{8}iG_{23}+\frac{1}{8}iG_{45},\quad H_{\alpha_2}=\frac{1}{24}iG_{23}-\frac{1}{12}iG_{45}+\frac{1}{24}iG_{67}.$                                 |
| 0902.0431_E00106 | 21   | ■ 1.000 | $\left(G_2\right)_{e_1}=\left\{\alpha\in G_2\mid\alpha e_1=e_1\right\}.$                                                                                              | $(G_2)_{e_1}=\{\alpha\in G_2\mid\alpha e_1=e_1\}.$                                                                                                                    | $(G_2)_{e_1}=\{\alpha\in G_2\mid\alpha e_1=e_1\}.$                                                                                                                    |
| 0902.0431_E00107 | 21   | ■ 1.000 | $\varphi(A)(a+\boldsymbol{m})=a+A\boldsymbol{m},\quad a+\boldsymbol{m}\in\boldsymbol{C}\oplus\boldsymbol{C}^3=\mathfrak{C}.$                                          | $\varphi(A)(a+\boldsymbol{m})=a+A\boldsymbol{m},\quad a+\boldsymbol{m}\in\boldsymbol{C}\oplus\boldsymbol{C}^3=\mathfrak{C}.$                                          | $\varphi(A)(a+\boldsymbol{m})=a+A\boldsymbol{m},\quad a+\boldsymbol{m}\in\boldsymbol{C}\oplus\boldsymbol{C}^3=\mathfrak{C}.$                                          |
| 0902.0431_E00109 | 21   | ■ 1.000 | $\alpha e_2=\boldsymbol{a}_1,\quad \alpha e_4=\boldsymbol{a}_2,\quad \alpha e_6=\boldsymbol{a}_3$                                                                     | $\alpha e_2=\boldsymbol{a}_1,\quad \alpha e_4=\boldsymbol{a}_2,\quad \alpha e_6=\boldsymbol{a}_3$                                                                     | $\alpha e_2=\boldsymbol{a}_1,\quad \alpha e_4=\boldsymbol{a}_2,\quad \alpha e_6=\boldsymbol{a}_3$                                                                     |
| 0902.0431_E00110 | 21   | ■ 1.000 | $\left\langle\boldsymbol{a}_1,\boldsymbol{a}_2\right\rangle=0,\quad \boldsymbol{a}_3=\overline{\boldsymbol{a}_1\times\boldsymbol{a}_2}.$                              | $\langle\boldsymbol{a}_1,\boldsymbol{a}_2\rangle=0,\quad \boldsymbol{a}_3=\overline{\boldsymbol{a}_1\times\boldsymbol{a}_2}.$                                         | $\langle\boldsymbol{a}_1,\boldsymbol{a}_2\rangle=0,\quad \boldsymbol{a}_3=\overline{\boldsymbol{a}_1\times\boldsymbol{a}_2}.$                                         |
| 0902.0431_E00111 | 22   | ■ 1.000 | $a_3=a_1a_2.$                                                                                                                                                         | $a_3=a_1a_2.$                                                                                                                                                         | $a_3=a_1a_2.$                                                                                                                                                         |
| 0902.0431_E00112 | 22   | ■ 1.000 | $a_5=a_1a_4,\quad a_6=a_4a_2,\quad a_7=a_3a_4.$                                                                                                                       | $a_5=a_1a_4,\quad a_6=a_4a_2,\quad a_7=a_3a_4.$                                                                                                                       | $a_5=a_1a_4,\quad a_6=a_4a_2,\quad a_7=a_3a_4.$                                                                                                                       |
| 0902.0431_E00114 | 22   | ■ 1.000 | $\alpha e_i=a_i,\quad i=0,1,\cdots,7$                                                                                                                                 | $\alpha e_i=a_i,\quad i=0,1,\cdots,7$                                                                                                                                 | $\alpha e_i=a_i,\quad i=0,1,\cdots,7$                                                                                                                                 |
| 0902.0431_E00115 | 22   | ■ 1.000 | $\alpha(xy)=(\alpha x)(\alpha y),\quad x,y\in\mathfrak{C}.$                                                                                                           | $\alpha(xy)=(\alpha x)(\alpha y),\quad x,y\in\mathfrak{C}.$                                                                                                           | $\alpha(xy)=(\alpha x)(\alpha y),\quad x,y\in\mathfrak{C}.$                                                                                                           |
| 0902.0431_E00116 | 22   | ■ 1.000 | $\alpha(e_ie_j)=(\alpha e_i)(\alpha e_j),\quad i,j=0,1,\cdots,7$                                                                                                      | $\alpha(e_ie_j)=(\alpha e_i)(\alpha e_j),\quad i,j=0,1,\cdots,7$                                                                                                      | $\alpha(e_ie_j)=(\alpha e_i)(\alpha e_j),\quad i,j=0,1,\cdots,7$                                                                                                      |
| 0902.0431_E00118 | 23   | ■ 1.000 | $w=\varphi(\operatorname{diag}(\omega_1,\omega_1,\omega_1))$                                                                                                          | $w=\varphi(\operatorname{diag}(\omega_1,\omega_1,\omega_1))$                                                                                                          | $w=\varphi(\operatorname{diag}(\omega_1,\omega_1,\omega_1))$                                                                                                          |
| 0902.0431_E00120 | 23   | ■ 1.000 | $(G_2)^w=\{\alpha\in G_2\mid w\alpha=\alpha w\}.$                                                                                                                     | $(G_2)^w=\{\alpha\in G_2\mid w\alpha=\alpha w\}.$                                                                                                                     | $(G_2)^w=\{\alpha\in G_2\mid w\alpha=\alpha w\}.$                                                                                                                     |
| 0902.0431_E00121 | 23   | ■ 1.000 | $w\varphi(A)(a+\boldsymbol{m})=w(a+A\boldsymbol{m})=a+\omega_1A\boldsymbol{m}=a+A\omega_1\boldsymbol{m}=\varphi(A)w(a+\boldsymbol{m}).$                               | $w\varphi(A)(a+\boldsymbol{m})=w(a+A\boldsymbol{m})=a+\omega_1A\boldsymbol{m}=a+A\omega_1\boldsymbol{m}=\varphi(A)w(a+\boldsymbol{m}).$                               | $w\varphi(A)(a+\boldsymbol{m})=w(a+A\boldsymbol{m})=a+\omega_1A\boldsymbol{m}=a+A\omega_1\boldsymbol{m}=\varphi(A)w(a+\boldsymbol{m}).$                               |
| 0902.0431_E00124 | 24   | ■ 1.000 | $\boldsymbol{H}=\{x_0+x_1e_1+x_2e_2+x_3e_3\mid x_i\in\boldsymbol{R}\}.$                                                                                               | $\boldsymbol{H}=\{x_0+x_1e_1+x_2e_2+x_3e_3\mid x_i\in\boldsymbol{R}\}.$                                                                                               | $\boldsymbol{H}=\{x_0+x_1e_1+x_2e_2+x_3e_3\mid x_i\in\boldsymbol{R}\}.$                                                                                               |
| 0902.0431_E00126 | 24   | ■ 1.000 | $x=m+ae_4,\quad m,a\in\boldsymbol{H}.$                                                                                                                                | $x=m+ae_4,\quad m,a\in\boldsymbol{H}.$                                                                                                                                | $x=m+ae_4,\quad m,a\in\boldsymbol{H}.$                                                                                                                                |
| 0902.0431_E00127 | 24   | ■ 1.000 | $(m+ae_4)(n+be_4)=(mn-\bar{b}a)+(\overline{a\bar{n}}+bm)e_4,$<br>$(m+ae_4,n+be_4)=(m,n)+(a,b),$<br>$\overline{m+ae_4}=\overline{m}-ae_4,$<br>$\gamma(m+ae_4)=m-ae_4.$ | $(m+ae_4)(n+be_4)=(mn-\bar{b}a)+(\overline{a\bar{n}}+bm)e_4,$<br>$(m+ae_4,n+be_4)=(m,n)+(a,b),$<br>$\overline{m+ae_4}=\overline{m}-ae_4,$<br>$\gamma(m+ae_4)=m-ae_4.$ | $(m+ae_4)(n+be_4)=(mn-\bar{b}a)+(\overline{a\bar{n}}+bm)e_4,$<br>$(m+ae_4,n+be_4)=(m,n)+(a,b),$<br>$\overline{m+ae_4}=\overline{m}-ae_4,$<br>$\gamma(m+ae_4)=m-ae_4.$ |
| 0902.0431_E00128 | 24   | ■ 1.000 | $\boldsymbol{H}\oplus\boldsymbol{H}e_4=\mathfrak{C}.$                                                                                                                 | $\boldsymbol{H}\oplus\boldsymbol{H}e_4=\mathfrak{C}.$                                                                                                                 | $\boldsymbol{H}\oplus\boldsymbol{H}e_4=\mathfrak{C}.$                                                                                                                 |
| 0902.0431_E00129 | 24   | ■ 1.000 | $(G_2)^\gamma=\{\alpha\in G_2\mid\gamma\alpha=\alpha\gamma\}.$                                                                                                        | $(G_2)^\gamma=\{\alpha\in G_2\mid\gamma\alpha=\alpha\gamma\}.$                                                                                                        | $(G_2)^\gamma=\{\alpha\in G_2\mid\gamma\alpha=\alpha\gamma\}.$                                                                                                        |
| 0902.0431_E00130 | 24   | ■ 1.000 | $\varphi(p,q)(m+ae_4)=qm\bar{q}+(pa\bar{q})e_4,\quad m+ae_4\in\boldsymbol{H}\oplus\boldsymbol{H}e_4=\mathfrak{C}.$                                                    | $\varphi(p,q)(m+ae_4)=qm\bar{q}+(pa\bar{q})e_4,\quad m+ae_4\in\boldsymbol{H}\oplus\boldsymbol{H}e_4=\mathfrak{C}.$                                                    | $\varphi(p,q)(m+ae_4)=qm\bar{q}+(pa\bar{q})e_4,\quad m+ae_4\in\boldsymbol{H}\oplus\boldsymbol{H}e_4=\mathfrak{C}.$                                                    |
| 0902.0431_E00132 | 25   | ■ 1.000 | $\alpha m=qm\bar{q},\quad m\in\boldsymbol{H}$                                                                                                                         | $\alpha m=qm\bar{q},\quad m\in\boldsymbol{H}$                                                                                                                         | $\alpha m=qm\bar{q},\quad m\in\boldsymbol{H}$                                                                                                                         |
| 0902.0431_E00135 | 25   | ■ 1.000 | $f(p,q)x=px\bar{q},\quad x\in\boldsymbol{H}$                                                                                                                          | $f(p,q)x=px\bar{q},\quad x\in\boldsymbol{H}$                                                                                                                          | $f(p,q)x=px\bar{q},\quad x\in\boldsymbol{H}$                                                                                                                          |
| 0902.0431_E00137 | 26   | ■ 1.000 | $z(G_2)=\{1\}.$                                                                                                                                                       | $z(G_2)=\{1\}.$                                                                                                                                                       | $z(G_2)=\{1\}.$                                                                                                                                                       |

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| 0902.0431_E00140     | 26   | ■ 1.000 | $(\alpha x, \alpha y) = (x, y), \quad x, y \in \mathfrak{C}$ .                                                                                                    | $(\alpha x, \alpha y) = (x, y), \quad x, y \in \mathfrak{C}^C.$                                                                          | $(\alpha x, \alpha y) = (x, y), \quad x, y \in \mathfrak{C}^C.$                                                                          |
| 0902.0431_E00141     | 26   | ■ 1.000 | $\overline{\alpha e_k} = -\alpha e_k, \quad k=1,2, \cdots, 7.$                                                                                                    | $\overline{\alpha e_k} = -\alpha e_k, \quad k = 1, 2, \cdots, 7.$                                                                        | $\overline{\alpha e_k} = -\alpha e_k, \quad k = 1, 2, \cdots, 7.$                                                                        |
| 0902.0431_E00144     | 26   | ■ 1.000 | $\langle x, y \rangle = (\tau x, y),$                                                                                                                             | $\langle x, y \rangle = (\tau x, y),$                                                                                                    | $\langle x, y \rangle = (\tau x, y),$                                                                                                    |
| 0902.0431_E00149     | 27   | ■ 1.000 | $G_{2(2)} = \left\{ \alpha \in \operatorname{Iso}_{\mathbf{R}}(\mathfrak{C}') \mid \alpha(xy) = (\alpha x)(\alpha y) \right\}.$                                   | $G_{2(2)} = \{ \alpha \in \operatorname{Iso}_{\mathbf{R}}(\mathfrak{C}') \mid \alpha(xy) = (\alpha x)(\alpha y) \}.$                     | $G_{2(2)} = \{ \alpha \in \operatorname{Iso}_{\mathbf{R}}(\mathfrak{C}') \mid \alpha(xy) = (\alpha x)(\alpha y) \}.$                     |
| 0902.0431_E00150     | 28   | ■ 1.000 | $G_{2(2)} = \left( G_{2\{ \}^{\{ \}} \right)^{\tau \gamma} = \left\{ \alpha \in G_{2^C} \mid \tau \gamma \alpha = \alpha \gamma \tau \right\}.$                   | $G_{2(2)} = \left( G_2^C \right)^{\tau \gamma} = \{ \alpha \in G_2^C \mid \tau \gamma \alpha = \alpha \gamma \tau \}.$                   | $G_{2(2)} = (G_2^C)^{\tau \gamma} = \{ \alpha \in G_2^C \mid \tau \gamma \alpha = \alpha \gamma \tau \}.$                                |
| 0902.0431_E00153     | 28   | ■ 1.000 | $z \left( G_{2(2)} \right) = \{ 1 \}.$                                                                                                                            | $z \left( G_{2(2)} \right) = \{ 1 \}.$                                                                                                   | $z(G_{2(2)}) = \{ 1 \}.$                                                                                                                 |
| 0902.0431_E00155     | 28   | ■ 1.000 | $A = D_{a_m} \cdots D_{a_2} D_{a_1}, \quad a_i \in S^{n-1}$                                                                                                       | $A = D_{a_m} \cdots D_{a_2} D_{a_1}, \quad a_i \in S^{n-1}$                                                                              | $A = D_{a_m} \cdots D_{a_2} D_{a_1}, \quad a_i \in S^{n-1}.$                                                                             |
| 0902.0431_E00156     | 28   | ■ 1.000 | $A = D_{a_{2m}} \cdots D_{a_2} D_{a_1}, \quad a_i \in S^{n-1}$                                                                                                    | $A = D_{a_{2m}} \cdots D_{a_2} D_{a_1}, \quad a_i \in S^{n-1}$                                                                           | $A = D_{a_{2m}} \cdots D_{a_2} D_{a_1}, \quad a_i \in S^{n-1}.$                                                                          |
| 0902.0431_E00157     | 28   | ■ 1.000 | $SO(\mathfrak{C}) = \left\{ \alpha \in \operatorname{Iso}_{\mathbf{R}}(\mathfrak{C}) \mid (\alpha x, \alpha y) = (x, y), \det \alpha = 1 \right\}.$               | $SO(\mathfrak{C}) = \{ \alpha \in \operatorname{Iso}_{\mathbf{R}}(\mathfrak{C}) \mid (\alpha x, \alpha y) = (x, y), \det \alpha = 1 \}.$ | $SO(\mathfrak{C}) = \{ \alpha \in \operatorname{Iso}_{\mathbf{R}}(\mathfrak{C}) \mid (\alpha x, \alpha y) = (x, y), \det \alpha = 1 \}.$ |
| 0902.0431_E00159     | 29   | ■ 1.000 | $D_a x = x - 2(x, a)a = x - (x \bar{a} + a \bar{x})a = -a \bar{x}a, \quad x \in \mathfrak{C},$                                                                    | $D_a x = x - 2(x, a)a = x - (x \bar{a} + a \bar{x})a = -a \bar{x}a, \quad x \in \mathfrak{C},$                                           | $D_a x = x - 2(x, a)a = x - (x \bar{a} + a \bar{x})a = -a \bar{x}a, \quad x \in \mathfrak{C},$                                           |
| 0902.0431_E00160     | 29   | ■ 1.000 | $\left( \alpha_1 x \right) \left( \alpha_2 y \right) = (b(\bar{a}x))((y \bar{a})b) = b(\bar{a}(xy) \bar{a})b = \alpha_3(xy), \quad x, y \in \mathfrak{C}.$        | $(\alpha_1 x)(\alpha_2 y) = (b(\bar{a}x))((y \bar{a})b) = b(\bar{a}(xy) \bar{a})b = \alpha_3(xy), \quad x, y \in \mathfrak{C}.$          | $(\alpha_1 x)(\alpha_2 y) = (b(\bar{a}x))((y \bar{a})b) = b(\bar{a}(xy) \bar{a})b = \alpha_3(xy), \quad x, y \in \mathfrak{C}.$          |
| 0902.0431_E00163     | 29   | ■ 1.000 | $(x p) y = x(p y), \quad x, y \in \mathfrak{C}.$                                                                                                                  | $(xp)y = x(py), \quad x, y \in \mathfrak{C}.$                                                                                            | $(xp)y = x(py), \quad x, y \in \mathfrak{C}.$                                                                                            |
| 0902.0431_E00167     | 30   | ■ 1.000 | $\left( \alpha_2 y \right) \left( \alpha_3 z \right) = \overline{\alpha_1(\overline{y z})}.$                                                                      | $(\alpha_2 y)(\alpha_3 z) = \overline{\alpha_1(\overline{y z})}.$                                                                        | $(\alpha_2 y)(\alpha_3 z) = \overline{\alpha_1(\overline{y z})}.$                                                                        |
| 0902.0431_E00168     | 30   | ■ 1.000 | $\left( \alpha_1 x \right) \left( \alpha_2 y \right) = \alpha_3(xy), \quad x, y \in \mathfrak{C},$                                                                | $(\alpha_1 x)(\alpha_2 y) = \alpha_3(xy), \quad x, y \in \mathfrak{C},$                                                                  | $(\alpha_1 x)(\alpha_2 y) = \alpha_3(xy), \quad x, y \in \mathfrak{C},$                                                                  |
| 0902.0431_E00169     | 30   | ■ 1.000 | $\left( \beta_1(\alpha_1 x) \right) \left( \beta_2(\alpha_2 y) \right) = \beta_3((\alpha_1 x)(\alpha_2 y)) = \beta_3(\alpha_3(xy)), \quad x, y \in \mathfrak{C}.$ | $(\beta_1(\alpha_1 x))(\beta_2(\alpha_2 y)) = \beta_3((\alpha_1 x)(\alpha_2 y)) = \beta_3(\alpha_3(xy)), \quad x, y \in \mathfrak{C}.$   | $(\beta_1(\alpha_1 x))(\beta_2(\alpha_2 y)) = \beta_3((\alpha_1 x)(\alpha_2 y)) = \beta_3(\alpha_3(xy)), \quad x, y \in \mathfrak{C}.$   |
| 0902.0431_E00170     | 30   | ■ 1.000 | $\bar{x}(\gamma_2 y) = \gamma_3(xy), \quad x, y \in \mathfrak{C}.$                                                                                                | $\bar{x}(\gamma_2 y) = \gamma_3(xy), \quad x, y \in \mathfrak{C}.$                                                                       | $\bar{x}(\gamma_2 y) = \gamma_3(xy), \quad x, y \in \mathfrak{C}.$                                                                       |
| 0902.0431_E00171 (i) | 30   | ■ 1.000 | $\bar{x}(\gamma_2 y) = \gamma_2(xy), \quad x, y \in \mathfrak{C}.$                                                                                                | $\bar{x}(\gamma_2 y) = \gamma_2(xy), \quad x, y \in \mathfrak{C}.$                                                                       | $\bar{x}(\gamma_2 y) = \gamma_2(xy), \quad x, y \in \mathfrak{C}.$                                                                       |
| 0902.0431_E00173     | 30   | ■ 1.000 | $p x = x p \quad \text{for all } x \in \mathfrak{C}.$                                                                                                             | $p x = x p \quad \text{for all } x \in \mathfrak{C}.$                                                                                    | $p x = x p \quad \text{for all } x \in \mathfrak{C}.$                                                                                    |
| 0902.0431_E00174     | 30   | ■ 1.000 | $x y = y x \quad \text{for all } x, y \in \mathfrak{C}.$                                                                                                          | $x y = y x \quad \text{for all } x, y \in \mathfrak{C}.$                                                                                 | $x y = y x \quad \text{for all } x, y \in \mathfrak{C}.$                                                                                 |
| 0902.0431_E00175     | 31   | ■ 1.000 | $\alpha_a x = a x a^{-1}, \quad x \in \mathfrak{C}$                                                                                                               | $\alpha_a x = a x a^{-1}, \quad x \in \mathfrak{C}$                                                                                      | $\alpha_a x = a x a^{-1}, \quad x \in \mathfrak{C}$                                                                                      |
| 0902.0431_E00177     | 31   | ■ 1.000 | $(a x \bar{a}) \left( a y a^2 \right) = a(x y) a^2, \quad x, y \in \mathfrak{C}.$                                                                                 | $(a x \bar{a}) \left( a y a^2 \right) = a(x y) a^2, \quad x, y \in \mathfrak{C}.$                                                        | $(a x \bar{a}) \left( a y a^2 \right) = a(x y) a^2, \quad x, y \in \mathfrak{C}.$                                                        |
| 0902.0431_E00179     | 31   | ■ 1.000 | $a y a^2 = \pm a y \bar{a}.$                                                                                                                                      | $a y a^2 = \pm a y \bar{a}.$                                                                                                             | $a y a^2 = \pm a y \bar{a}.$                                                                                                             |
| 0902.0431_E00184     | 32   | ■ 1.000 | $a_5 b_0 = a_1(a_4 b_0), \quad a_6 b_0 = a_2(a_4 b_0), \quad a_7 b_0 = a_6(a_1 b_0).$                                                                             | $a_5 b_0 = a_1(a_4 b_0), \quad a_6 b_0 = a_2(a_4 b_0), \quad a_7 b_0 = a_6(a_1 b_0).$                                                    | $a_5 b_0 = a_1(a_4 b_0), \quad a_6 b_0 = a_2(a_4 b_0), \quad a_7 b_0 = a_6(a_1 b_0).$                                                    |
| 0902.0431_E00186     | 32   | ■ 1.000 | $\alpha e_i = a_i, \quad i = 0, 1, \cdots, 7$                                                                                                                     | $\alpha e_i = a_i, \quad i = 0, 1, \cdots, 7$                                                                                            | $\alpha e_i = a_i, \quad i = 0, 1, \cdots, 7$                                                                                            |
| 0902.0431_E00188     | 32   | ■ 1.000 | $\left( \alpha e_i \right) \left( \left( \alpha e_j \right) b_0 \right) = \left( \alpha \left( e_i e_j \right) \right) b_0, \quad i, j = 0, 1, \cdots, 7.$        | $(\alpha e_i)((\alpha e_j) b_0) = (\alpha(e_i e_j)) b_0, \quad i, j = 0, 1, \cdots, 7.$                                                  | $(\alpha e_i)((\alpha e_j) b_0) = (\alpha(e_i e_j)) b_0, \quad i, j = 0, 1, \cdots, 7.$                                                  |

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| 0902.0431_E00190     | 33   | ■ 1.000 | $b_{\{i\}=a_{\{i\}}\,b_{\{0\}}}$ , $\quad i=0,1,\,\cdots,\,7$ ,                                                                                                                                                                            | $b_i = a_i b_0, \quad i = 0, 1, \cdots, 7,$                                                                                                                                                                      | $b_i = a_i b_0, \quad i = 0, 1, \cdots, 7,$                                                                                                                                                                           |
| 0902.0431_E00193     | 33   | ■ 1.000 | $\widetilde{B}_3 \cong \operatorname{Spin}(7)$ .                                                                                                                                                                                           | $\widetilde{B}_3 \cong \operatorname{Spin}(7).$                                                                                                                                                                  | $\widetilde{B}_3 \cong \operatorname{Spin}(7).$                                                                                                                                                                       |
| 0902.0431_E00196     | 33   | ■ 1.000 | $\widetilde{B}_3/\{1,-1\} \cong S\,0(7)$ .                                                                                                                                                                                                 | $\widetilde{B}_3/\{1,-1\} \cong SO(7).$                                                                                                                                                                          | $\widetilde{B}_3/\{1,-1\} \cong SO(7).$                                                                                                                                                                               |
| 0902.0431_E00198     | 34   | ■ 1.000 | $\operatorname{Spin}(7) \ni \widetilde{\alpha}$<br>$\xrightarrow{\quad \alpha, \widetilde{\alpha}, \kappa \widetilde{\alpha} \quad}$<br>$\in \widetilde{D}_4$ .                                                                            | $\operatorname{Spin}(7) \ni \widetilde{\alpha} \longleftrightarrow (\alpha, \widetilde{\alpha}, \kappa \widetilde{\alpha}) \in \widetilde{D}_4.$                                                                 | $\operatorname{Spin}(7) \ni \widetilde{\alpha} \longleftrightarrow (\alpha, \widetilde{\alpha}, \kappa \widetilde{\alpha}) \in \widetilde{D}_4.$                                                                      |
| 0902.0431_E00199     | 34   | ■ 1.000 | $\widetilde{D}_4/\operatorname{Spin}(7) \simeq S^7$ .                                                                                                                                                                                      | $\widetilde{D}_4/\operatorname{Spin}(7) \simeq S^7.$                                                                                                                                                             | $\widetilde{D}_4/\operatorname{Spin}(7) \simeq S^7.$                                                                                                                                                                  |
| 0902.0431_E00202     | 34   | ■ 1.000 | $\widetilde{D}_4 \cong \operatorname{Spin}(8)$ .                                                                                                                                                                                           | $\widetilde{D}_4 \cong \operatorname{Spin}(8).$                                                                                                                                                                  | $\widetilde{D}_4 \cong \operatorname{Spin}(8).$                                                                                                                                                                       |
| 0902.0431_E00203     | 34   | ■ 1.000 | $p(\alpha_1, \alpha_2, \alpha_3) = \alpha_1$ .                                                                                                                                                                                             | $p(\alpha_1, \alpha_2, \alpha_3) = \alpha_1.$                                                                                                                                                                    | $p(\alpha_1, \alpha_2, \alpha_3) = \alpha_1.$                                                                                                                                                                         |
| 0902.0431_E00204     | 34   | ■ 1.000 | $\widetilde{D}_4/\{(1,1,1),(1,-1,-1)\} \cong S\,0(8)$ .                                                                                                                                                                                    | $\widetilde{D}_4/\{(1,1,1),(1,-1,-1)\} \cong SO(8).$                                                                                                                                                             | $\widetilde{D}_4/\{(1,1,1),(1,-1,-1)\} \cong SO(8).$                                                                                                                                                                  |
| 0902.0431_E00205     | 34   | ■ 1.000 | $\begin{aligned} z(\operatorname{Spin}(8)) \, \&= \\ &\{(1,1,1),(1,-1,-1),(-1,-1,1),(-1,1,-1)\} \, \&= \\ &\{(1,1,1),(1,-1,-1)\} \, \times \{(1,1,1),(-1,-1,1)\} \\ &\cong \boldsymbol{Z}_2 \, \times \boldsymbol{Z}_2 \, . \end{aligned}$ | $\begin{aligned} z(\operatorname{Spin}(8)) = \{ &(1,1,1),(1,-1,-1),(-1,-1,1),(-1,1,-1)\} \\ &= \{(1,1,1),(1,-1,-1)\} \times \{(1,1,1),(-1,-1,1)\} \cong \boldsymbol{Z}_2 \times \boldsymbol{Z}_2. \end{aligned}$ | $\begin{aligned} z(\operatorname{Spin}(8)) \, = \, &\{(1,1,1),(1,-1,-1),(-1,-1,1),(-1,1,-1)\} \\ &= \{(1,1,1),(1,-1,-1)\} \times \{(1,1,1),(-1,-1,1)\} \cong \boldsymbol{Z}_2 \times \boldsymbol{Z}_2. \end{aligned}$ |
| 0902.0431_E00207     | 35   | ■ 1.000 | $\kappa^2=1$ , $\quad \pi^2=1$ , $\quad \nu^3=1$ , $\quad \nu=\pi\kappa$ .                                                                                                                                                                 | $\kappa^2 = 1, \quad \pi^2 = 1, \quad \nu^3 = 1, \quad \nu = \pi \kappa.$                                                                                                                                        | $\kappa^2 = 1, \quad \pi^2 = 1, \quad \nu^3 = 1, \quad \nu = \pi \kappa.$                                                                                                                                             |
| 0902.0431_E00210     | 35   | ■ 1.000 | $S\,0(8) \cong S\,s(8)$                                                                                                                                                                                                                    | $SO(8) \cong Ss(8)$                                                                                                                                                                                              | $SO(8) \cong Ss(8).$                                                                                                                                                                                                  |
| 0902.0431_E00212 (i) | 35   | ■ 1.000 | $(\alpha x)(\alpha y) = \kappa \alpha(x y)$ , $\quad x, y \in \mathfrak{C}$ ,                                                                                                                                                              | $(\alpha x)(\alpha y) = \kappa \alpha(xy), \quad x, y \in \mathfrak{C},$                                                                                                                                         | $(\alpha x)(\alpha y) = \kappa \alpha(xy), \quad x, y \in \mathfrak{C},$                                                                                                                                              |
| 0902.0431_E00213     | 35   | ■ 1.000 | $a(\alpha x) = (\alpha x) a$ $\quad$ for all $\quad x \in \mathfrak{C}$ .                                                                                                                                                                  | $a(\alpha x) = (\alpha x) a \quad$ for all $\quad x \in \mathfrak{C}.$                                                                                                                                           | $a(\alpha x) = (\alpha x) a \quad$ for all $\quad x \in \mathfrak{C}.$                                                                                                                                                |
| 0902.0431_E00214     | 35   | ■ 1.000 | $\nu^2(-1,-1,1) = (1,-1,-1)$ ,                                                                                                                                                                                                             | $\nu^2(-1,-1,1) = (1,-1,-1),$                                                                                                                                                                                    | $\nu^2(-1,-1,1) = (1,-1,-1),$                                                                                                                                                                                         |
| 0902.0431_E00215     | 36   | ■ 1.000 | $\mathfrak{J} = \left\{ X \in M(3, \mathfrak{C}) \mid X^* = X \right\}$ ,                                                                                                                                                                  | $\mathfrak{J} = \{ X \in M(3, \mathfrak{C}) \mid X^* = X \},$                                                                                                                                                    | $\mathfrak{J} = \{ X \in M(3, \mathfrak{C}) \mid X^* = X \},$                                                                                                                                                         |
| 0902.0431_E00217     | 36   | ■ 1.000 | $X \circ Y = \frac{1}{2}(XY + YX)$                                                                                                                                                                                                         | $X \circ Y = \frac{1}{2}(XY + YX)$                                                                                                                                                                               | $X \circ Y = \frac{1}{2}(XY + YX).$                                                                                                                                                                                   |
| 0902.0431_E00218     | 36   | ■ 1.000 | $\begin{aligned} \operatorname{tr}(X) &= \xi_1 + \xi_2 + \xi_3, \quad X = X(\xi, x), \\ (X, Y) &= \operatorname{tr}(X \circ Y), \quad \operatorname{tr}(X, Y, Z) = (X, Y \circ Z) . \end{aligned}$                                         | $\begin{aligned} \operatorname{tr}(X) &= \xi_1 + \xi_2 + \xi_3, \quad X = X(\xi, x), \\ (X, Y) &= \operatorname{tr}(X \circ Y), \quad \operatorname{tr}(X, Y, Z) = (X, Y \circ Z). \end{aligned}$                | $\begin{aligned} \operatorname{tr}(X) &= \xi_1 + \xi_2 + \xi_3, \quad X = X(\xi, x), \\ (X, Y) &= \operatorname{tr}(X \circ Y), \quad \operatorname{tr}(X, Y, Z) = (X, Y \circ Z). \end{aligned}$                     |
| 0902.0431_E00219     | 36   | ■ 1.000 | $X \times Y = \frac{1}{2}(2X \circ Y - \operatorname{tr}(X)Y - \operatorname{tr}(Y)X + (\operatorname{tr}(X)\operatorname{tr}(Y) - (X, Y))E)$ ,                                                                                            | $X \times Y = \frac{1}{2}(2X \circ Y - \operatorname{tr}(X)Y - \operatorname{tr}(Y)X + (\operatorname{tr}(X)\operatorname{tr}(Y) - (X, Y))E),$                                                                   | $X \times Y = \frac{1}{2}(2X \circ Y - \operatorname{tr}(X)Y - \operatorname{tr}(Y)X + (\operatorname{tr}(X)\operatorname{tr}(Y) - (X, Y))E),$                                                                        |
| 0902.0431_E00220     | 36   | ■ 1.000 | $(X, Y, Z) = (X, Y \times Z)$ , $\quad \det X = \frac{1}{3}(X, X, X)$ .                                                                                                                                                                    | $(X, Y, Z) = (X, Y \times Z), \quad \det X = \frac{1}{3}(X, X, X).$                                                                                                                                              | $(X, Y, Z) = (X, Y \times Z), \quad \det X = \frac{1}{3}(X, X, X).$                                                                                                                                                   |
| 0902.0431_E00227     | 38   | ■ 1.000 | $F_4 = \left\{ \alpha \in \operatorname{Iso}_{\boldsymbol{R}}(\mathfrak{J}) \mid \alpha(X \circ Y) = \alpha X \circ \alpha Y \right\}$ .                                                                                                   | $F_4 = \{ \alpha \in \operatorname{Iso}_{\boldsymbol{R}}(\mathfrak{J}) \mid \alpha(X \circ Y) = \alpha X \circ \alpha Y \}.$                                                                                     | $F_4 = \{ \alpha \in \operatorname{Iso}_{\boldsymbol{R}}(\mathfrak{J}) \mid \alpha(X \circ Y) = \alpha X \circ \alpha Y \}.$                                                                                          |
| 0902.0431_E00228 (i) | 38   | ■ 1.000 | $X \circ (X \circ X) - \operatorname{tr}(X)X^2 + \frac{1}{2}(\operatorname{tr}(X)^2 - \operatorname{tr}(X^2))X = (\det X)E$ .                                                                                                              | $X \circ (X \circ X) - \operatorname{tr}(X)X^2 + \frac{1}{2}(\operatorname{tr}(X)^2 - \operatorname{tr}(X^2))X = (\det X)E.$                                                                                     | $X \circ (X \circ X) - \operatorname{tr}(X)X^2 + \frac{1}{2}(\operatorname{tr}(X)^2 - \operatorname{tr}(X^2))X = (\det X)E.$                                                                                          |

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| 0902.0431_E08229 (ii) | 38   | ■ 1.000 | $X \circ (X \circ X) - \operatorname{tr}(\alpha X) X^2 + \frac{1}{2} (\operatorname{tr}(\alpha X)^2 - \operatorname{tr}((\alpha X^2)) X = (\det \alpha X) E.$                                                                         | $X \circ (X \circ X) - \operatorname{tr}(\alpha X) X^2 + \frac{1}{2} (\operatorname{tr}(\alpha X)^2 - \operatorname{tr}((\alpha X^2)) X = (\det \alpha X) E.$                                       | $X \circ (X \circ X) - \operatorname{tr}(\alpha X) X^2 + \frac{1}{2} (\operatorname{tr}(\alpha X)^2 - \operatorname{tr}((\alpha X^2)) X = (\det \alpha X) E.$                                       |
| 0902.0431_E08231      | 38   | ■ 1.000 | $\begin{aligned} & \operatorname{tr}(\alpha F_i(e_j)) (E_{i+1} + E_{i+2}) + \frac{1}{2} (-\operatorname{tr}(\alpha F_i(e_j))^2 + \operatorname{tr}((\alpha F_i(e_j))^2) - 2) F_i(e_j) \\ & = -\det(\alpha F_i(e_j)) E. \end{aligned}$ | $\operatorname{tr}(\alpha F_i(e_j)) (E_{i+1} + E_{i+2}) + \frac{1}{2} (-\operatorname{tr}(\alpha F_i(e_j))^2 + \operatorname{tr}((\alpha F_i(e_j))^2) - 2) F_i(e_j) \\ = -\det(\alpha F_i(e_j)) E.$ | $\operatorname{tr}(\alpha F_i(e_j)) (E_{i+1} + E_{i+2}) + \frac{1}{2} (-\operatorname{tr}(\alpha F_i(e_j))^2 + \operatorname{tr}((\alpha F_i(e_j))^2) - 2) F_i(e_j) \\ = -\det(\alpha F_i(e_j)) E.$ |
| 0902.0431_E08232      | 38   | ■ 1.000 | $\operatorname{tr}(\alpha F_i(e_j)) = 0 (= \operatorname{tr}(F_i(e_j)))$                                                                                                                                                              | $\operatorname{tr}(\alpha F_i(e_j)) = 0 (= \operatorname{tr}(F_i(e_j)))$                                                                                                                            | $\operatorname{tr}(\alpha F_i(e_j)) = 0 (= \operatorname{tr}(F_i(e_j)))$                                                                                                                            |
| 0902.0431_E08233      | 38   | ■ 1.000 | $\operatorname{tr}(\alpha E_i) = \operatorname{tr}(\alpha (E - F_i(1)^2)) = \operatorname{tr}(E) - \operatorname{tr}((\alpha F_i(1))^2) = 3 - 2 = 1 = \operatorname{tr}(E_i),$                                                        | $\operatorname{tr}(\alpha E_i) = \operatorname{tr}(\alpha (E - F_i(1)^2)) = \operatorname{tr}(E) - \operatorname{tr}((\alpha F_i(1))^2) = 3 - 2 = 1 = \operatorname{tr}(E_i),$                      | $\operatorname{tr}(\alpha E_i) = \operatorname{tr}(\alpha (E - F_i(1)^2)) = \operatorname{tr}(E) - \operatorname{tr}((\alpha F_i(1))^2) = 3 - 2 = 1 = \operatorname{tr}(E_i),$                      |
| 0902.0431_E08236      | 39   | ■ 1.000 | $\alpha^{-1}((\det X) X) \times {}^t \alpha^{-1}((\det X) X) = \alpha((\det X) X \times (\det X) X).$                                                                                                                                 | ${}^t \alpha^{-1}((\det X) X) \times {}^t \alpha^{-1}((\det X) X) = \alpha((\det X) X \times (\det X) X).$                                                                                          | ${}^t \alpha^{-1}((\det X) X) \times {}^t \alpha^{-1}((\det X) X) = \alpha((\det X) X \times (\det X) X).$                                                                                          |
| 0902.0431_E08238 (1)  | 40   | ■ 1.000 | $\alpha(X \circ Y) = \alpha X \circ \alpha Y.$                                                                                                                                                                                        | $\alpha(X \circ Y) = \alpha X \circ \alpha Y.$                                                                                                                                                      | $\alpha(X \circ Y) = \alpha X \circ \alpha Y.$                                                                                                                                                      |
| 0902.0431_E08239 (2)  | 40   | ■ 1.000 | $\operatorname{tr}(\alpha X, \alpha Y, \alpha Z) = \operatorname{tr}(X, Y, Z), (\alpha X, \alpha Y) = (X, Y).$                                                                                                                        | $\operatorname{tr}(\alpha X, \alpha Y, \alpha Z) = \operatorname{tr}(X, Y, Z), (\alpha X, \alpha Y) = (X, Y).$                                                                                      | $\operatorname{tr}(\alpha X, \alpha Y, \alpha Z) = \operatorname{tr}(X, Y, Z), (\alpha X, \alpha Y) = (X, Y).$                                                                                      |
| 0902.0431_E08241 (4)  | 40   | ■ 1.000 | $\det(\alpha X) = \det X, \alpha E = E.$                                                                                                                                                                                              | $\det(\alpha X) = \det X, \alpha E = E.$                                                                                                                                                            | $\det(\alpha X) = \det X, \alpha E = E.$                                                                                                                                                            |
| 0902.0431_E08242 (5)  | 40   | ■ 1.000 | $\alpha(X \times Y) = \alpha X \times \alpha Y.$                                                                                                                                                                                      | $\alpha(X \times Y) = \alpha X \times \alpha Y.$                                                                                                                                                    | $\alpha(X \times Y) = \alpha X \times \alpha Y.$                                                                                                                                                    |
| 0902.0431_E08245      | 40   | ■ 1.000 | $\alpha X \circ \alpha E = \alpha(X \times E) = \frac{1}{2} \alpha(\operatorname{tr}(X) E - X),$                                                                                                                                      | $\alpha X \circ \alpha E = \alpha(X \times E) = \frac{1}{2} \alpha(\operatorname{tr}(X) E - X),$                                                                                                    | $\alpha X \circ \alpha E = \alpha(X \times E) = \frac{1}{2} \alpha(\operatorname{tr}(X) E - X),$                                                                                                    |
| 0902.0431_E08246      | 40   | ■ 1.000 | $\alpha X \times P = \frac{1}{2} \operatorname{tr}(X) P - \frac{1}{2} \alpha X, \quad X \in \mathfrak{J}.$                                                                                                                            | $\alpha X \times P = \frac{1}{2} \operatorname{tr}(X) P - \frac{1}{2} \alpha X, \quad X \in \mathfrak{J}.$                                                                                          | $\alpha X \times P = \frac{1}{2} \operatorname{tr}(X) P - \frac{1}{2} \alpha X, \quad X \in \mathfrak{J}.$                                                                                          |
| 0902.0431_E08247      | 40   | ■ 1.000 | $\begin{aligned} & \frac{1}{2} (\rho_2 E_3 + \rho_3 E_2 - F_1(p_1)) \\ & = \frac{1}{2} (\lambda(\rho_1 E_1 + \rho_2 E_2 + \rho_3 E_3 + F_1(p_1) + F_2(p_2) + F_3(p_3)) - E_1) \end{aligned}$                                          | $\begin{aligned} & \frac{1}{2} (\rho_2 E_3 + \rho_3 E_2 - F_1(p_1)) \\ & = \frac{1}{2} (\lambda(\rho_1 E_1 + \rho_2 E_2 + \rho_3 E_3 + F_1(p_1) + F_2(p_2) + F_3(p_3)) - E_1) \end{aligned}$        | $\begin{aligned} & \frac{1}{2} (\rho_2 E_3 + \rho_3 E_2 - F_1(p_1)) \\ & = \frac{1}{2} (\lambda(\rho_1 E_1 + \rho_2 E_2 + \rho_3 E_3 + F_1(p_1) + F_2(p_2) + F_3(p_3)) - E_1), \end{aligned}$       |
| 0902.0431_E08248      | 41   | ■ 1.000 | $0 = \lambda \rho_1 - 1, \quad \rho_3 = \lambda \rho_2, \quad \rho_2 = \lambda \rho_3, \quad -p_1 = \lambda p_1, \quad 0 = \lambda p_2, \quad 0 = \lambda p_3.$                                                                       | $0 = \lambda \rho_1 - 1, \quad \rho_3 = \lambda \rho_2, \quad \rho_2 = \lambda \rho_3, \quad -p_1 = \lambda p_1, \quad 0 = \lambda p_2, \quad 0 = \lambda p_3.$                                     | $0 = \lambda \rho_1 - 1, \quad \rho_3 = \lambda \rho_2, \quad \rho_2 = \lambda \rho_3, \quad -p_1 = \lambda p_1, \quad 0 = \lambda p_2, \quad 0 = \lambda p_3.$                                     |
| 0902.0431_E08249      | 41   | ■ 1.000 | $-\frac{1}{2} \rho_1 F_1(1) = \frac{1}{2} (\mu(\rho_1 E_1 + \rho_2 E_2 + \rho_3 E_3) - F_1(1)),$                                                                                                                                      | $-\frac{1}{2} \rho_1 F_1(1) = \frac{1}{2} (\mu(\rho_1 E_1 + \rho_2 E_2 + \rho_3 E_3) - F_1(1)),$                                                                                                    | $-\frac{1}{2} \rho_1 F_1(1) = \frac{1}{2} (\mu(\rho_1 E_1 + \rho_2 E_2 + \rho_3 E_3) - F_1(1)),$                                                                                                    |
| 0902.0431_E08251      | 41   | ■ 1.000 | $(\alpha X, \alpha Y) = (X, Y).$                                                                                                                                                                                                      | $(\alpha X, \alpha Y) = (X, Y).$                                                                                                                                                                    | $(\alpha X, \alpha Y) = (X, Y).$                                                                                                                                                                    |

| Identifier       | Page | Conf.   | LaTeX source                                                                                                                                                                                                                                                                                                                                                        | Rendered                                                                                                                                                                                                                                                                      | Image                                                                                                                                                                                                                                                                         |
|------------------|------|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0902.0431_E00254 | 41   | ■ 1.000 | $\widetilde{\alpha}\left(\begin{array}{ccc}\xi_1 & x_3 & \bar{x}_2 \\ \bar{x}_3 & \xi_2 & x_1 \\ x_2 & \bar{x}_1 & \xi_3\end{array}\right)=\left(\begin{array}{ccc}\xi_1 & \alpha x_3 & \overline{\alpha x_2} \\ \overline{\alpha x_3} & \xi_2 & \alpha x_1 \\ \alpha x_2 & \overline{\alpha x_1} & \xi_3\end{array}\right).$                                       |                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                               |
| 0902.0431_E00258 | 42   | ■ 1.000 | $\mathfrak{M}^{\scriptscriptstyle -}=\left\{A\in M(3,\mathfrak{C})\mid A^{\scriptscriptstyle *}=-A\right\}.$                                                                                                                                                                                                                                                        | $\mathfrak{M}^{-}=\{A\in M(3,\mathfrak{C})\mid A^*=-A\}.$                                                                                                                                                                                                                     | $\mathfrak{M}^{-}=\{A\in M(3,\mathfrak{C})\mid A^*=-A\}.$                                                                                                                                                                                                                     |
| 0902.0431_E00261 | 42   | ■ 1.000 | $[X,\,X\,X]=a\,E\;.$                                                                                                                                                                                                                                                                                                                                                | $[X,XX]=aE.$                                                                                                                                                                                                                                                                  | $[X,XX]=aE.$                                                                                                                                                                                                                                                                  |
| 0902.0431_E00262 | 42   | ■ 1.000 | $a_{ij}=\sum_{k,l}\left(x_{ik}\left(x_{kl}x_{lj}\right)-\left(x_{ik}x_{kl}\right)x_{lj}\right)=-\sum_{k,l}\left\{x_{ik},x_{kl},x_{lj}\right\},\quad i,j=1,2,3.$                                                                                                                                                                                                     | $a_{ij}=\sum_{k,l}\left(x_{ik}\left(x_{kl}x_{lj}\right)-\left(x_{ik}x_{kl}\right)x_{lj}\right)=-\sum_{k,l}\left\{x_{ik},x_{kl},x_{lj}\right\},\quad i,j=1,2,3.$                                                                                                               | $a_{ij}=\sum_{k,l}\left(x_{ik}\left(x_{kl}x_{lj}\right)-\left(x_{ik}x_{kl}\right)x_{lj}\right)=-\sum_{k,l}\left\{x_{ik},x_{kl},x_{lj}\right\},\quad i,j=1,2,3.$                                                                                                               |
| 0902.0431_E00268 | 43   | ■ 1.000 | $([A,X],\,X\,X)=0,\quad X\in\mathfrak{J}\;.$                                                                                                                                                                                                                                                                                                                        | $([A,X],XX)=0,\quad X\in\mathfrak{J}.$                                                                                                                                                                                                                                        | $([A,X],XX)=0,\quad X\in\mathfrak{J}.$                                                                                                                                                                                                                                        |
| 0902.0431_E00270 | 44   | ■ 1.000 | $([A,X],\,Y\,Z+Z\,Y)+([A,Y],\,X\,Z+Z\,X)+([A,Z],\,X\,Y+Y\,X)=0,$                                                                                                                                                                                                                                                                                                    | $([A,X],YZ+ZY)+([A,Y],XZ+ZX)+([A,Z],XY+YX)=0,$                                                                                                                                                                                                                                | $([A,X],YZ+ZY)+([A,Y],XZ+ZX)+([A,Z],XY+YX)=0,$                                                                                                                                                                                                                                |
| 0902.0431_E00271 | 44   | ■ 1.000 | $A_{\scriptscriptstyle 1}(a)=\left(\begin{array}{ccc}\theta&\theta&\theta\\ \theta&-\bar{a}&\theta\end{array}\right),\quad A_{\scriptscriptstyle 2}(a)=\left(\begin{array}{ccc}\theta&\theta&-\bar{a}\\ \theta&\theta&\theta\end{array}\right),\quad A_{\scriptscriptstyle 3}(a)=\left(\begin{array}{ccc}\theta&a&\theta\\ \theta&\theta&\theta\end{array}\right).$ | $A_1(a)=\left(\begin{array}{ccc}0&0&0\\ 0&0&a\\ 0&-\bar{a}&0\end{array}\right),\quad A_2(a)=\left(\begin{array}{ccc}0&0&-\bar{a}\\ 0&0&0\\ a&0&0\end{array}\right),\quad A_3(a)=\left(\begin{array}{ccc}0&a&0\\ -\bar{a}&0&0\\ 0&0&0\end{array}\right).$                      | $A_1(a)=\left(\begin{array}{ccc}0&0&0\\ 0&0&a\\ 0&-\bar{a}&0\end{array}\right),\quad A_2(a)=\left(\begin{array}{ccc}0&0&-\bar{a}\\ 0&0&0\\ a&0&0\end{array}\right),\quad A_3(a)=\left(\begin{array}{ccc}0&a&0\\ -\bar{a}&0&0\\ 0&0&0\end{array}\right).$                      |
| 0902.0431_E00273 | 44   | ■ 1.000 | $\mathfrak{d}_{\scriptscriptstyle 4}=\left\{\delta\in\mathfrak{f}_4\mid\delta E_i=0,i=1,2,3\right\}$                                                                                                                                                                                                                                                                | $\mathfrak{d}_4=\{\delta\in\mathfrak{f}_4\mid\delta E_i=0,i=1,2,3\}$                                                                                                                                                                                                          | $\mathfrak{d}_4=\{\delta\in\mathfrak{f}_4\mid\delta E_i=0,i=1,2,3\}$                                                                                                                                                                                                          |
| 0902.0431_E00274 | 44   | ■ 1.000 | $\mathfrak{D}_{\scriptscriptstyle 4}=\left\{D\in\operatorname{Hom}_{\boldsymbol{\mathfrak{R}}}\left(\mathfrak{C}\right)\mid\left(Dx,y\right)+\left(x,Dy\right)=0\right\}$                                                                                                                                                                                           | $\mathfrak{D}_4=\{D\in\operatorname{Hom}_{\boldsymbol{\mathfrak{R}}}(\mathfrak{C})\mid(Dx,y)+(x,Dy)=0\}$                                                                                                                                                                      | $\mathfrak{D}_4=\{D\in\operatorname{Hom}_{\boldsymbol{\mathfrak{R}}}(\mathfrak{C})\mid(Dx,y)+(x,Dy)=0\}$                                                                                                                                                                      |
| 0902.0431_E00275 | 44   | ■ 1.000 | $\mathfrak{D}_{\scriptscriptstyle 4}\ni D_{\scriptscriptstyle 1}\longrightarrow\delta\in\mathfrak{d}_{\scriptscriptstyle 4}$                                                                                                                                                                                                                                        | $\mathfrak{D}_4\ni D_1\longrightarrow\delta\in\mathfrak{d}_4$                                                                                                                                                                                                                 | $\mathfrak{D}_4\ni D_1\longrightarrow\delta\in\mathfrak{d}_4$                                                                                                                                                                                                                 |
| 0902.0431_E00276 | 44   | ■ 1.000 | $\delta\left(\begin{array}{ccc}\xi_1 & x_3 & \bar{x}_2 \\ \bar{x}_3 & \xi_2 & x_1 \\ x_2 & \bar{x}_1 & \xi_3\end{array}\right)=\left(\begin{array}{ccc}0 & D_3x_3 & \overline{D_2x_2} \\ \overline{D_3x_3} & 0 & D_1x_1 \\ D_2x_2 & \overline{D_1x_1} & 0\end{array}\right),$                                                                                       | $\delta\left(\begin{array}{ccc}\xi_1 & x_3 & \bar{x}_2 \\ \bar{x}_3 & \xi_2 & x_1 \\ x_2 & \bar{x}_1 & \xi_3\end{array}\right)=\left(\begin{array}{ccc}0 & D_3x_3 & \overline{D_2x_2} \\ \overline{D_3x_3} & 0 & D_1x_1 \\ D_2x_2 & \overline{D_1x_1} & 0\end{array}\right),$ | $\delta\left(\begin{array}{ccc}\xi_1 & x_3 & \bar{x}_2 \\ \bar{x}_3 & \xi_2 & x_1 \\ x_2 & \bar{x}_1 & \xi_3\end{array}\right)=\left(\begin{array}{ccc}0 & D_3x_3 & \overline{D_2x_2} \\ \overline{D_3x_3} & 0 & D_1x_1 \\ D_2x_2 & \overline{D_1x_1} & 0\end{array}\right),$ |
| 0902.0431_E00277 | 44   | ■ 1.000 | $\left(D_{\scriptscriptstyle 1}x\right)y+x\left(D_{\scriptscriptstyle 2}y\right)=\overline{\left(D_{\scriptscriptstyle 3}\left(\overline{xy}\right)\right)},\quad x,y\in\mathfrak{C}.$                                                                                                                                                                              | $(D_1x)y+x(D_2y)=\overline{D_3(\overline{xy})},\quad x,y\in\mathfrak{C}.$                                                                                                                                                                                                     | $(D_1x)y+x(D_2y)=\overline{D_3(\overline{xy})},\quad x,y\in\mathfrak{C}.$                                                                                                                                                                                                     |
| 0902.0431_E00281 | 45   | ■ 1.000 | $\delta F_i(x)=F_i(D_ix),\quad x\in\mathfrak{C},$                                                                                                                                                                                                                                                                                                                   | $\delta F_i(x)=F_i(D_ix),\quad x\in\mathfrak{C},$                                                                                                                                                                                                                             | $\delta F_i(x)=F_i(D_ix),\quad x\in\mathfrak{C},$                                                                                                                                                                                                                             |
| 0902.0431_E00284 | 45   | ■ 1.000 | $\delta=D+\widetilde{A},\quad D\in\mathfrak{d},A\in\mathfrak{M}^{-},\operatorname{diag}A=0,$                                                                                                                                                                                                                                                                        | $\delta=D+\widetilde{A},\quad D\in\mathfrak{d}_4,A\in\mathfrak{M}^{-},\operatorname{diag}A=0,$                                                                                                                                                                                | $\delta=D+\widetilde{A},\quad D\in\mathfrak{d}_4,A\in\mathfrak{M}^{-},\operatorname{diag}A=0,$                                                                                                                                                                                |
| 0902.0431_E00285 | 45   | ■ 1.000 | $\operatorname{dim}\mathfrak{f}_{\scriptscriptstyle 4}=28+24=52\;.$                                                                                                                                                                                                                                                                                                 | $\dim\mathfrak{f}_4=28+24=52.$                                                                                                                                                                                                                                                | $\dim\mathfrak{f}_4=28+24=52.$                                                                                                                                                                                                                                                |

| Identifier       | Page | Conf.   | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Rendered                                                                                                                                                                                                                                                                                                            | Image                                                                                                                                                                                                                                                                                                               |
|------------------|------|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0902.0431_E00287 | 46   | ■ 1.000 | $\begin{array}{l} \backslash \mathrm{delta} \, E_{\{1\}} = \backslash \mathrm{left}(\backslash \mathrm{begin}\{array\}\{ccc\} \, 0 \, \& \, -a_{\{3\}} \, \& \\ \backslash \mathrm{bar}\{a\}_{\{2\}} \, \backslash \backslash \, -\backslash \mathrm{bar}\{a\}_{\{3\}} \, \& \, 0 \, \& \, 0 \, \backslash \backslash \, a_{\{2\}} \, \& \, 0 \, \& \, 0 \\ \backslash \mathrm{end}\{array\}\backslash \mathrm{right}), \, \backslash \mathrm{delta} \, E_{\{2\}} = \backslash \mathrm{left}(\backslash \mathrm{begin}\{array\}\{ccc\} \\ 0 \, \& \, a_{\{3\}} \, \& \, 0 \, \backslash \backslash \, \backslash \mathrm{bar}\{a\}_{\{3\}} \, \& \, 0 \, \& \, -a_{\{1\}} \, \backslash \backslash \, 0 \, \& \, - \\ \backslash \mathrm{bar}\{a\}_{\{1\}} \, \& \, 0 \, \backslash \mathrm{end}\{array\}\backslash \mathrm{right}), \, \backslash \mathrm{delta} \, E_{\{3\}} = \backslash \mathrm{left}( \\ \backslash \mathrm{begin}\{array\}\{ccc\} \, 0 \, \& \, 0 \, \& \, -\backslash \mathrm{bar}\{a\}_{\{2\}} \, \backslash \backslash \, 0 \, \& \, 0 \, \& \, a_{\{1\}} \\ \backslash \backslash \, -a_{\{2\}} \, \& \, \backslash \mathrm{bar}\{a\}_{\{1\}} \, \& \, 0 \, \backslash \mathrm{end}\{array\}\backslash \mathrm{right}), \end{array}$ | $\delta E_1 = \begin{pmatrix} 0 & -a_3 & \bar{a}_2 \\ -\bar{a}_3 & 0 & 0 \\ a_2 & 0 & 0 \end{pmatrix}, \delta E_2 = \begin{pmatrix} 0 & a_3 & 0 \\ \bar{a}_3 & 0 & -a_1 \\ 0 & -\bar{a}_1 & 0 \end{pmatrix}, \delta E_3 = \begin{pmatrix} 0 & 0 & -\bar{a}_2 \\ 0 & 0 & a_1 \\ -a_2 & \bar{a}_1 & 0 \end{pmatrix},$ | $\delta E_1 = \begin{pmatrix} 0 & -a_3 & \bar{a}_2 \\ -\bar{a}_3 & 0 & 0 \\ a_2 & 0 & 0 \end{pmatrix}, \delta E_2 = \begin{pmatrix} 0 & a_3 & 0 \\ \bar{a}_3 & 0 & -a_1 \\ 0 & -\bar{a}_1 & 0 \end{pmatrix}, \delta E_3 = \begin{pmatrix} 0 & 0 & -\bar{a}_2 \\ 0 & 0 & a_1 \\ -a_2 & \bar{a}_1 & 0 \end{pmatrix},$ |
| 0902.0431_E00288 | 46   | ■ 1.000 | $\backslash \mathrm{delta} \, E_{\{i\}} = \backslash \mathrm{widetilde}\{A\} \, E_{\{i\}}, \, \backslash \mathrm{quad} \, i = 1, 2, 3 \, .$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | $\delta E_i = \widetilde{A} E_i, \quad i = 1, 2, 3.$                                                                                                                                                                                                                                                                | $\delta E_i = \widetilde{A} E_i, \quad i = 1, 2, 3.$                                                                                                                                                                                                                                                                |
| 0902.0431_E00290 | 46   | ■ 1.000 | $\overline{\mathrm{X}_{\{1\}} + i \, \mathrm{X}_{\{2\}}} = \mathrm{bar}\{X\}_{\{1\}} + i \, \mathrm{bar}\{X\}_{\{2\}}, \, \backslash \mathrm{quad} \, \backslash \mathrm{tau} \backslash \mathrm{left}(\mathrm{X}_{\{1\}} + i \, \mathrm{X}_{\{2\}} \backslash \mathrm{right}) = \mathrm{X}_{\{1\}} - i \, \mathrm{X}_{\{2\}}, \, \backslash \mathrm{quad} \, \mathrm{X}_{\{i\}} \, \backslash \mathrm{in} \, \backslash \mathrm{mathfrak}\{J\} \, .$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | $\overline{X_1 + i X_2} = \bar{X}_1 + i \bar{X}_2, \quad \tau(X_1 + i X_2) = X_1 - i X_2, \quad X_i \in \mathfrak{J}.$                                                                                                                                                                                              | $\overline{X_1 + i X_2} = \bar{X}_1 + i \bar{X}_2, \quad \tau(X_1 + i X_2) = X_1 - i X_2, \quad X_i \in \mathfrak{J}.$                                                                                                                                                                                              |
| 0902.0431_E00291 | 46   | ■ 1.000 | $\backslash \mathrm{tau}(\mathrm{X} \, \mathrm{circ} \, \mathrm{Y}) = \backslash \mathrm{tau} \, \mathrm{X} \, \mathrm{circ} \, \backslash \mathrm{tau} \, \mathrm{Y}, \, \backslash \mathrm{quad} \, \backslash \mathrm{tau}(\mathrm{X} \, \mathrm{times} \, \mathrm{Y}) = \backslash \mathrm{tau} \, \mathrm{X} \, \mathrm{times} \, \backslash \mathrm{tau} \, \mathrm{Y}, \, \backslash \mathrm{quad} \, \mathrm{X}, \, \mathrm{Y} \, \backslash \mathrm{in} \, \backslash \mathrm{mathfrak}\{J\}^{\mathrm{C}}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | $\tau(X \circ Y) = \tau X \circ \tau Y, \quad \tau(X \times Y) = \tau X \times \tau Y, \quad X, Y \in \mathfrak{J}^C.$                                                                                                                                                                                              | $\tau(X \circ Y) = \tau X \circ \tau Y, \quad \tau(X \times Y) = \tau X \times \tau Y, \quad X, Y \in \mathfrak{J}^C.$                                                                                                                                                                                              |
| 0902.0431_E00294 | 47   | ■ 1.000 | $=(A,(\backslash \mathrm{operatorname}\{\det\} \, X) \, E)=(\backslash \mathrm{operatorname}\{\det\} \, X)(A, E)=(\backslash \mathrm{operatorname}\{\det\} \, X) \, \backslash \mathrm{operatorname}\{\mathrm{tr}\}(A)=0, \, \backslash \mathrm{quad} \, X \, \backslash \mathrm{in} \, \backslash \mathrm{mathfrak}\{J\}^{\mathrm{C}} \, .$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | $=(A,(\det X)E)=(\det X)(A,E)=(\det X)\mathrm{tr}(A)=0, \quad X \in \mathfrak{J}^C.$                                                                                                                                                                                                                                | $=(A,(\det X)E)=(\det X)(A,E)=(\det X)\mathrm{tr}(A)=0, \quad X \in \mathfrak{J}^C.$                                                                                                                                                                                                                                |
| 0902.0431_E00303 | 48   | ■ 1.000 | $\backslash \mathrm{widetilde}\{A\}_{\{1\}}(1) + \backslash \mathrm{widetilde}\{A\}_{\{2\}}(b) + \backslash \mathrm{widetilde}\{A\}_{\{3\}}(c) \, \backslash \mathrm{in} \, \backslash \mathrm{mathfrak}\{a\} \, .$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | $\widetilde{A}_1(1) + \widetilde{A}_2(b) + \widetilde{A}_3(c) \in \mathfrak{a}.$                                                                                                                                                                                                                                    | $\widetilde{A}_1(1) + \widetilde{A}_2(b) + \widetilde{A}_3(c) \in \mathfrak{a}.$                                                                                                                                                                                                                                    |
| 0902.0431_E00305 | 49   | ■ 1.000 | $-\backslash \mathrm{widetilde}\{A\}_{\{3\}}(\backslash \mathrm{bar}\{b\}) + \backslash \mathrm{widetilde}\{A\}_{\{2\}}(\backslash \mathrm{bar}\{c\}) \, \backslash \mathrm{in} \, \backslash \mathrm{mathfrak}\{a\} \, \backslash \mathrm{quad} \, \backslash \mathrm{text} \, \{ \, \mathrm{and} \, \mathrm{then} \, \} \, \backslash \mathrm{quad} \, \backslash \mathrm{widetilde}\{A\}_{\{3\}}(1) + \backslash \mathrm{widetilde}\{A\}_{\{2\}} \backslash \mathrm{left}(c^{\wedge} \backslash \mathrm{prime} \backslash \mathrm{right}) \, \backslash \mathrm{in} \, \backslash \mathrm{mathfrak}\{a\}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | $-\widetilde{A}_3(\bar{b}) + \widetilde{A}_2(\bar{c}) \in \mathfrak{a} \quad \text{and then} \quad \widetilde{A}_3(1) + \widetilde{A}_2(c') \in \mathfrak{a}$                                                                                                                                                       | $-\widetilde{A}_3(\bar{b}) + \widetilde{A}_2(\bar{c}) \in \mathfrak{a} \quad \text{and then} \quad \widetilde{A}_3(1) + \widetilde{A}_2(c') \in \mathfrak{a}$                                                                                                                                                       |
| 0902.0431_E00306 | 49   | ■ 1.000 | $[\backslash \mathrm{delta}, [\backslash \mathrm{widetilde}\{A\}, \backslash \mathrm{widetilde}\{B\}]] = [\backslash \mathrm{widetilde}\{ \backslash \mathrm{delta} \, A\}, \backslash \mathrm{widetilde}\{B\}] + [\backslash \mathrm{widetilde}\{A\}, \backslash \mathrm{widetilde}\{ \backslash \mathrm{delta} \, B\}] \, .$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | $[\delta, [\widetilde{A}, \widetilde{B}]] = [\widetilde{\delta A}, \widetilde{B}] + [\widetilde{A}, \widetilde{\delta B}].$                                                                                                                                                                                         | $[\delta, [\widetilde{A}, \widetilde{B}]] = [\widetilde{\delta A}, \widetilde{B}] + [\widetilde{A}, \widetilde{\delta B}].$                                                                                                                                                                                         |
| 0902.0431_E00308 | 49   | ■ 1.000 | $\mathrm{F}_{\{i\}}(1) \, \backslash \mathrm{xrightarrow}\{ \backslash \mathrm{widetilde}\{A\}_{\{i\}}(1) \} \, \mathrm{E}_{\{i+1\}} - \mathrm{E}_{\{i+2\}} \, \backslash \mathrm{xrightarrow}\{ \backslash \mathrm{widetilde}\{A\}_{\{i+2\}}(1) \} \, \mathrm{F}_{\{i+2\}}(1) \, \backslash \mathrm{longrightarrow} \, \backslash \mathrm{cdots} \, \backslash \mathrm{longrightarrow} \, \mathrm{F}_{\{i+1\}}(1),$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | $F_i(1) \xrightarrow{\widetilde{A}_i(1)} E_{i+1} - E_{i+2} \xrightarrow{\widetilde{A}_{i+2}(1)} F_{i+2}(1) \longrightarrow \cdots \longrightarrow F_{i+1}(1),$                                                                                                                                                      | $F_i(1) \xrightarrow{\widetilde{A}_i(1)} E_{i+1} - E_{i+2} \xrightarrow{\widetilde{A}_{i+2}(1)} F_{i+2}(1) \longrightarrow \cdots \longrightarrow F_{i+1}(1),$                                                                                                                                                      |
| 0902.0431_E00309 | 49   | ■ 1.000 | $0 \, \mathrm{neq} \, X = \backslash \mathrm{x}\mathrm{i} \backslash \mathrm{left}(\mathrm{E}_{\{1\}} - \mathrm{E}_{\{2\}} \backslash \mathrm{right}) + \backslash \mathrm{eta} \, \backslash \mathrm{left}(\mathrm{E}_{\{2\}} - \mathrm{E}_{\{3\}} \backslash \mathrm{right}) + \mathrm{F}_{\{1\}} \backslash \mathrm{left}(x_{\{1\}} \backslash \mathrm{right}) + \mathrm{F}_{\{2\}} \backslash \mathrm{left}(x_{\{2\}} \backslash \mathrm{right}) + \mathrm{F}_{\{3\}} \backslash \mathrm{left}(x_{\{3\}} \backslash \mathrm{right}) \, \backslash \mathrm{in} \, W \, .$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | $0 \neq X = \xi(E_1 - E_2) + \eta(E_2 - E_3) + F_1(x_1) + F_2(x_2) + F_3(x_3) \in W.$                                                                                                                                                                                                                               | $0 \neq X = \xi(E_1 - E_2) + \eta(E_2 - E_3) + F_1(x_1) + F_2(x_2) + F_3(x_3) \in W.$                                                                                                                                                                                                                               |
| 0902.0431_E00310 | 50   | ■ 1.000 | $\mathrm{F}_{\{1\}}(1) + \mathrm{F}_{\{2\}} \backslash \mathrm{left}(y_{\{2\}} \backslash \mathrm{right}) + \mathrm{F}_{\{3\}} \backslash \mathrm{left}(y_{\{3\}} \backslash \mathrm{right}) \, \backslash \mathrm{in} \, W \, .$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | $F_1(1) + F_2(y_2) + F_3(y_3) \in W.$                                                                                                                                                                                                                                                                               | $F_1(1) + F_2(y_2) + F_3(y_3) \in W.$                                                                                                                                                                                                                                                                               |
| 0902.0431_E00312 | 50   | ■ 1.000 | $X = \backslash \mathrm{x}\mathrm{i} \backslash \mathrm{left}(\mathrm{E}_{\{1\}} - \mathrm{E}_{\{2\}} \backslash \mathrm{right}) + \backslash \mathrm{eta} \backslash \mathrm{left}(\mathrm{E}_{\{2\}} - \mathrm{E}_{\{3\}} \backslash \mathrm{right}) \, \backslash \mathrm{in} \, W,$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | $X = \xi(E_1 - E_2) + \eta(E_2 - E_3) \in W,$                                                                                                                                                                                                                                                                       | $X = \xi(E_1 - E_2) + \eta(E_2 - E_3) \in W,$                                                                                                                                                                                                                                                                       |
| 0902.0431_E00313 | 50   | ■ 1.000 | $([\backslash \mathrm{widetilde}\{A\}, \backslash \mathrm{widetilde}\{B\}] \, C, D) = ([\backslash \mathrm{widetilde}\{C\}, \backslash \mathrm{widetilde}\{D\}] \, A, B) \, .$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | $([\widetilde{A}, \widetilde{B}]C, D) = ([\widetilde{C}, \widetilde{D}]A, B).$                                                                                                                                                                                                                                      | $([\widetilde{A}, \widetilde{B}]C, D) = ([\widetilde{C}, \widetilde{D}]A, B).$                                                                                                                                                                                                                                      |
| 0902.0431_E00316 | 50   | ■ 1.000 | $\backslash \mathrm{left}(\backslash \mathrm{delta}_{\{1\}}, \backslash \mathrm{delta}_{\{2\}} \backslash \mathrm{right})_{\{4\}} = \sum_{i,j} \backslash \mathrm{left}([\backslash \mathrm{widetilde}\{A\}_{\{i\}}, \backslash \mathrm{widetilde}\{B\}_{\{j\}} \backslash \mathrm{right}] \, C_{\{j\}}, D_{\{j\}} \backslash \mathrm{right}) = \sum_{i,j} \backslash \mathrm{left}(\backslash \mathrm{left}[\backslash \mathrm{widetilde}\{C\}_{\{j\}}, \backslash \mathrm{widetilde}\{D\}_{\{j\}} \backslash \mathrm{right}] \, A_{\{i\}}, B_{\{i\}} \backslash \mathrm{right}),$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | $(\delta_1, \delta_2)_4 = \sum_{i,j} \left( [\widetilde{A}_i, \widetilde{B}_j] C_j, D_j \right) = \sum_{i,j} \left( [\widetilde{C}_j, \widetilde{D}_j] A_i, B_i \right),$                                                                                                                                           | $(\delta_1, \delta_2)_4 = \sum_{i,j} ([\widetilde{A}_i, \widetilde{B}_i] C_j, D_j) = \sum_{i,j} ([\widetilde{C}_j, \widetilde{D}_j] A_i, B_i),$                                                                                                                                                                     |
| 0902.0431_E00326 | 52   | ■ 1.000 | $\backslash \mathrm{mathfrak}\{D\}_{\{4\}} \, \, \, \wedge^{\mathrm{C}} = \backslash \mathrm{left}\{D \, \backslash \mathrm{in} \, \backslash \mathrm{operatorname}\{\mathrm{Hom}\}_{\{C\}} \backslash \mathrm{left}(\backslash \mathrm{mathfrak}\{C\}^{\mathrm{C}} \backslash \mathrm{right}) \, \backslash \mathrm{mid}(D \, x, y) + (x, D \, y) = 0 \backslash \mathrm{right}\backslash \, .$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | $\mathfrak{D}_4^C = \{D \in \mathrm{Hom}_C(\mathfrak{C}^C) \mid (Dx, y) + (x, Dy) = 0\}.$                                                                                                                                                                                                                           | $\mathfrak{D}_4^C = \{D \in \mathrm{Hom}_C(\mathfrak{C}^C) \mid (Dx, y) + (x, Dy) = 0\}.$                                                                                                                                                                                                                           |
| 0902.0431_E00327 | 52   | ■ 1.000 | $\backslash \mathrm{mathfrak}\{h\} = \backslash \mathrm{left}\{H = \sum_{k=0}^3 \, \backslash \mathrm{lambda}_{\{i\}} \, H_{\{k\}} \, \backslash \mathrm{mid} \, \backslash \mathrm{lambda}_{\{k\}} \, \backslash \mathrm{in} \, C \backslash \mathrm{right}\backslash \}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | $\mathfrak{h} = \left\{ H = \sum_{k=0}^3 \lambda_i H_k \mid \lambda_k \in C \right\}$                                                                                                                                                                                                                               | $\mathfrak{h} = \{ H = \sum_{k=0}^3 \lambda_i H_k \mid \lambda_k \in C \}$                                                                                                                                                                                                                                          |

| Identifier       | Page | Conf.   | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Rendered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Image                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|------------------|------|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0902.0431_E00331 | 53   | ■ 1.000 | $\nu=\pi \kappa=\frac{1}{2}\left(\begin{array}{rrrr} -1 & -1 & 1 & -1 \\ 1 & 1 & 1 & -1 \\ -1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{array}\right), \quad \pi=\frac{1}{2}\left(\begin{array}{rrrr} 1 & -1 & 1 & -1 \\ -1 & 1 & 1 & -1 \\ 1 & 1 & 1 & 1 \\ -1 & -1 & 1 & 1 \end{array}\right)$                                                                                                                                                                                                                                                                                                                    | $\nu=\pi\kappa=\frac{1}{2}\left(\begin{array}{rrrr} -1 & -1 & 1 & -1 \\ 1 & 1 & 1 & -1 \\ -1 & 1 & 1 & 1 \\ 1 & -1 & 1 & 1 \end{array}\right), \quad \pi=\frac{1}{2}\left(\begin{array}{rrrr} 1 & -1 & 1 & -1 \\ -1 & 1 & 1 & -1 \\ 1 & 1 & 1 & 1 \\ -1 & -1 & 1 & 1 \end{array}\right)$                                                                                                                                                                                                                                                                                                                    | $\nu=\pi\kappa=\frac{1}{2}\left(\begin{array}{rrrr} -1 & -1 & 1 & -1 \\ 1 & 1 & 1 & -1 \\ -1 & 1 & 1 & 1 \\ 1 & -1 & 1 & 1 \end{array}\right), \quad \pi=\frac{1}{2}\left(\begin{array}{rrrr} 1 & -1 & 1 & -1 \\ -1 & 1 & 1 & -1 \\ 1 & 1 & 1 & 1 \\ -1 & -1 & 1 & 1 \end{array}\right)$                                                                                                                                                                                                                                                                                                                    |
| 0902.0431_E00334 | 53   | ■ 1.000 | $\pm\left(\lambda_k-\lambda_l\right), \quad \pm\left(\lambda_k+\lambda_l\right), \quad 0 \leq k < l \leq 3$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | $\pm\left(\lambda_k-\lambda_l\right), \quad \pm\left(\lambda_k+\lambda_l\right), \quad 0 \leq k < l \leq 3$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | $\pm\left(\lambda_k-\lambda_l\right), \quad \pm\left(\lambda_k+\lambda_l\right), \quad 0 \leq k < l \leq 3$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 0902.0431_E00336 | 54   | ■ 1.000 | $\begin{aligned} \nu H = & \nu\left(\sum_{k=0}^3 \lambda_k H_k\right) \\ & =\frac{1}{2}\left(\lambda_0\left(-H_0+H_1-H_2+H_3\right)+\lambda_1\left(-H_0+H_1+H_2-H_3\right)\right. \\ & \quad \left.+\lambda_2\left(H_0+H_1+H_2+H_3\right)+\lambda_3\left(-H_0-H_1+H_2+H_3\right)\right) \\ & =\frac{1}{2}\left(-\lambda_0-\lambda_1+\lambda_2-\lambda_3\right) H_0+\frac{1}{2}\left(\lambda_0+\lambda_1+\lambda_2-\lambda_3\right) H_1 \\ & \quad +\frac{1}{2}\left(-\lambda_0+\lambda_1+\lambda_2+\lambda_3\right) H_2+\frac{1}{2}\left(\lambda_0-\lambda_1+\lambda_2+\lambda_3\right) H_3, \end{aligned}$ | $\begin{aligned} \nu H = & \nu\left(\sum_{k=0}^3 \lambda_k H_k\right) \\ & =\frac{1}{2}\left(\lambda_0\left(-H_0+H_1-H_2+H_3\right)+\lambda_1\left(-H_0+H_1+H_2-H_3\right)\right. \\ & \quad \left.+\lambda_2\left(H_0+H_1+H_2+H_3\right)+\lambda_3\left(-H_0-H_1+H_2+H_3\right)\right) \\ & =\frac{1}{2}\left(-\lambda_0-\lambda_1+\lambda_2-\lambda_3\right) H_0+\frac{1}{2}\left(\lambda_0+\lambda_1+\lambda_2-\lambda_3\right) H_1 \\ & \quad +\frac{1}{2}\left(-\lambda_0+\lambda_1+\lambda_2+\lambda_3\right) H_2+\frac{1}{2}\left(\lambda_0-\lambda_1+\lambda_2+\lambda_3\right) H_3, \end{aligned}$ | $\begin{aligned} \nu H = & \nu\left(\sum_{k=0}^3 \lambda_k H_k\right) \\ & =\frac{1}{2}\left(\lambda_0\left(-H_0+H_1-H_2+H_3\right)+\lambda_1\left(-H_0+H_1+H_2-H_3\right)\right. \\ & \quad \left.+\lambda_2\left(H_0+H_1+H_2+H_3\right)+\lambda_3\left(-H_0-H_1+H_2+H_3\right)\right) \\ & =\frac{1}{2}\left(-\lambda_0-\lambda_1+\lambda_2-\lambda_3\right) H_0+\frac{1}{2}\left(\lambda_0+\lambda_1+\lambda_2-\lambda_3\right) H_1 \\ & \quad +\frac{1}{2}\left(-\lambda_0+\lambda_1+\lambda_2+\lambda_3\right) H_2+\frac{1}{2}\left(\lambda_0-\lambda_1+\lambda_2+\lambda_3\right) H_3, \end{aligned}$ |
| 0902.0431_E00338 | 54   | ■ 1.000 | $\alpha_1=\lambda_0-\lambda_1, \quad \alpha_2=\lambda_1-\lambda_2, \quad \alpha_3=\lambda_2, \quad \alpha_4=\frac{1}{2}\left(-\lambda_0-\lambda_1-\lambda_2+\lambda_3\right)$                                                                                                                                                                                                                                                                                                                                                                                                                               | $\alpha_1=\lambda_0-\lambda_1, \quad \alpha_2=\lambda_1-\lambda_2, \quad \alpha_3=\lambda_2, \quad \alpha_4=\frac{1}{2}\left(-\lambda_0-\lambda_1-\lambda_2+\lambda_3\right)$                                                                                                                                                                                                                                                                                                                                                                                                                               | $\alpha_1=\lambda_0-\lambda_1, \quad \alpha_2=\lambda_1-\lambda_2, \quad \alpha_3=\lambda_2, \quad \alpha_4=\frac{1}{2}\left(-\lambda_0-\lambda_1-\lambda_2+\lambda_3\right)$                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 0902.0431_E00339 | 54   | ■ 1.000 | $\mu=2 \alpha_1+3 \alpha_2+4 \alpha_3+2 \alpha_4$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | $\mu=2 \alpha_1+3 \alpha_2+4 \alpha_3+2 \alpha_4$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | $\mu=2 \alpha_1+3 \alpha_2+4 \alpha_3+2 \alpha_4$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 0902.0431_E00342 | 55   | ■ 1.000 | $\mathfrak{h}_{\boldsymbol{R}}=\left\{H=\sum_{k=0}^3 \lambda_k H_k \mid \lambda_k \in \boldsymbol{R}\right\}.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | $\mathfrak{h}_{\boldsymbol{R}}=\left\{H=\sum_{k=0}^3 \lambda_k H_k \mid \lambda_k \in \boldsymbol{R}\right\}.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | $\mathfrak{h}_{\boldsymbol{R}}=\left\{H=\sum_{k=0}^3 \lambda_k H_k \mid \lambda_k \in \boldsymbol{R}\right\}.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 0902.0431_E00347 | 56   | ■ 1.000 | $\left(\alpha_1, \alpha_1\right)=B_4\left(H_{\alpha_1}, H_{\alpha_1}\right)=18 \frac{1}{18} \frac{1}{18}(1+1)=\frac{1}{9}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | $\left(\alpha_1, \alpha_1\right)=B_4\left(H_{\alpha_1}, H_{\alpha_1}\right)=18 \frac{1}{18} \frac{1}{18}(1+1)=\frac{1}{9}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | $\left(\alpha_1, \alpha_1\right)=B_4\left(H_{\alpha_1}, H_{\alpha_1}\right)=18 \frac{1}{18} \frac{1}{18}(1+1)=\frac{1}{9}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 0902.0431_E00352 | 57   | ■ 1.000 | $\alpha F_i(x)=F_i\left(\alpha_i x\right), \quad x \in \mathfrak{C},$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | $\alpha F_i(x)=F_i\left(\alpha_i x\right), \quad x \in \mathfrak{C},$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | $\alpha F_i(x)=F_i\left(\alpha_i x\right), \quad x \in \mathfrak{C},$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 0902.0431_E00353 | 57   | ■ 1.000 | $\left(\alpha_i x, \alpha_i y\right)=(x, y), \quad x, y \in \mathfrak{C}.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | $\left(\alpha_i x, \alpha_i y\right)=(x, y), \quad x, y \in \mathfrak{C}.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | $\left(\alpha_i x, \alpha_i y\right)=(x, y), \quad x, y \in \mathfrak{C}.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 0902.0431_E00355 | 57   | ■ 1.000 | $\left(F_4\right)_{E_1}=\left\{\alpha \in F_4 \mid \alpha E_1=E_1\right\}.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | $\left(F_4\right)_{E_1}=\left\{\alpha \in F_4 \mid \alpha E_1=E_1\right\}.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | $\left(F_4\right)_{E_1}=\left\{\alpha \in F_4 \mid \alpha E_1=E_1\right\}.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

| Identifier       | Page | Conf.   | LaTeX source                                                                                                                                                                                                                                                             | Rendered                                                                                                                                                                                                         | Image                                                                                                                                                                                                            |
|------------------|------|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0902.0431_E00356 | 57   | ■ 1.000 | $\mathfrak{J}_{01}=\left\{X\in\mathfrak{J}\mid\mathfrak{J}\mid E_1\circ X=0,\mathrm{tr}(X)=0\right\}$<br>$\circ X=0$ , $\operatorname{tr}(X)=0$<br>$\left(\begin{array}{ccc}0&0&0\\0&\xi&x\\0&\bar{x}&-\xi\end{array}\right)$<br>$\xi\in\boldsymbol{R},x\in\mathfrak{C}$ | $\mathfrak{J}_{01}=\{X\in\mathfrak{J}\mid E_1\circ X=0,\mathrm{tr}(X)=0\}=\left\{\left(\begin{array}{ccc}0&0&0\\0&\xi&x\\0&\bar{x}&-\xi\end{array}\right)\middle \xi\in\boldsymbol{R},x\in\mathfrak{C}\right\},$ | $\mathfrak{J}_{01}=\{X\in\mathfrak{J}\mid E_1\circ X=0,\mathrm{tr}(X)=0\}=\left\{\left(\begin{array}{ccc}0&0&0\\0&\xi&x\\0&\bar{x}&-\xi\end{array}\right)\middle \xi\in\boldsymbol{R},x\in\mathfrak{C}\right\},$ |
| 0902.0431_E00357 | 58   | ■ 1.000 | $\mathfrak{J}_{23}=\left\{Y\in\mathfrak{J}\mid 2E_1\circ Y=Y\right\}$<br>$\circ Y=Y$<br>$\left(\begin{array}{ccc}0&y_3&\bar{y}_2\\y_3&0&0\\y_2&0&0\end{array}\right)$<br>$y_2,y_3\in\mathfrak{C}$                                                                        | $\mathfrak{J}_{23}=\{Y\in\mathfrak{J}\mid 2E_1\circ Y=Y\}=\left\{\left(\begin{array}{ccc}0&y_3&\bar{y}_2\\y_3&0&0\\y_2&0&0\end{array}\right)\middle y_2,y_3\in\mathfrak{C}\right\}.$                             | $\mathfrak{J}_{23}=\{Y\in\mathfrak{J}\mid 2E_1\circ Y=Y\}=\left\{\left(\begin{array}{ccc}0&y_3&\bar{y}_2\\y_3&0&0\\y_2&0&0\end{array}\right)\middle y_2,y_3\in\mathfrak{C}\right\}.$                             |
| 0902.0431_E00358 | 58   | ■ 1.000 | $Z_0=2X_0\circ Y_0$                                                                                                                                                                                                                                                      | $Z_0=2X_0\circ Y_0.$                                                                                                                                                                                             | $Z_0=2X_0\circ Y_0.$                                                                                                                                                                                             |
| 0902.0431_E00359 | 58   | ■ 1.000 | $Y_1=-2Z_0\circ X_1$ , $Z_2=-2X_2\circ Y_0$ , $X_3=-2Y_1\circ Z_2$                                                                                                                                                                                                       | $Y_1=-2Z_0\circ X_1$ , $Z_2=-2X_2\circ Y_0$ , $X_3=-2Y_1\circ Z_2.$                                                                                                                                              | $Y_1=-2Z_0\circ X_1$ , $Z_2=-2X_2\circ Y_0$ , $X_3=-2Y_1\circ Z_2.$                                                                                                                                              |
| 0902.0431_E00360 | 58   | ■ 1.000 | $Z_4=-2X_4\circ Y_0$ , $Y_2=-2Z_0\circ X_2$ , $Y_3=-2Z_0\circ X_3$ ,<br>$X_5=-2Y_1\circ Z_4$ , $X_6=2Y_2\circ Z_4$ , $X_7=-2Y_3\circ Z_4$                                                                                                                                | $Z_4=-2X_4\circ Y_0$ , $Y_2=-2Z_0\circ X_2$ , $Y_3=-2Z_0\circ X_3$ ,<br>$X_5=-2Y_1\circ Z_4$ , $X_6=2Y_2\circ Z_4$ , $X_7=-2Y_3\circ Z_4$                                                                        | $Z_4=-2X_4\circ Y_0$ , $Y_2=-2Z_0\circ X_2$ , $Y_3=-2Z_0\circ X_3$ ,<br>$X_5=-2Y_1\circ Z_4$ , $X_6=2Y_2\circ Z_4$ , $X_7=-2Y_3\circ Z_4$                                                                        |
| 0902.0431_E00363 | 58   | ■ 1.000 | $\alpha(X\circ Y)=\alpha X\circ\alpha Y$ , $X,Y\in\mathfrak{J}$                                                                                                                                                                                                          | $\alpha(X\circ Y)=\alpha X\circ\alpha Y$ , $X,Y\in\mathfrak{J}.$                                                                                                                                                 | $\alpha(X\circ Y)=\alpha X\circ\alpha Y$ , $X,Y\in\mathfrak{J}.$                                                                                                                                                 |
| 0902.0431_E00366 | 59   | ■ 1.000 | $(F_4)_{E_1}/\{1,\sigma\}\cong SO(9)$                                                                                                                                                                                                                                    | $(F_4)_{E_1}/\{1,\sigma\}\cong SO(9).$                                                                                                                                                                           | $(F_4)_{E_1}/\{1,\sigma\}\cong SO(9).$                                                                                                                                                                           |
| 0902.0431_E00367 | 60   | ■ 1.000 | $f(\alpha')=\varepsilon\alpha'\varepsilon^{-1}$ , $(\epsilon x=\bar{x},x\in\mathfrak{C})$                                                                                                                                                                                | $f(\alpha')=\varepsilon\alpha'\varepsilon^{-1}$ , $(\epsilon x=\bar{x},x\in\mathfrak{C})$                                                                                                                        | $f(\alpha')=\varepsilon\alpha'\varepsilon^{-1}$ , $(\epsilon x=\bar{x},x\in\mathfrak{C})$                                                                                                                        |
| 0902.0431_E00369 | 61   | ■ 1.000 | $\alpha X=\left(\begin{array}{ccc}\xi_1&0&0\\0&\xi_2&0\\0&0&\xi_3\end{array}\right)$ , $\xi_i\in\boldsymbol{R}$                                                                                                                                                          | $\alpha X=\left(\begin{array}{ccc}\xi_1&0&0\\0&\xi_2&0\\0&0&\xi_3\end{array}\right)$ , $\xi_i\in\boldsymbol{R}$                                                                                                  | $\alpha X=\left(\begin{array}{ccc}\xi_1&0&0\\0&\xi_2&0\\0&0&\xi_3\end{array}\right)$ , $\xi_i\in\boldsymbol{R}.$                                                                                                 |
| 0902.0431_E00373 | 62   | ■ 1.000 | $\mathfrak{C}P_2=\{X\in\mathfrak{J}\mid X^2=X,\mathrm{tr}(X)=1\}$                                                                                                                                                                                                        | $\mathfrak{C}P_2=\{X\in\mathfrak{J}\mid X^2=X,\mathrm{tr}(X)=1\}.$                                                                                                                                               | $\mathfrak{C}P_2=\{X\in\mathfrak{J}\mid X^2=X,\mathrm{tr}(X)=1\}.$                                                                                                                                               |
| 0902.0431_E00375 | 62   | ■ 1.000 | $\alpha X=\left(\begin{array}{ccc}\xi_1&0&0\\0&\xi_2&0\\0&0&\xi_3\end{array}\right)$ , $\xi_i\in\boldsymbol{R}$                                                                                                                                                          | $\alpha X=\left(\begin{array}{ccc}\xi_1&0&0\\0&\xi_2&0\\0&0&\xi_3\end{array}\right)$ , $\xi_i\in\boldsymbol{R}$                                                                                                  | $\alpha X=\left(\begin{array}{ccc}\xi_1&0&0\\0&\xi_2&0\\0&0&\xi_3\end{array}\right)$ , $\xi_i\in\boldsymbol{R}$                                                                                                  |
| 0902.0431_E00376 | 62   | ■ 1.000 | $\left(\begin{array}{ccc}\xi_1^2&0&0\\0&\xi_2^2&0\\0&0&\xi_3^2\end{array}\right)=\left(\begin{array}{ccc}\xi_1&0&0\\0&\xi_2&0\\0&0&\xi_3\end{array}\right).$                                                                                                             | $\left(\begin{array}{ccc}\xi_1^2&0&0\\0&\xi_2^2&0\\0&0&\xi_3^2\end{array}\right)=\left(\begin{array}{ccc}\xi_1&0&0\\0&\xi_2&0\\0&0&\xi_3\end{array}\right).$                                                     | $\left(\begin{array}{ccc}\xi_1^2&0&0\\0&\xi_2^2&0\\0&0&\xi_3^2\end{array}\right)=\left(\begin{array}{ccc}\xi_1&0&0\\0&\xi_2&0\\0&0&\xi_3\end{array}\right).$                                                     |
| 0902.0431_E00377 | 63   | ■ 1.000 | $\sigma\left(\begin{array}{ccc}\xi_1&x_3&\bar{x}_2\\x_3&\xi_2&x_1\\x_2&\bar{x}_1&\xi_3\end{array}\right)=\left(\begin{array}{ccc}\xi_1&-x_3&-\bar{x}_2\\-x_3&\xi_2&x_1\\-x_2&\bar{x}_1&\xi_3\end{array}\right).$                                                         | $\sigma\left(\begin{array}{ccc}\xi_1&x_3&\bar{x}_2\\x_3&\xi_2&x_1\\x_2&\bar{x}_1&\xi_3\end{array}\right)=\left(\begin{array}{ccc}\xi_1&-x_3&-\bar{x}_2\\-x_3&\xi_2&x_1\\-x_2&\bar{x}_1&\xi_3\end{array}\right).$ | $\sigma\left(\begin{array}{ccc}\xi_1&x_3&\bar{x}_2\\x_3&\xi_2&x_1\\x_2&\bar{x}_1&\xi_3\end{array}\right)=\left(\begin{array}{ccc}\xi_1&-x_3&-\bar{x}_2\\-x_3&\xi_2&x_1\\-x_2&\bar{x}_1&\xi_3\end{array}\right).$ |

| Identifier           | Page | Conf.   | LaTeX source                                                                                                                                                                                                                                                                                                                                                                    | Rendered                                                                                                                                                                                                                                                                                                                                                                        | Image                                                                                                                                                                                                                                                                                                                                                                           |
|----------------------|------|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0902.0431_E00378     | 63   | ■ 1.000 | $\left(F_4\right)^{\sigma}=\left\{\alpha \in F_4 \mid \alpha \sigma=\sigma \alpha\right\} .$                                                                                                                                                                                                                                                                                    | $\left(F_4\right)^{\sigma}=\left\{\alpha \in F_4 \mid \sigma \alpha=\alpha \sigma\right\} .$                                                                                                                                                                                                                                                                                    | $\left(F_4\right)^{\sigma}=\left\{\alpha \in F_4 \mid \sigma \alpha=\alpha \sigma\right\} .$                                                                                                                                                                                                                                                                                    |
| 0902.0431_E00380     | 63   | ■ 1.000 | $\alpha E_2, \alpha E_3 \in \mathfrak{J}(2, \mathfrak{C}) .$                                                                                                                                                                                                                                                                                                                    | $\alpha E_2, \alpha E_3 \in \mathfrak{J}(2, \mathfrak{C}) .$                                                                                                                                                                                                                                                                                                                    | $\alpha E_2, \alpha E_3 \in \mathfrak{J}(2, \mathfrak{C}) .$                                                                                                                                                                                                                                                                                                                    |
| 0902.0431_E00382     | 63   | ■ 1.000 | $\alpha E_1=E_1+\xi_2 E_2+\xi_3 E_3+F_1\left(x_1\right) .$                                                                                                                                                                                                                                                                                                                      | $\alpha E_1=E_1+\xi_2 E_2+\xi_3 E_3+F_1\left(x_1\right) .$                                                                                                                                                                                                                                                                                                                      | $\alpha E_1=E_1+\xi_2 E_2+\xi_3 E_3+F_1\left(x_1\right) .$                                                                                                                                                                                                                                                                                                                      |
| 0902.0431_E00385     | 64   | ■ 1.000 | $z\left(F_4\right)=\{1\} .$                                                                                                                                                                                                                                                                                                                                                     | $z\left(F_4\right)=\{1\} .$                                                                                                                                                                                                                                                                                                                                                     | $z\left(F_4\right)=\{1\} .$                                                                                                                                                                                                                                                                                                                                                     |
| 0902.0431_E00386     | 64   | ■ 1.000 | $\alpha=1 \quad \text { or } \quad \alpha=\sigma .$                                                                                                                                                                                                                                                                                                                             | $\alpha=1 \quad \text { or } \quad \alpha=\sigma .$                                                                                                                                                                                                                                                                                                                             | $\alpha=1 \quad \text { or } \quad \alpha=\sigma .$                                                                                                                                                                                                                                                                                                                             |
| 0902.0431_E00388     | 65   | ■ 1.000 | $\left(F_4\right)^{\gamma}=\left\{\alpha \in F_4 \mid \gamma \alpha=\alpha \gamma\right\} .$                                                                                                                                                                                                                                                                                    | $\left(F_4\right)^{\gamma}=\left\{\alpha \in F_4 \mid \gamma \alpha=\alpha \gamma\right\} .$                                                                                                                                                                                                                                                                                    | $\left(F_4\right)^{\gamma}=\left\{\alpha \in F_4 \mid \gamma \alpha=\alpha \gamma\right\} .$                                                                                                                                                                                                                                                                                    |
| 0902.0431_E00389     | 65   | ■ 1.000 | $X=\left(\begin{array}{ccc} \xi_1 & x_3 & \bar{x}_2 \\ \bar{x}_3 & \xi_2 & x_1 \\ x_2 & \bar{x}_1 & \xi_3 \end{array}\right)=\left(\begin{array}{ccc} \xi_1 & m_3 & \bar{m}_2 \\ \bar{m}_3 & \xi_2 & m_1 \\ m_2 & \bar{m}_1 & \xi_3 \end{array}\right)+\left(\begin{array}{ccc} 0 & a_3 e_4 & -a_2 e_4 \\ -a_3 e_4 & 0 & a_1 e_4 \\ a_2 e_4 & -a_1 e_4 & 0 \end{array}\right),$ | $X=\left(\begin{array}{ccc} \xi_1 & x_3 & \bar{x}_2 \\ \bar{x}_3 & \xi_2 & x_1 \\ x_2 & \bar{x}_1 & \xi_3 \end{array}\right)=\left(\begin{array}{ccc} \xi_1 & m_3 & \bar{m}_2 \\ \bar{m}_3 & \xi_2 & m_1 \\ m_2 & \bar{m}_1 & \xi_3 \end{array}\right)+\left(\begin{array}{ccc} 0 & a_3 e_4 & -a_2 e_4 \\ -a_3 e_4 & 0 & a_1 e_4 \\ a_2 e_4 & -a_1 e_4 & 0 \end{array}\right),$ | $X=\left(\begin{array}{ccc} \xi_1 & x_3 & \bar{x}_2 \\ \bar{x}_3 & \xi_2 & x_1 \\ x_2 & \bar{x}_1 & \xi_3 \end{array}\right)=\left(\begin{array}{ccc} \xi_1 & m_3 & \bar{m}_2 \\ \bar{m}_3 & \xi_2 & m_1 \\ m_2 & \bar{m}_1 & \xi_3 \end{array}\right)+\left(\begin{array}{ccc} 0 & a_3 e_4 & -a_2 e_4 \\ -a_3 e_4 & 0 & a_1 e_4 \\ a_2 e_4 & -a_1 e_4 & 0 \end{array}\right),$ |
| 0902.0431_E00390     | 65   | ■ 1.000 | $\left(\begin{array}{ccc} \xi_1 & m_3 & \bar{m}_2 \\ \bar{m}_3 & \xi_2 & m_1 \\ m_2 & \bar{m}_1 & \xi_3 \end{array}\right)+\left(a_1, a_2, a_3\right) .$                                                                                                                                                                                                                        | $\left(\begin{array}{ccc} \xi_1 & m_3 & \bar{m}_2 \\ \bar{m}_3 & \xi_2 & m_1 \\ m_2 & \bar{m}_1 & \xi_3 \end{array}\right)+\left(a_1, a_2, a_3\right) .$                                                                                                                                                                                                                        | $\left(\begin{array}{ccc} \xi_1 & m_3 & \bar{m}_2 \\ \bar{m}_3 & \xi_2 & m_1 \\ m_2 & \bar{m}_1 & \xi_3 \end{array}\right)+\left(a_1, a_2, a_3\right) .$                                                                                                                                                                                                                        |
| 0902.0431_E00391     | 65   | ■ 1.000 | $(M+\boldsymbol{a}) \times(N+\boldsymbol{b})=\left(M \times N-\frac{1}{2}\left(\boldsymbol{a}^* \boldsymbol{b}+\boldsymbol{b}^* \boldsymbol{a}\right)\right)-\frac{1}{2}(\boldsymbol{a} N+\boldsymbol{b} M),$<br>$(M+\boldsymbol{a}, N+\boldsymbol{b})=(M, N)+2(\boldsymbol{a}, \boldsymbol{b}),$<br>$\gamma(M+\boldsymbol{a})=M-\boldsymbol{a} .$                              | $(M+\boldsymbol{a}) \times(N+\boldsymbol{b})=\left(M \times N-\frac{1}{2}\left(\boldsymbol{a}^* \boldsymbol{b}+\boldsymbol{b}^* \boldsymbol{a}\right)\right)-\frac{1}{2}(\boldsymbol{a} N+\boldsymbol{b} M),$<br>$(M+\boldsymbol{a}, N+\boldsymbol{b})=(M, N)+2(\boldsymbol{a}, \boldsymbol{b}),$<br>$\gamma(M+\boldsymbol{a})=M-\boldsymbol{a} .$                              | $(M+\boldsymbol{a}) \times(N+\boldsymbol{b})=\left(M \times N-\frac{1}{2}\left(\boldsymbol{a}^* \boldsymbol{b}+\boldsymbol{b}^* \boldsymbol{a}\right)\right)-\frac{1}{2}(\boldsymbol{a} N+\boldsymbol{b} M),$<br>$(M+\boldsymbol{a}, N+\boldsymbol{b})=(M, N)+2(\boldsymbol{a}, \boldsymbol{b}),$<br>$\gamma(M+\boldsymbol{a})=M-\boldsymbol{a} .$                              |
| 0902.0431_E00392     | 65   | ■ 1.000 | $\mathfrak{J}(3, \boldsymbol{H}) \oplus \boldsymbol{H}^3=\mathfrak{J} .$                                                                                                                                                                                                                                                                                                        | $\mathfrak{J}(3, \boldsymbol{H}) \oplus \boldsymbol{H}^3=\mathfrak{J} .$                                                                                                                                                                                                                                                                                                        | $\mathfrak{J}(3, \boldsymbol{H}) \oplus \boldsymbol{H}^3=\mathfrak{J} .$                                                                                                                                                                                                                                                                                                        |
| 0902.0431_E00394     | 65   | ■ 1.000 | $\varphi(A) M=A M A^*, \quad M \in \mathfrak{J}_{\boldsymbol{H}} .$                                                                                                                                                                                                                                                                                                             | $\varphi(A) M=A M A^*, \quad M \in \mathfrak{J}_{\boldsymbol{H}} .$                                                                                                                                                                                                                                                                                                             | $\varphi(A) M=A M A^*, \quad M \in \mathfrak{J}_{\boldsymbol{H}} .$                                                                                                                                                                                                                                                                                                             |
| 0902.0431_E00395     | 66   | ■ 1.000 | $\alpha E_i=A E_i A^*, \quad i=1,2,3 .$                                                                                                                                                                                                                                                                                                                                         | $\alpha E_i=A E_i A^*, \quad i=1,2,3 .$                                                                                                                                                                                                                                                                                                                                         | $\alpha E_i=A E_i A^*, \quad i=1,2,3 .$                                                                                                                                                                                                                                                                                                                                         |
| 0902.0431_E00396     | 66   | ■ 1.000 | $\beta E_i=E_i, \quad i=1,2,3 .$                                                                                                                                                                                                                                                                                                                                                | $\beta E_i=E_i, \quad i=1,2,3 .$                                                                                                                                                                                                                                                                                                                                                | $\beta E_i=E_i, \quad i=1,2,3 .$                                                                                                                                                                                                                                                                                                                                                |
| 0902.0431_E00397 (i) | 66   | ■ 1.000 | $\left(\beta_1 m\right)\left(\beta_2 n\right)=\overline{\beta_3(\overline{m n})}, \quad m, n \in \boldsymbol{H} .$                                                                                                                                                                                                                                                              | $\left(\beta_1 m\right)\left(\beta_2 n\right)=\overline{\beta_3(\overline{m n})}, \quad m, n \in \boldsymbol{H} .$                                                                                                                                                                                                                                                              | $\left(\beta_1 m\right)\left(\beta_2 n\right)=\overline{\beta_3(\overline{m n})}, \quad m, n \in \boldsymbol{H} .$                                                                                                                                                                                                                                                              |
| 0902.0431_E00398     | 66   | ■ 1.000 | $\beta_2 m=\bar{p}\left(\beta_1 m\right) q, \quad \beta_3 m=\overline{\left(\beta_1 \bar{m}\right)} q .$                                                                                                                                                                                                                                                                        | $\beta_2 m=\bar{p}\left(\beta_1 m\right) q, \quad \beta_3 m=\overline{\left(\beta_1 \bar{m}\right)} q .$                                                                                                                                                                                                                                                                        | $\beta_2 m=\bar{p}\left(\beta_1 m\right) q, \quad \beta_3 m=\overline{\left(\beta_1 \bar{m}\right)} q .$                                                                                                                                                                                                                                                                        |
| 0902.0431_E00399     | 66   | ■ 1.000 | $(\zeta m)(\zeta n)=\zeta(m n), \quad m, n \in \boldsymbol{H},$                                                                                                                                                                                                                                                                                                                 | $(\zeta m)(\zeta n)=\zeta(m n), \quad m, n \in \boldsymbol{H},$                                                                                                                                                                                                                                                                                                                 | $(\zeta m)(\zeta n)=\zeta(m n), \quad m, n \in \boldsymbol{H},$                                                                                                                                                                                                                                                                                                                 |
| 0902.0431_E00401     | 66   | ■ 1.000 | $\beta M=B M B^*, \quad M \in \mathfrak{J}_{\boldsymbol{H}},$                                                                                                                                                                                                                                                                                                                   | $\beta M=B M B^*, \quad M \in \mathfrak{J}_{\boldsymbol{H}},$                                                                                                                                                                                                                                                                                                                   | $\beta M=B M B^*, \quad M \in \mathfrak{J}_{\boldsymbol{H}},$                                                                                                                                                                                                                                                                                                                   |

| Identifier       | Page | Conf.   | LaTeX source                                                                                                                                                                                                                                                                                                                                             | Rendered                                                                                                                                                                                                                                                                                                                                                 | Image                                                                                                                                                                                                                                                                                                                                                    |
|------------------|------|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0902.0431_E00403 | 67   | ■ 1.000 | $\varphi(p, A)(M+\boldsymbol{a})=A\,M\,A^*+p\boldsymbol{a}\,A^*, \quad M+\boldsymbol{a}\in\mathfrak{J}_H\oplus\boldsymbol{H}^3=\mathfrak{J}.$<br>$\boldsymbol{a}\in\mathfrak{J}_H\oplus\boldsymbol{H}^3=\mathfrak{J}.$                                                                                                                                   | $\varphi(p, A)(M+\boldsymbol{a})=A\,M\,A^*+p\boldsymbol{a}\,A^*, \quad M+\boldsymbol{a}\in\mathfrak{J}_H\oplus\boldsymbol{H}^3=\mathfrak{J}.$                                                                                                                                                                                                            | $\varphi(p, A)(M+\boldsymbol{a})=A\,M\,A^*+p\boldsymbol{a}\,A^*, \quad M+\boldsymbol{a}\in\mathfrak{J}_H\oplus\boldsymbol{H}^3=\mathfrak{J}.$                                                                                                                                                                                                            |
| 0902.0431_E00406 | 67   | ■ 1.000 | $\alpha\,M=A\,M\,A^*, \quad M\in\mathfrak{J}_H$                                                                                                                                                                                                                                                                                                          | $\alpha M=AMA^*, \quad M\in\mathfrak{J}_H$                                                                                                                                                                                                                                                                                                               | $\alpha M=AMA^*, \quad M\in\mathfrak{J}_H$                                                                                                                                                                                                                                                                                                               |
| 0902.0431_E00407 | 67   | ■ 1.000 | $\beta(m+ae_4)=m+(pa)e_4, \quad m+ae_4\in\boldsymbol{H}\oplus\boldsymbol{H}e_4=\mathfrak{C},$                                                                                                                                                                                                                                                            | $\beta(m+ae_4)=m+(pa)e_4, \quad m+ae_4\in\boldsymbol{H}\oplus\boldsymbol{H}e_4=\mathfrak{C},$                                                                                                                                                                                                                                                            | $\beta(m+ae_4)=m+(pa)e_4, \quad m+ae_4\in\boldsymbol{H}\oplus\boldsymbol{H}e_4=\mathfrak{C},$                                                                                                                                                                                                                                                            |
| 0902.0431_E00408 | 67   | ■ 1.000 | $\beta(M+\boldsymbol{a})=M+p\boldsymbol{a}=\varphi(p, E)(M+\boldsymbol{a}),$                                                                                                                                                                                                                                                                             | $\beta(M+\boldsymbol{a})=M+p\boldsymbol{a}=\varphi(p, E)(M+\boldsymbol{a}),$                                                                                                                                                                                                                                                                             | $\beta(M+\boldsymbol{a})=M+p\boldsymbol{a}=\varphi(p, E)(M+\boldsymbol{a}),$                                                                                                                                                                                                                                                                             |
| 0902.0431_E00411 | 68   | ■ 1.000 | $(F_4)^w=\{\alpha\in F_4\mid w\alpha=\alpha w\}.$                                                                                                                                                                                                                                                                                                        | $(F_4)^w=\{\alpha\in F_4\mid w\alpha=\alpha w\}.$                                                                                                                                                                                                                                                                                                        | $(F_4)^w=\{\alpha\in F_4\mid w\alpha=\alpha w\}.$                                                                                                                                                                                                                                                                                                        |
| 0902.0431_E00415 | 68   | ■ 1.000 | $P\,M\times P\,N=\widetilde{P}(M\times N), \quad M\,A\times N\,A=(M\times N)\widetilde{A},$                                                                                                                                                                                                                                                              | $P\,M\times P\,N=\widetilde{P}(M\times N), \quad M\,A\times N\,A=(M\times N)\widetilde{A},$                                                                                                                                                                                                                                                              | $P\,M\times P\,N=\widetilde{P}(M\times N), \quad M\,A\times N\,A=(M\times N)\widetilde{A},$                                                                                                                                                                                                                                                              |
| 0902.0431_E00416 | 68   | ■ 1.000 | $(M, N)=\frac{1}{2}\operatorname{tr}(M^*N+N^*M)=\sum_{i,j}(m_{ij}, n_{ij}),$                                                                                                                                                                                                                                                                             | $(M, N)=\frac{1}{2}\operatorname{tr}(M^*N+N^*M)=\sum_{i,j}(m_{ij}, n_{ij}),$                                                                                                                                                                                                                                                                             | $(M, N)=\frac{1}{2}\operatorname{tr}(M^*N+N^*M)=\sum_{i,j}(m_{ij}, n_{ij}),$                                                                                                                                                                                                                                                                             |
| 0902.0431_E00418 | 69   | ■ 1.000 | $\mathfrak{J}(3, \boldsymbol{C})\oplus M(3, \boldsymbol{C})=\mathfrak{J}$                                                                                                                                                                                                                                                                                | $\mathfrak{J}(3, \boldsymbol{C})\oplus M(3, \boldsymbol{C})=\mathfrak{J}$                                                                                                                                                                                                                                                                                | $\mathfrak{J}(3, \boldsymbol{C})\oplus M(3, \boldsymbol{C})=\mathfrak{J}.$                                                                                                                                                                                                                                                                               |
| 0902.0431_E00419 | 69   | ■ 1.000 | $\begin{aligned} F_{4, \boldsymbol{C}} &= \{\alpha \in \operatorname{Iso}_{\boldsymbol{R}}(\mathfrak{J}_{\boldsymbol{C}}) \mid \alpha(X \circ Y) = \alpha X \circ \alpha Y\} \\ &= \{\alpha \in \operatorname{Iso}_{\boldsymbol{R}}(\mathfrak{J}_{\boldsymbol{C}}) \mid \alpha(X \times Y) = \alpha X \times \alpha Y\}. \end{aligned}$                  | $\begin{aligned} F_{4, \boldsymbol{C}} &= \{\alpha \in \operatorname{Iso}_{\boldsymbol{R}}(\mathfrak{J}_{\boldsymbol{C}}) \mid \alpha(X \circ Y) = \alpha X \circ \alpha Y\} \\ &= \{\alpha \in \operatorname{Iso}_{\boldsymbol{R}}(\mathfrak{J}_{\boldsymbol{C}}) \mid \alpha(X \times Y) = \alpha X \times \alpha Y\}. \end{aligned}$                  | $\begin{aligned} F_{4, \boldsymbol{C}} &= \{\alpha \in \operatorname{Iso}_{\boldsymbol{R}}(\mathfrak{J}_{\boldsymbol{C}}) \mid \alpha(X \circ Y) = \alpha X \circ \alpha Y\} \\ &= \{\alpha \in \operatorname{Iso}_{\boldsymbol{R}}(\mathfrak{J}_{\boldsymbol{C}}) \mid \alpha(X \times Y) = \alpha X \times \alpha Y\}. \end{aligned}$                  |
| 0902.0431_E00420 | 69   | ■ 1.000 | $\epsilon A=\bar{A}, \quad A\in SU(3),$                                                                                                                                                                                                                                                                                                                  | $\epsilon A=\bar{A}, \quad A\in SU(3),$                                                                                                                                                                                                                                                                                                                  | $\epsilon A=\bar{A}, \quad A\in SU(3),$                                                                                                                                                                                                                                                                                                                  |
| 0902.0431_E00422 | 69   | ■ 1.000 | $\beta_1x=px, \beta_2x=xq, \beta_3x=\bar{q}x\bar{p} \quad \text{or} \quad \beta_1x=p\bar{x}, \beta_2x=\bar{x}q, \beta_3x=\bar{q}\bar{x}\bar{p}.$                                                                                                                                                                                                         | $\beta_1x=px, \beta_2x=xq, \beta_3x=\bar{q}x\bar{p} \quad \text{or} \quad \beta_1x=p\bar{x}, \beta_2x=\bar{x}q, \beta_3x=\bar{q}\bar{x}\bar{p}.$                                                                                                                                                                                                         | $\beta_1x=px, \beta_2x=xq, \beta_3x=\bar{q}x\bar{p} \quad \text{or} \quad \beta_1x=p\bar{x}, \beta_2x=\bar{x}q, \beta_3x=\bar{q}\bar{x}\bar{p}.$                                                                                                                                                                                                         |
| 0902.0431_E00424 | 69   | ■ 1.000 | $\varphi(P, A)(X+M)=A\,X\,A^*+P\,M\,A^*, \quad X+M\in\mathfrak{J}(3, \boldsymbol{C})\oplus M(3, \boldsymbol{C})=\mathfrak{J}.$                                                                                                                                                                                                                           | $\varphi(P, A)(X+M)=A\,X\,A^*+P\,M\,A^*, \quad X+M\in\mathfrak{J}(3, \boldsymbol{C})\oplus M(3, \boldsymbol{C})=\mathfrak{J}.$                                                                                                                                                                                                                           | $\varphi(P, A)(X+M)=A\,X\,A^*+P\,M\,A^*, \quad X+M\in\mathfrak{J}(3, \boldsymbol{C})\oplus M(3, \boldsymbol{C})=\mathfrak{J}.$                                                                                                                                                                                                                           |
| 0902.0431_E00427 | 70   | ■ 1.000 | $\alpha X=A\,X\,A^* \quad \text{or} \quad \alpha X=A\bar{X}A^*, \quad X\in\mathfrak{J}_C$                                                                                                                                                                                                                                                                | $\alpha X=A\,X\,A^* \quad \text{or} \quad \alpha X=A\bar{X}A^*, \quad X\in\mathfrak{J}_C$                                                                                                                                                                                                                                                                | $\alpha X=A\,X\,A^* \quad \text{or} \quad \alpha X=A\bar{X}A^*, \quad X\in\mathfrak{J}_C$                                                                                                                                                                                                                                                                |
| 0902.0431_E00428 | 70   | ■ 1.000 | $\beta(X+M)=X+P\,M=\varphi(P, E)(X+M), \quad X+M\in\mathfrak{J}_C\oplus M(3, \boldsymbol{C})=\mathfrak{J}.$                                                                                                                                                                                                                                              | $\beta(X+M)=X+P\,M=\varphi(P, E)(X+M), \quad X+M\in\mathfrak{J}_C\oplus M(3, \boldsymbol{C})=\mathfrak{J}.$                                                                                                                                                                                                                                              | $\beta(X+M)=X+P\,M=\varphi(P, E)(X+M), \quad X+M\in\mathfrak{J}_C\oplus M(3, \boldsymbol{C})=\mathfrak{J}.$                                                                                                                                                                                                                                              |
| 0902.0431_E00429 | 70   | ■ 1.000 | $\alpha=\varphi(E, A)\beta=\varphi(E, A)\varphi(P, E)=\varphi(P, A).$                                                                                                                                                                                                                                                                                    | $\alpha=\varphi(E, A)\beta=\varphi(E, A)\varphi(P, E)=\varphi(P, A).$                                                                                                                                                                                                                                                                                    | $\alpha=\varphi(E, A)\beta=\varphi(E, A)\varphi(P, E)=\varphi(P, A).$                                                                                                                                                                                                                                                                                    |
| 0902.0431_E00430 | 70   | ■ 1.000 | $\begin{aligned} &G_{01}, \quad G_{23}, \quad G_{45}, \quad G_{67}, \quad G_{26}+G_{37}, \quad -G_{27}+G_{36}, \\ &G_{24}+G_{35}, \quad -G_{25}+G_{34}, \quad G_{46}+G_{57}, \quad -G_{47}+G_{56}, \\ &\tilde{A}_1(1), \quad \tilde{A}_2(1), \quad \tilde{A}_3(1), \quad \tilde{A}_1(e_1), \quad \tilde{A}_2(e_1), \quad \tilde{A}_3(e_1) \end{aligned}$ | $\begin{aligned} &G_{01}, \quad G_{23}, \quad G_{45}, \quad G_{67}, \quad G_{26}+G_{37}, \quad -G_{27}+G_{36}, \\ &G_{24}+G_{35}, \quad -G_{25}+G_{34}, \quad G_{46}+G_{57}, \quad -G_{47}+G_{56}, \\ &\tilde{A}_1(1), \quad \tilde{A}_2(1), \quad \tilde{A}_3(1), \quad \tilde{A}_1(e_1), \quad \tilde{A}_2(e_1), \quad \tilde{A}_3(e_1) \end{aligned}$ | $\begin{aligned} &G_{01}, \quad G_{23}, \quad G_{45}, \quad G_{67}, \quad G_{26}+G_{37}, \quad -G_{27}+G_{36}, \\ &G_{24}+G_{35}, \quad -G_{25}+G_{34}, \quad G_{46}+G_{57}, \quad -G_{47}+G_{56}, \\ &\tilde{A}_1(1), \quad \tilde{A}_2(1), \quad \tilde{A}_3(1), \quad \tilde{A}_1(e_1), \quad \tilde{A}_2(e_1), \quad \tilde{A}_3(e_1) \end{aligned}$ |

| Identifier       | Page | Conf.   | LaTeX source                                                                                                                                                                                                                                                                                                                                                       | Rendered                                                                                                                                                                                                                                                                                                       | Image                                                                                                                                                                                                                                                                                                          |
|------------------|------|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0902.0431_E00432 | 71   | ■ 1.000 | $\backslash\langle X, Y\rangle=\backslash\tau X, Y) \ .$                                                                                                                                                                                                                                                                                                           | $\langle X, Y\rangle=(\tau X, Y).$                                                                                                                                                                                                                                                                             | $\langle X, Y\rangle=(\tau X, Y).$                                                                                                                                                                                                                                                                             |
| 0902.0431_E00437 | 72   | ■ 1.000 | $X \circ Y=\frac{1}{2}(X \, Y+Y \, X)$                                                                                                                                                                                                                                                                                                                             | $X \circ Y=\frac{1}{2}(XY+YX)$                                                                                                                                                                                                                                                                                 | $X \circ Y=\frac{1}{2}(XY+YX).$                                                                                                                                                                                                                                                                                |
| 0902.0431_E00439 | 72   | ■ 1.000 | $\begin{aligned} F_{4(4)} &= \left\{ \alpha \in \operatorname{Iso}_{\boldsymbol{R}}(\mathfrak{J}(3, \mathfrak{C}')) \mid \alpha(X \circ Y) = \alpha X \circ \alpha Y \right\}, \\ F_{4(-20)} &= \left\{ \alpha \in \operatorname{Iso}_{\boldsymbol{R}}(\mathfrak{J}(1, 2, \mathfrak{C})) \mid \alpha(X \circ Y) = \alpha X \circ \alpha Y \right\}. \end{aligned}$ | $F_{4(4)} = \{ \alpha \in \operatorname{Iso}_{\boldsymbol{R}}(\mathfrak{J}(3, \mathfrak{C}')) \mid \alpha(X \circ Y) = \alpha X \circ \alpha Y \},$<br>$F_{4(-20)} = \{ \alpha \in \operatorname{Iso}_{\boldsymbol{R}}(\mathfrak{J}(1, 2, \mathfrak{C})) \mid \alpha(X \circ Y) = \alpha X \circ \alpha Y \}.$ | $F_{4(4)} = \{ \alpha \in \operatorname{Iso}_{\boldsymbol{R}}(\mathfrak{J}(3, \mathfrak{C}')) \mid \alpha(X \circ Y) = \alpha X \circ \alpha Y \},$<br>$F_{4(-20)} = \{ \alpha \in \operatorname{Iso}_{\boldsymbol{R}}(\mathfrak{J}(1, 2, \mathfrak{C})) \mid \alpha(X \circ Y) = \alpha X \circ \alpha Y \}.$ |
| 0902.0431_E00442 | 72   | ■ 1.000 | $z\left(F_{4(4)}\right)=\{1\}, \quad z\left(F_{4(-20)}\right)=\{1\} \ .$                                                                                                                                                                                                                                                                                           | $z\left(F_{4(4)}\right)=\{1\}, \quad z\left(F_{4(-20)}\right)=\{1\}.$                                                                                                                                                                                                                                          | $z\left(F_{4(4)}\right)=\{1\}, \quad z\left(F_{4(-20)}\right)=\{1\}.$                                                                                                                                                                                                                                          |
| 0902.0431_E00444 | 73   | ■ 1.000 | $\lambda(\alpha)=\{ \ }^t\alpha^{-1}, \quad \alpha \in E_6^C$                                                                                                                                                                                                                                                                                                      | $\lambda(\alpha)=\,^t\alpha^{-1}, \quad \alpha \in E_6^C$                                                                                                                                                                                                                                                      | $\lambda(\alpha)=\,^t\alpha^{-1}, \quad \alpha \in E_6^C$                                                                                                                                                                                                                                                      |
| 0902.0431_E00445 | 73   | ■ 1.000 | $E_6=\left\{ \alpha \in E_6^C \mid \tau \lambda(\alpha) \tau = \alpha \right\} = \left( E_6^C \right)^{\tau \lambda}.$                                                                                                                                                                                                                                             | $E_6 = \left\{ \alpha \in E_6^C \mid \tau \lambda(\alpha) \tau = \alpha \right\} = \left( E_6^C \right)^{\tau \lambda}.$                                                                                                                                                                                       | $E_6 = \{ \alpha \in E_6^C \mid \tau \lambda(\alpha) \tau = \alpha \} = (E_6^C)^{\tau \lambda}.$                                                                                                                                                                                                               |
| 0902.0431_E00446 | 73   | ■ 1.000 | $U(27)=U\left(\mathfrak{J}^C\right)=\left\{ \alpha \in \operatorname{Iso}_C\left(\mathfrak{J}^C\right) \mid \langle \alpha X, \alpha Y\rangle=\langle X, Y\rangle\right\}.$                                                                                                                                                                                        | $U(27)=U\left(\mathfrak{J}^C\right)=\left\{ \alpha \in \operatorname{Iso}_C\left(\mathfrak{J}^C\right) \mid \langle \alpha X, \alpha Y\rangle=\langle X, Y\rangle\right\}.$                                                                                                                                    | $U(27)=U(\mathfrak{J}^C)=\{ \alpha \in \operatorname{Iso}_C(\mathfrak{J}^C) \mid \langle \alpha X, \alpha Y\rangle=\langle X, Y\rangle\}.$                                                                                                                                                                     |
| 0902.0431_E00451 | 74   | ■ 1.000 | $\left[\delta_1+\widetilde{T}_1,\delta_2+\widetilde{T}_2\right]=\left([\delta_1,\delta_2]+\left[\widetilde{T}_1,\widetilde{T}_2\right]\right)+\left(\widetilde{\delta_1T_2}-\widetilde{\delta_2T_1}\right),$                                                                                                                                                       | $\left[\delta_1+\widetilde{T}_1,\delta_2+\widetilde{T}_2\right]=\left([\delta_1,\delta_2]+\left[\widetilde{T}_1,\widetilde{T}_2\right]\right)+\left(\widetilde{\delta_1T_2}-\widetilde{\delta_2T_1}\right),$                                                                                                   | $[\delta_1+\widetilde{T}_1,\delta_2+\widetilde{T}_2]=([\delta_1,\delta_2]+[\widetilde{T}_1,\widetilde{T}_2])+(\widetilde{\delta_1T_2}-\widetilde{\delta_2T_1}),$                                                                                                                                               |
| 0902.0431_E00453 | 74   | ■ 1.000 | $\lambda(\phi)=-\{ \ }^t\phi=-\{ \ }^t(\delta+\widetilde{T})=\delta-\widetilde{T}.$                                                                                                                                                                                                                                                                                | $\lambda(\phi)=-\,^t\phi=-\,^t(\delta+\widetilde{T})=\delta-\widetilde{T}.$                                                                                                                                                                                                                                    | $\lambda(\phi)=-\,^t\phi=-\,^t(\delta+\widetilde{T})=\delta-\widetilde{T}.$                                                                                                                                                                                                                                    |
| 0902.0431_E00455 | 74   | ■ 1.000 | $\mathfrak{e}_6=\left\{ \phi \in \operatorname{Hom}_C\left(\mathfrak{J}^C\right) \mid \left(\phi X, X, X\right)=0, \left\langle \phi X, Y\right\rangle+\left\langle X, \phi Y\right\rangle=0\right\}.$                                                                                                                                                             | $\mathfrak{e}_6 = \left\{ \phi \in \operatorname{Hom}_C\left(\mathfrak{J}^C\right) \mid \left(\phi X, X, X\right)=0, \left\langle \phi X, Y\right\rangle+\left\langle X, \phi Y\right\rangle=0\right\}.$                                                                                                       | $\mathfrak{e}_6 = \{ \phi \in \operatorname{Hom}_C(\mathfrak{J}^C) \mid (\phi X, X, X) = 0, \langle \phi X, Y\rangle + \langle X, \phi Y\rangle = 0 \}.$                                                                                                                                                       |
| 0902.0431_E00456 | 74   | ■ 1.000 | $\phi=\delta+i\widetilde{T}, \quad \delta \in \mathfrak{f}_4, T \in \mathfrak{J}_0.$                                                                                                                                                                                                                                                                               | $\phi=\delta+i\widetilde{T}, \quad \delta \in \mathfrak{f}_4, T \in \mathfrak{J}_0.$                                                                                                                                                                                                                           | $\phi=\delta+i\widetilde{T}, \quad \delta \in \mathfrak{f}_4, T \in \mathfrak{J}_0.$                                                                                                                                                                                                                           |
| 0902.0431_E00461 | 76   | ■ 1.000 | $A \circ X=\left(A-\frac{1}{3}\operatorname{tr}(A)E\right)^{\sim}X+\frac{1}{3}\operatorname{tr}(A)X \in W.$                                                                                                                                                                                                                                                        | $A \circ X = \left(A - \frac{1}{3}\operatorname{tr}(A)E\right)^{\sim}X + \frac{1}{3}\operatorname{tr}(A)X \in W.$                                                                                                                                                                                              | $A \circ X = \left(A - \frac{1}{3}\operatorname{tr}(A)E\right)^{\sim}X + \frac{1}{3}\operatorname{tr}(A)X \in W.$                                                                                                                                                                                              |
| 0902.0431_E00462 | 76   | ■ 1.000 | $A \vee B=[\widetilde{A},\widetilde{B}]+\left(A \circ B-\frac{1}{3}(A, B)E\right)^{\sim}$                                                                                                                                                                                                                                                                          | $A \vee B = [\widetilde{A}, \widetilde{B}] + \left(A \circ B - \frac{1}{3}(A, B)E\right)^{\sim}$                                                                                                                                                                                                               | $A \vee B = [\widetilde{A}, \widetilde{B}] + \left(A \circ B - \frac{1}{3}(A, B)E\right)^{\sim}$                                                                                                                                                                                                               |
| 0902.0431_E00463 | 76   | ■ 1.000 | $(A \vee B) X=\frac{1}{2}(B, X) A+\frac{1}{6}(A, B) X-2 B \times(A \times X), \quad X \in \mathfrak{J}^C.$                                                                                                                                                                                                                                                         | $(A \vee B) X = \frac{1}{2}(B, X) A + \frac{1}{6}(A, B) X - 2 B \times (A \times X), \quad X \in \mathfrak{J}^C.$                                                                                                                                                                                              | $(A \vee B) X = \frac{1}{2}(B, X) A + \frac{1}{6}(A, B) X - 2 B \times (A \times X), \quad X \in \mathfrak{J}^C.$                                                                                                                                                                                              |
| 0902.0431_E00464 | 76   | ■ 1.000 | $X \circ(X \circ X)-\operatorname{tr}(X) X \circ X+\frac{1}{2}\left(\operatorname{tr}(X)^2-(X, X)\right) X=\frac{1}{3}(X, X, X) E.$                                                                                                                                                                                                                                | $X \circ (X \circ X) - \operatorname{tr}(X) X \circ X + \frac{1}{2}\left(\operatorname{tr}(X)^2 - (X, X)\right) X = \frac{1}{3}(X, X, X) E.$                                                                                                                                                                   | $X \circ (X \circ X) - \operatorname{tr}(X) X \circ X + \frac{1}{2}(\operatorname{tr}(X)^2 - (X, X)) X = \frac{1}{3}(X, X, X) E.$                                                                                                                                                                              |

| Identifier       | Page | Conf.   | LaTeX source                                                                                                                                                                                                                                                                                                                                         | Rendered                                                                                                                                                                                                                                                                                                                                             | Image                                                                                                                                                                                                                                                                                                                                                |
|------------------|------|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0902.0431_E00467 | 77   | ■ 1.000 | $\left[\left(E_1\vee E_1\right)\left(\begin{array}{ccc}\xi_1&x_3&\bar{x}_2\\\bar{x}_3&\xi_2&x_1\\x_2&\bar{x}_1&\xi_3\end{array}\right)\right]=\frac{1}{6}\left(\begin{array}{ccc}4\xi_1&x_3&\bar{x}_2\\\bar{x}_3&-2\xi_2&-2x_1\\x_2&-2\bar{x}_1&-2\xi_3\end{array}\right).$                                                                          | $(E_1\vee E_1)\left(\begin{array}{ccc}\xi_1&x_3&\bar{x}_2\\\bar{x}_3&\xi_2&x_1\\x_2&\bar{x}_1&\xi_3\end{array}\right)=\frac{1}{6}\left(\begin{array}{ccc}4\xi_1&x_3&\bar{x}_2\\\bar{x}_3&-2\xi_2&-2x_1\\x_2&-2\bar{x}_1&-2\xi_3\end{array}\right).$                                                                                                  | $(E_1\vee E_1)\left(\begin{array}{ccc}\xi_1&x_3&\bar{x}_2\\\bar{x}_3&\xi_2&x_1\\x_2&\bar{x}_1&\xi_3\end{array}\right)=\frac{1}{6}\left(\begin{array}{ccc}4\xi_1&x_3&\bar{x}_2\\\bar{x}_3&-2\xi_2&-2x_1\\x_2&-2\bar{x}_1&-2\xi_3\end{array}\right).$                                                                                                  |
| 0902.0431_E00470 | 77   | ■ 1.000 | $\left[(A\vee B)^\prime=-B\vee A\right]$                                                                                                                                                                                                                                                                                                             | $(A\vee B)^\prime=-B\vee A.$                                                                                                                                                                                                                                                                                                                         | $(A\vee B)^\prime=-B\vee A.$                                                                                                                                                                                                                                                                                                                         |
| 0902.0431_E00471 | 78   | ■ 1.000 | $[\phi,A\vee B]=\phi A\vee B+A\vee\phi^\prime B$                                                                                                                                                                                                                                                                                                     | $[\phi,A\vee B]=\phi A\vee B+A\vee\phi^\prime B.$                                                                                                                                                                                                                                                                                                    | $[\phi,A\vee B]=\phi A\vee B+A\vee\phi^\prime B.$                                                                                                                                                                                                                                                                                                    |
| 0902.0431_E00475 | 78   | ■ 1.000 | $\left[(\phi,A\vee B)_6=(\phi A,B)\right]$                                                                                                                                                                                                                                                                                                           | $(\phi,A\vee B)_6=(\phi A,B).$                                                                                                                                                                                                                                                                                                                       | $(\phi,A\vee B)_6=(\phi A,B).$                                                                                                                                                                                                                                                                                                                       |
| 0902.0431_E00476 | 78   | ■ 1.000 | $\begin{aligned}&\left(\left[\phi,\phi_1\right],\phi_2\right)_6\\&=\left(\left[\delta+\tilde{T},\delta_1+\tilde{T}_1\right],\delta_2+\tilde{T}_2\right)_6\end{aligned}$                                                                                                                                                                              | $\begin{aligned}&([\phi,\phi_1],\phi_2)_6\\&=\left(\left[\delta+\tilde{T},\delta_1+\tilde{T}_1\right],\delta_2+\tilde{T}_2\right)_6\end{aligned}$                                                                                                                                                                                                    | $\begin{aligned}&([\phi,\phi_1],\phi_2)_6\\&=(\left[\delta+\tilde{T},\delta_1+\tilde{T}_1\right],\delta_2+\tilde{T}_2)_6\end{aligned}$                                                                                                                                                                                                               |
| 0902.0431_E00481 | 79   | ■ 1.000 | $B_6\left(\phi_1,\phi_2\right)=k\left(\phi_1,\phi_2\right)_6=k'\operatorname{tr}\left(\phi_1\phi_2\right).$                                                                                                                                                                                                                                          | $B_6\left(\phi_1,\phi_2\right)=k\left(\phi_1,\phi_2\right)_6=k'\operatorname{tr}\left(\phi_1\phi_2\right).$                                                                                                                                                                                                                                          | $B_6\left(\phi_1,\phi_2\right)=k\left(\phi_1,\phi_2\right)_6=k'\operatorname{tr}\left(\phi_1\phi_2\right).$                                                                                                                                                                                                                                          |
| 0902.0431_E00482 | 79   | ■ 1.000 | $\left(\phi,\phi\right)_6=\left(\left(E_1-E_2\right)^\sim,\left(E_1-E_2\right)^\sim\right)_6=\left(E_1-E_2,E_1-E_2\right)=2.$                                                                                                                                                                                                                        | $(\phi,\phi)_6=((E_1-E_2)^\sim,(E_1-E_2)^\sim)_6=(E_1-E_2,E_1-E_2)=2.$                                                                                                                                                                                                                                                                               | $(\phi,\phi)_6=((E_1-E_2)^\sim,(E_1-E_2)^\sim)_6=(E_1-E_2,E_1-E_2)=2.$                                                                                                                                                                                                                                                                               |
| 0902.0431_E00486 | 80   | ■ 1.000 | $\begin{array}{l} \phi\phi E_1=\phi E_1=E_1, \quad \phi\phi\tilde{F}_1(e_i)=-\frac{1}{2}\phi\tilde{F}_1(e_i)=\frac{1}{4}\tilde{F}_1(e_i), \\ \phi\phi E_2=-\phi E_2=E_2, \quad \phi\phi\tilde{F}_2(e_i)=\frac{1}{2}\phi\tilde{F}_2(e_i)=\frac{1}{4}\tilde{F}_2(e_i), \\ \phi\phi E_3=\phi 0=0, \quad \phi\phi\tilde{F}_3(e_i)=\phi 0=0. \end{array}$ | $\begin{array}{l} \phi\phi E_1=\phi E_1=E_1, \quad \phi\phi\tilde{F}_1(e_i)=-\frac{1}{2}\phi\tilde{F}_1(e_i)=\frac{1}{4}\tilde{F}_1(e_i), \\ \phi\phi E_2=-\phi E_2=E_2, \quad \phi\phi\tilde{F}_2(e_i)=\frac{1}{2}\phi\tilde{F}_2(e_i)=\frac{1}{4}\tilde{F}_2(e_i), \\ \phi\phi E_3=\phi 0=0, \quad \phi\phi\tilde{F}_3(e_i)=\phi 0=0. \end{array}$ | $\begin{array}{l} \phi\phi E_1=\phi E_1=E_1, \quad \phi\phi\tilde{F}_1(e_i)=-\frac{1}{2}\phi\tilde{F}_1(e_i)=\frac{1}{4}\tilde{F}_1(e_i), \\ \phi\phi E_2=-\phi E_2=E_2, \quad \phi\phi\tilde{F}_2(e_i)=\frac{1}{2}\phi\tilde{F}_2(e_i)=\frac{1}{4}\tilde{F}_2(e_i), \\ \phi\phi E_3=\phi 0=0, \quad \phi\phi\tilde{F}_3(e_i)=\phi 0=0. \end{array}$ |
| 0902.0431_E00488 | 80   | ■ 1.000 | $A\vee(B\times C)+B\vee(C\times A)+C\vee(A\times B)=0, \quad A,B,C\in\mathfrak{J}^C.$                                                                                                                                                                                                                                                                | $A\vee(B\times C)+B\vee(C\times A)+C\vee(A\times B)=0, \quad A,B,C\in\mathfrak{J}^C.$                                                                                                                                                                                                                                                                | $A\vee(B\times C)+B\vee(C\times A)+C\vee(A\times B)=0, \quad A,B,C\in\mathfrak{J}^C.$                                                                                                                                                                                                                                                                |
| 0902.0431_E00490 | 80   | ■ 1.000 | $\mathfrak{M}^{\mathfrak{r}}=\{X\in M(3,\mathfrak{C})\mid \text{all diagonal elements of }X\text{ are real}\},$                                                                                                                                                                                                                                      | $\mathfrak{M}^{\mathfrak{r}}=\{X\in M(3,\mathfrak{C})\mid \text{all diagonal elements of }X\text{ are real}\},$                                                                                                                                                                                                                                      | $\mathfrak{M}^{\mathfrak{r}}=\{X\in M(3,\mathfrak{C})\mid \text{all diagonal elements of }X\text{ are real}\},$                                                                                                                                                                                                                                      |
| 0902.0431_E00491 | 81   | ■ 1.000 | $X\circ Y=\frac{1}{2}\left(XY+Y^*X^*\right)$                                                                                                                                                                                                                                                                                                         | $X\circ Y=\frac{1}{2}\left(XY+Y^*X^*\right)$                                                                                                                                                                                                                                                                                                         | $X\circ Y=\frac{1}{2}\left(XY+Y^*X^*\right),$                                                                                                                                                                                                                                                                                                        |
| 0902.0431_E00498 | 81   | ■ 1.000 | $\begin{aligned}&\sum_k\sigma_{ij}(x_{ik}y_{kj})+\sum_k\overline{\sigma_{ji}(x_{jk}y_{ki})}\\&=\sum_k\sigma_{ij}(x_{ik}y_{kj})+\sum_k\sigma_{ij}(\overline{x_{jk}y_{ki}})=(i,j)\text{-element of }\delta(X\circ Y).\end{aligned}$                                                                                                                    | $\begin{aligned}&=\sum_k\sigma_{ij}(x_{ik}y_{kj})+\sum_k\overline{\sigma_{ji}(x_{jk}y_{ki})}\\&=\sum_k\sigma_{ij}(x_{ik}y_{kj})+\sum_k\sigma_{ij}(\overline{x_{jk}y_{ki}})=(i,j)\text{-element of }\delta(X\circ Y).\end{aligned}$                                                                                                                   | $\begin{aligned}&=\sum_k\sigma_{ij}(x_{ik}y_{kj})+\sum_k\overline{\sigma_{ji}(x_{jk}y_{ki})}\\&=\sum_k\sigma_{ij}(x_{ik}y_{kj})+\sum_k\sigma_{ij}(\overline{x_{jk}y_{ki}})=(i,j)\text{-element of }\delta(X\circ Y).\end{aligned}$                                                                                                                   |
| 0902.0431_E00500 | 82   | ■ 1.000 | $[\delta,\widetilde{R}]=\widetilde{\delta R}.$                                                                                                                                                                                                                                                                                                       | $[\delta,\widetilde{R}]=\widetilde{\delta R}.$                                                                                                                                                                                                                                                                                                       | $[\delta,\widetilde{R}]=\widetilde{\delta R}.$                                                                                                                                                                                                                                                                                                       |
| 0902.0431_E00501 | 82   | ■ 1.000 | $\left[\widetilde{H},\widetilde{T}\right]=\frac{1}{2}[H,T]^\sim.$                                                                                                                                                                                                                                                                                    | $[\widetilde{H},\widetilde{T}]=\frac{1}{2}[H,T]^\sim.$                                                                                                                                                                                                                                                                                               | $[\widetilde{H},\widetilde{T}]=\frac{1}{2}[H,T]^\sim.$                                                                                                                                                                                                                                                                                               |

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|------------------|------|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0902.0431_E00503 | 82   | ■ 1.000 | $=\frac{1}{4}\left([H,T]X+X[H,T]^*\right)=\frac{1}{2}[H,T]\sim X,\quad X\in\mathfrak{J}^C.$                                                                                                                                                         | $=\frac{1}{4}([H,T]X+X[H,T]^*)=\frac{1}{2}[H,T]\sim X,\quad X\in\mathfrak{J}^C.$                                                                                                                                                                    | $=\frac{1}{4}([H,T]X+X[H,T]^*)=\frac{1}{2}[H,T]\sim X,\quad X\in\mathfrak{J}^C.$                                                                                                                                                                    |
| 0902.0431_E00510 | 84   | ■ 1.000 | $\left[h,\widetilde{F}_{32}(a)\right]=\left(-\lambda_k+\frac{1}{2}(\mu_3-\mu_2)\right)\widetilde{F}_{32}(a),\quad a=e_k+ie_{4+k},$                                                                                                                  | $\left[h,\widetilde{F}_{32}(a)\right]=\left(-\lambda_k+\frac{1}{2}(\mu_3-\mu_2)\right)\widetilde{F}_{32}(a),\quad a=e_k+ie_{4+k},$                                                                                                                  | $[h,\widetilde{F}_{32}(a)]=\left(-\lambda_k+\frac{1}{2}(\mu_3-\mu_2)\right)\widetilde{F}_{32}(a),\quad a=e_k+ie_{4+k},$                                                                                                                             |
| 0902.0431_E00514 | 84   | ■ 1.000 | $\mu=\alpha_1+2\alpha_2+3\alpha_3+2\alpha_4+2\alpha_5+\alpha_6$                                                                                                                                                                                     | $\mu=\alpha_1+2\alpha_2+3\alpha_3+2\alpha_4+2\alpha_5+\alpha_6$                                                                                                                                                                                     | $\mu=\alpha_1+2\alpha_2+3\alpha_3+2\alpha_4+2\alpha_5+\alpha_6$                                                                                                                                                                                     |
| 0902.0431_E00519 | 86   | ■ 1.000 | $\mathfrak{h}_{\boldsymbol{R}}=\left\{\sum_{k=0}^3\lambda_kH_k+\left(\sum_{j=1}^3\mu_jE_j\right)^{\sim}\mid\lambda_k,\mu_j\in\boldsymbol{R},\mu_1+\mu_2+\mu_3=0\right\}.$                                                                           | $\mathfrak{h}_{\boldsymbol{R}}=\left\{\sum_{k=0}^3\lambda_kH_k+\left(\sum_{j=1}^3\mu_jE_j\right)^{\sim}\mid\lambda_k,\mu_j\in\boldsymbol{R},\mu_1+\mu_2+\mu_3=0\right\}.$                                                                           | $\mathfrak{h}_{\boldsymbol{R}}=\left\{\sum_{k=0}^3\lambda_kH_k+\left(\sum_{j=1}^3\mu_jE_j\right)^{\sim}\mid\lambda_k,\mu_j\in\boldsymbol{R},\mu_1+\mu_2+\mu_3=0\right\}.$                                                                           |
| 0902.0431_E00523 | 86   | ■ 1.000 | $\left(\alpha_1,\alpha_1\right)=B_6\left(H_{\alpha_1},H_{\alpha_1}\right)=24\frac{1}{24}\frac{1}{24}2=\frac{1}{12},$                                                                                                                                | $\left(\alpha_1,\alpha_1\right)=B_6\left(H_{\alpha_1},H_{\alpha_1}\right)=24\frac{1}{24}\frac{1}{24}2=\frac{1}{12},$                                                                                                                                | $\left(\alpha_1,\alpha_1\right)=B_6\left(H_{\alpha_1},H_{\alpha_1}\right)=24\frac{1}{24}\frac{1}{24}2=\frac{1}{12},$                                                                                                                                |
| 0902.0431_E00525 | 87   | ■ 1.000 | $\begin{aligned} \left(E_6\right)^{\tau} &= \left\{\alpha\in E_6\mid\tau\alpha=\alpha\tau\right\} \\ &= \left\{\alpha\in E_6\mid\lambda(\alpha)=\alpha\right\} = (E_6)^{\lambda}. \end{aligned}$                                                    | $\begin{aligned} \left(E_6\right)^{\tau} &= \left\{\alpha\in E_6\mid\tau\alpha=\alpha\tau\right\} \\ &= \left\{\alpha\in E_6\mid\lambda(\alpha)=\alpha\right\} = (E_6)^{\lambda}. \end{aligned}$                                                    | $\begin{aligned} \left(E_6\right)^{\tau} &= \left\{\alpha\in E_6\mid\tau\alpha=\alpha\tau\right\} \\ &= \left\{\alpha\in E_6\mid\lambda(\alpha)=\alpha\right\} = (E_6)^{\lambda}. \end{aligned}$                                                    |
| 0902.0431_E00526 | 87   | ■ 1.000 | $\begin{aligned} \left(E_6\right)^{\tau} &= \left\{\alpha\in E_6\mid(\alpha X,\alpha Y)=(X,Y),X,Y\in\mathfrak{J}^C\right\} \\ &= \left\{\alpha\in E_6\mid\alpha E=E\right\} = (E_6)_E. \end{aligned}$                                               | $\begin{aligned} \left(E_6\right)^{\tau} &= \left\{\alpha\in E_6\mid(\alpha X,\alpha Y)=(X,Y),X,Y\in\mathfrak{J}^C\right\} \\ &= \left\{\alpha\in E_6\mid\alpha E=E\right\} = (E_6)_E. \end{aligned}$                                               | $\begin{aligned} \left(E_6\right)^{\tau} &= \left\{\alpha\in E_6\mid(\alpha X,\alpha Y)=(X,Y),X,Y\in\mathfrak{J}^C\right\} \\ &= \left\{\alpha\in E_6\mid\alpha E=E\right\} = (E_6)_E. \end{aligned}$                                               |
| 0902.0431_E00527 | 88   | ■ 1.000 | $F_4=\left\{\alpha\in\mathrm{Iso}_{\boldsymbol{R}}(\mathfrak{J})\mid\det(\alpha X)=\det X,(\alpha X,\alpha Y)=(X,Y)\right\} \\ =\left\{\alpha\in\mathrm{Iso}_{\boldsymbol{R}}(\mathfrak{J})\mid\alpha(X\times Y)=\alpha X\times\alpha Y\right\}.$   | $F_4=\left\{\alpha\in\mathrm{Iso}_{\boldsymbol{R}}(\mathfrak{J})\mid\det(\alpha X)=\det X,(\alpha X,\alpha Y)=(X,Y)\right\} \\ =\left\{\alpha\in\mathrm{Iso}_{\boldsymbol{R}}(\mathfrak{J})\mid\alpha(X\times Y)=\alpha X\times\alpha Y\right\}.$   | $F_4=\left\{\alpha\in\mathrm{Iso}_{\boldsymbol{R}}(\mathfrak{J})\mid\det(\alpha X)=\det X,(\alpha X,\alpha Y)=(X,Y)\right\} \\ =\left\{\alpha\in\mathrm{Iso}_{\boldsymbol{R}}(\mathfrak{J})\mid\alpha(X\times Y)=\alpha X\times\alpha Y\right\}.$   |
| 0902.0431_E00528 | 88   | ■ 1.000 | $\alpha_{12}(t)\begin{pmatrix}\xi_1&x_3&\bar{x}_2\\\bar{x}_3&\xi_2&x_1\\x_2&\bar{x}_1&\xi_3\end{pmatrix}=\begin{pmatrix}e^{it}\xi_1&x_3&e^{it/2}\bar{x}_2\\\bar{x}_3&e^{-it}\xi_2&e^{-it/2}x_1\\e^{it/2}x_2&e^{-it/2}\bar{x}_1&\xi_3\end{pmatrix},$ | $\alpha_{12}(t)\begin{pmatrix}\xi_1&x_3&\bar{x}_2\\\bar{x}_3&\xi_2&x_1\\x_2&\bar{x}_1&\xi_3\end{pmatrix}=\begin{pmatrix}e^{it}\xi_1&x_3&e^{it/2}\bar{x}_2\\\bar{x}_3&e^{-it}\xi_2&e^{-it/2}x_1\\e^{it/2}x_2&e^{-it/2}\bar{x}_1&\xi_3\end{pmatrix},$ | $\alpha_{12}(t)\begin{pmatrix}\xi_1&x_3&\bar{x}_2\\\bar{x}_3&\xi_2&x_1\\x_2&\bar{x}_1&\xi_3\end{pmatrix}=\begin{pmatrix}e^{it}\xi_1&x_3&e^{it/2}\bar{x}_2\\\bar{x}_3&e^{-it}\xi_2&e^{-it/2}x_1\\e^{it/2}x_2&e^{-it/2}\bar{x}_1&\xi_3\end{pmatrix},$ |
| 0902.0431_E00533 | 90   | ■ 1.000 | $EIV=\left\{X\in\mathfrak{J}^C\mid\det X=1,\langle X,X\rangle=3\right\}.$                                                                                                                                                                           | $EIV=\left\{X\in\mathfrak{J}^C\mid\det X=1,\langle X,X\rangle=3\right\}.$                                                                                                                                                                           | $EIV=\left\{X\in\mathfrak{J}^C\mid\det X=1,\langle X,X\rangle=3\right\}.$                                                                                                                                                                           |

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|------------------|------|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0902.0431_E00534 | 90   | ■ 1.000 | $\alpha X=\left(\begin{array}{ccc}\xi_1&\theta&\theta\\\xi_2&\theta&\theta\\\xi_3&\theta&\theta\end{array}\right),\quad \xi_1\in C,\xi_2\geq 0,\xi_3\geq 0$                                                                           | $\alpha X=\left(\begin{array}{ccc}\xi_1&0&0\\0&\xi_2&0\\0&0&\xi_3\end{array}\right),\quad \xi_1\in C,\xi_2\geq 0,\xi_3\geq 0$                                                                                                         | $\alpha X=\left(\begin{array}{ccc}\xi_1&0&0\\0&\xi_2&0\\0&0&\xi_3\end{array}\right),\quad \xi_1\in C,\xi_2\geq 0,\xi_3\geq 0$                                                                                                         |
| 0902.0431_E00536 | 90   | ■ 1.000 | $z\left(E_6\right)=\left\{1,\omega 1,\omega ^21\right\},\quad \omega =-\frac{1}{2}+\frac{\sqrt{3}}{2}i\in C.$                                                                                                                         | $z\left(E_6\right)=\left\{1,\omega 1,\omega ^21\right\},\quad \omega =-\frac{1}{2}+\frac{\sqrt{3}}{2}i\in C.$                                                                                                                         | $z\left(E_6\right)=\left\{1,\omega 1,\omega ^21\right\},\quad \omega =-\frac{1}{2}+\frac{\sqrt{3}}{2}i\in C.$                                                                                                                         |
| 0902.0431_E00537 | 90   | ■ 1.000 | $\beta Y=Y,\quad \text{for all }\beta\in F_4.$                                                                                                                                                                                        | $\beta Y=Y,\quad \text{for all }\beta\in F_4.$                                                                                                                                                                                        | $\beta Y=Y,\quad \text{for all }\beta\in F_4.$                                                                                                                                                                                        |
| 0902.0431_E00538 | 90   | ■ 1.000 | $\beta X=TXT^{-1},\quad X\in\mathfrak{J}^C,$                                                                                                                                                                                          | $\beta X=TXT^{-1},\quad X\in\mathfrak{J}^C,$                                                                                                                                                                                          | $\beta X=TXT^{-1},\quad X\in\mathfrak{J}^C,$                                                                                                                                                                                          |
| 0902.0431_E00539 | 91   | ■ 1.000 | $\alpha E=Y=\omega E,\quad \omega\in C,$                                                                                                                                                                                              | $\alpha E=Y=\omega E,\quad \omega\in C,$                                                                                                                                                                                              | $\alpha E=Y=\omega E,\quad \omega\in C,$                                                                                                                                                                                              |
| 0902.0431_E00540 | 91   | ■ 1.000 | $\left(E_6\right)^{\sigma}=\left\{\alpha\in E_6\mid\sigma\alpha=\alpha\sigma\right\}.$                                                                                                                                                | $\left(E_6\right)^{\sigma}=\left\{\alpha\in E_6\mid\sigma\alpha=\alpha\sigma\right\}.$                                                                                                                                                | $\left(E_6\right)^{\sigma}=\left\{\alpha\in E_6\mid\sigma\alpha=\alpha\sigma\right\}.$                                                                                                                                                |
| 0902.0431_E00542 | 91   | ■ 1.000 | $\alpha E_1=\xi E_1,\quad (\tau\xi)\xi=1.$                                                                                                                                                                                            | $\alpha E_1=\xi E_1,\quad (\tau\xi)\xi=1.$                                                                                                                                                                                            | $\alpha E_1=\xi E_1,\quad (\tau\xi)\xi=1.$                                                                                                                                                                                            |
| 0902.0431_E00543 | 91   | ■ 1.000 | $\alpha E_2,\alpha E_3\in\mathfrak{J}\left(2,\mathfrak{C}^C\right).$                                                                                                                                                                  | $\alpha E_2,\alpha E_3\in\mathfrak{J}\left(2,\mathfrak{C}^C\right).$                                                                                                                                                                  | $\alpha E_2,\alpha E_3\in\mathfrak{J}\left(2,\mathfrak{C}^C\right).$                                                                                                                                                                  |
| 0902.0431_E00545 | 92   | ■ 1.000 | $\alpha E_1\notin\mathfrak{J}\left(2,\mathfrak{C}^C\right).$                                                                                                                                                                          | $\alpha E_1\notin\mathfrak{J}\left(2,\mathfrak{C}^C\right).$                                                                                                                                                                          | $\alpha E_1\notin\mathfrak{J}\left(2,\mathfrak{C}^C\right).$                                                                                                                                                                          |
| 0902.0431_E00547 | 92   | ■ 1.000 | $\alpha E_1=\xi E_1+\xi_2E_2+\xi_3E_3+F_1\left(x_1\right),\quad \xi\neq 0.$                                                                                                                                                           | $\alpha E_1=\xi E_1+\xi_2E_2+\xi_3E_3+F_1\left(x_1\right),\quad \xi\neq 0.$                                                                                                                                                           | $\alpha E_1=\xi E_1+\xi_2E_2+\xi_3E_3+F_1\left(x_1\right),\quad \xi\neq 0.$                                                                                                                                                           |
| 0902.0431_E00549 | 92   | ■ 1.000 | $\alpha E_1=\xi E_1,\quad \xi\neq 0.$                                                                                                                                                                                                 | $\alpha E_1=\xi E_1,\quad \xi\neq 0.$                                                                                                                                                                                                 | $\alpha E_1=\xi E_1,\quad \xi\neq 0.$                                                                                                                                                                                                 |
| 0902.0431_E00550 | 92   | ■ 1.000 | $1=\left\langle E_1,E_1\right\rangle =\left\langle \alpha E_1,\alpha E_1\right\rangle =\left\langle \xi E_1,\xi E_1\right\rangle =\left(\tau\xi\right)\xi\left\langle E_1,E_1\right\rangle =\left(\tau\xi\right)\xi,$                 | $1=\left\langle E_1,E_1\right\rangle =\left\langle \alpha E_1,\alpha E_1\right\rangle =\left\langle \xi E_1,\xi E_1\right\rangle =\left(\tau\xi\right)\xi\left\langle E_1,E_1\right\rangle =\left(\tau\xi\right)\xi,$                 | $1=\left\langle E_1,E_1\right\rangle =\left\langle \alpha E_1,\alpha E_1\right\rangle =\left\langle \xi E_1,\xi E_1\right\rangle =\left(\tau\xi\right)\xi\left\langle E_1,E_1\right\rangle =\left(\tau\xi\right)\xi,$                 |
| 0902.0431_E00551 | 92   | ■ 1.000 | $\left(E_6\right)_{E_1}=\left\{\alpha\in E_6\mid\alpha E_1=E_1\right\}.$                                                                                                                                                              | $\left(E_6\right)_{E_1}=\left\{\alpha\in E_6\mid\alpha E_1=E_1\right\}.$                                                                                                                                                              | $\left(E_6\right)_{E_1}=\left\{\alpha\in E_6\mid\alpha E_1=E_1\right\}.$                                                                                                                                                              |
| 0902.0431_E00552 | 93   | ■ 1.000 | $V^{10}=\left\{X\in\mathfrak{J}^C\mid 2E_1\times X=-\tau X\right\}=\left\{\left(\begin{array}{ccc}0&0&0\\0&\xi&x\\0&\bar{x}&-\tau\xi\end{array}\right)\mid\xi\in C,x\in\mathfrak{C}\right\}.$                                         | $V^{10}=\left\{X\in\mathfrak{J}^C\mid 2E_1\times X=-\tau X\right\}=\left\{\left(\begin{array}{ccc}0&0&0\\0&\xi&x\\0&\bar{x}&-\tau\xi\end{array}\right)\mid\xi\in C,x\in\mathfrak{C}\right\}.$                                         | $V^{10}=\left\{X\in\mathfrak{J}^C\mid 2E_1\times X=-\tau X\right\}=\left\{\left(\begin{array}{ccc}0&0&0\\0&\xi&x\\0&\bar{x}&-\tau\xi\end{array}\right)\mid\xi\in C,x\in\mathfrak{C}\right\}.$                                         |
| 0902.0431_E00553 | 93   | ■ 1.000 | $2E_1\times\alpha X=2\alpha E_1\times\alpha X=2\tau\alpha\tau\left(E_1\times X\right)=\tau\alpha\tau\left(-\tau X\right)=-\tau\left(\alpha X\right),\quad\left\langle\alpha X,\alpha X\right\rangle=\left\langle X,X\right\rangle=2.$ | $2E_1\times\alpha X=2\alpha E_1\times\alpha X=2\tau\alpha\tau\left(E_1\times X\right)=\tau\alpha\tau\left(-\tau X\right)=-\tau\left(\alpha X\right),\quad\left\langle\alpha X,\alpha X\right\rangle=\left\langle X,X\right\rangle=2.$ | $2E_1\times\alpha X=2\alpha E_1\times\alpha X=2\tau\alpha\tau\left(E_1\times X\right)=\tau\alpha\tau\left(-\tau X\right)=-\tau\left(\alpha X\right),\quad\left\langle\alpha X,\alpha X\right\rangle=\left\langle X,X\right\rangle=2.$ |
| 0902.0431_E00554 | 93   | ■ 1.000 | $\alpha_{23}\left(t_0\right)X\in S^8=\left\{X\in V^9\mid\left\langle X,X\right\rangle=2\right\}$                                                                                                                                      | $\alpha_{23}\left(t_0\right)X\in S^8=\left\{X\in V^9\mid\left\langle X,X\right\rangle=2\right\}$                                                                                                                                      | $\alpha_{23}\left(t_0\right)X\in S^8=\left\{X\in V^9\mid\left\langle X,X\right\rangle=2\right\}$                                                                                                                                      |
| 0902.0431_E00556 | 93   | ■ 1.000 | $\alpha_{23}\left(\pi/2\right)\beta\alpha_{23}\left(t_0\right)X=i\left(E_2+E_3\right).$                                                                                                                                               | $\alpha_{23}\left(\pi/2\right)\beta\alpha_{23}\left(t_0\right)X=i\left(E_2+E_3\right).$                                                                                                                                               | $\alpha_{23}\left(\pi/2\right)\beta\alpha_{23}\left(t_0\right)X=i\left(E_2+E_3\right).$                                                                                                                                               |
| 0902.0431_E00557 | 93   | ■ 1.000 | $p:\left(E_6\right)_{E_1}\rightarrow SO\left(10\right)=SO\left(V^{10}\right)$                                                                                                                                                         | $p:\left(E_6\right)_{E_1}\rightarrow SO\left(10\right)=SO\left(V^{10}\right)$                                                                                                                                                         | $p:\left(E_6\right)_{E_1}\rightarrow SO\left(10\right)=SO\left(V^{10}\right)$                                                                                                                                                         |
| 0902.0431_E00559 | 94   | ■ 1.000 | $\left(E_6\right)_{E_1}/\left\{1,\sigma\right\}\cong SO\left(10\right).$                                                                                                                                                              | $\left(E_6\right)_{E_1}/\left\{1,\sigma\right\}\cong SO\left(10\right).$                                                                                                                                                              | $\left(E_6\right)_{E_1}/\left\{1,\sigma\right\}\cong SO\left(10\right).$                                                                                                                                                              |

| Identifier       | Page | Conf.   | LaTeX source                                                                                                                                                                                                                                                                                                                                                       | Rendered                                                                                                                                                                                                                                                                                                                                                           | Image                                                                                                                                                                                                                                                                                                                                                              |
|------------------|------|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0902.0431_E00561 | 94   | ■ 1.000 | $\begin{aligned} U(1) \, \&= \left\{ \alpha_{12}(t) \right. \\ \alpha_{13}(t) &= \exp i \, t \left( 2 \, E_1 - E_2 - E_3 \right) \\ \sim \mid t \, \in \, \boldsymbol{R} \, \right\} \, \&= \left\{ \phi(\theta) \right. \\ \theta &\in C, (\tau \theta) \theta = 1 \, \end{aligned}$                                                                              | $U(1) = \{ \alpha_{12}(t) \alpha_{13}(t) = \exp i t (2E_1 - E_2 - E_3)^\sim \mid t \in \boldsymbol{R} \}$ $= \{ \phi(\theta) \mid \theta \in C, (\tau \theta) \theta = 1 \}$                                                                                                                                                                                       | $U(1) = \{ \alpha_{12}(t) \alpha_{13}(t) = \exp i t (2E_1 - E_2 - E_3)^\sim \mid t \in \boldsymbol{R} \}$ $= \{ \phi(\theta) \mid \theta \in C, (\tau \theta) \theta = 1 \}$                                                                                                                                                                                       |
| 0902.0431_E00563 | 94   | ■ 1.000 | $\phi(\theta) \mid \mathfrak{E}_1^C = \theta^4 1, \quad \phi(\theta) \mid \mathfrak{J}(2, \mathfrak{C})^C = \theta^{-2} 1, \quad \phi(\theta) \mid (\mathfrak{J}^C)_{-\sigma} = \theta 1.$                                                                                                                                                                         | $\phi(\theta) \mid \mathfrak{E}_1^C = \theta^4 1, \quad \phi(\theta) \mid \mathfrak{J}(2, \mathfrak{C})^C = \theta^{-2} 1, \quad \phi(\theta) \mid (\mathfrak{J}^C)_{-\sigma} = \theta 1.$                                                                                                                                                                         | $\phi(\theta) \mid \mathfrak{E}_1^C = \theta^4 1, \quad \phi(\theta) \mid \mathfrak{J}(2, \mathfrak{C})^C = \theta^{-2} 1, \quad \phi(\theta) \mid (\mathfrak{J}^C)_{-\sigma} = \theta 1.$                                                                                                                                                                         |
| 0902.0431_E00564 | 95   | ■ 1.000 | $\varphi(\theta, \beta) = \phi(\theta) \beta.$                                                                                                                                                                                                                                                                                                                     | $\varphi(\theta, \beta) = \phi(\theta) \beta.$                                                                                                                                                                                                                                                                                                                     | $\varphi(\theta, \beta) = \phi(\theta) \beta.$                                                                                                                                                                                                                                                                                                                     |
| 0902.0431_E00565 | 95   | ■ 1.000 | $\alpha E_1 = \theta^4 E_1 = \phi(\theta) E_1$                                                                                                                                                                                                                                                                                                                     | $\alpha E_1 = \theta^4 E_1 = \phi(\theta) E_1$                                                                                                                                                                                                                                                                                                                     | $\alpha E_1 = \theta^4 E_1 = \phi(\theta) E_1$                                                                                                                                                                                                                                                                                                                     |
| 0902.0431_E00566 | 95   | ■ 1.000 | $\operatorname{Ker} \varphi = \{ (\theta, \phi(\theta)^{-1}) \mid \theta \in C, \theta^4 = 1 \} = \{ (\theta, \phi(\theta^{-1})) \mid \theta = \pm 1, \pm i \} = \boldsymbol{Z}_4.$                                                                                                                                                                                | $\operatorname{Ker} \varphi = \{ (\theta, \phi(\theta)^{-1}) \mid \theta \in C, \theta^4 = 1 \} = \{ (\theta, \phi(\theta^{-1})) \mid \theta = \pm 1, \pm i \} = \boldsymbol{Z}_4.$                                                                                                                                                                                | $\operatorname{Ker} \varphi = \{ (\theta, \phi(\theta)^{-1}) \mid \theta \in C, \theta^4 = 1 \} = \{ (\theta, \phi(\theta^{-1})) \mid \theta = \pm 1, \pm i \} = \boldsymbol{Z}_4.$                                                                                                                                                                                |
| 0902.0431_E00567 | 95   | ■ 1.000 | $(E_6)^\gamma = \{ \alpha \in E_6 \mid \gamma \alpha = \alpha \gamma \}.$                                                                                                                                                                                                                                                                                          | $(E_6)^\gamma = \{ \alpha \in E_6 \mid \gamma \alpha = \alpha \gamma \}.$                                                                                                                                                                                                                                                                                          | $(E_6)^\gamma = \{ \alpha \in E_6 \mid \gamma \alpha = \alpha \gamma \}.$                                                                                                                                                                                                                                                                                          |
| 0902.0431_E00568 | 95   | ■ 1.000 | $\mathfrak{J}^C = \mathfrak{J}(3, \boldsymbol{H})^C \oplus (\boldsymbol{H}^3)^C,$                                                                                                                                                                                                                                                                                  | $\mathfrak{J}^C = \mathfrak{J}(3, \boldsymbol{H})^C \oplus (\boldsymbol{H}^3)^C,$                                                                                                                                                                                                                                                                                  | $\mathfrak{J}^C = \mathfrak{J}(3, \boldsymbol{H})^C \oplus (\boldsymbol{H}^3)^C,$                                                                                                                                                                                                                                                                                  |
| 0902.0431_E00570 | 95   | ■ 1.000 | $k: M(3, \boldsymbol{H}) \rightarrow M(6, C) \quad \text{and} \quad k: \boldsymbol{H}^3 \rightarrow M(2, 6, C).$                                                                                                                                                                                                                                                   | $k: M(3, \boldsymbol{H}) \rightarrow M(6, C) \quad \text{and} \quad k: \boldsymbol{H}^3 \rightarrow M(2, 6, C).$                                                                                                                                                                                                                                                   | $k: M(3, \boldsymbol{H}) \rightarrow M(6, C) \quad \text{and} \quad k: \boldsymbol{H}^3 \rightarrow M(2, 6, C).$                                                                                                                                                                                                                                                   |
| 0902.0431_E00574 | 96   | ■ 1.000 | $k_J(M) = k(M) J.$                                                                                                                                                                                                                                                                                                                                                 | $k_J(M) = k(M) J.$                                                                                                                                                                                                                                                                                                                                                 | $k_J(M) = k(M) J.$                                                                                                                                                                                                                                                                                                                                                 |
| 0902.0431_E00576 | 96   | ■ 1.000 | $\langle S, T \rangle = \operatorname{tr}((\tau^t S) T), \quad \langle P, Q \rangle = \operatorname{tr}((\tau^t P) Q).$                                                                                                                                                                                                                                            | $\langle S, T \rangle = \operatorname{tr}((\tau^t S) T), \quad \langle P, Q \rangle = \operatorname{tr}((\tau^t P) Q).$                                                                                                                                                                                                                                            | $\langle S, T \rangle = \operatorname{tr}((\tau^t S) T), \quad \langle P, Q \rangle = \operatorname{tr}((\tau^t P) Q).$                                                                                                                                                                                                                                            |
| 0902.0431_E00577 | 96   | ■ 1.000 | $\langle k_J(M), k_J(N) \rangle = 2 \langle M, N \rangle, \quad M, N \in \mathfrak{J}(3, \boldsymbol{H})^C,$ $\langle k(\boldsymbol{a}), k(\boldsymbol{b}) \rangle = 2 \langle \boldsymbol{a}, \boldsymbol{b} \rangle, \quad \boldsymbol{a}, \boldsymbol{b} \in (\boldsymbol{H}^3)^C.$ $\det(k_J(M)) = (\det M)^2, \quad M \in \mathfrak{J}(3, \boldsymbol{H})^C.$ | $\langle k_J(M), k_J(N) \rangle = 2 \langle M, N \rangle, \quad M, N \in \mathfrak{J}(3, \boldsymbol{H})^C,$ $\langle k(\boldsymbol{a}), k(\boldsymbol{b}) \rangle = 2 \langle \boldsymbol{a}, \boldsymbol{b} \rangle, \quad \boldsymbol{a}, \boldsymbol{b} \in (\boldsymbol{H}^3)^C.$ $\det(k_J(M)) = (\det M)^2, \quad M \in \mathfrak{J}(3, \boldsymbol{H})^C.$ | $\langle k_J(M), k_J(N) \rangle = 2 \langle M, N \rangle, \quad M, N \in \mathfrak{J}(3, \boldsymbol{H})^C,$ $\langle k(\boldsymbol{a}), k(\boldsymbol{b}) \rangle = 2 \langle \boldsymbol{a}, \boldsymbol{b} \rangle, \quad \boldsymbol{a}, \boldsymbol{b} \in (\boldsymbol{H}^3)^C.$ $\det(k_J(M)) = (\det M)^2, \quad M \in \mathfrak{J}(3, \boldsymbol{H})^C.$ |
| 0902.0431_E00579 | 96   | ■ 1.000 | $\begin{pmatrix} a' & b' \\ -\tau b' & \tau a' \end{pmatrix} + i \begin{pmatrix} c' & d' \\ -\tau d' & \tau c' \end{pmatrix} = 0.$                                                                                                                                                                                                                                 | $\begin{pmatrix} a' & b' \\ -\tau b' & \tau a' \end{pmatrix} + i \begin{pmatrix} c' & d' \\ -\tau d' & \tau c' \end{pmatrix} = 0.$                                                                                                                                                                                                                                 | $\begin{pmatrix} a' & b' \\ -\tau b' & \tau a' \end{pmatrix} + i \begin{pmatrix} c' & d' \\ -\tau d' & \tau c' \end{pmatrix} = 0.$                                                                                                                                                                                                                                 |
| 0902.0431_E00580 | 97   | ■ 1.000 | $\langle k(m_1 + in_1), k(m_2 + in_2) \rangle = 2 \langle m_1 + in_1, m_2 + in_2 \rangle.$                                                                                                                                                                                                                                                                         | $\langle k(m_1 + in_1), k(m_2 + in_2) \rangle = 2 \langle m_1 + in_1, m_2 + in_2 \rangle.$                                                                                                                                                                                                                                                                         | $\langle k(m_1 + in_1), k(m_2 + in_2) \rangle = 2 \langle m_1 + in_1, m_2 + in_2 \rangle.$                                                                                                                                                                                                                                                                         |

| Identifier       | Page | Conf.   | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                                           | Rendered                                                                                                                                                                                                                                                                                                                                                                                  | Image                                                                                                                                                                                                                                                                                                                                                                                     |
|------------------|------|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0902.0431_E00583 | 97   | ■ 1.000 | $\begin{aligned} & \operatorname{det}\left(\begin{array}{ccc} \xi_1 & m_3+n_3e_2 & \tau(m_2+n_2e_2) \\ \tau(m_3+n_3e_2) & \xi_2 & m_1+n_1e_2 \\ m_2+n_2e_2 & \tau(m_1+n_1e_2) & \xi_3 \end{array}\right) \\ & = \xi_1\xi_2\xi_3 + (m_1+n_1e_2)(m_2+n_2e_2)(m_3+n_3e_2) \\ & \quad + \tau((m_1+n_1e_2)(m_2+n_2e_2)(m_3+n_3e_2)) - \sum_{i=1}^3 \xi_i(m_i+n_ie_2)\tau(m_i+n_ie_2) \\ & = \text{the contents of in the parenthesis above.} \end{aligned}$ | $\det\left(\begin{array}{ccc} \xi_1 & m_3+n_3e_2 & \tau(m_2+n_2e_2) \\ \tau(m_3+n_3e_2) & \xi_2 & m_1+n_1e_2 \\ m_2+n_2e_2 & \tau(m_1+n_1e_2) & \xi_3 \end{array}\right)$ $= \xi_1\xi_2\xi_3 + (m_1+n_1e_2)(m_2+n_2e_2)(m_3+n_3e_2)$ $+ \tau((m_1+n_1e_2)(m_2+n_2e_2)(m_3+n_3e_2)) - \sum_{i=1}^3 \xi_i(m_i+n_ie_2)\tau(m_i+n_ie_2)$ $= \text{the contents of in the parenthesis above.}$ | $\det\left(\begin{array}{ccc} \xi_1 & m_3+n_3e_2 & \tau(m_2+n_2e_2) \\ \tau(m_3+n_3e_2) & \xi_2 & m_1+n_1e_2 \\ m_2+n_2e_2 & \tau(m_1+n_1e_2) & \xi_3 \end{array}\right)$ $= \xi_1\xi_2\xi_3 + (m_1+n_1e_2)(m_2+n_2e_2)(m_3+n_3e_2)$ $+ \tau((m_1+n_1e_2)(m_2+n_2e_2)(m_3+n_3e_2)) - \sum_{i=1}^3 \xi_i(m_i+n_ie_2)\tau(m_i+n_ie_2)$ $= \text{the contents of in the parenthesis above.}$ |
| 0902.0431_E00584 | 98   | ■ 1.000 | $E_{\{6, \boldsymbol{H}\}} = \left\{ \alpha \in \operatorname{Iso}_C \left( (\mathfrak{H}_{\boldsymbol{H}})^C \right) \mid \det(\alpha M) = \det M, \langle \alpha M, \alpha N \rangle = \langle M, N \rangle \right\}.$                                                                                                                                                                                                                               | $E_{6,\boldsymbol{H}} = \left\{ \alpha \in \operatorname{Iso}_C \left( (\mathfrak{H}_{\boldsymbol{H}})^C \right) \mid \det(\alpha M) = \det M, \langle \alpha M, \alpha N \rangle = \langle M, N \rangle \right\}.$                                                                                                                                                                       | $E_{6,\boldsymbol{H}} = \left\{ \alpha \in \operatorname{Iso}_C \left( (\mathfrak{H}_{\boldsymbol{H}})^C \right) \mid \det(\alpha M) = \det M, \langle \alpha M, \alpha N \rangle = \langle M, N \rangle \right\}.$                                                                                                                                                                       |
| 0902.0431_E00585 | 98   | ■ 1.000 | $F_{\{4, \boldsymbol{H}\}} = \left\{ \alpha \in E_{6,\boldsymbol{H}} \mid \alpha E = E \right\},$                                                                                                                                                                                                                                                                                                                                                      | $F_{4,\boldsymbol{H}} = \left\{ \alpha \in E_{6,\boldsymbol{H}} \mid \alpha E = E \right\},$                                                                                                                                                                                                                                                                                              | $F_{4,\boldsymbol{H}} = \left\{ \alpha \in E_{6,\boldsymbol{H}} \mid \alpha E = E \right\},$                                                                                                                                                                                                                                                                                              |
| 0902.0431_E00589 | 98   | ■ 1.000 | $\varphi_*(C)M = CM + MC^* = [C, M], \quad M \in \mathfrak{H}_{\boldsymbol{H}}$                                                                                                                                                                                                                                                                                                                                                                        | $\varphi_*(C)M = CM + MC^* = [C, M], \quad M \in \mathfrak{H}_{\boldsymbol{H}}$                                                                                                                                                                                                                                                                                                           | $\varphi_*(C)M = CM + MC^* = [C, M], \quad M \in \mathfrak{H}_{\boldsymbol{H}}$                                                                                                                                                                                                                                                                                                           |
| 0902.0431_E00590 | 98   | ■ 1.000 | $E_{\{6, \boldsymbol{H}\}} / F_{\{4, \boldsymbol{H}\}} \simeq EIV_{\boldsymbol{H}} = \left\{ X \in (\mathfrak{H}_{\boldsymbol{H}})^C \mid \det M = 1, \langle M, M \rangle = 3 \right\}$                                                                                                                                                                                                                                                               | $E_{6,\boldsymbol{H}}/F_{4,\boldsymbol{H}} \simeq EIV_{\boldsymbol{H}} = \left\{ X \in (\mathfrak{H}_{\boldsymbol{H}})^C \mid \det M = 1, \langle M, M \rangle = 3 \right\}$                                                                                                                                                                                                              | $E_{6,\boldsymbol{H}}/F_{4,\boldsymbol{H}} \simeq EIV_{\boldsymbol{H}} = \left\{ X \in (\mathfrak{H}_{\boldsymbol{H}})^C \mid \det M = 1, \langle M, M \rangle = 3 \right\}$                                                                                                                                                                                                              |
| 0902.0431_E00592 | 98   | ■ 1.000 | $\operatorname{det}(\varphi(A)M) = \det M.$                                                                                                                                                                                                                                                                                                                                                                                                            | $\det(\varphi(A)M) = \det M.$                                                                                                                                                                                                                                                                                                                                                             | $\det(\varphi(A)M) = \det M.$                                                                                                                                                                                                                                                                                                                                                             |
| 0902.0431_E00596 | 99   | ■ 1.000 | $\alpha X \times \alpha Y = {}^t\alpha^{-1}(X \times Y), \quad X, Y \in \mathfrak{J}^C.$                                                                                                                                                                                                                                                                                                                                                               | $\alpha X \times \alpha Y = {}^t\alpha^{-1}(X \times Y), \quad X, Y \in \mathfrak{J}^C.$                                                                                                                                                                                                                                                                                                  | $\alpha X \times \alpha Y = {}^t\alpha^{-1}(X \times Y), \quad X, Y \in \mathfrak{J}^C.$                                                                                                                                                                                                                                                                                                  |
| 0902.0431_E00598 | 99   | ■ 1.000 | $k_J^{-1}(M) = -k^{-1}(MJ) = -k^{-1}(J(\tau M)), \quad \tau k^{-1}(M) = -k^{-1}(JMJ),$                                                                                                                                                                                                                                                                                                                                                                 | $k_J^{-1}(M) = -k^{-1}(MJ) = -k^{-1}(J(\tau M)), \quad \tau k^{-1}(M) = -k^{-1}(JMJ),$                                                                                                                                                                                                                                                                                                    | $k_J^{-1}(M) = -k^{-1}(MJ) = -k^{-1}(J(\tau M)), \quad \tau k^{-1}(M) = -k^{-1}(JMJ),$                                                                                                                                                                                                                                                                                                    |
| 0902.0431_E00600 | 100  | ■ 1.000 | $\alpha = \varphi(1, A)\beta = \varphi(1, A)\varphi(p, E) = \varphi(p, A).$                                                                                                                                                                                                                                                                                                                                                                            | $\alpha = \varphi(1, A)\beta = \varphi(1, A)\varphi(p, E) = \varphi(p, A).$                                                                                                                                                                                                                                                                                                               | $\alpha = \varphi(1, A)\beta = \varphi(1, A)\varphi(p, E) = \varphi(p, A).$                                                                                                                                                                                                                                                                                                               |
| 0902.0431_E00601 | 100  | ■ 1.000 | $(E_6)^{\tau\gamma} = \left\{ \alpha \in E_6 \mid \tau\gamma\alpha = \alpha\tau\gamma \right\}$ $= \left\{ \alpha \in E_6 \mid \gamma\lambda(\alpha)\gamma = \alpha \right\} = (E_6)^{\lambda\gamma}.$                                                                                                                                                                                                                                                 | $(E_6)^{\tau\gamma} = \left\{ \alpha \in E_6 \mid \tau\gamma\alpha = \alpha\tau\gamma \right\}$ $= \left\{ \alpha \in E_6 \mid \gamma\lambda(\alpha)\gamma = \alpha \right\} = (E_6)^{\lambda\gamma}.$                                                                                                                                                                                    | $(E_6)^{\tau\gamma} = \left\{ \alpha \in E_6 \mid \tau\gamma\alpha = \alpha\tau\gamma \right\}$ $= \left\{ \alpha \in E_6 \mid \gamma\lambda(\alpha)\gamma = \alpha \right\} = (E_6)^{\lambda\gamma}.$                                                                                                                                                                                    |
| 0902.0431_E00605 | 101  | ■ 1.000 | $\mathfrak{J}(4, \boldsymbol{H}) = \left\{ P \in M(4, \boldsymbol{H}) \mid P^* = P \right\},$                                                                                                                                                                                                                                                                                                                                                          | $\mathfrak{J}(4, \boldsymbol{H}) = \left\{ P \in M(4, \boldsymbol{H}) \mid P^* = P \right\},$                                                                                                                                                                                                                                                                                             | $\mathfrak{J}(4, \boldsymbol{H}) = \left\{ P \in M(4, \boldsymbol{H}) \mid P^* = P \right\},$                                                                                                                                                                                                                                                                                             |
| 0902.0431_E00606 | 101  | ■ 1.000 | $P \circ Q = \frac{1}{2}(PQ + QP), \quad (P, Q) = \operatorname{tr}(P \circ Q).$                                                                                                                                                                                                                                                                                                                                                                       | $P \circ Q = \frac{1}{2}(PQ + QP), \quad (P, Q) = \operatorname{tr}(P \circ Q).$                                                                                                                                                                                                                                                                                                          | $P \circ Q = \frac{1}{2}(PQ + QP), \quad (P, Q) = \operatorname{tr}(P \circ Q).$                                                                                                                                                                                                                                                                                                          |
| 0902.0431_E00609 | 101  | ■ 1.000 | $\langle P, Q \rangle = (\tau P, Q).$                                                                                                                                                                                                                                                                                                                                                                                                                  | $\langle P, Q \rangle = (\tau P, Q).$                                                                                                                                                                                                                                                                                                                                                     | $\langle P, Q \rangle = (\tau P, Q).$                                                                                                                                                                                                                                                                                                                                                     |

| Identifier       | Page | Conf.   | LaTeX source                                                                                                                                                                                                                                                                                                                     | Rendered                                                                                                                                                                                                                                                                                                           | Image                                                                                                                                                                                                                                                                                                              |
|------------------|------|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0902.0431_E00610 | 101  | ■ 1.000 | $\backslash\left(X_{\{1\}}+i X_{\{2\}}\right) A^{*}=A X_{\{1\}} A^{*}+i A X_{\{2\}} A^{*}, \quad \backslash\quad A \in \operatorname{Sp}(4), X_{\{1\}}, X_{\{2\}} \in \mathfrak{J}(4, \boldsymbol{H})$ .                                                                                                                         | $A\left(X_1+i X_2\right) A^{*}=A X_1 A^{*}+i A X_2 A^{*}, \quad A \in \operatorname{Sp}(4), X_1, X_2 \in \mathfrak{J}(4, \boldsymbol{H})$ .                                                                                                                                                                        | $A\left(X_1+i X_2\right) A^{*}=A X_1 A^{*}+i A X_2 A^{*}, \quad A \in \operatorname{Sp}(4), X_1, X_2 \in \mathfrak{J}(4, \boldsymbol{H})$ .                                                                                                                                                                        |
| 0902.0431_E00611 | 101  | ■ 1.000 | $\backslash\left\langle\right\rangle A P A^{*}, A Q A^{*} \backslash\right\rangle=\left\langle P, Q\right\rangle, \quad P, Q \in \mathfrak{J}(4, \boldsymbol{H})^C$ .                                                                                                                                                            | $\langle A P A^{*}, A Q A^{*}\rangle=\langle P, Q\rangle, \quad P, Q \in \mathfrak{J}(4, \boldsymbol{H})^C$ .                                                                                                                                                                                                      | $\langle A P A^{*}, A Q A^{*}\rangle=\langle P, Q\rangle, \quad P, Q \in \mathfrak{J}(4, \boldsymbol{H})^C$ .                                                                                                                                                                                                      |
| 0902.0431_E00612 | 102  | ■ 1.000 | $g(M+\boldsymbol{a})=\left(\begin{array}{cc} \frac{1}{2} \operatorname{tr}(M) & i \boldsymbol{a} \\ i \boldsymbol{a}^{*} & M-\frac{1}{2} \operatorname{tr}(M) E \end{array}\right), \quad M+\boldsymbol{a} \in\left(\mathfrak{J}_{\boldsymbol{H}}\right)^C \oplus\left(\boldsymbol{H}^3\right)^C=\mathfrak{J}^C$ .               | $g(M+\boldsymbol{a})=\left(\begin{array}{cc} \frac{1}{2} \operatorname{tr}(M) & i \boldsymbol{a} \\ i \boldsymbol{a}^{*} & M-\frac{1}{2} \operatorname{tr}(M) E \end{array}\right), \quad M+\boldsymbol{a} \in\left(\mathfrak{J}_{\boldsymbol{H}}\right)^C \oplus\left(\boldsymbol{H}^3\right)^C=\mathfrak{J}^C$ . | $g(M+\boldsymbol{a})=\left(\begin{array}{cc} \frac{1}{2} \operatorname{tr}(M) & i \boldsymbol{a} \\ i \boldsymbol{a}^{*} & M-\frac{1}{2} \operatorname{tr}(M) E \end{array}\right), \quad M+\boldsymbol{a} \in\left(\mathfrak{J}_{\boldsymbol{H}}\right)^C \oplus\left(\boldsymbol{H}^3\right)^C=\mathfrak{J}^C$ . |
| 0902.0431_E00613 | 102  | ■ 1.000 | $g(M+i \boldsymbol{a})=\left(\begin{array}{cc} \operatorname{tr}(M) & -\boldsymbol{a} \\ -\boldsymbol{a}^{*} & M \end{array}\right)-\frac{1}{2} \operatorname{tr}(M) E, \quad M+i \boldsymbol{a} \in \mathfrak{J}_{\boldsymbol{H}} \oplus i \boldsymbol{H}^3=\left(\mathfrak{J}^C\right)_{\tau \gamma}$ .                        | $g(M+i \boldsymbol{a})=\left(\begin{array}{cc} \operatorname{tr}(M) & -\boldsymbol{a} \\ -\boldsymbol{a}^{*} & M \end{array}\right)-\frac{1}{2} \operatorname{tr}(M) E, \quad M+i \boldsymbol{a} \in \mathfrak{J}_{\boldsymbol{H}} \oplus i \boldsymbol{H}^3=\left(\mathfrak{J}^C\right)_{\tau \gamma}$ .          | $g(M+i \boldsymbol{a})=\left(\begin{array}{cc} \operatorname{tr}(M) & -\boldsymbol{a} \\ -\boldsymbol{a}^{*} & M \end{array}\right)-\frac{1}{2} \operatorname{tr}(M) E, \quad M+i \boldsymbol{a} \in \mathfrak{J}_{\boldsymbol{H}} \oplus i \boldsymbol{H}^3=\left(\mathfrak{J}^C\right)_{\tau \gamma}$ .          |
| 0902.0431_E00618 | 103  | ■ 1.000 | $\backslash\left\langle\right\rangle g\left(X_{\{1\}}+i X_{\{2\}}\right), g\left(Y_{\{1\}}+i Y_{\{2\}}\right) \backslash\right\rangle=\left\langle\right\rangle X_{\{1\}}+i X_{\{2\}}, Y_{\{1\}}+i Y_{\{2\}} \backslash\right\rangle, \quad \backslash\quad X_{\{i\}}, Y_{\{i\}} \in\left(\mathfrak{J}^C\right)_{\tau \gamma}$ . | $\langle g\left(X_1+i X_2\right), g\left(Y_1+i Y_2\right)\rangle=\left\langle X_1+i X_2, Y_1+i Y_2\right\rangle, \quad X_i, Y_i \in\left(\mathfrak{J}^C\right)_{\tau \gamma}$ .                                                                                                                                    | $\langle g\left(X_1+i X_2\right), g\left(Y_1+i Y_2\right)\rangle=\left\langle X_1+i X_2, Y_1+i Y_2\right\rangle, \quad X_i, Y_i \in\left(\mathfrak{J}^C\right)_{\tau \gamma}$ .                                                                                                                                    |
| 0902.0431_E00619 | 103  | ■ 1.000 | $\varphi(A) X=g^{-1}\left(A(g X) A^{*}\right), \quad X \in \mathfrak{J}^C$ .                                                                                                                                                                                                                                                     | $\varphi(A) X=g^{-1}\left(A(g X) A^{*}\right), \quad X \in \mathfrak{J}^C$ .                                                                                                                                                                                                                                       | $\varphi(A) X=g^{-1}\left(A(g X) A^{*}\right), \quad X \in \mathfrak{J}^C$ .                                                                                                                                                                                                                                       |
| 0902.0431_E00622 | 103  | ■ 1.000 | $\tau \gamma \varphi(A) \tau \gamma X=\varphi(A) X, \quad X \in\left(\mathfrak{J}^C\right)_{\tau \gamma}$ ,                                                                                                                                                                                                                      | $\tau \gamma \varphi(A) \tau \gamma X=\varphi(A) X, \quad X \in\left(\mathfrak{J}^C\right)_{\tau \gamma}$ ,                                                                                                                                                                                                        | $\tau \gamma \varphi(A) \tau \gamma X=\varphi(A) X, \quad X \in\left(\mathfrak{J}^C\right)_{\tau \gamma}$ ,                                                                                                                                                                                                        |
| 0902.0431_E00623 | 103  | ■ 1.000 | $(g(\alpha E))^2=g(\alpha E)+\frac{3}{4} E$ .                                                                                                                                                                                                                                                                                    | $(g(\alpha E))^2=g(\alpha E)+\frac{3}{4} E$ .                                                                                                                                                                                                                                                                      | $(g(\alpha E))^2=g(\alpha E)+\frac{3}{4} E$ .                                                                                                                                                                                                                                                                      |
| 0902.0431_E00625 | 104  | ■ 1.000 | $P=\frac{1}{4}(2 g(\alpha E)+E),$                                                                                                                                                                                                                                                                                                | $P=\frac{1}{4}(2 g(\alpha E)+E),$                                                                                                                                                                                                                                                                                  | $P=\frac{1}{4}(2 g(\alpha E)+E),$                                                                                                                                                                                                                                                                                  |
| 0902.0431_E00626 | 104  | ■ 1.000 | $\begin{aligned} P^2 &=\frac{1}{16}\left(4(g(\alpha E))^2+4 g(\alpha E)+E\right)=\frac{1}{4}(2 g(\alpha E)+E)=P, \\ \operatorname{tr}(P) &=\frac{1}{4} \operatorname{tr}(2 g(\alpha E))+\frac{1}{4} \operatorname{tr}(E)=0+1=1. \end{aligned}$                                                                                   | $\begin{aligned} P^2 &=\frac{1}{16}\left(4(g(\alpha E))^2+4 g(\alpha E)+E\right)=\frac{1}{4}(2 g(\alpha E)+E)=P, \\ \operatorname{tr}(P) &=\frac{1}{4} \operatorname{tr}(2 g(\alpha E))+\frac{1}{4} \operatorname{tr}(E)=0+1=1. \end{aligned}$                                                                     | $\begin{aligned} P^2 &=\frac{1}{16}\left(4(g(\alpha E))^2+4 g(\alpha E)+E\right)=\frac{1}{4}(2 g(\alpha E)+E)=P, \\ \operatorname{tr}(P) &=\frac{1}{4} \operatorname{tr}(2 g(\alpha E))+\frac{1}{4} \operatorname{tr}(E)=0+1=1. \end{aligned}$                                                                     |
| 0902.0431_E00627 | 104  | ■ 1.000 | $P=A E_1 A^{*} .$                                                                                                                                                                                                                                                                                                                | $P=A E_1 A^{*} .$                                                                                                                                                                                                                                                                                                  | $P=A E_1 A^{*} .$                                                                                                                                                                                                                                                                                                  |
| 0902.0431_E00629 | 104  | ■ 1.000 | $\beta(M+\boldsymbol{a})=D M D^{*}+p \boldsymbol{a} D^{*}, \quad M+\boldsymbol{a} \in \mathfrak{J}_{\boldsymbol{H}} \oplus \boldsymbol{H}^3=\mathfrak{J} .$                                                                                                                                                                      | $\beta(M+\boldsymbol{a})=D M D^{*}+p \boldsymbol{a} D^{*}, \quad M+\boldsymbol{a} \in \mathfrak{J}_{\boldsymbol{H}} \oplus \boldsymbol{H}^3=\mathfrak{J} .$                                                                                                                                                        | $\beta(M+\boldsymbol{a})=D M D^{*}+p \boldsymbol{a} D^{*}, \quad M+\boldsymbol{a} \in \mathfrak{J}_{\boldsymbol{H}} \oplus \boldsymbol{H}^3=\mathfrak{J} .$                                                                                                                                                        |
| 0902.0431_E00630 | 104  | ■ 1.000 | $\beta=\varphi(B) .$                                                                                                                                                                                                                                                                                                             | $\beta=\varphi(B) .$                                                                                                                                                                                                                                                                                               | $\beta=\varphi(B) .$                                                                                                                                                                                                                                                                                               |
| 0902.0431_E00633 | 105  | ■ 1.000 | $\left(E_6\right)^w=\left\{\alpha \in E_6 \mid w \alpha=\alpha w\right\} .$                                                                                                                                                                                                                                                      | $\left(E_6\right)^w=\left\{\alpha \in E_6 \mid w \alpha=\alpha w\right\} .$                                                                                                                                                                                                                                        | $\left(E_6\right)^w=\left\{\alpha \in E_6 \mid w \alpha=\alpha w\right\} .$                                                                                                                                                                                                                                        |
| 0902.0431_E00634 | 105  | ■ 1.000 | $\mathfrak{J}(3, \boldsymbol{C})^C \oplus M(3, \boldsymbol{C})^C=\mathfrak{J}^C .$                                                                                                                                                                                                                                               | $\mathfrak{J}(3, \boldsymbol{C})^C \oplus M(3, \boldsymbol{C})^C=\mathfrak{J}^C .$                                                                                                                                                                                                                                 | $\mathfrak{J}(3, \boldsymbol{C})^C \oplus M(3, \boldsymbol{C})^C=\mathfrak{J}^C .$                                                                                                                                                                                                                                 |
| 0902.0431_E00636 | 105  | ■ 1.000 | $F_{4, \boldsymbol{C}}=\left\{\alpha \in E_{6, \boldsymbol{C}} \mid \alpha E=E\right\},$                                                                                                                                                                                                                                         | $F_{4, \boldsymbol{C}}=\left\{\alpha \in E_{6, \boldsymbol{C}} \mid \alpha E=E\right\},$                                                                                                                                                                                                                           | $F_{4, \boldsymbol{C}}=\left\{\alpha \in E_{6, \boldsymbol{C}} \mid \alpha E=E\right\},$                                                                                                                                                                                                                           |

| Identifier       | Page | Conf.   | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Rendered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Image                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|------------------|------|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0902.0431_E00639 | 105  | ■ 1.000 | $\mathbf{E} \mathbf{I} \mathbf{V}_{\{\boldsymbol{C}\}} = \left\{ X \in (\mathfrak{J}_C)^C \mid \det X = 1, \langle X, X \rangle = 3 \right\}$                                                                                                                                                                                                                                                                                                                                                                                                              | $EIV_C = \left\{ X \in (\mathfrak{J}_C)^C \mid \det X = 1, \langle X, X \rangle = 3 \right\}$                                                                                                                                                                                                                                                                                                                                                                                                                                                              | $EIV_C = \{ X \in (\mathfrak{J}_C)^C \mid \det X = 1, \langle X, X \rangle = 3 \}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 0902.0431_E00644 | 106  | ■ 1.000 | $\phi_{\boldsymbol{C}}(C, D)X = h(C, D)X + Xh(C, D)^*, \quad X \in (\mathfrak{J}_C)^C$                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | $\phi_{\boldsymbol{C}}(C, D)X = h(C, D)X + Xh(C, D)^*, \quad X \in (\mathfrak{J}_C)^C$                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | $\phi_{\boldsymbol{C}}(C, D)X = h(C, D)X + Xh(C, D)^*, \quad X \in (\mathfrak{J}_C)^C$                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 0902.0431_E00650 | 107  | ■ 1.000 | $h(A, B)Xh(A, B)^* = \bar{X}, \quad X \in (\mathfrak{J}_C)^C$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | $h(A, B)Xh(A, B)^* = \bar{X}, \quad X \in (\mathfrak{J}_C)^C$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | $h(A, B)Xh(A, B)^* = \bar{X}, \quad X \in (\mathfrak{J}_C)^C$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 0902.0431_E00653 | 108  | ■ 1.000 | $\begin{aligned} \tau \alpha \tau((X + M) \times (Y + N)) &= \alpha(X + M) \times \alpha(Y + N), \\ \langle \alpha(X + M), \alpha(Y + N) \rangle &= \langle X + M, Y + M \rangle. \end{aligned}$                                                                                                                                                                                                                                                                                                                                                           | $\tau \alpha \tau((X + M) \times (Y + N)) = \alpha(X + M) \times \alpha(Y + N),$ $\langle \alpha(X + M), \alpha(Y + N) \rangle = \langle X + M, Y + M \rangle.$                                                                                                                                                                                                                                                                                                                                                                                            | $\tau \alpha \tau((X + M) \times (Y + N)) = \alpha(X + M) \times \alpha(Y + N),$ $\langle \alpha(X + M), \alpha(Y + N) \rangle = \langle X + M, Y + M \rangle.$                                                                                                                                                                                                                                                                                                                                                                                            |
| 0902.0431_E00654 | 108  | ■ 1.000 | $\tau \alpha \tau(X \times Y) = \alpha X \times \alpha Y, \quad \langle \alpha X, \alpha Y \rangle = \langle X, Y \rangle$                                                                                                                                                                                                                                                                                                                                                                                                                                 | $\tau \alpha \tau(X \times Y) = \alpha X \times \alpha Y, \quad \langle \alpha X, \alpha Y \rangle = \langle X, Y \rangle$                                                                                                                                                                                                                                                                                                                                                                                                                                 | $\tau \alpha \tau(X \times Y) = \alpha X \times \alpha Y, \quad \langle \alpha X, \alpha Y \rangle = \langle X, Y \rangle$                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 0902.0431_E00657 | 108  | ■ 1.000 | $\alpha X = h(A, B)Xh(A, B)^* \quad \text{or} \quad \alpha X = h(A, B)\bar{X}h(A, B)^*, \quad X \in (\mathfrak{J}_C)^C$                                                                                                                                                                                                                                                                                                                                                                                                                                    | $\alpha X = h(A, B)Xh(A, B)^* \quad \text{or} \quad \alpha X = h(A, B)\bar{X}h(A, B)^*, \quad X \in (\mathfrak{J}_C)^C$                                                                                                                                                                                                                                                                                                                                                                                                                                    | $\alpha X = h(A, B)Xh(A, B)^* \quad \text{or} \quad \alpha X = h(A, B)\bar{X}h(A, B)^*, \quad X \in (\mathfrak{J}_C)^C$                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 0902.0431_E00659 | 109  | ■ 1.000 | $\alpha = \varphi(E, A, B)\beta = \varphi(E, A, B)\varphi(P, E, E) = \varphi(P, A, B).$                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | $\alpha = \varphi(E, A, B)\beta = \varphi(E, A, B)\varphi(P, E, E) = \varphi(P, A, B).$                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | $\alpha = \varphi(E, A, B)\beta = \varphi(E, A, B)\varphi(P, E, E) = \varphi(P, A, B).$                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 0902.0431_E00660 | 109  | ■ 1.000 | $\begin{aligned} &G_{01}, \quad G_{23}, \quad G_{45}, \quad G_{67}, \quad G_{26} + G_{37}, \quad -G_{27} + G_{36}, \\ &G_{24} + G_{35}, \quad -G_{25} + G_{34}, \quad G_{46} + G_{57}, \quad -G_{47} + G_{56}, \\ &\tilde{A}_1(1), \quad \tilde{A}_2(1), \quad \tilde{A}_3(1), \quad \tilde{A}_1(e_1), \quad \tilde{A}_2(e_1), \quad \tilde{A}_3(e_1), \\ &(E_1 - E_2)^\sim, \quad \tilde{F}_1(1), \quad \tilde{F}_2(1), \quad \tilde{F}_3(1), \\ &(E_2 - E_3)^\sim, \quad \tilde{F}_1(e_1), \quad \tilde{F}_2(e_1), \quad \tilde{F}_3(e_1) \end{aligned}$ | $\begin{aligned} &G_{01}, \quad G_{23}, \quad G_{45}, \quad G_{67}, \quad G_{26} + G_{37}, \quad -G_{27} + G_{36}, \\ &G_{24} + G_{35}, \quad -G_{25} + G_{34}, \quad G_{46} + G_{57}, \quad -G_{47} + G_{56}, \\ &\tilde{A}_1(1), \quad \tilde{A}_2(1), \quad \tilde{A}_3(1), \quad \tilde{A}_1(e_1), \quad \tilde{A}_2(e_1), \quad \tilde{A}_3(e_1), \\ &(E_1 - E_2)^\sim, \quad \tilde{F}_1(1), \quad \tilde{F}_2(1), \quad \tilde{F}_3(1), \\ &(E_2 - E_3)^\sim, \quad \tilde{F}_1(e_1), \quad \tilde{F}_2(e_1), \quad \tilde{F}_3(e_1) \end{aligned}$ | $\begin{aligned} &G_{01}, \quad G_{23}, \quad G_{45}, \quad G_{67}, \quad G_{26} + G_{37}, \quad -G_{27} + G_{36}, \\ &G_{24} + G_{35}, \quad -G_{25} + G_{34}, \quad G_{46} + G_{57}, \quad -G_{47} + G_{56}, \\ &\tilde{A}_1(1), \quad \tilde{A}_2(1), \quad \tilde{A}_3(1), \quad \tilde{A}_1(e_1), \quad \tilde{A}_2(e_1), \quad \tilde{A}_3(e_1), \\ &(E_1 - E_2)^\sim, \quad \tilde{F}_1(1), \quad \tilde{F}_2(1), \quad \tilde{F}_3(1), \\ &(E_2 - E_3)^\sim, \quad \tilde{F}_1(e_1), \quad \tilde{F}_2(e_1), \quad \tilde{F}_3(e_1) \end{aligned}$ |
| 0902.0431_E00661 | 109  | ■ 1.000 | $E_{\{6\}} \simeq E_{\{6\}} \times \mathbf{R}^{78}.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | $E_6^C \simeq E_6 \times \mathbf{R}^{78}.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | $E_6^C \simeq E_6 \times \mathbf{R}^{78}.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 0902.0431_E00662 | 109  | ■ 1.000 | $E_{\{6\}} \simeq \left( E_{\{6\}} \cap U(\mathfrak{J}^C) \right) \times \mathbf{R}^d = E_6 \times \mathbf{R}^d, \quad d = 78.$                                                                                                                                                                                                                                                                                                                                                                                                                            | $E_6^C \simeq (E_6^C \cap U(\mathfrak{J}^C)) \times \mathbf{R}^d = E_6 \times \mathbf{R}^d, \quad d = 78.$                                                                                                                                                                                                                                                                                                                                                                                                                                                 | $E_6^C \simeq (E_6^C \cap U(\mathfrak{J}^C)) \times \mathbf{R}^d = E_6 \times \mathbf{R}^d, \quad d = 78.$                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 0902.0431_E00663 | 110  | ■ 1.000 | $\langle X, Y \rangle_\gamma = \langle \gamma X, Y \rangle, \quad \langle X, Y \rangle_\sigma = \langle \sigma X, Y \rangle.$                                                                                                                                                                                                                                                                                                                                                                                                                              | $\langle X, Y \rangle_\gamma = \langle \gamma X, Y \rangle, \quad \langle X, Y \rangle_\sigma = \langle \sigma X, Y \rangle.$                                                                                                                                                                                                                                                                                                                                                                                                                              | $\langle X, Y \rangle_\gamma = \langle \gamma X, Y \rangle, \quad \langle X, Y \rangle_\sigma = \langle \sigma X, Y \rangle.$                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 0902.0431_E00667 | 110  | ■ 1.000 | $z(E_{6(6)}) = \{1\}, \quad z(E_{6(2)}) = \mathbf{Z}_3, \quad z(E_{6(-14)}) = \mathbf{Z}_3, \quad z(E_{6(-26)}) = \{1\}.$                                                                                                                                                                                                                                                                                                                                                                                                                                  | $z(E_{6(6)}) = \{1\}, \quad z(E_{6(2)}) = \mathbf{Z}_3, \quad z(E_{6(-14)}) = \mathbf{Z}_3, \quad z(E_{6(-26)}) = \{1\}.$                                                                                                                                                                                                                                                                                                                                                                                                                                  | $z(E_{6(6)}) = \{1\}, \quad z(E_{6(2)}) = \mathbf{Z}_3, \quad z(E_{6(-14)}) = \mathbf{Z}_3, \quad z(E_{6(-26)}) = \{1\}.$                                                                                                                                                                                                                                                                                                                                                                                                                                  |

| Identifier            | Page | Conf.  | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Rendered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Image                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-----------------------|------|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0902.0431_E00670      | 111  | ■1.000 | $\begin{aligned} & \Phi(\phi, A, B, \nu) \left( \begin{array}{l} X \\ Y \\ \xi \\ \eta \end{array} \right) = \left( \begin{array}{cccc} \phi - \frac{1}{3}\nu & 2B & 0 & A \\ 2A & -{}^t\phi + \frac{1}{3}\nu & B & 0 \\ 0 & A & \nu & 0 \\ B & 0 & 0 & -\nu \end{array} \right) \left( \begin{array}{l} X \\ Y \\ \xi \\ \eta \end{array} \right) \\ & = \left( \begin{array}{l} \phi X - \frac{1}{3}\nu X + 2B \times Y + \eta A \\ 2A \times X - {}^t\phi Y + \frac{1}{3}\nu Y + \xi B \\ (A, Y) + \nu \xi \\ (B, X) - \nu \eta \end{array} \right). \end{aligned}$ | $\Phi(\phi, A, B, \nu) \left( \begin{array}{l} X \\ Y \\ \xi \\ \eta \end{array} \right) = \left( \begin{array}{cccc} \phi - \frac{1}{3}\nu & 2B & 0 & A \\ 2A & -{}^t\phi + \frac{1}{3}\nu & B & 0 \\ 0 & A & \nu & 0 \\ B & 0 & 0 & -\nu \end{array} \right) \left( \begin{array}{l} X \\ Y \\ \xi \\ \eta \end{array} \right) \\ = \left( \begin{array}{l} \phi X - \frac{1}{3}\nu X + 2B \times Y + \eta A \\ 2A \times X - {}^t\phi Y + \frac{1}{3}\nu Y + \xi B \\ (A, Y) + \nu \xi \\ (B, X) - \nu \eta \end{array} \right).$ | $\Phi(\phi, A, B, \nu) \left( \begin{array}{l} X \\ Y \\ \xi \\ \eta \end{array} \right) = \left( \begin{array}{cccc} \phi - \frac{1}{3}\nu & 2B & 0 & A \\ 2A & -{}^t\phi + \frac{1}{3}\nu & B & 0 \\ 0 & A & \nu & 0 \\ B & 0 & 0 & -\nu \end{array} \right) \left( \begin{array}{l} X \\ Y \\ \xi \\ \eta \end{array} \right) \\ = \left( \begin{array}{l} \phi X - \frac{1}{3}\nu X + 2B \times Y + \eta A \\ 2A \times X - {}^t\phi Y + \frac{1}{3}\nu Y + \xi B \\ (A, Y) + \nu \xi \\ (B, X) - \nu \eta \end{array} \right).$ |
| 0902.0431_E00676      | 112  | ■1.000 | $(P \times R)Q - (Q \times R)P + \frac{1}{8}\{Q, R\}P - \frac{1}{8}\{P, R\}Q - \frac{1}{4}\{P, Q\}R = 0.$                                                                                                                                                                                                                                                                                                                                                                                                                                                              | $(P \times R)Q - (Q \times R)P + \frac{1}{8}\{Q, R\}P - \frac{1}{8}\{P, R\}Q - \frac{1}{4}\{P, Q\}R = 0.$                                                                                                                                                                                                                                                                                                                                                                                                                            | $(P \times R)Q - (Q \times R)P + \frac{1}{8}\{Q, R\}P - \frac{1}{8}\{P, R\}Q - \frac{1}{4}\{P, Q\}R = 0.$                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 0902.0431_E00680      | 113  | ■1.000 | $U(56) = U(\mathfrak{P}^C) = \{\alpha \in \text{Iso}_C(\mathfrak{P}^C) \mid \langle \alpha P, \alpha Q \rangle = \langle P, Q \rangle\}.$                                                                                                                                                                                                                                                                                                                                                                                                                              | $U(56) = U(\mathfrak{P}^C) = \{\alpha \in \text{Iso}_C(\mathfrak{P}^C) \mid \langle \alpha P, \alpha Q \rangle = \langle P, Q \rangle\}.$                                                                                                                                                                                                                                                                                                                                                                                            | $U(56) = U(\mathfrak{P}^C) = \{\alpha \in \text{Iso}_C(\mathfrak{P}^C) \mid \langle \alpha P, \alpha Q \rangle = \langle P, Q \rangle\}.$                                                                                                                                                                                                                                                                                                                                                                                            |
| 0902.0431_E00681 (1)  | 113  | ■1.000 | $\alpha \mathfrak{M}^C = \mathfrak{M}^C.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | $\alpha \mathfrak{M}^C = \mathfrak{M}^C.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | $\alpha \mathfrak{M}^C = \mathfrak{M}^C.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 0902.0431_E00697      | 115  | ■1.000 | $l(Y) = 2B \times Y, \quad Y \in \mathfrak{J}^C.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | $l(Y) = 2B \times Y, \quad Y \in \mathfrak{J}^C.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | $l(Y) = 2B \times Y, \quad Y \in \mathfrak{J}^C.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 0902.0431_E00698      | 115  | ■1.000 | $2(B \times (X \times X), X \times X) + (\det X)(X, B) = 3(\det X)b(X),$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | $2(B \times (X \times X), X \times X) + (\det X)(X, B) = 3(\det X)b(X),$                                                                                                                                                                                                                                                                                                                                                                                                                                                             | $2(B \times (X \times X), X \times X) + (\det X)(X, B) = 3(\det X)b(X),$                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 0902.0431_E00699      | 115  | ■1.000 | $b(X) = (B, X), \quad X \in \mathfrak{J}^C.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | $b(X) = (B, X), \quad X \in \mathfrak{J}^C.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | $b(X) = (B, X), \quad X \in \mathfrak{J}^C.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 0902.0431_E00700 (i)  | 116  | ■1.000 | $k(X) = 2A \times X, \quad a(Y) = (A, Y), \quad X, Y \in \mathfrak{J}^C.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | $k(X) = 2A \times X, \quad a(Y) = (A, Y), \quad X, Y \in \mathfrak{J}^C.$                                                                                                                                                                                                                                                                                                                                                                                                                                                            | $k(X) = 2A \times X, \quad a(Y) = (A, Y), \quad X, Y \in \mathfrak{J}^C.$                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 0902.0431_E00701 (ii) | 116  | ■1.000 | $\begin{aligned} & 2g(X) \times X = \mu X \times X + h(X \times X), \\ & 2h(X \times X) \times (X \times X) = (\det X)(\nu X + g(X)), \\ & (g(X), X \times X) + (h(X \times X), X) = 3(\nu + \mu) \det X. \end{aligned}$                                                                                                                                                                                                                                                                                                                                               | $\begin{aligned} & 2g(X) \times X = \mu X \times X + h(X \times X), \\ & 2h(X \times X) \times (X \times X) = (\det X)(\nu X + g(X)), \\ & (g(X), X \times X) + (h(X \times X), X) = 3(\nu + \mu) \det X. \end{aligned}$                                                                                                                                                                                                                                                                                                             | $\begin{aligned} & 2g(X) \times X = \mu X \times X + h(X \times X), \\ & 2h(X \times X) \times (X \times X) = (\det X)(\nu X + g(X)), \\ & (g(X), X \times X) + (h(X \times X), X) = 3(\nu + \mu) \det X. \end{aligned}$                                                                                                                                                                                                                                                                                                             |
| 0902.0431_E00703      | 116  | ■1.000 | $\phi \in \mathfrak{e}_6^C.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | $\phi \in \mathfrak{e}_6^C.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | $\phi \in \mathfrak{e}_6^C.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 0902.0431_E00705      | 116  | ■1.000 | $\psi'^{\prime} = \phi.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | $\psi' = \phi.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | $\psi' = \phi.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 0902.0431_E00707      | 117  | ■1.000 | $[\Phi, P \times Q] = \Phi P \times Q + P \times \Phi Q.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | $[\Phi, P \times Q] = \Phi P \times Q + P \times \Phi Q.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | $[\Phi, P \times Q] = \Phi P \times Q + P \times \Phi Q.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 0902.0431_E00710      | 117  | ■1.000 | $\lambda(X, Y, \xi, \eta) = (Y, -X, \eta, -\xi).$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | $\lambda(X, Y, \xi, \eta) = (Y, -X, \eta, -\xi).$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | $\lambda(X, Y, \xi, \eta) = (Y, -X, \eta, -\xi).$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 0902.0431_E00711      | 117  | ■1.000 | $(P, Q) = \{\lambda P, Q\} = -\{P, \lambda Q\}, \quad \langle P, Q \rangle = \{\tau \lambda P, Q\}.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | $(P, Q) = \{\lambda P, Q\} = -\{P, \lambda Q\}, \quad \langle P, Q \rangle = \{\tau \lambda P, Q\}.$                                                                                                                                                                                                                                                                                                                                                                                                                                 | $(P, Q) = \{\lambda P, Q\} = -\{P, \lambda Q\}, \quad \langle P, Q \rangle = \{\tau \lambda P, Q\}.$                                                                                                                                                                                                                                                                                                                                                                                                                                 |

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|----------------------|------|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0902.0431_E00716     | 118  | ■ 1.000 | $\Phi(\phi, A, -\tau A, \nu) = \left( \begin{array}{cccc} \phi - \frac{1}{3}\nu & -2\tau A & 0 & A \\ 2A & \tau\phi\tau + \frac{1}{3}\nu & -\tau A & 0 \\ 0 & A & \nu & 0 \\ -\tau A & 0 & 0 & -\nu \end{array} \right).$                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                            |
| 0902.0431_E00719     | 118  | ■ 1.000 | $\operatorname{dim}(\left(\frac{e}{7}\right))=78+54+1=133.$                                                                                                                                                                                                                                                                                                | $\dim(\mathfrak{e}_7) = 78 + 54 + 1 = 133.$                                                                                                                                                                                                                                                                                                                | $\dim(\mathfrak{e}_7) = 78 + 54 + 1 = 133.$                                                                                                                                                                                                                                                                                                                |
| 0902.0431_E00721     | 119  | ■ 1.000 | $\Phi = \frac{\Phi - \Phi^*}{2} + i \frac{\Phi + \Phi^*}{2i}, \quad \frac{\Phi - \Phi^*}{2}, \frac{\Phi + \Phi^*}{2i} \in \mathfrak{e}_7.$                                                                                                                                                                                                                 | $\Phi = \frac{\Phi - \Phi^*}{2} + i \frac{\Phi + \Phi^*}{2i}, \quad \frac{\Phi - \Phi^*}{2}, \frac{\Phi + \Phi^*}{2i} \in \mathfrak{e}_7.$                                                                                                                                                                                                                 | $\Phi = \frac{\Phi - \Phi^*}{2} + i \frac{\Phi + \Phi^*}{2i}, \quad \frac{\Phi - \Phi^*}{2}, \frac{\Phi + \Phi^*}{2i} \in \mathfrak{e}_7.$                                                                                                                                                                                                                 |
| 0902.0431_E00723     | 119  | ■ 1.000 | $\mathfrak{a} \ni \left( \Phi(\phi_1, 0, 0, 0), \Phi(\phi, A, B, \nu) \right) = \Phi([\phi_1, \phi], \phi_1 A, \phi_1' B, 0)$                                                                                                                                                                                                                              | $\mathfrak{a} \ni [\Phi(\phi_1, 0, 0, 0), \Phi(\phi, A, B, \nu)] = \Phi([\phi_1, \phi], \phi_1 A, \phi_1' B, 0)$                                                                                                                                                                                                                                           | $\mathfrak{a} \ni [\Phi(\phi_1, 0, 0, 0), \Phi(\phi, A, B, \nu)] = \Phi([\phi_1, \phi], \phi_1 A, \phi_1' B, 0)$                                                                                                                                                                                                                                           |
| 0902.0431_E00725     | 119  | ■ 1.000 | $\mathfrak{a} \ni \left( \Phi(0, E_1, 0, 0), \Phi(0, 0, E_1, 0) \right) = \Phi(2E_1 \vee E_1, 0, 0, 1),$                                                                                                                                                                                                                                                   | $\mathfrak{a} \ni [\Phi(0, E_1, 0, 0), \Phi(0, 0, E_1, 0)] = \Phi(2E_1 \vee E_1, 0, 0, 1),$                                                                                                                                                                                                                                                                | $\mathfrak{a} \ni [\Phi(0, E_1, 0, 0), \Phi(0, 0, E_1, 0)] = \Phi(2E_1 \vee E_1, 0, 0, 1),$                                                                                                                                                                                                                                                                |
| 0902.0431_E00727     | 120  | ■ 1.000 | $\mathfrak{a} \ni [\Phi(0, 0, 0, \nu), \Phi(0, A, 0, 0)] = \Phi\left(0, \frac{2}{3}\nu A, 0, 0\right).$                                                                                                                                                                                                                                                    | $\mathfrak{a} \ni [\Phi(0, 0, 0, \nu), \Phi(0, A, 0, 0)] = \Phi\left(0, \frac{2}{3}\nu A, 0, 0\right).$                                                                                                                                                                                                                                                    | $\mathfrak{a} \ni [\Phi(0, 0, 0, \nu), \Phi(0, A, 0, 0)] = \Phi\left(0, \frac{2}{3}\nu A, 0, 0\right).$                                                                                                                                                                                                                                                    |
| 0902.0431_E00729 (c) | 120  | ■ 1.000 | $W \ni \Phi(0, 0, 0, 3)(X, Y, \xi, \eta) = (-X, Y, 3\xi, -3\eta),$<br>$W \ni \Phi(0, 0, 0, 3)(-X, Y, 3\xi, -3\eta) = (X, Y, 9\xi, 9\eta).$                                                                                                                                                                                                                 | $W \ni \Phi(0, 0, 0, 3)(X, Y, \xi, \eta) = (-X, Y, 3\xi, -3\eta),$<br>$W \ni \Phi(0, 0, 0, 3)(-X, Y, 3\xi, -3\eta) = (X, Y, 9\xi, 9\eta).$                                                                                                                                                                                                                 | $W \ni \Phi(0, 0, 0, 3)(X, Y, \xi, \eta) = (-X, Y, 3\xi, -3\eta),$<br>$W \ni \Phi(0, 0, 0, 3)(-X, Y, 3\xi, -3\eta) = (X, Y, 9\xi, 9\eta).$                                                                                                                                                                                                                 |
| 0902.0431_E00731     | 121  | ■ 1.000 | $W \ni \Phi(0, 0, B, 0)(0, Y, \xi, \eta) = (2B \times Y, \xi B, 0, 0).$                                                                                                                                                                                                                                                                                    | $W \ni \Phi(0, 0, B, 0)(0, Y, \xi, \eta) = (2B \times Y, \xi B, 0, 0).$                                                                                                                                                                                                                                                                                    | $W \ni \Phi(0, 0, B, 0)(0, Y, \xi, \eta) = (2B \times Y, \xi B, 0, 0).$                                                                                                                                                                                                                                                                                    |
| 0902.0431_E00732     | 121  | ■ 1.000 | $W \ni \Phi(0, 0, B, 0)(0, 0, \xi, \eta) = (0, \xi B, 0, 0).$                                                                                                                                                                                                                                                                                              | $W \ni \Phi(0, 0, B, 0)(0, 0, \xi, \eta) = (0, \xi B, 0, 0).$                                                                                                                                                                                                                                                                                              | $W \ni \Phi(0, 0, B, 0)(0, 0, \xi, \eta) = (0, \xi B, 0, 0).$                                                                                                                                                                                                                                                                                              |
| 0902.0431_E00733     | 121  | ■ 1.000 | $(\Phi_1, \Phi_2)_7 = -2(\phi_1, \phi_2)_6 - 4(A_1, B_2) - 4(A_2, B_1) - \frac{8}{3}\nu_1\nu_2,$                                                                                                                                                                                                                                                           | $(\Phi_1, \Phi_2)_7 = -2(\phi_1, \phi_2)_6 - 4(A_1, B_2) - 4(A_2, B_1) - \frac{8}{3}\nu_1\nu_2$                                                                                                                                                                                                                                                            | $(\Phi_1, \Phi_2)_7 = -2(\phi_1, \phi_2)_6 - 4(A_1, B_2) - 4(A_2, B_1) - \frac{8}{3}\nu_1\nu_2,$                                                                                                                                                                                                                                                           |
| 0902.0431_E00735     | 121  | ■ 1.000 | $(\Phi, P \times Q)_7 = \{\Phi P, Q\}.$                                                                                                                                                                                                                                                                                                                    | $(\Phi, P \times Q)_7 = \{\Phi P, Q\}.$                                                                                                                                                                                                                                                                                                                    | $(\Phi, P \times Q)_7 = \{\Phi P, Q\}.$                                                                                                                                                                                                                                                                                                                    |
| 0902.0431_E00736     | 121  | ■ 1.000 | $= \left( \Phi \left( \begin{array}{c} [\phi, \phi_1] + 2A \vee B_1 - 2A_1 \vee B \\ (\phi + \frac{2}{3}\nu)A_1 - (\phi_1 + \frac{2}{3}\nu_1)A \\ -({}^t\phi + \frac{2}{3}\nu)B_1 + ({}^t\phi_1 + \frac{2}{3}\nu_1)B \\ (A, B_1) - (B, A_1) \end{array} \right), \Phi \left( \begin{array}{c} \phi_2 \\ A_2 \\ B_2 \\ \nu_2 \end{array} \right) \right)_7$ | $= \left( \Phi \left( \begin{array}{c} [\phi, \phi_1] + 2A \vee B_1 - 2A_1 \vee B \\ (\phi + \frac{2}{3}\nu)A_1 - (\phi_1 + \frac{2}{3}\nu_1)A \\ -({}^t\phi + \frac{2}{3}\nu)B_1 + ({}^t\phi_1 + \frac{2}{3}\nu_1)B \\ (A, B_1) - (B, A_1) \end{array} \right), \Phi \left( \begin{array}{c} \phi_2 \\ A_2 \\ B_2 \\ \nu_2 \end{array} \right) \right)_7$ | $= \left( \Phi \left( \begin{array}{c} [\phi, \phi_1] + 2A \vee B_1 - 2A_1 \vee B \\ (\phi + \frac{2}{3}\nu)A_1 - (\phi_1 + \frac{2}{3}\nu_1)A \\ -({}^t\phi + \frac{2}{3}\nu)B_1 + ({}^t\phi_1 + \frac{2}{3}\nu_1)B \\ (A, B_1) - (B, A_1) \end{array} \right), \Phi \left( \begin{array}{c} \phi_2 \\ A_2 \\ B_2 \\ \nu_2 \end{array} \right) \right)_7$ |

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|------------------|------|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0902.0431_E00739 | 122  | ■ 1.000 | $\begin{aligned} & \backslash begin{aligned} B_{\{7\}} \backslash left(\backslash Phi_{\{1\}}, \backslash Phi_{\{2\}} \backslash right) \& \\ & = -9 \backslash left(\backslash Phi_{\{1\}}, \backslash Phi_{\{2\}} \backslash right)_{\{7\}} \backslash \& = 18 \backslash left( \\ & \backslash phi_{\{1\}}, \backslash phi_{\{2\}} \backslash right)_{\{6\}} + 36 \backslash left(A_{\{1\}}, B_{\{2\}} \\ & \backslash right) + 36 \backslash left(A_{\{2\}}, B_{\{1\}} \backslash right) + 24 \backslash nu_{\{1\}} \backslash nu_{\{2\}} \backslash \\ & \& = \frac{3}{2} B_6(\phi_1, \phi_2) + 36(A_1, B_2) + 36(A_2, B_1) + 24 \nu_1 \nu_2 \\ & \backslash left(A_{\{1\}}, B_{\{2\}} \backslash right) + 36 \backslash left(A_{\{2\}}, B_{\{1\}} \backslash right) + 24 \\ & \backslash nu_{\{1\}} \backslash nu_{\{2\}} \backslash \& = 3 \operatorname{tr}(\Phi_1 \Phi_2) \\ & \backslash Phi_{\{2\}} \backslash right) \end{aligned}$                                                                                                                                                                                                                                                                                                                              | $\begin{aligned} B_7(\Phi_1, \Phi_2) &= -9(\Phi_1, \Phi_2)_7 \\ &= 18(\phi_1, \phi_2)_6 + 36(A_1, B_2) + 36(A_2, B_1) + 24\nu_1\nu_2 \\ &= \frac{3}{2}B_6(\phi_1, \phi_2) + 36(A_1, B_2) + 36(A_2, B_1) + 24\nu_1\nu_2 \\ &= 3\operatorname{tr}(\Phi_1\Phi_2) \end{aligned}$                                                                                                                                                                            | $\begin{aligned} B_7(\Phi_1, \Phi_2) &= -9(\Phi_1, \Phi_2)_7 \\ &= 18(\phi_1, \phi_2)_6 + 36(A_1, B_2) + 36(A_2, B_1) + 24\nu_1\nu_2 \\ &= \frac{3}{2}B_6(\phi_1, \phi_2) + 36(A_1, B_2) + 36(A_2, B_1) + 24\nu_1\nu_2 \\ &= 3\operatorname{tr}(\Phi_1\Phi_2), \end{aligned}$                                                                                                                                                                           |
| 0902.0431_E00742 | 122  | ■ 1.000 | $\begin{aligned} & \backslash left[\backslash Phi_{\{0\}}, \backslash left[\backslash Phi_{\{0\}}, \backslash Phi(\backslash phi, A, B, \backslash nu) \\ & \backslash right] \backslash right] = \backslash left[\backslash Phi_{\{0\}}, \backslash Phi \backslash left(0, \frac{2}{3} \{3\} \\ & A, -\frac{2}{3} \{3\} B, 0 \backslash right) \backslash right] = \backslash Phi \backslash left(0, \\ & \frac{4}{9} A, \frac{4}{9} B, 0 \backslash right) \end{aligned}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | $[\Phi_0, [\Phi_0, \Phi(\phi, A, B, \nu)]] = \left[ \Phi_0, \Phi\left(0, \frac{2}{3}A, -\frac{2}{3}B, 0\right) \right] = \Phi\left(0, \frac{4}{9}A, \frac{4}{9}B, 0\right).$                                                                                                                                                                                                                                                                            | $[\Phi_0, [\Phi_0, \Phi(\phi, A, B, \nu)]] = \left[ \Phi_0, \Phi\left(0, \frac{2}{3}A, -\frac{2}{3}B, 0\right) \right] = \Phi\left(0, \frac{4}{9}A, \frac{4}{9}B, 0\right).$                                                                                                                                                                                                                                                                            |
| 0902.0431_E00752 | 124  | ■ 1.000 | $\begin{aligned} & \backslash begin{aligned} \& \{ \backslash left[\backslash Phi \backslash left(h_{\{ \delta \}} + \\ & \backslash widetilde{H} \}, 0, 0, \backslash nu \backslash right), \backslash Phi \backslash left(0, F_{\{1\}}(a), \\ & 0, 0 \backslash right) \backslash right] \quad \backslash quad a = e_k \pm i e_{4+k} \} \backslash \& \\ & \quad \backslash quad = \backslash Phi \backslash left(0, \backslash left(h_{\{ \delta \}} + \backslash widetilde{H} \} + \\ & \frac{2}{3} \{3\} \backslash nu \backslash right) F_{\{1\}}(a), 0, 0 \backslash right) \backslash \& \quad \backslash quad = \\ & \backslash Phi \backslash left(0, F_{\{1\}} \backslash left(h_{\{ \delta \}} a \backslash right) + \frac{1}{2} \{2\} \\ & \backslash left(\backslash mu_{\{2\}} + \backslash mu_{\{3\}} \backslash right) F_{\{1\}}(a) + \frac{2}{3} \backslash nu \\ & F_{\{1\}}(a), 0, 0 \backslash right) \end{aligned}$                                                                                                                                                                                                                                                                                                                                    | $\begin{aligned} & \left[ \Phi\left(h_\delta + \tilde{H}, 0, 0, \nu\right), \Phi\left(0, F_1(a), 0, 0\right) \right] \quad a = e_k \pm ie_{4+k} \\ &= \Phi\left(0, \left(h_\delta + \tilde{H} + \frac{2}{3}\nu\right)F_1(a), 0, 0\right) \\ &= \Phi\left(0, F_1(h_\delta a) + \frac{1}{2}(\mu_2 + \mu_3)F_1(a) + \frac{2}{3}\nu F_1(a), 0, 0\right) \end{aligned}$                                                                                      | $\begin{aligned} & [\Phi(h_\delta + \tilde{H}, 0, 0, \nu), \Phi(0, F_1(a), 0, 0)] \quad a = e_k \pm ie_{4+k} \\ &= \Phi\left(0, \left(h_\delta + \tilde{H} + \frac{2}{3}\nu\right)F_1(a), 0, 0\right) \\ &= \Phi\left(0, F_1(h_\delta a) + \frac{1}{2}(\mu_2 + \mu_3)F_1(a) + \frac{2}{3}\nu F_1(a), 0, 0\right) \end{aligned}$                                                                                                                         |
| 0902.0431_E00753 | 125  | ■ 1.000 | $\begin{aligned} & = \backslash left(\backslash pm \backslash lambda_{\{k\}} - \frac{1}{2} \{2\} \backslash mu_{\{1\}} + \frac{2}{3} \{3\} \\ & \backslash nu \backslash right) \backslash Phi \backslash left(0, F_{\{1\}}(a), 0, 0 \backslash right) \end{aligned}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | $= \left(\pm \lambda_k - \frac{1}{2}\mu_1 + \frac{2}{3}\nu\right)\Phi(0, F_1(a), 0, 0).$                                                                                                                                                                                                                                                                                                                                                                | $= \left(\pm \lambda_k - \frac{1}{2}\mu_1 + \frac{2}{3}\nu\right)\Phi(0, F_1(a), 0, 0).$                                                                                                                                                                                                                                                                                                                                                                |
| 0902.0431_E00754 | 125  | ■ 1.000 | $\begin{aligned} & \backslash begin{aligned} \& \backslash alpha_{\{1\}} = \backslash lambda_{\{0\}} - \backslash lambda_{\{1\}}, \\ & \quad \backslash quad \backslash alpha_{\{2\}} = \backslash lambda_{\{1\}} - \backslash lambda_{\{2\}}, \quad \backslash quad \\ & \backslash alpha_{\{3\}} = \backslash lambda_{\{2\}} - \backslash lambda_{\{3\}}, \quad \backslash \& \backslash alpha_{\{4\}} = \\ & \frac{1}{2} \{2\} \backslash left(-\backslash lambda_{\{0\}} - \backslash lambda_{\{1\}} - \backslash lambda_{\{2\}} + \\ & \backslash lambda_{\{3\}} \backslash right) + \frac{1}{2} \{2\} \backslash left(\backslash mu_{\{3\}} - \backslash mu_{\{1\}} \\ & \backslash right), \quad \backslash \& \backslash alpha_{\{5\}} = \frac{1}{2} \{2\} \backslash left(\backslash lambda_{\{0\}} + \\ & \backslash lambda_{\{1\}} + \backslash lambda_{\{2\}} + \backslash lambda_{\{3\}} \backslash right) + \frac{1}{2} \{2\} \\ & \backslash left(\backslash mu_{\{1\}} - \backslash mu_{\{2\}} \backslash right), \quad \backslash \& \backslash alpha_{\{6\}} = \backslash mu_{\{2\}} + \\ & \frac{3}{2} \{2\} \backslash nu, \quad \backslash quad \backslash alpha_{\{7\}} = -\backslash mu_{\{3\}} - \frac{3}{2} \{2\} \\ & \backslash nu \end{aligned}$ | $\begin{aligned} & \alpha_1 = \lambda_0 - \lambda_1, \quad \alpha_2 = \lambda_1 - \lambda_2, \quad \alpha_3 = \lambda_2 - \lambda_3, \\ & \alpha_4 = \frac{1}{2}(-\lambda_0 - \lambda_1 - \lambda_2 + \lambda_3) + \frac{1}{2}(\mu_3 - \mu_1), \\ & \alpha_5 = \frac{1}{2}(\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3) + \frac{1}{2}(\mu_1 - \mu_2), \\ & \alpha_6 = \mu_2 + \frac{3}{2}\nu, \quad \alpha_7 = -\mu_3 - \frac{3}{2}\nu \end{aligned}$ | $\begin{aligned} & \alpha_1 = \lambda_0 - \lambda_1, \quad \alpha_2 = \lambda_1 - \lambda_2, \quad \alpha_3 = \lambda_2 - \lambda_3, \\ & \alpha_4 = \frac{1}{2}(-\lambda_0 - \lambda_1 - \lambda_2 + \lambda_3) + \frac{1}{2}(\mu_3 - \mu_1), \\ & \alpha_5 = \frac{1}{2}(\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3) + \frac{1}{2}(\mu_1 - \mu_2), \\ & \alpha_6 = \mu_2 + \frac{3}{2}\nu, \quad \alpha_7 = -\mu_3 - \frac{3}{2}\nu \end{aligned}$ |
| 0902.0431_E00755 | 125  | ■ 1.000 | $\begin{aligned} & \backslash mu = \backslash alpha_{\{1\}} + 2 \backslash alpha_{\{2\}} + 3 \backslash alpha_{\{3\}} + 4 \backslash alpha_{\{4\}} + 3 \\ & \backslash alpha_{\{5\}} + 2 \backslash alpha_{\{6\}} + 2 \backslash alpha_{\{7\}} \end{aligned}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | $\mu = \alpha_1 + 2\alpha_2 + 3\alpha_3 + 4\alpha_4 + 3\alpha_5 + 2\alpha_6 + 2\alpha_7$                                                                                                                                                                                                                                                                                                                                                                | $\mu = \alpha_1 + 2\alpha_2 + 3\alpha_3 + 4\alpha_4 + 3\alpha_5 + 2\alpha_6 + 2\alpha_7$                                                                                                                                                                                                                                                                                                                                                                |
| 0902.0431_E00763 | 128  | ■ 1.000 | $\begin{aligned} & B_{\{7\}} \backslash left(h, h^{\backslash prime} \backslash right) = 6 \backslash left(6 \sum_{k=0}^3 \backslash lambda_{\{k\}} \backslash lambda_{\{k\}}^{\backslash prime} + 3 \sum_{j=1}^3 \backslash mu_{\{j\}} \\ & \backslash mu_{\{j\}}^{\backslash prime} + 4 \backslash nu \backslash nu^{\backslash prime} \backslash right) \end{aligned}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | $B_7(h, h') = 6\left(6\sum_{k=0}^3\lambda_k\lambda'_k + 3\sum_{j=1}^3\mu_j\mu'_j + 4\nu\nu'\right)$                                                                                                                                                                                                                                                                                                                                                     | $B_7(h, h') = 6\left(6\sum_{k=0}^3\lambda_k\lambda'_k + 3\sum_{j=1}^3\mu_j\mu'_j + 4\nu\nu'\right)$                                                                                                                                                                                                                                                                                                                                                     |

| Identifier       | Page | Conf.   | LaTeX source                                                                                                                                                                                                                                                                                                                       | Rendered                                                                                                                                                                                                                                                                                                               | Image                                                                                                                                                                                                                                                                                                                  |
|------------------|------|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0902.0431_E00766 | 128  | ■ 1.000 | $\left(\alpha_{\{1\}}, \alpha_{\{1\}}\right)=B_{\{7\}}\left(H_{\{\alpha_{\{1\}}\}}, H_{\{\alpha_{\{1\}}\}}\right)=36 \frac{1}{36} \frac{1}{36} 2=\frac{1}{18}$                                                                                                                                                                     | $(\alpha_1, \alpha_1) = B_7(H_{\alpha_1}, H_{\alpha_1}) = 36 \frac{1}{36} \frac{1}{36} 2 = \frac{1}{18}$                                                                                                                                                                                                               | $(\alpha_1, \alpha_1) = B_7(H_{\alpha_1}, H_{\alpha_1}) = 36 \frac{1}{36} \frac{1}{36} 2 = \frac{1}{18}$                                                                                                                                                                                                               |
| 0902.0431_E00768 | 129  | ■ 1.000 | $\left(E_{\{7\}}\right)_{\{(0,0,1,0)\}}=\left\{\alpha \in E_{\{7\}} \mid \alpha(0,0,1,0)=(0,0,1,0)\right\} .$                                                                                                                                                                                                                      | $(E_7)_{(0,0,1,0)} = \{\alpha \in E_7 \mid \alpha(0,0,1,0) = (0,0,1,0)\} .$                                                                                                                                                                                                                                            | $(E_7)_{(0,0,1,0)} = \{\alpha \in E_7 \mid \alpha(0,0,1,0) = (0,0,1,0)\} .$                                                                                                                                                                                                                                            |
| 0902.0431_E00769 | 129  | ■ 1.000 | $\alpha(0,0,0,1)=\alpha(-\tau \lambda(0,0,1,0))=-\tau \lambda \alpha(0,0,1,0)=-\tau \lambda(0,0,1,0)=(0,0,0,1) .$                                                                                                                                                                                                                  | $\alpha(0,0,0,1) = \alpha(-\tau \lambda(0,0,1,0)) = -\tau \lambda \alpha(0,0,1,0) = -\tau \lambda(0,0,1,0) = (0,0,0,1).$                                                                                                                                                                                               | $\alpha(0,0,0,1) = \alpha(-\tau \lambda(0,0,1,0)) = -\tau \lambda \alpha(0,0,1,0) = -\tau \lambda(0,0,1,0) = (0,0,0,1).$                                                                                                                                                                                               |
| 0902.0431_E00770 | 129  | ■ 1.000 | $\left(E_{\{7\}}\right)_{\{(0,0,1,0)\}} \cong E_{\{6\}} .$                                                                                                                                                                                                                                                                         | $(E_7)_{(0,0,1,0)} \cong E_6 .$                                                                                                                                                                                                                                                                                        | $(E_7)_{(0,0,1,0)} \cong E_6 .$                                                                                                                                                                                                                                                                                        |
| 0902.0431_E00771 | 130  | ■ 1.000 | $\widetilde{\alpha}=\left(\begin{array}{cccc} \alpha & 0 & 0 & 0 \\ 0 & \tau \alpha \tau & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{array}\right) \in\left(E_7\right)_{(0,0,1,0)} \subset E_7 .$                                                                                                                               | $\tilde{\alpha} = \begin{pmatrix} \alpha & 0 & 0 & 0 \\ 0 & \tau \alpha \tau & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \in (E_7)_{(0,0,1,0)} \subset E_7 .$                                                                                                                                              | $\tilde{\alpha} = \begin{pmatrix} \alpha & 0 & 0 & 0 \\ 0 & \tau \alpha \tau & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \in (E_7)_{(0,0,1,0)} \subset E_7 .$                                                                                                                                              |
| 0902.0431_E00775 | 130  | ■ 1.000 | $\mathfrak{M}^{\wedge\{C\}} \ni \alpha \left(\begin{array}{c} X \\ \frac{1}{\eta} X \times X \\ \frac{1}{\eta^2} \det X \\ \eta \end{array}\right) = \left(\begin{array}{c} \beta X + \frac{1}{\eta} \epsilon(X \times X) \\ \delta X + \frac{1}{\eta} \beta_1(X \times X) \\ \frac{1}{\eta^2} \det X \\ \eta \end{array}\right),$ | $\mathfrak{M}^C \ni \alpha \left(\begin{array}{c} X \\ \frac{1}{\eta} X \times X \\ \frac{1}{\eta^2} \det X \\ \eta \end{array}\right) = \left(\begin{array}{c} \beta X + \frac{1}{\eta} \epsilon(X \times X) \\ \delta X + \frac{1}{\eta} \beta_1(X \times X) \\ \frac{1}{\eta^2} \det X \\ \eta \end{array}\right),$ | $\mathfrak{M}^C \ni \alpha \left(\begin{array}{c} X \\ \frac{1}{\eta} X \times X \\ \frac{1}{\eta^2} \det X \\ \eta \end{array}\right) = \left(\begin{array}{c} \beta X + \frac{1}{\eta} \epsilon(X \times X) \\ \delta X + \frac{1}{\eta} \beta_1(X \times X) \\ \frac{1}{\eta^2} \det X \\ \eta \end{array}\right),$ |
| 0902.0431_E00776 | 130  | ■ 1.000 | $\left(\beta X+\frac{1}{\eta} \epsilon(X \times X)\right) \times\left(\beta X+\frac{1}{\eta} \epsilon(X \times X)\right)=\eta\left(\delta X+\frac{1}{\eta} \beta_1(X \times X)\right)$                                                                                                                                             | $\left(\beta X + \frac{1}{\eta} \epsilon(X \times X)\right) \times \left(\beta X + \frac{1}{\eta} \epsilon(X \times X)\right) = \eta \left(\delta X + \frac{1}{\eta} \beta_1(X \times X)\right)$                                                                                                                       | $\left(\beta X + \frac{1}{\eta} \epsilon(X \times X)\right) \times \left(\beta X + \frac{1}{\eta} \epsilon(X \times X)\right) = \eta \left(\delta X + \frac{1}{\eta} \beta_1(X \times X)\right)$                                                                                                                       |
| 0902.0431_E00778 | 130  | ■ 1.000 | $\beta X \times \beta X=\beta_1(X \times X), \quad\left(\beta X, \beta_1(X \times X)\right)=3 \det X .$                                                                                                                                                                                                                            | $\beta X \times \beta X = \beta_1(X \times X), \quad (\beta X, \beta_1(X \times X)) = 3 \det X .$                                                                                                                                                                                                                      | $\beta X \times \beta X = \beta_1(X \times X), \quad (\beta X, \beta_1(X \times X)) = 3 \det X .$                                                                                                                                                                                                                      |
| 0902.0431_E00779 | 131  | ■ 1.000 | $\operatorname{det}(\beta X)=\frac{1}{3}(\beta X, \beta X \times \beta X)=\frac{1}{3}(\beta X, \beta_1(X \times X))=\det X,$                                                                                                                                                                                                       | $\det(\beta X) = \frac{1}{3}(\beta X, \beta X \times \beta X) = \frac{1}{3}(\beta X, \beta_1(X \times X)) = \det X,$                                                                                                                                                                                                   | $\det(\beta X) = \frac{1}{3}(\beta X, \beta X \times \beta X) = \frac{1}{3}(\beta X, \beta_1(X \times X)) = \det X,$                                                                                                                                                                                                   |
| 0902.0431_E00781 | 131  | ■ 1.000 | $(\operatorname{det} X) \beta_1 X=(\operatorname{det} X) \tau \beta \tau X .$                                                                                                                                                                                                                                                      | $(\det X) \beta_1 X = (\det X) \tau \beta \tau X .$                                                                                                                                                                                                                                                                    | $(\det X) \beta_1 X = (\det X) \tau \beta \tau X .$                                                                                                                                                                                                                                                                    |
| 0902.0431_E00782 | 131  | ■ 1.000 | $\varphi_1(\theta)(X, Y, \xi, \eta)=\left(\theta^{-1} X, \theta Y, \theta^3 \xi, \theta^{-3} \eta\right) .$                                                                                                                                                                                                                        | $\varphi_1(\theta)(X, Y, \xi, \eta) = (\theta^{-1} X, \theta Y, \theta^3 \xi, \theta^{-3} \eta) .$                                                                                                                                                                                                                     | $\varphi_1(\theta)(X, Y, \xi, \eta) = (\theta^{-1} X, \theta Y, \theta^3 \xi, \theta^{-3} \eta) .$                                                                                                                                                                                                                     |
| 0902.0431_E00783 | 131  | ■ 1.000 | $U(1)=\left\{\varphi_1(\theta) \mid \theta \in C,(\tau \theta) \theta=1\right\}$                                                                                                                                                                                                                                                   | $U(1) = \{\varphi_1(\theta) \mid \theta \in C, (\tau \theta) \theta = 1\}$                                                                                                                                                                                                                                             | $U(1) = \{\varphi_1(\theta) \mid \theta \in C, (\tau \theta) \theta = 1\}$                                                                                                                                                                                                                                             |

| Identifier       | Page | Conf.   | LaTeX source                                                                                                                                                                                                                                                                                                                                  | Rendered                                                                                                                                                                                                                                                                                                                                | Image                                                                                                                                                                                                                                                                                                                                   |
|------------------|------|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0902.0431_E00784 | 131  | ■ 1.000 | $\alpha_{\{i\}}(a)=\left(\begin{array}{cccc} 1+(\cos  a -1)p_i & 2a\frac{\sin  a }{ a }E_i & 0 & -\tau a\frac{\sin  a }{ a }E_i \\ -2\tau a\frac{\sin  a }{ a }E_i & 1+(\cos  a -1)p_i & a\frac{\sin  a }{ a }E_i & 0 \\ 0 & -\tau a\frac{\sin  a }{ a }E_i & \cos  a  & 0 \\ a\frac{\sin  a }{ a }E_i & 0 & 0 & \cos  a  \end{array}\right)$ | $\alpha_i(a)=\left(\begin{array}{cccc} 1+(\cos  a -1)p_i & 2a\frac{\sin  a }{ a }E_i & 0 & -\tau a\frac{\sin  a }{ a }E_i \\ -2\tau a\frac{\sin  a }{ a }E_i & 1+(\cos  a -1)p_i & a\frac{\sin  a }{ a }E_i & 0 \\ 0 & -\tau a\frac{\sin  a }{ a }E_i & \cos  a  & 0 \\ a\frac{\sin  a }{ a }E_i & 0 & 0 & \cos  a  \end{array}\right)$ | $\alpha_i(a)=\left(\begin{array}{cccc} 1+(\cos  a -1)p_i & 2a\frac{\sin  a }{ a }E_i & 0 & -\tau a\frac{\sin  a }{ a }E_i \\ -2\tau a\frac{\sin  a }{ a }E_i & 1+(\cos  a -1)p_i & a\frac{\sin  a }{ a }E_i & 0 \\ 0 & -\tau a\frac{\sin  a }{ a }E_i & \cos  a  & 0 \\ a\frac{\sin  a }{ a }E_i & 0 & 0 & \cos  a  \end{array}\right)$ |
| 0902.0431_E00785 | 132  | ■ 1.000 | $p_{\{i\}}\left(\begin{array}{ccc} \xi_1 & x_3 & \bar{x}_2 \\ \bar{x}_3 & \xi_2 & x_1 \\ x_2 & \bar{x}_1 & \xi_3 \end{array}\right)=\left(\begin{array}{ccc} \xi_1 & \delta_{i3}x_3 & \delta_{i2}\bar{x}_2 \\ \delta_{i3}\bar{x}_3 & \xi_2 & \delta_{i1}x_1 \\ \delta_{i2}x_2 & \delta_{i1}\bar{x}_1 & \xi_3 \end{array}\right),$             | $p_i\left(\begin{array}{ccc} \xi_1 & x_3 & \bar{x}_2 \\ \bar{x}_3 & \xi_2 & x_1 \\ x_2 & \bar{x}_1 & \xi_3 \end{array}\right)=\left(\begin{array}{ccc} \xi_1 & \delta_{i3}x_3 & \delta_{i2}\bar{x}_2 \\ \delta_{i3}\bar{x}_3 & \xi_2 & \delta_{i1}x_1 \\ \delta_{i2}x_2 & \delta_{i1}\bar{x}_1 & \xi_3 \end{array}\right),$             | $p_i\left(\begin{array}{ccc} \xi_1 & x_3 & \bar{x}_2 \\ \bar{x}_3 & \xi_2 & x_1 \\ x_2 & \bar{x}_1 & \xi_3 \end{array}\right)=\left(\begin{array}{ccc} \xi_1 & \delta_{i3}x_3 & \delta_{i2}\bar{x}_2 \\ \delta_{i3}\bar{x}_3 & \xi_2 & \delta_{i1}x_1 \\ \delta_{i2}x_2 & \delta_{i1}\bar{x}_1 & \xi_3 \end{array}\right),$             |
| 0902.0431_E00786 | 132  | ■ 1.000 | $\Phi_{\{i\}}(a)=\Phi\left(0,-\tau a E_{\{i\}},\tau a E_{\{i\}},0\right)=\left(\begin{array}{cccc} 0 & 2a E_{\{i\}} & 0 & -\tau a E_{\{i\}} \\ -2\tau a E_{\{i\}} & 0 & a E_{\{i\}} & 0 \\ 0 & -\tau a E_{\{i\}} & 0 & 0 \\ a E_{\{i\}} & 0 & 0 & 0 \end{array}\right)\in\mathfrak{e}_7$                                                      | $\Phi_i(a)=\Phi(0,-\tau a E_i,\tau a E_i,0)=\left(\begin{array}{cccc} 0 & 2a E_i & 0 & -\tau a E_i \\ -2\tau a E_i & 0 & a E_i & 0 \\ 0 & -\tau a E_i & 0 & 0 \\ a E_i & 0 & 0 & 0 \end{array}\right)\in\mathfrak{e}_7$                                                                                                                 | $\Phi_i(a)=\Phi(0,-\tau a E_i,\tau a E_i,0)=\left(\begin{array}{cccc} 0 & 2a E_i & 0 & -\tau a E_i \\ -2\tau a E_i & 0 & a E_i & 0 \\ 0 & -\tau a E_i & 0 & 0 \\ a E_i & 0 & 0 & 0 \end{array}\right)\in\mathfrak{e}_7$                                                                                                                 |
| 0902.0431_E00788 | 132  | ■ 1.000 | $\beta X=\frac{1}{\xi}\beta(Y\times Y)=\frac{1}{\xi}\tau\beta\tau Y\times\tau\beta\tau Y$                                                                                                                                                                                                                                                     | $\beta X=\frac{1}{\xi}\beta(Y\times Y)=\frac{1}{\xi}\tau\beta\tau Y\times\tau\beta\tau Y$                                                                                                                                                                                                                                               | $\beta X=\frac{1}{\xi}\beta(Y\times Y)=\frac{1}{\xi}\tau\beta\tau Y\times\tau\beta\tau Y$                                                                                                                                                                                                                                               |
| 0902.0431_E00789 | 132  | ■ 1.000 | $\tau\beta\tau Y=\left(\begin{array}{ccc} \eta_1 & 0 & 0 \\ 0 & \eta_2 & 0 \\ 0 & 0 & \eta_3 \end{array}\right),\quad \eta_i\in C$                                                                                                                                                                                                            | $\tau\beta\tau Y=\left(\begin{array}{ccc} \eta_1 & 0 & 0 \\ 0 & \eta_2 & 0 \\ 0 & 0 & \eta_3 \end{array}\right),\quad \eta_i\in C$                                                                                                                                                                                                      | $\tau\beta\tau Y=\left(\begin{array}{ccc} \eta_1 & 0 & 0 \\ 0 & \eta_2 & 0 \\ 0 & 0 & \eta_3 \end{array}\right),\quad \eta_i\in C$                                                                                                                                                                                                      |
| 0902.0431_E00790 | 132  | ■ 1.000 | $\alpha_{\{i\}}(-\pi/2)\beta P=\left(\begin{array}{cccc} 1-p_i & -2E_i & 0 & E_i \\ 2E_i & 1-p_i & -E_i & 0 \\ 0 & E_i & 0 & 0 \\ -E_i & 0 & 0 & 0 \end{array}\right)\left(\begin{array}{c} \beta X \\ \tau\beta\tau Y \\ 0 \\ \eta \end{array}\right)=\left(\begin{array}{c} * \\ * \\ \eta_i \\ * \end{array}\right),\quad \eta_i\neq 0,$   | $\alpha_i(-\pi/2)\beta P=\left(\begin{array}{cccc} 1-p_i & -2E_i & 0 & E_i \\ 2E_i & 1-p_i & -E_i & 0 \\ 0 & E_i & 0 & 0 \\ -E_i & 0 & 0 & 0 \end{array}\right)\left(\begin{array}{c} \beta X \\ \tau\beta\tau Y \\ 0 \\ \eta \end{array}\right)=\left(\begin{array}{c} * \\ * \\ \eta_i \\ * \end{array}\right),\quad \eta_i\neq 0,$   | $\alpha_i(-\pi/2)\beta P=\left(\begin{array}{cccc} 1-p_i & -2E_i & 0 & E_i \\ 2E_i & 1-p_i & -E_i & 0 \\ 0 & E_i & 0 & 0 \\ -E_i & 0 & 0 & 0 \end{array}\right)\left(\begin{array}{c} \beta X \\ \tau\beta\tau Y \\ 0 \\ \eta \end{array}\right)=\left(\begin{array}{c} * \\ * \\ \eta_i \\ * \end{array}\right),\quad \eta_i\neq 0,$   |
| 0902.0431_E00791 | 132  | ■ 1.000 | $\alpha_{i+1}(-\pi/2)\beta P=(*,\xi_i E_{i+2}+\xi_{i+2} E_i,0,*),\quad \xi_i\neq 0,$                                                                                                                                                                                                                                                          | $\alpha_{i+1}(-\pi/2)\beta P=(*,\xi_i E_{i+2}+\xi_{i+2} E_i,0,*),\quad \xi_i\neq 0,$                                                                                                                                                                                                                                                    | $\alpha_{i+1}(-\pi/2)\beta P=(*,\xi_i E_{i+2}+\xi_{i+2} E_i,0,*),\quad \xi_i\neq 0,$                                                                                                                                                                                                                                                    |
| 0902.0431_E00792 | 133  | ■ 1.000 | $\alpha_1(-\pi/2)P=(\eta E_1,0,0,0),\quad \eta\neq 0,$                                                                                                                                                                                                                                                                                        | $\alpha_1(-\pi/2)P=(\eta E_1,0,0,0),\quad \eta\neq 0,$                                                                                                                                                                                                                                                                                  | $\alpha_1(-\pi/2)P=(\eta E_1,0,0,0),\quad \eta\neq 0,$                                                                                                                                                                                                                                                                                  |
| 0902.0431_E00793 | 133  | ■ 1.000 | $\mathfrak{M}_1=\{P\in\mathfrak{P}^C\mid P\times P=0,\langle P,P\rangle=1\}.$                                                                                                                                                                                                                                                                 | $\mathfrak{M}_1=\{P\in\mathfrak{P}^C\mid P\times P=0,\langle P,P\rangle=1\}.$                                                                                                                                                                                                                                                           | $\mathfrak{M}_1=\{P\in\mathfrak{P}^C\mid P\times P=0,\langle P,P\rangle=1\}.$                                                                                                                                                                                                                                                           |
| 0902.0431_E00794 | 133  | ■ 1.000 | $E_{\{7\}}/E_{\{6\}}\simeq\mathfrak{M}_1.$                                                                                                                                                                                                                                                                                                    | $E_7/E_6\simeq\mathfrak{M}_1.$                                                                                                                                                                                                                                                                                                          | $E_7/E_6\simeq\mathfrak{M}_1.$                                                                                                                                                                                                                                                                                                          |

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|----------------------|------|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-------|
| 0902.0431_E00795     | 133  | ■1.000 | $\alpha P = \left( \frac{1}{\xi} \left( \begin{array}{ccc} \eta_2 \eta_3 & 0 & 0 \\ 0 & \eta_3 \eta_1 & 0 \\ 0 & 0 & \eta_1 \eta_2 \end{array} \right), \left( \begin{array}{ccc} \eta_1 & 0 & 0 \\ 0 & \eta_2 & 0 \\ 0 & 0 & \eta_3 \end{array} \right), \xi, \frac{1}{\xi^2} \eta_1 \eta_2 \eta_3 \right), \quad \xi > 0$ |          |       |
| 0902.0431_E00796     | 133  | ■1.000 | $\frac{1}{\xi^2} \left(  \eta_2 \eta_3 ^2 +  \eta_3 \eta_1 ^2 +  \eta_1 \eta_2 ^2 \right) + \left(  \eta_1 ^2 +  \eta_2 ^2 +  \eta_3 ^2 \right) + \xi^2 + \frac{1}{\xi^4}  \eta_1 \eta_2 \eta_3 ^2 = 1,$                                                                                                                    |          |       |
| 0902.0431_E00797 (i) | 133  | ■1.000 | $\left( 1 + \frac{ \eta_1 ^2}{\xi^2} \right) \left( 1 + \frac{ \eta_2 ^2}{\xi^2} \right) \left( 1 + \frac{ \eta_3 ^2}{\xi^2} \right) = \frac{1}{\xi^2}.$                                                                                                                                                                    |          |       |
| 0902.0431_E00798     | 133  | ■1.000 | $\tan r_i = \frac{ \eta_i }{\xi}, \quad i = 1, 2, 3,$                                                                                                                                                                                                                                                                       |          |       |
| 0902.0431_E00800     | 134  | ■1.000 | $a_i = \frac{\eta_i}{ \eta_i } r_i, \quad i = 1, 2, 3$                                                                                                                                                                                                                                                                      |          |       |
| 0902.0431_E00801     | 134  | ■1.000 | $r_i =  a_i , \quad \eta_i = \frac{1}{ a_i } \frac{\eta_i}{ \eta_i } r_i \frac{ \eta_i }{\xi} \xi = \frac{a_i}{ a_i } \tan r_i \cos r_1 \cos r_2 \cos r_3.$                                                                                                                                                                 |          |       |
| 0902.0431_E00803     | 134  | ■1.000 | $\alpha_1 (a_1)^{-1} \alpha_2 (a_2)^{-1} \alpha_3 (a_3)^{-1} \alpha P = (0, 0, 1, 0).$                                                                                                                                                                                                                                      |          |       |
| 0902.0431_E00804     | 134  | ■1.000 | $z(E_7) = \{1, -1\}.$                                                                                                                                                                                                                                                                                                       |          |       |
| 0902.0431_E00805     | 134  | ■1.000 | $\beta X = X, \quad \tau \beta \tau Y = Y \quad \text{for all } \beta \in E_6.$                                                                                                                                                                                                                                             |          |       |
| 0902.0431_E00806     | 135  | ■1.000 | $\alpha(0, 0, 1, 0) = (0, 0, \xi, \eta).$                                                                                                                                                                                                                                                                                   |          |       |
| 0902.0431_E00808     | 135  | ■1.000 | $\alpha(0, 0, 1, 0) = (0, 0, \xi, 0), \quad \alpha(0, 0, 0, 1) = (0, 0, 0, \xi^{-1}).$                                                                                                                                                                                                                                      |          |       |
| 0902.0431_E00814     | 136  | ■1.000 | $(E_7)^\iota = \{ \alpha \in E_7 \mid \iota \alpha = \alpha \iota \} \\ \cong \{ \alpha \in E_7 \mid \lambda \alpha = \alpha \lambda \} = (E_7)^\lambda.$                                                                                                                                                                   |          |       |



| Identifier       | Page | Conf.  | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                | Rendered                                                                                                                                                                                                                                                                                                                                                                        | Image                                                                                                                                                                                                                                                                                                                                                                           |
|------------------|------|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0902.0431_E00844 | 140  | ■1.000 | $\begin{aligned} & \left[\Phi(\phi(\nu), aE_1, -\tau aE_1, \nu), \Phi(\phi(\rho), bE_1, -\tau bE_1, \rho)\right] \\ & = \Phi(\phi(b\tau a) - a(\tau b)), 2(b\nu - a\rho)E_1, -2\tau(b\nu - a\rho)E_1, (\tau a)b - a(\tau b)). \end{aligned}$                                                                                                                                                | $[\Phi(\phi(\nu), aE_1, -\tau aE_1, \nu), \Phi(\phi(\rho), bE_1, -\tau bE_1, \rho)]$ $= \Phi(\phi(b\tau a) - a(\tau b)), 2(b\nu - a\rho)E_1, -2\tau(b\nu - a\rho)E_1, (\tau a)b - a(\tau b)).$                                                                                                                                                                                  | $\left[\Phi(\phi(\nu), aE_1, -\tau aE_1, \nu), \Phi(\phi(\rho), bE_1, -\tau bE_1, \rho)\right]$ $= \Phi(\phi(b\tau a) - a(\tau b)), 2(b\nu - a\rho)E_1, -2\tau(b\nu - a\rho)E_1, (\tau a)b - a(\tau b)).$                                                                                                                                                                       |
| 0902.0431_E00845 | 140  | ■1.000 | $\left(\frac{e_7}{\nu}\right)^{\sigma} \ni \Phi\left(\begin{array}{c} \phi \\ A \\ -\tau A \\ \nu \end{array}\right) = \Phi\left(\begin{array}{c} \phi(\nu') \\ aE_1 \\ -\tau aE_1 \\ \nu' \end{array}\right) + \Phi\left(\begin{array}{c} \phi - \phi(\nu') \\ A - aE_1 \\ -\tau A + \tau aE_1 \\ \nu - \nu' \end{array}\right) \in \mathfrak{a}_1 \oplus (\mathfrak{e}_7)^{\kappa, \mu},$ | $(\mathfrak{e}_7)^{\sigma} \ni \Phi\left(\begin{array}{c} \phi \\ A \\ -\tau A \\ \nu \end{array}\right) = \Phi\left(\begin{array}{c} \phi(\nu') \\ aE_1 \\ -\tau aE_1 \\ \nu' \end{array}\right) + \Phi\left(\begin{array}{c} \phi - \phi(\nu') \\ A - aE_1 \\ -\tau A + \tau aE_1 \\ \nu - \nu' \end{array}\right) \in \mathfrak{a}_1 \oplus (\mathfrak{e}_7)^{\kappa, \mu},$ | $(\mathfrak{e}_7)^{\sigma} \ni \Phi\left(\begin{array}{c} \phi \\ A \\ -\tau A \\ \nu \end{array}\right) = \Phi\left(\begin{array}{c} \phi(\nu') \\ aE_1 \\ -\tau aE_1 \\ \nu' \end{array}\right) + \Phi\left(\begin{array}{c} \phi - \phi(\nu') \\ A - aE_1 \\ -\tau A + \tau aE_1 \\ \nu - \nu' \end{array}\right) \in \mathfrak{a}_1 \oplus (\mathfrak{e}_7)^{\kappa, \mu},$ |
| 0902.0431_E00846 | 141  | ■1.000 | $\alpha_{23}(a) = \alpha_2(a)\alpha_3(\tau a) \in ((E_7)^{\kappa, \mu})_{(0, E_1, 0, 1)},$                                                                                                                                                                                                                                                                                                  | $\alpha_{23}(a) = \alpha_2(a)\alpha_3(\tau a) \in ((E_7)^{\kappa, \mu})_{(0, E_1, 0, 1)},$                                                                                                                                                                                                                                                                                      | $\alpha_{23}(a) = \alpha_2(a)\alpha_3(\tau a) \in ((E_7)^{\kappa, \mu})_{(0, E_1, 0, 1)},$                                                                                                                                                                                                                                                                                      |
| 0902.0431_E00847 | 141  | ■1.000 | $\begin{aligned} & \alpha_{23}(a) = \alpha_2(a)\alpha_3(\tau a) \\ & = \exp \Phi(0, -\tau aE_2 - aE_3, aE_2 + \tau aE_3, 0). \end{aligned}$                                                                                                                                                                                                                                                 | $\alpha_{23}(a) = \alpha_2(a)\alpha_3(\tau a)$ $= \exp \Phi(0, -\tau aE_2 - aE_3, aE_2 + \tau aE_3, 0).$                                                                                                                                                                                                                                                                        | $\alpha_{23}(a) = \alpha_2(a)\alpha_3(\tau a)$ $= \exp \Phi(0, -\tau aE_2 - aE_3, aE_2 + \tau aE_3, 0).$                                                                                                                                                                                                                                                                        |
| 0902.0431_E00848 | 141  | ■1.000 | $\text{Spin}(10) = \{\alpha \in E_6 \mid \alpha E_1 = E_1\}$ $= \{\alpha \in E_6 \mid \sigma \alpha = \alpha \sigma, \alpha E_1 = E_1\} \subset E_7$                                                                                                                                                                                                                                        | $\text{Spin}(10) = \{\alpha \in E_6 \mid \alpha E_1 = E_1\}$ $= \{\alpha \in E_6 \mid \sigma \alpha = \alpha \sigma, \alpha E_1 = E_1\} \subset E_7$                                                                                                                                                                                                                            | $\text{Spin}(10) = \{\alpha \in E_6 \mid \alpha E_1 = E_1\}$ $= \{\alpha \in E_6 \mid \sigma \alpha = \alpha \sigma, \alpha E_1 = E_1\} \subset E_7$                                                                                                                                                                                                                            |
| 0902.0431_E00849 | 141  | ■1.000 | $S^9 = \left\{ \left( \begin{pmatrix} 0 & 0 & 0 \\ 0 & \xi & x \\ 0 & \bar{x} & -\tau \xi \end{pmatrix}, 0, 0, 0 \right) \mid \xi \in C, x \in \mathfrak{C}, \bar{x}x + (\tau \xi)\xi = 1 \right\}.$                                                                                                                                                                                        | $S^9 = \left\{ \left( \begin{pmatrix} 0 & 0 & 0 \\ 0 & \xi & x \\ 0 & \bar{x} & -\tau \xi \end{pmatrix}, 0, 0, 0 \right) \mid \xi \in C, x \in \mathfrak{C}, \bar{x}x + (\tau \xi)\xi = 1 \right\}.$                                                                                                                                                                            | $S^9 = \left\{ \left( \begin{pmatrix} 0 & 0 & 0 \\ 0 & \xi & x \\ 0 & \bar{x} & -\tau \xi \end{pmatrix}, 0, 0, 0 \right) \mid \xi \in C, x \in \mathfrak{C}, \bar{x}x + (\tau \xi)\xi = 1 \right\}.$                                                                                                                                                                            |
| 0902.0431_E00850 | 141  | ■1.000 | $\alpha(0, -E_1, 0, 1) = (0, -E_1, 0, 1) \quad \text{if and only if} \quad \alpha(0, 0, 1, 0) = (0, 0, 1, 0).$                                                                                                                                                                                                                                                                              | $\alpha(0, -E_1, 0, 1) = (0, -E_1, 0, 1) \quad \text{if and only if} \quad \alpha(0, 0, 1, 0) = (0, 0, 1, 0).$                                                                                                                                                                                                                                                                  | $\alpha(0, -E_1, 0, 1) = (0, -E_1, 0, 1) \quad \text{if and only if} \quad \alpha(0, 0, 1, 0) = (0, 0, 1, 0).$                                                                                                                                                                                                                                                                  |
| 0902.0431_E00851 | 141  | ■1.000 | $\left\{ \alpha \in ((E_7)^{\kappa, \mu})_{(0, E_1, 0, 1)} \mid \alpha(0, -E_1, 0, 1) = (0, -E_1, 0, 1) \right\} \cong \text{Spin}(10).$                                                                                                                                                                                                                                                    | $\left\{ \alpha \in ((E_7)^{\kappa, \mu})_{(0, E_1, 0, 1)} \mid \alpha(0, -E_1, 0, 1) = (0, -E_1, 0, 1) \right\} \cong \text{Spin}(10).$                                                                                                                                                                                                                                        | $\left\{ \alpha \in ((E_7)^{\kappa, \mu})_{(0, E_1, 0, 1)} \mid \alpha(0, -E_1, 0, 1) = (0, -E_1, 0, 1) \right\} \cong \text{Spin}(10).$                                                                                                                                                                                                                                        |
| 0902.0431_E00854 | 142  | ■1.000 | $P = \left( \begin{pmatrix} 0 & 0 & 0 \\ 0 & \xi & x \\ 0 & \bar{x} & -\tau \xi \end{pmatrix}, \begin{pmatrix} \eta & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, 0, \tau \eta \right) \in S^{10},$                                                                                                                                                                                      | $P = \left( \begin{pmatrix} 0 & 0 & 0 \\ 0 & \xi & x \\ 0 & \bar{x} & -\tau \xi \end{pmatrix}, \begin{pmatrix} \eta & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, 0, \tau \eta \right) \in S^{10},$                                                                                                                                                                          | $P = \left( \begin{pmatrix} 0 & 0 & 0 \\ 0 & \xi & x \\ 0 & \bar{x} & -\tau \xi \end{pmatrix}, \begin{pmatrix} \eta & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, 0, \tau \eta \right) \in S^{10},$                                                                                                                                                                          |
| 0902.0431_E00858 | 142  | ■1.000 | $\beta \alpha_{23}(a)P = (i(E_2 + E_3), 0, 0, 0).$                                                                                                                                                                                                                                                                                                                                          | $\beta \alpha_{23}(a)P = (i(E_2 + E_3), 0, 0, 0).$                                                                                                                                                                                                                                                                                                                              | $\beta \alpha_{23}(a)P = (i(E_2 + E_3), 0, 0, 0).$                                                                                                                                                                                                                                                                                                                              |
| 0902.0431_E00859 | 142  | ■1.000 | $\alpha_{23}(-\pi/4)\beta \alpha_{23}(a)P = (0, -iE_1, 0, i).$                                                                                                                                                                                                                                                                                                                              | $\alpha_{23}(-\pi/4)\beta \alpha_{23}(a)P = (0, -iE_1, 0, i).$                                                                                                                                                                                                                                                                                                                  | $\alpha_{23}(-\pi/4)\beta \alpha_{23}(a)P = (0, -iE_1, 0, i).$                                                                                                                                                                                                                                                                                                                  |

| Identifier      | Page | Conf.   | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Rendered                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Image                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-----------------|------|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0902.0431_E0865 | 144  | ■ 1.000 | $P=\left(\left(\begin{array}{ccc} 0 & 0 & 0 \\ 0 & \xi & x \\ 0 & \bar{x} & -\tau\xi \end{array}\right),\left(\begin{array}{ccc} \eta & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{array}\right),0,\tau\eta\right)\in S^{11},$                                                                                                                                                                                                                                                                                                                                             | $P=\left(\left(\begin{array}{ccc} 0 & 0 & 0 \\ 0 & \xi & x \\ 0 & \bar{x} & -\tau\xi \end{array}\right),\left(\begin{array}{ccc} \eta & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{array}\right),0,\tau\eta\right)\in S^{11},$                                                                                                                                                                                                                                                            | $P=\left(\left(\begin{array}{ccc} 0 & 0 & 0 \\ 0 & \xi & x \\ 0 & \bar{x} & -\tau\xi \end{array}\right),\left(\begin{array}{ccc} \eta & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{array}\right),0,\tau\eta\right)\in S^{11},$                                                                                                                                                                                                                                                            |
| 0902.0431_E0866 | 144  | ■ 1.000 | $\alpha(t) \, P \, \in S^{\wedge 10} \, .$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | $\alpha(t)P\in S^{10}.$                                                                                                                                                                                                                                                                                                                                                                                                                                                             | $\alpha(t)P\in S^{10}.$                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 0902.0431_E0867 | 144  | ■ 1.000 | $\beta\alpha(t) \, P=\left(0,-i\,E_1,0,i\right) \, .$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | $\beta\alpha(t)P=(0,-iE_1,0,i) \, .$                                                                                                                                                                                                                                                                                                                                                                                                                                                | $\beta\alpha(t)P=(0,-iE_1,0,i).$                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 0902.0431_E0868 | 144  | ■ 1.000 | $\alpha(-\pi/4) \, \beta\alpha(t) \, P=\left(0,E_1,0,1\right) \, .$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | $\alpha(-\pi/4)\beta\alpha(t)P=(0,E_1,0,1) \, .$                                                                                                                                                                                                                                                                                                                                                                                                                                    | $\alpha(-\pi/4)\beta\alpha(t)P=(0,E_1,0,1).$                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 0902.0431_E0869 | 144  | ■ 1.000 | $p\colon\left(E_7\right)^{\kappa,\mu}\rightarrow S\,0(12)=S\,0\left(V^{\wedge 12}\right)$                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | $p:(E_7)^{\kappa,\mu}\rightarrow SO(12)=SO\left(V^{12}\right)$                                                                                                                                                                                                                                                                                                                                                                                                                      | $p:(E_7)^{\kappa,\mu}\rightarrow SO(12)=SO(V^{12})$                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 0902.0431_E0871 | 144  | ■ 1.000 | $\left(E_7\right)^{\kappa,\mu}/\{1,\sigma\}\cong S\,0(12) \, .$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | $(E_7)^{\kappa,\mu}/\{1,\sigma\}\cong SO(12).$                                                                                                                                                                                                                                                                                                                                                                                                                                      | $(E_7)^{\kappa,\mu}/\{1,\sigma\}\cong SO(12).$                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 0902.0431_E0874 | 145  | ■ 1.000 | $\varphi_2(S\,U(2))=\left\{\varphi_2(A)\in E_7\mid A\in SU(2)\right\}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | $\varphi_2(SU(2))=\{\varphi_2(A)\in E_7\mid A\in SU(2)\}$                                                                                                                                                                                                                                                                                                                                                                                                                           | $\varphi_2(SU(2))=\{\varphi_2(A)\in E_7\mid A\in SU(2)\}$                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 0902.0431_E0877 | 145  | ■ 1.000 | $\Phi\bigl(\phi(\nu),aE_1,-\tau aE_1,\nu\bigr)(X,Y,\xi,\eta)=(X',Y',\xi',\eta')$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | $\Phi(\phi(\nu),aE_1,-\tau aE_1,\nu)(X,Y,\xi,\eta)=(X',Y',\xi',\eta')$                                                                                                                                                                                                                                                                                                                                                                                                              | $\Phi(\phi(\nu),aE_1,-\tau aE_1,\nu)(X,Y,\xi,\eta)=(X',Y',\xi',\eta')$                                                                                                                                                                                                                                                                                                                                                                                                              |
| 0902.0431_E0879 | 145  | ■ 1.000 | $\exp\left(\Phi\bigl(\phi(\nu),aE_1,-\tau aE_1,\nu\bigr)\right)=\varphi_2(A)\in\varphi_2(SU(2))\subset(E_7)^\sigma \, .$                                                                                                                                                                                                                                                                                                                                                                                                                                             | $\exp\left(\Phi(\phi(\nu),aE_1,-\tau aE_1,\nu)\right)=\varphi_2(A)\in\varphi_2(SU(2))\subset(E_7)^\sigma \, .$                                                                                                                                                                                                                                                                                                                                                                      | $\exp(\Phi(\phi(\nu),aE_1,-\tau aE_1,\nu))=\varphi_2(A)\in\varphi_2(SU(2))\subset(E_7)^\sigma.$                                                                                                                                                                                                                                                                                                                                                                                     |
| 0902.0431_E0880 | 146  | ■ 1.000 | $\varphi(A,\beta)=\varphi_2(A)\beta \, .$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | $\varphi(A,\beta)=\varphi_2(A)\beta.$                                                                                                                                                                                                                                                                                                                                                                                                                                               | $\varphi(A,\beta)=\varphi_2(A)\beta.$                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 0902.0431_E0884 | 147  | ■ 1.000 | $\begin{aligned} \left(E_7\right)^{\tau\gamma} &= \left\{\alpha\in E_7\mid \tau\gamma\alpha=\alpha\tau\gamma\right\} \\ &= \left\{\alpha\in E_7\mid \lambda\gamma\alpha=\alpha\lambda\gamma\right\} = \left(E_7\right)^{\lambda\gamma} \, . \end{aligned}$                                                                                                                                                                                                                                                                                                           | $\begin{aligned} (E_7)^{\tau\gamma} &= \{\alpha\in E_7\mid \tau\gamma\alpha=\alpha\tau\gamma\} \\ &= \{\alpha\in E_7\mid \lambda\gamma\alpha=\alpha\lambda\gamma\} = (E_7)^{\lambda\gamma} \, . \end{aligned}$                                                                                                                                                                                                                                                                      | $\begin{aligned} (E_7)^{\tau\gamma} &= \{\alpha\in E_7\mid \tau\gamma\alpha=\alpha\tau\gamma\} \\ &= \{\alpha\in E_7\mid \lambda\gamma\alpha=\alpha\lambda\gamma\} = (E_7)^{\lambda\gamma} \, . \end{aligned}$                                                                                                                                                                                                                                                                      |
| 0902.0431_E0885 | 147  | ■ 1.000 | $\begin{aligned} \left(\mathfrak{P}^C\right)_{\tau\gamma} &= \left\{P\in\mathfrak{P}^C\mid \tau\gamma P=P\right\} \\ &= \left\{(X,Y,\xi,\eta)\in\mathfrak{P}^C\mid X,Y\in\left(\mathfrak{J}^C\right)_{\tau\gamma},\xi,\eta\in\boldsymbol{R}\right\}, \\ \left(\mathfrak{P}^C\right)_{-\tau\gamma} &= \left\{P\in\mathfrak{P}^C\mid \tau\gamma P=-P\right\} \\ &= \left\{(X,Y,\xi,\eta)\in\mathfrak{P}^C\mid X,Y\in\left(\mathfrak{J}^C\right)_{-\tau\gamma},\xi,\eta\in i\boldsymbol{R}\right\}, \\ &= i\left(\mathfrak{P}^C\right)_{\tau\gamma} \, . \end{aligned}$ | $\begin{aligned} \left(\mathfrak{P}^C\right)_{\tau\gamma} &= \{P\in\mathfrak{P}^C\mid \tau\gamma P=P\} \\ &= \{(X,Y,\xi,\eta)\in\mathfrak{P}^C\mid X,Y\in(\mathfrak{J}^C)_{\tau\gamma},\xi,\eta\in\boldsymbol{R}\}, \\ \left(\mathfrak{P}^C\right)_{-\tau\gamma} &= \{P\in\mathfrak{P}^C\mid \tau\gamma P=-P\} \\ &= \{(X,Y,\xi,\eta)\in\mathfrak{P}^C\mid X,Y\in(\mathfrak{J}^C)_{-\tau\gamma},\xi,\eta\in i\boldsymbol{R}\}, \\ &= i(\mathfrak{P}^C)_{\tau\gamma}. \end{aligned}$ | $\begin{aligned} \left(\mathfrak{P}^C\right)_{\tau\gamma} &= \{P\in\mathfrak{P}^C\mid \tau\gamma P=P\} \\ &= \{(X,Y,\xi,\eta)\in\mathfrak{P}^C\mid X,Y\in(\mathfrak{J}^C)_{\tau\gamma},\xi,\eta\in\boldsymbol{R}\}, \\ \left(\mathfrak{P}^C\right)_{-\tau\gamma} &= \{P\in\mathfrak{P}^C\mid \tau\gamma P=-P\} \\ &= \{(X,Y,\xi,\eta)\in\mathfrak{P}^C\mid X,Y\in(\mathfrak{J}^C)_{-\tau\gamma},\xi,\eta\in i\boldsymbol{R}\}, \\ &= i(\mathfrak{P}^C)_{\tau\gamma}. \end{aligned}$ |
| 0902.0431_E0887 | 147  | ■ 1.000 | $k\bigl(\left(a+be_2\right)\bigr)=\left(\left(\begin{array}{cc} a & b \\ -\bar{b} & \bar{a} \end{array}\right)\right),\quad a,b\in\boldsymbol{C}.$                                                                                                                                                                                                                                                                                                                                                                                                                   | $k((a+be_2))=\left(\left(\begin{array}{cc} a & b \\ -\bar{b} & \bar{a} \end{array}\right)\right),\quad a,b\in\boldsymbol{C}.$                                                                                                                                                                                                                                                                                                                                                       | $k\left(\left(a+be_2\right)\right)=\left(\left(\begin{array}{cc} a & b \\ -\bar{b} & \bar{a} \end{array}\right)\right),\quad a,b\in\boldsymbol{C}.$                                                                                                                                                                                                                                                                                                                                 |
| 0902.0431_E0888 | 147  | ■ 1.000 | $B=k(D)+e_1\,k(T),\quad D\in\mathfrak{sp}(4),\,T\in\mathfrak{J}(4,\boldsymbol{H})_0.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | $B=k(D)+e_1k(T),\quad D\in\mathfrak{sp}(4),T\in\mathfrak{J}(4,\boldsymbol{H})_0.$                                                                                                                                                                                                                                                                                                                                                                                                   | $B=k(D)+e_1k(T),\quad D\in\mathfrak{sp}(4),T\in\mathfrak{J}(4,\boldsymbol{H})_0.$                                                                                                                                                                                                                                                                                                                                                                                                   |

| Identifier       | Page | Conf.  | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                 | Rendered                                                                                                                                                                                                                                                                                                                                                                                                          | Image                                                                                                                                                                                                                                                                                                                                                                                       |
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| 0902.0431_E00890 | 147  | ■1.000 | $D_{\{1\}}+e_{\{1\}}T_{\{1\}}=0, \ D_{\{1\}}\in k(\mathfrak{sp}(4)), \ T_{\{1\}}\in k(\mathfrak{J}(4,\boldsymbol{H})_0) \quad \text{\texttt{implies}} \quad C_{\{1\}}=T_{\{1\}}=0 \ .$                                                                                                                                                                                                                       | $D_1+e_1T_1=0, D_1\in k(\mathfrak{sp}(4)), T_1\in k(\mathfrak{J}(4,\boldsymbol{H})_0) \quad \text{\texttt{implies}} \quad C_1=T_1=0.$                                                                                                                                                                                                                                                                             | $D_1+e_1T_1=0, D_1\in k(\mathfrak{sp}(4)), T_1\in k(\mathfrak{J}(4,\boldsymbol{H})_0) \quad \text{\texttt{implies}} \quad C_1=T_1=0.$                                                                                                                                                                                                                                                       |
| 0902.0431_E00891 | 148  | ■1.000 | $\begin{aligned} \left(\mathfrak{e}_{\{7\}}\right)^{\tau\gamma}&=\left(\Phi\left(\phi,A,-\gamma A,0\right)\in\mathfrak{e}_7\mid\phi\in\left(\mathfrak{e}_6\right)^{\tau\gamma},A\in\left(\mathfrak{J}^C\right)_{\tau\gamma}\right)\\&=\left(\Phi\left(\varphi_*(D),g^{-1}(T),-\gamma g^{-1}(T),0\right)\in\mathfrak{e}_7\mid D\in\mathfrak{sp}(4),T\in\mathfrak{J}(4,\boldsymbol{H})_0\right).\end{aligned}$ | $\begin{aligned} (\mathfrak{e}_7)^{\tau\gamma}&=\{\Phi\in\mathfrak{e}_7\mid\tau\gamma\Phi=\Phi\tau\gamma\}\\&=\left\{\Phi(\phi,A,-\gamma A,0)\in\mathfrak{e}_7\mid\phi\in(\mathfrak{e}_6)^{\tau\gamma},A\in(\mathfrak{J}^C)_{\tau\gamma}\right\}\\&=\left\{\Phi(\varphi_*(D),g^{-1}(T),-\gamma g^{-1}(T),0)\in\mathfrak{e}_7\mid D\in\mathfrak{sp}(4),T\in\mathfrak{J}(4,\boldsymbol{H})_0\right\}.\end{aligned}$ | $\begin{aligned} (\mathfrak{e}_7)^{\tau\gamma}&=\{\Phi\in\mathfrak{e}_7\mid\tau\gamma\Phi=\Phi\tau\gamma\}\\&=\{\Phi(\phi,A,-\gamma A,0)\in\mathfrak{e}_7\mid\phi\in(\mathfrak{e}_6)^{\tau\gamma},A\in(\mathfrak{J}^C)_{\tau\gamma}\}\\&=\{\Phi(\varphi_*(D),g^{-1}(T),-\gamma g^{-1}(T),0)\in\mathfrak{e}_7\mid D\in\mathfrak{sp}(4),T\in\mathfrak{J}(4,\boldsymbol{H})_0\}.\end{aligned}$ |
| 0902.0431_E00895 | 148  | ■1.000 | $k_{\{J\}}\left(M_{\{1\}}+iM_{\{2\}}\right)=k\left(M_{\{1\}}\right)J+i\,k\left(M_{\{2\}}\right)J, \quad \text{\texttt{quad}} \ M_{\{1\}},M_{\{2\}}\in\mathfrak{J}(4,\boldsymbol{H}),$                                                                                                                                                                                                                        | $k_J(M_1+iM_2)=k(M_1)J+ik(M_2)J, \quad M_1,M_2\in\mathfrak{J}(4,\boldsymbol{H}),$                                                                                                                                                                                                                                                                                                                                 | $k_J(M_1+iM_2)=k(M_1)J+ik(M_2)J, \quad M_1,M_2\in\mathfrak{J}(4,\boldsymbol{H}),$                                                                                                                                                                                                                                                                                                           |
| 0902.0431_E00896 | 148  | ■1.000 | $\chi(X,Y,\xi,\eta)=k_{\{J\}}\left(gX-\frac{\xi}{2}E\right)+e_1k_J\left(g(\gamma Y)-\frac{\eta}{2}E\right).$                                                                                                                                                                                                                                                                                                 | $\chi(X,Y,\xi,\eta)=k_J\left(gX-\frac{\xi}{2}E\right)+e_1k_J\left(g(\gamma Y)-\frac{\eta}{2}E\right).$                                                                                                                                                                                                                                                                                                            | $\chi(X,Y,\xi,\eta)=k_J\left(gX-\frac{\xi}{2}E\right)+e_1k_J\left(g(\gamma Y)-\frac{\eta}{2}E\right).$                                                                                                                                                                                                                                                                                      |
| 0902.0431_E00897 | 148  | ■1.000 | $\varphi_*(B)P=\chi^{-1}\left(B(\chi P)+(\chi P)^tB\right), \quad P\in\mathfrak{P}^C.$                                                                                                                                                                                                                                                                                                                       | $\varphi_*(B)P=\chi^{-1}\left(B(\chi P)+(\chi P)^tB\right), \quad P\in\mathfrak{P}^C.$                                                                                                                                                                                                                                                                                                                            | $\varphi_*(B)P=\chi^{-1}(B(\chi P)+(\chi P)^tB), \quad P\in\mathfrak{P}^C.$                                                                                                                                                                                                                                                                                                                 |
| 0902.0431_E00901 | 150  | ■1.000 | $\varphi(A)P=\chi^{-1}\left(A(\chi P)^tA\right), \quad P\in\mathfrak{P}^C.$                                                                                                                                                                                                                                                                                                                                  | $\varphi(A)P=\chi^{-1}\left(A(\chi P)^tA\right), \quad P\in\mathfrak{P}^C.$                                                                                                                                                                                                                                                                                                                                       | $\varphi(A)P=\chi^{-1}(A(\chi P)^tA), \quad P\in\mathfrak{P}^C.$                                                                                                                                                                                                                                                                                                                            |
| 0902.0431_E00902 | 150  | ■1.000 | $\varphi_*(D)P=\chi^{-1}\left(D(\chi P)+(\chi P)^tD\right), \quad P\in\mathfrak{P}^C,$                                                                                                                                                                                                                                                                                                                       | $\varphi_*(D)P=\chi^{-1}\left(D(\chi P)+(\chi P)^tD\right), \quad P\in\mathfrak{P}^C,$                                                                                                                                                                                                                                                                                                                            | $\varphi_*(D)P=\chi^{-1}(D(\chi P)+(\chi P)^tD), \quad P\in\mathfrak{P}^C,$                                                                                                                                                                                                                                                                                                                 |
| 0902.0431_E00904 | 150  | ■1.000 | $\tau\gamma Y=Y, \quad X=\frac{1}{\xi}(Y\times Y), \quad \tau\xi=\xi, \quad \tau\eta=\eta.$                                                                                                                                                                                                                                                                                                                  | $\tau\gamma Y=Y, \quad X=\frac{1}{\xi}(Y\times Y), \quad \tau\xi=\xi, \quad \tau\eta=\eta.$                                                                                                                                                                                                                                                                                                                       | $\tau\gamma Y=Y, \quad X=\frac{1}{\xi}(Y\times Y), \quad \tau\xi=\xi, \quad \tau\eta=\eta.$                                                                                                                                                                                                                                                                                                 |
| 0902.0431_E00910 | 151  | ■1.000 | $\left(E_{\{7\}}\right)^w=\left\{\alpha\in E_{\{7\}}\mid w\alpha=\alpha w\right\}.$                                                                                                                                                                                                                                                                                                                          | $(E_7)^w=\{\alpha\in E_7\mid w\alpha=\alpha w\}.$                                                                                                                                                                                                                                                                                                                                                                 | $(E_7)^w=\{\alpha\in E_7\mid w\alpha=\alpha w\}.$                                                                                                                                                                                                                                                                                                                                           |
| 0902.0431_E00912 | 151  | ■1.000 | $E_{\{6,\boldsymbol{C}\}}=\left\{\alpha\in E_{\{7,\boldsymbol{C}\}}\mid\alpha(0,0,1,0)=(0,0,1,0)\right\},$                                                                                                                                                                                                                                                                                                   | $E_{6,C}=\{\alpha\in E_{7,C}\mid\alpha(0,0,1,0)=(0,0,1,0)\},$                                                                                                                                                                                                                                                                                                                                                     | $E_{6,C}=\{\alpha\in E_{7,C}\mid\alpha(0,0,1,0)=(0,0,1,0)\},$                                                                                                                                                                                                                                                                                                                               |
| 0902.0431_E00913 | 152  | ■1.000 | $\mathfrak{e}_{\{7,\boldsymbol{C}\}}=\left\{\Phi(\phi,A,-\tau A,\nu)\mid\phi\in\mathfrak{e}_{6,C},A\in(\mathfrak{J}_C)^C,\nu\in i\boldsymbol{R}\right\}$                                                                                                                                                                                                                                                     | $\mathfrak{e}_{7,C}=\left\{\Phi(\phi,A,-\tau A,\nu)\mid\phi\in\mathfrak{e}_{6,C},A\in(\mathfrak{J}_C)^C,\nu\in i\boldsymbol{R}\right\}$                                                                                                                                                                                                                                                                           | $\mathfrak{e}_{7,C}=\{\Phi(\phi,A,-\tau A,\nu)\mid\phi\in\mathfrak{e}_{6,C},A\in(\mathfrak{J}_C)^C,\nu\in i\boldsymbol{R}\}$                                                                                                                                                                                                                                                                |
| 0902.0431_E00915 | 152  | ■1.000 | $\left(\mathfrak{M}_{\{\boldsymbol{C}\}}\right)_1=\left\{P\in(\mathfrak{M}_C)^C\mid P\times P=0,\langle P,P\rangle=1\right\}$                                                                                                                                                                                                                                                                                | $(\mathfrak{M}_C)_1=\left\{P\in(\mathfrak{M}_C)^C\mid P\times P=0,\langle P,P\rangle=1\right\}$                                                                                                                                                                                                                                                                                                                   | $(\mathfrak{M}_C)_1=\{P\in(\mathfrak{M}_C)^C\mid P\times P=0,\langle P,P\rangle=1\}$                                                                                                                                                                                                                                                                                                        |
| 0902.0431_E00917 | 152  | ■1.000 | $h^{\prime}(a+bi)=a+be_1, \quad a,b\in\boldsymbol{R}.$                                                                                                                                                                                                                                                                                                                                                       | $h'(a+bi)=a+be_1, \quad a,b\in\boldsymbol{R}.$                                                                                                                                                                                                                                                                                                                                                                    | $h'(a+bi)=a+be_1, \quad a,b\in\boldsymbol{R}.$                                                                                                                                                                                                                                                                                                                                              |
| 0902.0431_E00918 | 152  | ■1.000 | $f(a\,v)=h^{\prime}(a)f(v), \quad a\in C,v\in V.$                                                                                                                                                                                                                                                                                                                                                            | $f(av)=h'(a)f(v), \quad a\in C,v\in V.$                                                                                                                                                                                                                                                                                                                                                                           | $f(av)=h'(a)f(v), \quad a\in C,v\in V.$                                                                                                                                                                                                                                                                                                                                                     |

| Identifier       | Page | Conf.   | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Rendered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Image                                                                                                                                                                                                                                                   |
|------------------|------|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0902.0431_E00921 | 153  | ■ 1.000 | $\begin{array}{lll} x_{156}=h^{\{\prime\}}\left(\xi_1\right), & x_{164}=h^{\{\prime\}}\left(x_3\right), & x_{145}=h^{\{\prime\}}\left(\bar{x}_2\right), \\ x_{256}=h^{\{\prime\}}\left(\bar{x}_3\right), & x_{264}=h^{\{\prime\}}\left(\xi_2\right), & x_{245}=h^{\{\prime\}}\left(x_1\right), \\ x_{356}=h^{\{\prime\}}\left(x_2\right), & x_{364}=h^{\{\prime\}}\left(\bar{x}_1\right), & x_{345}=h^{\{\prime\}}\left(\xi_3\right), \\ x_{423}=h^{\{\prime\}}\left(\eta_1\right), & x_{431}=h^{\{\prime\}}\left(y_3\right), & x_{412}=h^{\{\prime\}}\left(\bar{y}_2\right), \\ x_{523}=h^{\{\prime\}}\left(\bar{y}_3\right), & x_{531}=h^{\{\prime\}}\left(\eta_2\right), & x_{512}=h^{\{\prime\}}\left(y_1\right), \\ x_{623}=h^{\{\prime\}}\left(y_2\right), & x_{631}=h^{\{\prime\}}\left(\bar{y}_1\right), & x_{612}=h^{\{\prime\}}\left(\eta_3\right), \\ & x_{123}=h^{\{\prime\}}\left(\xi\right), & \\ & x_{456}=h^{\{\prime\}}\left(\eta\right). & \end{array}$ | $\begin{array}{lll} x_{156}=h^{\prime}\left(\xi_1\right), & x_{164}=h^{\prime}\left(x_3\right), & x_{145}=h^{\prime}\left(\bar{x}_2\right), \\ x_{256}=h^{\prime}\left(\bar{x}_3\right), & x_{264}=h^{\prime}\left(\xi_2\right), & x_{245}=h^{\prime}\left(x_1\right), \\ x_{356}=h^{\prime}\left(x_2\right), & x_{364}=h^{\prime}\left(\bar{x}_1\right), & x_{345}=h^{\prime}\left(\xi_3\right), \\ x_{423}=h^{\prime}\left(\eta_1\right), & x_{431}=h^{\prime}\left(y_3\right), & x_{412}=h^{\prime}\left(\bar{y}_2\right), \\ x_{523}=h^{\prime}\left(\bar{y}_3\right), & x_{531}=h^{\prime}\left(\eta_2\right), & x_{512}=h^{\prime}\left(y_1\right), \\ x_{623}=h^{\prime}\left(y_2\right), & x_{631}=h^{\prime}\left(\bar{y}_1\right), & x_{612}=h^{\prime}\left(\eta_3\right), \\ & x_{123}=h^{\prime}\left(\xi\right), & \\ & x_{456}=h^{\prime}\left(\eta\right). & \end{array}$ |                                                                                                                                                                                                                                                         |
| 0902.0431_E00925 | 153  | ■ 1.000 | $A(\boldsymbol{a} \wedge \boldsymbol{b} \wedge \boldsymbol{c})=A \boldsymbol{a} \wedge \boldsymbol{A} \boldsymbol{b} \wedge \boldsymbol{A} \boldsymbol{c}.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | $A(\boldsymbol{a} \wedge \boldsymbol{b} \wedge \boldsymbol{c})=A \boldsymbol{a} \wedge \boldsymbol{A} \boldsymbol{b} \wedge \boldsymbol{A} \boldsymbol{c}.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | $A(\boldsymbol{a} \wedge \boldsymbol{b} \wedge \boldsymbol{c})=A \boldsymbol{a} \wedge \boldsymbol{A} \boldsymbol{b} \wedge \boldsymbol{A} \boldsymbol{c}.$                                                                                             |
| 0902.0431_E00926 | 153  | ■ 1.000 | $D(\boldsymbol{a} \wedge \boldsymbol{b} \wedge \boldsymbol{c})=D \boldsymbol{a} \wedge \boldsymbol{b} \wedge \boldsymbol{c}+\boldsymbol{a} \wedge D \boldsymbol{b} \wedge \boldsymbol{c}+\boldsymbol{a} \wedge \boldsymbol{b} \wedge D \boldsymbol{c}.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | $D(\boldsymbol{a} \wedge \boldsymbol{b} \wedge \boldsymbol{c})=D \boldsymbol{a} \wedge \boldsymbol{b} \wedge \boldsymbol{c}+\boldsymbol{a} \wedge D \boldsymbol{b} \wedge \boldsymbol{c}+\boldsymbol{a} \wedge \boldsymbol{b} \wedge D \boldsymbol{c}.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | $D(\boldsymbol{a} \wedge \boldsymbol{b} \wedge \boldsymbol{c})=D \boldsymbol{a} \wedge \boldsymbol{b} \wedge \boldsymbol{c}+\boldsymbol{a} \wedge D \boldsymbol{b} \wedge \boldsymbol{c}+\boldsymbol{a} \wedge \boldsymbol{b} \wedge D \boldsymbol{c}.$ |
| 0902.0431_E00929 | 154  | ■ 1.000 | $\varphi_{\boldsymbol{C}}\left(\left(\begin{array}{cc} B & L \\ -L^* & C \end{array}\right)+\frac{\nu}{3}\left(\begin{array}{cc} E & 0 \\ 0 & -E \end{array}\right)\right)=\Phi\left(\phi_{\boldsymbol{C}}(B, C), h(L),-\tau h(L),-i \nu e_1\right)$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | $\varphi_{\boldsymbol{C}}\left(\left(\begin{array}{cc} B & L \\ -L^* & C \end{array}\right)+\frac{\nu}{3}\left(\begin{array}{cc} E & 0 \\ 0 & -E \end{array}\right)\right)=\Phi\left(\phi_{\boldsymbol{C}}(B, C), h(L),-\tau h(L),-i \nu e_1\right)$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | $\varphi_{\boldsymbol{C}}\left(\left(\begin{array}{cc} B & L \\ -L^* & C \end{array}\right)+\frac{\nu}{3}\left(\begin{array}{cc} E & 0 \\ 0 & -E \end{array}\right)\right)=\Phi\left(\phi_{\boldsymbol{C}}(B, C), h(L),-\tau h(L),-i \nu e_1\right)$    |
| 0902.0431_E00930 | 154  | ■ 1.000 | $\nu=\operatorname{tr}\left(B^{\{\prime\}}\right)=-\operatorname{tr}\left(C^{\{\prime\}}\right), \quad B=B^{\{\prime\}}-\frac{\nu}{3} E, \quad C=C^{\{\prime\}}+\frac{\nu}{3} E,$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | $\nu=\operatorname{tr}\left(B^{\prime}\right)=-\operatorname{tr}\left(C^{\prime}\right), \quad B=B^{\prime}-\frac{\nu}{3} E, \quad C=C^{\prime}+\frac{\nu}{3} E,$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | $\nu=\operatorname{tr}\left(B^{\prime}\right)=-\operatorname{tr}\left(C^{\prime}\right), \quad B=B^{\prime}-\frac{\nu}{3} E, \quad C=C^{\prime}+\frac{\nu}{3} E,$                                                                                       |
| 0902.0431_E00934 | 154  | ■ 1.000 | $\psi_{*}(D) P=f^{-1}(D(f P)), \quad P \in\left(\mathfrak{P}_C\right)^C$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | $\psi_{*}(D) P=f^{-1}(D(f P)), \quad P \in\left(\mathfrak{P}_C\right)^C$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | $\psi_{*}(D) P=f^{-1}(D(f P)), \quad P \in\left(\mathfrak{P}_C\right)^C$                                                                                                                                                                                |
| 0902.0431_E00935 | 154  | ■ 1.000 | $D=\left(\begin{array}{cc} B & L \\ -L^* & C \end{array}\right)+\frac{\nu}{3}\left(\begin{array}{cc} E & 0 \\ 0 & -E \end{array}\right)$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | $D=\left(\begin{array}{cc} B & L \\ -L^* & C \end{array}\right)+\frac{\nu}{3}\left(\begin{array}{cc} E & 0 \\ 0 & -E \end{array}\right)$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | $D=\left(\begin{array}{cc} B & L \\ -L^* & C \end{array}\right)+\frac{\nu}{3}\left(\begin{array}{cc} E & 0 \\ 0 & -E \end{array}\right)$                                                                                                                |

| Identifier       | Page | Conf.   | LaTeX source                                                                                                                                                                                                                                                                      | Rendered                                                                                                                                                                                                                                                                             | Image                                                                                                                                                                                                                                                                                |
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| 0902.0431_E00942 | 157  | ■ 1.000 | $=\left(\begin{array}{c} \boldsymbol{\phi}(B, C) \\ F_1(1)+\frac{i}{3}\nu e_1 F_1(1) \\ 2h(L)\times F_1(1) \\ 0 \\ -(h(L), F_1(1)) \end{array}\right)=\Phi(\boldsymbol{\phi}(B, C), h(L), -\tau h(L), -i\nu e_1)\left(\begin{array}{c} F_1(1) \\ 0 \\ 0 \\ 0 \end{array}\right).$ | $=\left(\begin{array}{c} \phi_{\boldsymbol{C}}(B, C)F_1(1)+\frac{i\nu e_1}{3}F_1(1) \\ 2h(L)\times F_1(1) \\ 0 \\ -(h(L), F_1(1)) \end{array}\right)=\Phi(\phi_{\boldsymbol{C}}(B, C), h(L), -\tau h(L), -i\nu e_1)\left(\begin{array}{c} F_1(1) \\ 0 \\ 0 \\ 0 \end{array}\right).$ | $=\left(\begin{array}{c} \phi_{\boldsymbol{C}}(B, C)F_1(1)+\frac{i\nu e_1}{3}F_1(1) \\ 2h(L)\times F_1(1) \\ 0 \\ -(h(L), F_1(1)) \end{array}\right)=\Phi(\phi_{\boldsymbol{C}}(B, C), h(L), -\tau h(L), -i\nu e_1)\left(\begin{array}{c} F_1(1) \\ 0 \\ 0 \\ 0 \end{array}\right).$ |
| 0902.0431_E00943 | 157  | ■ 1.000 | $f^{-1}(D(fP))=\Phi(\boldsymbol{\phi}(B, C), h(L), -\tau h(L), -i\nu e_1)P.$                                                                                                                                                                                                      | $f^{-1}(D(fP))=\Phi(\phi_{\boldsymbol{C}}(B, C), h(L), -\tau h(L), -i\nu e_1)P.$                                                                                                                                                                                                     | $f^{-1}(D(fP))=\Phi(\phi_{\boldsymbol{C}}(B, C), h(L), -\tau h(L), -i\nu e_1)P.$                                                                                                                                                                                                     |
| 0902.0431_E00950 | 158  | ■ 1.000 | $\mu^{-1}(M_1+iM_2, N_1+iN_2)=\left(\begin{array}{cc} -N_2-N_1e_1 & M_2+M_1e_1 \\ M_2-M_1e_1 & N_2-N_1e_1 \end{array}\right), \quad M_i, N_i \in M(3, \boldsymbol{C}).$                                                                                                           | $\mu^{-1}(M_1+iM_2, N_1+iN_2)=\left(\begin{array}{cc} -N_2-N_1e_1 & M_2+M_1e_1 \\ M_2-M_1e_1 & N_2-N_1e_1 \end{array}\right), \quad M_i, N_i \in M(3, \boldsymbol{C}).$                                                                                                              | $\mu^{-1}(M_1+iM_2, N_1+iN_2)=\left(\begin{array}{cc} -N_2-N_1e_1 & M_2+M_1e_1 \\ M_2-M_1e_1 & N_2-N_1e_1 \end{array}\right), \quad M_i, N_i \in M(3, \boldsymbol{C}).$                                                                                                              |
| 0902.0431_E00951 | 158  | ■ 1.000 | $\mu(\widetilde{MD}^*)=\psi_*(D)(\mu\widetilde{M}).$                                                                                                                                                                                                                              | $\mu(\widetilde{MD}^*)=\psi_*(D)(\mu\widetilde{M}).$                                                                                                                                                                                                                                 | $\mu(\widetilde{MD}^*)=\psi_*(D)(\mu\widetilde{M}).$                                                                                                                                                                                                                                 |
| 0902.0431_E00954 | 159  | ■ 1.000 | $=\left(\begin{array}{cc} -N_2-N_1e_1 & M_2+M_1e_1 \\ M_2+M_1e_1 & N_2+N_1e_1 \end{array}\right)\left(\left(\begin{array}{cc} -B & -L \\ L^* & -C \end{array}\right)-\frac{\nu}{3}\left(\begin{array}{cc} E & 0 \\ 0 & -E \end{array}\right)\right)=\widetilde{MD}^*.$            | $=\left(\begin{array}{cc} -N_2-N_1e_1 & M_2+M_1e_1 \\ M_2+M_1e_1 & N_2+N_1e_1 \end{array}\right)\left(\left(\begin{array}{cc} -B & -L \\ L^* & -C \end{array}\right)-\frac{\nu}{3}\left(\begin{array}{cc} E & 0 \\ 0 & -E \end{array}\right)\right)=\widetilde{MD}^*.$               | $=\left(\begin{array}{cc} -N_2-N_1e_1 & M_2+M_1e_1 \\ M_2+M_1e_1 & N_2+N_1e_1 \end{array}\right)\left(\left(\begin{array}{cc} -B & -L \\ L^* & -C \end{array}\right)-\frac{\nu}{3}\left(\begin{array}{cc} E & 0 \\ 0 & -E \end{array}\right)\right)=\widetilde{MD}^*.$               |
| 0902.0431_E00956 | 159  | ■ 1.000 | $(Q, A)\left(\sum(\boldsymbol{a}\wedge\boldsymbol{b}\wedge\boldsymbol{c})+\widetilde{M}\right)=\sum(A\boldsymbol{a}\wedge A\boldsymbol{b}\wedge A\boldsymbol{c})+Q\widetilde{M}A^*,$                                                                                              | $(Q, A)\left(\sum(\boldsymbol{a}\wedge\boldsymbol{b}\wedge\boldsymbol{c})+\widetilde{M}\right)=\sum(A\boldsymbol{a}\wedge A\boldsymbol{b}\wedge A\boldsymbol{c})+Q\widetilde{M}A^*,$                                                                                                 | $(Q, A)(\sum(\boldsymbol{a}\wedge\boldsymbol{b}\wedge\boldsymbol{c})+\widetilde{M})=\sum(A\boldsymbol{a}\wedge A\boldsymbol{b}\wedge A\boldsymbol{c})+Q\widetilde{M}A^*,$                                                                                                            |
| 0902.0431_E00958 | 159  | ■ 1.000 | $\Phi(\boldsymbol{\phi}(B, C), h(L), -\tau h(L), -i\nu e_1)\in\mathfrak{e}_7.$                                                                                                                                                                                                    | $\Phi(\phi_{\boldsymbol{C}}(B, C), h(L), -\tau h(L), -i\nu e_1)\in\mathfrak{e}_7.$                                                                                                                                                                                                   | $\Phi(\phi_{\boldsymbol{C}}(B, C), h(L), -\tau h(L), -i\nu e_1)\in\mathfrak{e}_7.$                                                                                                                                                                                                   |
| 0902.0431_E00961 | 160  | ■ 1.000 | $\alpha=\psi(E, A)\beta=\psi(E, A)\psi(Q, E)=\psi(Q, A).$                                                                                                                                                                                                                         | $\alpha=\psi(E, A)\beta=\psi(E, A)\psi(Q, E)=\psi(Q, A).$                                                                                                                                                                                                                            | $\alpha=\psi(E, A)\beta=\psi(E, A)\psi(Q, E)=\psi(Q, A).$                                                                                                                                                                                                                            |
| 0902.0431_E00966 | 161  | ■ 1.000 | $\mathfrak{P}=\mathfrak{J}(3, \mathfrak{C})\oplus\mathfrak{J}(3, \mathfrak{C})\oplus\boldsymbol{R}\oplus\boldsymbol{R},$ $\mathfrak{P}'=\mathfrak{J}(3, \mathfrak{C}')\oplus\mathfrak{J}(3, \mathfrak{C}')\oplus\boldsymbol{R}\oplus\boldsymbol{R}.$                              | $\mathfrak{P}=\mathfrak{J}(3, \mathfrak{C})\oplus\mathfrak{J}(3, \mathfrak{C})\oplus\boldsymbol{R}\oplus\boldsymbol{R},$ $\mathfrak{P}'=\mathfrak{J}(3, \mathfrak{C}')\oplus\mathfrak{J}(3, \mathfrak{C}')\oplus\boldsymbol{R}\oplus\boldsymbol{R}.$                                 | $\mathfrak{P}=\mathfrak{J}(3, \mathfrak{C})\oplus\mathfrak{J}(3, \mathfrak{C})\oplus\boldsymbol{R}\oplus\boldsymbol{R},$ $\mathfrak{P}'=\mathfrak{J}(3, \mathfrak{C}')\oplus\mathfrak{J}(3, \mathfrak{C}')\oplus\boldsymbol{R}\oplus\boldsymbol{R}.$                                 |
| 0902.0431_E00967 | 161  | ■ 1.000 | $\langle P, Q\rangle_\sigma=\langle\sigma P, Q\rangle.$                                                                                                                                                                                                                           | $\langle P, Q\rangle_\sigma=\langle\sigma P, Q\rangle.$                                                                                                                                                                                                                              | $\langle P, Q\rangle_\sigma=\langle\sigma P, Q\rangle.$                                                                                                                                                                                                                              |
| 0902.0431_E00970 | 161  | ■ 1.000 | $E_{7(7)}\simeq SU(8)/\boldsymbol{Z}_2\times\boldsymbol{R}^{70},$ $E_{7(-5)}\simeq (SU(2)\times Spin(12))/\boldsymbol{Z}_2\times\boldsymbol{R}^{64},$ $E_{7(-25)}\simeq (U(1)\times E_6)/\boldsymbol{Z}_3\times\boldsymbol{R}^{54}.$                                              | $E_{7(7)}\simeq SU(8)/\boldsymbol{Z}_2\times\boldsymbol{R}^{70},$ $E_{7(-5)}\simeq (SU(2)\times Spin(12))/\boldsymbol{Z}_2\times\boldsymbol{R}^{64},$ $E_{7(-25)}\simeq (U(1)\times E_6)/\boldsymbol{Z}_3\times\boldsymbol{R}^{54}.$                                                 | $E_{7(7)}\simeq SU(8)/\boldsymbol{Z}_2\times\boldsymbol{R}^{70},$ $E_{7(-5)}\simeq (SU(2)\times Spin(12))/\boldsymbol{Z}_2\times\boldsymbol{R}^{64},$ $E_{7(-25)}\simeq (U(1)\times E_6)/\boldsymbol{Z}_3\times\boldsymbol{R}^{54}.$                                                 |

| Identifier       | Page | Conf.   | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                                        | Rendered                                                                                                                                                                                                                                                                                                                                                                                                                                      | Image                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|------------------|------|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0902.0431_E09971 | 161  | ■ 1.000 | $z\left(E_{7(7)}\right)=\boldsymbol{Z}_{\{2\}}, \quad z\left(E_{7(-5)}\right)=\boldsymbol{Z}_{\{2\}}, \quad z\left(E_{7(-25)}\right)=\boldsymbol{Z}_{\{2\}}.$                                                                                                                                                                                                                                                                                       | $z\left(E_{7(7)}\right)=\boldsymbol{Z}_2, \quad z\left(E_{7(-5)}\right)=\boldsymbol{Z}_2, \quad z\left(E_{7(-25)}\right)=\boldsymbol{Z}_2.$                                                                                                                                                                                                                                                                                                   | $z(E_{7(7)})=\boldsymbol{Z}_2, \quad z(E_{7(-5)})=\boldsymbol{Z}_2, \quad z(E_{7(-25)})=\boldsymbol{Z}_2.$                                                                                                                                                                                                                                                                                                                                    |
| 0902.0431_E09972 | 162  | ■ 1.000 | $\mathfrak{e}_{\{8\}}^{\{C\}}=\mathfrak{e}_{\{7\}}^{\{C\}} \oplus \mathfrak{e}_{\{P\}}^{\{C\}} \oplus \mathfrak{e}_{\{C\}} \oplus \mathfrak{e}_{\{C\}} \oplus \mathfrak{e}_{\{C\}},$                                                                                                                                                                                                                                                                | $\mathfrak{e}_8^C = \mathfrak{e}_7^C \oplus \mathfrak{P}^C \oplus \mathfrak{P}^C \oplus C \oplus C \oplus C,$                                                                                                                                                                                                                                                                                                                                 | $\mathfrak{e}_8^C = \mathfrak{e}_7^C \oplus \mathfrak{P}^C \oplus \mathfrak{P}^C \oplus C \oplus C \oplus C,$                                                                                                                                                                                                                                                                                                                                 |
| 0902.0431_E09975 | 162  | ■ 1.000 | $\begin{aligned} & \left[R_1, R_2+R_3\right]=\left[R_1, R_2\right]+\left[R_1, R_3\right], \\ & \left[k R_1, R_2\right]=k\left[R_1, R_2\right], \quad k \in C, \\ & \left[R_1, R_2\right]=-\left[R_2, R_1\right] \end{aligned}$                                                                                                                                                                                                                      | $\begin{aligned} \left[R_1, R_2+R_3\right] &= \left[R_1, R_2\right] + \left[R_1, R_3\right], \\ \left[k R_1, R_2\right] &= k \left[R_1, R_2\right], \quad k \in C, \\ \left[R_1, R_2\right] &= -\left[R_2, R_1\right] \end{aligned}$                                                                                                                                                                                                          | $\begin{aligned} \left[R_1, R_2+R_3\right] &= \left[R_1, R_2\right] + \left[R_1, R_3\right], \\ \left[k R_1, R_2\right] &= k \left[R_1, R_2\right], \quad k \in C, \\ \left[R_1, R_2\right] &= -\left[R_2, R_1\right] \end{aligned}$                                                                                                                                                                                                          |
| 0902.0431_E09978 | 163  | ■ 1.000 | $\mathfrak{e}_{\{8\}}^{\{C\}}=\mathfrak{e}_{\{7\}}^{\{C\}} \oplus \mathfrak{e}_{\{K\}}^{\{C\}},$                                                                                                                                                                                                                                                                                                                                                    | $\mathfrak{e}_8^C = \mathfrak{e}_7^C \oplus \mathfrak{K}^C,$                                                                                                                                                                                                                                                                                                                                                                                  | $\mathfrak{e}_8^C = \mathfrak{e}_7^C \oplus \mathfrak{K}^C,$                                                                                                                                                                                                                                                                                                                                                                                  |
| 0902.0431_E09980 | 163  | ■ 1.000 | $\begin{aligned} & \mathfrak{a} \ni \left[\Phi(0,0,0,1), (0,0,1,0)^-\right] = (0,0,1,0)^-, \\ & \mathfrak{a} \ni \left[\Phi(0,0,0,1), (0,0,0,-1)_-\right] = (0,0,0,1)_-, \\ & \mathfrak{a} \ni \left[(0,0,1,0)^-, (0,0,0,4)^-\right] = 1^-, \\ & \mathfrak{a} \ni \left[(0,0,0,1)_-, (0,0,4,0)_-\right] = 1_-, \\ & \mathfrak{a} \ni \left[1^-, 1_-\right] = 1, \\ & \mathfrak{a} \ni \left[1^- + 1_-, Q^- + P_-\right] = P^- + Q_-, \end{aligned}$ | $\begin{aligned} \mathfrak{a} \ni \left[\Phi(0,0,0,1), (0,0,1,0)^-\right] &= (0,0,1,0)^-, \\ \mathfrak{a} \ni \left[\Phi(0,0,0,1), (0,0,0,-1)_-\right] &= (0,0,0,1)_-, \\ \mathfrak{a} \ni \left[(0,0,1,0)^-, (0,0,0,4)^-\right] &= 1^-, \\ \mathfrak{a} \ni \left[(0,0,0,1)_-, (0,0,4,0)_-\right] &= 1_-, \\ \mathfrak{a} \ni \left[1^-, 1_-\right] &= 1, \\ \mathfrak{a} \ni \left[1^- + 1_-, Q^- + P_-\right] &= P^- + Q_-, \end{aligned}$ | $\begin{aligned} \mathfrak{a} \ni \left[\Phi(0,0,0,1), (0,0,1,0)^-\right] &= (0,0,1,0)^-, \\ \mathfrak{a} \ni \left[\Phi(0,0,0,1), (0,0,0,-1)_-\right] &= (0,0,0,1)_-, \\ \mathfrak{a} \ni \left[(0,0,1,0)^-, (0,0,0,4)^-\right] &= 1^-, \\ \mathfrak{a} \ni \left[(0,0,0,1)_-, (0,0,4,0)_-\right] &= 1_-, \\ \mathfrak{a} \ni \left[1^-, 1_-\right] &= 1, \\ \mathfrak{a} \ni \left[1^- + 1_-, Q^- + P_-\right] &= P^- + Q_-, \end{aligned}$ |
| 0902.0431_E09984 | 164  | ■ 1.000 | $\mathfrak{a} \ni \left[1_-, R\right] = (0,0,0,-s,0,0).$                                                                                                                                                                                                                                                                                                                                                                                            | $\mathfrak{a} \ni \left[1_-, R\right] = (0,0,0,-s,0,0).$                                                                                                                                                                                                                                                                                                                                                                                      | $\mathfrak{a} \ni \left[1_-, R\right] = (0,0,0,-s,0,0).$                                                                                                                                                                                                                                                                                                                                                                                      |
| 0902.0431_E09990 | 165  | ■ 1.000 | $\begin{aligned} & B_8\left(R_1, R_2\right) \\ & =-15\left(R_1, R_2\right)_8 \\ & =-15\left(\Phi_1, \Phi_2\right)_7+15\left\{Q_1, P_2\right\}-15\left\{P_1, Q_2\right\}+120 r_1 r_2+60 t_1 s_2+60 s_1 t_2 \\ & =\frac{5}{3} B_7\left(\Phi_1, \Phi_2\right)+15\left\{Q_1, P_2\right\}-15\left\{P_1, Q_2\right\}+120 r_1 r_2+60 t_1 s_2+60 s_1 t_2 \end{aligned}$                                                                                     | $\begin{aligned} B_8\left(R_1, R_2\right) &= -15\left(R_1, R_2\right)_8 \\ &= -15\left(\Phi_1, \Phi_2\right)_7+15\left\{Q_1, P_2\right\}-15\left\{P_1, Q_2\right\}+120 r_1 r_2+60 t_1 s_2+60 s_1 t_2 \\ &= \frac{5}{3} B_7\left(\Phi_1, \Phi_2\right)+15\left\{Q_1, P_2\right\}-15\left\{P_1, Q_2\right\}+120 r_1 r_2+60 t_1 s_2+60 s_1 t_2 \end{aligned}$                                                                                    | $\begin{aligned} B_8\left(R_1, R_2\right) &= -15\left(R_1, R_2\right)_8 \\ &= -15\left(\Phi_1, \Phi_2\right)_7+15\left\{Q_1, P_2\right\}-15\left\{P_1, Q_2\right\}+120 r_1 r_2+60 t_1 s_2+60 s_1 t_2 \\ &= \frac{5}{3} B_7\left(\Phi_1, \Phi_2\right)+15\left\{Q_1, P_2\right\}-15\left\{P_1, Q_2\right\}+120 r_1 r_2+60 t_1 s_2+60 s_1 t_2, \end{aligned}$                                                                                   |
| 0902.0431_E09991 | 165  | ■ 1.000 | $B_{\{8\}}\left(R_1, R_2\right)=k\left(R_1, R_2\right)_8, \quad R_i \in \mathfrak{e}_8^C.$                                                                                                                                                                                                                                                                                                                                                          | $B_8\left(R_1, R_2\right)=k\left(R_1, R_2\right)_8, \quad R_i \in \mathfrak{e}_8^C.$                                                                                                                                                                                                                                                                                                                                                          | $B_8\left(R_1, R_2\right)=k\left(R_1, R_2\right)_8, \quad R_i \in \mathfrak{e}_8^C.$                                                                                                                                                                                                                                                                                                                                                          |
| 0902.0431_E09992 | 165  | ■ 1.000 | $(1,1)_{\{8\}}=-8.$                                                                                                                                                                                                                                                                                                                                                                                                                                 | $(1,1)_8=-8.$                                                                                                                                                                                                                                                                                                                                                                                                                                 | $(1,1)_8=-8.$                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 0902.0431_E09993 | 165  | ■ 1.000 | $\left[1,\left[1,\left(\Phi, P, Q, r, s, t\right)\right]\right]=\left[1,\left(0, P,-Q, 0,2 s,-2 t\right)\right]=\left(0, P, Q, 0,4 s, 4 t\right),$                                                                                                                                                                                                                                                                                                  | $\left[1,\left[1,\left(\Phi, P, Q, r, s, t\right)\right]\right]=\left[1,\left(0, P,-Q, 0,2 s,-2 t\right)\right]=\left(0, P, Q, 0,4 s, 4 t\right),$                                                                                                                                                                                                                                                                                            | $\left[1,\left[1,\left(\Phi, P, Q, r, s, t\right)\right]\right]=\left[1,\left(0, P,-Q, 0,2 s,-2 t\right)\right]=\left(0, P, Q, 0,4 s, 4 t\right),$                                                                                                                                                                                                                                                                                            |
| 0902.0431_E01002 | 167  | ■ 1.000 | $\tau(\Phi, P, Q, r, s, t)=\left(\tau \Phi \tau, \tau P, \tau Q, \tau r, \tau s, \tau t\right),$                                                                                                                                                                                                                                                                                                                                                    | $\tau(\Phi, P, Q, r, s, t)=\left(\tau \Phi \tau, \tau P, \tau Q, \tau r, \tau s, \tau t\right),$                                                                                                                                                                                                                                                                                                                                              | $\tau(\Phi, P, Q, r, s, t)=\left(\tau \Phi \tau, \tau P, \tau Q, \tau r, \tau s, \tau t\right),$                                                                                                                                                                                                                                                                                                                                              |

| Identifier       | Page | Conf.   | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                     | Rendered                                                                                                                                                                                                                                                                                                                                                                                         | Image                                                                                                                                                                                                                                                                                                                                                                                            |
|------------------|------|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0902.0431_E01005 | 167  | ■ 1.000 | $\left(\tau\lambda\lambda^{-1}\tau,\Phi_2\right)_7=2\left(\tau^t\phi_1\tau,\phi_2\right)_6+4\left\langle A_1,A_2\right\rangle +4\left\langle B_1,B_2\right\rangle +\frac{8}{3}\left(\tau\nu_1\right)\nu_2.$                                                                                                                                                                                      | $(\tau\lambda\Phi_1\lambda^{-1}\tau,\Phi_2)_7=2\left(\tau^t\phi_1\tau,\phi_2\right)_6+4\left\langle A_1,A_2\right\rangle +4\left\langle B_1,B_2\right\rangle +\frac{8}{3}\left(\tau\nu_1\right)\nu_2.$                                                                                                                                                                                           | $(\tau\lambda\Phi_1\lambda^{-1}\tau,\Phi_2)_7=2(\tau^t\phi_1\tau,\phi_2)_6+4\langle A_1,A_2\rangle +4\langle B_1,B_2\rangle +\frac{8}{3}(\tau\nu_1)\nu_2.$                                                                                                                                                                                                                                       |
| 0902.0431_E01006 | 167  | ■ 1.000 | $\left(\tau\lambda\lambda^{-1}\tau,\Phi_2\right)_7=2\left(\tau^t\phi_1\tau,\phi_2\right)_6+4\left\langle A_1,A_2\right\rangle +4\left\langle B_1,B_2\right\rangle +\frac{8}{3}\left(\tau\nu_1\right)\nu_2.$                                                                                                                                                                                      | $(\tau^t\phi_1\tau,\phi_2)_6=-\left(\tau\delta_1\tau,\delta_2\right)_4+\left\langle T_1,T_2\right\rangle .$                                                                                                                                                                                                                                                                                      | $(\tau^t\phi_1\tau,\phi_2)_6=-\left(\tau\delta_1\tau,\delta_2\right)_4+\left\langle T_1,T_2\right\rangle .$                                                                                                                                                                                                                                                                                      |
| 0902.0431_E01011 | 168  | ■ 1.000 | $\tau\widetilde{\lambda}R=\left(\tau\lambda\Phi\lambda^{-1}\tau,\tau\lambda Q,-\tau\lambda P,-\tau r,-\tau t,-\tau s\right).$                                                                                                                                                                                                                                                                    | $\tau\widetilde{\lambda}R=\left(\tau\lambda\Phi\lambda^{-1}\tau,\tau\lambda Q,-\tau\lambda P,-\tau r,-\tau t,-\tau s\right).$                                                                                                                                                                                                                                                                    | $\tau\widetilde{\lambda}R=\left(\tau\lambda\Phi\lambda^{-1}\tau,\tau\lambda Q,-\tau\lambda P,-\tau r,-\tau t,-\tau s\right).$                                                                                                                                                                                                                                                                    |
| 0902.0431_E01012 | 168  | ■ 1.000 | $R=\frac{R-R^*}{2}+i\frac{R+R^*}{2i},\quad\frac{R-R^*}{2},\frac{R+R^*}{2i}\in\mathfrak{e}_8.$                                                                                                                                                                                                                                                                                                    | $R=\frac{R-R^*}{2}+i\frac{R+R^*}{2i},\quad\frac{R-R^*}{2},\frac{R+R^*}{2i}\in\mathfrak{e}_8.$                                                                                                                                                                                                                                                                                                    | $R=\frac{R-R^*}{2}+i\frac{R+R^*}{2i},\quad\frac{R-R^*}{2},\frac{R+R^*}{2i}\in\mathfrak{e}_8.$                                                                                                                                                                                                                                                                                                    |
| 0902.0431_E01017 | 170  | ■ 1.000 | $\mathfrak{h}=\left\{\left(\Phi(h),0,0,r,0,0\right)\mid\Phi(h)\in\mathfrak{h}_7,r\in C\right\}$                                                                                                                                                                                                                                                                                                  | $\mathfrak{h}=\left\{\left(\Phi(h),0,0,r,0,0\right)\mid\Phi(h)\in\mathfrak{h}_7,r\in C\right\}$                                                                                                                                                                                                                                                                                                  | $\mathfrak{h}=\left\{\left(\Phi(h),0,0,r,0,0\right)\mid\Phi(h)\in\mathfrak{h}_7,r\in C\right\}$                                                                                                                                                                                                                                                                                                  |
| 0902.0431_E01018 | 170  | ■ 1.000 | $\left[\left(\Phi(h),0,0,r,0,0\right),\left(\Phi,0,0,0,0,0\right)\right]=\left[\left(\Phi(h),\Phi\right],0,0,0,0,0\right).$                                                                                                                                                                                                                                                                      | $\left[\left(\Phi(h),0,0,r,0,0\right),\left(\Phi,0,0,0,0,0\right)\right]=\left[\left(\Phi(h),\Phi\right],0,0,0,0,0\right).$                                                                                                                                                                                                                                                                      | $\left[\left(\Phi(h),0,0,r,0,0\right),\left(\Phi,0,0,0,0,0\right)\right]=\left[\left(\Phi(h),\Phi\right],0,0,0,0,0\right).$                                                                                                                                                                                                                                                                      |
| 0902.0431_E01020 | 170  | ■ 1.000 | $\begin{aligned} & \left(\Phi\left(h_\delta+\widetilde{H},0,0,\nu\right)+r1\right)(X,Y,\xi,\eta) \\ & =\left(\left(h_\delta+\widetilde{H}-\frac{1}{3}\nu+r\right)X,\left(h_\delta-\widetilde{H}+\frac{1}{3}\nu+r\right)Y,(\nu+r)\xi,(-\nu+r)\eta\right). \end{aligned}$                                                                                                                          | $\begin{aligned} & \left(\Phi\left(h_\delta+\widetilde{H},0,0,\nu\right)+r1\right)(X,Y,\xi,\eta) \\ & =\left(\left(h_\delta+\widetilde{H}-\frac{1}{3}\nu+r\right)X,\left(h_\delta-\widetilde{H}+\frac{1}{3}\nu+r\right)Y,(\nu+r)\xi,(-\nu+r)\eta\right). \end{aligned}$                                                                                                                          | $\begin{aligned} & (\Phi(h_\delta+\widetilde{H},0,0,\nu)+r1)(X,Y,\xi,\eta) \\ & =\left(\left(h_\delta+\widetilde{H}-\frac{1}{3}\nu+r\right)X,\left(h_\delta-\widetilde{H}+\frac{1}{3}\nu+r\right)Y,(\nu+r)\xi,(-\nu+r)\eta\right). \end{aligned}$                                                                                                                                                |
| 0902.0431_E01024 | 171  | ■ 1.000 | $\begin{aligned} & \alpha_1=\frac{1}{2}\left(\lambda_0-\lambda_1-\lambda_2-\lambda_3\right)+\frac{1}{2}\left(\mu_3-\mu_1\right), \\ & \alpha_2=\mu_1-\frac{1}{3}\nu-r,\quad\alpha_3=2r, \\ & \alpha_4=\mu_2-\frac{1}{3}\nu-r,\quad\alpha_5=\lambda_3-\frac{1}{2}\left(\mu_2-\mu_3\right), \\ & \alpha_6=\lambda_2-\lambda_3,\quad\alpha_7=\lambda_1-\lambda_2,\quad\alpha_8=\nu-r \end{aligned}$ | $\begin{aligned} & \alpha_1=\frac{1}{2}\left(\lambda_0-\lambda_1-\lambda_2-\lambda_3\right)+\frac{1}{2}\left(\mu_3-\mu_1\right), \\ & \alpha_2=\mu_1-\frac{1}{3}\nu-r,\quad\alpha_3=2r, \\ & \alpha_4=\mu_2-\frac{1}{3}\nu-r,\quad\alpha_5=\lambda_3-\frac{1}{2}\left(\mu_2-\mu_3\right), \\ & \alpha_6=\lambda_2-\lambda_3,\quad\alpha_7=\lambda_1-\lambda_2,\quad\alpha_8=\nu-r \end{aligned}$ | $\begin{aligned} & \alpha_1=\frac{1}{2}\left(\lambda_0-\lambda_1-\lambda_2-\lambda_3\right)+\frac{1}{2}\left(\mu_3-\mu_1\right), \\ & \alpha_2=\mu_1-\frac{1}{3}\nu-r,\quad\alpha_3=2r, \\ & \alpha_4=\mu_2-\frac{1}{3}\nu-r,\quad\alpha_5=\lambda_3-\frac{1}{2}\left(\mu_2-\mu_3\right), \\ & \alpha_6=\lambda_2-\lambda_3,\quad\alpha_7=\lambda_1-\lambda_2,\quad\alpha_8=\nu-r \end{aligned}$ |
| 0902.0431_E01025 | 171  | ■ 1.000 | $\mu=2\alpha_1+4\alpha_2+6\alpha_3+5\alpha_4+4\alpha_5+3\alpha_6+2\alpha_7+3\alpha_8$                                                                                                                                                                                                                                                                                                            | $\mu=2\alpha_1+4\alpha_2+6\alpha_3+5\alpha_4+4\alpha_5+3\alpha_6+2\alpha_7+3\alpha_8$                                                                                                                                                                                                                                                                                                            | $\mu=2\alpha_1+4\alpha_2+6\alpha_3+5\alpha_4+4\alpha_5+3\alpha_6+2\alpha_7+3\alpha_8$                                                                                                                                                                                                                                                                                                            |
| 0902.0431_E01035 | 176  | ■ 1.000 | $\mathfrak{h}_{\boldsymbol{R}}=\left\{\left(\Phi(h),0,0,r,0\right)\mid\Phi(h)\in\left(\mathfrak{h}_7\right)_{\boldsymbol{R}},r\in\boldsymbol{R}\right\},$                                                                                                                                                                                                                                        | $\mathfrak{h}_{\boldsymbol{R}}=\left\{\left(\Phi(h),0,0,r,0\right)\mid\Phi(h)\in\left(\mathfrak{h}_7\right)_{\boldsymbol{R}},r\in\boldsymbol{R}\right\},$                                                                                                                                                                                                                                        | $\mathfrak{h}_{\boldsymbol{R}}=\left\{\left(\Phi(h),0,0,r,0\right)\mid\Phi(h)\in\left(\mathfrak{h}_7\right)_{\boldsymbol{R}},r\in\boldsymbol{R}\right\},$                                                                                                                                                                                                                                        |
| 0902.0431_E01040 | 177  | ■ 1.000 | $\left(\alpha_1,\alpha_1\right)=B_7\left(H_{\alpha_1},H_{\alpha_1}\right)=60\frac{1}{120}\frac{1}{120}4+30\frac{1}{120}\frac{1}{120}8=\frac{1}{30},$                                                                                                                                                                                                                                             | $(\alpha_1,\alpha_1)=B_7(H_{\alpha_1},H_{\alpha_1})=60\frac{1}{120}\frac{1}{120}4+30\frac{1}{120}\frac{1}{120}8=\frac{1}{30},$                                                                                                                                                                                                                                                                   | $(\alpha_1,\alpha_1)=B_7(H_{\alpha_1},H_{\alpha_1})=60\frac{1}{120}\frac{1}{120}4+30\frac{1}{120}\frac{1}{120}8=\frac{1}{30},$                                                                                                                                                                                                                                                                   |



| Identifier       | Page | Conf.   | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Rendered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Image                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|------------------|------|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0902.0431_EQ1068 | 181  | ■ 1.000 | $\left(\left(\left(x_{kl}+iy_{kl}\right)\right)\right)=\left(\left(\begin{array}{cc}x_{kl}&y_{kl}\\\hline-y_{kl}&x_{kl}\end{array}\right)\right),\quad x_{kl},y_{kl}\in\boldsymbol{R}.$                                                                                                                                                                                                                                                                                                                                   | $l\left(\left(x_{kl}+iy_{kl}\right)\right)=\left(\left(\begin{array}{cc}x_{kl}&y_{kl}\\\hline-y_{kl}&x_{kl}\end{array}\right)\right),\quad x_{kl},y_{kl}\in\boldsymbol{R}.$                                                                                                                                                                                                                                                                                                                                               | $l\left(\left(x_{kl}+iy_{kl}\right)\right)=\left(\left(\begin{array}{cc}x_{kl}&y_{kl}\\\hline-y_{kl}&x_{kl}\end{array}\right)\right),\quad x_{kl},y_{kl}\in\boldsymbol{R}.$                                                                                                                                                                                                                                                                                                                                               |
| 0902.0431_EQ1069 | 182  | ■ 1.000 | $I=\operatorname{diag}(I,\cdots,I),\quad I=\left(\begin{array}{cc}1&0\\\hline0&-1\end{array}\right),\quad J=\operatorname{diag}(J,\cdots,J),\quad J=\left(\begin{array}{cc}0&1\\\hline-1&0\end{array}\right),$                                                                                                                                                                                                                                                                                                            | $I=\operatorname{diag}(I,\cdots,I),\quad I=\left(\begin{array}{cc}1&0\\\hline0&-1\end{array}\right),\quad J=\operatorname{diag}(J,\cdots,J),\quad J=\left(\begin{array}{cc}0&1\\\hline-1&0\end{array}\right),$                                                                                                                                                                                                                                                                                                            | $I=\operatorname{diag}(I,\cdots,I),\quad I=\left(\begin{array}{cc}1&0\\\hline0&-1\end{array}\right),\quad J=\operatorname{diag}(J,\cdots,J),\quad J=\left(\begin{array}{cc}0&1\\\hline-1&0\end{array}\right),$                                                                                                                                                                                                                                                                                                            |
| 0902.0431_EQ1070 | 182  | ■ 1.000 | $l(\frac{S}{8},C))\,I=\{B\in\mathfrak{so}(16)\mid JB=-BJ\}.$                                                                                                                                                                                                                                                                                                                                                                                                                                                              | $l(\mathfrak{S}(8,C))I=\{B\in\mathfrak{so}(16)\mid JB=-BJ\}.$                                                                                                                                                                                                                                                                                                                                                                                                                                                             | $l(\mathfrak{S}(8,C))I=\{B\in\mathfrak{so}(16)\mid JB=-BJ\}.$                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 0902.0431_EQ1072 | 182  | ■ 1.000 | $l(-D)=-B=\left\{ \right\}^t B=\left\{ \right\}^t l(D)=l\left(\tau^t D\right)$                                                                                                                                                                                                                                                                                                                                                                                                                                            | $l(-D)=-B=\left\{ \right\}^t B=\left\{ \right\}^t l(D)=l\left(\tau^t D\right)$                                                                                                                                                                                                                                                                                                                                                                                                                                            | $l(-D)=-B=\left\{ \right\}^t B=\left\{ \right\}^t l(D)=l\left(\tau^t D\right)$                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 0902.0431_EQ1074 | 182  | ■ 1.000 | $l(-S)\,I=-B=\left\{ \right\}^t B=\left\{ \right\}^t I^t(l(S))=Il\left(\tau^t S\right)=l\left(^t S\right)I$                                                                                                                                                                                                                                                                                                                                                                                                               | $l(-S)I=-B=\left\{ \right\}^t B=\left\{ \right\}^t I^t(l(S))=Il\left(\tau^t S\right)=l\left(^t S\right)I$                                                                                                                                                                                                                                                                                                                                                                                                                 | $l(-S)I=-B=\left\{ \right\}^t B=\left\{ \right\}^t I^t(l(S))=Il\left(\tau^t S\right)=l\left(^t S\right)I$                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 0902.0431_EQ1075 | 182  | ■ 1.000 | $\varphi_*([gX_1,gX_2])=2(X_1\vee\gamma X_2-X_2\vee\gamma X_1),\quad X_1,X_2\in\mathfrak{J}^C.$                                                                                                                                                                                                                                                                                                                                                                                                                           | $\varphi_*([gX_1,gX_2])=2(X_1\vee\gamma X_2-X_2\vee\gamma X_1),\quad X_1,X_2\in\mathfrak{J}^C.$                                                                                                                                                                                                                                                                                                                                                                                                                           | $\varphi_*([gX_1,gX_2])=2(X_1\vee\gamma X_2-X_2\vee\gamma X_1),\quad X_1,X_2\in\mathfrak{J}^C.$                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 0902.0431_EQ1084 | 184  | ■ 1.000 | $\begin{aligned}&\&\left.A=2\,\gamma\,X_1\,\times\,Y_2-2\,\gamma\,X_2\,\times\,Y_1+\eta_1\,X_2-\eta_2\,X_1-\xi_1\,\gamma\,X_2+\xi_2\,\gamma\,Y_1\in\mathfrak{J}^C\right)\\&=kD+ik(gA)+\frac{i}{2}\{P_1,P_2\}E.\end{aligned}$                                                                                                                                                                                                                                                                                              | $\begin{aligned}A&=2\gamma X_1\times Y_2-2\gamma X_2\times Y_1+\eta_1X_2-\eta_2X_1-\xi_1\gamma X_2+\xi_2\gamma Y_1\in\mathfrak{J}^C)\\&=kD+ik(gA)+\frac{i}{2}\{P_1,P_2\}E.\end{aligned}$                                                                                                                                                                                                                                                                                                                                  | $\begin{aligned}A&=2\gamma X_1\times Y_2-2\gamma X_2\times Y_1+\eta_1X_2-\eta_2X_1-\xi_1\gamma X_2+\xi_2\gamma Y_1\in\mathfrak{J}^C)\\&=kD+ik(gA)+\frac{i}{2}\{P_1,P_2\}E.\end{aligned}$                                                                                                                                                                                                                                                                                                                                  |
| 0902.0431_EQ1085 | 184  | ■ 1.000 | $\operatorname{tr}\left(S_1\tau S_2-S_2\tau S_1\right)=4i\left\{P_1,P_2\right\}=4i\left\{\chi^{-1}S_1,\chi^{-1}S_2\right\}.$                                                                                                                                                                                                                                                                                                                                                                                              | $\operatorname{tr}\left(S_1\tau S_2-S_2\tau S_1\right)=4i\left\{P_1,P_2\right\}=4i\left\{\chi^{-1}S_1,\chi^{-1}S_2\right\}.$                                                                                                                                                                                                                                                                                                                                                                                              | $\operatorname{tr}\left(S_1\tau S_2-S_2\tau S_1\right)=4i\left\{P_1,P_2\right\}=4i\left\{\chi^{-1}S_1,\chi^{-1}S_2\right\}.$                                                                                                                                                                                                                                                                                                                                                                                              |
| 0902.0431_EQ1086 | 184  | ■ 1.000 | $S_1\tau S_2-S_2\tau S_1-\frac{1}{8}\operatorname{tr}\left(S_1\tau S_2-S_2\tau S_1\right)=kD+ik(gA)$                                                                                                                                                                                                                                                                                                                                                                                                                      | $S_1\tau S_2-S_2\tau S_1-\frac{1}{8}\operatorname{tr}\left(S_1\tau S_2-S_2\tau S_1\right)=kD+ik(gA)$                                                                                                                                                                                                                                                                                                                                                                                                                      | $S_1\tau S_2-S_2\tau S_1-\frac{1}{8}\operatorname{tr}\left(S_1\tau S_2-S_2\tau S_1\right)=kD+ik(gA),$                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 0902.0431_EQ1087 | 184  | ■ 1.000 | $\begin{aligned}&\lambda\gamma P_1\times P_2=\left(\begin{array}{c}\gamma Y_1\\-\gamma X_1\\\eta_1\\-\xi_1\end{array}\right)\times\left(\begin{array}{c}X_2\\Y_2\\\xi_2\\\eta_2\end{array}\right)\\&=\Phi\left(\begin{array}{c}-\frac{1}{2}(\gamma Y_1\vee Y_2-X_2\vee\gamma X_1)\\\frac{1}{4}(2\gamma X_1\times Y_2+\eta_1X_2+\xi_2\gamma Y_1)\\\frac{1}{4}(2\gamma Y_1\times X_2+\xi_1Y_2+\eta_2\gamma X_1)\\\frac{1}{8}((\gamma Y_1,Y_2)-(X_2,\gamma X_1)-3(\eta_1\eta_2-\xi_2\xi_1))\end{array}\right).\end{aligned}$ | $\begin{aligned}&\lambda\gamma P_1\times P_2=\left(\begin{array}{c}\gamma Y_1\\-\gamma X_1\\\eta_1\\-\xi_1\end{array}\right)\times\left(\begin{array}{c}X_2\\Y_2\\\xi_2\\\eta_2\end{array}\right)\\&=\Phi\left(\begin{array}{c}-\frac{1}{2}(\gamma Y_1\vee Y_2-X_2\vee\gamma X_1)\\\frac{1}{4}(2\gamma X_1\times Y_2+\eta_1X_2+\xi_2\gamma Y_1)\\\frac{1}{4}(2\gamma Y_1\times X_2+\xi_1Y_2+\eta_2\gamma X_1)\\\frac{1}{8}((\gamma Y_1,Y_2)-(X_2,\gamma X_1)-3(\eta_1\eta_2-\xi_2\xi_1))\end{array}\right).\end{aligned}$ | $\begin{aligned}\lambda\gamma P_1\times P_2&=\left(\begin{array}{c}\gamma Y_1\\-\gamma X_1\\\eta_1\\-\xi_1\end{array}\right)\times\left(\begin{array}{c}X_2\\Y_2\\\xi_2\\\eta_2\end{array}\right)\\&=\Phi\left(\begin{array}{c}-\frac{1}{2}(\gamma Y_1\vee Y_2-X_2\vee\gamma X_1)\\\frac{1}{4}(2\gamma X_1\times Y_2+\eta_1X_2+\xi_2\gamma Y_1)\\\frac{1}{4}(2\gamma Y_1\times X_2+\xi_1Y_2+\eta_2\gamma X_1)\\\frac{1}{8}((\gamma Y_1,Y_2)-(X_2,\gamma X_1)-3(\eta_1\eta_2-\xi_2\xi_1))\end{array}\right).\end{aligned}$ |

| Identifier       | Page | Conf.   | LaTeX source                                                                                                                                                                                                                                                                                                                                                                    | Rendered                                                                                                                                                                                                                                                                                                                           | Image                                                                                                                                                                                                                                                                                         |
|------------------|------|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0902.0431_E01094 | 186  | ■ 1.000 | =\left[\backslash\Theta\left(\backslash\varphi_{*}\,D,\,0,0,0,0,0\right),\backslash\Theta(0,0,0,0,2\,c,-2\,c)=[\backslash\zeta\,l(D),\backslash\zeta\,l(i\,c\,E)]\,.\backslash\right].                                                                                                                                                                                          | $= [\Theta(\varphi_*D, 0, 0, 0, 0, 0), \Theta(0, 0, 0, 0, 2c, -2c) = [\zeta l(D), \zeta l(icE)].$                                                                                                                                                                                                                                  | $= [\Theta(\varphi_*D, 0, 0, 0, 0, 0), \Theta(0, 0, 0, 0, 2c, -2c) = [\zeta l(D), \zeta l(icE)].$                                                                                                                                                                                             |
| 0902.0431_E01111 | 188  | ■ 1.000 | \sigma(\backslash\Phi,\,P,\,Q,\,r,\,s,\,t)=(\backslash\sigma\,\backslash\Phi\,\backslash\sigma\sigma,\,\backslash\sigma\sigma\,P,\,\backslash\sigma\sigma\,Q,\,r,\,s,\,t)\,.                                                                                                                                                                                                    | $\sigma(\Phi, P, Q, r, s, t) = (\sigma\Phi\sigma, \sigma P, \sigma Q, r, s, t).$                                                                                                                                                                                                                                                   | $\sigma(\Phi, P, Q, r, s, t) = (\sigma\Phi\sigma, \sigma P, \sigma Q, r, s, t).$                                                                                                                                                                                                              |
| 0902.0431_E01112 | 188  | ■ 1.000 | \left(E_{\mathrm{8}}\right)^{\backslash\sigma}=\left\{\alpha\in E_{\mathrm{8}}\,\backslash\mathrm{mid}\,\backslash\sigma\alpha=\alpha\sigma\right\}\,.                                                                                                                                                                                                                          | $(E_8)^\sigma = \{\alpha \in E_8 \mid \sigma\alpha = \alpha\sigma\}.$                                                                                                                                                                                                                                                              | $(E_8)^\sigma = \{\alpha \in E_8 \mid \sigma\alpha = \alpha\sigma\}.$                                                                                                                                                                                                                         |
| 0902.0431_E01114 | 189  | ■ 1.000 | z\left(E_{\mathrm{8}}\right)=\{1\}\,.                                                                                                                                                                                                                                                                                                                                           | $z(E_8) = \{1\}.$                                                                                                                                                                                                                                                                                                                  | $z(E_8) = \{1\}.$                                                                                                                                                                                                                                                                             |
| 0902.0431_E01115 | 189  | ■ 1.000 | \alpha=\varphi(E,\,1)=1\,\mathrm{quad}\,\mathrm{or}\,\,\mathrm{quad}\,\alpha=\varphi(E,\,-1)=v\,.                                                                                                                                                                                                                                                                               | $\alpha = \varphi(E, 1) = 1 \quad \text{or} \quad \alpha = \varphi(E, -1) = v.$                                                                                                                                                                                                                                                    | $\alpha = \varphi(E, 1) = 1 \quad \text{or} \quad \alpha = \varphi(E, -1) = v.$                                                                                                                                                                                                               |
| 0902.0431_E01120 | 190  | ■ 1.000 | \left[\left(D_1,\phi_1,\boldsymbol{X}_1,\boldsymbol{Y}_1\right),\left(D_2,\phi_2,\boldsymbol{X}_2,\boldsymbol{Y}_2\right)\right]=\left(D,\phi,\boldsymbol{X},\boldsymbol{Y}\right),                                                                                                                                                                                             | $[(D_1, \phi_1, \boldsymbol{X}_1, \boldsymbol{Y}_1), (D_2, \phi_2, \boldsymbol{X}_2, \boldsymbol{Y}_2)] = (D, \phi, \boldsymbol{X}, \boldsymbol{Y}),$                                                                                                                                                                              | $[(D_1, \phi_1, \boldsymbol{X}_1, \boldsymbol{Y}_1), (D_2, \phi_2, \boldsymbol{X}_2, \boldsymbol{Y}_2)] = (D, \phi, \boldsymbol{X}, \boldsymbol{Y}),$                                                                                                                                         |
| 0902.0431_E01123 | 190  | ■ 1.000 | B_{\mathrm{8}}\left(R_{\mathrm{1}},R_{\mathrm{2}}\right)=60\,\mathrm{operatorname{tr}}\left(D_{\mathrm{1}}D_{\mathrm{2}}\right)+\frac{5}{2}\,B_{\mathrm{6}}\left(\phi_1,\phi_2\right)+15\left(\boldsymbol{X}_1,\boldsymbol{Y}_2\right)+15\left(\boldsymbol{X}_2,\boldsymbol{Y}_1\right)                                                                                         | $B_8(R_1, R_2) = 60 \operatorname{tr}(D_1 D_2) + \frac{5}{2} B_6(\phi_1, \phi_2) + 15(\boldsymbol{X}_1, \boldsymbol{Y}_2) + 15(\boldsymbol{X}_2, \boldsymbol{Y}_1)$                                                                                                                                                                | $B_8(R_1, R_2) = 60 \operatorname{tr}(D_1 D_2) + \frac{5}{2} B_6(\phi_1, \phi_2) + 15(\boldsymbol{X}_1, \boldsymbol{Y}_2) + 15(\boldsymbol{X}_2, \boldsymbol{Y}_1)$                                                                                                                           |
| 0902.0431_E01125 | 190  | ■ 1.000 | \left\langle R_1,R_2\right\rangle=-B_8\left(R_1,\tau\widetilde{\lambda}R_2\right).                                                                                                                                                                                                                                                                                              | $\langle R_1, R_2 \rangle = -B_8\left(R_1, \tau \widetilde{\lambda} R_2\right).$                                                                                                                                                                                                                                                   | $\langle R_1, R_2 \rangle = -B_8(R_1, \tau \widetilde{\lambda} R_2).$                                                                                                                                                                                                                         |
| 0902.0431_E01127 | 191  | ■ 1.000 | E_{\mathrm{8}}=\left\{\alpha\in\operatorname{Aut}\left(\mathfrak{e}_{\mathrm{8}}^C\right)\,\backslash\mathrm{mid}\left\langle\alpha R_1,\alpha R_2\right\rangle=\left\langle R_1,R_2\right\rangle\right\}                                                                                                                                                                       | $E_8 = \{\alpha \in \operatorname{Aut}(\mathfrak{e}_8^C) \mid \langle \alpha R_1, \alpha R_2 \rangle = \langle R_1, R_2 \rangle\}$                                                                                                                                                                                                 | $E_8 = \{\alpha \in \operatorname{Aut}(\mathfrak{e}_8^C) \mid \langle \alpha R_1, \alpha R_2 \rangle = \langle R_1, R_2 \rangle\}$                                                                                                                                                            |
| 0902.0431_E01129 | 191  | ■ 1.000 | \left(E_{\mathrm{8}}\right)^w=\left\{\alpha\in E_{\mathrm{8}}\,\backslash\mathrm{mid}\,w\alpha=\alpha w\right\}\,.                                                                                                                                                                                                                                                              | $(E_8)^w = \{\alpha \in E_8 \mid w\alpha = \alpha w\}.$                                                                                                                                                                                                                                                                            | $(E_8)^w = \{\alpha \in E_8 \mid w\alpha = \alpha w\}.$                                                                                                                                                                                                                                       |
| 0902.0431_E01130 | 191  | ■ 1.000 | \varphi_1(A)(D,\phi,\boldsymbol{X},\boldsymbol{Y})=\left(ADA^{-1},\phi,A\boldsymbol{X},{}^tA^{-1}\boldsymbol{Y}\right).                                                                                                                                                                                                                                                         | $\varphi_1(A)(D, \phi, \boldsymbol{X}, \boldsymbol{Y}) = (ADA^{-1}, \phi, A\boldsymbol{X}, {}^tA^{-1}\boldsymbol{Y}).$                                                                                                                                                                                                             | $\varphi_1(A)(D, \phi, \boldsymbol{X}, \boldsymbol{Y}) = (ADA^{-1}, \phi, A\boldsymbol{X}, {}^tA^{-1}\boldsymbol{Y}).$                                                                                                                                                                        |
| 0902.0431_E01133 | 191  | ■ 1.000 | \varphi_2(\alpha)(D,\phi,\boldsymbol{X},\boldsymbol{Y})=\left(D,\alpha\phi\alpha^{-1},\alpha\boldsymbol{X},{}^t\alpha^{-1}\boldsymbol{Y}\right).                                                                                                                                                                                                                                | $\varphi_2(\alpha)(D, \phi, \boldsymbol{X}, \boldsymbol{Y}) = (D, \alpha\phi\alpha^{-1}, \alpha\boldsymbol{X}, {}^t\alpha^{-1}\boldsymbol{Y}).$                                                                                                                                                                                    | $\varphi_2(\alpha)(D, \phi, \boldsymbol{X}, \boldsymbol{Y}) = (D, \alpha\phi\alpha^{-1}, \alpha\boldsymbol{X}, {}^t\alpha^{-1}\boldsymbol{Y}).$                                                                                                                                               |
| 0902.0431_E01135 | 192  | ■ 1.000 | \begin{aligned} B_6\left(\alpha\phi_1\alpha^{-1},\tau^t\left(\alpha\phi_2\alpha^{-1}\right)\tau\right)&=B_6\left(\alpha\phi_1\alpha^{-1},\alpha\tau^t\phi_2\tau\alpha^{-1}\right)=B_6\left(\phi_1,\tau^t\phi_2\tau\right),\\ \left\langle\alpha\boldsymbol{X},{}^t\alpha^{-1}\boldsymbol{Y}\right\rangle&=\left\langle\boldsymbol{X},\boldsymbol{Y}\right\rangle, \end{aligned} | $B_6\left(\alpha\phi_1\alpha^{-1},\tau^t\left(\alpha\phi_2\alpha^{-1}\right)\tau\right)=B_6\left(\alpha\phi_1\alpha^{-1},\alpha\tau^t\phi_2\tau\alpha^{-1}\right)=B_6\left(\phi_1,\tau^t\phi_2\tau\right),$<br>$\langle\alpha\boldsymbol{X}, {}^t\alpha^{-1}\boldsymbol{Y}\rangle = \langle\boldsymbol{X}, \boldsymbol{Y}\rangle,$ | $B_6(\alpha\phi_1\alpha^{-1}, \tau^t(\alpha\phi_2\alpha^{-1})\tau) = B_6(\alpha\phi_1\alpha^{-1}, \alpha\tau^t\phi_2\tau\alpha^{-1}) = B_6(\phi_1, \tau^t\phi_2\tau),$<br>$\langle\alpha\boldsymbol{X}, {}^t\alpha^{-1}\boldsymbol{Y}\rangle = \langle\boldsymbol{X}, \boldsymbol{Y}\rangle,$ |
| 0902.0431_E01139 | 192  | ■ 1.000 | (\boldsymbol{*}\,\boldsymbol{u},\boldsymbol{v})=(\boldsymbol{u}\wedge\boldsymbol{v},\boldsymbol{e}_1\wedge\cdots\wedge\boldsymbol{e}_n),\quad\boldsymbol{v}\in\Lambda^{n-k}(C^m).                                                                                                                                                                                               | $(\boldsymbol{*}\boldsymbol{u}, \boldsymbol{v}) = (\boldsymbol{u} \wedge \boldsymbol{v}, \boldsymbol{e}_1 \wedge \cdots \wedge \boldsymbol{e}_n), \quad \boldsymbol{v} \in \Lambda^{n-k}(C^m).$                                                                                                                                    | $(\boldsymbol{*}\boldsymbol{u}, \boldsymbol{v}) = (\boldsymbol{u} \wedge \boldsymbol{v}, \boldsymbol{e}_1 \wedge \cdots \wedge \boldsymbol{e}_n), \quad \boldsymbol{v} \in \Lambda^{n-k}(C^m).$                                                                                               |
| 0902.0431_E01140 | 192  | ■ 1.000 | *:\Lambda^k(C^m)\rightarrow\Lambda^{n-k}(C^m)                                                                                                                                                                                                                                                                                                                                   | $* : \Lambda^k(C^m) \rightarrow \Lambda^{n-k}(C^m)$                                                                                                                                                                                                                                                                                | $* : \Lambda^k(C^m) \rightarrow \Lambda^{n-k}(C^m)$                                                                                                                                                                                                                                           |
| 0902.0431_E01141 | 192  | ■ 1.000 | *^2\boldsymbol{u}=(-1)^{k(n-k)}\boldsymbol{u},\quad\boldsymbol{u}\in\Lambda^k(C^m).                                                                                                                                                                                                                                                                                             | $*^2\boldsymbol{u} = (-1)^{k(n-k)}\boldsymbol{u}, \quad \boldsymbol{u} \in \Lambda^k(C^m).$                                                                                                                                                                                                                                        | $*^2\boldsymbol{u} = (-1)^{k(n-k)}\boldsymbol{u}, \quad \boldsymbol{u} \in \Lambda^k(C^m).$                                                                                                                                                                                                   |

| Identifier       | Page | Conf.   | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Rendered                                                                                                                                                                                                                                                                                                                                                                                                                                   | Image                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|------------------|------|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0902.0431_E01142 | 192  | ■ 1.000 | $A\left(\boldsymbol{x}_{\{1\}}\wedge\cdots\wedge\boldsymbol{x}_{\{k\}}\right)=A\boldsymbol{x}_{\{1\}}\wedge\cdots\wedge A\boldsymbol{x}_{\{k\}},\quad A\mathbb{1}=1.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | $A\left(\boldsymbol{x}_1\wedge\cdots\wedge\boldsymbol{x}_k\right)=A\boldsymbol{x}_1\wedge\cdots\wedge A\boldsymbol{x}_k,\quad A\mathbb{1}=1.$                                                                                                                                                                                                                                                                                              | $A\left(\boldsymbol{x}_1\wedge\cdots\wedge\boldsymbol{x}_k\right)=A\boldsymbol{x}_1\wedge\cdots\wedge A\boldsymbol{x}_k,\quad A\mathbb{1}=1.$                                                                                                                                                                                                                                                                                              |
| 0902.0431_E01143 | 193  | ■ 1.000 | $D\left(\boldsymbol{x}_{\{1\}}\wedge\cdots\wedge\boldsymbol{x}_{\{k\}}\right)=\sum_{j=1}^k\boldsymbol{x}_1\wedge\cdots\wedge D\boldsymbol{x}_j\wedge\cdots\wedge\boldsymbol{x}_k,\quad D\mathbb{1}=0.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | $D\left(\boldsymbol{x}_1\wedge\cdots\wedge\boldsymbol{x}_k\right)=\sum_{j=1}^k\boldsymbol{x}_1\wedge\cdots\wedge D\boldsymbol{x}_j\wedge\cdots\wedge\boldsymbol{x}_k,\quad D\mathbb{1}=0.$                                                                                                                                                                                                                                                 | $D\left(\boldsymbol{x}_1\wedge\cdots\wedge\boldsymbol{x}_k\right)=\sum_{j=1}^k\boldsymbol{x}_1\wedge\cdots\wedge D\boldsymbol{x}_j\wedge\cdots\wedge\boldsymbol{x}_k,\quad D\mathbb{1}=0.$                                                                                                                                                                                                                                                 |
| 0902.0431_E01146 | 193  | ■ 1.000 | $\left(\left(D_{\{1\}},\boldsymbol{u}_{\{1\}},\boldsymbol{v}_{\{1\}}\right),\left(D_{\{2\}},\boldsymbol{u}_{\{2\}},\boldsymbol{v}_{\{2\}}\right)\right)=\left(D,\boldsymbol{u},\boldsymbol{v}\right),$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | $\left[\left(D_1,\boldsymbol{u}_1,\boldsymbol{v}_1\right),\left(D_2,\boldsymbol{u}_2,\boldsymbol{v}_2\right)\right]=\left(D,\boldsymbol{u},\boldsymbol{v}\right),$                                                                                                                                                                                                                                                                         | $\left[\left(D_1,\boldsymbol{u}_1,\boldsymbol{v}_1\right),\left(D_2,\boldsymbol{u}_2,\boldsymbol{v}_2\right)\right]=\left(D,\boldsymbol{u},\boldsymbol{v}\right),$                                                                                                                                                                                                                                                                         |
| 0902.0431_E01156 | 195  | ■ 1.000 | $\boldsymbol{e}_{\{I\}}=\boldsymbol{e}_{\{i\}}\wedge\boldsymbol{e}_{\{j\}}\wedge\boldsymbol{e}_{\{k\}}\in\Lambda^3\left(C^9\right).$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | $\boldsymbol{e}_I=\boldsymbol{e}_i\wedge\boldsymbol{e}_j\wedge\boldsymbol{e}_k\in\Lambda^3\left(C^9\right).$                                                                                                                                                                                                                                                                                                                               | $\boldsymbol{e}_I=\boldsymbol{e}_i\wedge\boldsymbol{e}_j\wedge\boldsymbol{e}_k\in\Lambda^3(C^9).$                                                                                                                                                                                                                                                                                                                                          |
| 0902.0431_E01158 | 195  | ■ 1.000 | $D=\frac{1}{3}\left(E_{ii}+E_{jj}+E_{kk}\right)-E_{ll},\quad l\neq i,j,k$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | $D=\frac{1}{3}\left(E_{ii}+E_{jj}+E_{kk}\right)-E_{ll},\quad l\neq i,j,k$                                                                                                                                                                                                                                                                                                                                                                  | $D=\frac{1}{3}\left(E_{ii}+E_{jj}+E_{kk}\right)-E_{ll},\quad l\neq i,j,k.$                                                                                                                                                                                                                                                                                                                                                                 |
| 0902.0431_E01159 | 195  | ■ 1.000 | $\begin{aligned} &\left(\boldsymbol{0},\boldsymbol{e}_{\{i\}}\wedge\boldsymbol{e}_{\{j\}}\wedge\boldsymbol{e}_{\{k\}},\boldsymbol{0}\right)=\left[\left(D,\boldsymbol{0},\boldsymbol{0}\right),\left(\boldsymbol{0},\boldsymbol{e}_{\{i\}}\wedge\boldsymbol{e}_{\{j\}}\wedge\boldsymbol{e}_{\{k\}},\boldsymbol{0}\right)\right]\in\mathfrak{a},\\ &\left(\boldsymbol{0},\boldsymbol{0},\boldsymbol{e}_{\{i\}}\wedge\boldsymbol{e}_{\{j\}}\wedge\boldsymbol{e}_{\{k\}}\right)=\left[\left(D,\boldsymbol{0},\boldsymbol{0}\right),\left(\boldsymbol{0},\boldsymbol{0},-\boldsymbol{e}_{\{i\}}\wedge\boldsymbol{e}_{\{j\}}\wedge\boldsymbol{e}_{\{k\}}\right)\right]\in\mathfrak{a}. \end{aligned}$ | $\begin{aligned} (0,\boldsymbol{e}_i\wedge\boldsymbol{e}_j\wedge\boldsymbol{e}_k,0) &= \left[\left(D,0,0\right),\left(0,\boldsymbol{e}_i\wedge\boldsymbol{e}_j\wedge\boldsymbol{e}_k,0\right)\right]\in\mathfrak{a},\\ (0,0,\boldsymbol{e}_i\wedge\boldsymbol{e}_j\wedge\boldsymbol{e}_k) &= \left[\left(D,0,0\right),\left(0,0,-\boldsymbol{e}_i\wedge\boldsymbol{e}_j\wedge\boldsymbol{e}_k\right)\right]\in\mathfrak{a}. \end{aligned}$ | $\begin{aligned} (0,\boldsymbol{e}_i\wedge\boldsymbol{e}_j\wedge\boldsymbol{e}_k,0) &= \left[\left(D,0,0\right),\left(0,\boldsymbol{e}_i\wedge\boldsymbol{e}_j\wedge\boldsymbol{e}_k,0\right)\right]\in\mathfrak{a},\\ (0,0,\boldsymbol{e}_i\wedge\boldsymbol{e}_j\wedge\boldsymbol{e}_k) &= \left[\left(D,0,0\right),\left(0,0,-\boldsymbol{e}_i\wedge\boldsymbol{e}_j\wedge\boldsymbol{e}_k\right)\right]\in\mathfrak{a}. \end{aligned}$ |
| 0902.0431_E01161 | 196  | ■ 1.000 | $B_{\{8\}}\left(\left(D_{\{1\}},\boldsymbol{u}_{\{1\}},\boldsymbol{v}_{\{1\}}\right),\left(D_{\{2\}},\boldsymbol{u}_{\{2\}},\boldsymbol{v}_{\{2\}}\right)\right)=60\left(\operatorname{tr}\left(D_{\{1\}}D_{\{2\}}\right)+\left(\boldsymbol{u}_1,\boldsymbol{v}_2\right)+\left(\boldsymbol{u}_2,\boldsymbol{v}_1\right)\right).$                                                                                                                                                                                                                                                                                                                                                                 | $B_8\left(\left(D_1,\boldsymbol{u}_1,\boldsymbol{v}_1\right),\left(D_2,\boldsymbol{u}_2,\boldsymbol{v}_2\right)\right)=60\left(\operatorname{tr}\left(D_1D_2\right)+\left(\boldsymbol{u}_1,\boldsymbol{v}_2\right)+\left(\boldsymbol{u}_2,\boldsymbol{v}_1\right)\right).$                                                                                                                                                                 | $B_8\left(\left(D_1,\boldsymbol{u}_1,\boldsymbol{v}_1\right),\left(D_2,\boldsymbol{u}_2,\boldsymbol{v}_2\right)\right)=60\left(\operatorname{tr}\left(D_1D_2\right)+\left(\boldsymbol{u}_1,\boldsymbol{v}_2\right)+\left(\boldsymbol{u}_2,\boldsymbol{v}_1\right)\right).$                                                                                                                                                                 |
| 0902.0431_E01162 | 196  | ■ 1.000 | $B\left(\left(D_{\{1\}},\boldsymbol{u}_{\{1\}},\boldsymbol{v}_{\{1\}}\right),\left(D_{\{2\}},\boldsymbol{u}_{\{2\}},\boldsymbol{v}_{\{2\}}\right)\right)=\operatorname{tr}\left(D_{\{1\}}D_{\{2\}}\right)+\left(\boldsymbol{u}_1,\boldsymbol{v}_2\right)+\left(\boldsymbol{u}_2,\boldsymbol{v}_1\right).$                                                                                                                                                                                                                                                                                                                                                                                        | $B\left(\left(D_1,\boldsymbol{u}_1,\boldsymbol{v}_1\right),\left(D_2,\boldsymbol{u}_2,\boldsymbol{v}_2\right)\right)=\operatorname{tr}\left(D_1D_2\right)+\left(\boldsymbol{u}_1,\boldsymbol{v}_2\right)+\left(\boldsymbol{u}_2,\boldsymbol{v}_1\right).$                                                                                                                                                                                  | $B\left(\left(D_1,\boldsymbol{u}_1,\boldsymbol{v}_1\right),\left(D_2,\boldsymbol{u}_2,\boldsymbol{v}_2\right)\right)=\operatorname{tr}\left(D_1D_2\right)+\left(\boldsymbol{u}_1,\boldsymbol{v}_2\right)+\left(\boldsymbol{u}_2,\boldsymbol{v}_1\right).$                                                                                                                                                                                  |
| 0902.0431_E01163 | 196  | ■ 1.000 | $B_{\{8\}}(R,R)=120,\quad B(R,R)=2.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | $B_8(R,R)=120,\quad B(R,R)=2.$                                                                                                                                                                                                                                                                                                                                                                                                             | $B_8(R,R)=120,\quad B(R,R)=2.$                                                                                                                                                                                                                                                                                                                                                                                                             |
| 0902.0431_E01164 | 196  | ■ 1.000 | $\tau\widetilde{\lambda}(D,\boldsymbol{u},\boldsymbol{v})=\left(-\tau^tD,-\tau\boldsymbol{v},-\tau\boldsymbol{u}\right).$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | $\tau\widetilde{\lambda}(D,\boldsymbol{u},\boldsymbol{v})=\left(-\tau^tD,-\tau\boldsymbol{v},-\tau\boldsymbol{u}\right).$                                                                                                                                                                                                                                                                                                                  | $\tau\widetilde{\lambda}(D,\boldsymbol{u},\boldsymbol{v})=\left(-\tau^tD,-\tau\boldsymbol{v},-\tau\boldsymbol{u}\right).$                                                                                                                                                                                                                                                                                                                  |
| 0902.0431_E01166 | 196  | ■ 1.000 | $\left\langle R_1,R_2\right\rangle =60\left(\operatorname{tr}\left(D_1\tau^tD_2\right)+\left(\boldsymbol{u}_1,\tau\boldsymbol{u}_2\right)+\left(\boldsymbol{v}_2,\tau\boldsymbol{v}_1\right)\right).$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | $\left\langle R_1,R_2\right\rangle =60\left(\operatorname{tr}\left(D_1\tau^tD_2\right)+\left(\boldsymbol{u}_1,\tau\boldsymbol{u}_2\right)+\left(\boldsymbol{v}_2,\tau\boldsymbol{v}_1\right)\right).$                                                                                                                                                                                                                                      | $\left\langle R_1,R_2\right\rangle =60\left(\operatorname{tr}\left(D_1\tau^tD_2\right)+\left(\boldsymbol{u}_1,\tau\boldsymbol{u}_2\right)+\left(\boldsymbol{v}_2,\tau\boldsymbol{v}_1\right)\right).$                                                                                                                                                                                                                                      |
| 0902.0431_E01167 | 196  | ■ 1.000 | $E_{\{8\}}=\left\{\alpha\in\operatorname{Aut}\left(\mathfrak{e}_{\{8\}}^C\right)\mid\left\langle\alpha R_1,\alpha R_2\right\rangle=\left\langle R_1,R_2\right\rangle\right\}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | $E_8=\left\{\alpha\in\operatorname{Aut}\left(\mathfrak{e}_8^C\right)\mid\left\langle\alpha R_1,\alpha R_2\right\rangle=\left\langle R_1,R_2\right\rangle\right\}$                                                                                                                                                                                                                                                                          | $E_8=\left\{\alpha\in\operatorname{Aut}(\mathfrak{e}_8^C)\mid\left\langle\alpha R_1,\alpha R_2\right\rangle=\left\langle R_1,R_2\right\rangle\right\}$                                                                                                                                                                                                                                                                                     |

| Identifier           | Page | Conf.   | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Rendered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Image                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|----------------------|------|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0902.0431_E01169     | 197  | ■ 1.000 | $\left(E_8\right)^{\wedge\{w_3\}}=\left\{\alpha\in E_8\mid w_3\alpha=\alpha w_3\right\}.$                                                                                                                                                                                                                                                                                                                                                                                                                              | $(E_8)^{w_3} = \{\alpha \in E_8 \mid w_3\alpha = \alpha w_3\}.$                                                                                                                                                                                                                                                                                                                                                                                                                                                               | $(E_8)^{w_3} = \{\alpha \in E_8 \mid w_3\alpha = \alpha w_3\}.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 0902.0431_E01170     | 197  | ■ 1.000 | $\varphi(A)(D,\boldsymbol{u},\boldsymbol{v})=\left(A\,D\,A^{-1},\,A\,\boldsymbol{u},\{ \}^{\{t\}}\,A^{-1}\,\boldsymbol{v}\right).$                                                                                                                                                                                                                                                                                                                                                                                     | $\varphi(A)(D,\boldsymbol{u},\boldsymbol{v}) = (ADA^{-1}, Au, {}^tA^{-1}\boldsymbol{v}).$                                                                                                                                                                                                                                                                                                                                                                                                                                     | $\varphi(A)(D,\boldsymbol{u},\boldsymbol{v}) = (ADA^{-1}, Au, {}^tA^{-1}\boldsymbol{v}).$                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 0902.0431_E01177     | 198  | ■ 1.000 | $*(\boldsymbol{a}\wedge\boldsymbol{b}\wedge\boldsymbol{c})+(\boldsymbol{b}\wedge\boldsymbol{c}\wedge\boldsymbol{a})+(\boldsymbol{c}\wedge\boldsymbol{a}\wedge\boldsymbol{b})$                                                                                                                                                                                                                                                                                                                                          | $*(\boldsymbol{a}\wedge(\boldsymbol{b}\wedge\boldsymbol{c})+*\boldsymbol{b}\wedge*(\boldsymbol{c}\wedge\boldsymbol{a})+*\boldsymbol{c}\wedge*(\boldsymbol{a}\wedge\boldsymbol{b}))$                                                                                                                                                                                                                                                                                                                                           | $*(\boldsymbol{a}\wedge*(\boldsymbol{b}\wedge\boldsymbol{c})+*\boldsymbol{b}\wedge*(\boldsymbol{c}\wedge\boldsymbol{a})+*\boldsymbol{c}\wedge*(\boldsymbol{a}\wedge\boldsymbol{b}))$                                                                                                                                                                                                                                                                                                                                           |
| 0902.0431_E01184     | 199  | ■ 1.000 | $((\boldsymbol{x}\times\boldsymbol{y})\boldsymbol{z},\boldsymbol{v})=-(\boldsymbol{x}\wedge\boldsymbol{z},\boldsymbol{y}\wedge\boldsymbol{v})+\frac{4}{5}(\boldsymbol{x},\boldsymbol{y})(\boldsymbol{z},\boldsymbol{v})-(\boldsymbol{y},\boldsymbol{z})(\boldsymbol{x},\boldsymbol{v})-\frac{1}{5}(\boldsymbol{x},\boldsymbol{y})(\boldsymbol{z},\boldsymbol{v}),$                                                                                                                                                     | $((\boldsymbol{x}\times\boldsymbol{y})\boldsymbol{z},\boldsymbol{v}) = -(\boldsymbol{x}\wedge\boldsymbol{z},\boldsymbol{y}\wedge\boldsymbol{v}) + \frac{4}{5}(\boldsymbol{x},\boldsymbol{y})(\boldsymbol{z},\boldsymbol{v}) = (\boldsymbol{y},\boldsymbol{z})(\boldsymbol{x},\boldsymbol{v}) - \frac{1}{5}(\boldsymbol{x},\boldsymbol{y})(\boldsymbol{z},\boldsymbol{v}),$                                                                                                                                                    | $((\boldsymbol{x}\times\boldsymbol{y})\boldsymbol{z},\boldsymbol{v}) = -(\boldsymbol{x}\wedge\boldsymbol{z},\boldsymbol{y}\wedge\boldsymbol{v}) + \frac{4}{5}(\boldsymbol{x},\boldsymbol{y})(\boldsymbol{z},\boldsymbol{v}) = (\boldsymbol{y},\boldsymbol{z})(\boldsymbol{x},\boldsymbol{v}) - \frac{1}{5}(\boldsymbol{x},\boldsymbol{y})(\boldsymbol{z},\boldsymbol{v}),$                                                                                                                                                     |
| 0902.0431_E01185 (i) | 199  | ■ 1.000 | $(\boldsymbol{x}\times\boldsymbol{y})\boldsymbol{z}=(\boldsymbol{y},\boldsymbol{z})\boldsymbol{x}-\frac{1}{5}(\boldsymbol{x},\boldsymbol{y})\boldsymbol{z}.$                                                                                                                                                                                                                                                                                                                                                           | $(\boldsymbol{x}\times\boldsymbol{y})\boldsymbol{z} = (\boldsymbol{y},\boldsymbol{z})\boldsymbol{x} - \frac{1}{5}(\boldsymbol{x},\boldsymbol{y})\boldsymbol{z}.$                                                                                                                                                                                                                                                                                                                                                              | $(\boldsymbol{x}\times\boldsymbol{y})\boldsymbol{z} = (\boldsymbol{y},\boldsymbol{z})\boldsymbol{x} - \frac{1}{5}(\boldsymbol{x},\boldsymbol{y})\boldsymbol{z}.$                                                                                                                                                                                                                                                                                                                                                               |
| 0902.0431_E01191     | 200  | ■ 1.000 | $\begin{aligned} & ((\boldsymbol{a}\times\boldsymbol{b})\boldsymbol{v},\boldsymbol{w})=(\boldsymbol{a}\wedge\boldsymbol{v},\boldsymbol{b}\wedge\boldsymbol{w})-\frac{3}{5}(\boldsymbol{a},\boldsymbol{b})(\boldsymbol{v},\boldsymbol{w}) \\ & =-(\boldsymbol{a}_1,\boldsymbol{w})(\boldsymbol{x}\wedge\boldsymbol{a}_2,\boldsymbol{b})+(\boldsymbol{a}_2,\boldsymbol{w})(\boldsymbol{x}\wedge\boldsymbol{a}_1,\boldsymbol{b})+\frac{2}{5}(\boldsymbol{a},\boldsymbol{b})(\boldsymbol{v},\boldsymbol{w}) \end{aligned}$ | $\begin{aligned} ((\boldsymbol{a}\times\boldsymbol{b})\boldsymbol{v},\boldsymbol{w}) &= (\boldsymbol{a}\wedge\boldsymbol{v},\boldsymbol{b}\wedge\boldsymbol{w}) - \frac{3}{5}(\boldsymbol{a},\boldsymbol{b})(\boldsymbol{v},\boldsymbol{w}) \\ &= -(\boldsymbol{a}_1,\boldsymbol{w})(\boldsymbol{x}\wedge\boldsymbol{a}_2,\boldsymbol{b}) + (\boldsymbol{a}_2,\boldsymbol{w})(\boldsymbol{x}\wedge\boldsymbol{a}_1,\boldsymbol{b}) + \frac{2}{5}(\boldsymbol{a},\boldsymbol{b})(\boldsymbol{v},\boldsymbol{w}) \end{aligned}$ | $\begin{aligned} ((\boldsymbol{a}\times\boldsymbol{b})\boldsymbol{v},\boldsymbol{w}) &= (\boldsymbol{a}\wedge\boldsymbol{v},\boldsymbol{b}\wedge\boldsymbol{w}) - \frac{3}{5}(\boldsymbol{a},\boldsymbol{b})(\boldsymbol{v},\boldsymbol{w}) \\ &= -(\boldsymbol{a}_1,\boldsymbol{w})(\boldsymbol{x}\wedge\boldsymbol{a}_2,\boldsymbol{b}) + (\boldsymbol{a}_2,\boldsymbol{w})(\boldsymbol{x}\wedge\boldsymbol{a}_1,\boldsymbol{b}) + \frac{2}{5}(\boldsymbol{a},\boldsymbol{b})(\boldsymbol{v},\boldsymbol{w}), \end{aligned}$ |
| 0902.0431_E01194     | 201  | ■ 1.000 | $(\boldsymbol{x}\times\boldsymbol{y})\boldsymbol{a}=(\boldsymbol{y},\boldsymbol{a}_1)\boldsymbol{x}\wedge\boldsymbol{a}_2-(\boldsymbol{y},\boldsymbol{a}_2)\boldsymbol{x}\wedge\boldsymbol{a}_1-\frac{2}{5}(\boldsymbol{x},\boldsymbol{y})\boldsymbol{a}.$                                                                                                                                                                                                                                                             | $(\boldsymbol{x}\times\boldsymbol{y})\boldsymbol{a} = (\boldsymbol{y},\boldsymbol{a}_1)\boldsymbol{x}\wedge\boldsymbol{a}_2 - (\boldsymbol{y},\boldsymbol{a}_2)\boldsymbol{x}\wedge\boldsymbol{a}_1 - \frac{2}{5}(\boldsymbol{x},\boldsymbol{y})\boldsymbol{a}.$                                                                                                                                                                                                                                                              | $(\boldsymbol{x}\times\boldsymbol{y})\boldsymbol{a} = (\boldsymbol{y},\boldsymbol{a}_1)\boldsymbol{x}\wedge\boldsymbol{a}_2 - (\boldsymbol{y},\boldsymbol{a}_2)\boldsymbol{x}\wedge\boldsymbol{a}_1 - \frac{2}{5}(\boldsymbol{x},\boldsymbol{y})\boldsymbol{a}.$                                                                                                                                                                                                                                                               |
| 0902.0431_E01198     | 201  | ■ 1.000 | $\left[(C,D),\left(E_{45},0\right)\right]=\left(5E_{45},0\right)\in\mathfrak{g}_{01}\cap\mathfrak{a},$                                                                                                                                                                                                                                                                                                                                                                                                                 | $[(C,D),(E_{45},0)] = (5E_{45},0) \in \mathfrak{g}_{01} \cap \mathfrak{a},$                                                                                                                                                                                                                                                                                                                                                                                                                                                   | $[(C,D),(E_{45},0)] = (5E_{45},0) \in \mathfrak{g}_{01} \cap \mathfrak{a},$                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 0902.0431_E01199     | 202  | ■ 1.000 | $\mathfrak{a}\supset\left[\mathfrak{g}_1,\mathfrak{g}_{-1}\right]\ni\left[\boldsymbol{e}_1\otimes\left(\boldsymbol{e}_1\wedge\boldsymbol{e}_2\right),\boldsymbol{e}_1\otimes\left(\boldsymbol{e}_1\wedge\boldsymbol{e}_3\right)\right]=\left(0,-E_{23}\right),$                                                                                                                                                                                                                                                        | $\mathfrak{a} \supset [\mathfrak{g}_1, \mathfrak{g}_{-1}] \ni [\boldsymbol{e}_1 \otimes (\boldsymbol{e}_1 \wedge \boldsymbol{e}_2), \boldsymbol{e}_1 \otimes (\boldsymbol{e}_1 \wedge \boldsymbol{e}_3)] = (0, -E_{23}),$                                                                                                                                                                                                                                                                                                     | $\mathfrak{a} \supset [\mathfrak{g}_1, \mathfrak{g}_{-1}] \ni [\boldsymbol{e}_1 \otimes (\boldsymbol{e}_1 \wedge \boldsymbol{e}_2), \boldsymbol{e}_1 \otimes (\boldsymbol{e}_1 \wedge \boldsymbol{e}_3)] = (0, -E_{23}),$                                                                                                                                                                                                                                                                                                      |

| Identifier       | Page | Conf.   | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Rendered                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Image                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|------------------|------|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0902.0431_E01200 | 202  | ■ 1.000 | $\operatorname{ad}(T) \operatorname{ad}(S_{1523}) \operatorname{ad}(S_{1415}) \operatorname{ad}(S_{1314}) \operatorname{ad}(S_{1213}) R = (-E_{12}, 0) \in \mathfrak{g}_{01} \cap \mathfrak{a}.$                                                                                                                                                                                                                                                                                                             | $\operatorname{ad}(T) \operatorname{ad}(S_{1523}) \operatorname{ad}(S_{1415}) \operatorname{ad}(S_{1314}) \operatorname{ad}(S_{1213}) R = (-E_{12}, 0) \in \mathfrak{g}_{01} \cap \mathfrak{a}.$                                                                                                                                                                                                                                                                                                             | $\operatorname{ad}(T) \operatorname{ad}(S_{1523}) \operatorname{ad}(S_{1415}) \operatorname{ad}(S_{1314}) \operatorname{ad}(S_{1213}) R = (-E_{12}, 0) \in \mathfrak{g}_{01} \cap \mathfrak{a}.$                                                                                                                                                                                                                                                                                                             |
| 0902.0431_E01203 | 202  | ■ 1.000 | $B_8(R, R) = 120, \quad B(R, R) = 2.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | $B_8(R, R) = 120, \quad B(R, R) = 2.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | $B_8(R, R) = 120, \quad B(R, R) = 2.$                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 0902.0431_E01204 | 202  | ■ 1.000 | $\begin{aligned} & \tau \widetilde{\lambda}(C, D, \mathbf{x} \otimes \mathbf{a}, \mathbf{b} \otimes \mathbf{y}, \mathbf{c} \otimes \mathbf{z}, \mathbf{w} \otimes \mathbf{d}) \\ &= (-\tau^t C, -\tau^t D, \tau \mathbf{w} \otimes \tau \mathbf{d}, \tau \mathbf{c} \otimes \tau \mathbf{z}, \tau \mathbf{b} \otimes \tau \mathbf{y}, \tau \mathbf{x} \otimes \tau \mathbf{a}). \end{aligned}$                                                                                                               | $\tau \widetilde{\lambda}(C, D, \mathbf{x} \otimes \mathbf{a}, \mathbf{b} \otimes \mathbf{y}, \mathbf{c} \otimes \mathbf{z}, \mathbf{w} \otimes \mathbf{d}) \\ = (-\tau^t C, -\tau^t D, \tau \mathbf{w} \otimes \tau \mathbf{d}, \tau \mathbf{c} \otimes \tau \mathbf{z}, \tau \mathbf{b} \otimes \tau \mathbf{y}, \tau \mathbf{x} \otimes \tau \mathbf{a}).$                                                                                                                                                | $\tau \widetilde{\lambda}(C, D, \mathbf{x} \otimes \mathbf{a}, \mathbf{b} \otimes \mathbf{y}, \mathbf{c} \otimes \mathbf{z}, \mathbf{w} \otimes \mathbf{d}) \\ = (-\tau^t C, -\tau^t D, \tau \mathbf{w} \otimes \tau \mathbf{d}, \tau \mathbf{c} \otimes \tau \mathbf{z}, \tau \mathbf{b} \otimes \tau \mathbf{y}, \tau \mathbf{x} \otimes \tau \mathbf{a}).$                                                                                                                                                |
| 0902.0431_E01205 | 203  | ■ 1.000 | $\langle R_1, R_2 \rangle = -B_8(R_1, \tau \widetilde{\lambda} R_2)$                                                                                                                                                                                                                                                                                                                                                                                                                                         | $\langle R_1, R_2 \rangle = -B_8(R_1, \tau \widetilde{\lambda} R_2)$                                                                                                                                                                                                                                                                                                                                                                                                                                         | $\langle R_1, R_2 \rangle = -B_8(R_1, \tau \widetilde{\lambda} R_2).$                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 0902.0431_E01207 | 203  | ■ 1.000 | $E_8 = \{ \alpha \in \operatorname{Aut}(\mathfrak{e}_8^C) \mid \langle \alpha R_1, \alpha R_2 \rangle = \langle R_1, R_2 \rangle \}$                                                                                                                                                                                                                                                                                                                                                                         | $E_8 = \{ \alpha \in \operatorname{Aut}(\mathfrak{e}_8^C) \mid \langle \alpha R_1, \alpha R_2 \rangle = \langle R_1, R_2 \rangle \}$                                                                                                                                                                                                                                                                                                                                                                         | $E_8 = \{ \alpha \in \operatorname{Aut}(\mathfrak{e}_8^C) \mid \langle \alpha R_1, \alpha R_2 \rangle = \langle R_1, R_2 \rangle \}$                                                                                                                                                                                                                                                                                                                                                                         |
| 0902.0431_E01208 | 203  | ■ 1.000 | $z_5(C, D, g_1, g_2, g_{-2}, g_{-1}) = (C, D, \zeta(g_1), \zeta^2(g_2), \zeta^3(g_{-2}), \zeta^4(g_{-1})).$                                                                                                                                                                                                                                                                                                                                                                                                  | $z_5(C, D, g_1, g_2, g_{-2}, g_{-1}) = (C, D, \zeta(g_1), \zeta^2(g_2), \zeta^3(g_{-2}), \zeta^4(g_{-1})).$                                                                                                                                                                                                                                                                                                                                                                                                  | $z_5(C, D, g_1, g_2, g_{-2}, g_{-1}) = (C, D, \zeta(g_1), \zeta^2(g_2), \zeta^3(g_{-2}), \zeta^4(g_{-1})).$                                                                                                                                                                                                                                                                                                                                                                                                  |
| 0902.0431_E01209 | 203  | ■ 1.000 | $(E_8)^{z_5} = \{ \alpha \in E_8 \mid z_5 \alpha = \alpha z_5 \}.$                                                                                                                                                                                                                                                                                                                                                                                                                                           | $(E_8)^{z_5} = \{ \alpha \in E_8 \mid z_5 \alpha = \alpha z_5 \}.$                                                                                                                                                                                                                                                                                                                                                                                                                                           | $(E_8)^{z_5} = \{ \alpha \in E_8 \mid z_5 \alpha = \alpha z_5 \}.$                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 0902.0431_E01212 | 203  | ■ 1.000 | $\langle \varphi(A) R_1, \varphi_1(A) R_2 \rangle = \langle R_1, R_2 \rangle.$                                                                                                                                                                                                                                                                                                                                                                                                                               | $\langle \varphi(A) R_1, \varphi_1(A) R_2 \rangle = \langle R_1, R_2 \rangle.$                                                                                                                                                                                                                                                                                                                                                                                                                               | $\langle \varphi(A) R_1, \varphi_1(A) R_2 \rangle = \langle R_1, R_2 \rangle.$                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 0902.0431_E01213 | 204  | ■ 1.000 | $\varphi(A, B) = \varphi_1(A) \varphi_2(B).$                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | $\varphi(A, B) = \varphi_1(A) \varphi_2(B).$                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | $\varphi(A, B) = \varphi_1(A) \varphi_2(B).$                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 0902.0431_E01215 | 204  | ■ 1.000 | $\begin{aligned} & \mathfrak{e}_{8(8)} = \mathfrak{e}_{7(7)} \oplus \mathfrak{P}' \oplus \mathfrak{P}' \oplus \mathbf{R} \oplus \mathbf{R} \oplus \mathbf{R}, \quad \left( \text{where } \mathfrak{e}_{7(7)} = (\mathfrak{e}_7^C)^{\tau\gamma} \right), \\ & \mathfrak{e}_{8(-24)} = \mathfrak{e}_{7(-25)} \oplus \mathfrak{P} \oplus \mathfrak{P} \oplus \mathbf{R} \oplus \mathbf{R} \oplus \mathbf{R}, \quad \left( \text{where } \mathfrak{e}_{7(-25)} = (\mathfrak{e}_7^C)^\tau \right). \end{aligned}$ | $\begin{aligned} & \mathfrak{e}_{8(8)} = \mathfrak{e}_{7(7)} \oplus \mathfrak{P}' \oplus \mathfrak{P}' \oplus \mathbf{R} \oplus \mathbf{R} \oplus \mathbf{R}, \quad \left( \text{where } \mathfrak{e}_{7(7)} = (\mathfrak{e}_7^C)^{\tau\gamma} \right), \\ & \mathfrak{e}_{8(-24)} = \mathfrak{e}_{7(-25)} \oplus \mathfrak{P} \oplus \mathfrak{P} \oplus \mathbf{R} \oplus \mathbf{R} \oplus \mathbf{R}, \quad \left( \text{where } \mathfrak{e}_{7(-25)} = (\mathfrak{e}_7^C)^\tau \right). \end{aligned}$ | $\begin{aligned} & \mathfrak{e}_{8(8)} = \mathfrak{e}_{7(7)} \oplus \mathfrak{P}' \oplus \mathfrak{P}' \oplus \mathbf{R} \oplus \mathbf{R} \oplus \mathbf{R}, \quad \left( \text{where } \mathfrak{e}_{7(7)} = (\mathfrak{e}_7^C)^{\tau\gamma} \right), \\ & \mathfrak{e}_{8(-24)} = \mathfrak{e}_{7(-25)} \oplus \mathfrak{P} \oplus \mathfrak{P} \oplus \mathbf{R} \oplus \mathbf{R} \oplus \mathbf{R}, \quad \left( \text{where } \mathfrak{e}_{7(-25)} = (\mathfrak{e}_7^C)^\tau \right). \end{aligned}$ |
| 0902.0431_E01219 | 204  | ■ 1.000 | $z(E_{8(8)}) = \{1\}, \quad z(E_{8(-24)}) = \{1\}.$                                                                                                                                                                                                                                                                                                                                                                                                                                                          | $z(E_{8(8)}) = \{1\}, \quad z(E_{8(-24)}) = \{1\}.$                                                                                                                                                                                                                                                                                                                                                                                                                                                          | $z(E_{8(8)}) = \{1\}, \quad z(E_{8(-24)}) = \{1\}.$                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

| Identifier       | Page | Conf. | LaTeX source | Rendered | Image                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|------------------|------|-------|--------------|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0902.0431_E00341 | 55   | —     | —            | —        | $\begin{aligned} \lambda_0 - \lambda_1 &= \alpha_1 \\ \lambda_0 - \lambda_2 &= \alpha_1 + \alpha_2 \\ -\lambda_0 + \lambda_3 &= \alpha_2 + 2\alpha_3 + 2\alpha_4 \\ \lambda_1 - \lambda_2 &= \alpha_2 \\ -\lambda_1 + \lambda_3 &= \alpha_1 + \alpha_2 + 2\alpha_3 + 2\alpha_4 \\ -\lambda_2 + \lambda_3 &= \alpha_1 + 2\alpha_2 + 2\alpha_3 + 2\alpha_4 \\ \lambda_0 + \lambda_1 &= \alpha_1 + 2\alpha_2 + 2\alpha_3 \\ \lambda_0 + \lambda_2 &= \alpha_1 + \alpha_2 + 2\alpha_3 \\ \lambda_0 + \lambda_3 &= 2\alpha_1 + 3\alpha_2 + 4\alpha_3 + 2\alpha_4 \\ \lambda_1 + \lambda_2 &= \alpha_2 + 2\alpha_3 \\ \lambda_1 + \lambda_3 &= \alpha_1 + 3\alpha_2 + 4\alpha_3 + 2\alpha_4 \\ \lambda_2 + \lambda_3 &= \alpha_1 + 2\alpha_2 + 4\alpha_3 + 4\alpha_4, \\ \frac{1}{2}(\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3) &= \alpha_1 + 2\alpha_2 + 3\alpha_3 + \alpha_4 \\ \frac{1}{2}(-\lambda_0 - \lambda_1 - \lambda_2 + \lambda_3) &= \alpha_4 \\ \frac{1}{2}(\lambda_0 + \lambda_1 - \lambda_2 + \lambda_3) &= \alpha_1 + 2\alpha_2 + 2\alpha_3 + \alpha_4 \\ \frac{1}{2}(\lambda_0 - \lambda_1 + \lambda_2 + \lambda_3) &= \alpha_1 + \alpha_2 + 2\alpha_3 + \alpha_4 \\ \frac{1}{2}(-\lambda_0 - \lambda_1 + \lambda_2 + \lambda_3) &= \alpha_3 + \alpha_4 \\ \frac{1}{2}(-\lambda_0 + \lambda_1 - \lambda_2 + \lambda_3) &= \alpha_2 + \alpha_3 + \alpha_4 \\ \frac{1}{2}(\lambda_0 - \lambda_1 - \lambda_2 + \lambda_3) &= \alpha_1 + \alpha_2 + \alpha_3 + \alpha_4 \\ \frac{1}{2}(-\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3) &= \alpha_2 + 2\alpha_3 + \alpha_4. \end{aligned}$ |

| Identifier       | Page | Conf. | LaTeX source | Rendered | Image                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|------------------|------|-------|--------------|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0902.0431_EQ0758 | 126  | —     | —            | —        | $\begin{aligned} \frac{1}{2}(-\lambda_0 - \lambda_1 + \lambda_2 - \lambda_3) + \frac{1}{2}(\mu_3 - \mu_1) &= 0 & 0 & 1 & 1 & 0 & 0 & 0 \\ \frac{1}{2}(\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3) + \frac{1}{2}(\mu_3 - \mu_1) &= 1 & 2 & 3 & 3 & 2 & 1 & 1 \\ \frac{1}{2}(-\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3) + \frac{1}{2}(\mu_3 - \mu_1) &= 0 & 1 & 2 & 3 & 2 & 1 & 1 \\ \frac{1}{2}(\lambda_0 - \lambda_1 + \lambda_2 + \lambda_3) + \frac{1}{2}(\mu_3 - \mu_1) &= 1 & 1 & 2 & 3 & 2 & 1 & 1 \\ \frac{1}{2}(\lambda_0 + \lambda_1 - \lambda_2 + \lambda_3) + \frac{1}{2}(\mu_3 - \mu_1) &= 1 & 2 & 2 & 3 & 2 & 1 & 1 \\ \frac{1}{2}(-\lambda_0 - \lambda_1 - \lambda_2 + \lambda_3) + \frac{1}{2}(\mu_3 - \mu_1) &= 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ \frac{1}{2}(\lambda_0 - \lambda_1 - \lambda_2 - \lambda_3) + \frac{1}{2}(\mu_3 - \mu_1) &= 1 & 1 & 1 & 1 & 0 & 0 & 0 \\ \frac{1}{2}(-\lambda_0 + \lambda_1 - \lambda_2 - \lambda_3) + \frac{1}{2}(\mu_3 - \mu_1) &= 0 & 1 & 1 & 1 & 0 & 0 & 0 \\ \frac{1}{2}(\lambda_0 - \lambda_1 + \lambda_2 - \lambda_3) - \frac{1}{2}(\mu_1 - \mu_2) &= 1 & 1 & 2 & 2 & 1 & 1 & 1 \\ \frac{1}{2}(\lambda_0 - \lambda_1 - \lambda_2 + \lambda_3) - \frac{1}{2}(\mu_1 - \mu_2) &= 1 & 1 & 1 & 2 & 1 & 1 & 1 \\ \frac{1}{2}(\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3) + \frac{1}{2}(\mu_1 - \mu_2) &= 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ \frac{1}{2}(\lambda_0 + \lambda_1 - \lambda_2 - \lambda_3) - \frac{1}{2}(\mu_1 - \mu_2) &= 1 & 2 & 2 & 2 & 1 & 1 & 1 \\ \frac{1}{2}(-\lambda_0 + \lambda_1 - \lambda_2 + \lambda_3) - \frac{1}{2}(\mu_1 - \mu_2) &= 0 & 1 & 1 & 2 & 1 & 1 & 1 \\ \frac{1}{2}(-\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3) - \frac{1}{2}(\mu_1 - \mu_2) &= 0 & 1 & 2 & 2 & 1 & 1 & 1 \\ \frac{1}{2}(\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3) - \frac{1}{2}(\mu_1 - \mu_2) &= 1 & 2 & 3 & 4 & 3 & 2 & 2 \\ \frac{1}{2}(-\lambda_0 - \lambda_1 + \lambda_2 + \lambda_3) - \frac{1}{2}(\mu_1 - \mu_2) &= 0 & 0 & 1 & 2 & 1 & 1 & 1 \\ -\mu_1 - \frac{2}{3}\nu &= 1 & 2 & 3 & 4 & 2 & 1 & 2 \\ \mu_2 + \frac{2}{3}\nu &= 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ -\mu_3 - \frac{2}{3}\nu &= 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{aligned}$ |

| Identifier       | Page | Conf. | LaTeX source | Rendered | Image                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|------------------|------|-------|--------------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0902.0431_E00940 | 156  | —     | —            | —        | $ \begin{aligned} &P = (F_1(1), 0, 0, 0) \\ &\xrightarrow{f} \mathbf{e}_2 \wedge \mathbf{e}_4 \wedge \mathbf{e}_5 + \mathbf{e}_3 \wedge \mathbf{e}_6 \wedge \mathbf{e}_4 \\ &\xrightarrow{D} (D\mathbf{e}_2 \wedge \mathbf{e}_4 \wedge \mathbf{e}_5 + \mathbf{e}_2 \wedge D\mathbf{e}_4 \wedge \mathbf{e}_5 + \mathbf{e}_2 \wedge \mathbf{e}_4 \wedge D\mathbf{e}_5) \\ &\quad + (D\mathbf{e}_3 \wedge \mathbf{e}_6 \wedge \mathbf{e}_4 + \mathbf{e}_3 \wedge D\mathbf{e}_6 \wedge \mathbf{e}_4 + \mathbf{e}_3 \wedge \mathbf{e}_6 \wedge D\mathbf{e}_4) \\ &= \left( b_{12}\mathbf{e}_1 \wedge \mathbf{e}_4 \wedge \mathbf{e}_5 + \left( b_{22} + \frac{\nu}{3} \right) \mathbf{e}_2 \wedge \mathbf{e}_4 \wedge \mathbf{e}_5 - \bar{b}_{23}\mathbf{e}_3 \wedge \mathbf{e}_4 \wedge \mathbf{e}_5 - \bar{l}_{23}\mathbf{e}_6 \wedge \mathbf{e}_4 \wedge \mathbf{e}_5 \right. \\ &\quad + l_{11}\mathbf{e}_2 \wedge \mathbf{e}_1 \wedge \mathbf{e}_5 + l_{31}\mathbf{e}_2 \wedge \mathbf{e}_3 \wedge \mathbf{e}_5 + \left( c_{11} - \frac{\nu}{3} \right) \mathbf{e}_2 \wedge \mathbf{e}_4 \wedge \mathbf{e}_5 - \bar{c}_{13}\mathbf{e}_2 \wedge \mathbf{e}_6 \wedge \mathbf{e}_5 \\ &\quad + l_{12}\mathbf{e}_2 \wedge \mathbf{e}_4 \wedge \mathbf{e}_1 + l_{32}\mathbf{e}_2 \wedge \mathbf{e}_4 \wedge \mathbf{e}_3 + \left( c_{22} - \frac{\nu}{3} \right) \mathbf{e}_3 \wedge \mathbf{e}_6 \wedge \mathbf{e}_4 - \bar{c}_{23}\mathbf{e}_2 \wedge \mathbf{e}_4 \wedge \mathbf{e}_6 \Big) \\ &\quad + \left( b_{13}\mathbf{e}_1 \wedge \mathbf{e}_6 \wedge \mathbf{e}_4 + b_{23}\mathbf{e}_2 \wedge \mathbf{e}_6 \wedge \mathbf{e}_4 + \left( b_{33} + \frac{\nu}{3} \right) \mathbf{e}_3 \wedge \mathbf{e}_6 \wedge \mathbf{e}_4 - \bar{l}_{32}\mathbf{e}_5 \wedge \mathbf{e}_6 \wedge \mathbf{e}_4 \right. \\ &\quad + l_{13}\mathbf{e}_3 \wedge \mathbf{e}_1 \wedge \mathbf{e}_4 + l_{23}\mathbf{e}_3 \wedge \mathbf{e}_2 \wedge \mathbf{e}_4 + c_{23}\mathbf{e}_3 \wedge \mathbf{e}_5 \wedge \mathbf{e}_4 + \left( c_{33} - \frac{\nu}{3} \right) \mathbf{e}_3 \wedge \mathbf{e}_6 \wedge \mathbf{e}_4 \\ &\quad \left. + l_{11}\mathbf{e}_3 \wedge \mathbf{e}_6 \wedge \mathbf{e}_1 + l_{21}\mathbf{e}_3 \wedge \mathbf{e}_6 \wedge \mathbf{e}_2 + \left( c_{11} - \frac{\nu}{3} \right) \mathbf{e}_3 \wedge \mathbf{e}_6 \wedge \mathbf{e}_4 - \bar{c}_{12}\mathbf{e}_3 \wedge \mathbf{e}_6 \wedge \mathbf{e}_5 \right) \\ &\xrightarrow{f^{-1}} \left( \begin{array}{ccc} 0 & h(b_{13}, c_{13}) & 0 \\ * & h(b_{23} + \bar{c}_{23}) & h(b_{22} - c_{33} - \frac{\nu}{3}, -b_{33} + c_{33} + \frac{\nu}{3}) \\ h(\bar{c}_{12}, \bar{b}_{12}) & * & -h(\bar{b}_{23} + c_{23}) \end{array} \right) \\ &\quad \left( \begin{array}{ccc} -h(l_{23} + l_{32}) & h(l_{13}, \bar{l}_{31}) & * \\ * & 0 & -h(l_{11}) \\ h(l_{21}, \bar{l}_{12}) & * & 0 \end{array} \right) \\ &\quad \left( \begin{array}{ccc} 0 & & \\ & & \\ & & -h(\bar{l}_{23} + \bar{l}_{32}) \end{array} \right) \end{aligned} $ |

| Identifier       | Page | Conf. | LaTeX source | Rendered | Image                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|------------------|------|-------|--------------|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0902.0431_E01028 | 172  | —     | —            | —        | $\begin{aligned} \lambda_2 + \frac{1}{2}(\mu_2 - \mu_3) &= 0 & 1 & 2 & 2 & 1 & 1 & 0 & 1 \\ \lambda_3 + \frac{1}{2}(\mu_2 - \mu_3) &= 0 & 1 & 2 & 2 & 1 & 0 & 0 & 1 \\ \lambda_0 - \frac{1}{2}(\mu_2 - \mu_3) &= 2 & 3 & 4 & 3 & 3 & 2 & 1 & 2 \\ \lambda_1 - \frac{1}{2}(\mu_2 - \mu_3) &= 0 & 0 & 0 & 0 & 1 & 1 & 1 & 0 \\ \lambda_2 - \frac{1}{2}(\mu_2 - \mu_3) &= 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 \\ \lambda_3 - \frac{1}{2}(\mu_2 - \mu_3) &= 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ \frac{1}{2}(\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3) + \frac{1}{2}(\mu_3 - \mu_1) &= 1 & 1 & 2 & 2 & 2 & 2 & 1 & 1 \\ \frac{1}{2}(\lambda_0 + \lambda_1 - \lambda_2 + \lambda_3) + \frac{1}{2}(\mu_3 - \mu_1) &= 1 & 1 & 2 & 2 & 2 & 1 & 1 & 1 \\ \frac{1}{2}(\lambda_0 - \lambda_1 + \lambda_2 + \lambda_3) + \frac{1}{2}(\mu_3 - \mu_1) &= 1 & 1 & 2 & 2 & 2 & 1 & 0 & 1 \\ \frac{1}{2}(\lambda_0 - \lambda_1 - \lambda_2 - \lambda_3) + \frac{1}{2}(\mu_3 - \mu_1) &= 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{1}{2}(\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3) - \frac{1}{2}(\mu_3 - \mu_1) &= 1 & 3 & 4 & 3 & 2 & 2 & 1 & 2 \\ \frac{1}{2}(\lambda_0 + \lambda_1 - \lambda_2 + \lambda_3) - \frac{1}{2}(\mu_3 - \mu_1) &= 1 & 3 & 4 & 3 & 2 & 1 & 1 & 2 \\ \frac{1}{2}(\lambda_0 - \lambda_1 + \lambda_2 + \lambda_3) - \frac{1}{2}(\mu_3 - \mu_1) &= 1 & 3 & 4 & 3 & 2 & 1 & 0 & 2 \\ \frac{1}{2}(\lambda_0 - \lambda_1 - \lambda_2 - \lambda_3) - \frac{1}{2}(\mu_3 - \mu_1) &= 1 & 2 & 2 & 1 & 0 & 0 & 0 & 1 \\ \frac{1}{2}(\lambda_0 - \lambda_1 + \lambda_2 - \lambda_3) + \frac{1}{2}(\mu_1 - \mu_2) &= 1 & 2 & 2 & 1 & 1 & 1 & 0 & 1 \\ \frac{1}{2}(\lambda_0 - \lambda_1 - \lambda_2 + \lambda_3) + \frac{1}{2}(\mu_1 - \mu_2) &= 1 & 2 & 2 & 1 & 1 & 0 & 0 & 1 \\ \frac{1}{2}(\lambda_0 + \lambda_1 - \lambda_2 - \lambda_3) + \frac{1}{2}(\mu_1 - \mu_2) &= 1 & 2 & 2 & 1 & 1 & 1 & 1 & 1 \\ \frac{1}{2}(\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3) + \frac{1}{2}(\mu_1 - \mu_2) &= 1 & 3 & 4 & 3 & 3 & 2 & 1 & 2 \\ \frac{1}{2}(\lambda_0 - \lambda_1 + \lambda_2 - \lambda_3) - \frac{1}{2}(\mu_1 - \mu_2) &= 1 & 1 & 2 & 2 & 1 & 1 & 0 & 1 \\ \frac{1}{2}(\lambda_0 - \lambda_1 - \lambda_2 + \lambda_3) - \frac{1}{2}(\mu_1 - \mu_2) &= 1 & 1 & 2 & 2 & 1 & 0 & 0 & 1 \\ \frac{1}{2}(\lambda_0 + \lambda_1 - \lambda_2 - \lambda_3) - \frac{1}{2}(\mu_1 - \mu_2) &= 1 & 1 & 2 & 2 & 1 & 1 & 1 & 1 \\ \frac{1}{2}(\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3) - \frac{1}{2}(\mu_1 - \mu_2) &= 1 & 2 & 4 & 4 & 3 & 2 & 1 & 2 \\ \mu_1 + \frac{2}{3}\nu &= 0 & 1 & 1 & 0 & 0 & 0 & 0 & 1 \\ \mu_2 + \frac{2}{3}\nu &= 0 & 0 & 1 & 1 & 0 & 0 & 0 & 1 \\ -\mu_3 - \frac{2}{3}\nu &= 0 & 1 & 1 & 1 & 0 & 0 & 0 & 0 \end{aligned}$ |